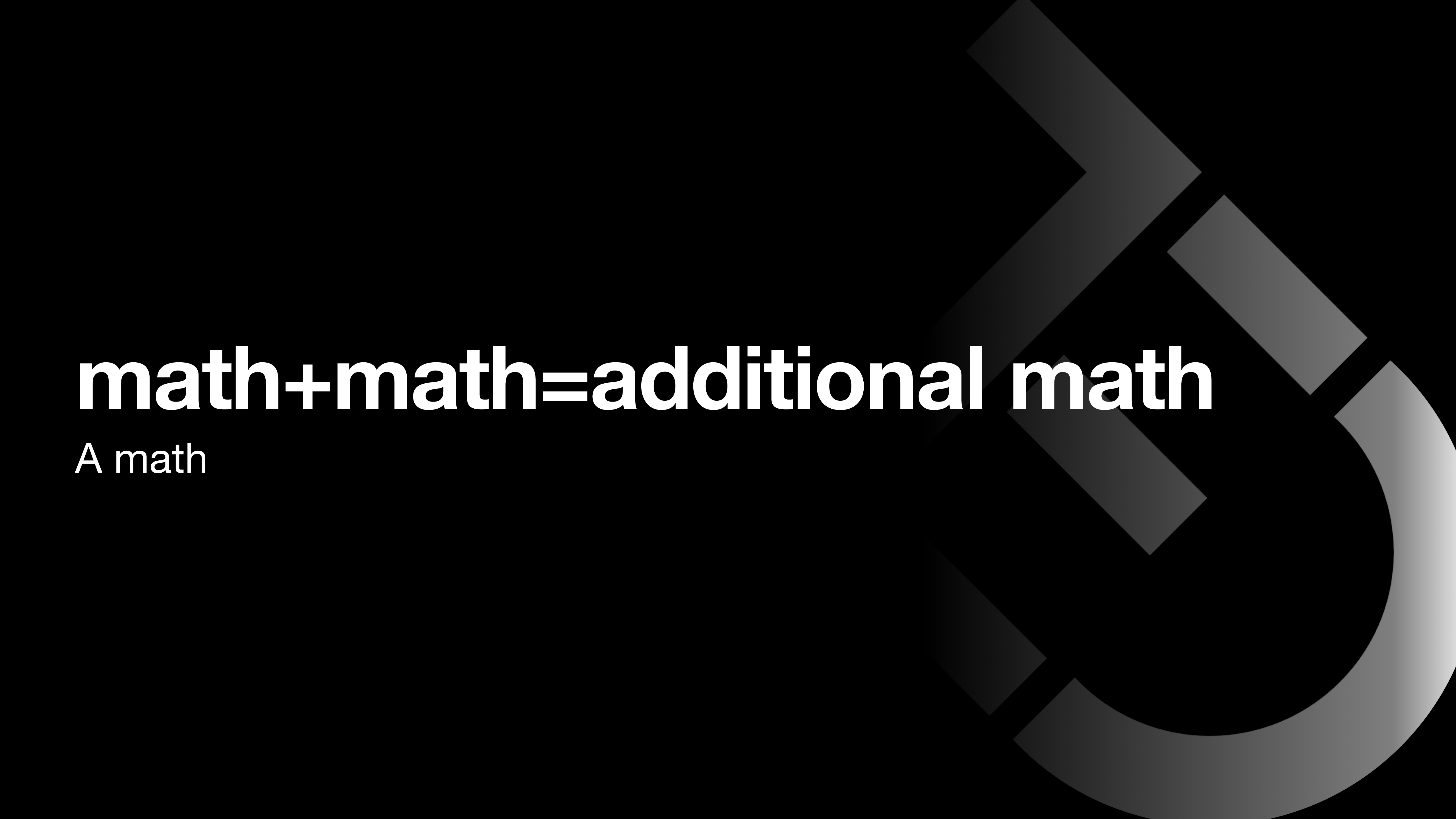




math+math=additional math

A math

Quadratics, Inequalities, & Simultaneous equations

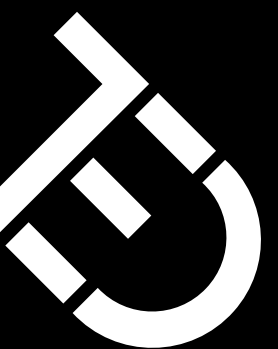


Functions

fun

$$f(x) = x^2$$

$$f(3) = 3^2 \quad f(-5) = (-5)^2 \quad f(0) = (0)^2$$



Quadratic Functions

linearnt

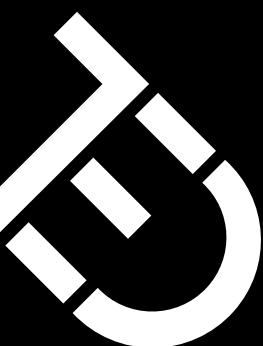
$$y = ax^2 + bx + c$$

Determines y-
intercept

Determines shape of curve
+a gives a min curve, -a
gives a max curve

Determines position
of curve

The lower a is, the wider
The higher a is, the thinner



Completing The Square

Egg

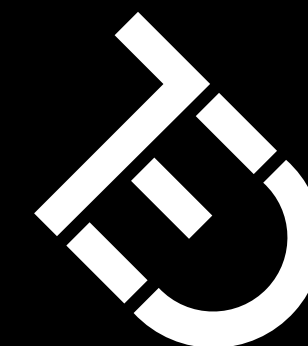
$$\begin{aligned} y &= ax^2 + bx + c \\ &= a\left[x^2 + \frac{b}{a}x + \frac{c}{a}\right] \end{aligned}$$

Factorise a from each term

$$= a \left[\left(x + \frac{b}{2a}\right)^2 - \left(\frac{b}{2a}\right)^2 + \frac{c}{a} \right]$$

$x^2 + 2(x)\left(\frac{b}{2}\right) + \left(\frac{b}{2}\right)^2 - \left(\frac{b}{2}\right)^2$
 $x^2 + bx$

$$= a(x - h)^2 + k, \text{ where } (h, k) \text{ is the turning point}$$



Completing The Square

Egg

$$y = x^2 + 2x + 4$$

$$= \left(x + \frac{2}{2} \right)^2 - \left(\frac{2}{2} \right)^2 + 4$$

sign inside stays the same

sign outside is always negative

$$= \left(x + \frac{2}{2} \right)^2 - 1 + 4$$

$$= (x + 1)^2 + 3$$

$\therefore$ turning point is $(-1, 3)$

$$y = 2x^2 + 5x + 3$$

$$= 2 \left(x^2 + \frac{5}{2}x + \frac{3}{2} \right) \text{ factorise 2 throughout}$$

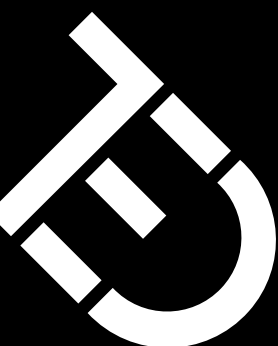
$$= 2 \left(\left(x + \frac{5}{4} \right)^2 - \left(\frac{5}{4} \right)^2 + \frac{3}{2} \right) \text{ apply complete the square inside the brackets}$$

$$= 2 \left(\left(x + \frac{5}{4} \right)^2 - 0.0625 \right)$$

$$= 2 (x + 1.25)^2 - 2(0.0625) \text{ multiply 2 throughout}$$

$$= 2 (x + 1.25)^2 - 0.125$$

$\therefore$ turning point is $(-1.25, -0.125)$



Completing The Square

Egg

Show that $a(x - h)^2 + k \geq k$ for all real values of x

$$\text{As } (x - h)^2 \geq 0$$

$$a(x - h)^2 \geq 0$$

$$a(x - h)^2 + k \geq k$$

Change the inequality correctly if your a is negative!!!!!!!!!!!!

Show that $-2(x - 2)^2 - 3 \leq -1$ for all real values of x

$$\text{As } (x - 2)^2 \geq 0$$

anything squared is always positive

$$-2(x - 2)^2 \leq 0$$

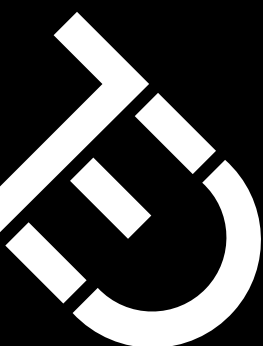
multiply both sides by -2 and swap inequality

$$-2(x - 2)^2 - 3 \leq -3$$

subtract both sides by 3

$$\therefore -2(x - 2)^2 - 3 \leq -1 \text{ as } -3 < -1$$

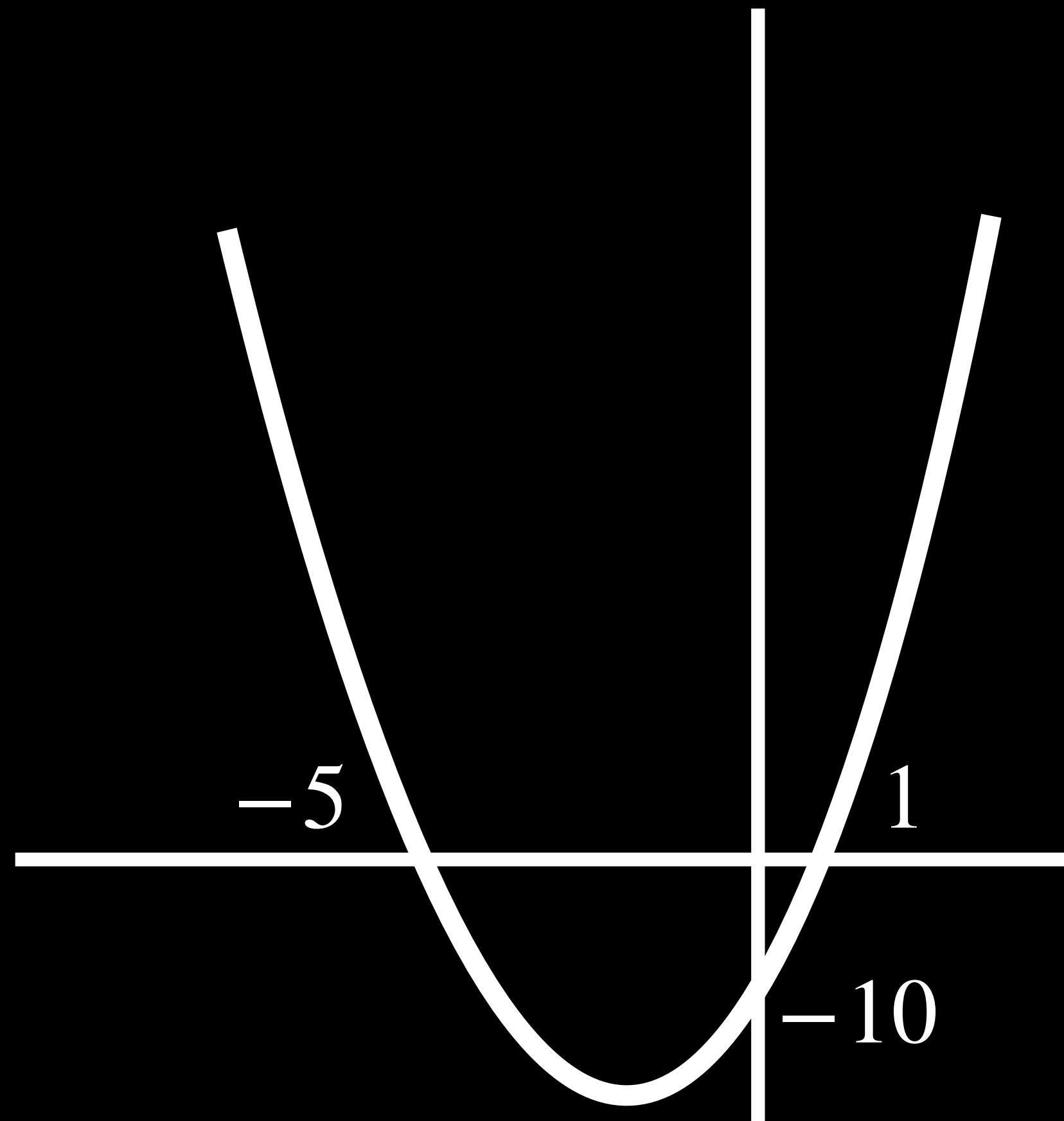
adapt inequality to qn if needed



Equations From Sketches

worms

From the sketch, find the equation of the graph in the form $a(x - p)(x - q)$



$$\begin{aligned} a(x - p)(x - q) &= a(x - (-5))(x - 1) && \text{sub in roots} \\ &= a(x + 5)(x - 1) \\ &= a(x^2 - x + 5x - 5) \\ &= a(x^2 + 4x - 5) \\ &= ax^2 + 4ax - 5a \end{aligned}$$

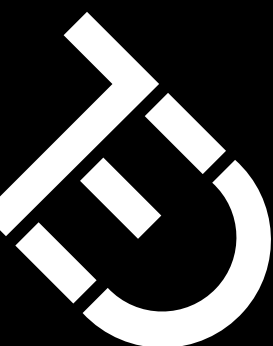
$$-5a = -10$$

$$a = 2$$

expand and compare y intercept to y intercept on graph

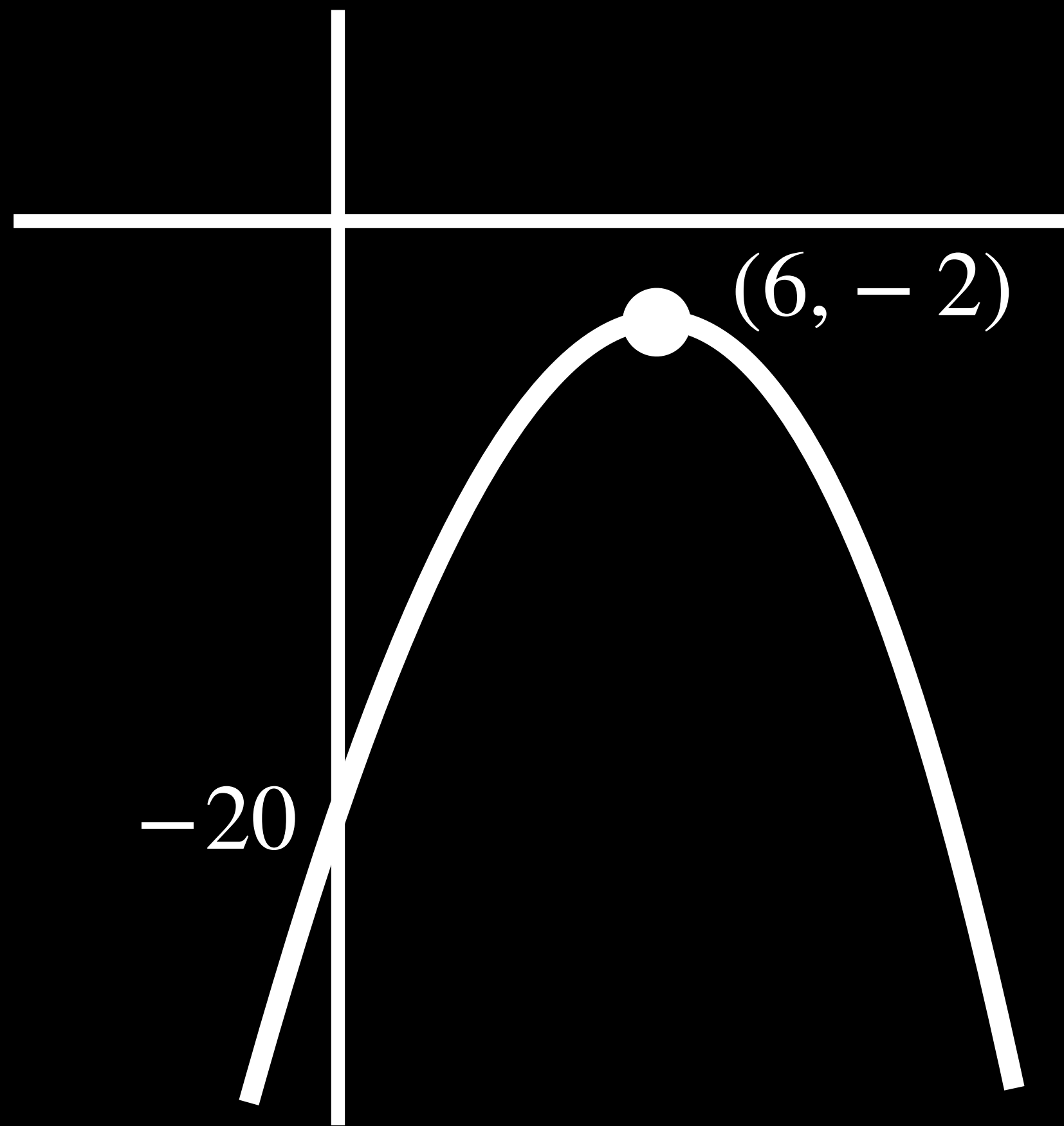
$$y = 2(x + 5)(x - 1)$$

replace back a



Equations From Sketches

worms



From the sketch, find the equation of the graph in the form $a(x - h)^2 + k$

$$\begin{aligned} a(x - h)^2 + k &= \overset{\text{sub in turning point}}{a(x - 6)^2 - 2} \\ &= a[x^2 - 2(x)(6) + 6^2] - 2 \\ &= a[x^2 - 12x + 36] - 2 \\ &= ax^2 - 12ax + 36a - 2 \end{aligned}$$

expand and compare y intercept to y intercept on graph

$$36a - 2 = -20$$

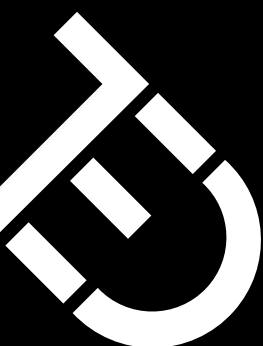
$$36a = -18$$

$$a = \frac{-18}{36}$$

$$a = -0.5$$

$$y = -0.5(x - 6)^2 - 2$$

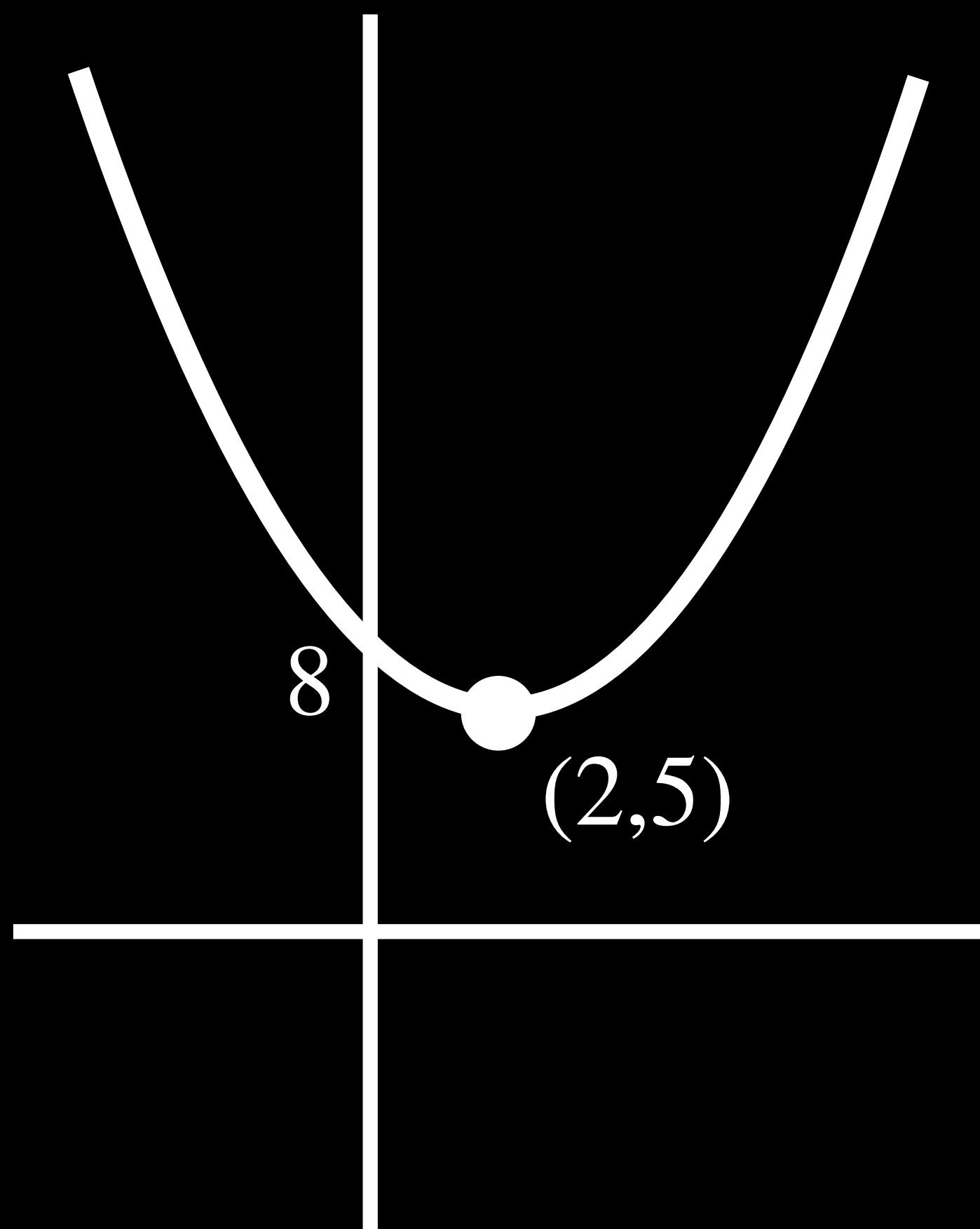
replace back a



Equations From Sketches

worms

From the sketch, find the equation of the graph in the form $ax^2 + bx + c$



sub in turning point

$$\begin{aligned}a(x - h)^2 + k &= a(x - 2)^2 + 5 \\&= a[x^2 - 2(x)(2) + 2^2] + 5 \\&= a[x^2 - 4x + 4] + 5 \\&= ax^2 - 4ax + 4a + 5\end{aligned}$$

expand and compare y intercept to y intercept on graph

$$4a + 5 = 8$$

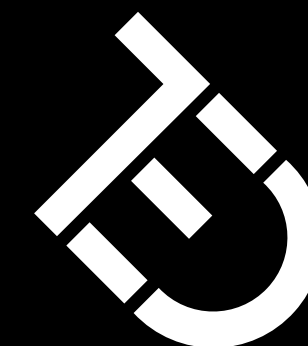
$$4a = 8 - 5$$

$$4a = 3$$

$$a = \frac{3}{4} = 0.75$$

replace back a

$$\begin{aligned}y &= ax^2 - 4ax + 4a + 5 \\&= (0.75)x^2 - 4(0.75)x + 4(0.75) + 5 \\&= 0.75x^2 - 3x + 8\end{aligned}$$



Number Lines

iimmesne

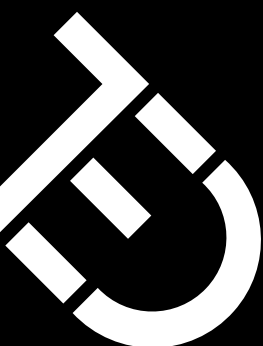
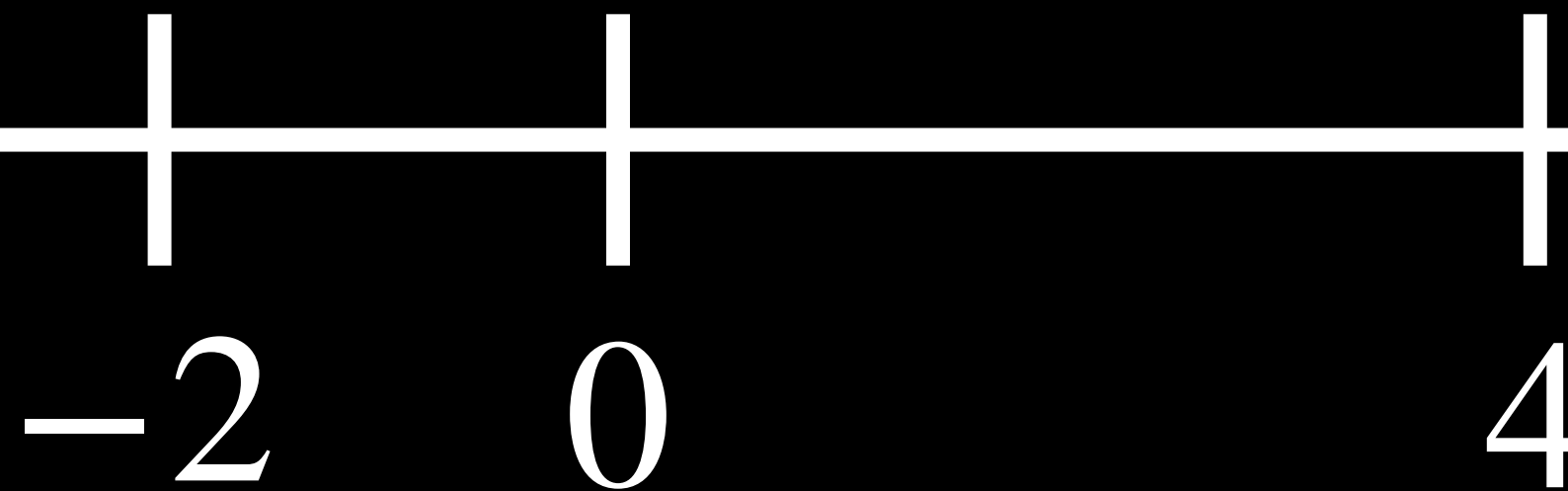
*tip: the shape of the arrowhead is the same as the inequality sign for an equality of $x > n$ or $x < n$ (aka when x is on left side)

Shade in circle if inequality includes number itself

$$x \geq -2$$

$$x < 4$$

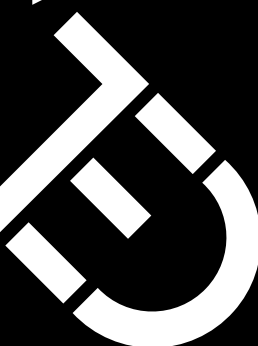
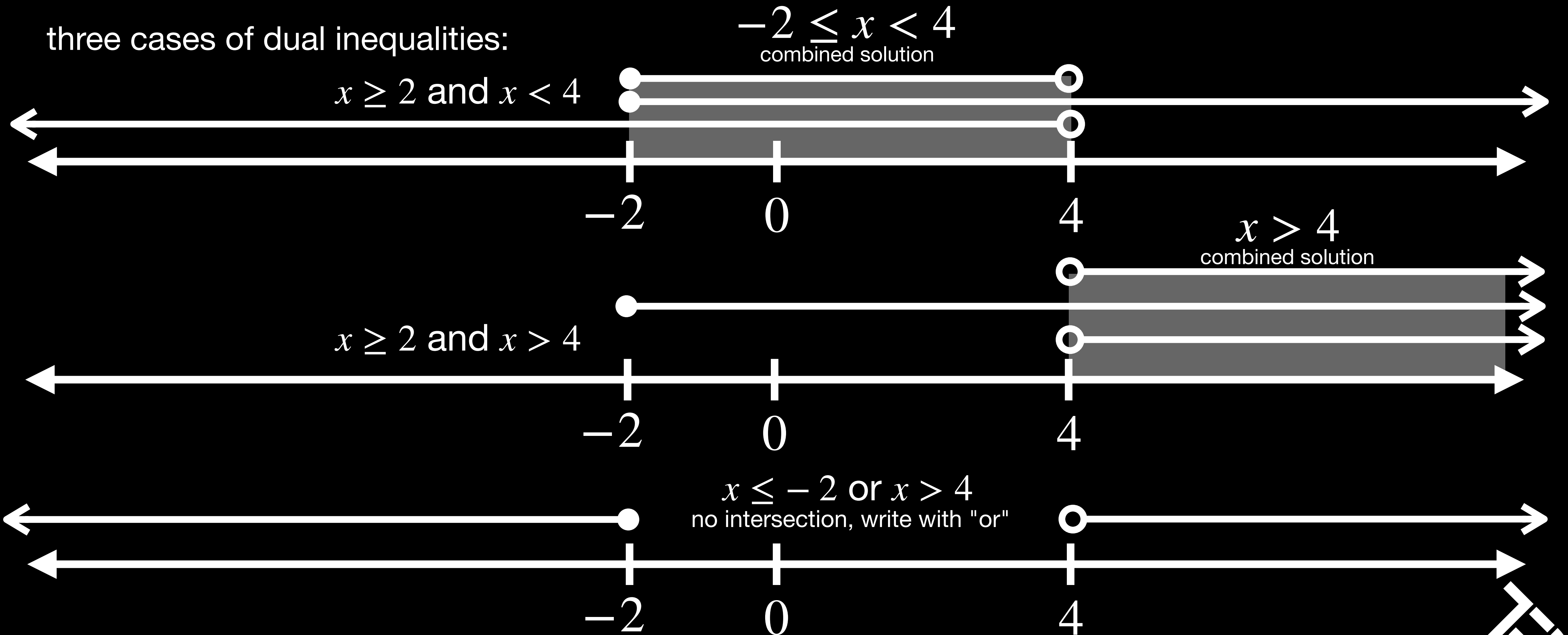
Arrow should be drawn beyond line to show it goes to infinity



Number Lines

Simmesne

three cases of dual inequalities:



Linear Inequalities

Easy is relative

$$-11 - 6x > 4x + 19$$

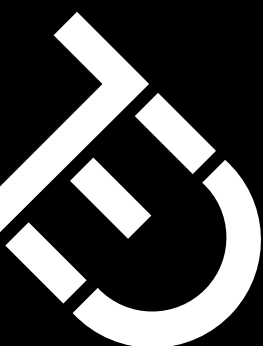
$$-4x - 6x > 19 + 11$$

$$-10x > 30$$

$$x < \frac{30}{-10}$$

$$x < -3$$

Inequality changes
whenever multiplying/
dividing by negative



Fractional Inequalities

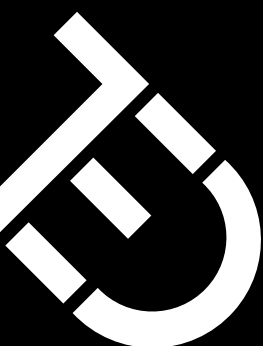
Easy is relative

$$\frac{4x + 5}{5} \geq \frac{6x - 3}{7}$$

$$\begin{aligned} 4x + 5 &\geq 5\left(\frac{6x - 3}{7}\right) \\ 7(4x + 5) &\geq 5(6x - 3) \end{aligned}$$

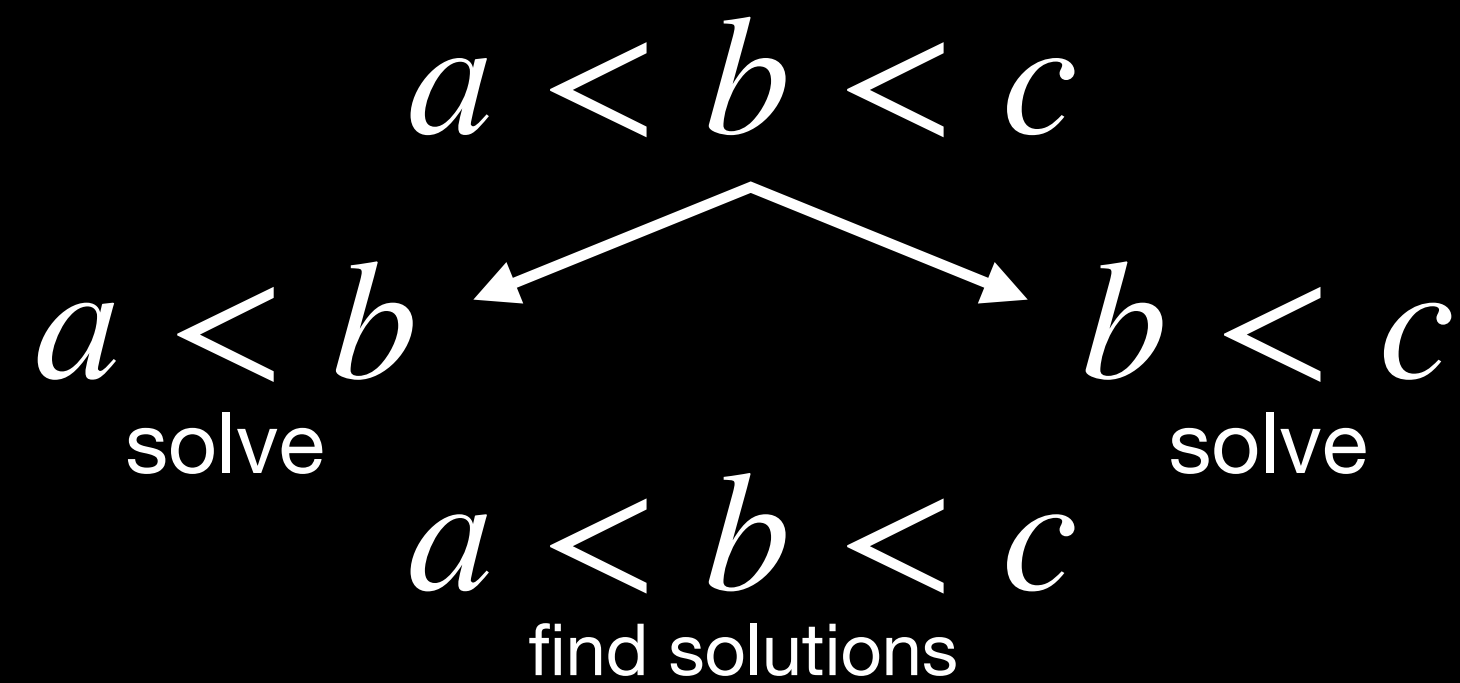
*Cross multiplication cannot be used directly, can only be written like this

$$\begin{aligned} 28x + 35 &\geq 30x - 15 \\ 28x - 30x &\geq -35 - 15 \\ -2x &\geq -50 \\ x &\leq \frac{-50}{-2} \\ x &\leq 25 \end{aligned}$$



Dual Inequalities

Easy is relative

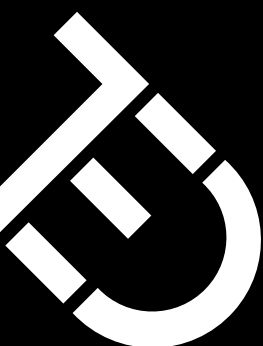
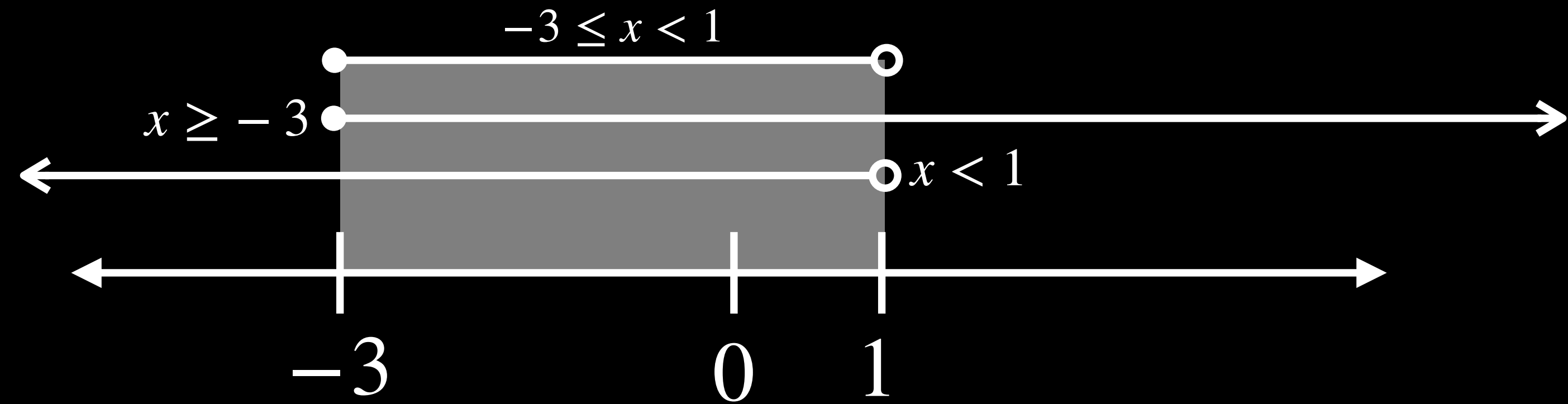


Solve $3x + 6 < 2x + 7 \leq 3x + 10$

$$\begin{aligned} 3x + 6 &< 2x + 7 \\ 3x - 2x &< 7 - 6 \\ x &< 1 \end{aligned}$$

$$\begin{aligned} 2x + 7 &\leq 3x + 10 \\ 2x - 3x &\leq 10 - 7 \\ -x &\leq 3 \\ x &\geq -3 \end{aligned}$$

$$\therefore -3 \leq x < 1$$



Quadratic Inequalities

c

$$x^2 - 2x - 3 < 5$$

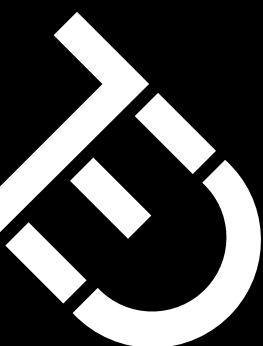
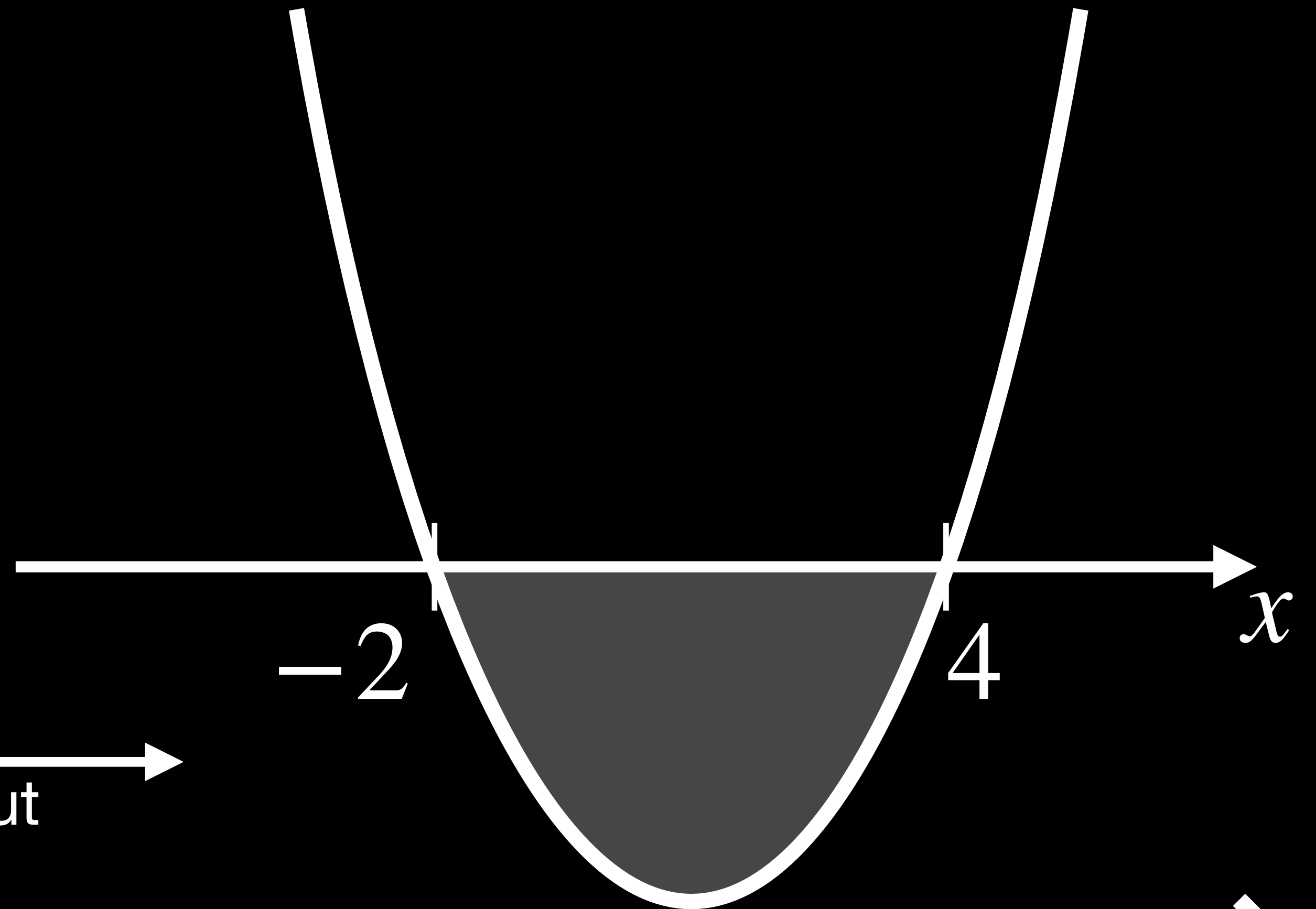
$$x^2 - 2x - 8 < 0$$

$$(x + 2)(x - 4) < 0$$

$$-2 < x < 4$$

write inequality based on shaded region

sketch graph out
shade region



Linear Simultaneous Eqns - Substitution

The better one

$4x - 52 = 2y \rightarrow (1)$ Subsitute (3) into (2) : Subsitute $x = 11$ into (3) :

$$2x + 3y = 10 \rightarrow (2) \quad 2x + 3y = 10$$
$$2x + 3\left(\frac{4x - 52}{2}\right) = 10$$

From (1) :

$$4x - 52 = 2y$$

$$y = \frac{4x - 52}{2} \rightarrow (3)$$

$$2x + 3(2x - 26) = 10$$

$$2x + 6x - 78 = 10$$

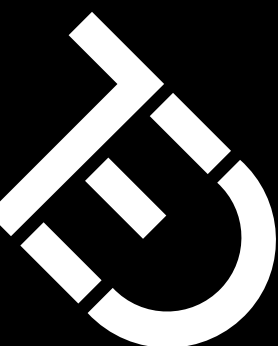
$$8x - 78 = 10$$

$$8x = 10 + 78$$

$$x = 11$$

$$y = \frac{4x - 52}{2}$$
$$= \frac{4(11) - 52}{2}$$
$$= -4$$

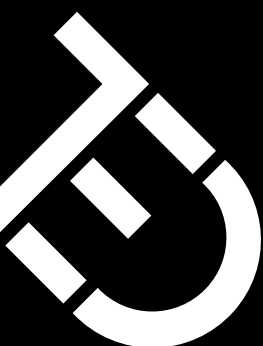
$$x = 11, y = -4$$



Linear Simultaneous Eqns - Elimination

dotn

Please do not use elimination it is not good and only works on specific cases instead of substitution which can be used for almost all cases



Quadratic Simultaneous Eqns

q

$$\begin{aligned} y &= x^2 - 5x + 4 \rightarrow (1) \\ y &= -4x + 10 \rightarrow (2) \end{aligned} \quad \left. \vphantom{\begin{aligned} y &= x^2 - 5x + 4 \rightarrow (1) \\ y &= -4x + 10 \rightarrow (2) \end{aligned}} \right\}$$

Make sure both equations are in the form $y =$
If they are not in the correct form, change them accordingly

$$(x - 3) = 0, (x + 2) = 0$$

$$x = 3 \text{ or } x = -2$$

Substitute (2) into (1) :

$$-4x + 10 = x^2 - 5x + 4$$

$$x^2 - 5x + 4 + 4x - 10 = 0$$

$$x^2 - x - 6 = 0 \quad \text{move all terms to one side to factorise quadratic}$$

$$(x - 3)(x + 2) = 0$$

Substitute $x = 3$ or $x = -2$ into (2) :

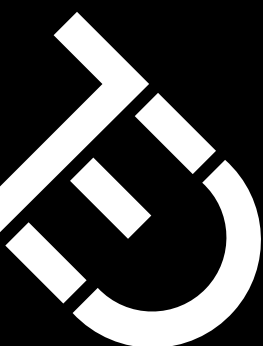
$$y = -4x + 10$$

$$y = -4(3) + 10, -4(-2) + 10$$

$$y = 2, 18$$

$$x = 3, y = 2$$

$$x = -2, y = 18$$



Discriminant

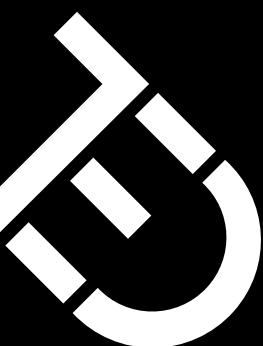
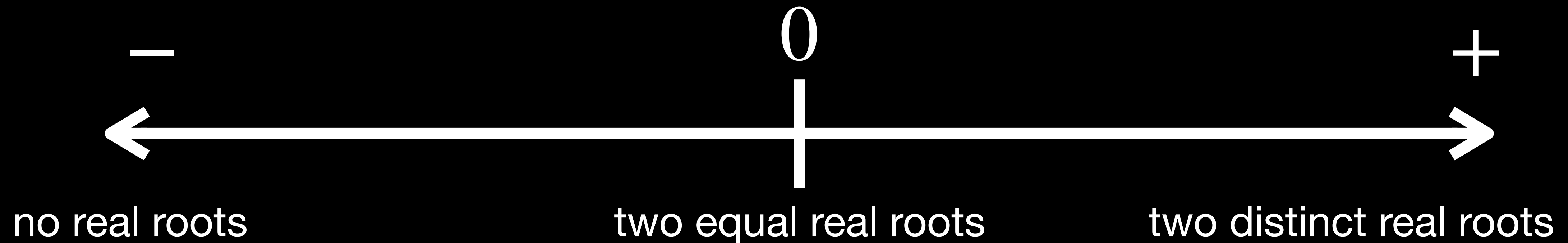
Good

$$b^2 - 4ac$$

discriminant

finds nature of roots

where a , b and c are constants in the quadratic $ax^2 + bx + c$

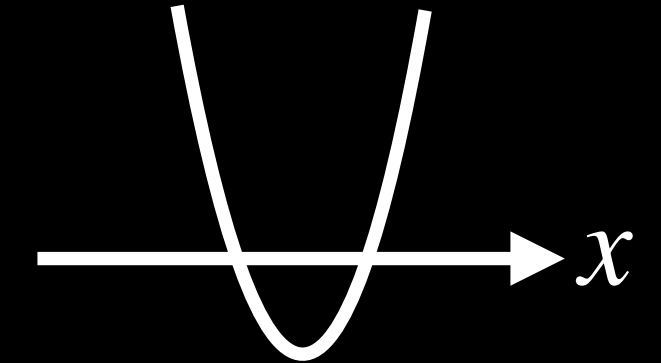


Discriminant

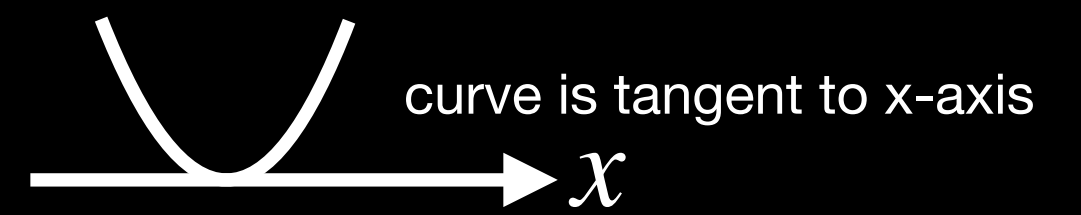
Good

$b^2 - 4ac > 0$ check for two distinct roots

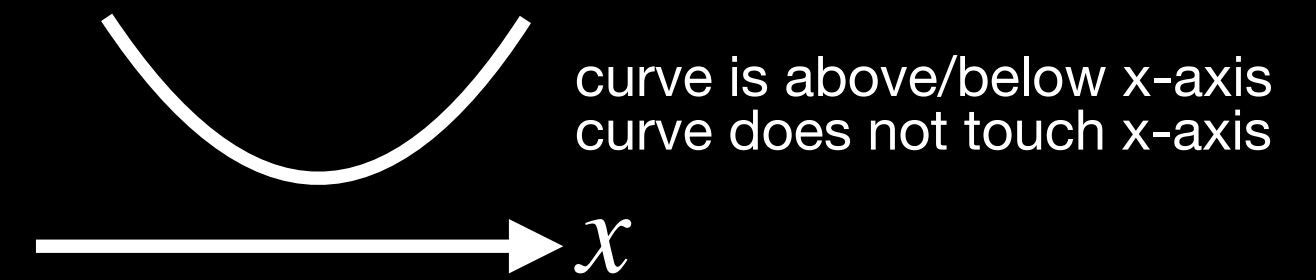
curve intersects x-axis at 2 distinct points



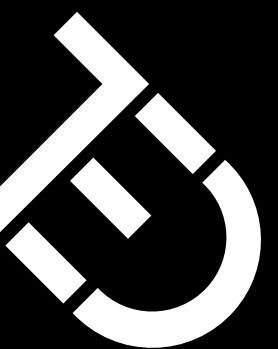
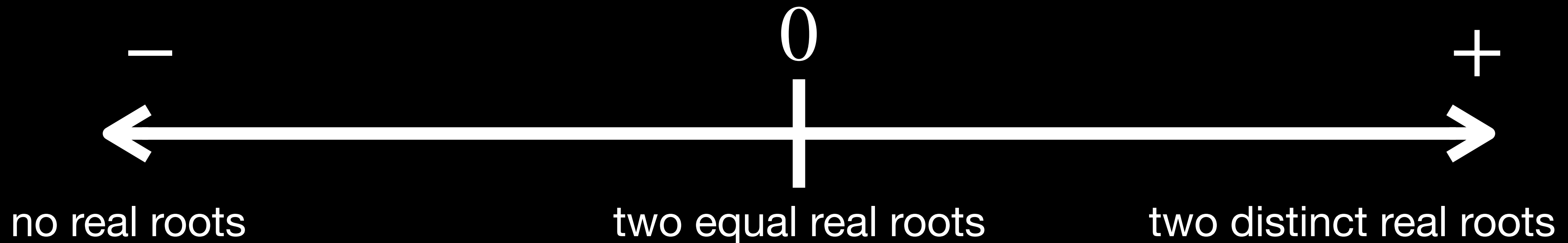
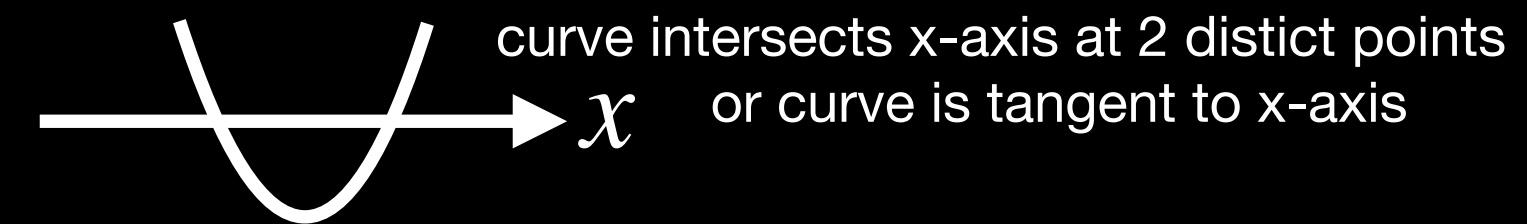
$b^2 - 4ac = 0$ check for two equal roots



$b^2 - 4ac < 0$ check for no real roots



$b^2 - 4ac \geq 0$ check for real roots



Discriminant

Good

$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

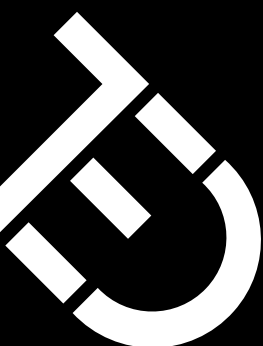
To remember discriminant:

Quadratic formula finds roots

Expression in square root must give a real number

Thus expression in square root must be positive for square root to be a real number

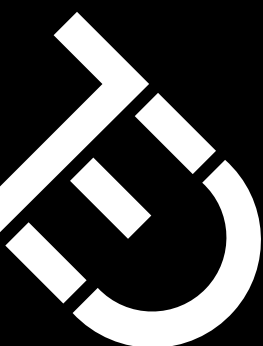
Therefore when discriminant is positive , there will be real roots



Discriminant

Good

- a) Show that $2x^2 - 8x + 8 = 0$ has equal roots.
- b) Given the equation $x^2 + (2h + 3)x + h^2 = 0$, where h is a constant, find the values of h such that the equation has no real roots
- c) The curve $y = kx^2 + (3 - k)x + k - 3$, where k is a constant, has a minimum point. Find the range of values of k such that $kx^2 + (3 - k)x + k - 3 = 0$ intersects the x axis.
- d) Show that $ax^2 + 3x - a + 1 = 0$ has real solutions for all values of a , where a is a constant



Discriminant

Good

- a) Show that $2x^2 - 8x + 8 = 0$ has equal roots.
- b) Given the equation $x^2 + (2h + 3)x + h^2 = 0$, where h is a constant, find the values of h such that the equation has no real roots
- c) The curve $y = kx^2 + (3 - k)x + k - 3$, where k is a constant, has a minimum point. Find the range of values of k such that $kx^2 + (3 - k)x + k - 3 = 0$ intersects the x axis.
- d) Show that $ax^2 + 3x - a + 1 = 0$ has real solutions for all values of a , where a is a constant

$$\begin{aligned} \text{a) } 2x^2 - 8x + 8 &= 0 \\ \text{discriminant} &= (-8)^2 - 4(2)(8) \\ &= 0 \\ \text{As discriminant} &= 0, \text{ there are repeated roots} \end{aligned}$$

$$\begin{aligned} \text{b) } x^2 + (2h + 3)x + h^2 &= 0 \\ \text{discriminant} &= (2h + 3)^2 - 4(1)(h^2) \\ &= (2h + 3)^2 - 4h^2 \\ &= (2h)^2 + 2(2h)(3) + 3^2 - 4h^2 \\ &= 4h^2 + 12h + 9 - 4h^2 \\ &= 12h + 9 \quad \text{expand and simplify} \end{aligned}$$

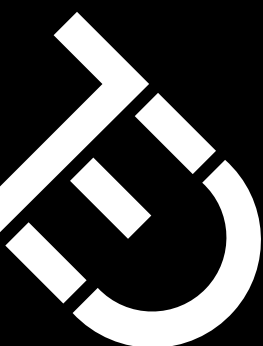
As no real roots, let discriminant < 0

$$12h + 9 < 0$$

$$12h < -9$$

$$h < \frac{-9}{12}$$

$$h < -\frac{3}{4}$$



Discriminant

Good

- a) Show that $2x^2 - 8x + 8 = 0$ has equal roots.
- b) Given the equation $x^2 + (2h + 3)x + h^2 = 0$, where h is a constant, find the values of h such that the equation has no real roots
- c) The curve $y = kx^2 + (3 - k)x + k - 3$, where k is a constant, has a minimum point. Find the range of values of k such that $kx^2 + (3 - k)x + k - 3 = 0$ intersects the x axis.
- d) Show that $ax^2 + 3x - a + 1 = 0$ has real solutions for all values of a , where a is a constant

c) $kx^2 + (3 - k)x + k - 3 = 0$

$$\begin{aligned}\text{discriminant} &= (3 - k)^2 - 4(k)(k - 3) \\ &= 3^2 - 2(3)(k) + k^2 - 4k(k - 3) \\ &= 3^2 - 2(3)(k) + k^2 - 4k^2 + 3(4)(k) \\ &= 9 - 6k + k^2 - 4k^2 + 12k \\ \text{expand and simplify} \quad &= k^2 - 4k^2 + 12k - 6k + 9 \\ &= -3k^2 + 6k + 9 \\ &= -3(k^2 - 2k - 3) \\ &= -3(k + 1)(k - 3)\end{aligned}$$

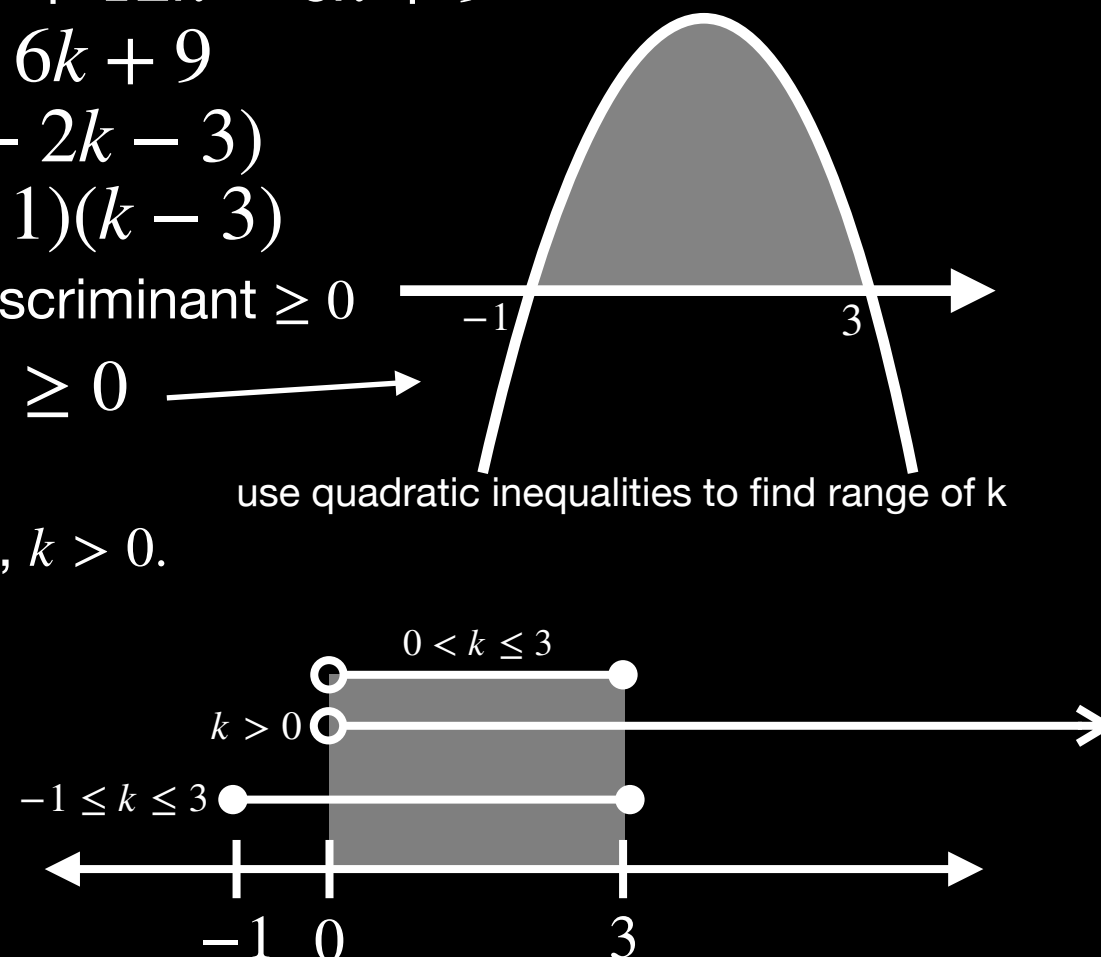
As there are real roots, let discriminant ≥ 0

$$\begin{aligned}-3(k + 1)(k - 3) &\geq 0 \\ -1 &\leq k \leq 3\end{aligned}$$

As the curve has a minimum, $k > 0$.

$$\therefore 0 < k \leq 3$$

curve has minimum
 $\therefore x^2$ coefficient must be positive
thus include $k > 0$



$$\begin{aligned}d) \quad ax^2 + 3x - a + 1 &= 0 \\ ax^2 + 3x + (-a + 1) &= 0 \\ \text{discriminant} &= 3^2 - 4(a)(-a + 1) \\ &= 9 - 4a(-a + 1) \\ &= 9 + 4a^2 - 4a \\ &= 4a^2 - 4a + 9\end{aligned}$$

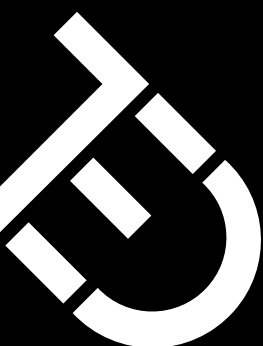
we know that this discriminant will be always positive
as there are always real solutions

thus $4a^2 - 4a + 9 = 0$ will have no real solutions (which we have to show)

$$\begin{aligned}\text{discriminant of } 4a^2 - 4a + 9 &= (-4)^2 - 4(4)(9) \\ &= -128\end{aligned}$$

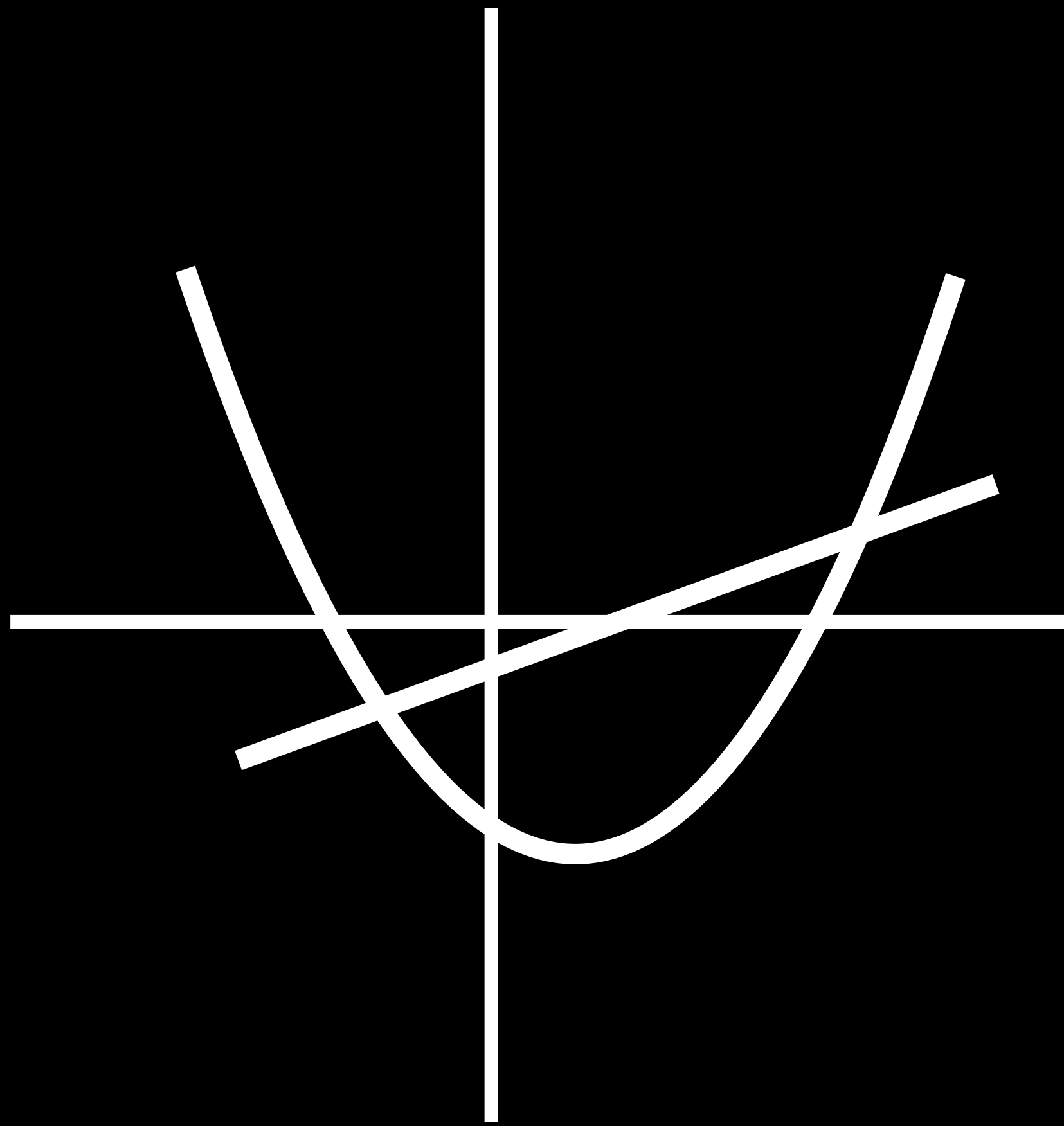
As discriminant of $4a^2 - 4a + 9 < 0$, and coeff. of a^2 is positive,
 $4a^2 - 4a + 9$ is always positive

$\therefore$ discriminant of $ax^2 + 3x + (-a + 1)$ is always positive
thus, $ax^2 + 3x + (-a + 1) = 0$ will always have real solutions



Discriminant - Intersecting Lines

Good



To find if a quadratic and a line intersect/are tangent/don't intersect:

Substitute line equation into quadratic to form new quadratic

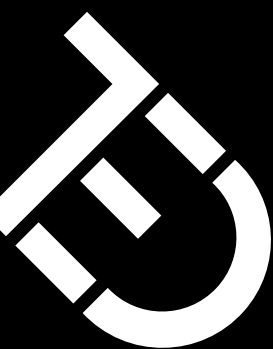
Use discriminant to check new quadratic roots

Type of roots determines type of intersections

Real roots/ distinct roots: intersects

Equal roots: tangent/intersects once

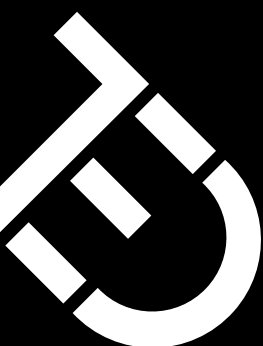
No real roots: don't intersect



Discriminant - Intersecting Lines

Good

- a) The curve $y = x^2 + 3x + 1$ and the line $y = 2x + k$, where k is a constant, are tangent to each other. Solve for k .
- b) Given that the curve $y = px^2 + 3px + 5$ and the line $y = px + 4p$ intersect each other, with p being a constant, find the range of values of p .



Discriminant - Intersecting Lines

Good

- a) The curve $y = x^2 + 3x + 1$ and the line $y = 2x + k$, where k is a constant, are tangent to each other. Solve for k .
- b) Given that the curve $y = px^2 + 3px + 5$ and the line $y = px + 4p$ intersect each other, with p being a constant, find the range of values of p .

$$\begin{aligned}
 \text{a) } x^2 + 3x + 1 &= 2x + k && \text{let curve=line} \\
 x^2 + 3x + 1 - 2x - k &= 0 && \text{move all terms to left side} \\
 x^2 + x + 1 - k &= 0 \\
 \text{discriminant} &= 1^2 - 4(1)(1 - k) \\
 &= 1 - 4 + 4k \\
 &= 4k - 3
 \end{aligned}$$

As line is tangent to curve, discriminant = 0. Thus,

$$4k - 3 = 0 \quad \text{let discriminant} = 0 \text{ and solve for } k$$

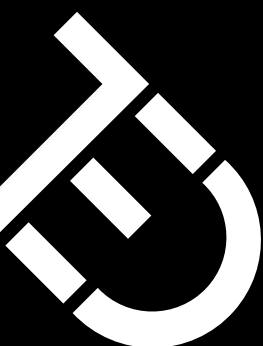
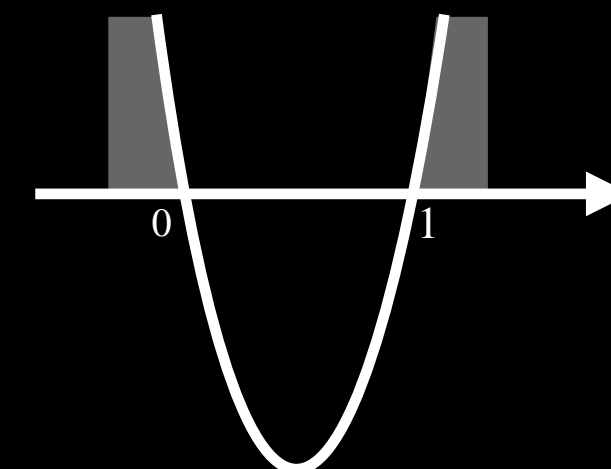
$$4k = 3$$

$$k = \frac{3}{4} = 0.75$$

$$\begin{aligned}
 \text{b) } px^2 + 3px + 5 &= px + 4p && \text{let curve=line} \\
 px^2 + 3px + 5 - px - 4p &= 0 \\
 px^2 + 3px - px + 5 - 4p &= 0 \\
 px^2 + 2px + 5 - 4p &= 0 && \text{move all terms to left side} \\
 \text{discriminant} &= (2p)^2 - 4(p)(5 - 4p) && \text{move all terms to left side} \\
 &= 4p^2 - 4p(5 - 4p) \\
 &= 4p^2 - 20p + 16p^2 \\
 &= 20p^2 - 20p \\
 &= 20p(p - 1)
 \end{aligned}$$

As line intersects curve, discriminant ≥ 0

$$\begin{aligned}
 \text{let discriminant } \geq 0 \text{ and solve for } p \quad &20p(p - 1) \geq 0 \\
 &p \leq 0 \text{ or } p \geq 1 \\
 &\text{use "or", not "and"}
 \end{aligned}$$



Surds



Surd Rules

indices part ii

$$\sqrt{a} \times \sqrt{a} = (\sqrt{a})^2 = a$$

$$\sqrt{a} \times \sqrt{b} = \sqrt{ab}$$

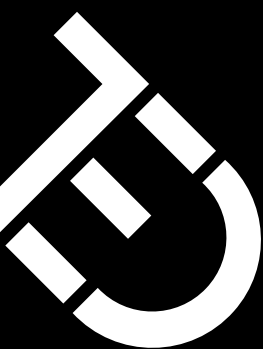
$$\frac{\sqrt{a}}{\sqrt{b}} = \sqrt{\frac{a}{b}}$$

$$(a\sqrt{b})(c\sqrt{d}) = ac\sqrt{bd}$$

Surds are usually be expressed as

$$a + b\sqrt{c}$$

where a, b and c are constants



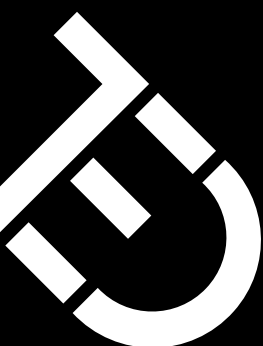
Surd Simplification

Roo beer

$$\begin{aligned}\sqrt{162} &= \sqrt{3^4 \times 2} \\ &= \sqrt{3^4} \times \sqrt{2} \\ &= 3^2 \times \sqrt{2} \\ &= 9\sqrt{2}\end{aligned}$$

$$\begin{array}{r|l} 3 & 162 \\ \hline & 54 \\ 3 & 54 \\ \hline & 18 \\ 3 & 18 \\ \hline & 6 \\ 3 & 6 \\ \hline & 2 \end{array}$$

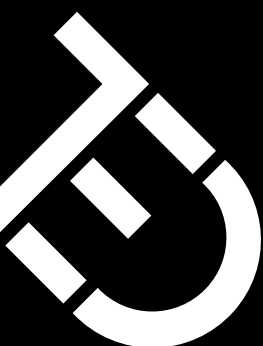
use prime factorisation to simplify down surds into their primes



Surd Operations

Roo beer

$$\begin{aligned}\sqrt{80} - \sqrt{320} + \sqrt{45} &= \sqrt{4^2 \times 5} - \sqrt{2^6 \times 5} + \sqrt{3^2 \times 5} \\&= \sqrt{4^2} \sqrt{5} - \sqrt{2^6} \sqrt{5} + \sqrt{3^2} \sqrt{5} \quad \leftarrow \text{simplify first} \\&= 4\sqrt{5} - 2^3\sqrt{5} + 3\sqrt{5} \\&= (4 - 2^3 + 3) \sqrt{5} \quad \leftarrow \text{factorise common terms} \\&= (4 - 8 + 3) \sqrt{5} \\&= -1\sqrt{5}\end{aligned}$$

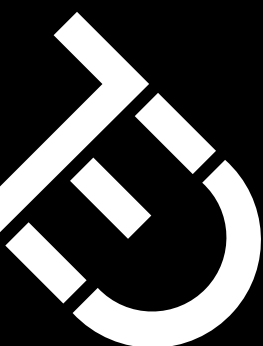


Surd Operations

Roo beer

$$\begin{aligned}\left(2 + \sqrt{5}\right)\left(3 - \sqrt{5}\right) &= 2(3) + 2\left(-\sqrt{5}\right) + 3\sqrt{5} + \sqrt{5}\left(-\sqrt{5}\right) \\ &= 6 - 2\sqrt{5} + 3\sqrt{5} - 5 \\ &= 3\sqrt{5} - 2\sqrt{5} + 6 - 5 \\ &= \sqrt{5} + 1\end{aligned}$$

$$\begin{aligned}\left(2 + \sqrt{5}\right)^2 &= 2^2 + 2(2)\left(\sqrt{5}\right) + \left(\sqrt{5}\right)^2 \\ &= 4 + 4\sqrt{5} + 5 \\ &= 4\sqrt{5} + 9\end{aligned}$$



Conjugate Surds

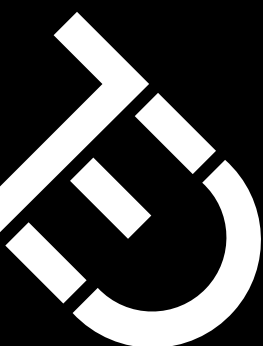
j

a pair of surds when multiplied together gives a rational number

$$\begin{aligned}(\sqrt{3} + \sqrt{2})(\sqrt{3} - \sqrt{2}) &= \sqrt{3}^2 - \sqrt{2}^2 \\ &= 3 - 2 \\ &= 1\end{aligned}$$

$$\begin{aligned}(\sqrt{m} + \sqrt{n})(\sqrt{m} - \sqrt{n}) &= \sqrt{m}^2 - \sqrt{n}^2 \\ &= m - n\end{aligned}$$

$$\therefore (\sqrt{m} + \sqrt{n}) \text{ is conjugate to } (\sqrt{m} - \sqrt{n})$$



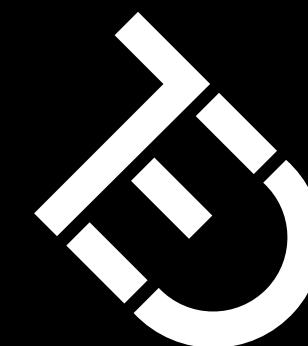
Rationalising Surds

Re

simplifying a fraction with a surd as the denominator into a fraction with no surd in the denominator

$$\begin{aligned}\frac{8}{\sqrt{2}} &= \frac{8}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} \\ &= \frac{8\sqrt{2}}{\sqrt{2} \times \sqrt{2}} \\ &= \frac{8\sqrt{2}}{2} = 4\sqrt{2}\end{aligned}$$

$$\begin{aligned}\frac{24}{4\sqrt{3}} &= \frac{24}{4\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} \\ &= \frac{24\sqrt{3}}{4\sqrt{3} \times \sqrt{3}} \\ &= \frac{24\sqrt{3}}{12} = 2\sqrt{3}\end{aligned}$$



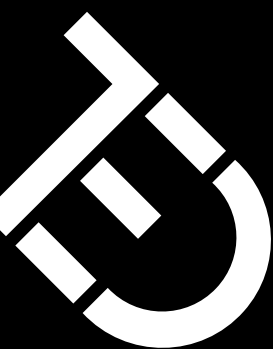
Rationalising Surds

Re

simplifying a fraction with a surd as the denominator into a fraction with no surd in the denominator

$$\begin{aligned}\frac{11}{5 - \sqrt{3}} &= \frac{11}{5 - \sqrt{3}} * \frac{\overbrace{5 + \sqrt{3}}^{=1}}{5 + \sqrt{3}} &= \frac{55 + 11\sqrt{3}}{22} \\ &= \frac{11(5 + \sqrt{3})}{(5 - \sqrt{3})(5 + \sqrt{3})} &= \frac{55}{22} + \frac{11\sqrt{3}}{22} \\ &= \frac{55 + 11\sqrt{3}}{5^2 - \sqrt{3}^2} &= 0.5\sqrt{3} + 2.5\end{aligned}$$

In A Math, you are never supposed to leave a fraction with a surd in the denominator



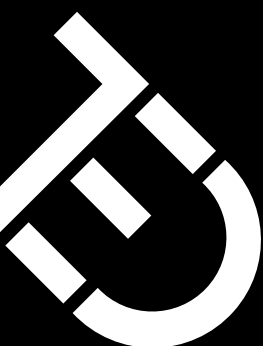
Rationalising Surds

Re

a) A rectangle has an area $6 + \sqrt{2}$ and a base of $2 - \sqrt{2}$. Find the exact value of its height in the form of $a + b\sqrt{2}$, where a and b are integers.

b) A square base pyramid has side length $\frac{3 + 5\sqrt{3}}{2\sqrt{3} - 1}$ m and height of $\frac{1}{3 - \sqrt{3}}$ m. Find the volume of the pyramid in the form of $a + b\sqrt{3}$, where a and b are rational numbers.

c) An isosceles triangle ABC has $\angle ABC = 60^\circ$ and $AB = BC$. If the area of the triangle is $\frac{3\sqrt{3} + 3}{4\sqrt{3} - 4}$ m², find AB in the form of $\sqrt{a + b\sqrt{3}}$.



Rationalising Surds

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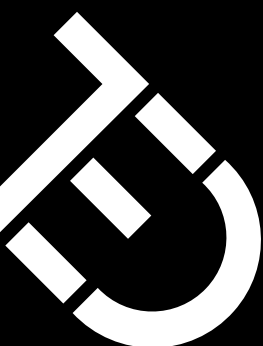
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$$\begin{aligned} a) \text{ length} &= \frac{6 + \sqrt{2}}{2 - \sqrt{2}} \quad l = \frac{A}{b} \\ &= \frac{(6 + \sqrt{2})(2 + \sqrt{2})}{(2 - \sqrt{2})(2 + \sqrt{2})} \\ &= \frac{12 + 6\sqrt{2} + 2\sqrt{2} + 2}{2^2 - 2} \\ &= \frac{14 + 8\sqrt{2}}{2} \\ &= 7 + 4\sqrt{2} \end{aligned}$$

$$\begin{aligned} b) \text{ side length} &= \frac{3 + 5\sqrt{3}}{2\sqrt{3} - 1} \\ &= \frac{(3 + 5\sqrt{3})(2\sqrt{3} + 1)}{(2\sqrt{3} - 1)(2\sqrt{3} + 1)} \quad \text{rationalise surd first to simplify} \\ &= \frac{6\sqrt{3} + 3 + 30 + 5\sqrt{3}}{(2\sqrt{3})^2 - 1^2} \\ &= \frac{33 + 11\sqrt{3}}{11} \\ &= 3 + \sqrt{3} \end{aligned}$$

$$\begin{aligned} \text{volume of pyramid} &= \left(\frac{1}{3}\right) (3 + \sqrt{3})^2 \left(\frac{1}{3 - \sqrt{3}}\right) \quad V = \frac{1}{3}Bh = \frac{1}{3}b^2h \\ &= \left(\frac{1}{3}\right) (3^2 + 2(3)(\sqrt{3}) + 3) \left(\frac{1}{3 - \sqrt{3}}\right) \\ &= \left(\frac{1}{3}\right) (12 + 6\sqrt{3}) \left(\frac{1}{3 - \sqrt{3}}\right) \\ &= (4 + 2\sqrt{3}) \left(\frac{1}{3 - \sqrt{3}}\right) \\ &= (4 + 2\sqrt{3}) \left(\frac{1}{3 - \sqrt{3}}\right) \\ &= \frac{4 + 2\sqrt{3}}{3 - \sqrt{3}} \quad \text{rationalise surd again} \\ &= \frac{(4 + 2\sqrt{3})(3 + \sqrt{3})}{(3 - \sqrt{3})(3 + \sqrt{3})} \\ &= \frac{12 + 4\sqrt{3} + 6\sqrt{3} + 6}{3^2 - 3} \\ &= \frac{18 + 10\sqrt{3}}{6} = 3 + \frac{10}{6}\sqrt{3} \end{aligned}$$



Rationalising Surds

Re

a) A rectangle has an area $6 + \sqrt{2}$ and a base of $2 - \sqrt{2}$. Find the exact value of its height in the form of $a + b\sqrt{2}$, where a and b are integers.

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$$\begin{aligned} c) \quad \frac{3\sqrt{3} + 3}{4\sqrt{3} - 4} &= \frac{(3\sqrt{3} + 3)(4\sqrt{3} + 4)}{(4\sqrt{3} - 4)(4\sqrt{3} + 4)} \quad \text{rationalise surd first to simplify} \\ &= \frac{36 + 12\sqrt{3} + 12\sqrt{3} + 12}{(4\sqrt{3})^2 - 4^2} \\ &= \frac{48 + 24\sqrt{3}}{32} \\ &= \frac{6 + 3\sqrt{3}}{4} \\ &= \frac{3(2 + \sqrt{3})}{4} \end{aligned}$$

$$\therefore \frac{3(2 + \sqrt{3})}{4} = \frac{1}{2}(AB)(BC)\sin(60^\circ)$$

$$\frac{3(2 + \sqrt{3})}{4} = \frac{1}{2}(AB^2)\frac{\sqrt{3}}{2} \quad (AB=BC)$$

$$\frac{3(2 + \sqrt{3})}{4} = \frac{\sqrt{3}}{4}(AB^2)$$

$$\frac{3(2 + \sqrt{3})}{4} \cancel{\sqrt{3}} \frac{\cancel{4}}{\sqrt{3}} = AB^2$$

$$\frac{3(2 + \sqrt{3})}{\sqrt{3}} = AB^2$$

$$AB^2 = \frac{3(2 + \sqrt{3})\sqrt{3}}{\sqrt{3}\sqrt{3}}$$

$$AB^2 = \frac{\cancel{3}(2 + \sqrt{3})\sqrt{3}}{\cancel{3}}$$

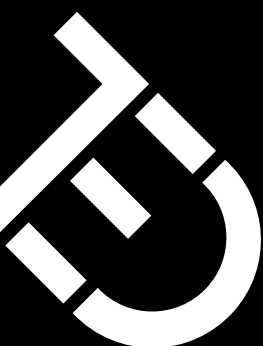
$$AB^2 = (2 + \sqrt{3})\sqrt{3}$$

special angle; $\sin(60^\circ) = \frac{\sqrt{3}}{2}$

$$AB^2 = 2\sqrt{3} + \sqrt{3}\sqrt{3}$$

$$AB^2 = 2\sqrt{3} + 3$$

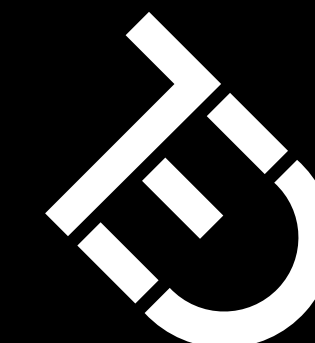
$$AB = \sqrt{3 + 2\sqrt{3}} \quad (\text{rej negative value as lengths} > 0)$$



Surd Equations

su(rd)s

Find the solutions of $\sqrt{2x-5} = -\sqrt{x+2} + 6$

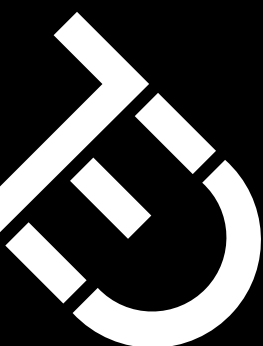


Surd Equations

su(rd)s

Find the solutions of $\sqrt{2x-5} = -\sqrt{x+2} + 6$

$$\begin{aligned} \sqrt{2x-5} &= -\sqrt{x+2} + 6 \\ 2x-5 &= \boxed{(-\sqrt{x+2})^2} + 2(-\sqrt{x+2})(6) + 6^2 \quad \left. \begin{array}{l} \text{square both sides} \\ (a+b)^2 = a^2 + 2ab + b^2 \end{array} \right\} \\ 2x-5 &= \boxed{x+2} - 12\sqrt{x+2} + 36 \\ 12\sqrt{x+2} &= x+2+36-2x+5 \quad \left. \begin{array}{l} \text{Make surd the subject} \\ \text{keep making surd subject until all surds are gone} \end{array} \right\} \\ 12\sqrt{x+2} &= -x+43 \\ (12\sqrt{x+2})^2 &= (-x+43)^2 \quad \left. \begin{array}{l} \text{square both sides} \\ (a+b)^2 = a^2 + 2ab + b^2 \end{array} \right\} \\ 144(x+2) &= \boxed{(-x)^2} + 2(-x)(43) + 43^2 \\ 144x+288 &= \boxed{x^2} - 86x + 1849 \\ x^2 - 86x + 1849 - 144x - 288 &= 0 \\ x^2 - 230x + 1561 &= 0 \\ (x-7)(x-223) &= 0 \end{aligned}$$



Surd Equations

su(rd)s

Substitute for all solutions

At $x = 7$,

$$\text{for } \sqrt{2x - 5} = -\sqrt{x + 2} + 6$$

$$LHS = \sqrt{2x - 5}$$

$$= \sqrt{9}$$

$$= 3$$

$$RHS = -\sqrt{x + 2} + 6$$

$$= -\sqrt{7 + 2} + 6$$

$$= -3 + 6 = 3$$

At $x = 223$,

$$\text{for } \sqrt{2x - 5} = -\sqrt{x + 2} + 6$$

$$LHS = \sqrt{2x - 5}$$

$$= \sqrt{2(223) + 9}$$

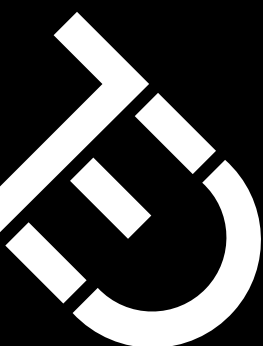
$$= \sqrt{455}$$

$$RHS = -\sqrt{x + 2} + 6$$

$$= -\sqrt{223 + 2} + 6$$

$$= -9$$

$$\therefore x = 7$$



Polynomials & Partial Fractions



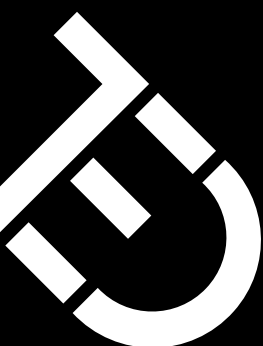
Polynomials

Polyroly

Order/degree of polynomial = highest power of variable
= 5

$$x^5 + 3x^4 + 6x^3 + x^2 + 2x + 4$$

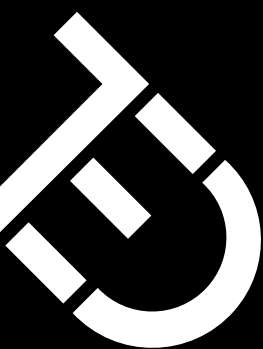
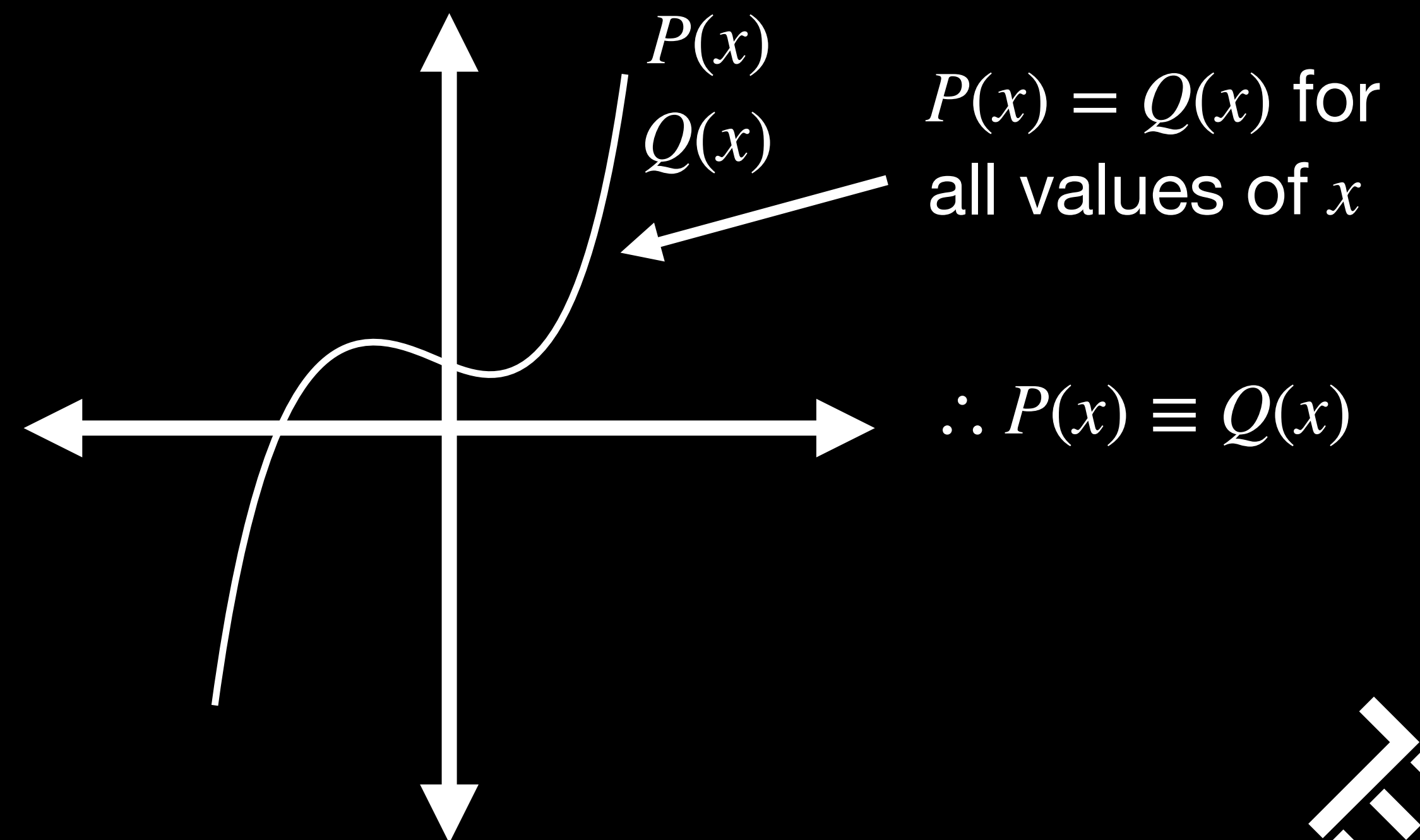
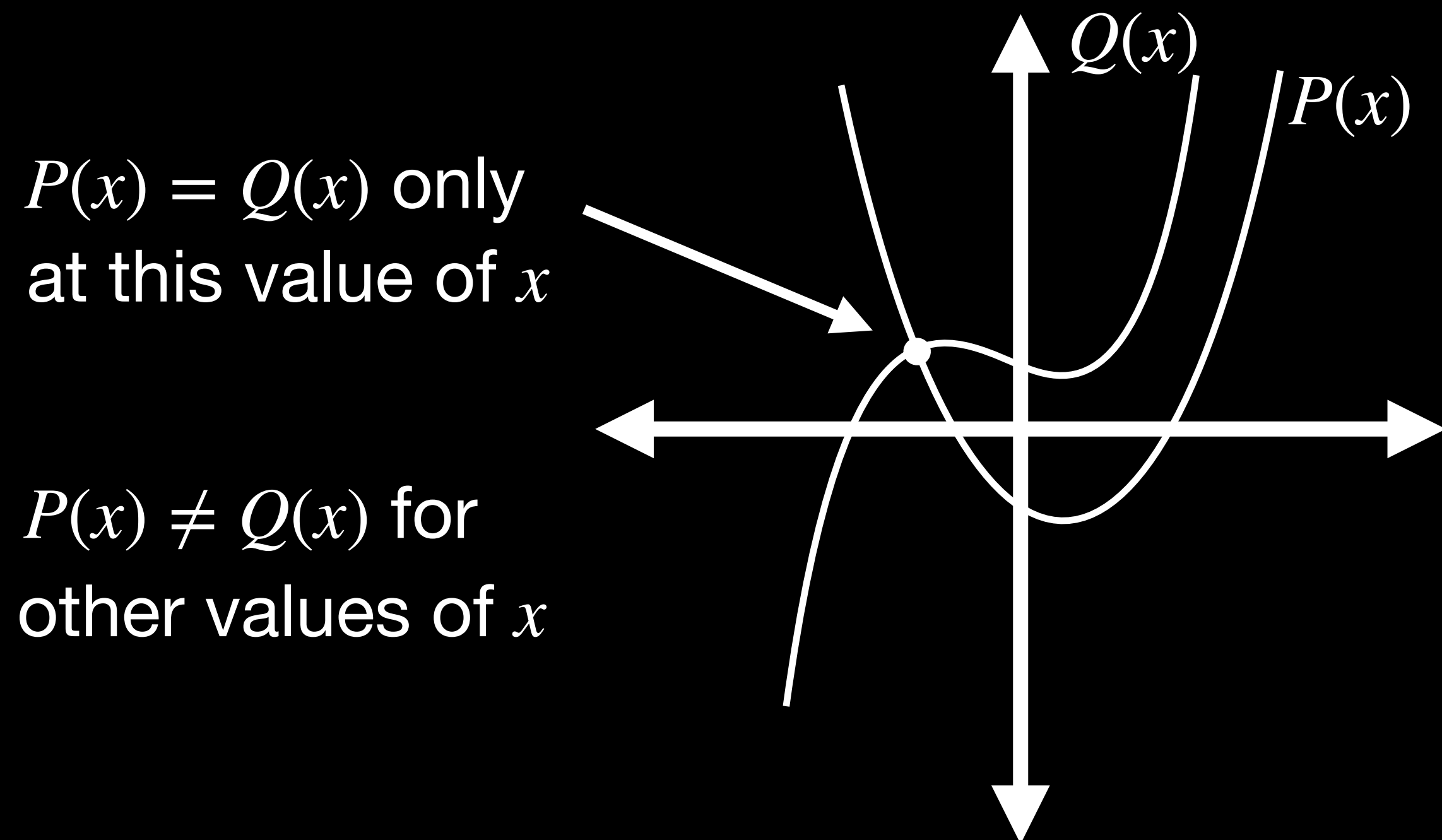
an algebraic expression of the sum of terms x^n ,
where n is a positive integer



Identities

i

For two functions $P(x)$ and $Q(x)$:
When $P(x) = Q(x)$ for all values of x ,
 $P(x) \equiv Q(x)$



Unknowns and Identities - Coefficients

bruh

$$2x^2 - 11x - 40 \equiv A(x - 5)(x + 2) - B(x + 4)$$

$$2x^2 - 11x - 40 = A(x^2 - 3x - 10) - B(x + 4)$$

$$2x^2 - 11x - 40 = Ax^2 - 3Ax - 10A - Bx + 4B$$

$$2x^2 - 11x - 40 = Ax^2 - 3Ax - Bx - 10A - 4B$$

$$2x^2 - 11x - 40 = Ax^2 + (-3A - B)x - 10A - 4B$$

comparing coefficients,

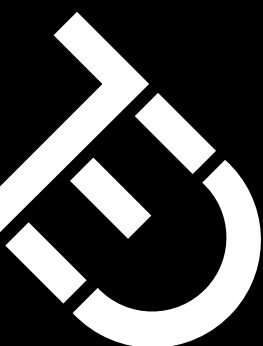
$$A = 2 \quad -3A - B = -11$$

$$-3(2) - B = -11$$

$$-B = -11 + 6$$

$$-B = -5$$

$$B = 5$$



Unknowns and Identities - Substitution

bruh

Given that $2x^2 - 11x - 40 \equiv A(x - 5)(x + 2) - B(x + 4)$, find A and B.

let $x = -4$

$$2(-4)^2 - 11(-4) - 40 = A((-4) - 5)((-4) + 2) - B((-4) + 4)$$

$$36 = A(-9)(-2)$$

$$36 = 18A$$

$$A = 2$$

Number chosen should let one bracket equal 0

$$-4 + 4 = 0$$

All values multiplied by it will be 0

$$0 \times B = 0, \text{ thus entire term} = 0$$

let $x = -2$

$$2(-2)^2 - 11(-2) - 40 = A((-2) - 5)((-2) + 2) - B((-2) + 4)$$

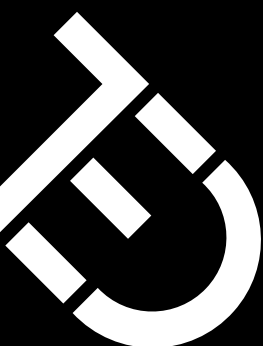
$$-10 = -2B$$

$$B = 5$$

$$-2 + 2 = 0$$

All values multiplied by it will be 0

$$\text{Entire term} = 0$$



Polynomial Long Division

Look long division is back

For $\frac{3x^2 - 12x + 10}{x - 3},$

divisor

$$\begin{array}{r} \text{quotient} \\ \hline \text{dividend} \\ \vdots \\ \hline \text{remainder} \end{array}$$

recall that:

$$\frac{\text{dividend}}{\text{divisor}} = \text{quotient} + \frac{\text{remainder}}{\text{divisor}}$$

multiply by divisor

$$\text{dividend} = (\text{quotient})(\text{divisor}) + \text{remainder}$$

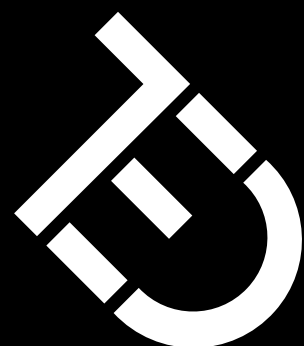
$$\begin{array}{r} 3x \quad -3 \\ \hline x-3 \overline{) 3x^2 - 12x + 10} \\ \underline{-(3x^2 - 9x)} \\ -3x + 10 \\ \underline{-(-3x + 9)} \\ 1 \end{array}$$

1st term of divisor should be the same as dividend

Stop when degree here is less than degree of divisor ($x^0 > x^1$)

$$\frac{3x^2 - 12x + 10}{x - 3} = 3x - 3 + \frac{1}{x - 3}$$

$$3x^2 - 12x + 10 = (x - 3)(3x - 3) + 1$$



Polynomial Synthetic Division

I enjoy

For $\frac{3x^2 - 12x + 10}{x - 3}$,

2. value here makes linear factor = 0
 $x - 3 = 0$
 $x = 3$

1. write polynomial coeffs in order of highest power to lowest power (using 0 for any terms not shown)

3. bring down first coeff

4. multiply value to first coeff (3×3)

5. add together

6. multiply value with added term (aka 3×-3) bring back up and repeat until all coeffs are used

Last value is remainder

3	3	-12	10
	9		
		-9	
	3	-3	1

$\therefore$ quotient is $3x - 3$

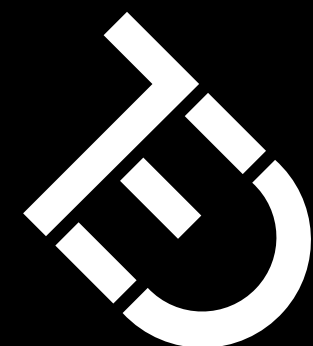
ONLY WORKS FOR LINEAR FACTORS
OF $(x \pm a)$

Quotient is always 1 degree lower

For dividing polynomials like $3x^3 - 2x^2 + 4$, we write the coefficients as so:

3	3	-2	0	4
		9	21	63
	3	7	21	67

Quotient = $3x^2 + 7x + 21$
Remainder = 67



Division Algorithm

Slide Subtitle

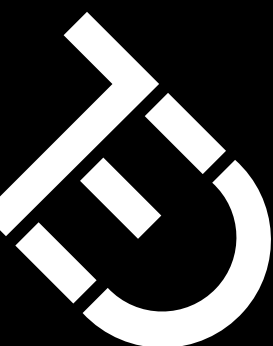
$$\begin{array}{r} \text{quotient} \\ \text{divisor} \overline{) \text{dividend}} \\ \vdots \\ \hline \text{remainder} \end{array}$$

Dividend

= Divisor $\times$ Quotient + Remainder

Dividend degree = Divisor degree + Quotient degree

Divisor degree $>$ Remainder degree



Remainder Theorem

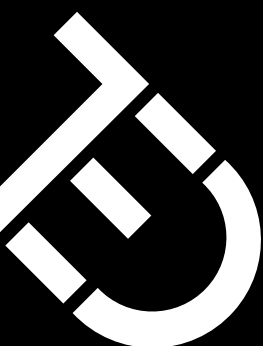
cloud is here now

For a polynomial $f(x)$, Value of x where expr. = 0

$$\text{Remainder of } \frac{f(x)}{(ax - b)} = f\left(\frac{b}{a}\right)$$

$$f(x) = 2x^3 - x^2 - 3x - 2$$

$$\begin{aligned}\text{Remainder of } \frac{f(x)}{2x - 1} &= f(0.5) \\ &= -3.5\end{aligned}$$



Remainder Theorem - Unknowns

cloud is here now

Given $f(x) = x^4 + 3x^3 + px^2 + 30$, When $f(x)$ is divided by $(x + 3)$, its remainder is 72
Find p .

By remainder theorem, $f(-3) = 72$

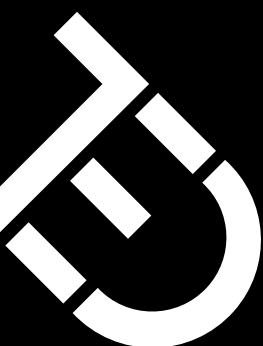
$$(-3)^4 + 3(-3)^3 + p(-3)^2 + 30 = 72$$

$$81 - 81 + 9p + 30 = 72$$

$$9p + 30 = 72$$

$$9p = 42$$

$$p = \frac{14}{3}$$



Remainder Theorem - Unknowns

cloud is here now

Given $f(x) = x^4 + px^3 + qx^2 + 30$, When $f(x)$ is divided by $x^2 + x - 6$, its remainder is $11x$. Find p and q .

For polynomials with 2 unknowns,
form two simultaneous equations and solve

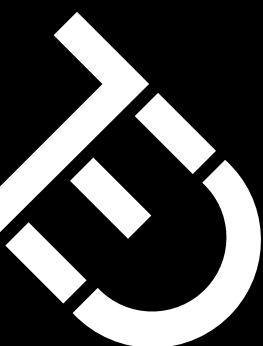
$$x^2 + x - 6 = (x - 3)(x + 2)$$

$$f(3) = 11(3) \text{ (Sub in same value for } x, \text{ then simplify)}$$

$$f(-2) = 11(-2) \text{ (Sub in same value for } x, \text{ then simplify)}$$

Solve as simultaneous equations

$$\text{Ans: } p = 2, q = -10$$



Factor Theorem

My favourite pasttime is to

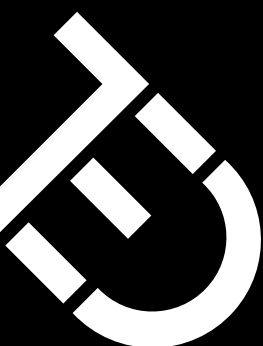
For a polynomial $f(x)$,

When remainder of $\frac{f(x)}{(ax - b)} = 0$

$(ax - b)$ is a factor of $f(x)$, $f(x)$ is exactly divisible by $(ax - b)$

$\therefore$ when $f\left(\frac{b}{a}\right) = 0$

$(ax - b)$ is a factor of $f(x)$, $f(x)$ is exactly divisible by $(ax - b)$



Factor Theorem - Full Factorisation

My favourite pasttime is to

Given that $f(x) = x^3 - 2x^2 - 5x + 6$, show that one of its factors is $(x - 1)$ and factorise $f(x)$

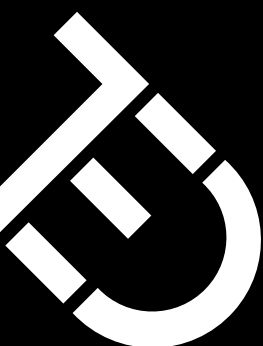
$$f(1) = (1)^3 - 2(1)^2 - 5(1) + 6$$

$$f(1) = 1 - 2 - 5 + 6$$

$$f(1) = 0$$

By factor theorem, $(x - 1)$ is a factor of $f(x)$

$$\begin{aligned} x^3 - 2x^2 - 5x + 6 &= (x - 1)(x^2 - x - 6) \\ &= (x - 3)(x + 2)(x - 1) \end{aligned}$$



Factor Theorem - Common Factors

My favourite pasttime is to

Given that $x^3 - px^2 - x + 30$ and $x^2 - 5x + 6$ both have a common factor, find the possible values of p .

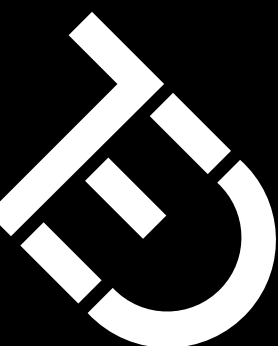
Since one of the polynomials is a quadratic, it has 2 factors max

Thus, quadratic can be factorised to get $(x - 2)(x - 3)$

Let $f(2) = 0$, then solve for p

Let $f(3) = 0$, then solve for p

$\therefore p = 6 \text{ or } 9$



Factor Theorem - Common Factors

My favourite pasttime is to

Given that $ax^2 + (a - k)x + 3$ and $kx^2 + (a + k)x + 3$ both have a common factor of $(x - 1)$, find the values of a and k .

Let each function be $f(x)$ and $g(x)$ respectively:

$$f(1) = 0 \quad 2a - k + 3 = 0 \rightarrow (1)$$

$$g(1) = 0 \quad 2k + a + 3 = 0 \rightarrow (2)$$

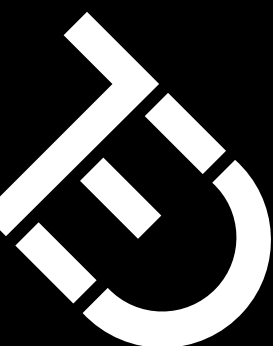
Solve as simultaneous equations

$$\therefore a = -1.8, k = -0.6$$

In summary,

Use factor/remainder theorem to form equations based on different values of x .

Then simplify and solve them accordingly



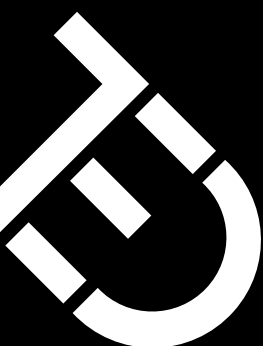
Factorising Cubics - Identities

Thank you for purchasing the quadratic+ package

$$a^3 + b^3 = (a + b)(a^2 - ab + b^2)$$

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

sum/difference of cubes
NOT IN FORMULA SHEET



Factorising Cubics - Algorithm

Thank you for purchasing the quadratic+ package

Given a cubic polynomial $f(x)$ (where degree is 3)

1. Find out one factor of the cubic

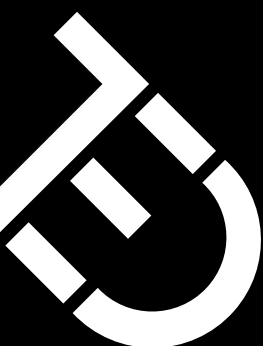
Use calculator to factorise and pick an integer factor to use

2. Divide said factor by cubic

Synthetic division/Long division

3. Factorise quadratic quotient

cross factorisation



Factorising Cubics - Algorithm

Thank you for purchasing the quadratic+ package

Factorise the cubic polynomial $x^3 + 6x^2 + 5x - 12$

$$\text{Let } f(x) = x^3 + 6x^2 + 5x - 12$$

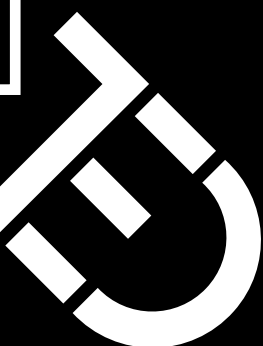
$$\begin{aligned} \text{let } x &= -3, \\ f(-3) &= (-3)^3 + 6(-3)^2 + 5(-3) - 12 \\ &= 0 \end{aligned}$$

$\therefore$ by the factor theorem, $(x+3)$ is a factor of $f(x)$

$$\begin{aligned} x^3 + 6x^2 + 5x - 12 &= (x + 3)(x^2 + 3x - 4) \\ &= (x + 3)(x + 4)(x - 1) \end{aligned}$$

To pick what value to sub in,
use calculator to factorise

refer to exam tips section



Factorising Cubics - Algorithm

Thank you for purchasing the quadratic+ package

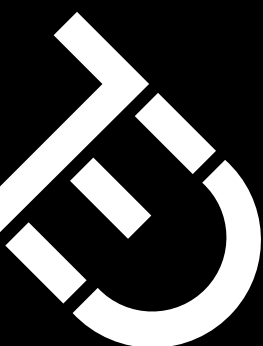
Thus, solve $x^3 + 6x^2 + 5x - 12 = 0$

$$x^3 + 6x^2 + 5x - 12 = 0$$

$$(x + 3)(x + 4)(x - 1) = 0$$

$$(x + 3) = 0, (x + 4) = 0, (x - 1) = 0$$

$$x = -3, x = -4, x = 1$$



Factorising Cubics - Algorithm

Thank you for purchasing the quadratic+ package

Factorise the cubic polynomial $2x^3 - 9x^2 + 6x + 8$

$$\text{Let } f(x) = 2x^3 - 9x^2 + 6x + 8$$

$$\text{let } x = 2,$$

$$f(2) = 2(2)^3 - 9(2)^2 + 6(2) + 8$$

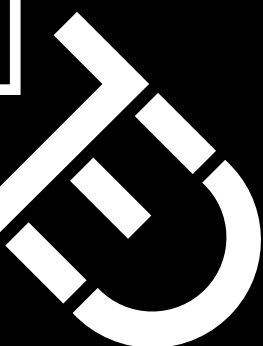
$$f(2) = 0$$

$\therefore$ by the factor theorem, $(x-2)$ is a factor of $f(x)$

$$2x^3 - 9x^2 + 6x + 8 = (x - 2) \underbrace{(2x^2 - 5x - 4)}_{\text{Not factorisable}}$$

To pick what value to sub in,
use calculator to factorise

refer to exam tips section



Factorising Cubics - Substitution

Thank you for purchasing the quadratic+ package

Factorise the cubic polynomial $x^3 - 4x^2 + x + 6$,
and thus, solve for $(y - 3)^3 - 4(y - 3)^2 + (y - 3) = -6$

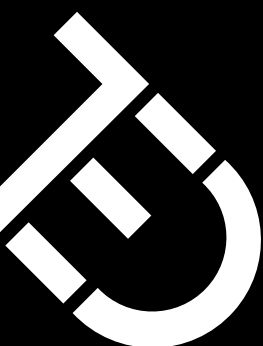
$$x^3 - 4x^2 + x + 6 = (x + 1)(x - 2)(x - 3)$$

$$(y - 3)^3 - 4(y - 3)^2 + (y - 3) = -6$$
$$(y - 3)^3 - 4(y - 3)^2 + (y - 3) + 6 = 0 \quad \left. \begin{array}{l} \nearrow \\ \searrow \end{array} \right\} \text{convert to given cubic}$$

$$((y - 3) + 1)((y - 3) - 2)((y - 3) - 3) = 0$$

$$(y - 2)(y - 5)(y - 6) = 0$$

$$y = 2, y = 5, y = 6$$



Partial Fractions - Improper to Proper

Parrtial fractions more like partial fractions haha

$$\frac{3x^3 - 6x^2 - 5x + 6}{x^2 - 2x + 3} = 3x + \frac{-14x + 6}{x^2 - 2x + 3}$$

divide

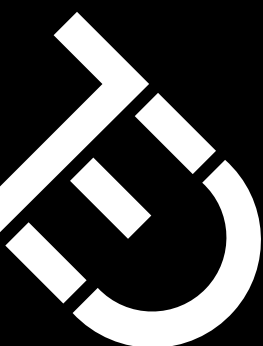
$$\begin{array}{r} x^2 - 2x + 3 \overline{) 3x^3 - 6x^2 - 5x + 6} \\ \underline{(3x^3 - 6x^2 + 9x)} \\ -14x + 6 \end{array}$$

3x

Finding Partial Fractions - Algorithm

Parrtial fractions more like partial fractions haha

1. Check is fraction is proper or improper (if improper, simplify to get a proper fraction)
2. Factorise the fraction and simplify it if possible
3. Form an identity with unknowns (refer to next slide)
4. Solve each unknown with substitution and/or comparing coefficients
5. Substitute unknowns back into original identity as answer and check answer



Finding Partial Fractions - Algorithm

Parrtial fractions more like partial fractions haha

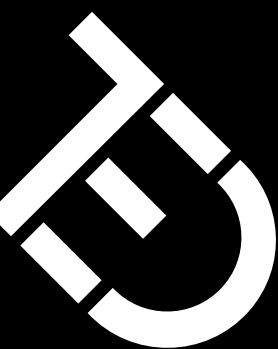
$$\frac{px + q}{(ax + b)(cx + d)} \equiv \overset{\text{1 degree less}}{\left(\frac{A}{ax + b}\right)} + \overset{\text{1 degree less}}{\left(\frac{B}{cx + d}\right)} \quad (\text{Distinct Linear Factors})$$

$$\frac{px + q}{(ax + b)^2} \equiv \overset{\text{1 degree less}}{\left(\frac{A}{ax + b}\right)} + \frac{B}{(ax + b)^2} \quad (\text{Repeated Linear Factors})$$

$$\frac{px + q}{(ax^2 + bx + c)(\text{something})} \equiv \overset{\text{1 degree less}}{\left(\frac{Ax + B}{ax^2 + bx + c}\right)} + \text{something}$$

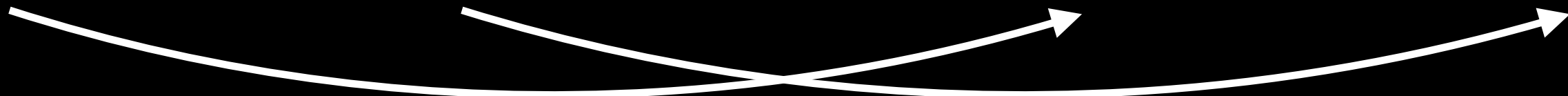
**Clarification on next slide

(Non-Factorisable Quadratic)

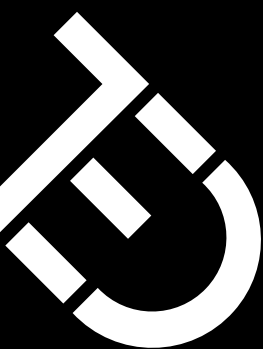


Finding Partial Fractions - Algorithm

Parrtial fractions more like partial fractions haha

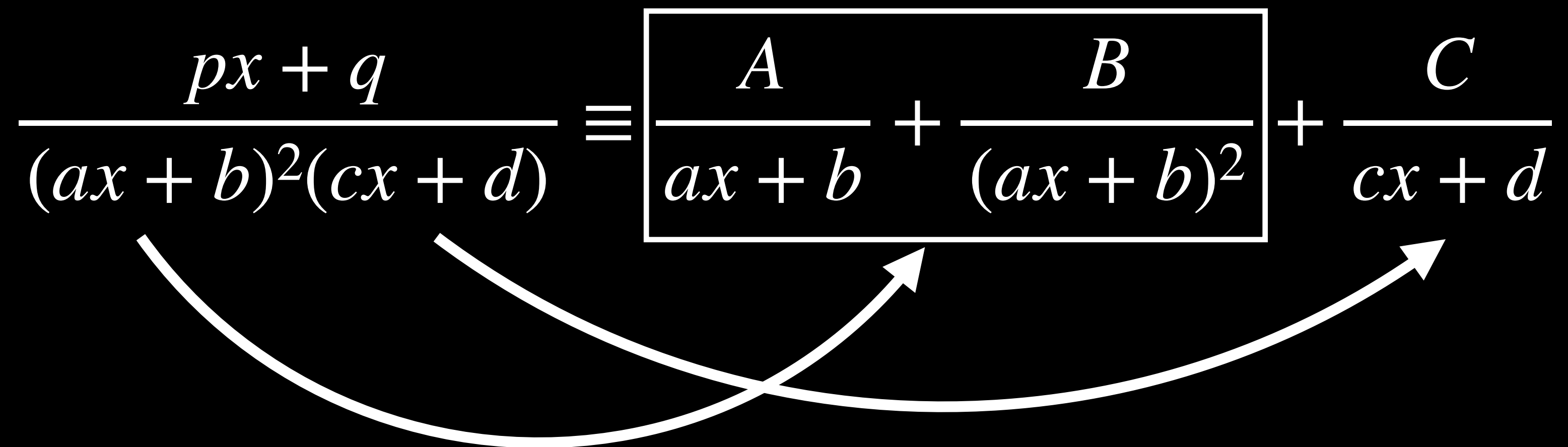
$$\frac{px + q}{(ax^2 + bx + c)(dx + e)} \equiv \frac{Ax + B}{ax^2 + bx + c} + \frac{C}{dx + e}$$


$$\frac{px + q}{(ax^2 + bx + c)^2} \equiv \frac{Ax + B}{ax^2 + bx + c}$$

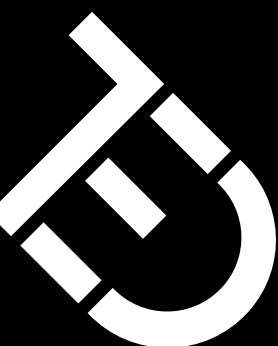


Finding Partial Fractions - Algorithm

Parrtial fractions more like partial fractions haha

$$\frac{px + q}{(ax + b)^2(cx + d)} \equiv \boxed{\frac{A}{ax + b} + \frac{B}{(ax + b)^2}} + \frac{C}{cx + d}$$


Different factors could be combined to form different partial fractions



Finding Partial Fractions - Algorithm

Parrtial fractions more like partial fractions haha

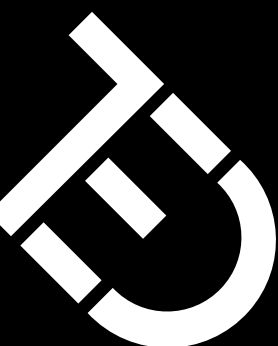
$$\frac{px + q}{(ax + b)(cx + d)} \equiv \frac{A}{ax + b} + \frac{B}{cx + d}$$

use identity symbol at start, use equals for other operations

$$px + q = \frac{A}{\cancel{ax + b}} ((\cancel{ax + b})(cx + d)) + \frac{B}{\cancel{cx + d}} ((ax + b)(\cancel{cx + d}))$$

$$px + q = A(cx + d) + B(ax + b)$$

Then use comparing coefficients/subsitution to solve for A and B



Finding Partial Fractions

Parrtial fractions more like partial fractions haha

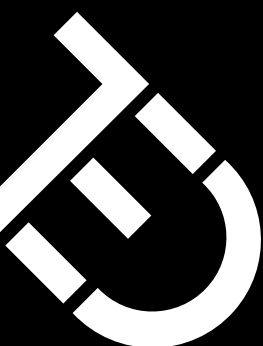
Express $\frac{18}{x^3 + 3x^2 - 4}$ as a sum of partial fractions

$$\frac{18}{x^3 + 3x^2 - 4} = \frac{18}{(x - 1)(x + 2)(x + 2)}$$

1. Check is fraction is proper or improper [PROPER]
2. Factorise the fraction to simplify it

$$= \frac{18}{\underset{\text{Linear Factor}}{(x - 1)}\underset{\text{Repeated Factor}}{(x + 2)^2}}$$

Thus, Let $\frac{18}{(x - 1)(x + 2)^2} \equiv \frac{A}{x - 1} + \frac{B}{(x + 2)} + \frac{C}{(x + 2)^2}$ 3. Form an identity with unknowns

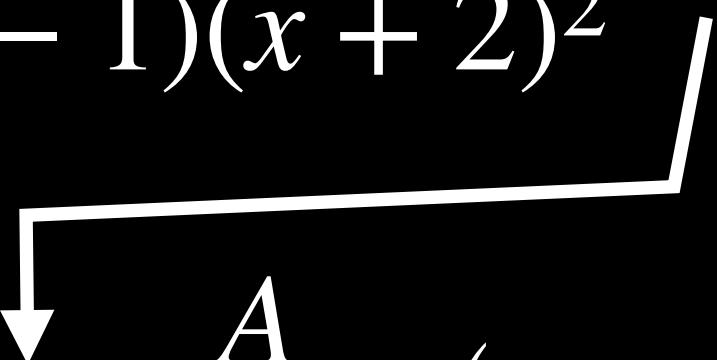


Finding Partial Fractions

Parrtial fractions more like partial fractions haha

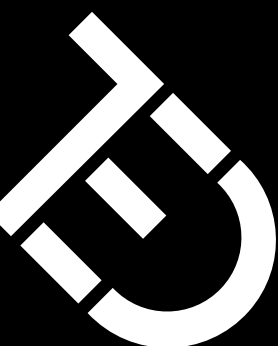
Express $\frac{18}{x^3 + 3x^2 - 4}$ as a sum of partial fractions

$$\frac{18}{(x-1)(x+2)^2} \equiv \frac{A}{x-1} + \frac{B}{(x+2)} + \frac{C}{(x+2)^2} \quad 3. \text{ Form an identity with unknowns}$$


$$18 = \frac{A}{\cancel{x-1}} ((\cancel{x-1})(x+2)^2) + \frac{B}{(\cancel{x+2})} ((x-1)(\cancel{x+2})^2) + \frac{C}{(\cancel{x+2})^2} ((x-1)(\cancel{x+2})^2)$$

this step does not need to be written, only for better understanding

$$18 = A(x+2)^2 + B(x-1)(x+2) + C(x-1)$$



Finding Partial Fractions

Parrtial fractions more like partial fractions haha

Express $\frac{18}{x^3 + 3x^2 - 4}$ as a sum of partial fractions

$$18 = A(x + 2)^2 + B(x - 1)(x + 2) + C(x - 1) \quad \text{4. Solve each identity with substitution and/or comparing coefficients}$$

Solve via substitution:

Let $x = -2$: removes any terms with $(x+2)$, isolates C

$$18 = A(-2 + 2)^2 + B(-2 - 1)(-2 + 2) + C(-2 - 1)$$

$$18 = C(-2 - 1)$$

$$18 = -3C$$

$$C = -6$$

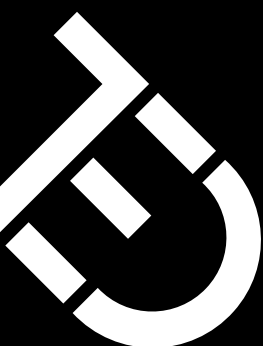
Let $x = 1$: removes any terms with $(x-1)$, isolates A

$$18 = A(1 + 2)^2 + B(1 - 1)(-2 + 2) + C(1 - 1)$$

$$18 = A(1 + 2)^2$$

$$18 = 9A$$

$$A = 2$$



Finding Partial Fractions

Parrtial fractions more like partial fractions haha

Express $\frac{18}{x^3 + 3x^2 - 4}$ as a sum of partial fractions

4. Solve each identity with substitution and/or comparing coefficients

Solve via substitution:

$$18 = A(x + 2)^2 + B(x - 1)(x + 2) + C(x - 1)$$

$$18 = 2(x + 2)^2 + B(x - 1)(x + 2) - 6(x - 1)$$

Let $x = 0$: If substitutions of x cannot isolate other unknowns, use $-1, 0, 1$ and sub in unknowns previously found to find remaining unknowns

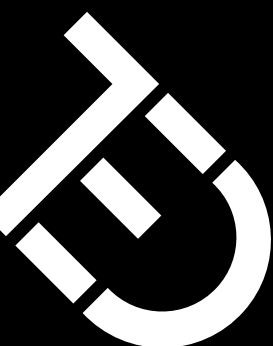
$$18 = 2(0 + 2)^2 + B(0 - 1)(0 + 2) - 6(0 - 1)$$

$$18 = 8 - 2B + 6$$

$$B = -2$$

5. Substitute unknowns back into original identity as answer and check answer

$$\therefore \frac{18}{x^3 + 3x^2 - 4} = \frac{2}{x - 1} + \frac{-2}{(x + 2)} + \frac{6}{(x + 2)^2}$$



Finding Partial Fractions

Parrtial fractions more like partial fractions haha

Express $\frac{18}{x^3 + 3x^2 - 4}$ as a sum of partial fractions [full working with intermediate steps]

$$\begin{aligned}\frac{18}{x^3 + 3x^2 - 4} &= \frac{18}{(x-1)(x+2)(x+2)} \\ &= \frac{18}{(x-1)(x+2)^2} \\ \frac{18}{(x-1)(x+2)^2} &\equiv \frac{A}{x-1} + \frac{B}{(x+2)} + \frac{C}{(x+2)^2} \\ 18 &= A(x+2)^2 + B(x-1)(x+2) + C(x-1)\end{aligned}$$

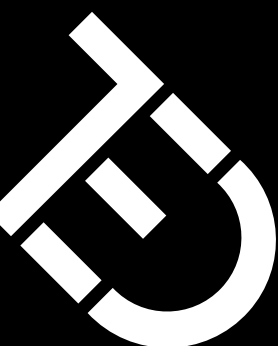
$$\begin{aligned}\text{Let } x &= -2 : \\ 18 &= A(-2+2)^2 + B(-2-1)(-2+2) + C(-2-1) \\ 18 &= C(-2-1) \\ 18 &= -3C \\ C &= -6\end{aligned}$$

$$\begin{aligned}\text{Let } x &= 1 : \\ 18 &= A(1+2)^2 + B(1-1)(-2+2) + C(1-1) \\ 18 &= A(1+2)^2 \\ 18 &= 9A \\ A &= 2\end{aligned}$$

$$\begin{aligned}18 &= A(x+2)^2 + B(x-1)(x+2) + C(x-1) \\ 18 &= 2(x+2)^2 + B(x-1)(x+2) - 6(x-1)\end{aligned}$$

$$\begin{aligned}\text{Let } x &= 0 : \\ 18 &= 2(0+2)^2 + B(0-1)(0+2) - 6(0-1) \\ 18 &= 8 - 2B + 6 \\ 18 &= 14 - 2B \\ 4 &= -2B \\ B &= -2\end{aligned}$$

$$\therefore \frac{18}{x^3 + 3x^2 - 4} = \frac{2}{x-1} + \frac{-2}{(x+2)} + \frac{6}{(x+2)^2}$$



Finding Partial Fractions

Parrtial fractions more like partial fractions haha

Express $\frac{2x^4 - 12x^3 + 8x^2 - 40x + 4}{(x^2 + 4)(x - 6)}$ as a sum of partial fractions

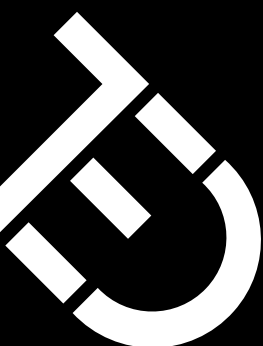
$$\frac{2x^4 - 12x^3 + 8x^2 - 40x + 4}{(x^2 + 4)(x - 6)} = 2x + \frac{8x + 4}{(x^2 + 4)(x - 6)}$$
$$\begin{array}{r} x^3 - 6x^2 + 4x - 24 \overline{) 2x^4 - 12x^3 + 8x^2 - 40x + 4} \\ \underline{-(2x^4 - 12x^3 + 8x^2 - 48x)} \\ 8x + 4 \end{array}$$

only express fractional portion into partial fractions

Let $\frac{8x + 4}{(x^2 + 4)(x - 6)} \equiv \frac{Ax + B}{x^2 + 4} + \frac{C}{x - 6}$

1. Check is fraction is proper or improper [IMPROPER, DO LONG DIVISION]
2. Factorise the fraction to simplify it [ALREADY SIMPLIFIED]

3. Form an identity with unknowns



Finding Partial Fractions

Parrtial fractions more like partial fractions haha

Express $\frac{2x^4 - 12x^3 + 8x^2 - 40x + 4}{(x^2 + 4)(x - 6)}$ as a sum of partial fractions

$$\frac{8x + 4}{(x^2 + 4)(x - 6)} \equiv \frac{Ax + B}{x^2 + 4} + \frac{C}{x - 6}$$

$$8x + 4 = (Ax + B)(x - 6) + C(x^2 + 4)$$

3. Form an identity with unknowns
4. Solve each identity with substitution and/or comparing coefficients

Let $x = 6$:

$$8(6) + 4 = (6A + B)(6 - 6) + C(6^2 + 4)$$

$$52 = 40C$$

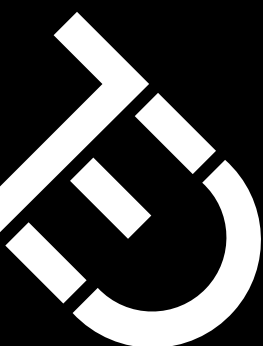
$$C = 1.3$$

Let $x = 0$:

$$4 = (B)(-6) + C(4)$$

$$4 = -6B + 1.3(4)$$

$$B = 0.2$$



Finding Partial Fractions

Partial fractions more like partial fractions haha

Express $\frac{2x^4 - 12x^3 + 8x^2 - 40x + 4}{(x^2 + 4)(x - 6)}$ as a sum of partial fractions

Let $x = 1$: 4. Solve each identity with substitution and/or comparing coefficients

$$8 + 4 = (A + B)(1 - 6) + C(1 + 4)$$

$$12 = -5(A + B) + 5C$$

$$12 = -5(A + 0.2) + 5(1.3)$$

$$12 = -5A - 1 + 6.5$$

$$12 = -5A + 5.5$$

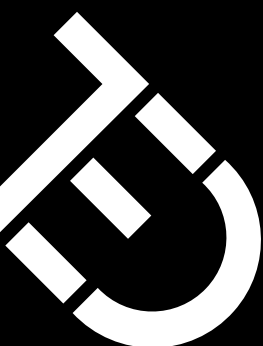
$$12 - 5.5 = -5A$$

$$6.5 = -5A$$

$$A = -1.3$$

5. Substitute unknowns back into original identity as answer and check answer

$$\begin{aligned} \therefore \frac{2x^4 - 12x^3 + 8x^2 - 40x + 4}{(x^2 + 4)(x - 6)} &= 2x + \frac{-1.3x + 0.2}{x^2 + 4} + \frac{1.3}{x - 6} \\ &= 2x + \frac{-\frac{13x}{10} + \frac{2}{10}}{x^2 + 4} + \frac{\frac{13}{10}}{x - 6} \\ &= 2x + \frac{-13x + 2}{10(x^2 + 4)} + \frac{13}{10(x - 6)} \end{aligned}$$



Finding Partial Fractions

Parrtial fractions more like partial fractions haha

Express $\frac{2x^4 - 12x^3 + 8x^2 - 40x + 4}{(x^2 + 4)(x - 6)}$ as a sum of partial fractions [Full solution with intermediate steps]

$$\frac{2x^4 - 12x^3 + 8x^2 - 40x + 4}{(x^2 + 4)(x - 6)} = 2x + \frac{8x + 4}{(x^2 + 4)(x - 6)}$$

$$\text{Let } \frac{8x + 4}{(x^2 + 4)(x - 6)} \equiv \frac{Ax + B}{x^2 + 4} + \frac{C}{x - 6}$$

$$8x + 4 = (Ax + B)(x - 6) + C(x^2 + 4)$$

Let $x = 6$:

$$8(6) + 4 = (6A + B)(6 - 6) + C(6^2 + 4)$$

$$48 + 4 = C(36 + 4)$$

$$52 = 40C$$

$$C = 1.3$$

Let $x = 0$:

$$8(0) + 4 = (0A + B)(0 - 6) + C(0^2 + 4)$$

$$4 = -6B + 4C$$

$$4 = -6B + 4(1.3)$$

$$4 - 5.2 = -6B$$

$$-1.2 = -6B \quad B = 0.2$$

Let $x = 1$:

$$8 + 4 = (A + B)(1 - 6) + C(1 + 4)$$

$$12 = -5(A + B) + 5C$$

$$12 = -5(A + 0.2) + 5(1.3)$$

$$12 = -5A - 1 + 6.5$$

$$12 = -5A + 5.5$$

$$12 - 5.5 = -5A$$

$$6.5 = -5A$$

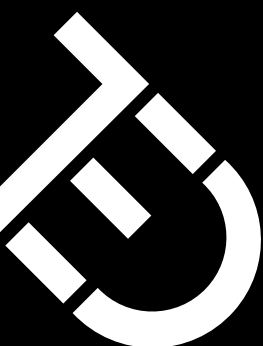
$$A = -1.3$$

$$\therefore \frac{2x^4 - 12x^3 + 8x^2 - 40x + 4}{(x^2 + 4)(x - 6)} = 2x + \frac{-1.3x + 0.2}{x^2 + 4} + \frac{1.3}{x - 6}$$

$$= 2x + \frac{-\frac{13x}{10} + \frac{2}{10}}{x^2 + 4} + \frac{\frac{13}{10}}{x - 6}$$

$$= 2x + \frac{-13x + 2}{10(x^2 + 4)} + \frac{13}{10(x - 6)}$$

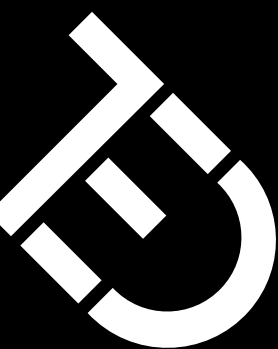
$$\begin{array}{r} x^3 - 6x^2 + 4x - 24 \overline{) 2x^4 - 12x^3 + 8x^2 - 40x + 4} \\ \underline{-(2x^4 - 12x^3 + 8x^2 - 48x)} \\ 8x + 4 \end{array}$$



Partial Fractions - Notes

Parrtial fractions more like partial fractions haha

1. They are most likely going to ask you to find three unknowns, it is very unlikely that they will ask for four, but it is not impossible
2. Remember to substitute the values of the unknowns back into the identity, and back into the proper fraction form if the given fraction was improper



Binomial Theorem



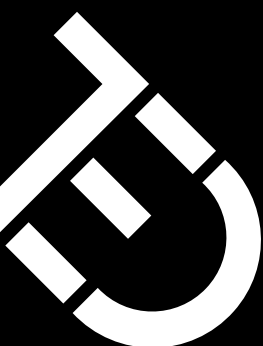
Binomial Expansion

Beep

where n is a pos integer

$$(a + b)^n = (a + b)(a + b) \dots (a + b)$$

Binomial Expansion



Pascals Triangle

pasece

			1			
		1	1			
	1	2	1			
	1	3	3	1		
1	4	6	4	1		
1	5	10	10	5	1	
						⋮

*Should only be used for $0 \leq n \leq 6$

0 Represents coeff of binomial expansion

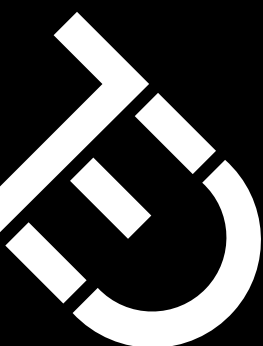
1 $(a + b)^3 = 1a^3 + 3a^2b + 3b^2a + 1b^3$

2 $\xrightarrow{\text{power of 1st term decreases}}$

3 $\xrightarrow{\text{power of 2nd term increases}}$

4 $(a - b)^3 = 1a^3 - 3a^2b + 3b^2a - 1b^3$

5 $\xleftarrow{\text{alternating sign}} \quad \uparrow \quad \nearrow$



Factorials

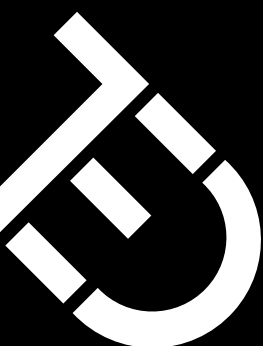
!!!!!!!!!!

$$n! = n(n - 1)(n - 2) \dots (1)$$

$$4! = 4 \times 3 \times 2 \times 1$$

factorials

note: $0! = 1$
 $(-n)! = \text{undefined}$

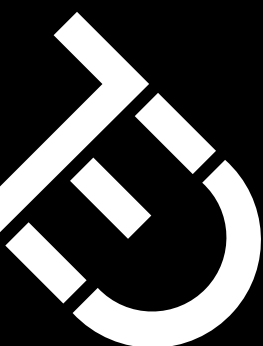


Combinatorial Numbers

C

$${}_nC_r = \binom{n}{r} = \frac{\overbrace{n!}^{\text{O level formula}}}{r!(n-r)!} = \frac{\overbrace{n(n-1)(n-2)\dots(n-r+1)}^{\text{in practice formula; numerator will have } r \text{ terms decreasing from } n}}{r!}$$

combinatorial numbers, where $n \geq r$ and n, r are positive integers



Combinatorial Numbers - Unknowns

C

solve $\binom{n+1}{2} = 3$

numerator has 2 terms that decrease by 1 starting from $(n+1)$

$$\frac{(n+1)(n)}{2!} = 3$$

$$\frac{n^2 - n}{2} = 3$$

$$n^2 - n = 6$$

$$n^2 - n - 6 = 0$$

$$(n+2)(n-3) = 0$$

$$n = -2, n = 3$$

reject $n = -2$ as $n+1 > 0$

$$\therefore n = 3$$

solve $\binom{n}{5} = \binom{n}{7}$

numerator has 5 terms

$$\frac{n(n-1)(n-2)(n-3)(n-4)}{5!} = \frac{n(n-1)(n-2)(n-3)(n-4)(n-5)(n-6)}{7!}$$

$$\cancel{n(n-1)(n-2)(n-3)(n-4)} 7! = \cancel{n(n-1)(n-2)(n-3)(n-4)} (n-5)(n-6) 5!$$

terms cancel out (no need to worry abt dividing by 0 as $n > 0$)

$$7! = (n-5)(n-6) 5!$$

$$(n-5)(n-6) = \frac{7!}{5!}$$

$$n^2 - 6n - 5n + 30 = 42$$

$$n^2 - 11n + 30 = 42$$

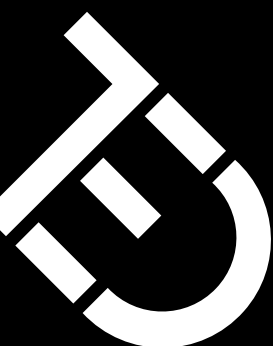
$$n^2 - 11n - 12 = 0$$

$$(n+1)(n-12) = 0$$

$$n = -1, 12$$

reject $n = -1$ as $x > 0$

$$\therefore n = 12$$

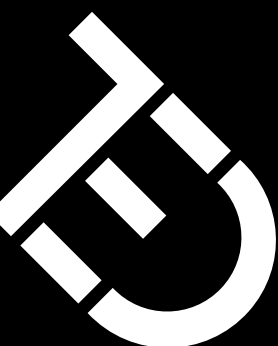


Full Binomial Theorem

real

$$(a + b)^n = a^n + \binom{n}{1} a^{n-1} b^1 + \binom{n}{2} a^{n-2} b^2 + \dots + \binom{n}{r} a^{n-r} b^r + \dots + b^n$$

Full Binomial Expansion

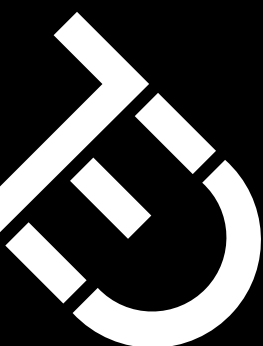


Full Binomial Theorem

real

Expand $(x^2 - 3x)^7$ to the first 5 terms

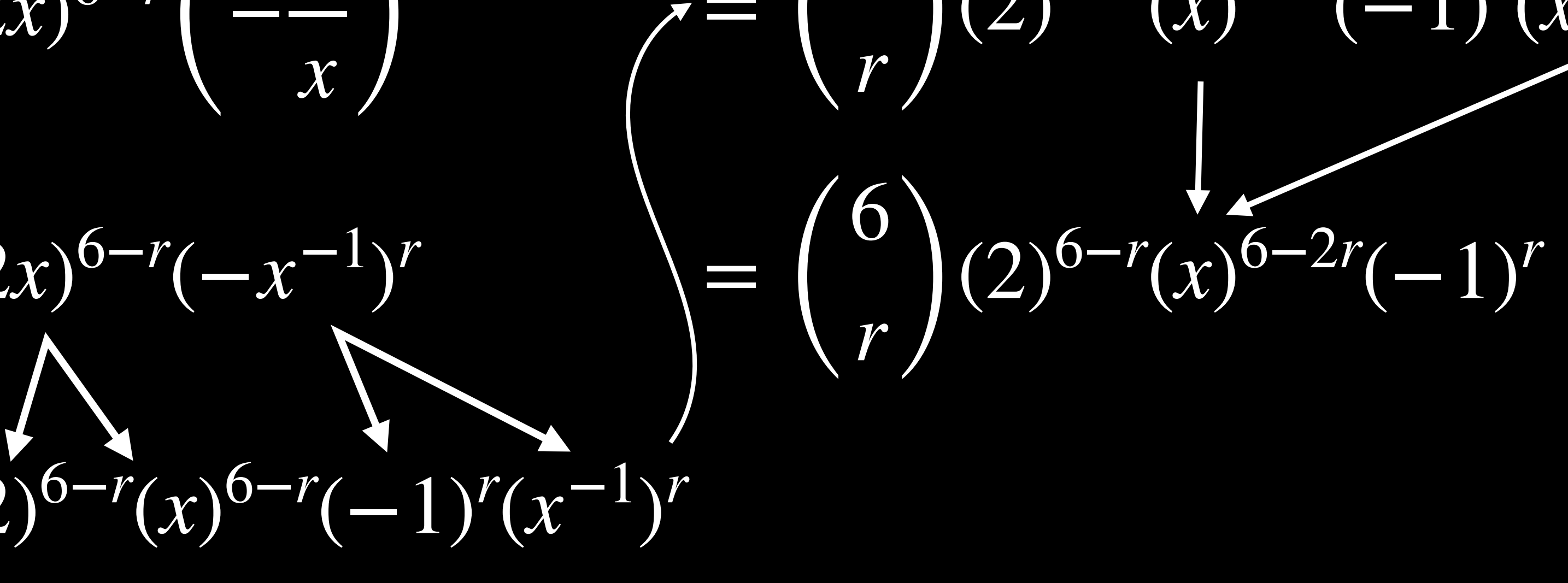
$$\begin{aligned}(x^2 - 3x)^7 &= (x^2)^7 + \binom{7}{1} (x^2)^6(-3x) + \binom{7}{2} (x^2)^5(-3x)^2 + \binom{7}{3} (x^2)^4(-3x)^3 + \binom{7}{4} (x^2)^3(-3x)^4 + \dots \\&= (x^2)^7 + 7 (x^2)^6(-3x) + 21 (x^2)^5(-3x)^2 + 35 (x^2)^4(-3x)^3 + 35 (x^2)^3(-3x)^4 + \dots \\&= x^{14} + 7 (x^{12})(-3x) + 21 (x^{10})(-3)^2x^2 + 35 (x^8)(-3)^3x^3 + 35 (x^6)(-3)^4x^4 + \dots \\&= x^{14} - 21x^{13} + 189x^{12} - 945x^{11} + 2835x^{10} + \dots\end{aligned}$$

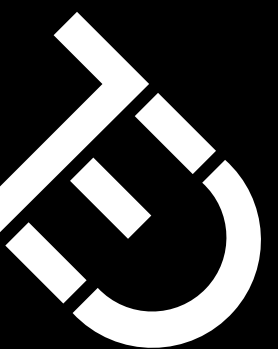


General Form

Wow

$$\left(2x - \frac{1}{x}\right)^6$$

$$\begin{aligned}\text{General Term} &= \binom{6}{r} (2x)^{6-r} \left(-\frac{1}{x}\right)^r \\ &= \binom{6}{r} (2x)^{6-r} (-x^{-1})^r \\ &= \binom{6}{r} (2)^{6-r} (x)^{6-r} (-1)^r (x^{-1})^r \\ &= \binom{6}{r} (2)^{6-r} (x)^{6-2r} (-1)^r\end{aligned}$$




General Form - Independent Term

Wow

Independent term: a term of a binomial expansion where it is only numbers
aka term where x^0

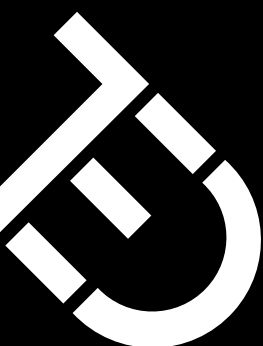
Thus, to find the constant term of $\left(2x - \frac{1}{x}\right)^6$,

$$\text{General Term} = \binom{6}{r} (2)^{6-r} (x)^{6-2r} (-1)^r$$

$$6 - 2r = 0$$

$$r = 3$$

$$\text{Thus, Constant term} = \binom{6}{3} (2)^{6-3} (-1)^3 = -160$$



Applications

n

Given $(3.05)^7$,

$$(3.05)^7 = (3 + 0.05)^7$$

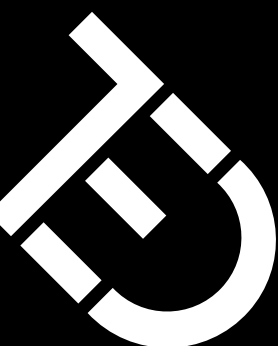
$$\text{Compound Interest} = P \left(1 + \frac{r}{n} \right)^{nt}$$

where P = initial balance

r = interest rate, e.g for 5% increase, $r=0.05$

t = elapsed time in years

n = how many times interest is applied per year



Binomial Theorem Techniques

oops

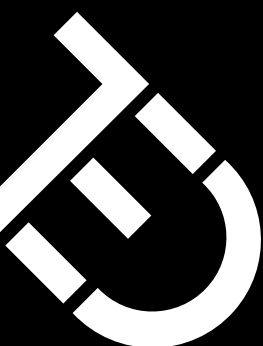
Substitution:

Given you know the expansion of $(x^2 + 3x)^5$:

Find $(x^2 + 3x + x^{-1})^5$

$$(x^2 + 3x + x^{-1})^5 = \left(x^2 + (3x + x^{-1}) \right)^5$$

Substitute $3x$ as $3x + x^{-1}$ in expansion



Binomial Theorem Techniques

oops

Extra terms:

find the coeff. of the x^2 term in the expansion of $(1 - 2x)^2(2 + 3x)^7$

$$(1 - 2x)^2(2 + 3x)^7 = (1 - 4x + 4x^2)(2 + 3x)^7$$

$$= (1 - 4x + 4x^2) \left(2^7 + \binom{7}{1} 2^6(3x) + \binom{7}{2} 2^5(3x)^2 + \dots \right)$$

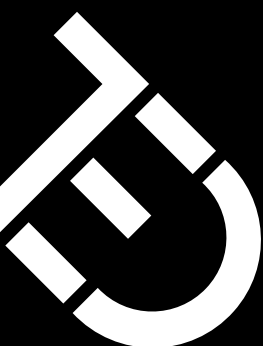
$$= (1 - 4x + 4x^2)(128 + 1344x + 6048x^2 + \dots)$$

only include terms that give x^2 when multiplied

$$= \dots + 6048x^2 + (-4x)(1344x) + 4x^2(128) + \dots$$

$$= \dots + 6048x^2 - 5376x^2 + 512x^2 + \dots$$

$$= \dots + 1184x^2 + \dots$$



Binomial Theorem Techniques

oops

Unknowns:

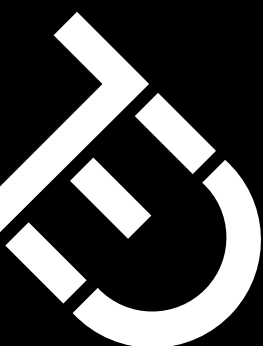
The expansion of $(1 + kx)^5$ is $1 + 20x + 160x^2 + 640x^3 + \dots$, where k is a constant. Find k .

$$\begin{aligned}(1 + kx)^5 &= 1^5 + \binom{5}{1}kx + \binom{5}{2}(kx)^2 + \dots \\ &= 1^5 + 5kx + 10(kx)^2 + \dots\end{aligned}$$

Form equations based on common terms and coefficients and solve accordingly

$$\therefore 5k = 20 \text{ (coefficients of } x \text{ term)}$$

$$k = 4$$



Binomial Theorem Techniques

oops

Unknowns:

The expansion of $(1 + kx)^n$ is $1 + 45x + 810x^2 + \dots$, where k and n are constants. Find k and n .

$$(1 + kx)^n = 1 + \binom{n}{1} 1 (kx) + \binom{n}{2} 1 (kx)^2 + \dots$$

$$= 1 + nkx + \binom{n}{2} k^2 x^2$$

$$\therefore nk = 45$$

$$k = \frac{45}{n} \rightarrow (1)$$

$$\therefore \binom{n}{2} k^2 = 810$$

$$\frac{n(n-1)}{2} k^2 = 810$$

$$n(n-1)k^2 = 1620$$

Sub in (1):

$$n(n-1) \left(\frac{45}{n} \right)^2 = 1620$$

$$n(n-1) \left(\frac{45^2}{n^2} \right) = 1620$$

$$(n^2 - n) \left(\frac{1}{n^2} \right) = \frac{1620}{45^2}$$

$$\frac{n^2}{n^2} - \frac{n}{n^2} = 0.8$$

$$1 - n^{-1} = 0.8$$

$$-n^{-1} = -0.2$$

$$n^{-1} = 0.2$$

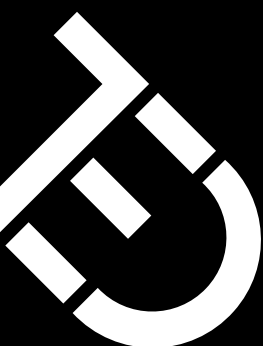
$$n = 0.2^{-1} = 5$$

Sub in $n = 5$ into (1)

$$k = \frac{45}{5} \\ = 9$$

$$\therefore k = 9, n = 5$$

Form equations based on common terms and coefficients and solve accordingly



Exponentials and Logarithms

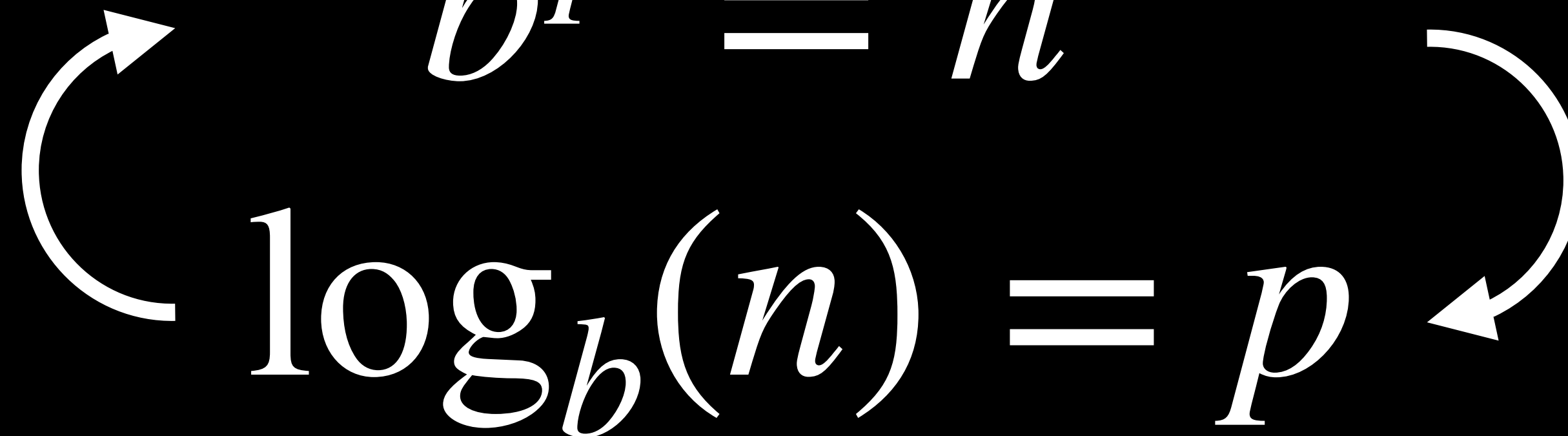


Logarithmic Form

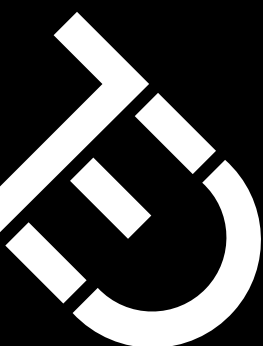
Indices on crack part 3

simple exponential (not an official term)

$$b^p = n$$


$$\log_b(n) = p$$

simple logarithm (not an official term)



Logarithmic Validity

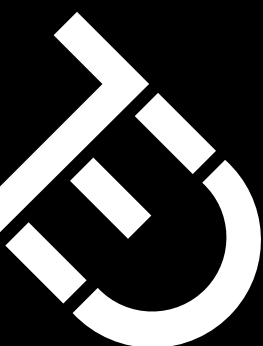
Indices on crack part 3

$$\log_b(n)^{n > 0} = p$$

$$b \neq 1, > 0$$

p will be a real number

$$(p \in \mathbb{R})$$

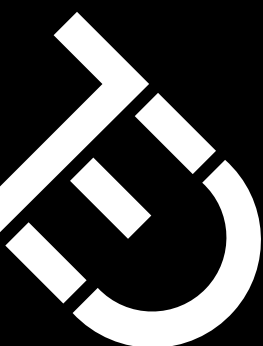


Logarithmic Bases

Indices on crack part 3

$$\log_{10}(x) = \lg(x)$$

$$\log_e(x) = \ln(x)$$



Logarithmic Laws

Indices laws on crack part 3

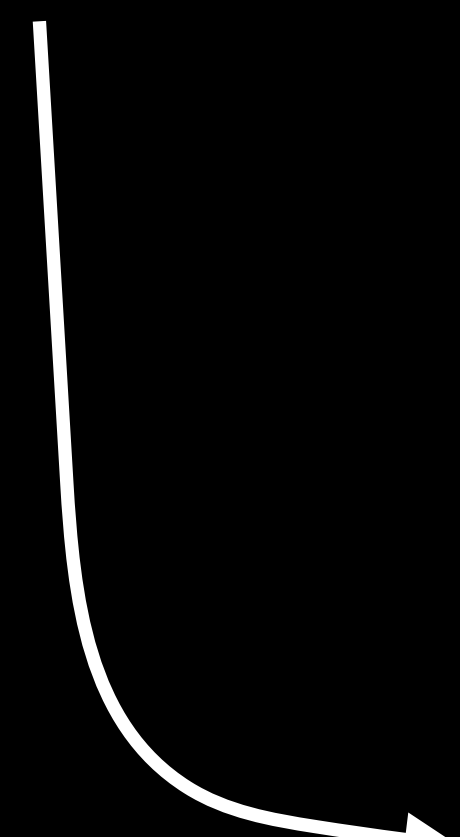
$$\log_a(b) + \log_a(c) = \log_a(bc)$$

$$\log_a(b) - \log_a(c) = \log_a(b \div c)$$

$$\log_a(b^c) = c \log_a(b)$$

$$\log_a(a) = 1$$

$$\log_a(1) = 0$$

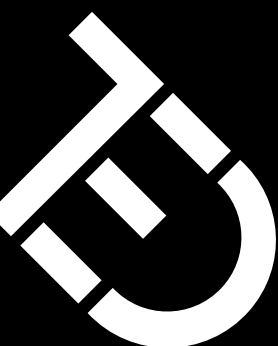

$$\begin{aligned} (\log_a(b))^c &\neq c \log_a(b) \\ (\log_a(b))^c &\neq \log_a(b^c) \end{aligned}$$

****Remember that equal signs are two way; given $2 \log_a(b)$, you can convert to $\log_a(b^2)$**

Logarithm rules only apply when logarithm is valid

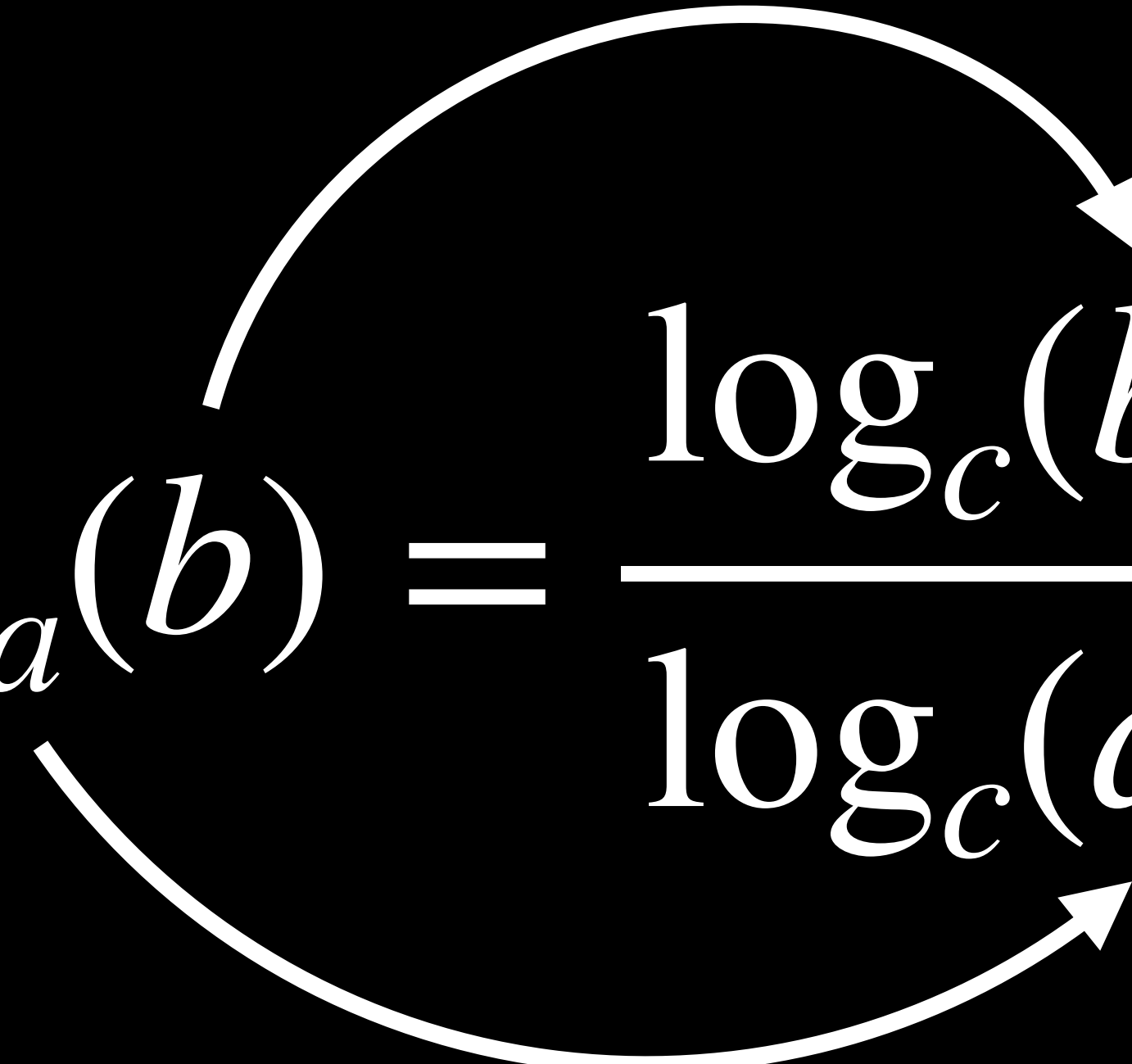
If logarithm is not valid, the logarithm is undefined

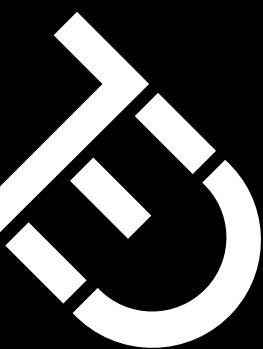
e.g. $\log_1(1) \neq 1$



Logarithmic Laws [Base Change]

Indices laws on crack part 3

$$\log_a(b) = \frac{\log_c(b)}{\log_c(a)}$$




Solving Exponentials - Logarithms

I love non-integers

$$e^{x-5} = 12$$

$$\log_e(12) = x - 5$$

$$\log_e(12) + 5 = x$$

$$\ln(12) + 5 = x$$

$$2e^{5x} = 22$$

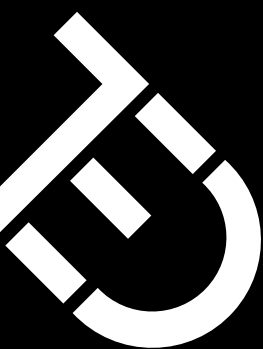
$$e^{5x} = 22 \div 2$$

$$e^{5x} = 11$$

convert to
simple exponential
($b^p = n$)

$$\log_e(11) = 5x$$

$$x = \frac{\ln(11)}{5}$$



Solving Exponentials - Substitution

Good

$$2(3^{2x}) - 5(3^x) = 3$$

$$2(3^{2x}) - 5(3^x) - 3 = 0$$

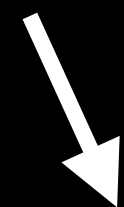
$$2(3^x)^2 - 5(3^x) - 3 = 0$$

find common term

let $3^x = a$:

$$2a^2 - 5a - 3 = 0 \text{ solve quadratic}$$

$$(2a + 1)(a - 3) = 0$$



**MAY NOT BE QUADRATIC, CAN BE CUBIC

$$2a + 1 = 0, a - 3 = 0$$

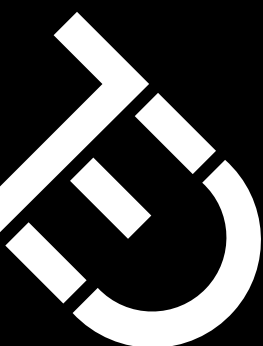
$$a = -0.5, 3$$

$$3^x = -0.5, 3^x = 3$$

(rej. as $3^x > 0$)

$$x = 1$$

$$\therefore x = 1$$



Solving Logarithms - Exponents

Good

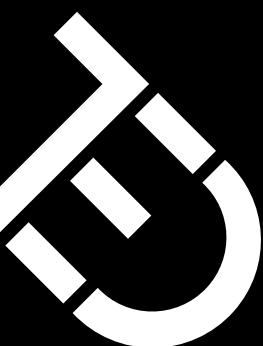
$$\ln(x + 1) = \lg(3e - 1)$$

$$\log_e(x + 1) = \lg(3e - 1)$$

convert to index form

$$e^{\lg(3e-1)} = x + 1$$

$$e^{\lg(3e-1)} - 1 = x$$



Solving Logarithms - Cancellation

cancel

$$\text{solve } \log_5(2x^2 - 3) = \log_5(x - 1) + \log_5(3)$$

$$\log_5(2x^2 - 3) = \log_5(3(x - 1))$$

$$\log_5(2x^2 - 3) = \log_5(3x - 3)$$

$$2x^2 - 3 = 3x - 3$$

$$2x^2 - 3x = 0$$

$$x(2x - 3) = 0$$

$$x = 0, x = 1.5$$

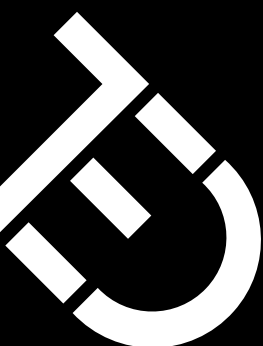
(rej $x=0$. as $\log_5(0 - 1)$ is invalid)

$$\therefore x = 1.5$$

Substitute values into original equation and see if any logarithm is invalid

Both sides should be simplified to simple logarithms before removing logarithm function

****Logarithms can only be removed when both logarithms are of the same base.**



Solving Logarithms - Substitution

cancel

$$\text{Solve } \ln(x)^3 - \ln(x^{\sqrt{2}})^2 = \ln(x^3)$$

$$\ln(x)^3 - \left(\ln(x^{\sqrt{2}})\right)^2 - \ln(x^3) = 0$$

$$\ln(x)^3 - (\sqrt{2} \ln(x))^2 - 3 \ln(x) = 0$$

$$\ln(x)^3 - \sqrt{2}^2 \ln(x)^2 - 3 \ln(x) = 0$$

$$\ln(x)^3 - 2 \ln(x)^2 - 3 \ln(x) = 0$$

find common term

$$\text{Let } \ln(x) = a$$

$$a^3 - 2a^2 - 3a = 0$$

$$a(a^2 - 2a - 3) = 0$$

if there is no constant term,
expression can just be factorised

$$a(a + 1)(a - 3) = 0$$

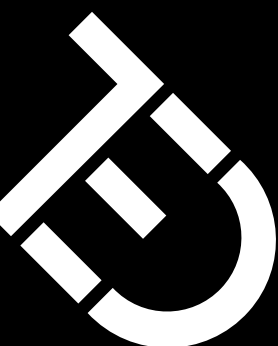
$$a = 0, a = -1, a = 3$$

all solutions are usually valid
as output of log is always real

$$\ln(x) = 0, \ln(x) = -1, \ln(x) = 3$$

$$e^0 = x, e^{-1} = x, e^3 = x$$

$$x = 1, x = 0.368, x = 20.1(3.s.f)$$



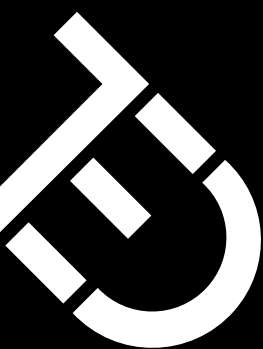
Simultaneous Logarithmic Equations

Simultaneous equations part 2

Solve the following equations simultaneously:

$$\log_2(9 - y) = 1 + \frac{1}{\log_x(2)} \rightarrow (1)$$

$$\frac{(\sqrt{2})^x}{2^y} = \sqrt{8} \rightarrow (2)$$



Simultaneous Logarithmic Equations

Simultaneous equations part 2

$$\log_2(9 - y) = 1 + \frac{1}{\log_x(2)} \rightarrow (1)$$

From (1) :

$$\log_2(9 - y) = 1 + \frac{1}{\log_x(2)} \rightarrow (1)$$

$$\log_2(9 - y) = 1 + (\log_x(2))^{-1}$$

$$\log_2(9 - y) = 1 + \left(\frac{\log_2(2)}{\log_2(x)}\right)^{-1}$$

$$\log_2(9 - y) = 1 + \left(\frac{1}{\log_2(x)}\right)^{-1}$$

$$\log_2(9 - y) = 1 + \log_2(x)$$

$$\log_2(9 - y) - \log_2(x) = 1$$

$$\log_2\left(\frac{9 - y}{x}\right) = 1 \quad \text{convert to simple logarithm}$$

$$2^1 = \frac{9 - y}{x}$$

simplify logarithms to smallest prime base
when unknowns are in log base, use change of base to remove

Alternatively:

$$\log_2(9 - y) = 1 + \log_2(x)$$

$$\log_2(9 - y) = \log_2(2) + \log_2(x)$$

$$\log_2(9 - y) = \log_2(2x)$$

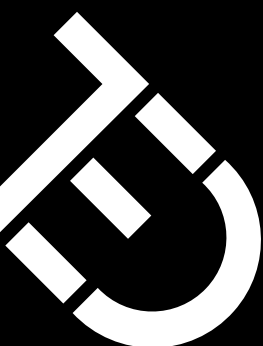
$$2x = 9 - y$$

$$2x = 9 - y$$

$$2x - 9 = -y$$

$$y = 9 - 2x \rightarrow (3)$$

usually will be linear



Simultaneous Logarithmic Equations

Simultaneous equations part 2

$$\frac{(\sqrt{2})^x}{2^y} = \sqrt{8} \rightarrow (2)$$

Main idea: simplify to smallest base, simplify down to linear(usually is linear, though could be quadratic), then substitute

simplify exponentials to smallest prime base

From (2) :

$$\frac{(\sqrt{2})^x}{2^y} = \sqrt{8}$$

$$\frac{(2^{0.5})^x}{2^y} = \sqrt{8}$$

$$\frac{2^{0.5x}}{2^y} = 8^{0.5}$$

$$\frac{2^{0.5x}}{2^y} = (2^3)^{0.5}$$

$$\frac{2^{0.5x}}{2^y} = 2^{1.5}$$

$$2^{0.5x-y} = 2^{1.5}$$

$$0.5x - y = 1.5 \rightarrow (4)$$

Substitute (3) into (4):

$$0.5x - (9 - 2x) = 1.5$$

$$0.5x - 9 + 2x = 1.5$$

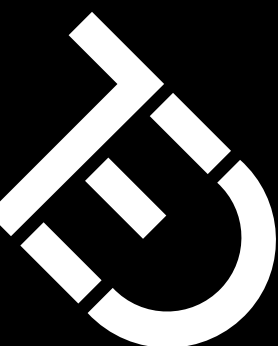
$$2.5x = 10.5$$

$$x = 4.2$$

Substitute $x = 4.2$ into (3):

$$y = 0.6$$

$$\therefore x = 4.2, y = 0.6$$



Simultaneous Logarithmic Equations

Simultaneous equations part 2

Solve the following equations simultaneously:

$$\log_2(9 - y) = 1 + \frac{1}{\log_x(2)} \rightarrow (1)$$

$$\frac{(\sqrt{2})^x}{2^y} = \sqrt{8} \rightarrow (2)$$

From (1) :

$$\log_2(9 - y) = 1 + \frac{1}{\log_x(2)} \rightarrow (1)$$

$$\log_2(9 - y) = 1 + (\log_x(2))^{-1}$$

$$\log_2(9 - y) = 1 + \left(\frac{\log_2(2)}{\log_2(x)}\right)^{-1}$$

$$\log_2(9 - y) = 1 + \left(\frac{1}{\log_2(x)}\right)^{-1}$$

$$\log_2(9 - y) = 1 + \log_2(x)$$

$$\log_2(9 - y) - \log_2(x) = 1$$

$$\log_2\left(\frac{9 - y}{x}\right) = 1$$

$$2^1 = \frac{9 - y}{x}$$

$$2x = 9 - y$$

$$2x - 9 = -y$$

$$y = 9 - 2x \rightarrow (3)$$

From (2) :

$$\frac{(\sqrt{2})^x}{2^y} = \sqrt{8}$$

$$\frac{(2^{0.5})^x}{2^y} = \sqrt{8}$$

$$\frac{2^{0.5x}}{2^y} = 8^{0.5}$$

$$\frac{2^{0.5x}}{2^y} = (2^3)^{0.5}$$

$$\frac{2^{0.5x}}{2^y} = 2^{1.5}$$

$$2^{0.5x-y} = 2^{1.5}$$

$$0.5x - y = 1.5 \rightarrow (4)$$

Substitute (3) into (4):

$$0.5x - (9 - 2x) = 1.5$$

$$0.5x - 9 + 2x = 1.5$$

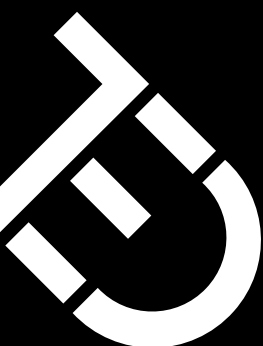
$$2.5x = 10.5$$

$$x = 4.2$$

Substitute $x = 4.2$ into (3):

$$y = 9 - 2(4.2) = 0.6$$

$$\therefore x = 4.2, y = 0.6$$

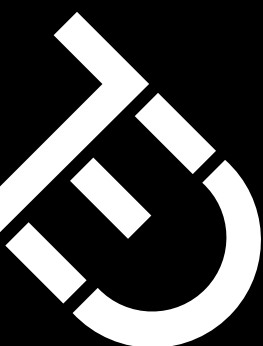


General Rule of Thumb

Simultaneous equations part 2

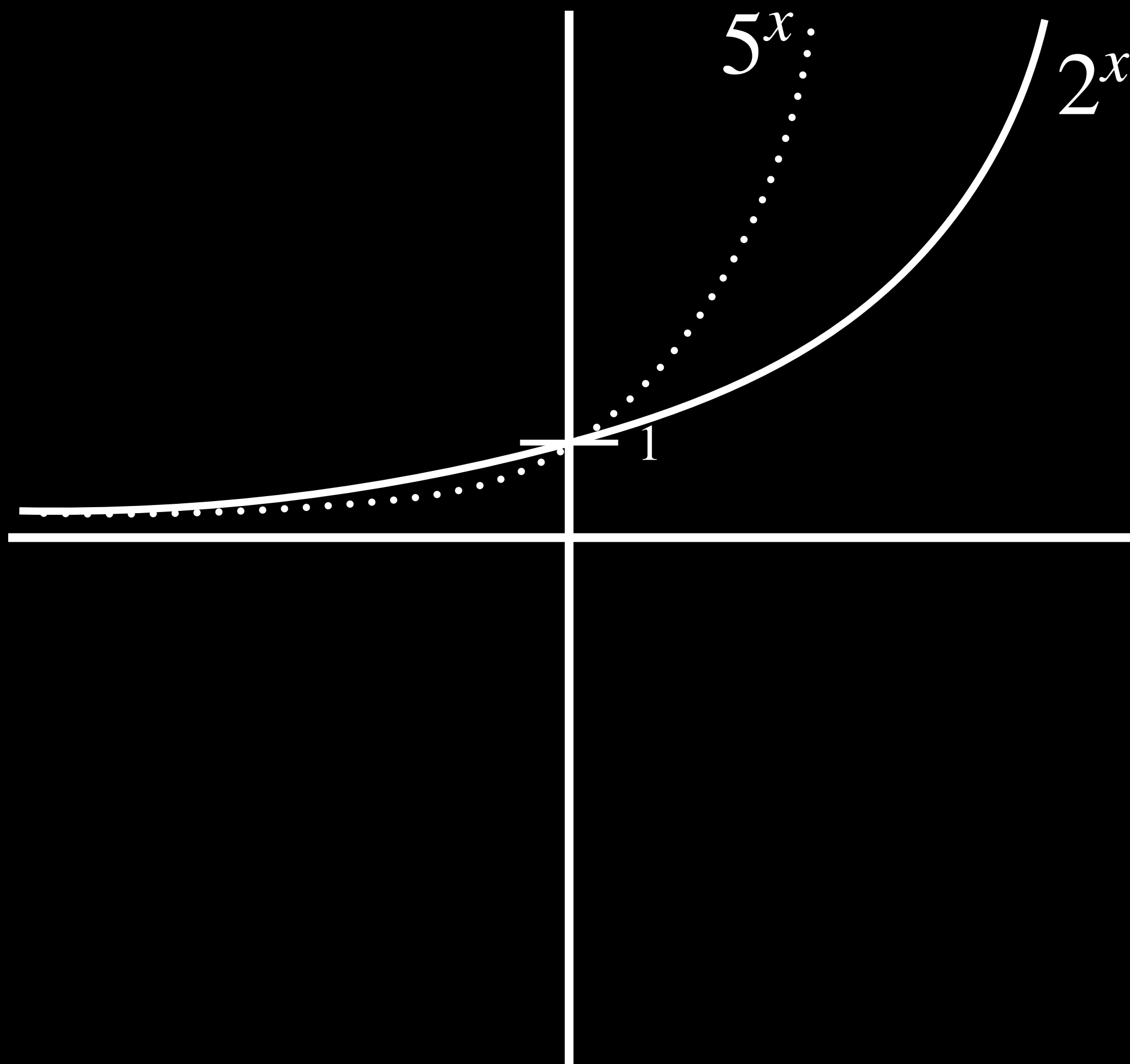
****Try and make the base of the index/logarithm the smallest possible prime**

i.e if your question involves bases like 2, 4 and 8, you should try and make all of the bases 2 (likewise if they involve 3, 9, 27, try and make base 3)

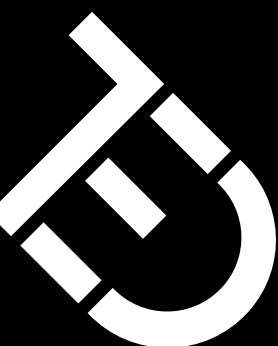


Graphs of Exponentials and Logarithms

Yo I heard you like graphs

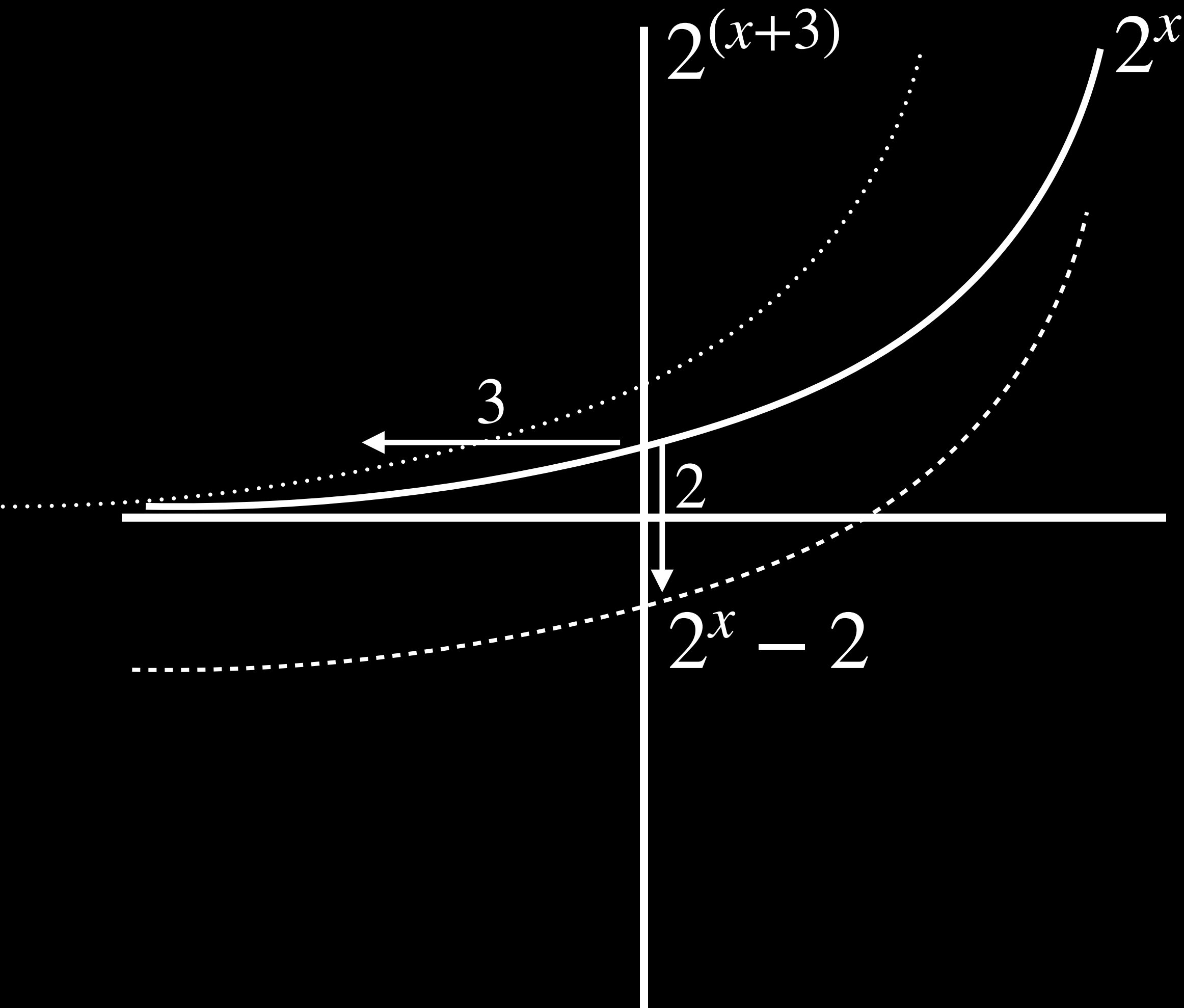


higher base, more extreme graph
cuts y axis at 1
horizontal asymptote of $x = 0$



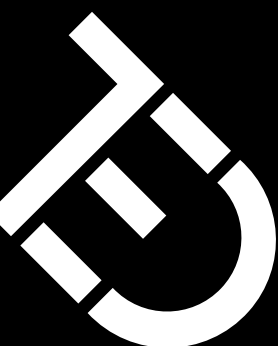
Graphs of Exponentials and Logarithms

Yo I heard you like graphs



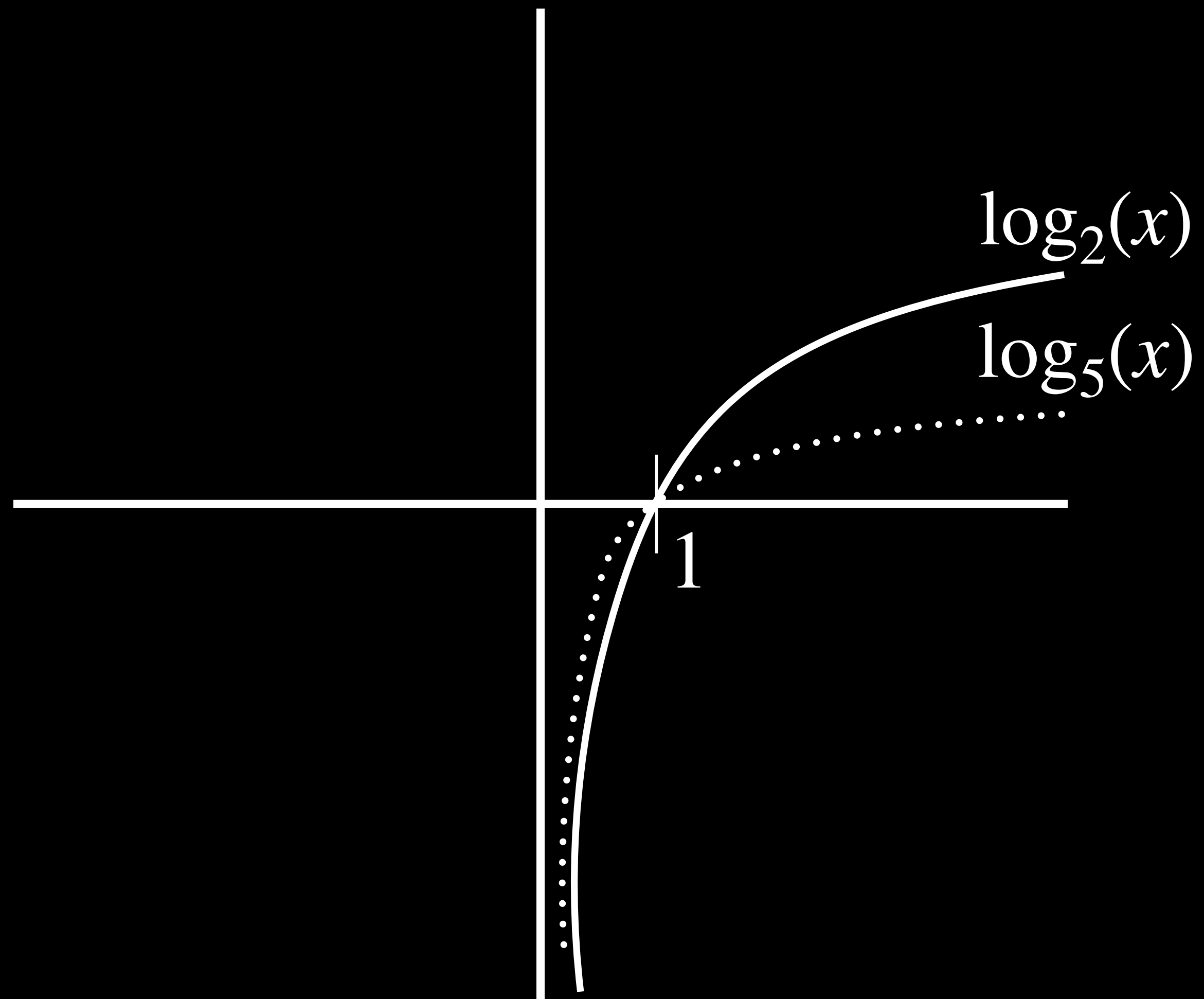
graph transformations still apply

**note that when an exponential is multiplied by a constant k , the y intercept will be k

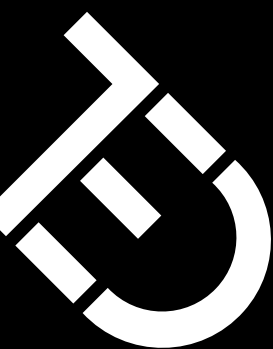


Graphs of Exponentials and Logarithms

Yo I heard you like graphs

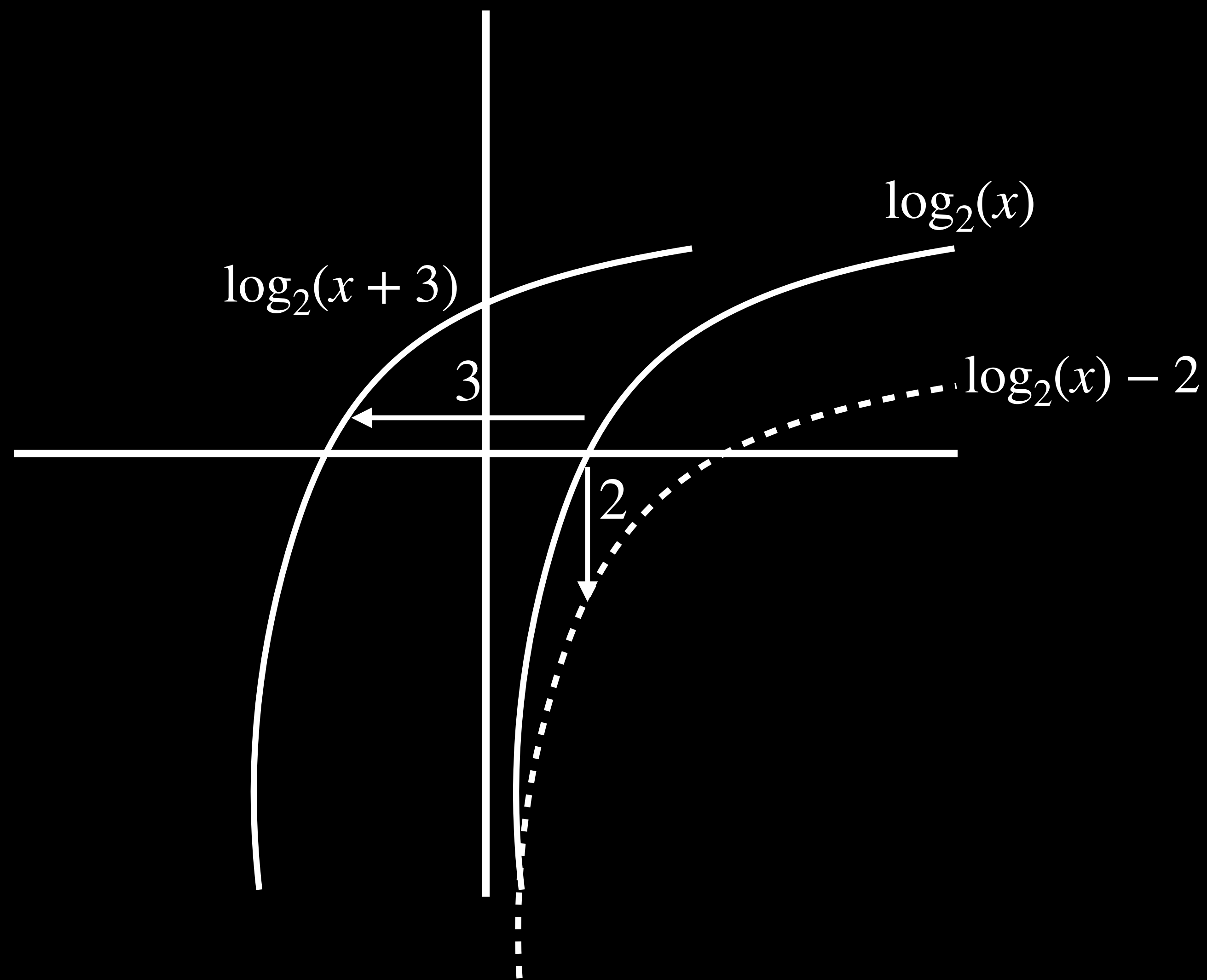


higher base, more extreme graph
cuts x-axis at 1
vertical asymptote at $y = 0$

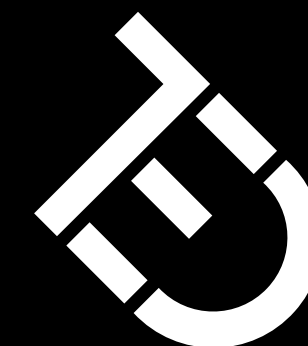


Graphs of Exponentials and Logarithms

Yo I heard you like graphs



graph transformations still apply



Applied Problems (Exponential Growth)

a

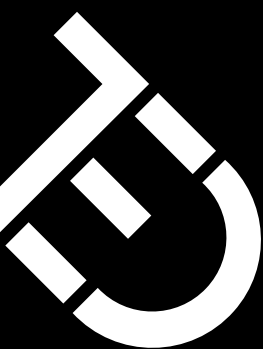
growth rate constant, can be negative

$$y = Ce^{kt}$$

original amount
amount when $t=0$

time

**Does not need to be memorised, just for general purpose

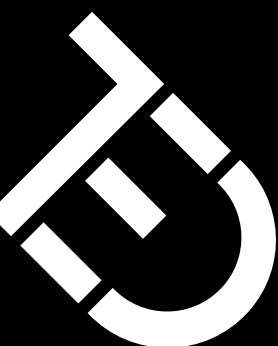


Applied Problems (Exponential Growth)

A

An ice cube is taken out of a freezer of temperature -10°C . The temperature, $T^{\circ}\text{C}$, is defined by the formula $T = 25 - Ae^{-kt}$, where t is the elapsed time in minutes, and A and k are constants.

- a) Show that $A = 35$
- b) After 1 minutes the ice cube's temperature was 1.5°C . Find k .
- c) How much time has elapsed when the ice cube is at 0°C
- d) Will the ice cube ever reach 25°C ? Explain why or why not.



Applied Problems (Exponential Growth)

a

An ice cube is taken out of a freezer of temperature -10°C . The temperature, $T^{\circ}\text{C}$, is defined by the formula $T = 25 - Ae^{-kt}$, where t is the elapsed time in minutes, and A and k are constants.

a) Show that $A = 35$

$$T = 25 - Ae^{-kt}$$

$$-10 = 25 - Ae^{-k(0)}$$

$$-10 = 25 - Ae^0$$

$$-10 = 25 - 1A$$

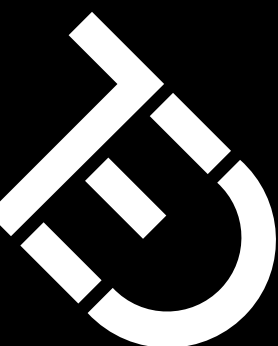
$$-10 = 25 - A$$

$$-10 - 25 = -A$$

$$-A = -35$$

$$A = 35$$

Usually if they ask for the coefficient of the exponential, let $t=0$ and the output be the original value



Applied Problems (Exponential Growth)

a

An ice cube is taken out of a freezer of temperature -10°C . The temperature, $T^{\circ}\text{C}$, is defined by the formula $T = 25 - Ae^{-kt}$, where t is the elapsed time in minutes, and A and k are constants.

b) After 1 minutes the ice cube's temperature was 1.5°C . Find k .

$$T = 25 - Ae^{-kt}$$

$$1.5 = 25 - 35e^{-k(1)}$$

$$1.5 = 25 - 35e^{-k}$$

$$1.5 = 25 - 35e^{-k}$$

$$1.5 - 25 = -35e^{-k}$$

$$-23.5 = -35e^{-k}$$

$$\frac{-23.5}{-35} = e^{-k}$$

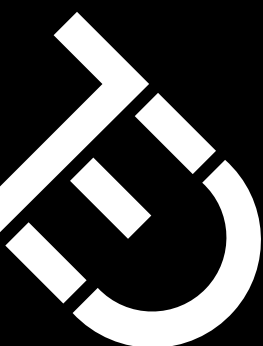
$$\frac{47}{70} = e^{-k}$$

$$\frac{47}{70} = e^{-k}$$

$$\ln\left(\frac{47}{70}\right) = -k$$

Substitute values, then make
unknown the subject

$$k = -\ln\left(\frac{47}{70}\right) \\ = 0.398(3.s.f)$$



Applied Problems (Exponential Growth)

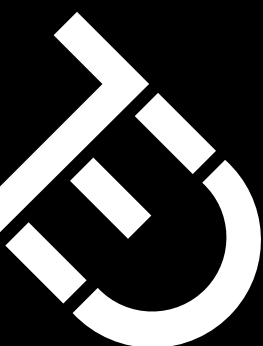
a

An ice cube is taken out of a freezer of temperature -10°C . The temperature, $T^{\circ}\text{C}$, is defined by the formula $T = 25 - Ae^{-kt}$, where t is the elapsed time in minutes, and A and k are constants.

c) How much time has elapsed when the ice cube is at 0°C

$$\begin{aligned}
 T &= 25 - Ae^{-kt} \\
 0 &= 25 - 35e^{-\left(-\ln\left(\frac{47}{70}\right)\right)t} & \frac{5}{7} &= e^{-\left(-\ln\left(\frac{47}{70}\right)\right)t} & \ln\left(\frac{5}{7}\right) \div \ln\left(\frac{47}{70}\right) &= t \\
 -25 &= -35e^{-\left(-\ln\left(\frac{47}{70}\right)\right)t} & \ln\left(\frac{5}{7}\right) &= -\left(-\ln\left(\frac{47}{70}\right)\right)t & t &= 0.845 \text{ minutes (3.s.f)} \\
 \frac{-25}{-35} &= e^{-\left(-\ln\left(\frac{47}{70}\right)\right)t} & \ln\left(\frac{5}{7}\right) &= \ln\left(\frac{47}{70}\right)t
 \end{aligned}$$

Use exact value or 5.s.f value for non rational values to avoid s.f. errors



Applied Problems (Exponential Growth)

a

An ice cube is taken out of a freezer of temperature -10°C . The temperature, $T^{\circ}\text{C}$, is defined by the formula $T = 25 - Ae^{-kt}$, where t is the elapsed time in minutes, and A and k are constants.

d) Will the ice cube ever reach 25°C ? Explain why or why not.

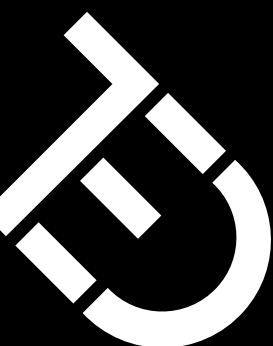
$e^{-kt} > 0$ graph lies above x axis, therefore greater than 0

$-Ae^{-kt} < 0$ multiply both sides by $-A$, flip inequality as negative

$25 - Ae^{-kt} < 25$ add 25

$T < 25$

Thus, the ice will never reach 25°C .



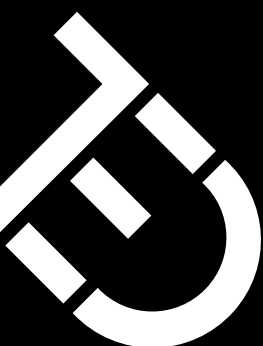
Linear Law



General Linear Equation

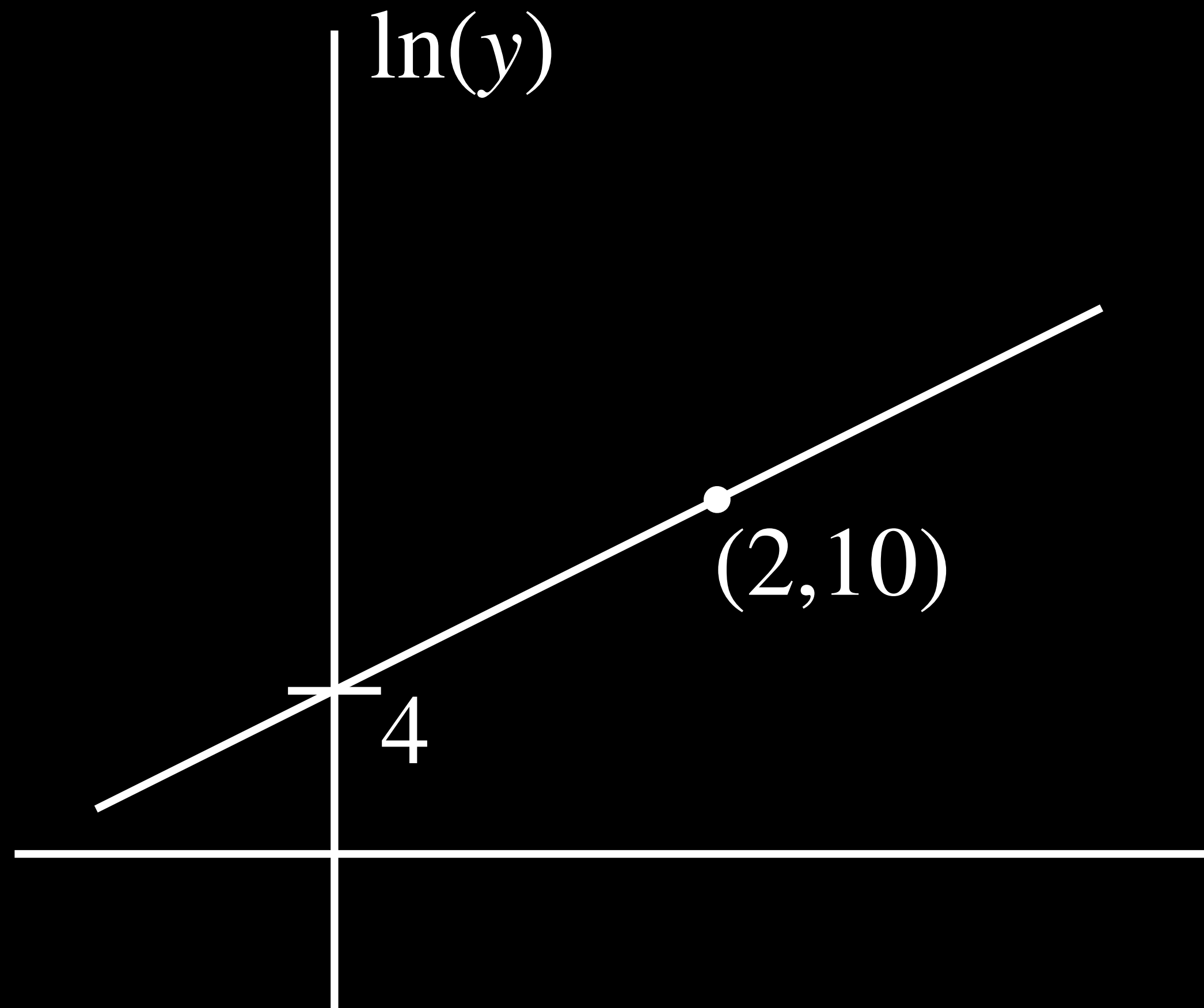
;

$$\begin{array}{ccccccc} & & \text{gradient} & & \text{vertical intercept} & & \\ & & m & & c & & \\ Y & = & X & + & & & \\ \text{vertical variable} & & \text{horizontal variable} & & & & \end{array}$$



Linear to Nonlinear

;



Find gradient:

$$\frac{10 - 4}{2 - 0} = 3$$

Find another point/y-intercept:

$Y - Y_1 = m(X - X_1)$ (Write in caps for linear law)

$$Y - 10 = 3(X - 2)$$

$$Y - 10 = 3X - 6$$

$$Y = 3X + 4$$

$$Y = mX + c$$

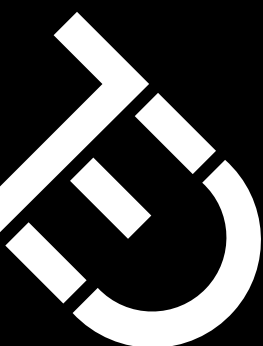
$$Y = 3X + 4 \text{ or just substitute in y intercept}$$

Substitute in X and Y variables:

$$x^2 \quad Y = 3X + 4$$

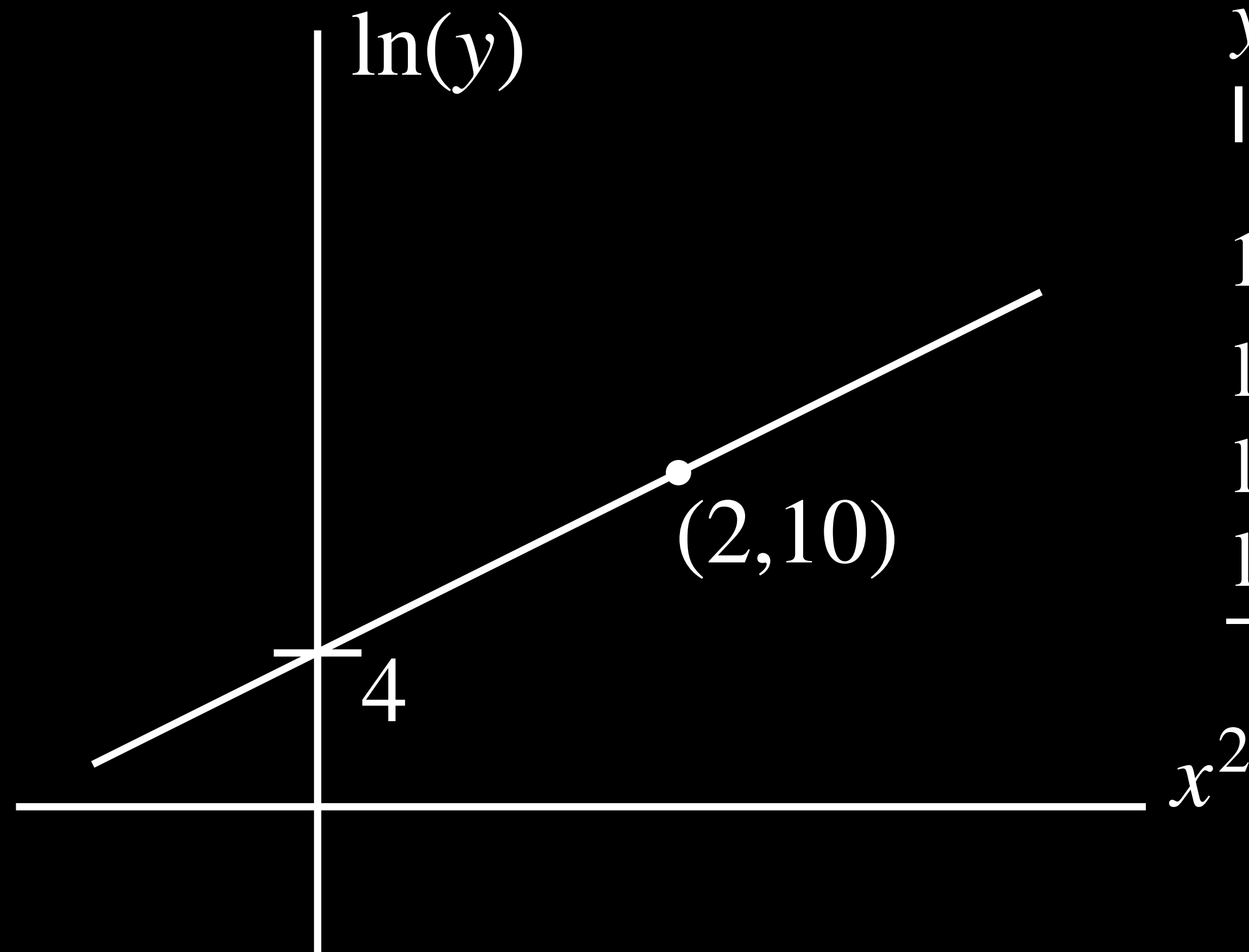
$$\ln(y) = 3x^2 + 4$$

$$y = e^{3x^2+4} \text{ Make y the subj.}$$



Linear to Nonlinear

;



$$y = e^{3x^2+4}$$

If the point $(h, 100)$ lies on the line, find h .

$$100 = e^{3h^2+4} \text{ (Sub in point into equation)}$$

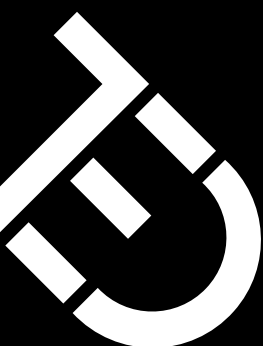
$$\ln(100) = 3h^2 + 4$$

$$\ln(100) - 4 = 3h^2$$

$$\frac{\ln(100) - 4}{3} = h^2$$

make h the subj.

$$h = \pm \sqrt{\frac{\ln(100) - 4}{3}}$$
$$= \pm 0.449$$



Nonlinear to Linear

H moment

Single Term Equations:

$$y = 3e^{2x+1}$$

$$\ln(y) = \ln(3e^{2x+1})$$

$$\ln(y) = \ln(3) + \ln(e^{2x+1})$$

$$\ln(y) = \ln(3) + (2x + 1)\ln(e)$$

$$\ln(y) = \ln(3) + (2x + 1)$$

$$\ln(y) = 2x + 1 + \ln(3)$$

$$\underbrace{\ln(y)}_Y = \underbrace{2x + 1}_{mX} + \underbrace{\ln(3)}_c$$

$$y = ax^n$$

$$\lg(y) = \lg(ax^n)$$

$$\lg(y) = \lg(a) + \lg(x^n)$$

$$\lg(y) = n \lg(x) + \lg(a)$$

$$\underbrace{\lg(y)}_Y = \underbrace{n}_{m} \underbrace{\lg(x)}_X + \underbrace{\lg(a)}_c$$

Take lg() on both sides [or ln() if base is e]
split out the coeff of exponent
Bring down exponent

$$y = ax^n$$

$$\frac{y}{x^{(n-1)}} = \frac{ax^n}{x^{(n-1)}}$$

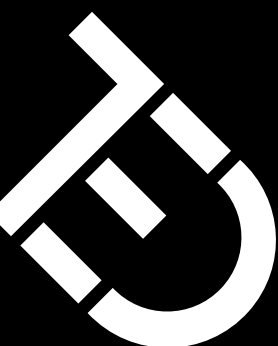
$$\frac{y}{x^{(n-1)}} = ax$$

$$y = 2x^3$$

$$\frac{y}{x^2} = \frac{2x^3}{x^2}$$

$$\frac{y}{x^2} = \frac{2x^3}{x^2} = 2x$$

!! DO NOT DO THIS !!
Y intercept should not
be 0, try using
logarithms for this single
term equations first



Nonlinear to Linear

H moment

Multi Term Equations:

Polynomial Like

$$y = 6x^2 - 4x$$

$$y = x(6x - 4)$$

$$\frac{y}{x} = 6x - 4$$

$$\underbrace{\frac{y}{x}}_Y \quad \underbrace{\quad \quad \quad}_{mX} \quad \underbrace{\quad}_{c}$$

$$y = 6x^5 - 4x^4$$

$$y = x^4(6x - 4)$$

$$\frac{y}{x^4} = 6x - 4$$

$$\underbrace{\frac{y}{x^4}}_Y \quad \underbrace{\quad \quad \quad}_{mX} \quad \underbrace{\quad}_{c}$$

Terms usually have a difference for the degree/power

- Find common term
- Factor out said common term (Remaining terms should be linear)
- Divide out common term to make linear form the subj.

Fractional

$$y = 3\sqrt{x} + \frac{4}{\sqrt{x}}$$

$$y\sqrt{x} = 3x + 4$$

$$\underbrace{\quad}_Y \quad \underbrace{\quad \quad \quad}_{mX} \quad \underbrace{\quad}_{c}$$

$$y = 3x + \frac{4}{x}$$

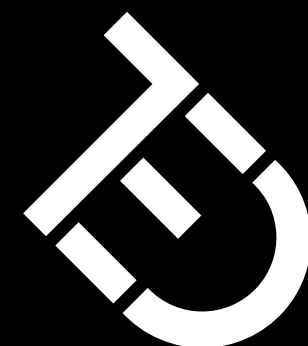
$$xy = 3x^2 + 4$$

$$\underbrace{\quad}_Y \quad \underbrace{\quad \quad \quad}_{mX} \quad \underbrace{\quad}_{c}$$

Terms usually are a+b/a

- Find common term
- Multiply fraction denominator on both sides to get rid of fraction

X variable can still be x^2, despite not seeming like linear form



Nonlinear to Linear

H moment

Multi Term Equations:

Swapped

$$y = \frac{a}{x + b}$$

X and Y variables do not necessarily have to contain x and y respectively

$$y(x + b) = a$$

Get rid of fraction, expand, then remove by from left side to get linear form

$$xy + by = a$$

Try to avoid having constants in the X and Y variables, thus we let xy be the Y variable

$$\underbrace{xy}_Y = \underbrace{-by}_{mX} + \underbrace{a}_c$$

Polynomial+Constant

$$y = 5x^2 + 3x - 4$$

$$y + 4 = 5x^2 + 3x$$

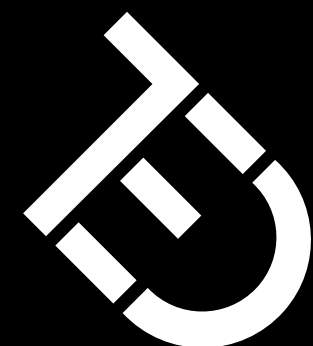
$$y + 4 = x(5x + 3)$$

$$\frac{y + 4}{x} = 5x + 3$$

$$\underbrace{\quad}_Y \quad \underbrace{\quad}_{mX} \quad \underbrace{\quad}_c$$

Equation here contains the polynomial case shown earlier
Thus, the constant -4 can be removed from the right side to get the polynomial case

For this case, having a constant in the Y variable is inevitable, thus we leave it as so



Nonlinear to Linear

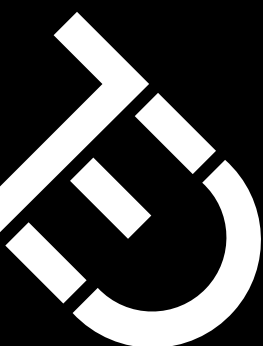
H moment

Multi Term Equations:

Do note that these Multi Term Equations are not exhaustively listed

There may be other forms that were not listed

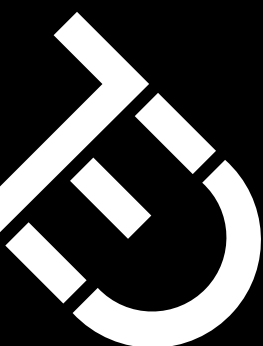
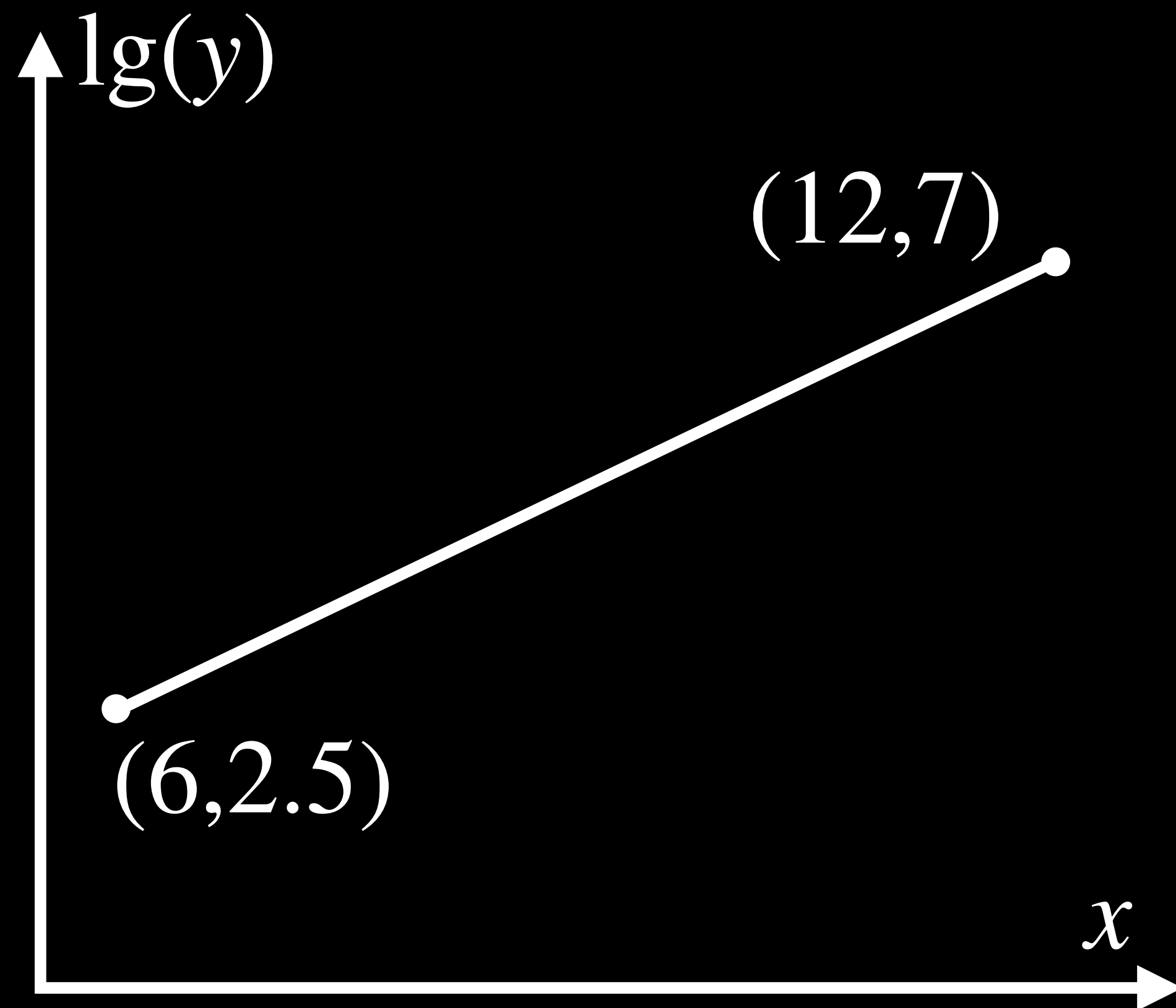
For MTEs, Y and X are usually given which helps to hint you to the answer



Finding Unknowns

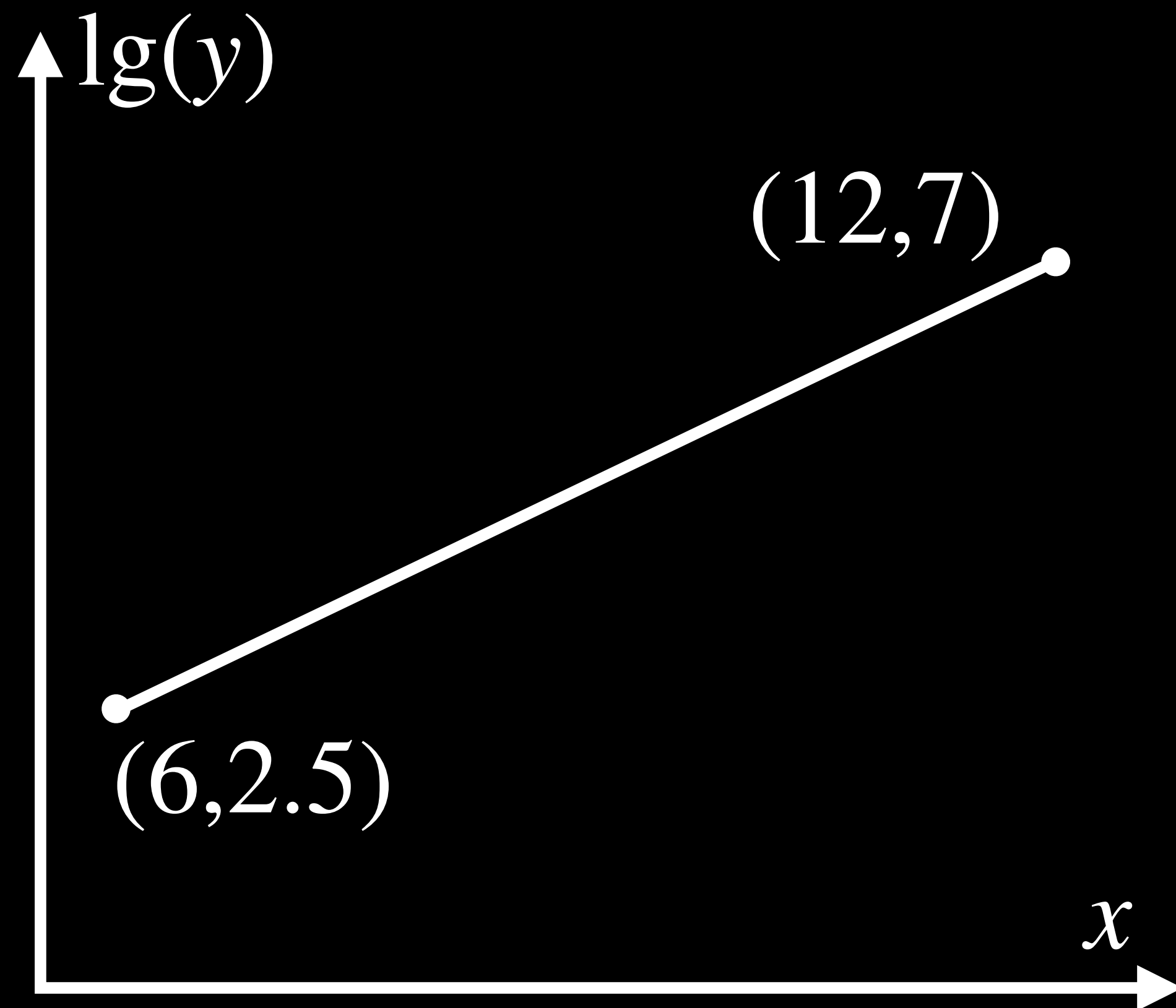
H moment

The diagram shows part of the straight line graph that represents $y = ab^x$, where a and b are constants. Find the values of a and b .



Finding Unknowns

H moment



The diagram shows part of the straight line graph that represents $y = ab^x$, where a and b are constants. Find the values of a and b .

$$y = ab^x$$

$$\lg(y) = \lg(ab^x)$$

$$\lg(y) = \lg(a) + \lg(b^x)$$

$$\lg(y) = \lg(a) + x \lg(b)$$

$$\underbrace{\lg(y)}_Y = \underbrace{\lg(b)}_m x + \underbrace{\lg(a)}_c \quad \text{find linear form of eqn}$$

$$m = \frac{7 - 2.5}{12 - 6}$$

$$m = 0.75$$

$$\begin{aligned} Y - Y_1 &= m(X - X_1) \\ Y - 7 &= 0.75(X - 12) \\ Y &= 0.75(X - 12) + 7 \\ Y &= 0.75X - 9 + 7 \\ Y &= 0.75X - 2 \end{aligned}$$

Find eqn of line in terms of Y and X

$$Y = 0.75X - 2$$

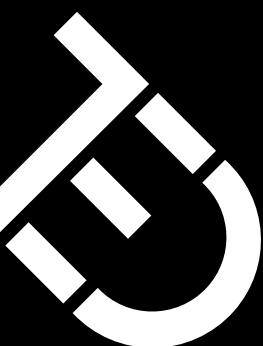
$$\lg(y) = \lg(b)x + \lg(a)$$

By comparison,

$$\lg(a) = -2 \quad \lg(b) = 0.75$$

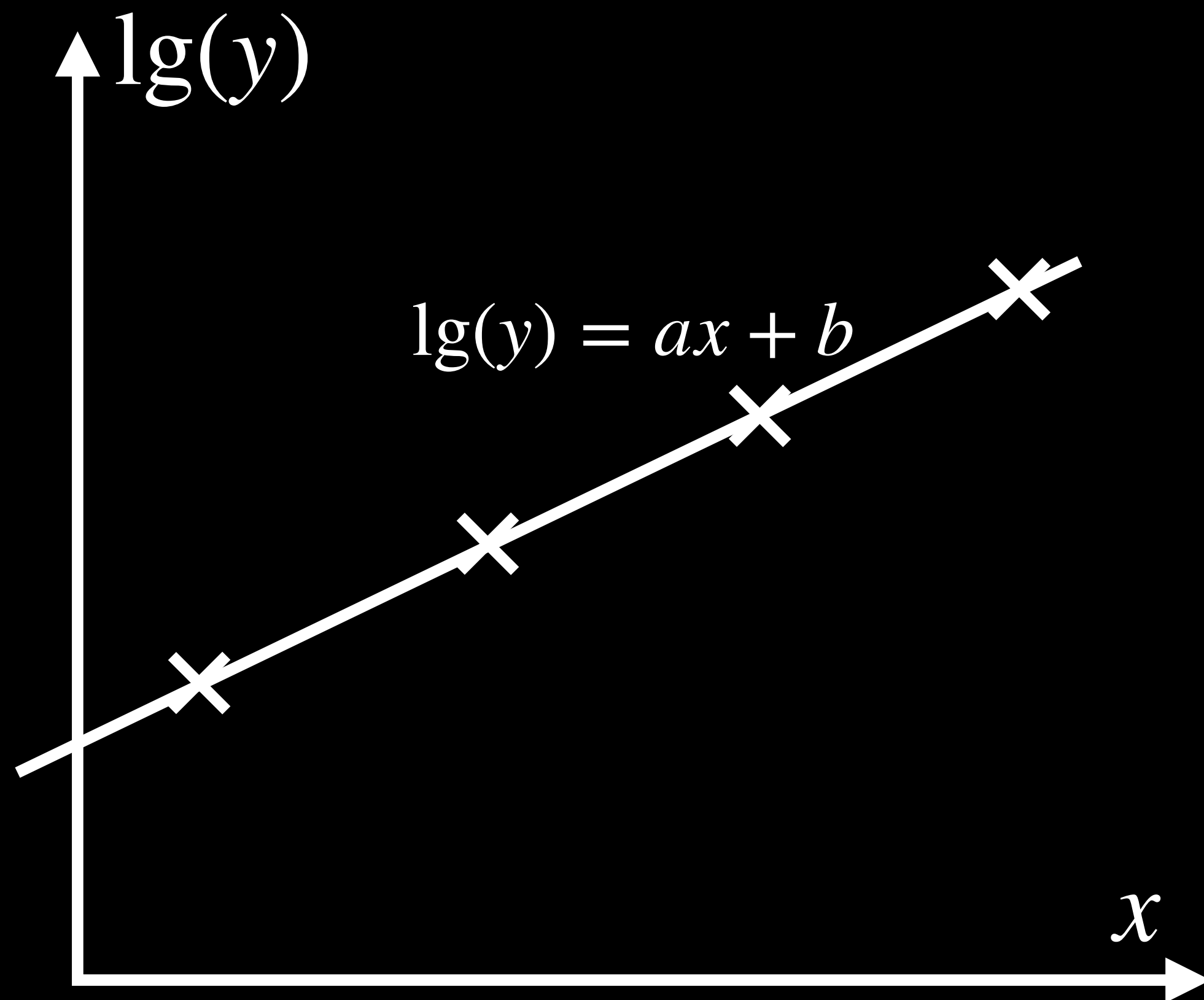
$$10^{-2} = a \quad 10^{0.75} = b$$

$$a = 0.01 \quad b = 5.62(3.s.f)$$



Plotting

Line supremacy



Plotting:

Scale is of correct size

usually stated in qn, if not, make appropriate scale

Line is linear and continuous

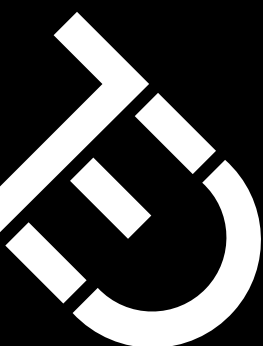
no double line, line MUST cut y axis
and have line labelled with its function

Axes are labelled properly

x and y labelled, intervals are spaced evenly

Points are clearly plotted

Points are plotted with a clear "x", at the right coordinates



Applied Questions

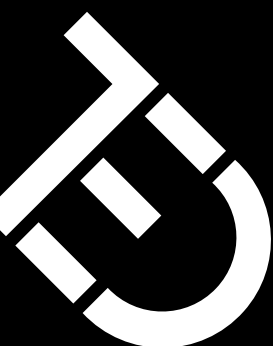
Sst reflection cycle

The price of a very nice thing P is connected to the number of weeks t by the following relationship of $P = Ae^{\frac{t}{k}}$, where A and k are constants.

Below is a table of values of t and P .

t	2	4	6	8
P	8.90	10.56	12.52	14.86

- Plot a graph of $\ln(P)$ plotted against t
- From your graph, estimate the values of A and k .
- From your graph, find the price of the thing at 3 weeks
- When is the price of the object is 20, how many weeks have elapsed



Applied Questions

Sst reflection cycle

The price of a very nice thing P is connected to the number of weeks t by the following relationship of $P = Ae^{\frac{t}{k}}$, where A and k are constants.

Below is a table of values of t and P .

t	2	4	6	8
P	8.90	10.56	12.52	14.86
$\ln(P)$	2.19	2.36	2.53	2.70

a) Plot a graph of $\ln(P)$ plotted against t

$$P = Ae^{\frac{t}{k}}$$

$$\ln(P) = \ln(Ae^{\frac{t}{k}})$$

$$\ln(P) = \ln(A) + \ln(e^{\frac{t}{k}})$$

$$\ln(P) = \frac{t}{k} \ln(e) + \ln(A)$$

$$\ln(P) = \frac{1}{k}t + \ln(A)$$

Convert each value in the table into the newly transformed X and Y variables by writing a new row of $\ln(P)$.

To decide a scale,
For X axis: find
largest value of X – 0 [smallest value of X]

number of big squares, aka 10
up to nearest clean number (e.g. 1, 2, 5, 0.1, 0.2, 0.5) to
find how much 1 big square should be.

In this case, $8/10 = 0.8 \approx 1$

For Y axis: We need to make sure that our vertical
intercept fits in the scale we use, so we have to
calculate it first:

$$m = \frac{2.36 - 2.19}{4 - 2} = 0.085$$

find gradient w/ two points

$$Y = mX + C$$

$$2.19 = (0.085)(2) + C$$

$$C = 2.19 - (0.085)(2)$$

$$C = 2.02$$

sub in gradient and one point into eqn
solve for C

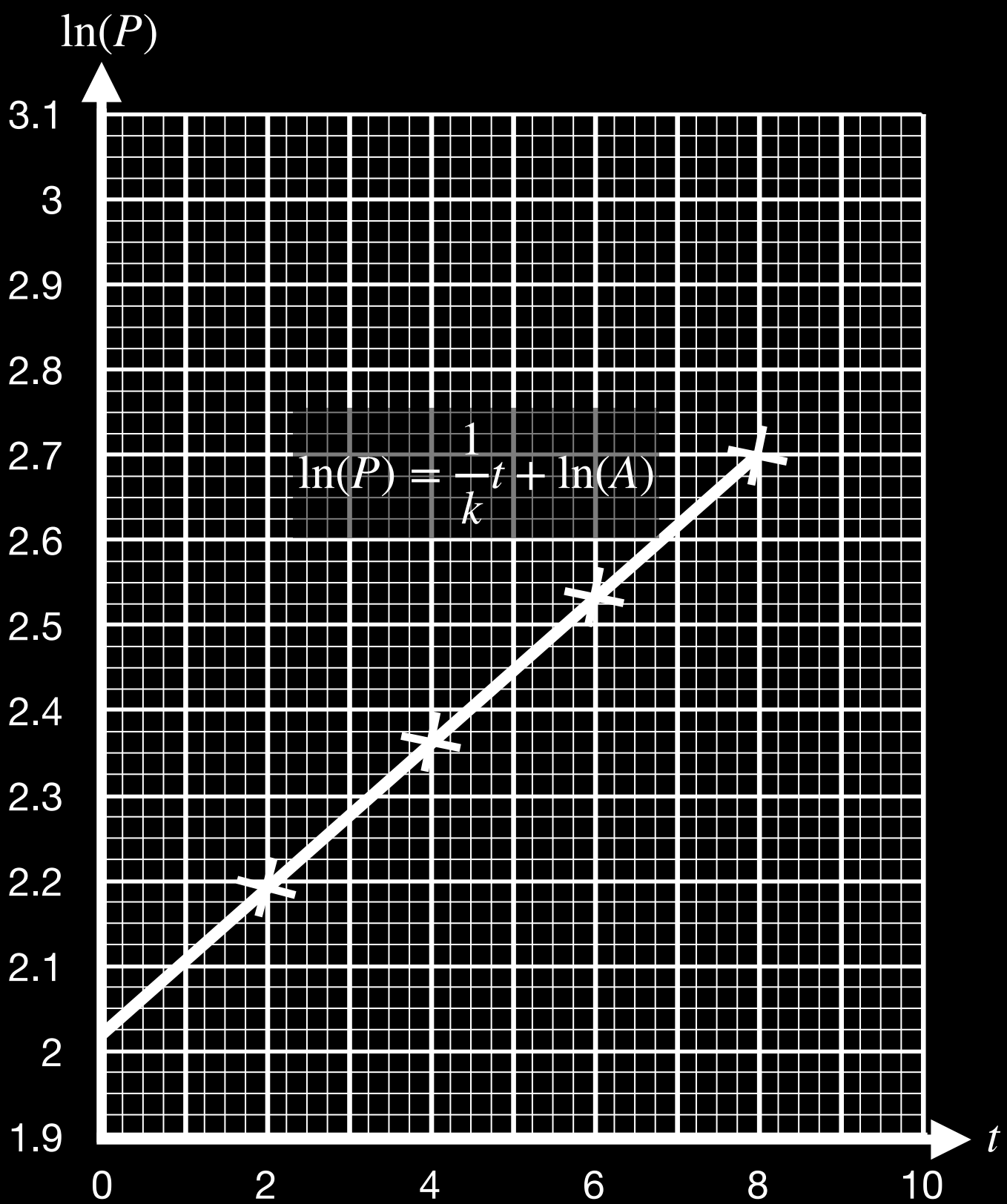
do not write this down !!!!!

We then find $\frac{\text{largest value of Y} - \text{smallest value of Y}}{\text{number of big squares, aka 12}}$, where the

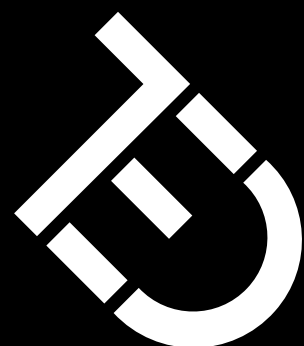
vertical intercept found may be either the largest or smallest of Y.
We then round up to the nearest clean number, which gives us:

$$\frac{2.70 - 2.02}{12} = 0.0567 = 0.1.$$

For the Y-axis, we do not need to start from 0, and can start from
a suitable value, like 1.9 (add/subtract 1 big square value from Y
intercept, $2.02 - 0.1 = 1.92$)



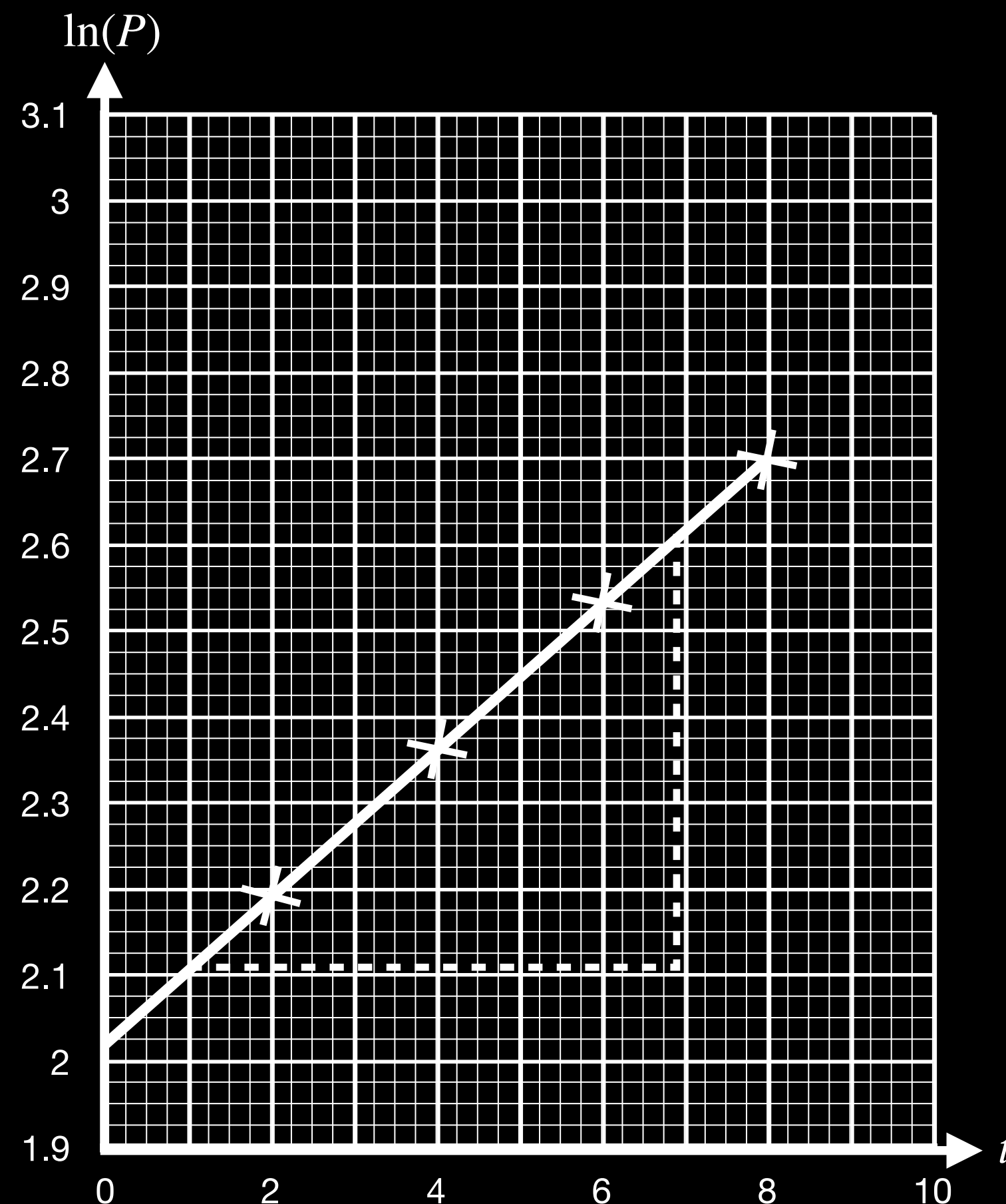
graph should cover at least half of paper
label your axes and line (can use linear form or nonlinear)



Applied Questions

Sst reflection cycle

The price of a very nice thing P is connected to the number of weeks t by the following relationship of $P = Ae^{\frac{t}{k}}$, where A and k are constants.



b) From your graph, estimate the values of A and k .

$$P = Ae^{\frac{t}{k}}$$

$$\ln(P) = \underbrace{\frac{1}{k}t}_m + \underbrace{\ln(A)}_c$$

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$0.0855 = \frac{1}{k}$$

$$k = \frac{1}{0.0855} = 11.7(3.s.f)$$

Gradient should be measured with 2 points that are spaced far apart and are not part of original values

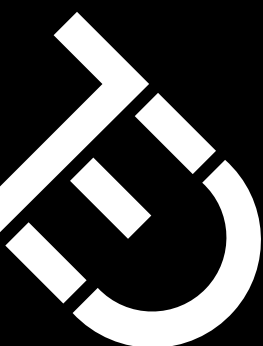
Make sure to extend the line to cut with the y axis to find vertical intercept

$$c = \ln(A)$$

$$A = e^c$$

$$A = e^{2.02}$$

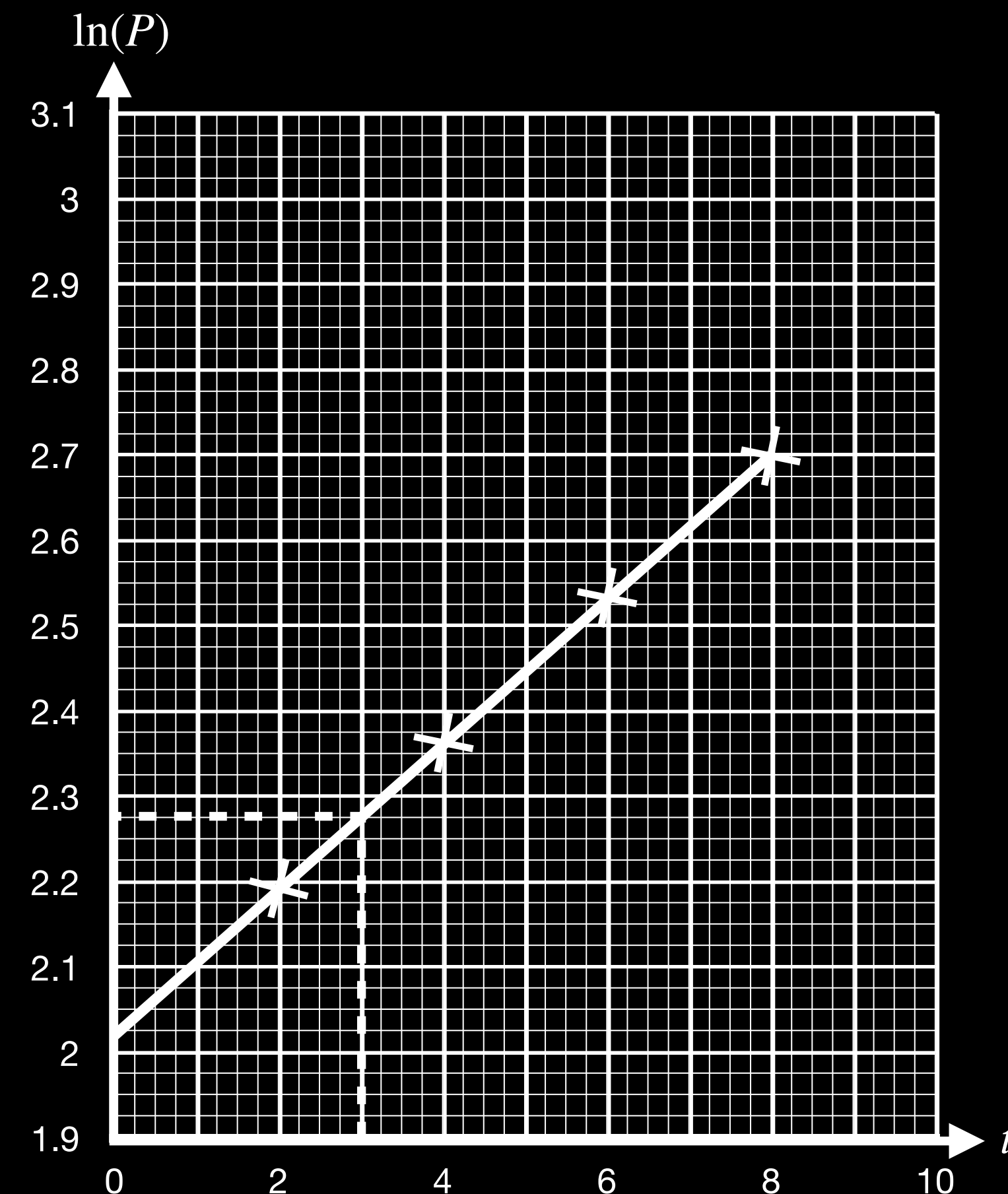
$$A = 7.54(3.s.f)$$



Applied Questions

Sst reflection cycle

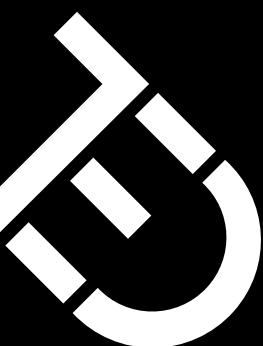
The price of a very nice thing P is connected to the number of weeks t by the following relationship of $P = Ae^{\frac{t}{k}}$, where A and k are constants.



c) From your graph, find the price of the thing at 3 weeks

Since the question asks you to find the answer from graph, you have to read from the graph and cannot use the given equation to find the price

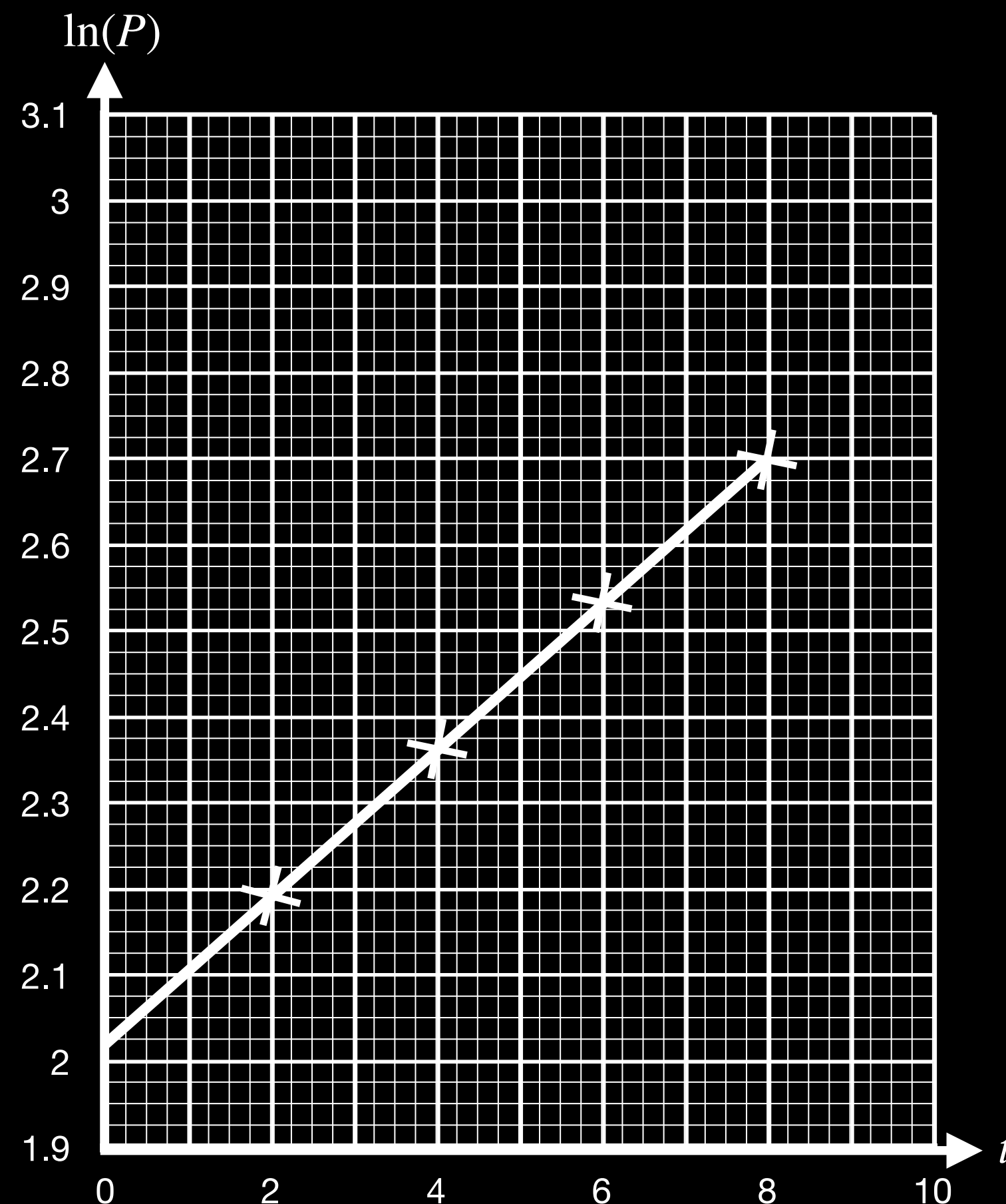
$$t = 2.27$$



Applied Questions

Sst rflevito cjdnsdlfkjnsfkjnljsnfelknsdfkljnsksrllks

The price of a very nice thing P is connected to the number of weeks t by the following relationship of $P = Ae^{\frac{t}{k}}$, where A and k are constants.



d) When is the price of the object is 20, how many weeks have elapsed

With this, you are allowed to use the equation to find the time elapsed as they did not tell you to use the graph

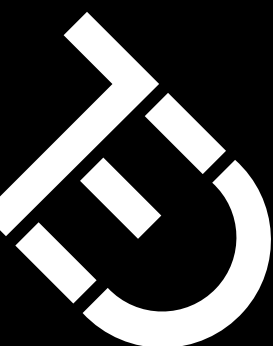
$$P = Ae^{\frac{t}{k}}$$

$$20 = 7.54e^{\frac{t}{11.7}}$$

$$\frac{20}{7.54} = e^{\frac{t}{11.7}}$$

$$\ln\left(\frac{20}{7.54}\right) = \frac{t}{11.7}$$

$$11.7 \ln\left(\frac{20}{7.54}\right) = t \quad t = 11.4$$



Coordinate Geometry and Circles

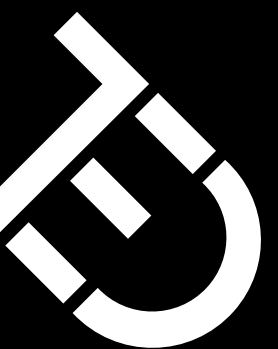
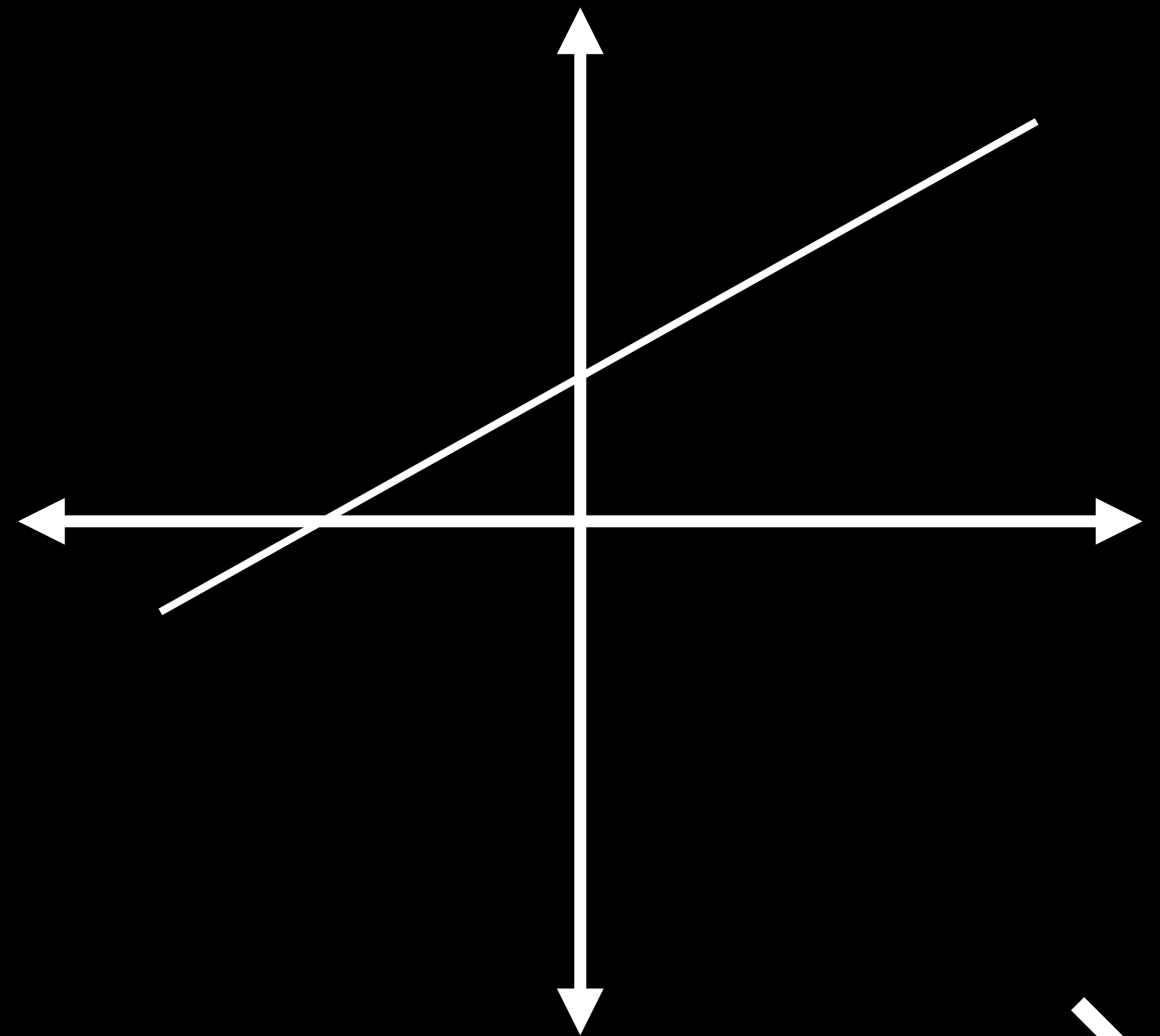


Equations and Formulas

t

$$y = mx + c$$

gradient-intercept form

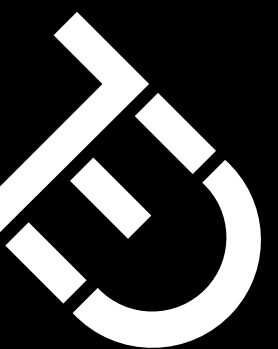
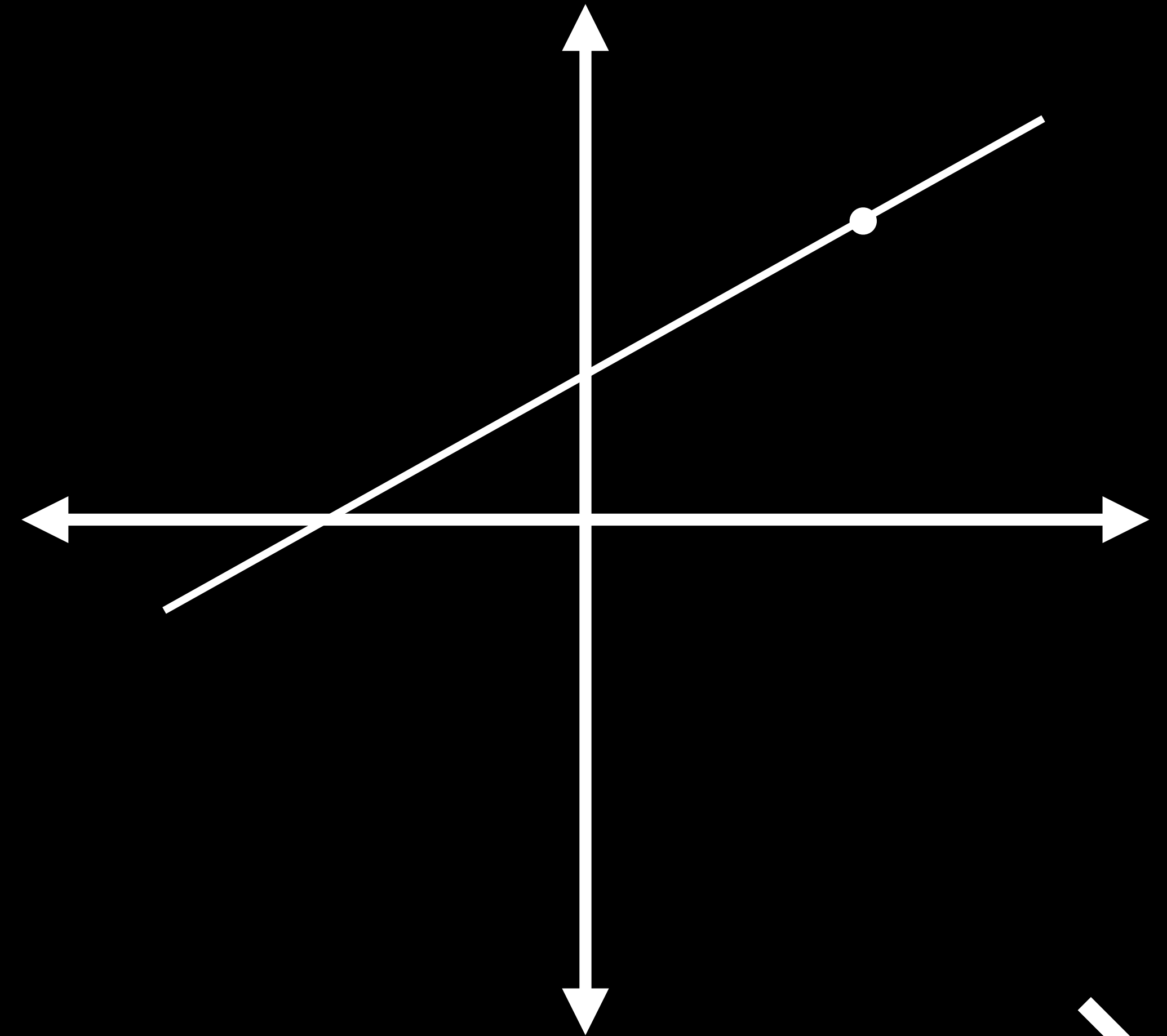


Equations and Formulas

h

$$y - y_1 = m(x - x_1)$$

gradient-point form

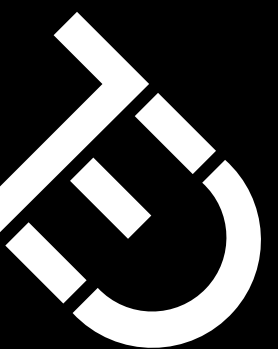
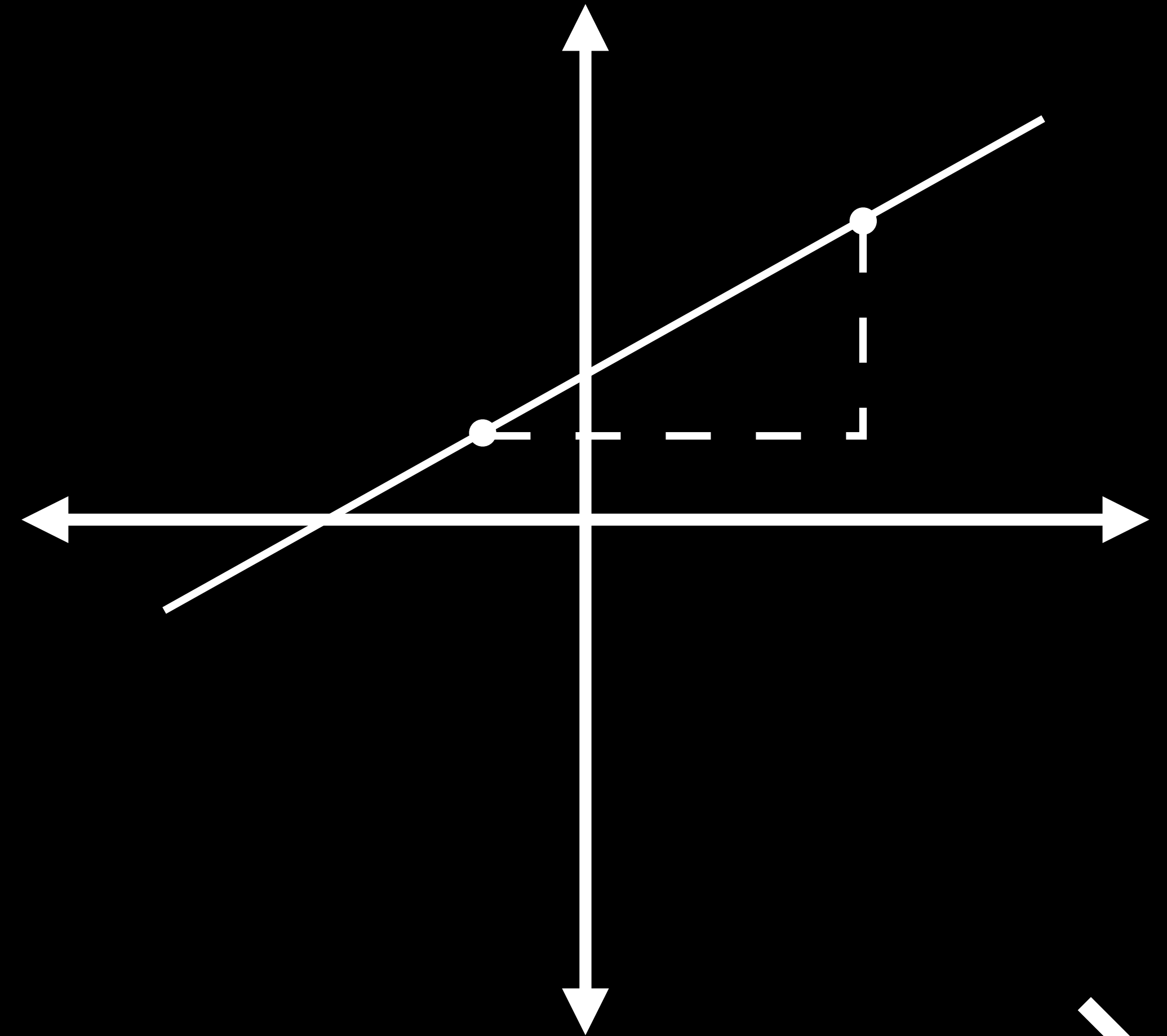


Equations and Formulas

a

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

gradient formula

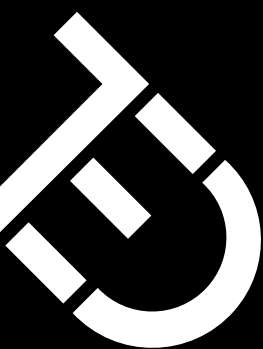
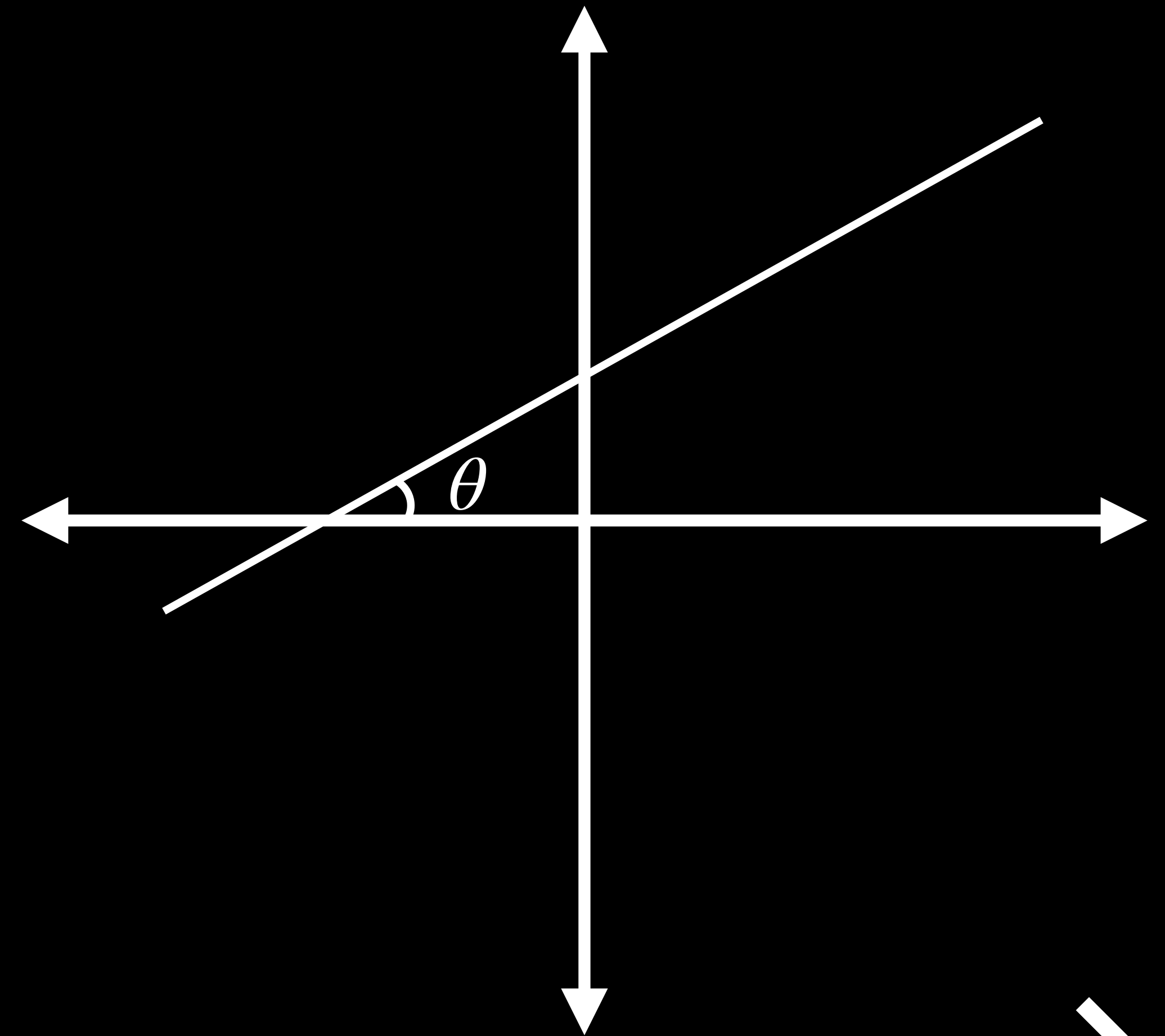


Equations and Formulas

t

$$m = \tan(\theta)$$

gradient angle conversion
finds acute angle line makes with x-axis

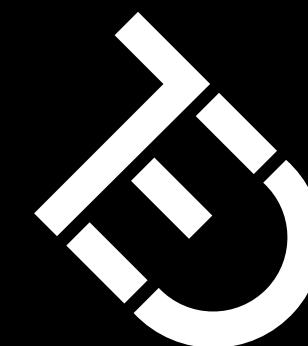
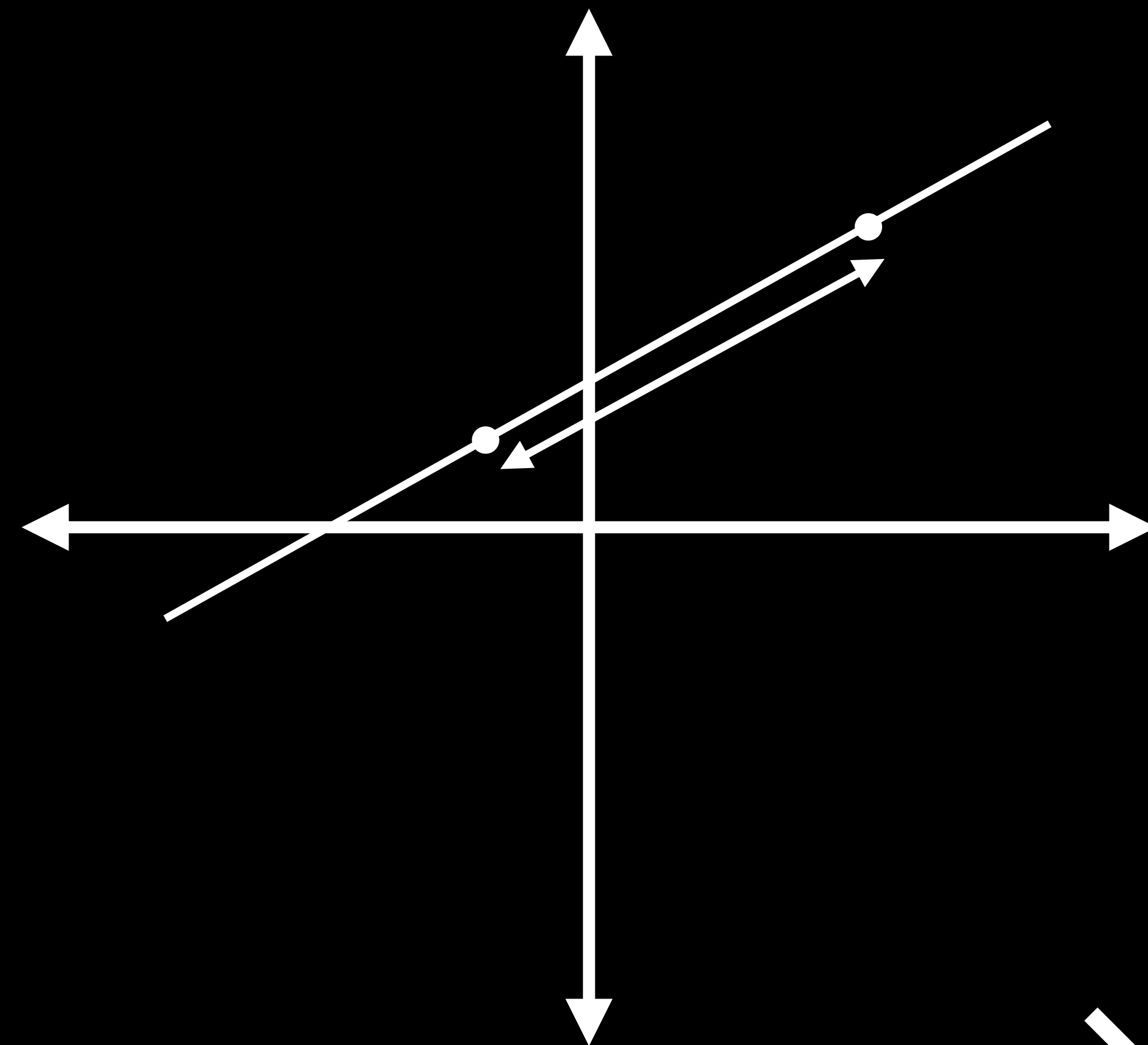


Equations and Formulas

s

$$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

distance formula

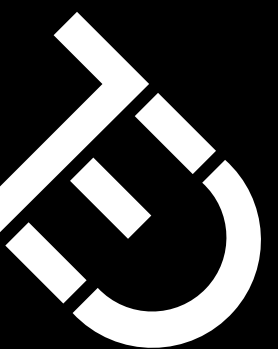
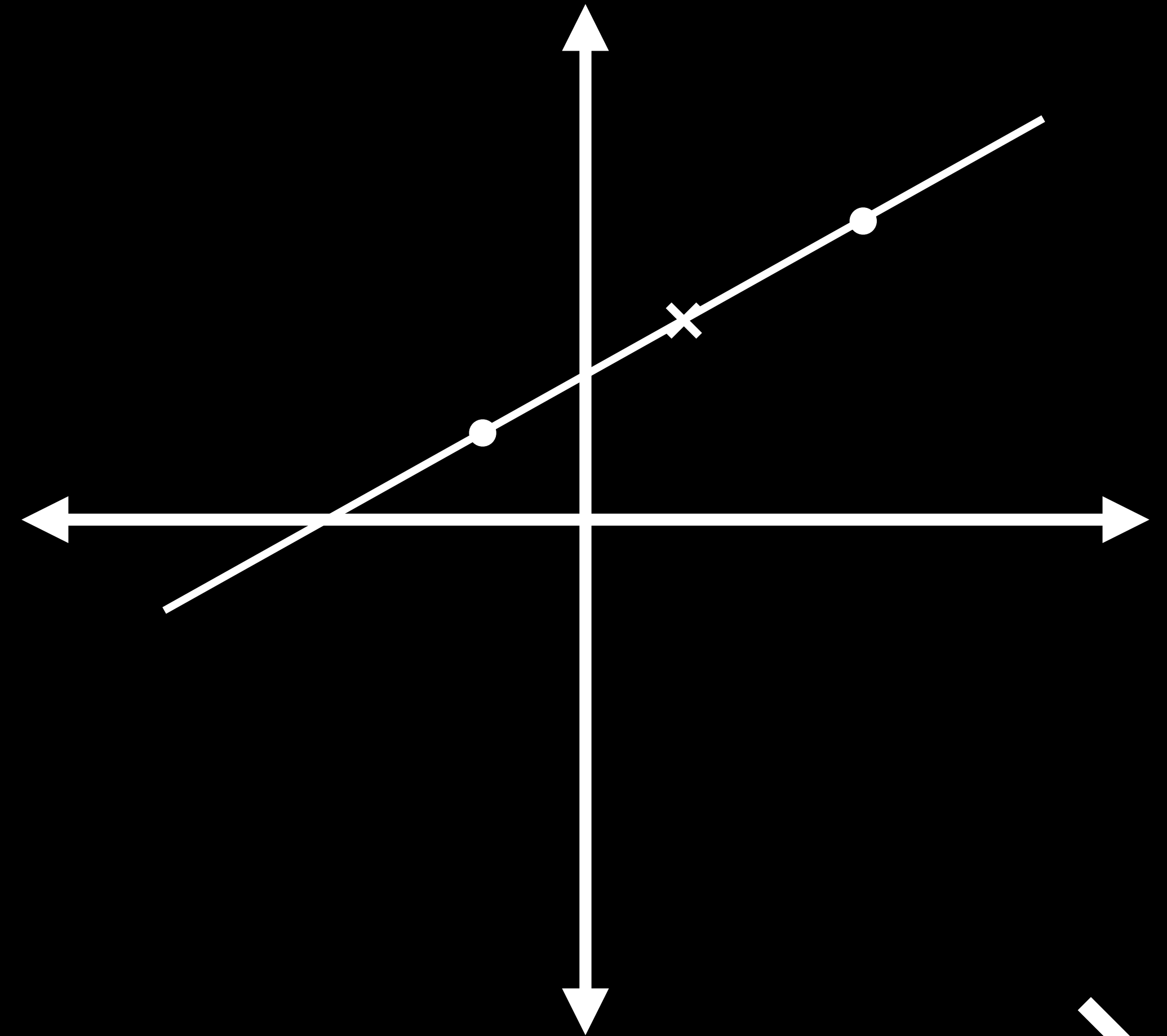


Equations and Formulas

c

$$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

midpoint formula

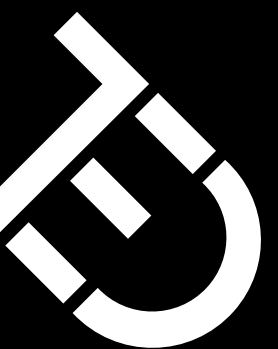
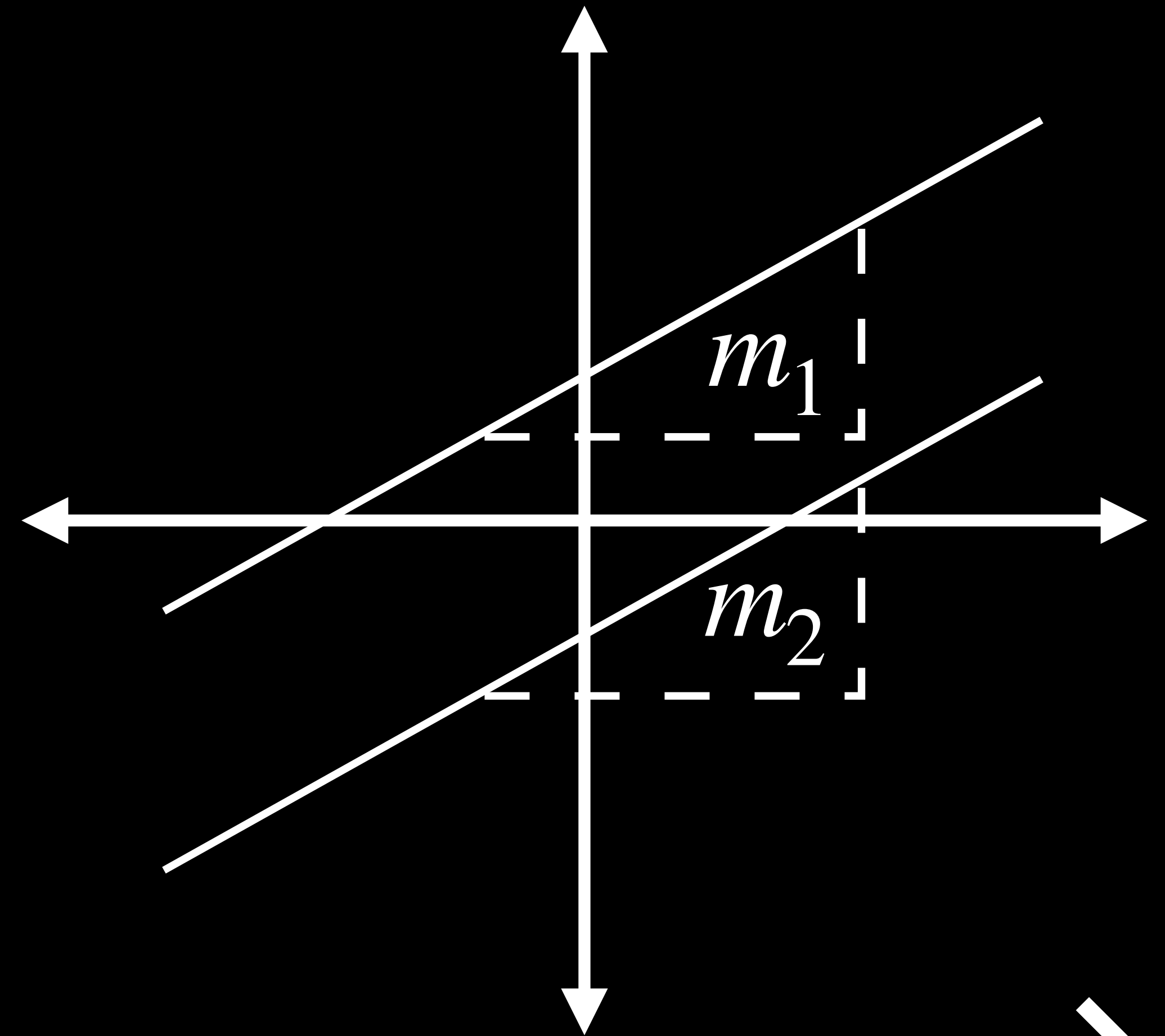


Equations and Formulas

r

$$m_1 = m_2$$

parallel lines



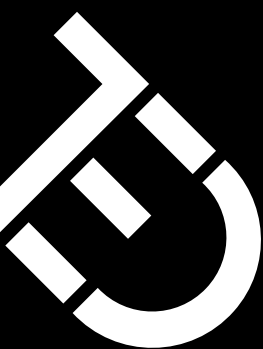
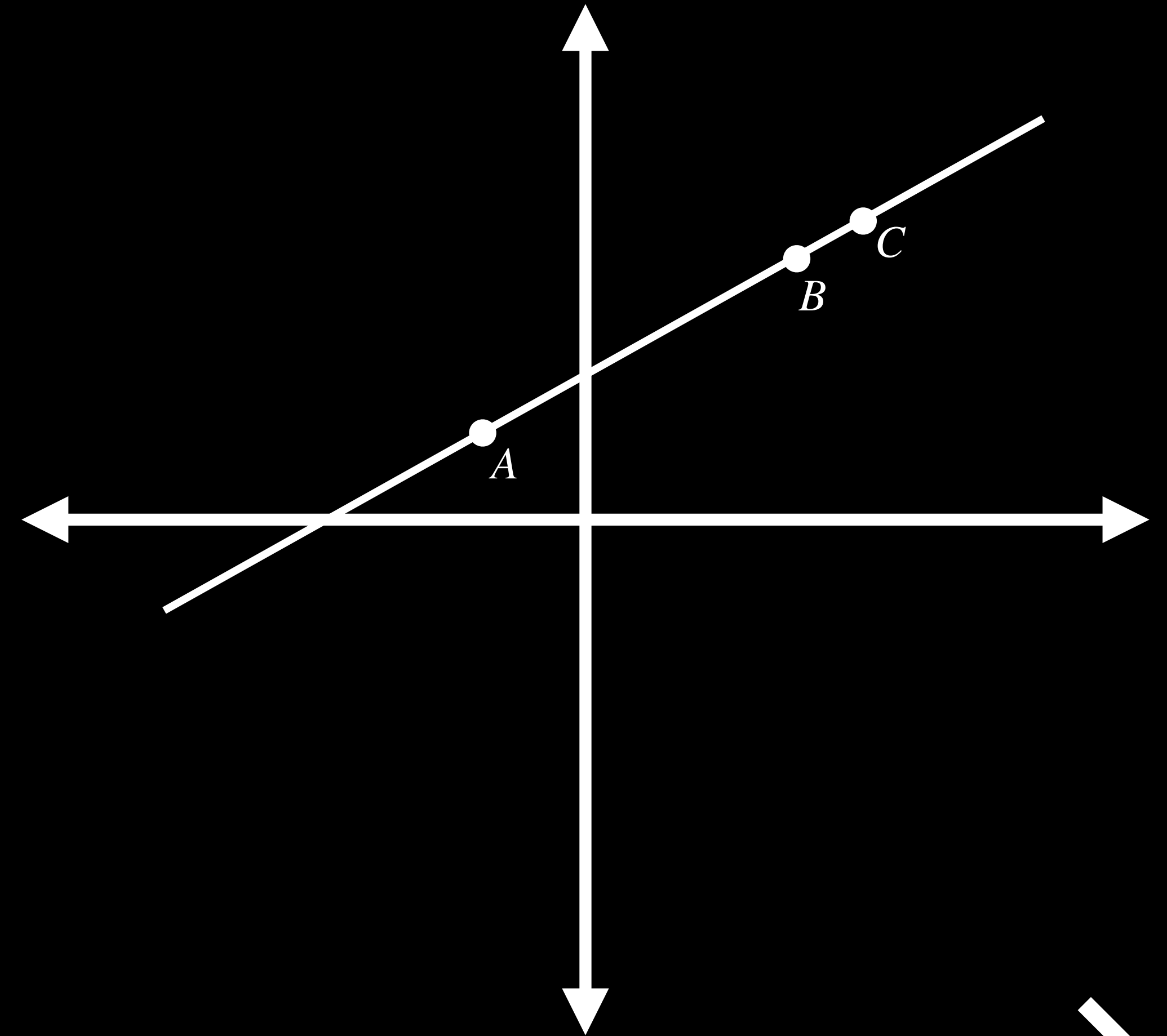
Equations and Formulas

a

$$m_{AB} = m_{BC} = m_{AC}$$

collinear points

$m_{AB} = m_{BC} = m_{AC}$ and [POINT] is a common point, it is collinear



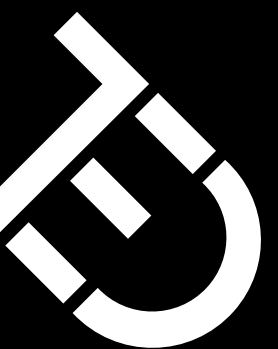
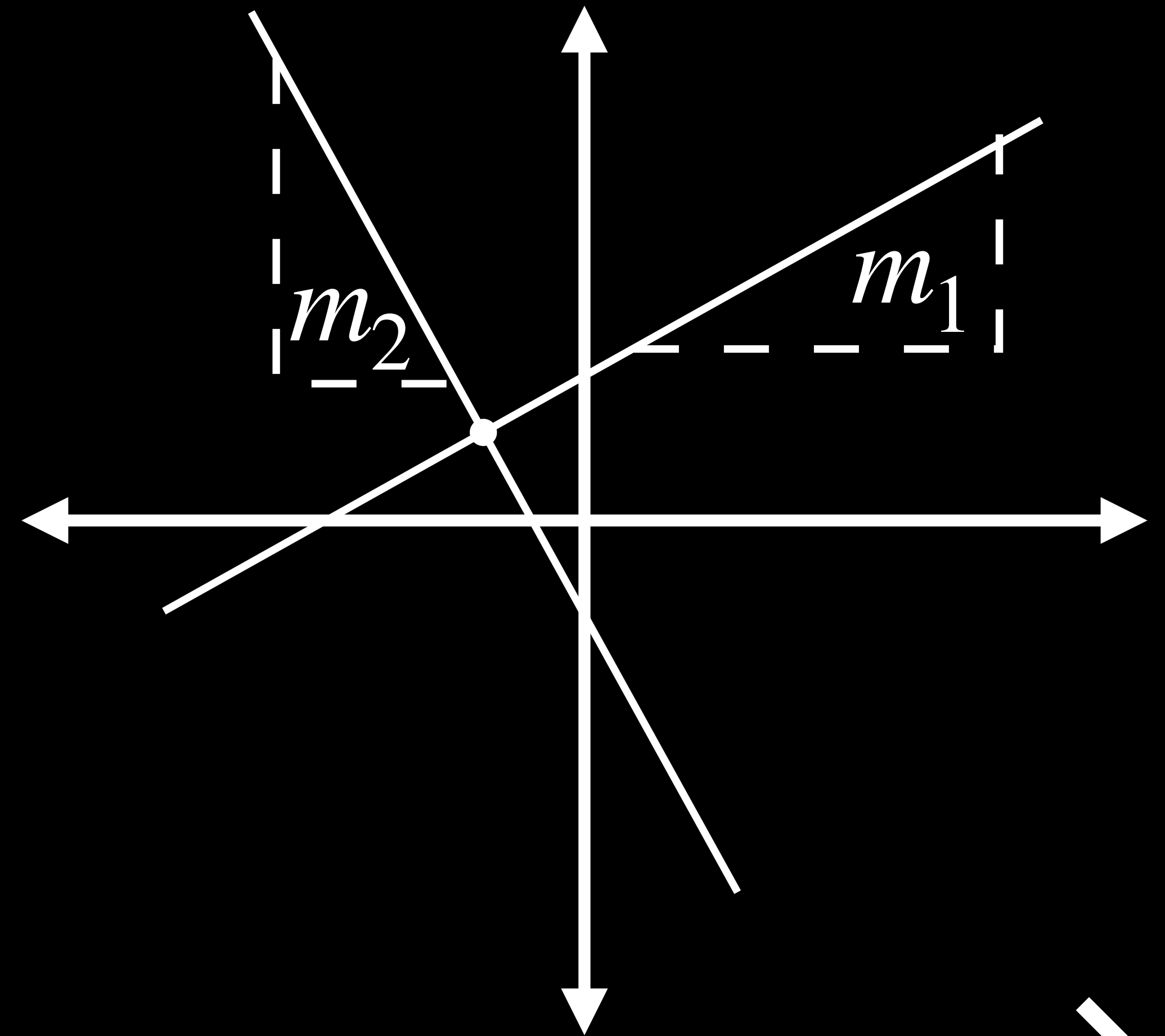
Equations and Formulas

z

$$m_1 m_2 = -1$$
$$m_2 = -(m_1)^{-1}$$

when m2 is the subj.

perpendicular lines



Equations and Formulas

y

Find midpoint of two points/line segment

$$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Find gradient of line segment and perpendicular gradient

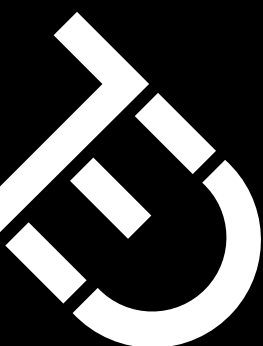
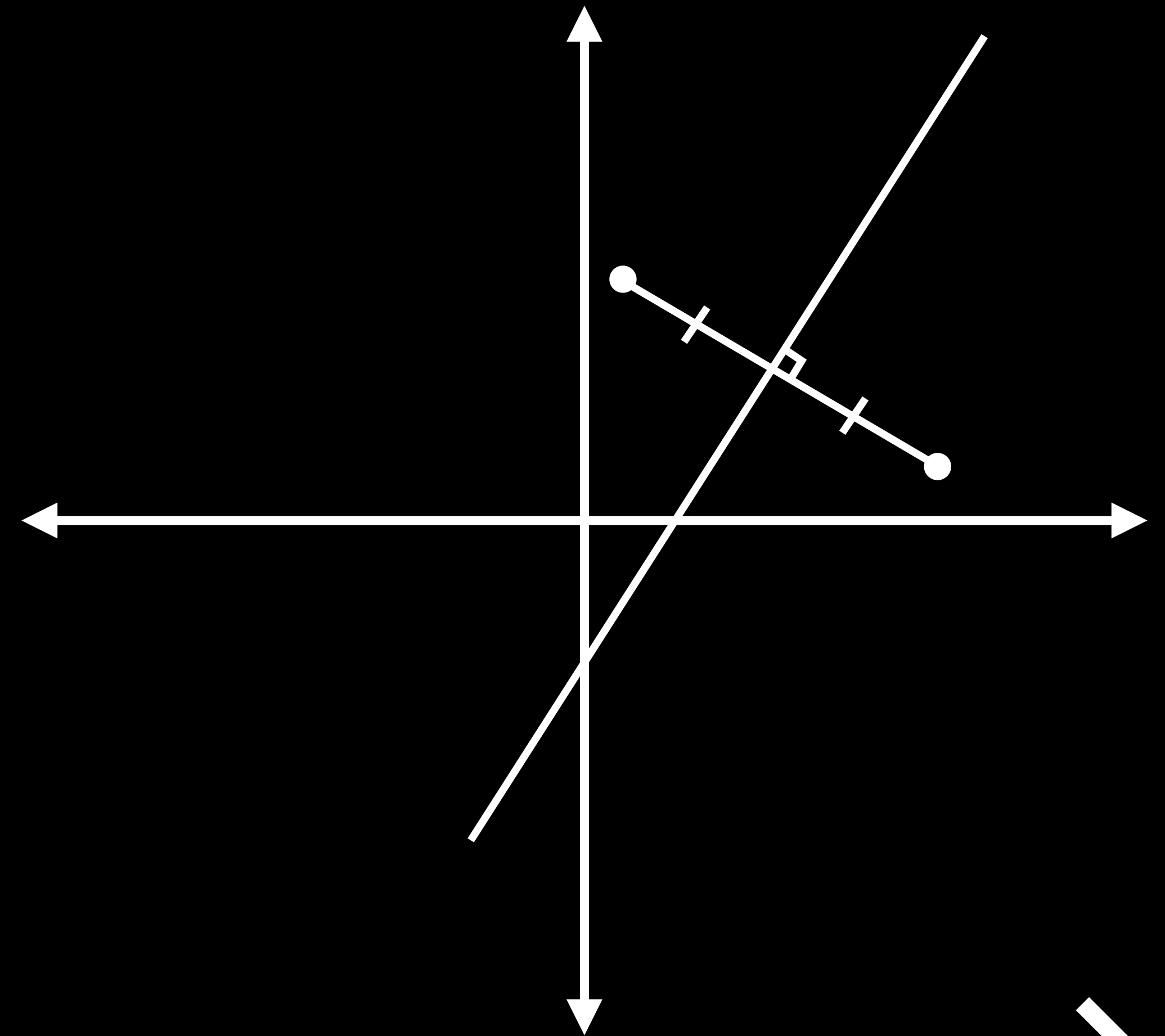
$$m_1 = \frac{y_2 - y_1}{x_2 - x_1}$$

$$m_2 = -(m_1)^{-1}$$

Use gradient & point to find eqn of line

$$y - y_1 = m_2(x - x_1)$$

perpendicular bisector



Equations and Formulas

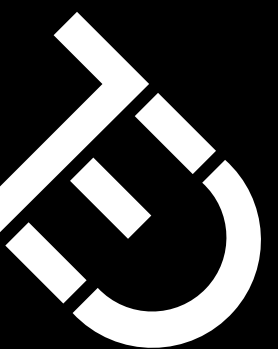
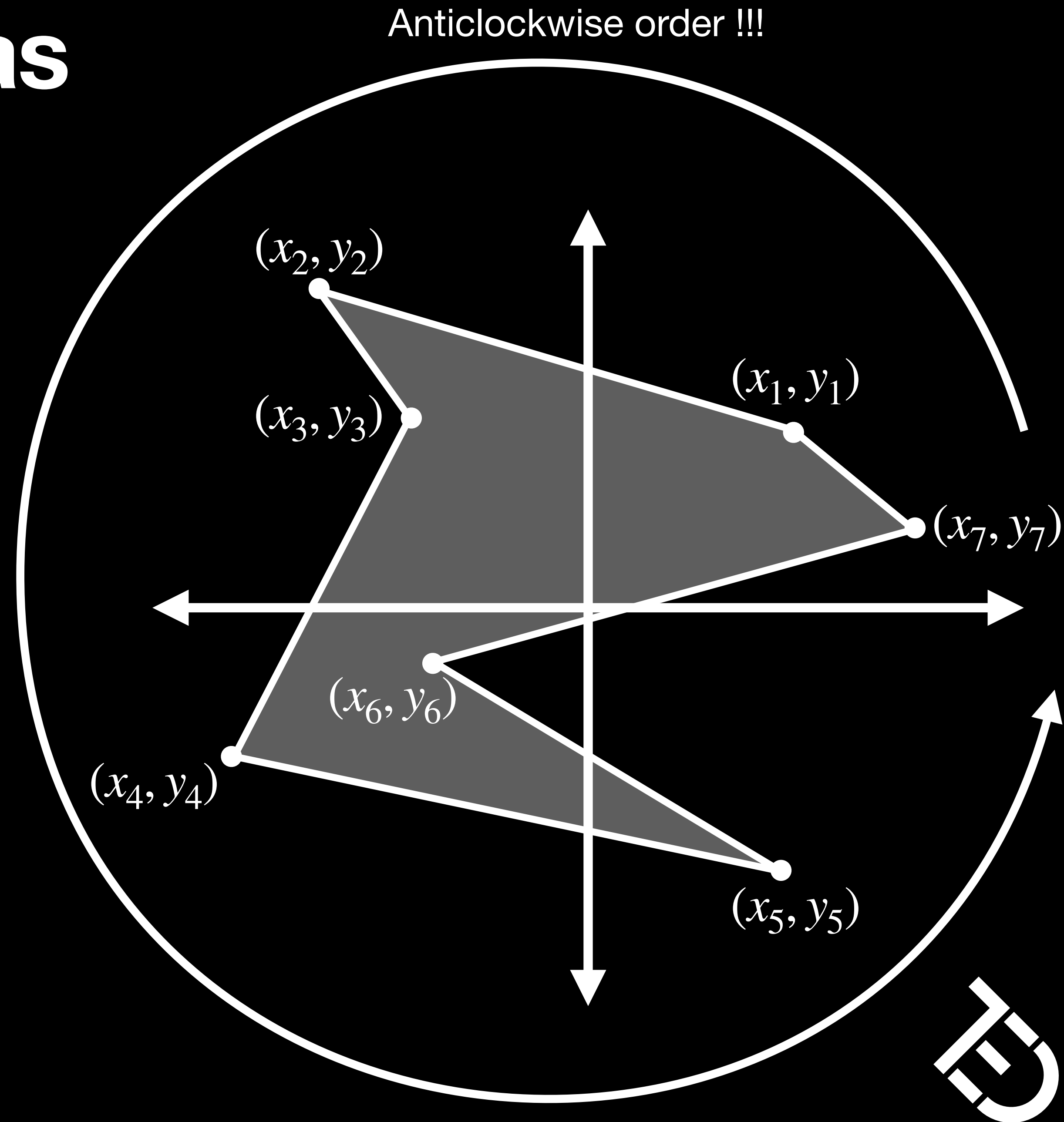
b

$$\frac{1}{2} \left| \begin{array}{ccccccc} x_1 & x_2 & x_3 & \dots & x_n & x_1 \\ y_1 & y_2 & y_3 & \dots & y_n & y_1 \end{array} \right|$$

$\begin{array}{c} + \longrightarrow \\ - \cdots \longrightarrow \end{array}$

$$= \frac{1}{2} |x_1y_2 + x_2y_3 + \dots + x_ny_1 - x_2y_1 - x_3y_2 - \dots - x_1y_n|$$

shoelace method (area of polygons)

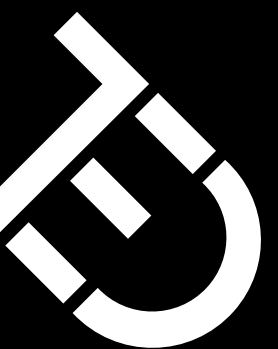
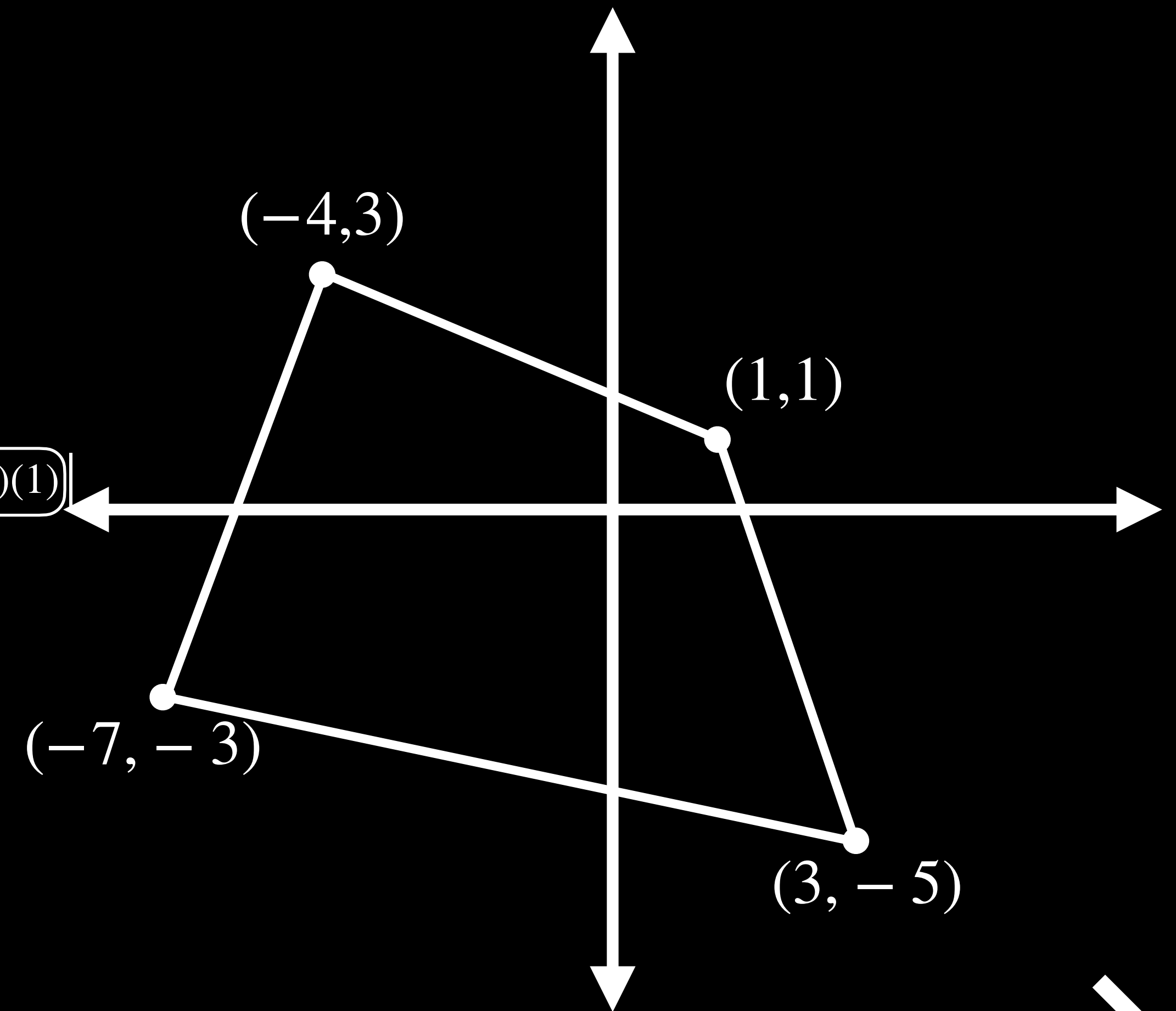


Equations and Formulas

r

shoelace method (area of polygons)

$$\begin{aligned}
 & \frac{1}{2} \left| \begin{array}{cccc} -4 & -7 & 3 & 1 \\ 3 & -3 & -5 & 1 \\ -4 & 1 & 3 & -4 \end{array} \right| \begin{array}{l} + \longrightarrow \\ - \cdots \longrightarrow \end{array} \\
 &= \frac{1}{2} \left| \boxed{+(-4)(-3)} + (-7)(-5) + (3)(1) + (1)(3) - (3)(-7) - (-3)(3) - (-5)(1) - \boxed{(-4)(1)} \right| \\
 &= \frac{1}{2} | 12 + 35 + 3 + 3 + 21 + 9 + 5 + 4 | \\
 &= \frac{1}{2} \times 92 \\
 &= 46 \text{ units}^2
 \end{aligned}$$

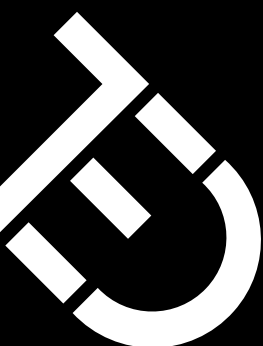
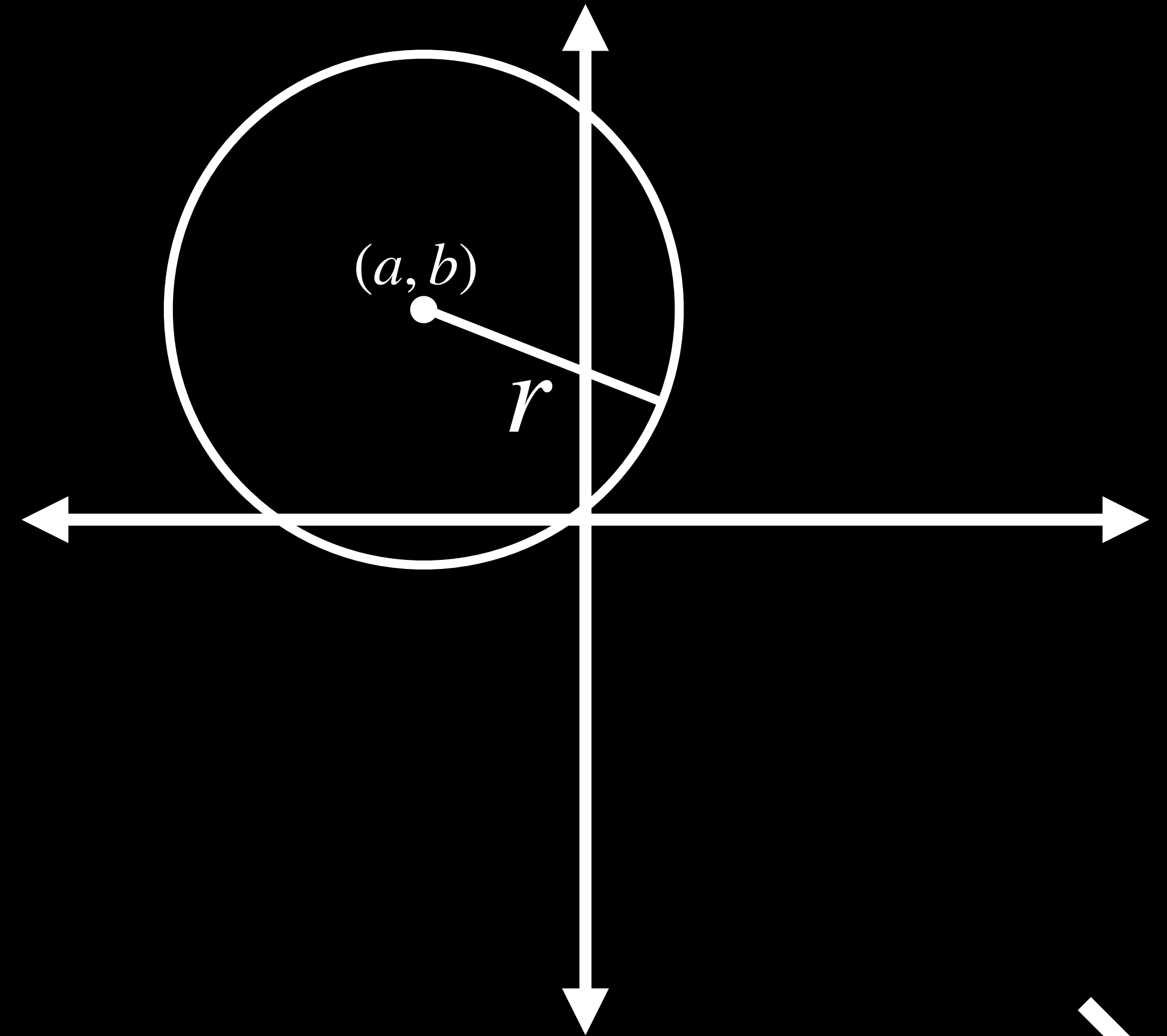


Equations and Formulas

o

$$(x - a)^2 + (y - b)^2 = r^2$$

Standard equation of circle

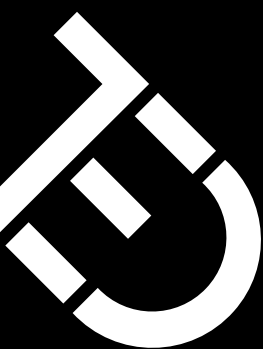
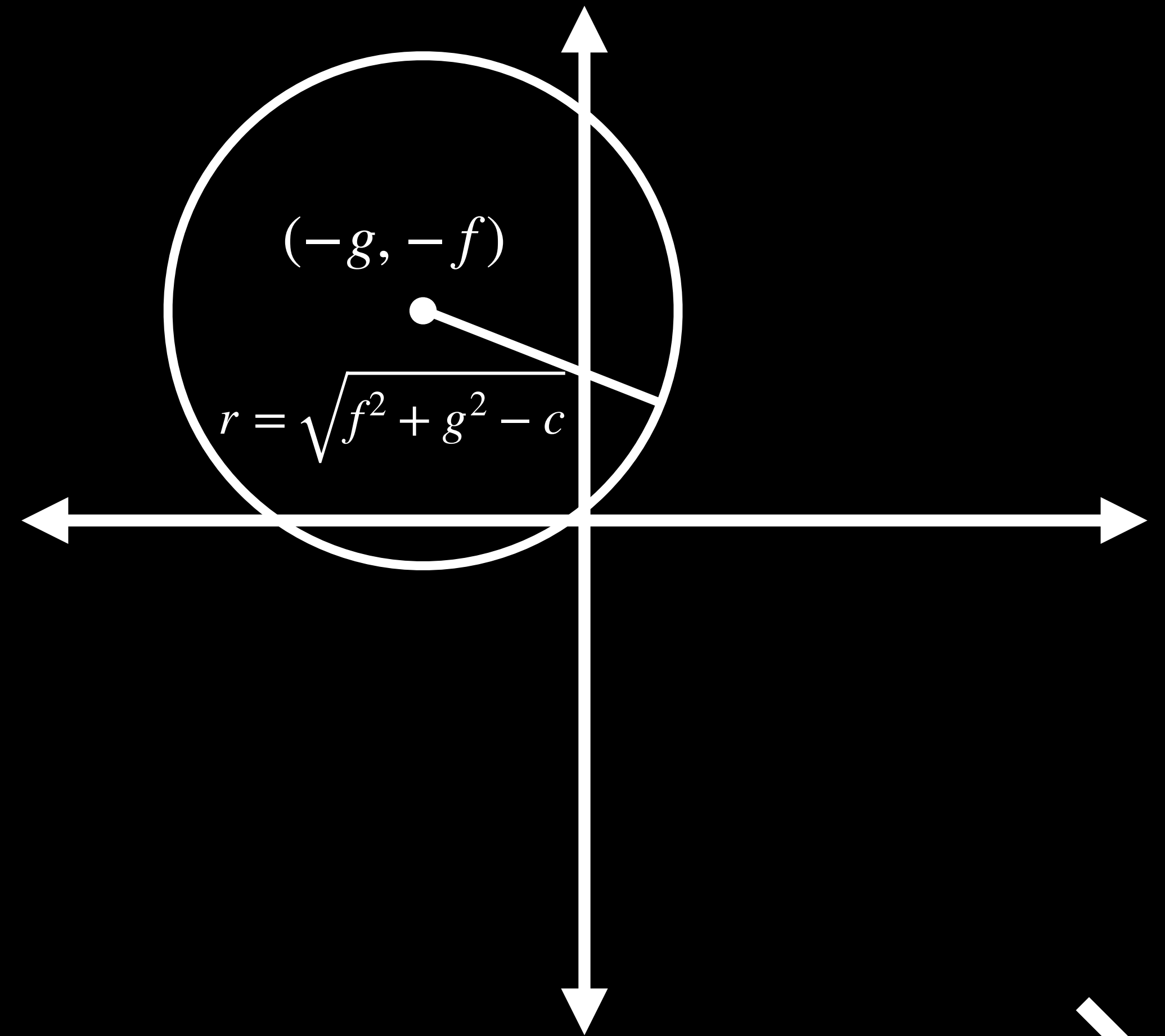


Equations and Formulas

c

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

General equation of circle



Equations and Formulas

r

Alternatively,

$$x^2 + y^2 - 4x + 6y - 23 = 0$$

$$x^2 - 4x + y^2 + 6y - 23 = 0$$

Use complete the square method

$$\left(x - \frac{4}{2}\right)^2 - \left(\frac{4}{2}\right)^2 + \left(y + \frac{6}{2}\right)^2 - \left(\frac{6}{2}\right)^2 - 23 = 0$$

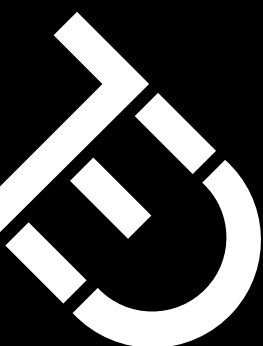
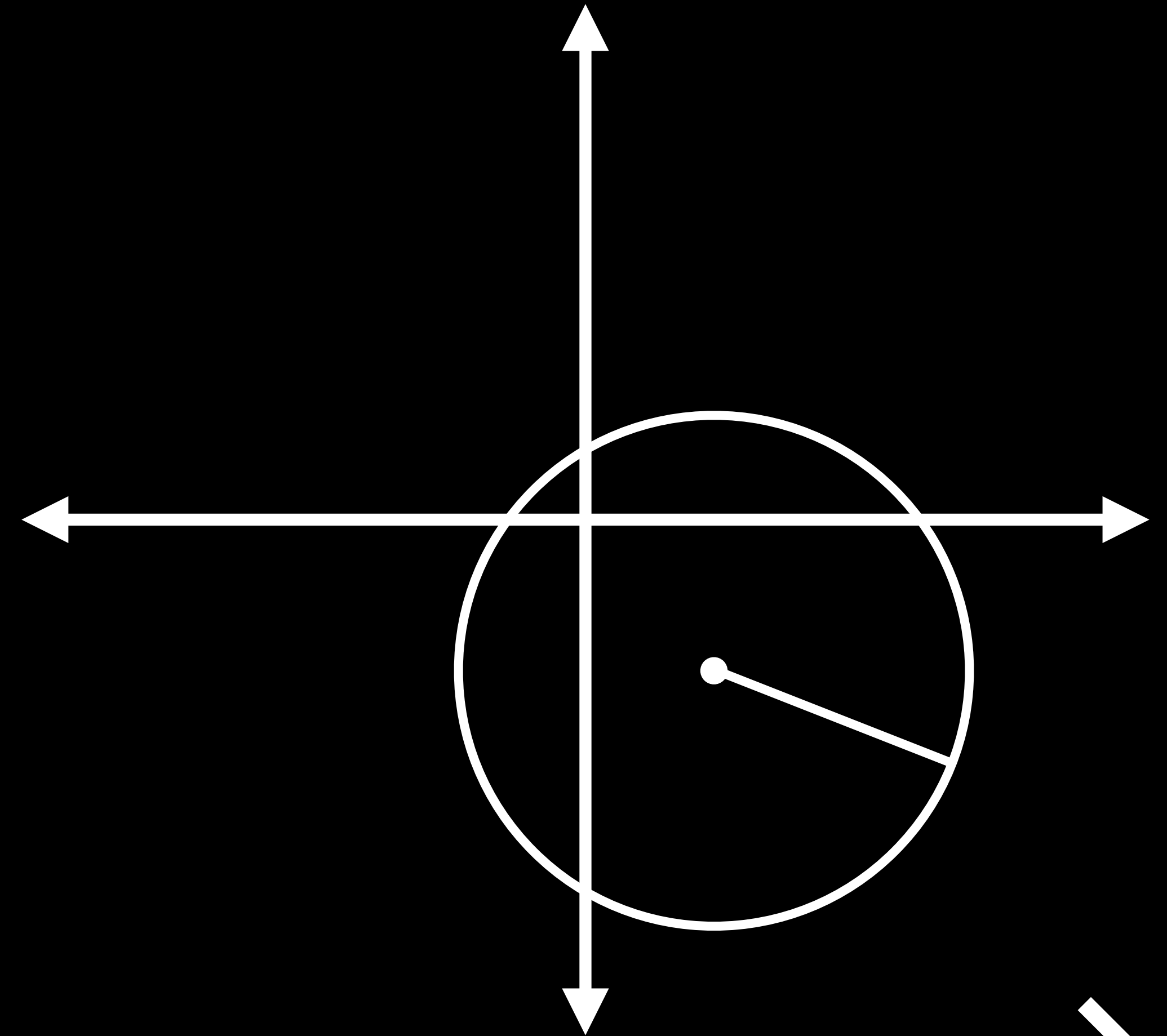
$$(x - 2)^2 - 2^2 + (y + 3)^2 - 3^2 - 23 = 0$$

$$(x - 2)^2 - 4 + (y + 3)^2 - 9 - 23 = 0$$

$$(x - 2)^2 + (y + 3)^2 = 23 + 4 + 9$$

$$(x - 2)^2 + (y + 3)^2 = 36$$

$$(x - 2)^2 + (y + 3)^2 = 6^2$$



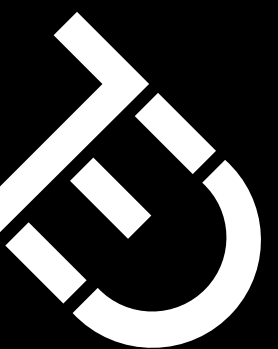
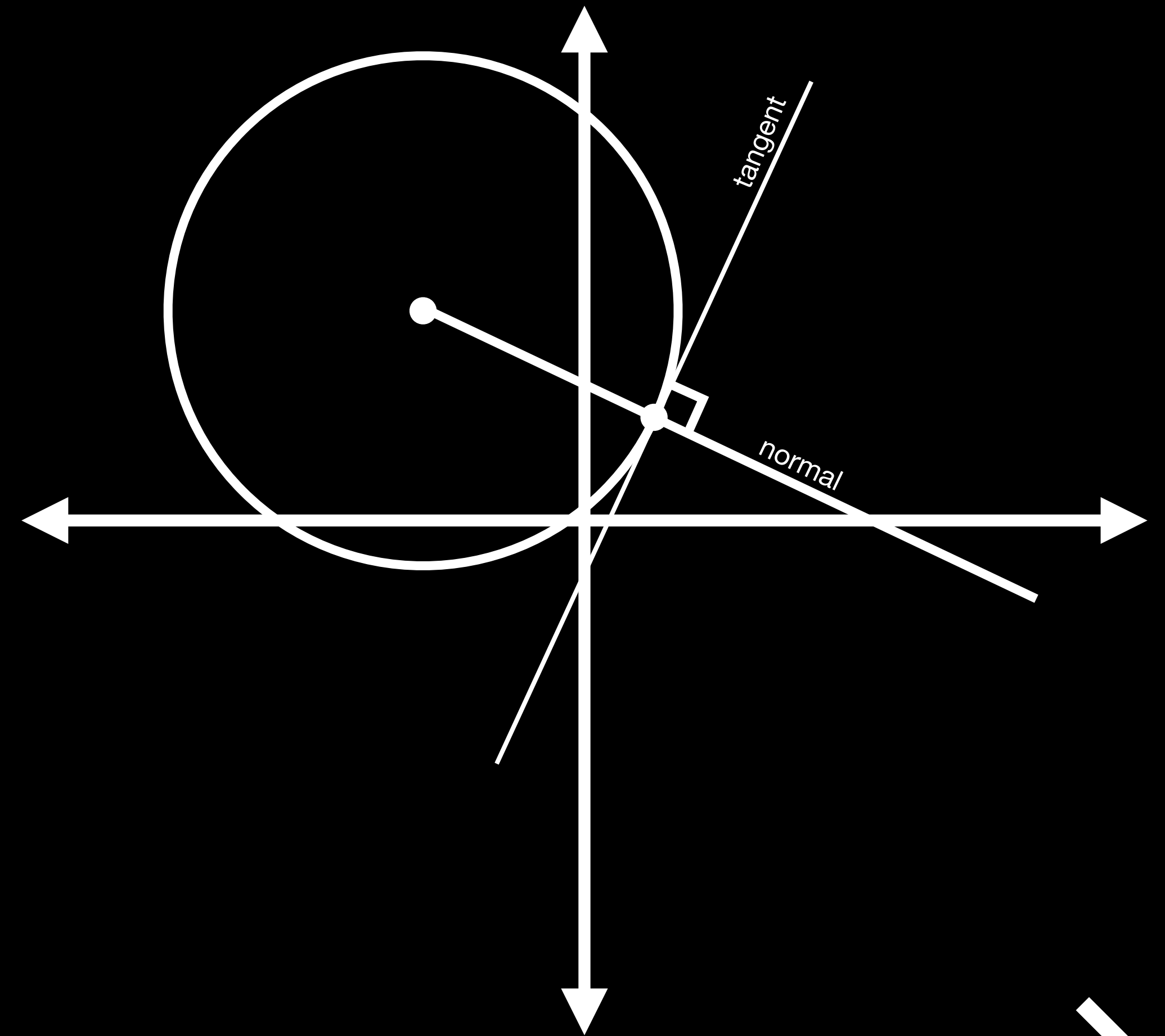
Equations and Formulas

a

Tangent & Normals

Eqn of normal can be found from
the centre of circle and point on circle

tangent is always perpendicular to normal



Equations and Formulas

z

Points & Circles

Given an equation of a circle,

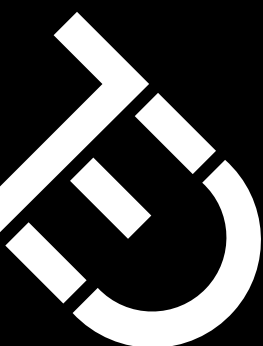
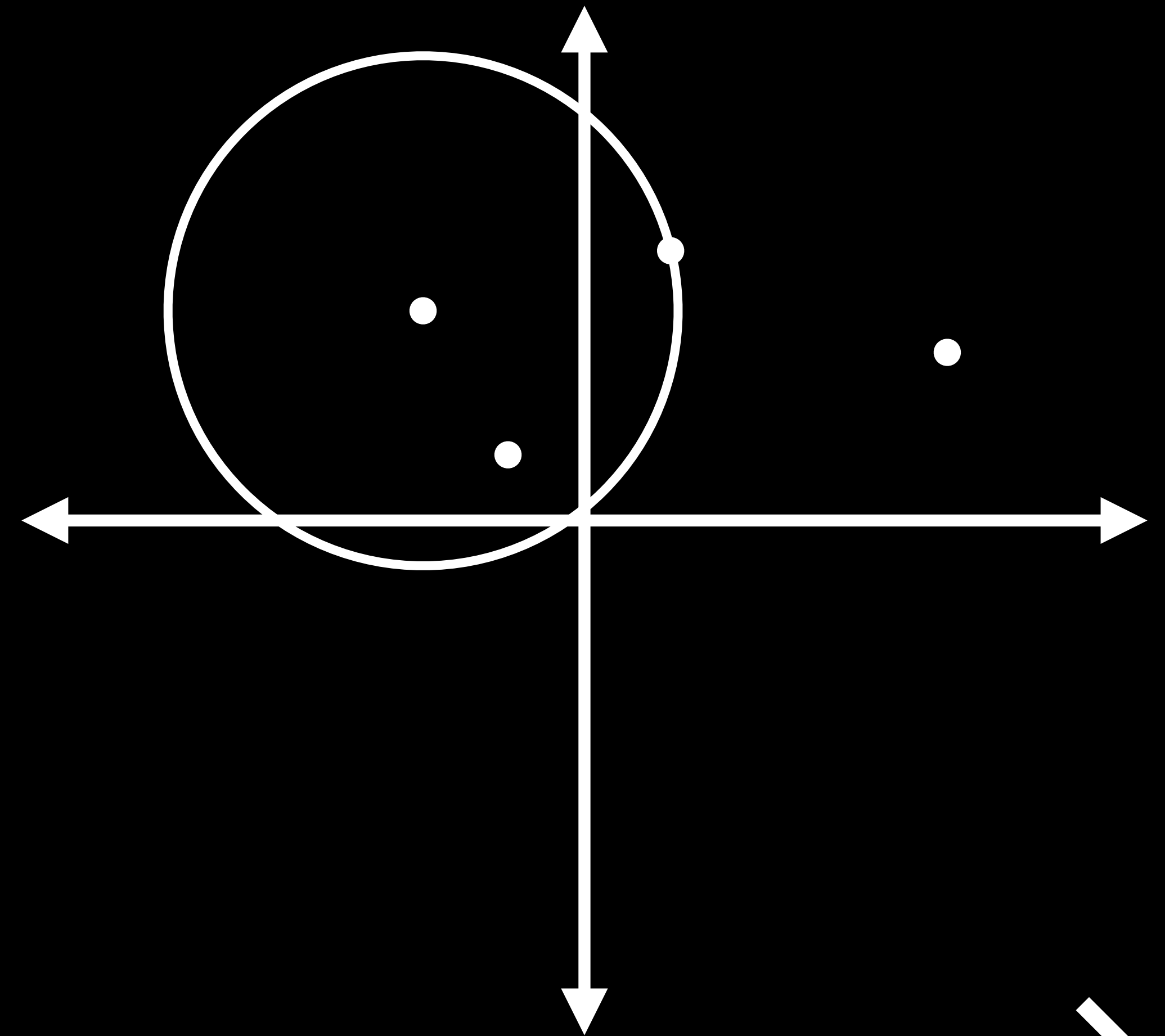
$$(x - a)^2 + (y - b)^2 = r^2$$

To find how a point (h, k) is related to the circle, find the distance of the point from the center

If the distance $<$ radius, it lies in the circle

If the distance $=$ radius, it lies on the circle

If the distance $>$ radius, it lies outside of the circle



Equations and Formulas

y

Points & Circles

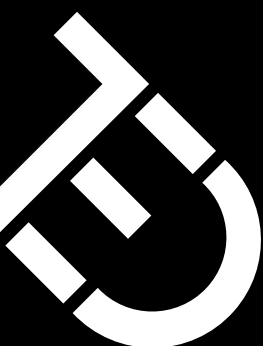
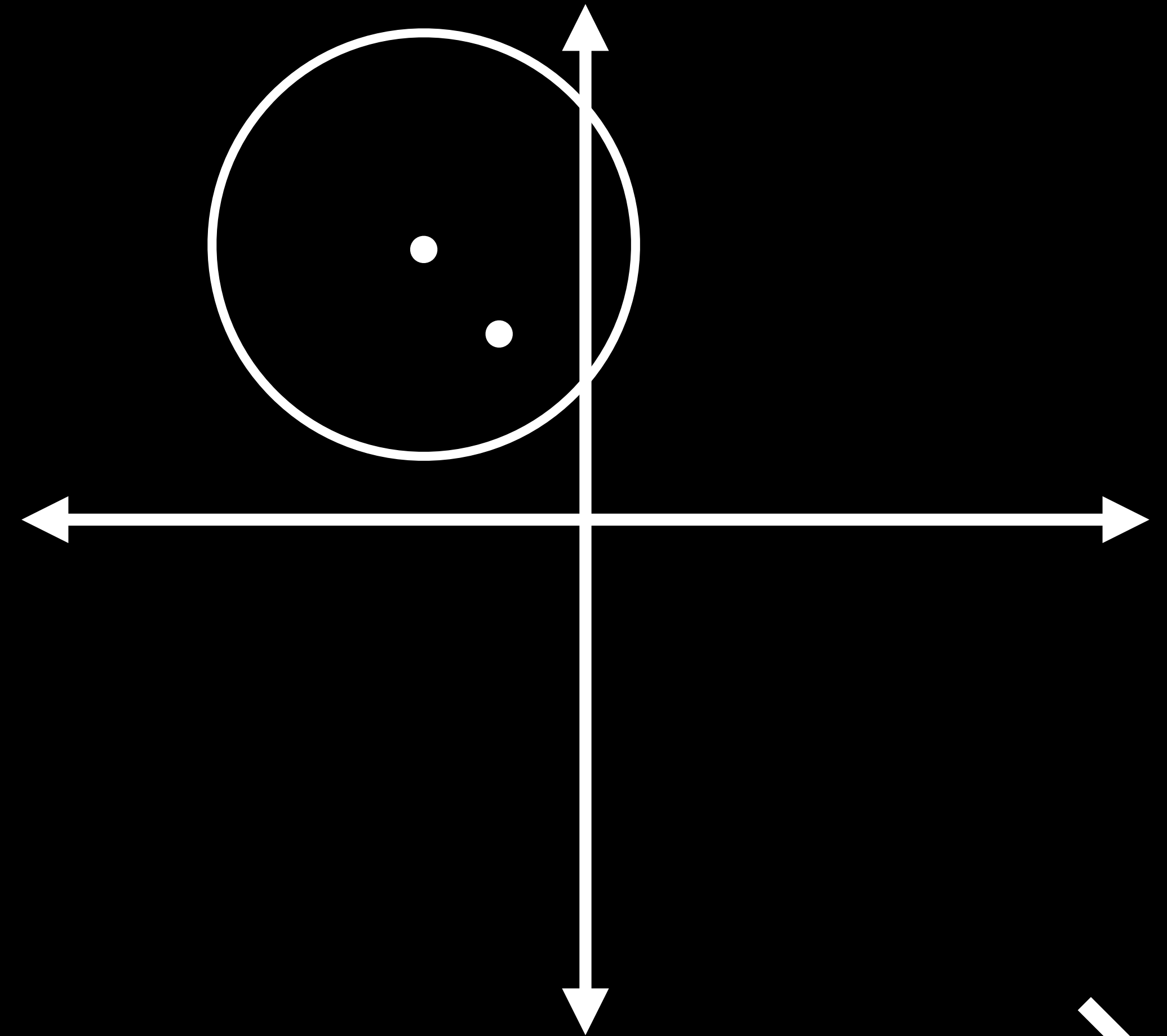
Show that the point $(-2,4)$ lies in the circle

$$(x + 3)^2 + (y - 5)^2 = 16$$

$$\begin{aligned} \text{distance of point from center} &= \sqrt{(-2 + 3)^2 + (4 - 5)^2} \\ &= \sqrt{2} \end{aligned}$$

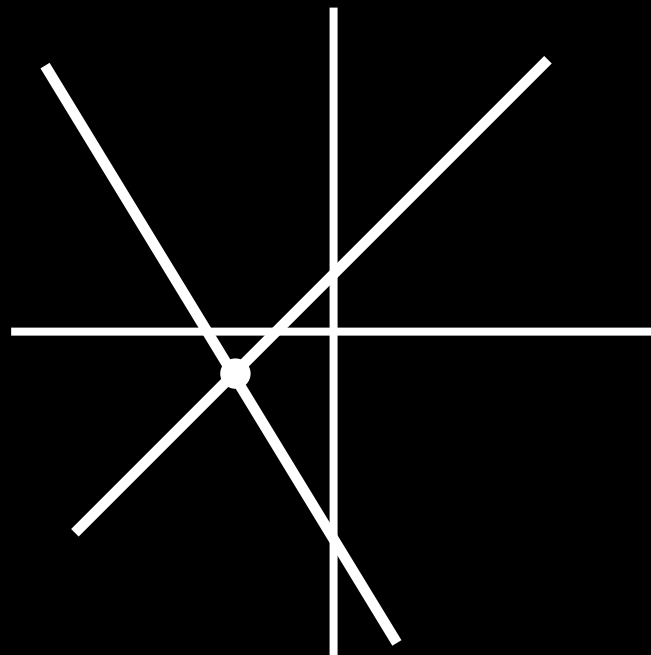
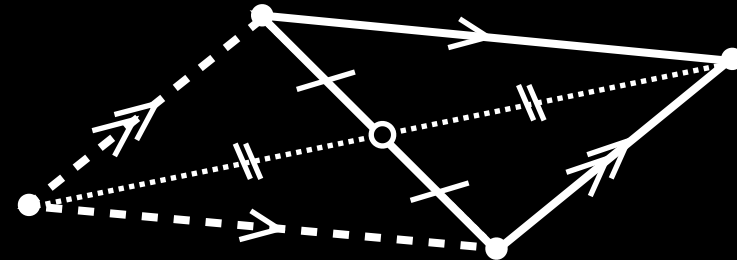
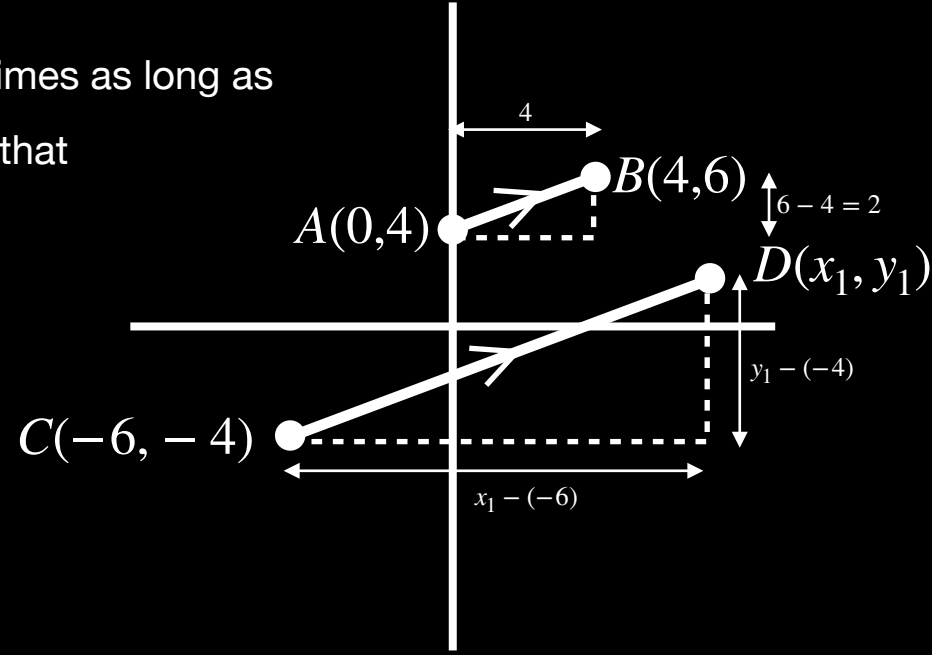
As $\sqrt{2} < \sqrt{16}$, distance from point is less than radius

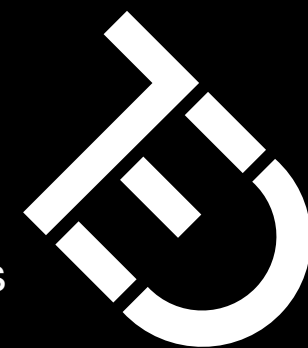
$\therefore (-2,4)$ lies in the circle



Finding Points?

Questions may ask you to find the coordinates of points, which there are 3 main methods of doing so:

Simul. Eqns	Midpoint	Similarity
solve two linear equations simultaneously to find intersection pt. $y = x + 3 \rightarrow (1)$ $y = -2x - 9 \rightarrow (2)$ Sub (1) into (2): $x + 3 = -2x - 9$ $x + 2x = -9 - 3$ $3x = -12$ $x = -4$ Sub $x = -4$ into (1): $y = -4 + 3$ $y = -1$ $\therefore$ intersection point is $(-4, -1)$ 	If two points form to have a known midpoint, we can find one of the unknown points Given that $C(-6,1)$ and the midpoint of CD is $(-2,-1)$, find the coordinates of D. Let $D(a, b)$ $\left(\frac{-6 + a}{2}, \frac{1 + b}{2}\right) = (-2, -1)$ $\therefore \frac{-6 + a}{2} = -2 \quad \therefore \frac{1 + b}{2} = -1$ $\begin{array}{l} -6 + a = -4 \\ a = -4 + 6 \\ a = 2 \end{array} \quad \begin{array}{l} 1 + b = -2 \\ b = -2 - 1 \\ b = -3 \end{array}$ $\therefore D(2, -3)$ Commonly used for finding the last point of a parallelogram/rhombus, where qns ask you to find midpoint of two opposite points, and then ask you to find last point of figure by making use of the common midpoint of the quadrilateral 	If two lines are given to be similar and are parallel, we can use similarity ratios to find points. The line AB where $A(0,4)$ and $B(4,6)$ is $\frac{1}{3}$ times as long as the line CD and is also parallel to CD. Given that $C(-6, -4)$, find D. Let $D(x_1, y_1)$ By similar triangles, $\frac{x_1 - (-6)}{4 - 0} = \frac{y_1 - (-4)}{6 - 4} = \frac{CD}{AB}$ $\frac{x_1 + 6}{4} = \frac{y_1 + 4}{2} = 3$ $\therefore \frac{x_1 + 6}{4} = 3 \quad \therefore \frac{y_1 + 4}{2} = 3$ $\begin{array}{l} x_1 + 6 = 12 \\ x_1 = 6 \end{array} \quad \begin{array}{l} y_1 + 4 = 6 \\ y_1 = 2 \end{array} \quad \therefore D(6,2)$ Form right triangles along each line, and form similarity ratios to find coordinates of D Make sure to have corresponding sides on the same fraction: i.e. $\frac{\text{horizontal length of } CD}{\text{horizontal length of } AB}$ Can also be used for finding points on extended lines, as points will be collinear and thus lines will be parallel (and ratio of lengths will usually be stated) 

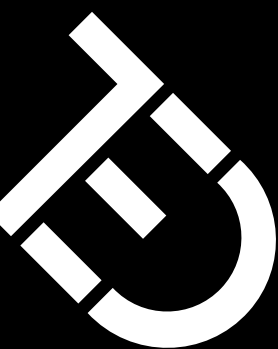
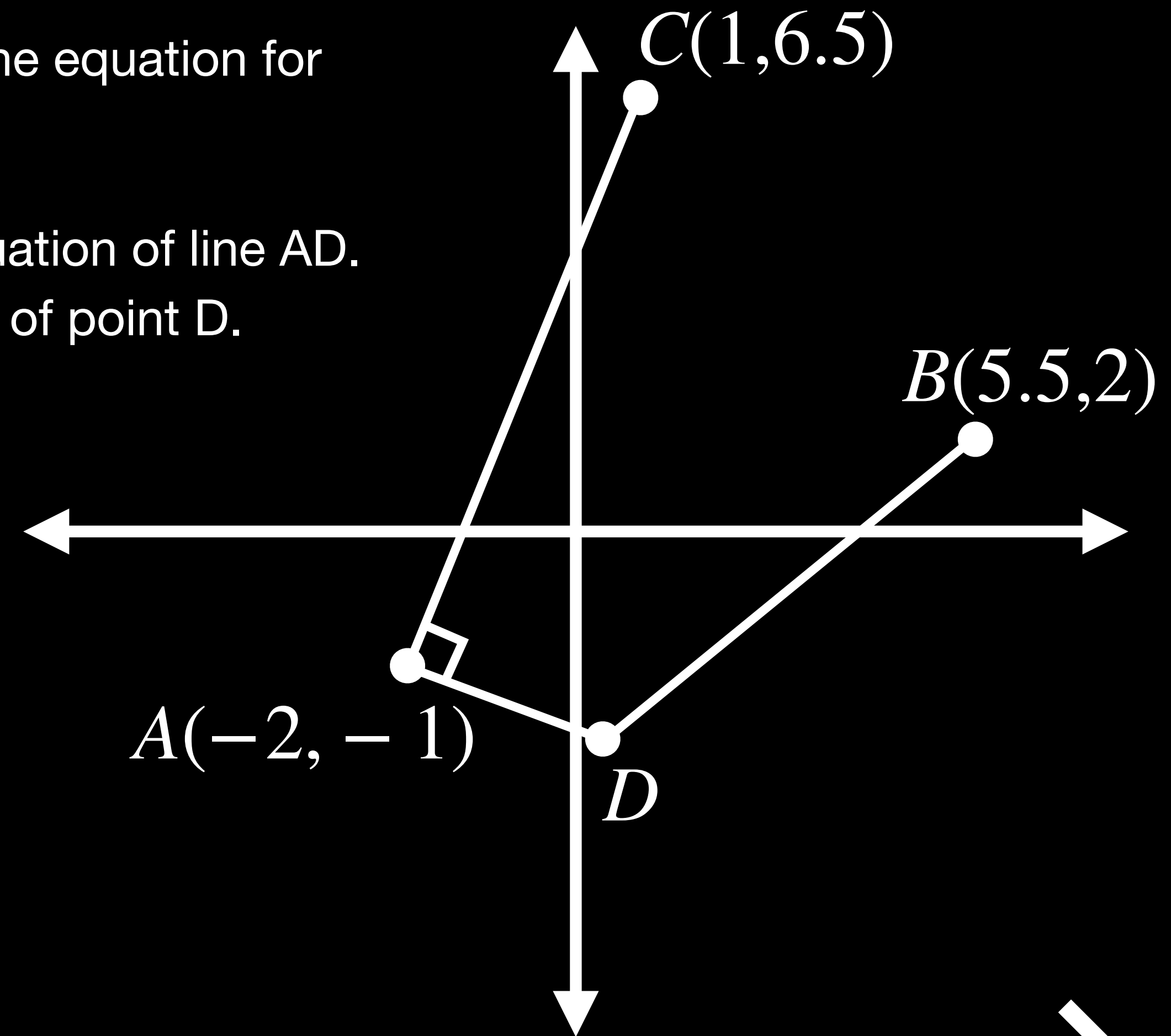


Solving Questions

i

The diagram shows the points $A(-2,-1)$, $B(5.5,2)$, $C(1,6.5)$ and D . The equation for line AC is $2y - 8 = 5x$.

- a) find the equation of the line BC
- b) Given that AC and AD are perpendicular at point A , find the equation of line AD .
- c) The equation of line BD is $y = 0.8x - 2.4$. Find the coordinates of point D .
- d) Find the area of the quadrilateral $ACBD$.



Solving Questions

W

The diagram shows the points $A(-2,-1)$, $B(5.5,2)$, $C(1,6.5)$ and D . The equation for line AC is $2y - 8 = 5x$.

a) find the equation of the line BC

b) Given that AC and AD are perpendicular at point A , find the equation of line AD .

c) The equation of line BD is $y = 0.8x - 2.4$. Find the coordinates of point D .

d) Find the area of the quadrilateral $ACBD$.

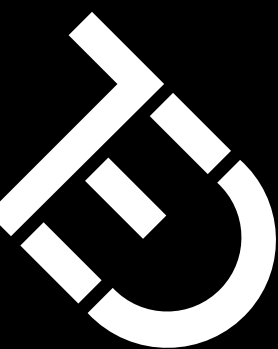
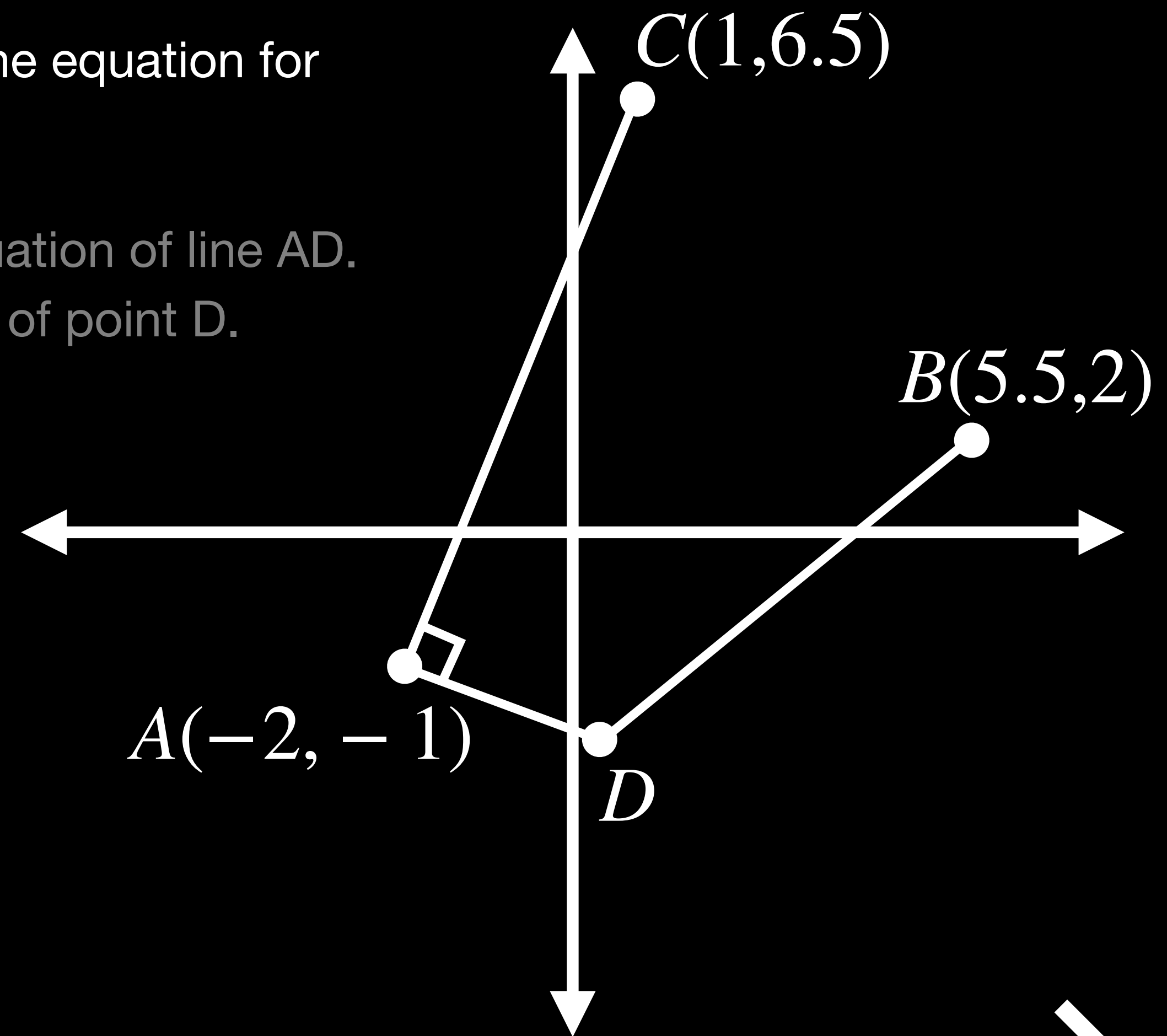
$$\begin{aligned} m_{BC} &= \frac{2 - 6.5}{5.5 - 1} \\ &= -1 \end{aligned}$$

Eqn of BC is

$$y - 2 = -(x - 5.5)$$

$$y - 2 = -x + 5.5$$

$$y = -x + 7.5$$



Solving Questions

a

The diagram shows the points $A(-2,-1)$, $B(5.5,2)$, $C(1,6.5)$ and D . The equation for line AC is $2y - 8 = 5x$.

a) find the equation of the line BC

b) Given that AC and AD are perpendicular at point A , find the equation of line AD .

c) The equation of line BD is $y = 0.8x - 2.4$. Find the coordinates of point D .

d) Find the area of the quadrilateral $ACBD$.

$$2y - 8 = 5x$$

$$y - 4 = 2.5x$$

$$y = 2.5x + 4$$

$$\therefore m_{AC} = 2.5$$

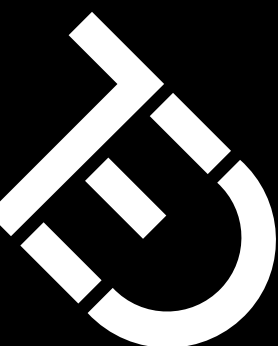
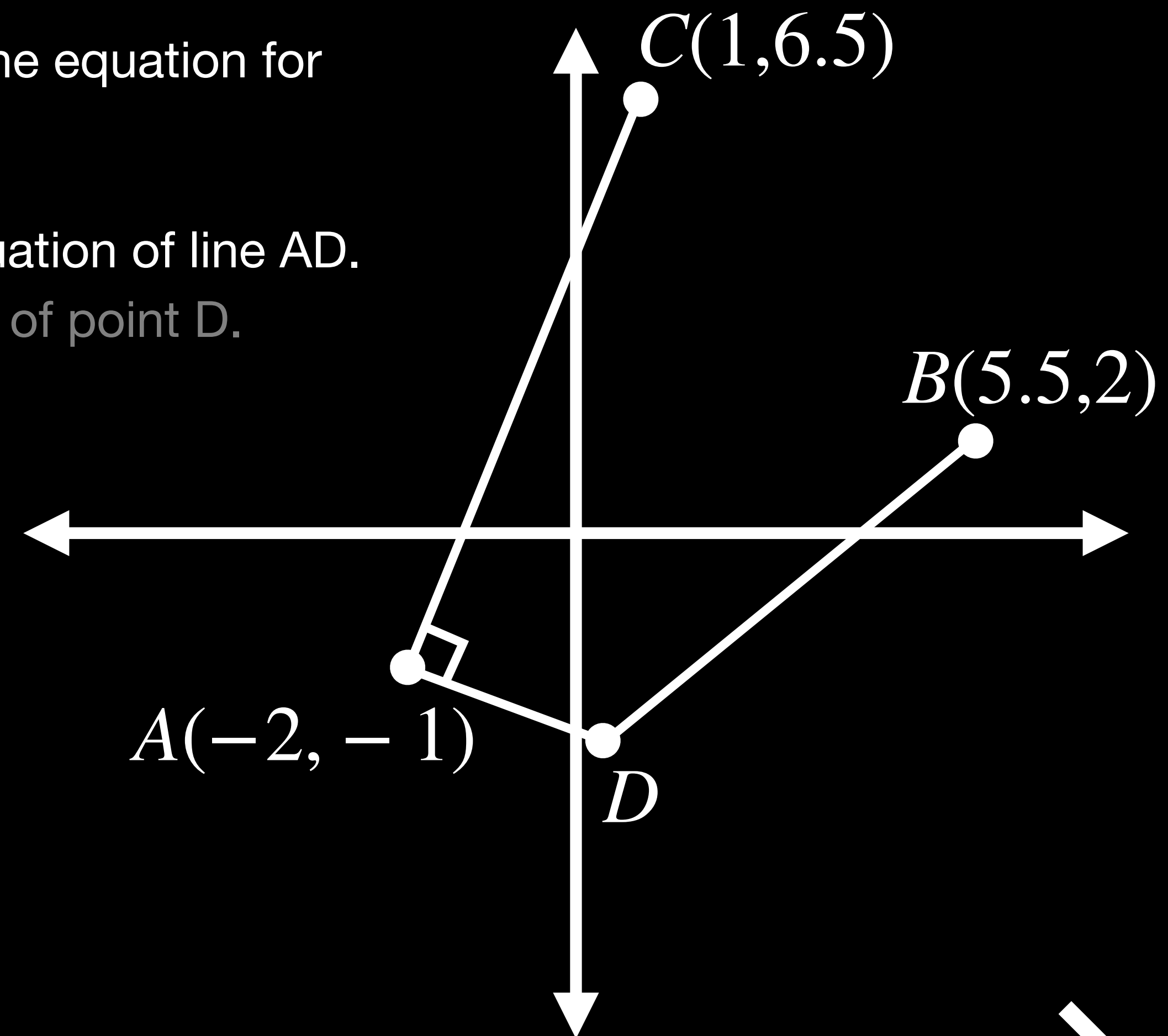
$$m_{AD} = -(2.5)^{-1} \\ = -0.4$$

Thus, eqn of AD is

$$y - (-1) = -0.4(x - (-2))$$

$$y + 1 = -0.4x - 0.8$$

$$y = -0.4x - 1.8$$



Solving Questions

S

The diagram shows the points $A(-2,-1)$, $B(5.5,2)$, $C(1,6.5)$ and D . The equation for line AC is $2y - 8 = 5x$.

a) find the equation of the line BC

b) Given that AC and AD are perpendicular at point A , find the equation of line AD .

c) The equation of line BD is $y = 0.8x - 2.4$. Find the coordinates of point D .

d) Find the area of the quadrilateral $ACBD$.

$$y = -0.4x - 1.8 \rightarrow (1)$$

$$y = 0.8x - 2.4 \rightarrow (2)$$

sub (1) into (2):

$$-0.4x - 1.8 = 0.8x - 2.4$$

$$-1.8 + 2.4 = 0.8x + 0.4x$$

$$0.6 = 1.2x$$

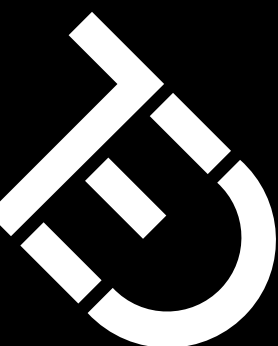
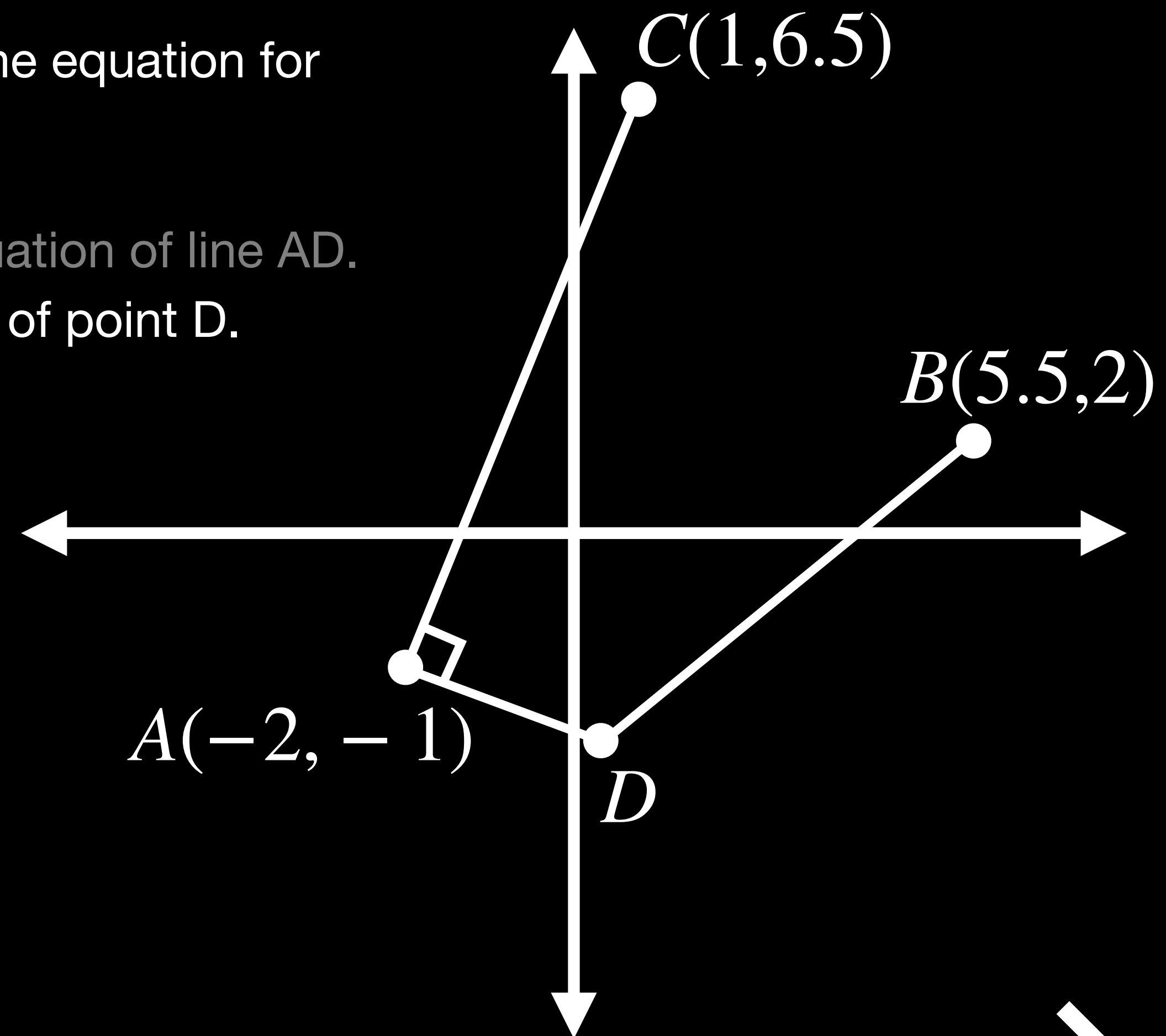
$$x = 0.5$$

sub $x=0.5$ into (2):

$$y = 0.8(0.5) - 2.4$$

$$y = -2$$

$$\therefore D(0.5, -2)$$



Solving Questions

C

The diagram shows the points $A(-2,-1)$, $B(5.5,2)$, $C(1,6.5)$ and D . The equation for line AC is $2y - 8 = 5x$.

a) find the equation of the line BC

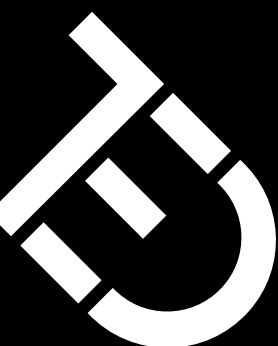
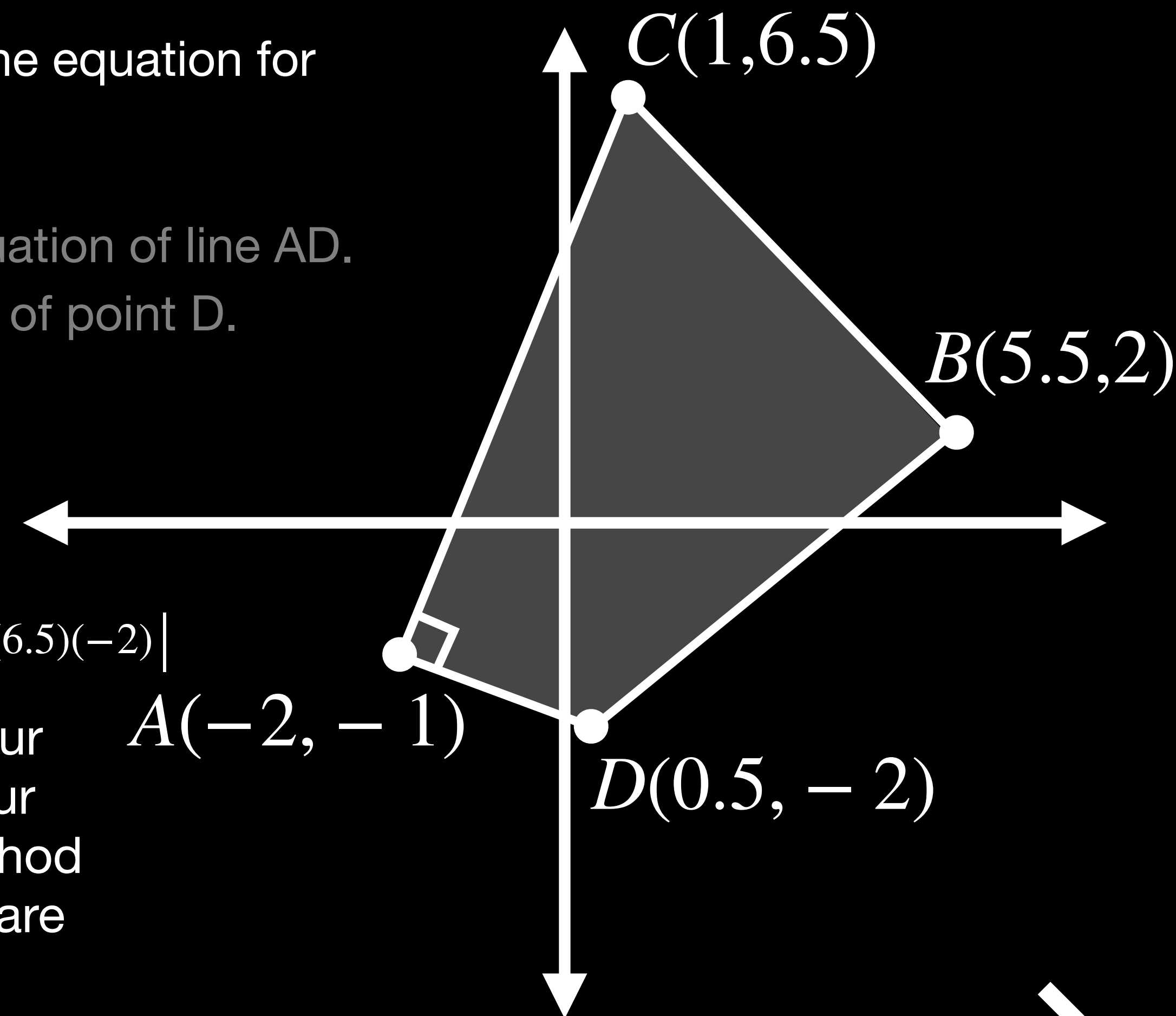
b) Given that AC and AD are perpendicular at point A , find the equation of line AD .

c) The equation of line BD is $y = 0.8x - 2.4$. Find the coordinates of point D .

d) Find the area of the quadrilateral $ACBD$.

$$\begin{aligned} & \frac{1}{2} \begin{vmatrix} -2 & 0.5 & 5.5 & 1 & -2 \\ -1 & -2 & 2 & 6.5 & -1 \end{vmatrix} \\ &= \frac{1}{2} |(-2)(-2) + (0.5)(2) + (5.5)(6.5) + (1)(-1) - (-1)(0.5) - (-2)(5.5) - (2)(1) - (6.5)(-2)| \\ &= \frac{1}{2} |62.25| \\ &= 31.125 \text{ units}^2 \end{aligned}$$

****When doing shoelace method, make sure that your points are “continuous”, aka they go in order of your quadrilateral. For example, using the shoelace method on $ABCD$ will give a different answer as the points are not in order.**



Solving Questions

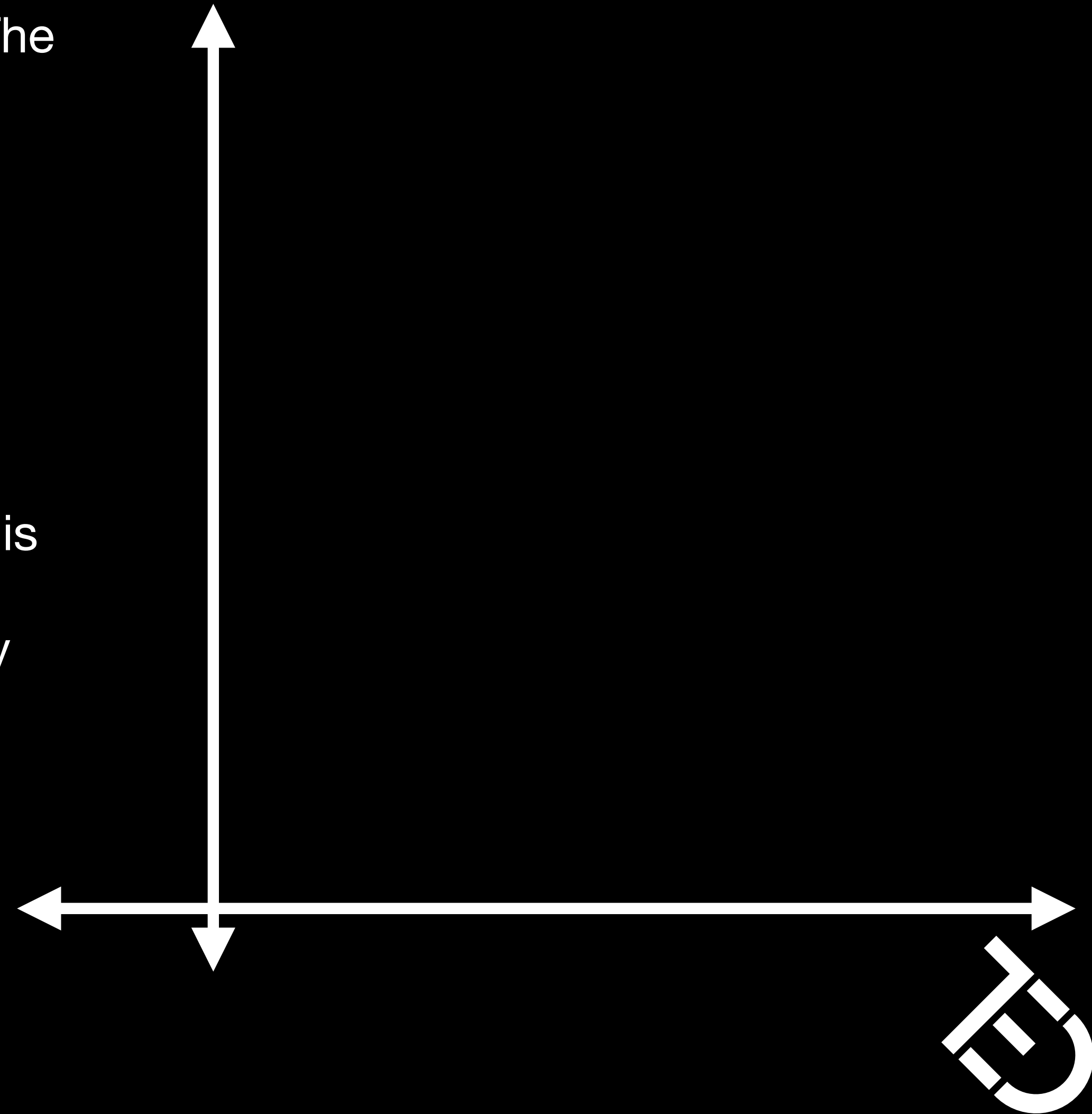
r

Two points A(2,3) and B(3,5) lie on the circumference of a circle. The center of the circle lies on the line $y = 3x - 0.875$

- a) Find the equation of the circle.
- b) Find the equation of the tangent at A
- c) Does the point (3,4) lie within the circle? Show why or why not.

Typically, for coordinate geometry of circles questions, a diagram is not provided, thus it is useful to sketch a diagram

As all of the points provided all have positive coordinates, we only need to focus on the first quadrant.



Solving Questions

a

Two points A(2,3) and B(3,5) lie on the circumference of a circle. The center of the circle lies on the line $y = 3x - 0.875$

a) Find the equation of the circle.

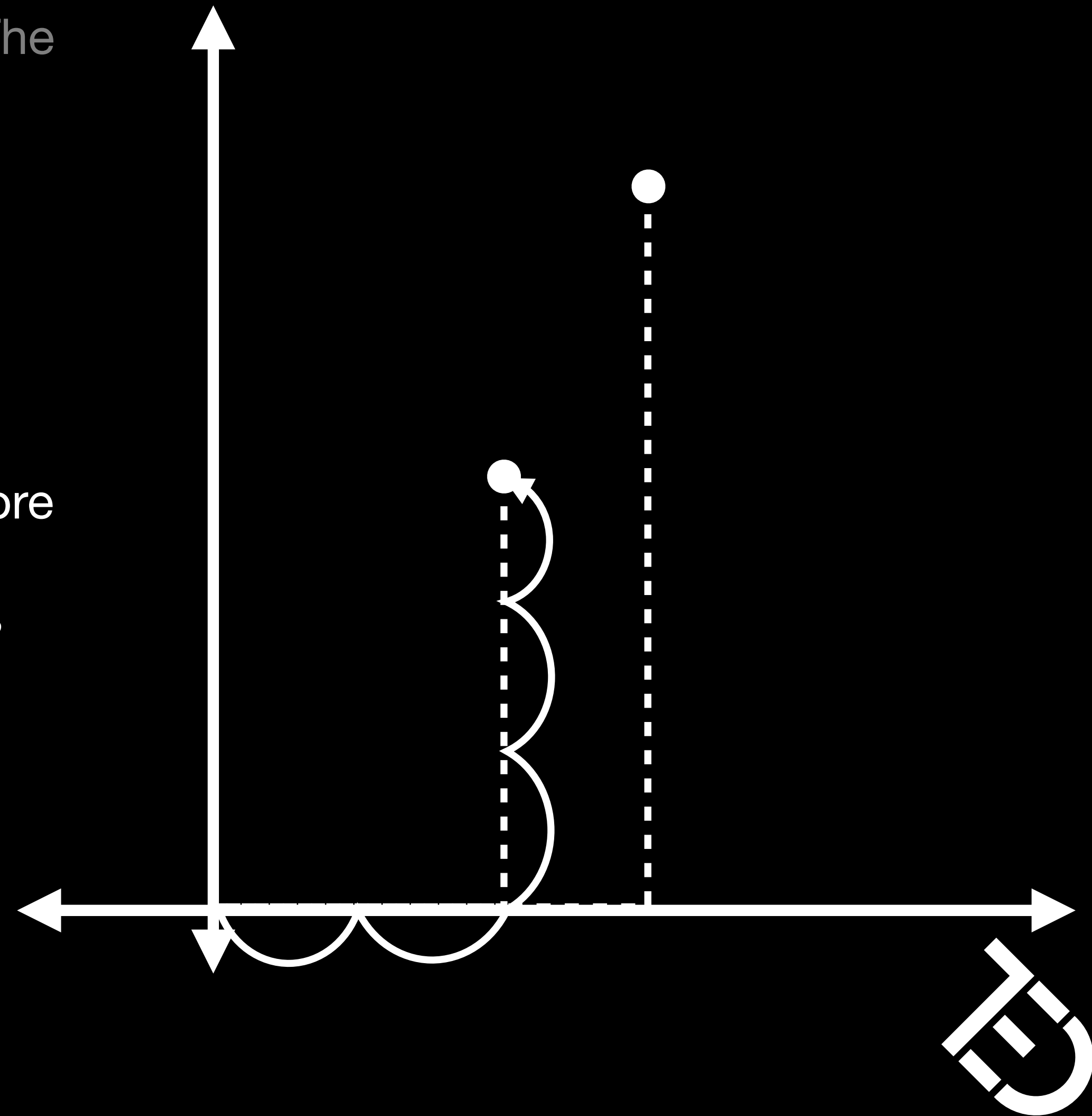
b) Find the equation of the tangent at A

c) Does the point (3,4) lie within the circle? Show why or why not.

Make a rough plot of the points on your axes,

For example, to plot (2,3), we know that it has to be moved up more than it is moved right.

The point doesn't need to be fully accurately plotted, it just needs to be more or less in the right place



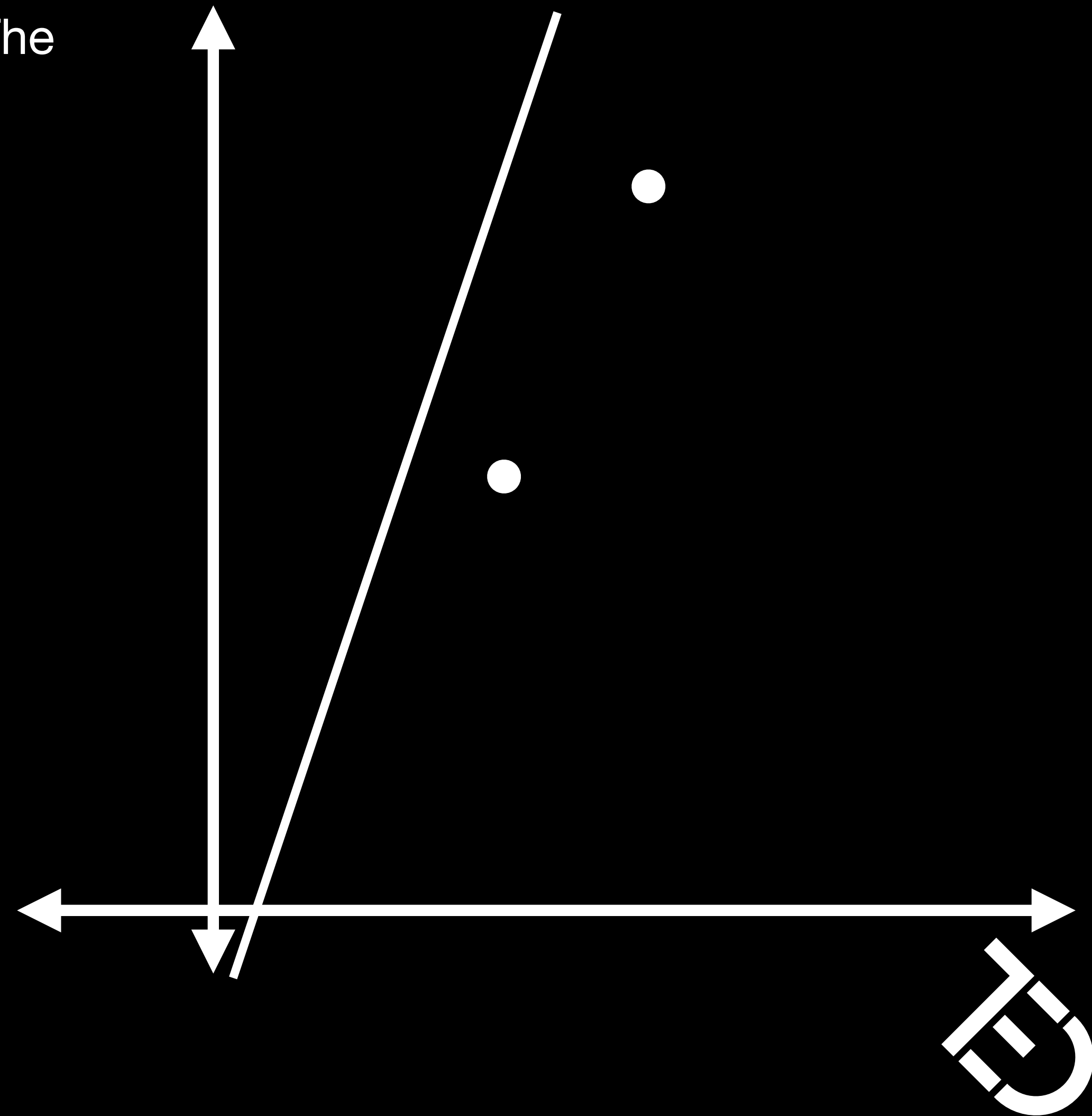
Solving Questions

Z

Two points A(2,3) and B(3,5) lie on the circumference of a circle. The center of the circle lies on the line $y = 3x - 0.875$

- a) Find the equation of the circle.
- b) Find the equation of the tangent at A
- c) Does the point (3,4) lie within the circle? Show why or why not.

To sketch lines, check the gradient of the line (gradient is 3, thus line is sloping up) and the y intercept, (y intercept is -0.875, line starts from the (0,-0.875) point)



Solving Questions

y

Two points A(2,3) and B(3,5) lie on the circumference of a circle. The center of the circle lies on the line $y = 3x - 0.875$

a) Find the equation of the circle.

b) Find the equation of the tangent at A

c) Does the point (3,4) lie within the circle? Show why or why not.

**whenever there are two points that lie on a circle, we can take the perpendicular bisector of the points to find the equation of the diameter.

$$\text{midpoint} = \left(\frac{2+3}{2}, \frac{3+5}{2} \right) \\ = (2.5, 4)$$

$$m_{AB} = \frac{5-3}{3-2} \\ = 2$$

$$m_{\text{perpendicular}} = -(2)^{-1} \\ = -0.5$$

$$y - 4 = -0.5(x - 2.5) \\ y - 4 = -0.5x + 1.25 \\ y = -0.5x + 5.25 \rightarrow (1)$$

$$y = 3x - 0.875 \rightarrow (2)$$

Sub (1) into (2):

$$\begin{aligned} -0.5x + 5.25 &= 3x - 0.875 \\ 5.25 + 0.875 &= 3x + 0.5x \\ 6.125 &= 3.5x \\ x &= 1.75 \end{aligned}$$

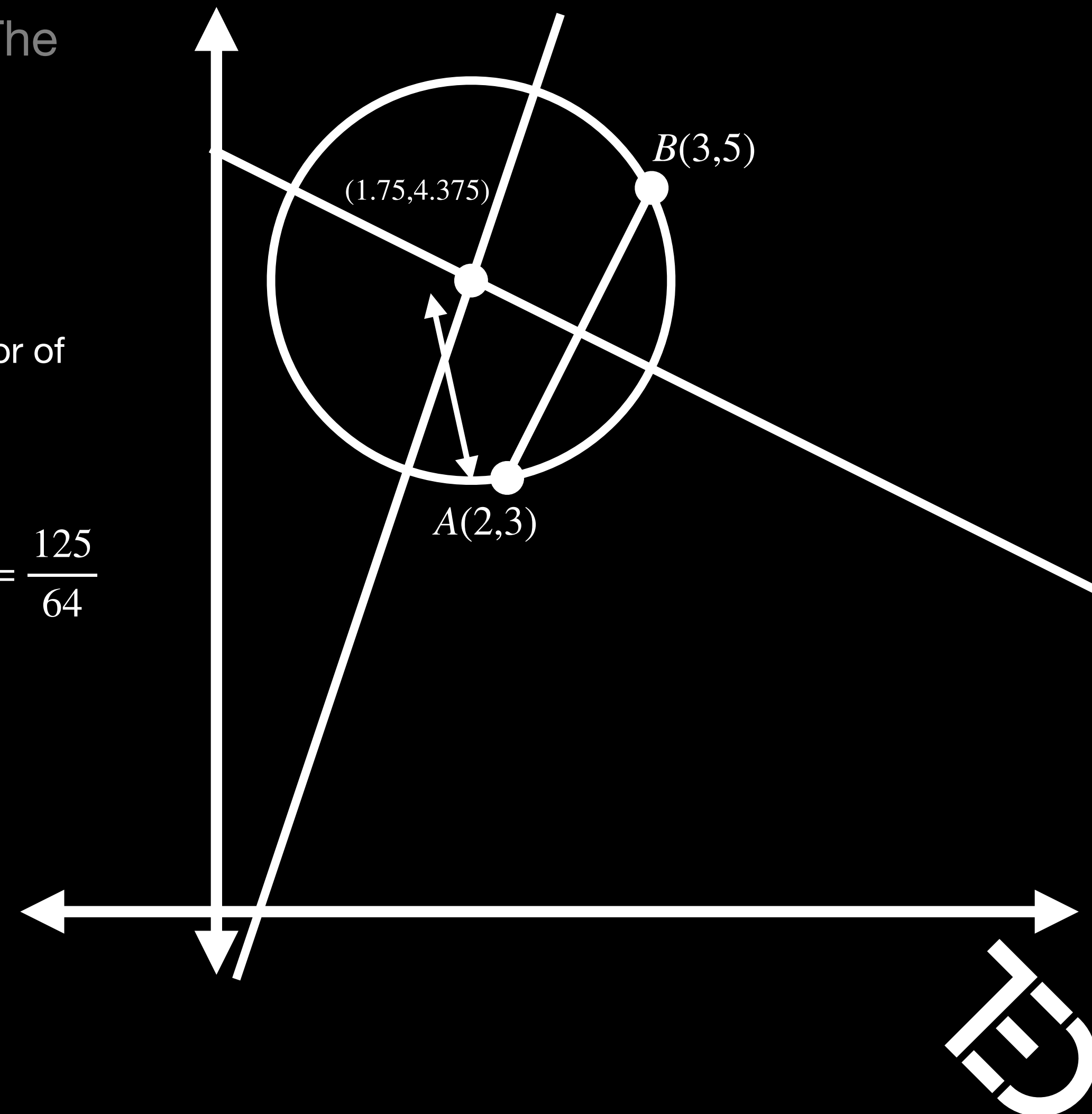
Sub $x=1.75$ into (1):

$$\begin{aligned} y &= -0.5(1.75) + 5.25 \\ y &= 4.375 \end{aligned}$$

$\therefore$ center is (1.75, 4.375)

$$\begin{aligned} \text{radius} &= \sqrt{(3 - 1.75)^2 + (5 - 4.375)^2} \\ &= \sqrt{\frac{125}{64}} \end{aligned}$$

$$\begin{aligned} \therefore \text{Eqn of circle is} \\ (x - 1.75)^2 + (y - 4.375)^2 &= \frac{125}{64} \end{aligned}$$



Solving Questions

O

Two points A(2,3) and B(3,5) lie on the circumference of a circle. The center of the circle lies on the line $y = 3x - 0.875$

a) Find the equation of the circle.

b) Find the equation of the tangent at A

c) Does the point (3,4) lie within the circle? Show why or why not.

$$\begin{aligned} m_{\text{normal}} &= \frac{3 - 4.375}{2 - 1.75} \\ &= -5.5 \end{aligned}$$

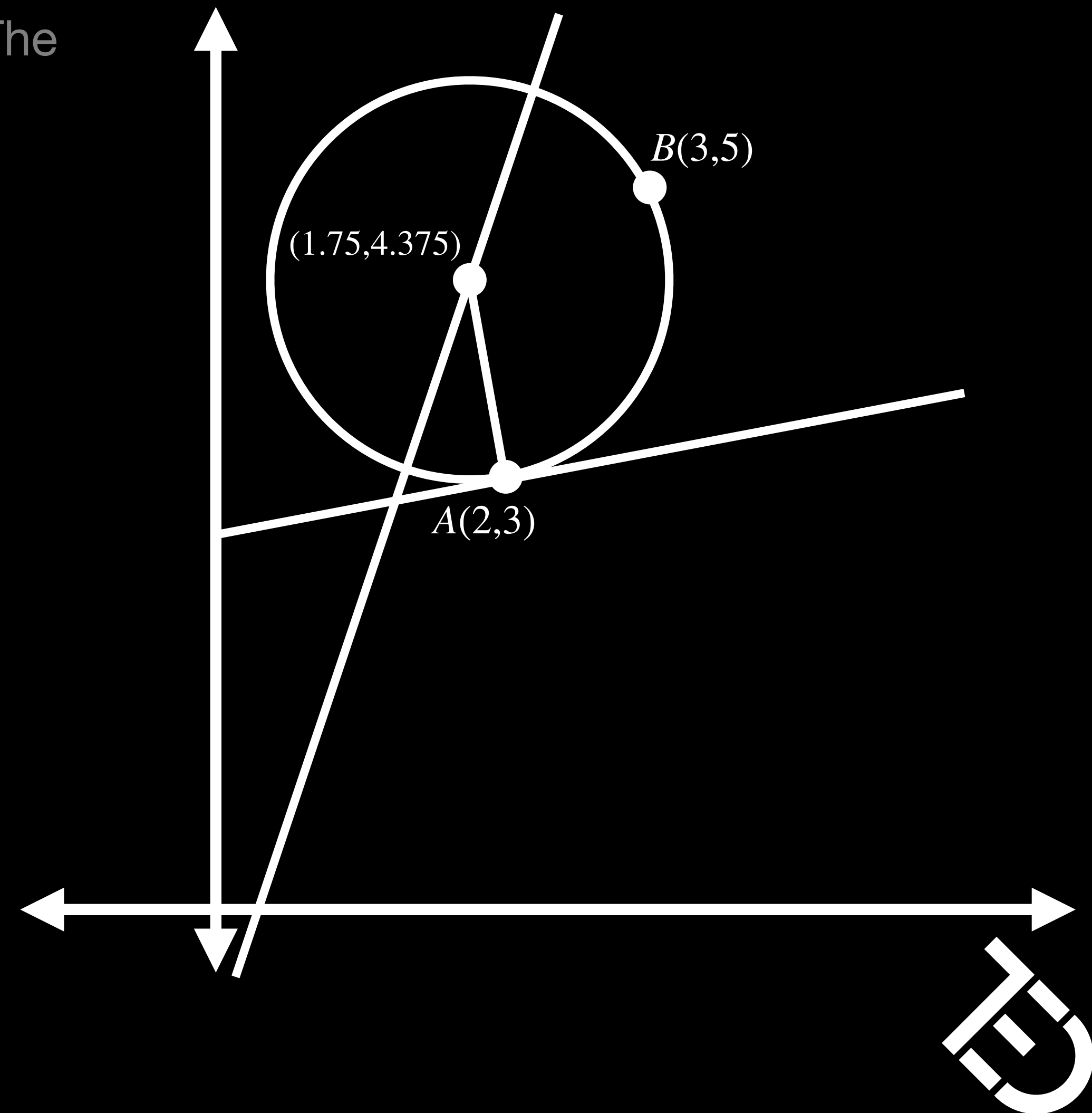
$$\begin{aligned} m_{\text{tangent}} &= -(-5.5)^{-1} \\ &= \frac{2}{11} \end{aligned}$$

$$y - 3 = \frac{2}{11}(x - 2)$$

$$y - 3 = \frac{2}{11}x - \frac{4}{11}$$

$$y = \frac{2}{11}x - \frac{4}{11} + 3$$

$$y = \frac{2}{11}x - \frac{29}{11}$$



Solving Questions

n

Two points A(2,3) and B(3,5) lie on the circumference of a circle. The center of the circle lies on the line $y = 3x - 0.875$

a) Find the equation of the circle.

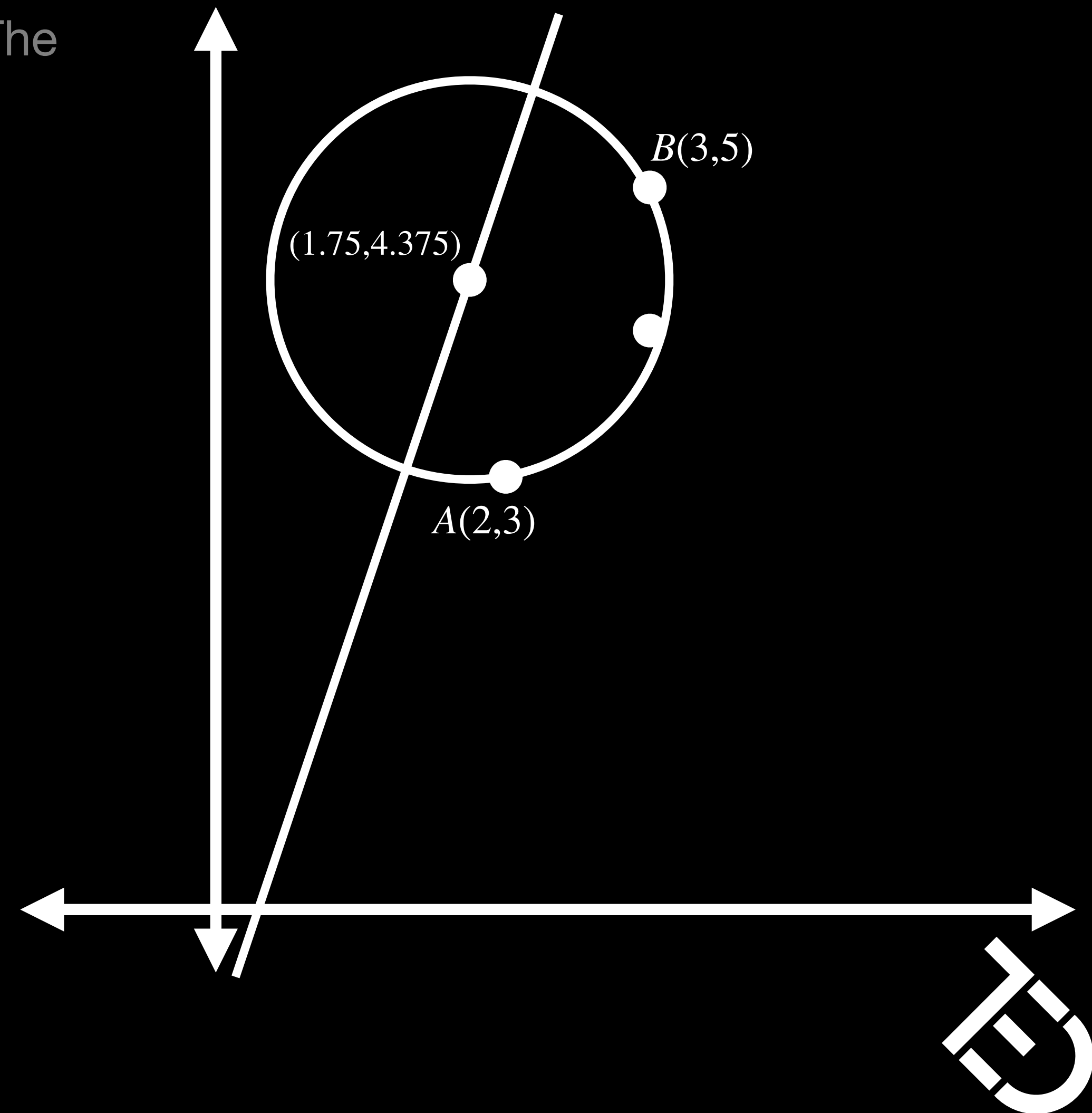
b) Find the equation of the tangent at A

c) Does the point (3,4) lie within the circle? Show why or why not.

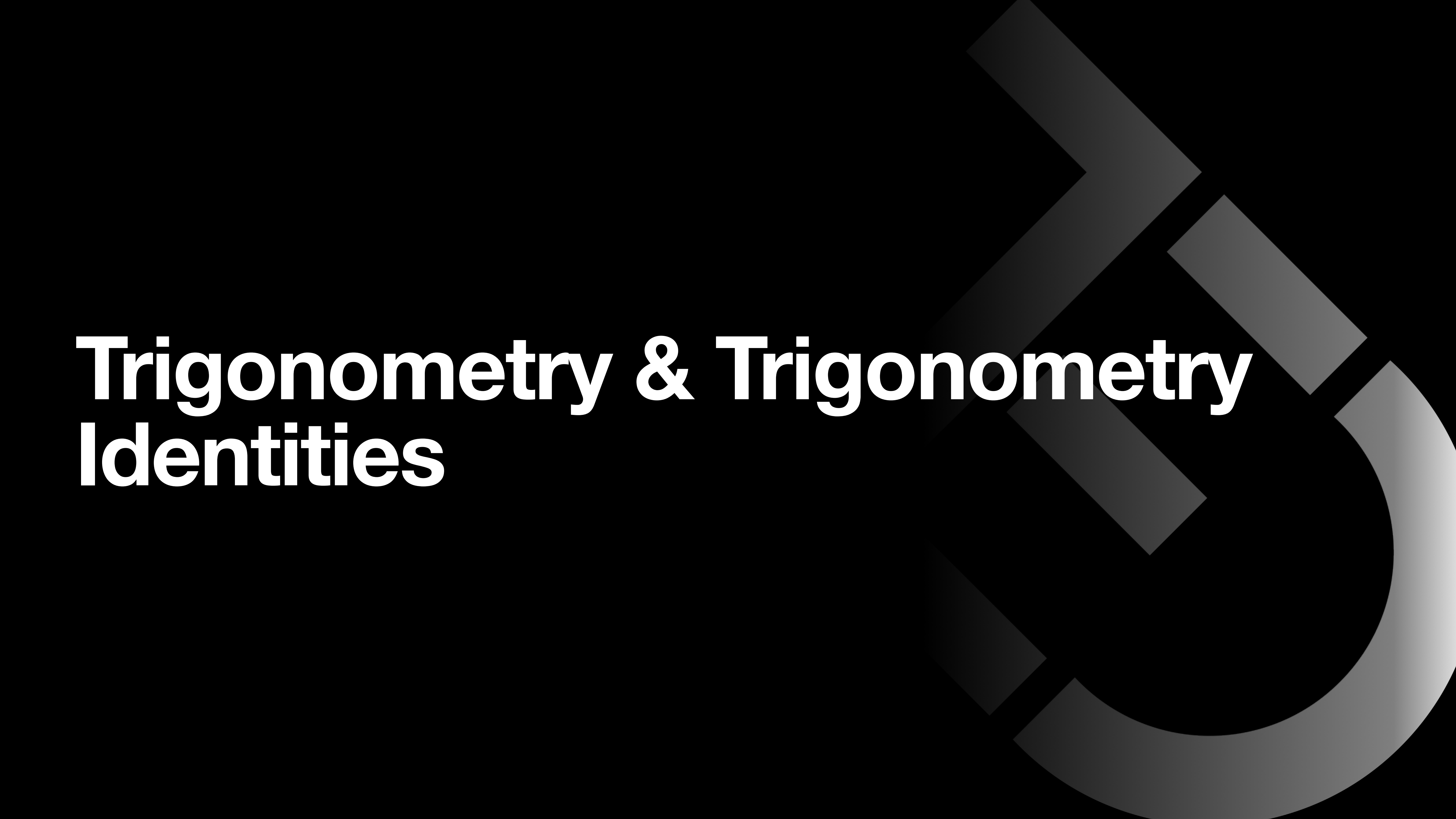
$$(x - 1.75)^2 + (y - 4.375)^2 = \frac{125}{64}$$

$$\begin{aligned}\text{distance of point to center} &= \sqrt{(3 - 1.75)^2 + (4 - 4.375)^2} \\ &= \sqrt{\frac{109}{64}}\end{aligned}$$

As $\sqrt{\frac{109}{64}} < \sqrt{\frac{125}{64}}$, distance from point is less than radius
thus the point (3,4) lies inside the circle



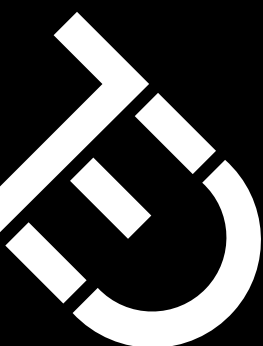
Trigonometry & Trigonometry Identities



Angle Presentation

cambrige

If the angle is in degrees, you are REQUIRED to put the degrees symbol
Else, it will be regarded as radians



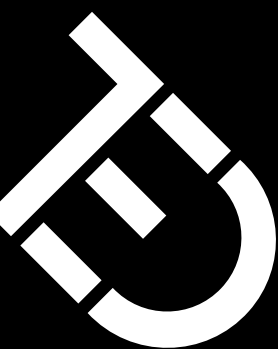
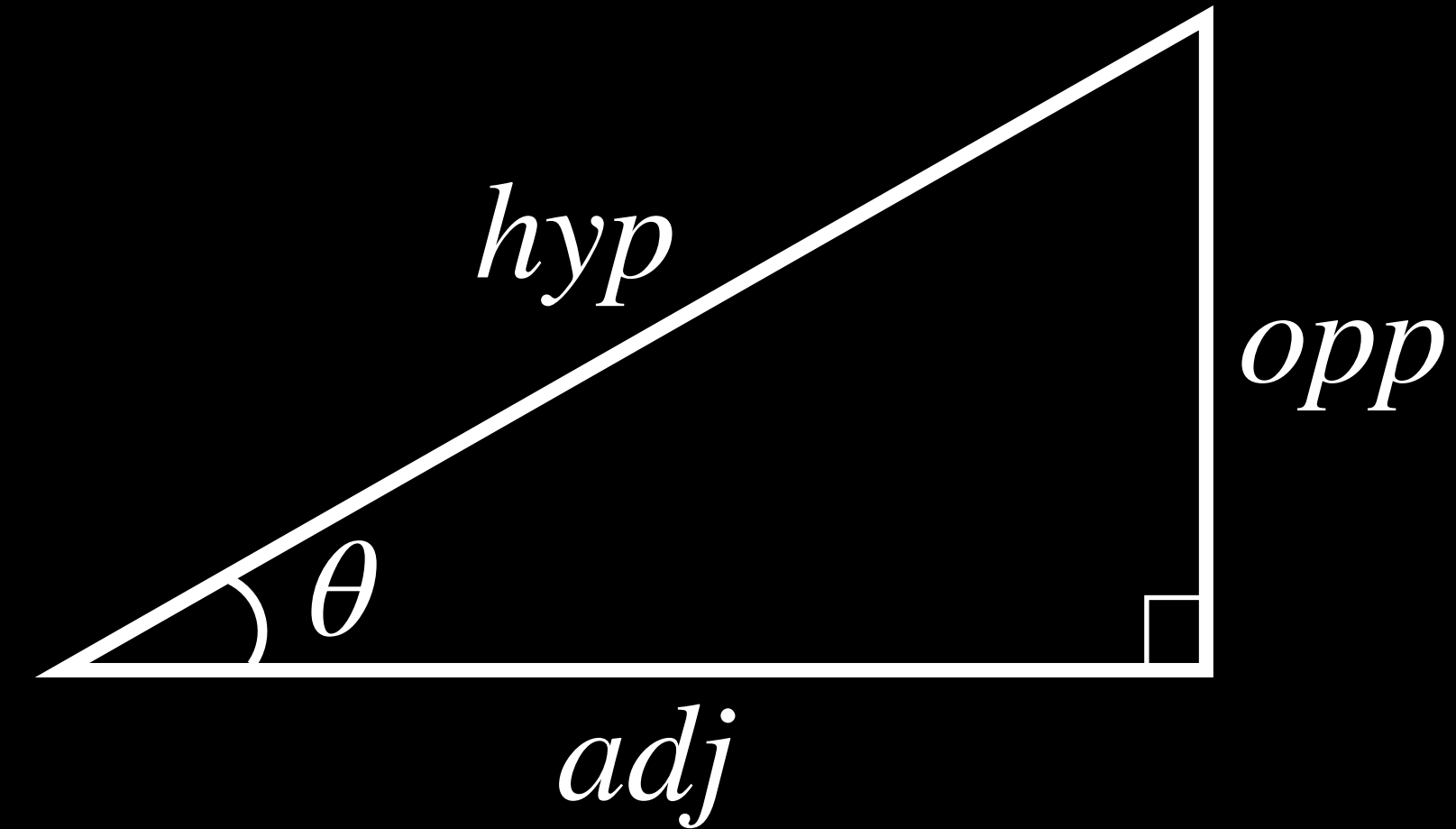
Trigonometric Ratios - SOH CAH TOA

Classic

$$\sin(\theta) = \frac{opp}{hyp}$$

$$\cos(\theta) = \frac{adj}{hyp}$$

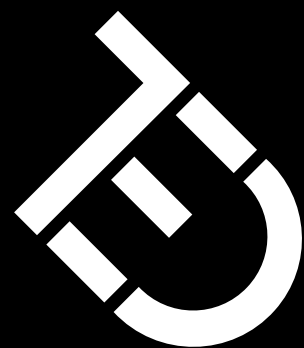
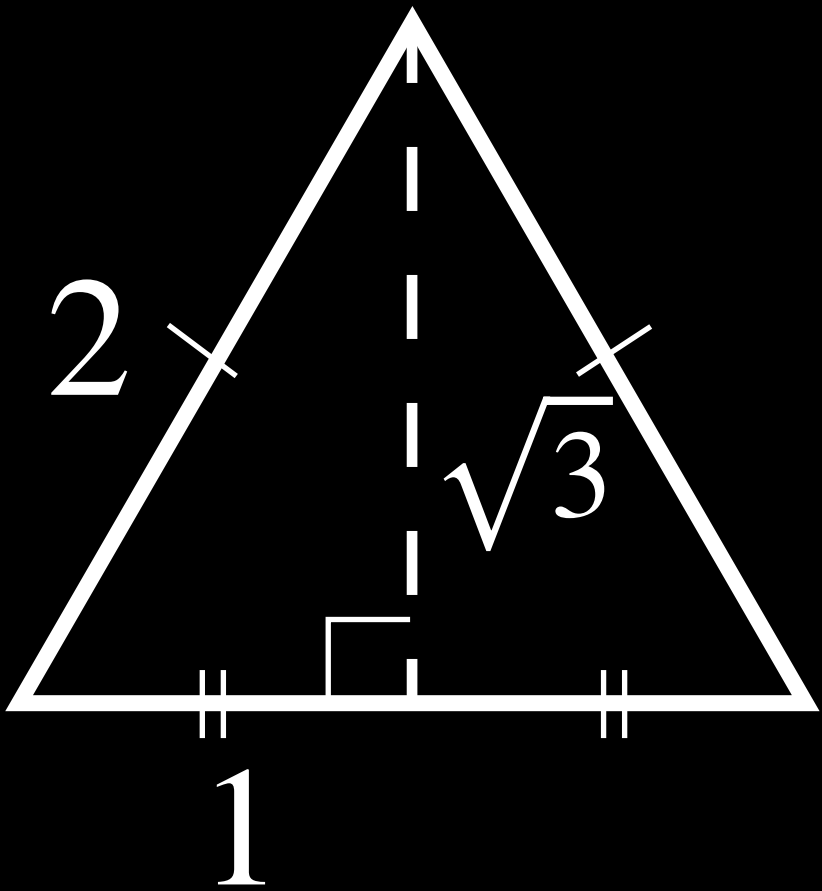
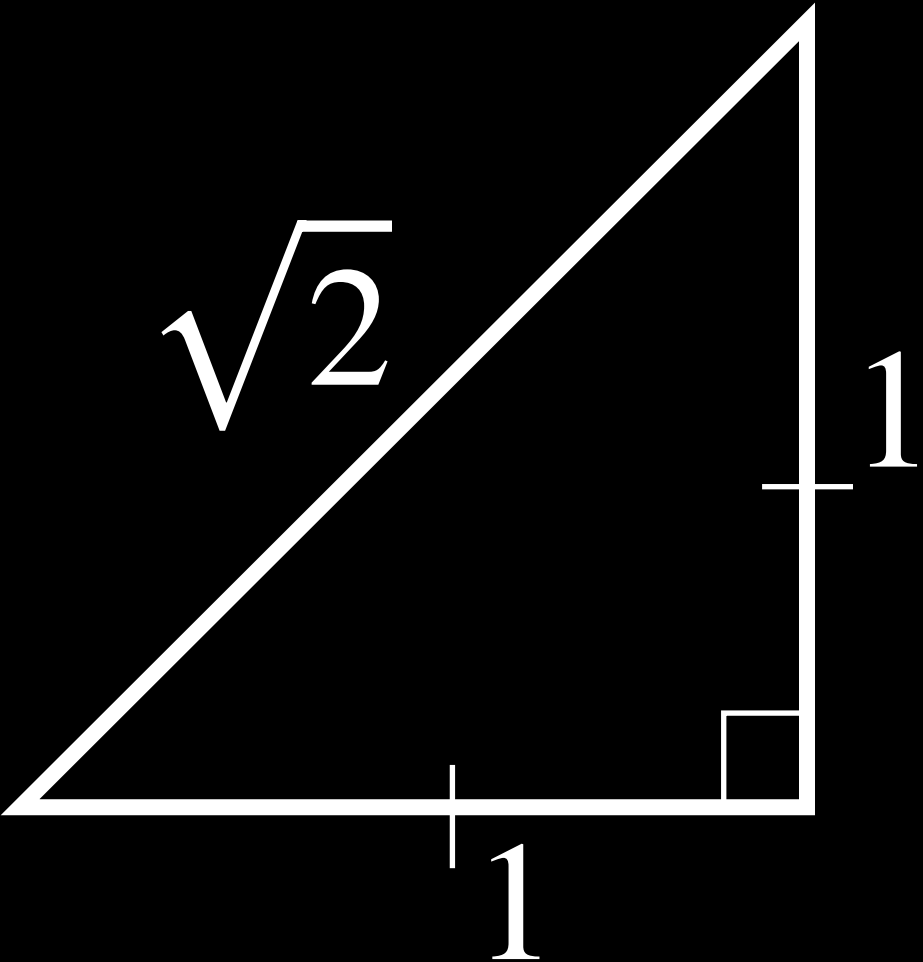
$$\tan(\theta) = \frac{opp}{adj} = \frac{\sin(\theta)}{\cos(\theta)}$$



Special Angles

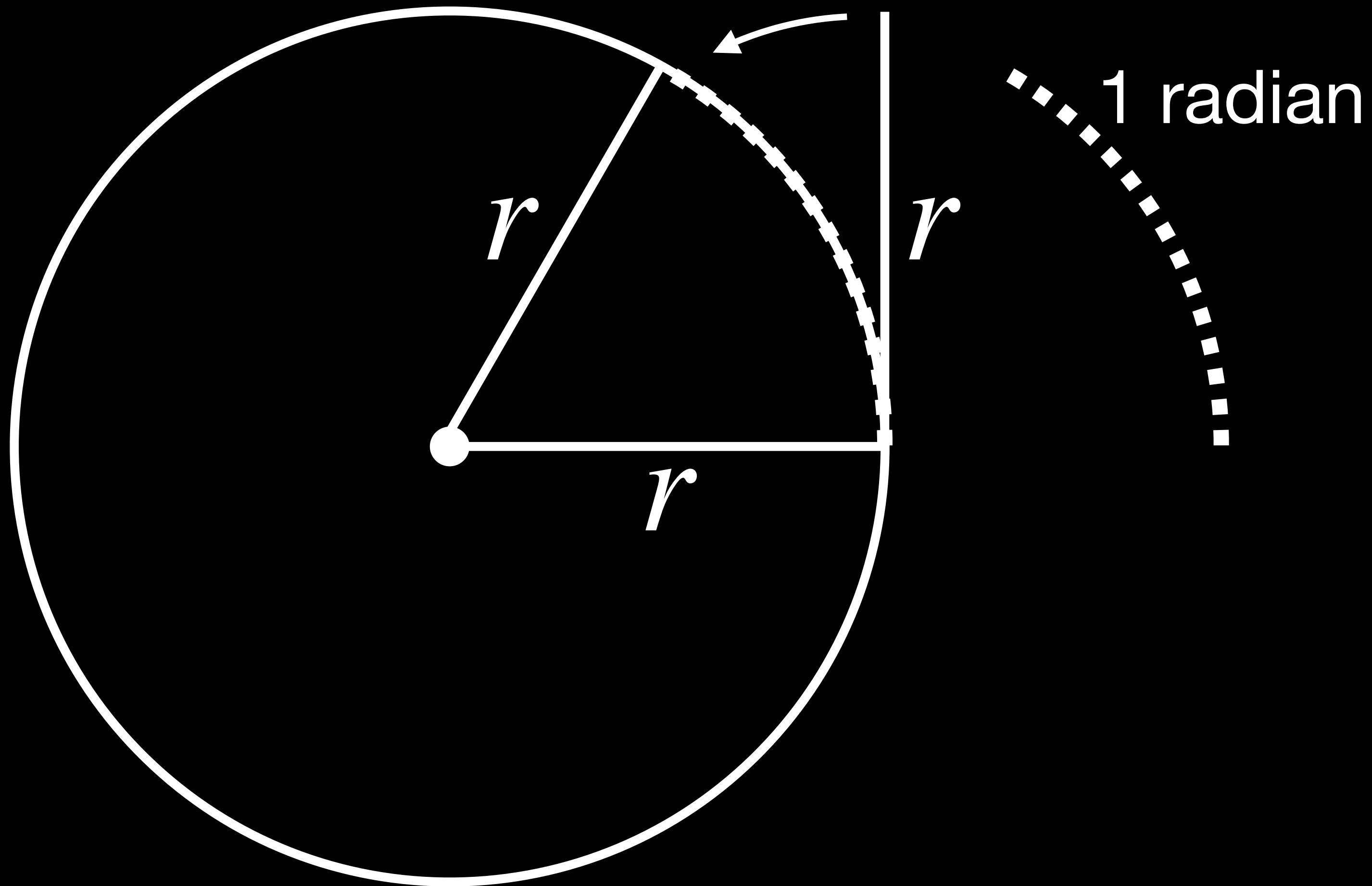
Classic

θ	0°	30°	45°	60°	90°
	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
$\sin(\theta)$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
$\cos(\theta)$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
$\tan(\theta)$	0	$\frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3}$	1	$\sqrt{3}$	undefined



Radians

(Hey guys radians are not really that cool)



just another way to measure angles

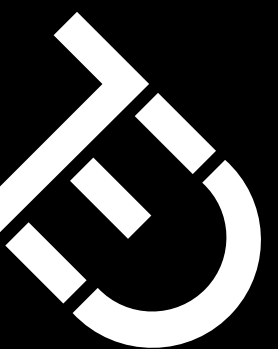
$$2\pi \text{ rad} = 360^\circ$$

$$\pi \text{ rad} = 180^\circ$$

$$\therefore 1 \text{ rad} = \frac{180}{\pi}$$

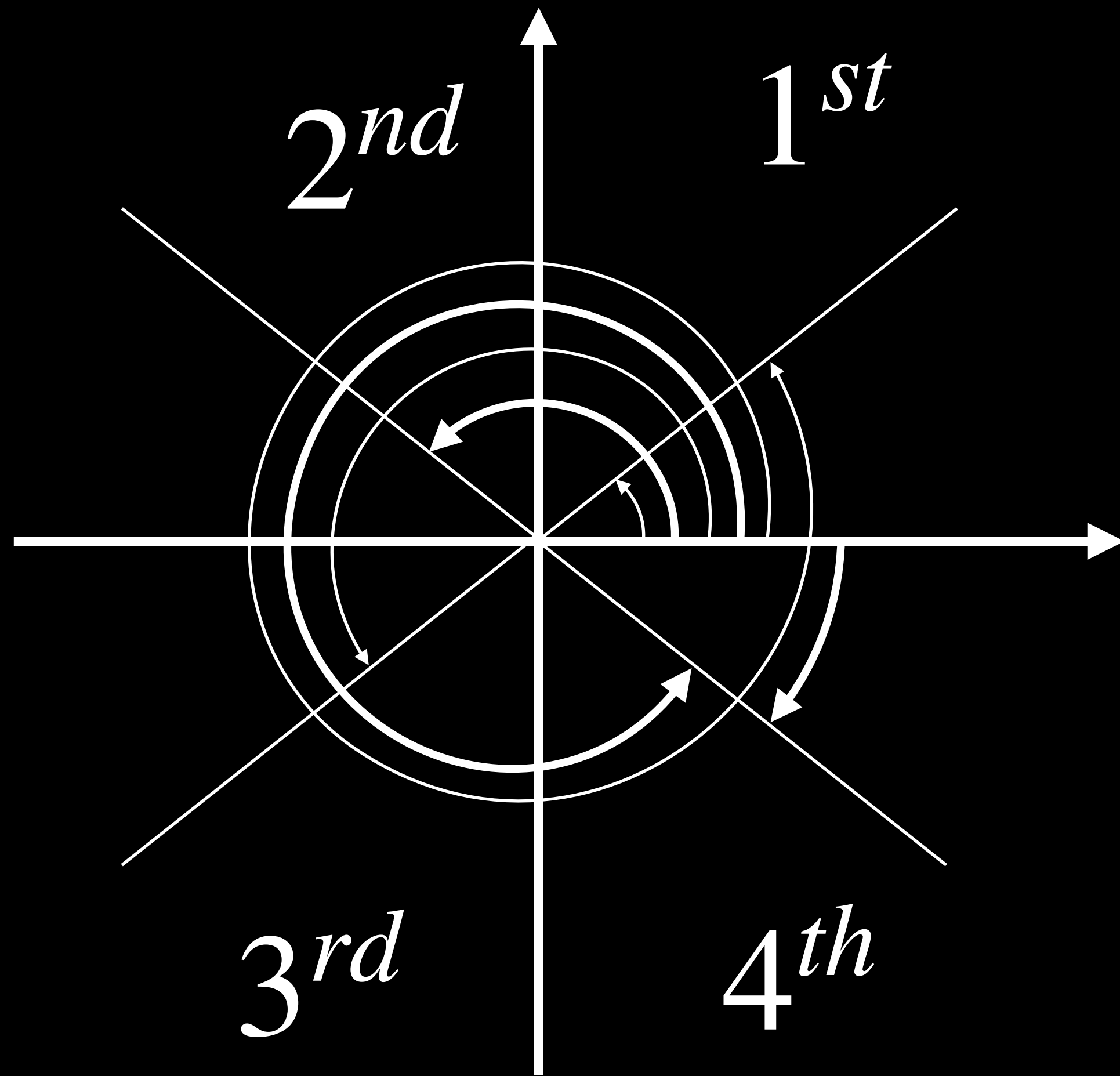
$$1^\circ = \frac{\pi}{180}$$

you do not need to understand this visual



Quadrants

The beginning



From inside to outside:

$\theta = 30^\circ$, 1st quadrant

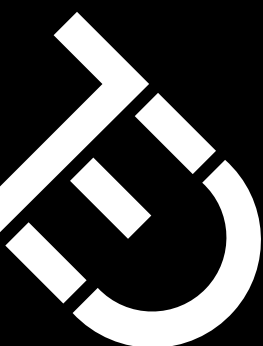
$\theta = 150^\circ$, 2nd quadrant

$\theta = 210^\circ$, 3rd quadrant

$\theta = 330^\circ$, 4th quadrant

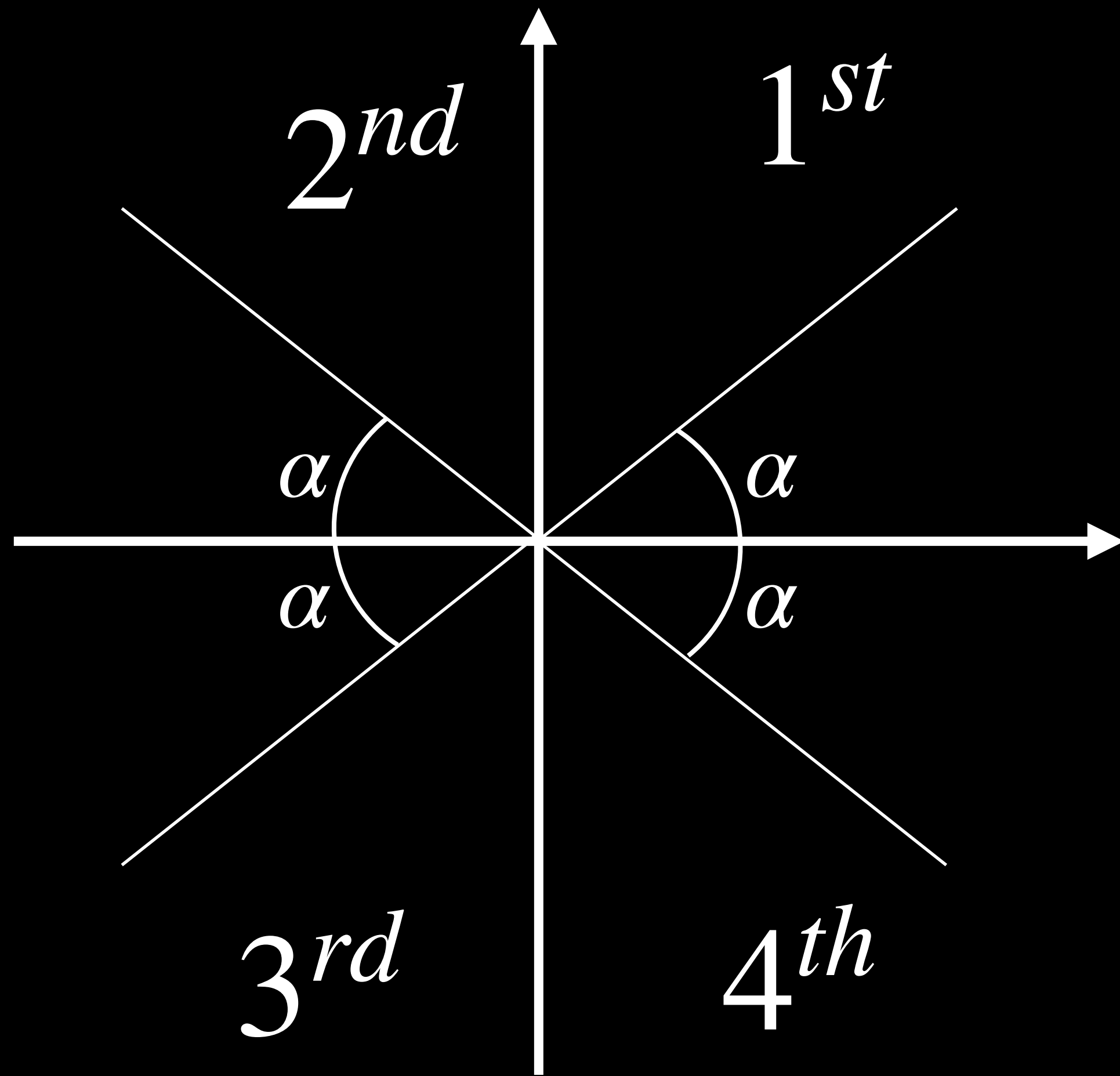
$\theta = 390^\circ$, 1st quadrant

$\theta = -30^\circ$, 4th quadrant



Quadrants

The beginning



Basic angle: α

positive angle between x axis and angle

$$\theta = 30^\circ, \alpha = 30^\circ$$

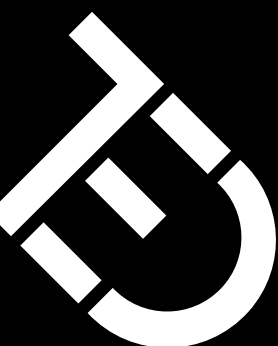
$$\theta = 150^\circ, \alpha = 30^\circ(180^\circ - 150^\circ)$$

$$\theta = 210^\circ, \alpha = 30^\circ(210^\circ - 180^\circ)$$

$$\theta = 330^\circ, \alpha = 30^\circ(360^\circ - 330^\circ)$$

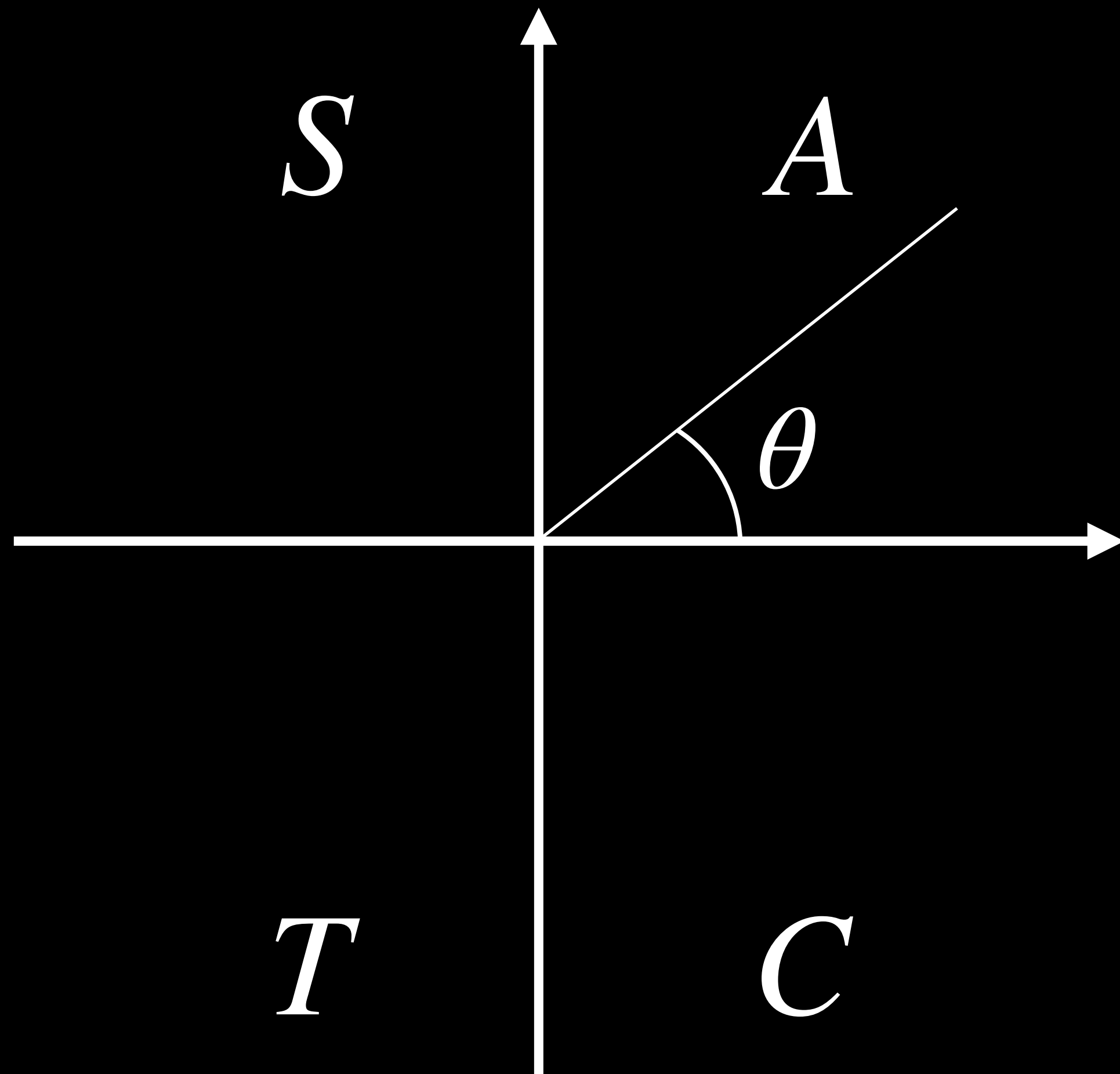
$$\theta = 390^\circ, \alpha = 30^\circ(390^\circ - 360^\circ)$$

$$\theta = -150^\circ, \alpha = 30^\circ(180^\circ - 150^\circ)$$



Quadrants

The beginning



For an angle θ ,

If in 1st quadrant:

$$\begin{aligned}\sin(\theta) &= \sin(\alpha) \\ \cos(\theta) &= \cos(\alpha) \\ \tan(\theta) &= \tan(\alpha) \\ \text{all values } &+\end{aligned}$$

If in 3rd quadrant:

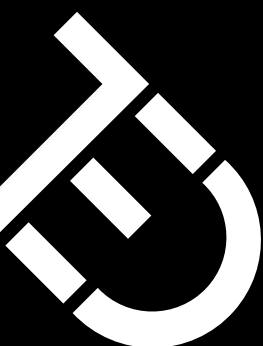
$$\begin{aligned}\sin(\theta) &= -\sin(\alpha) \\ \cos(\theta) &= -\cos(\alpha) \\ \tan(\theta) &= \tan(\alpha) \\ \text{only tangent } &+\end{aligned}$$

If in 2nd quadrant:

$$\begin{aligned}\sin(\theta) &= \sin(\alpha) \\ \cos(\theta) &= -\cos(\alpha) \\ \tan(\theta) &= -\tan(\alpha) \\ \text{only sine } &+\end{aligned}$$

If in 4th quadrant:

$$\begin{aligned}\sin(\theta) &= -\sin(\alpha) \\ \cos(\theta) &= \cos(\alpha) \\ \tan(\theta) &= -\tan(\alpha) \\ \text{only cosine } &+\end{aligned}$$



Complementary Angles

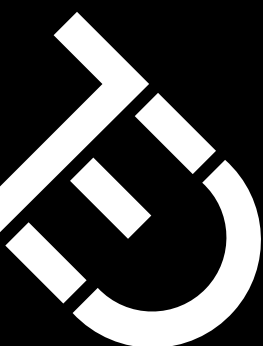
Who knows

$$\sin(90 - \theta) = \cos(\theta)$$

$$\cos(90 - \theta) = \sin(\theta)$$

$$\tan(90 - \theta) = \frac{1}{\tan(\theta)}$$

NOT IN FORMULA SHEET



Trigonometry Unknowns

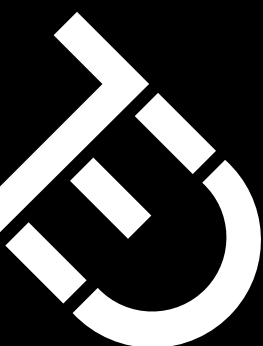
un knows

Given that $\tan(A) = \frac{3}{4}$, and that $180^\circ \leq A \leq 360^\circ$, find:

$\cos(A)$

$\tan(90 - A)$

$\frac{\sin(A)}{\tan^2(A)}$



Trigonometry Unknowns

un knows

Given that $\tan(A) = \frac{3}{4}$, and that $180^\circ \leq A \leq 360^\circ$, find:

$\cos(A)$

$\tan(90 - A)$

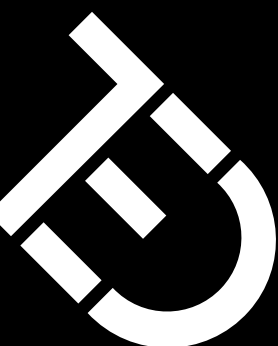
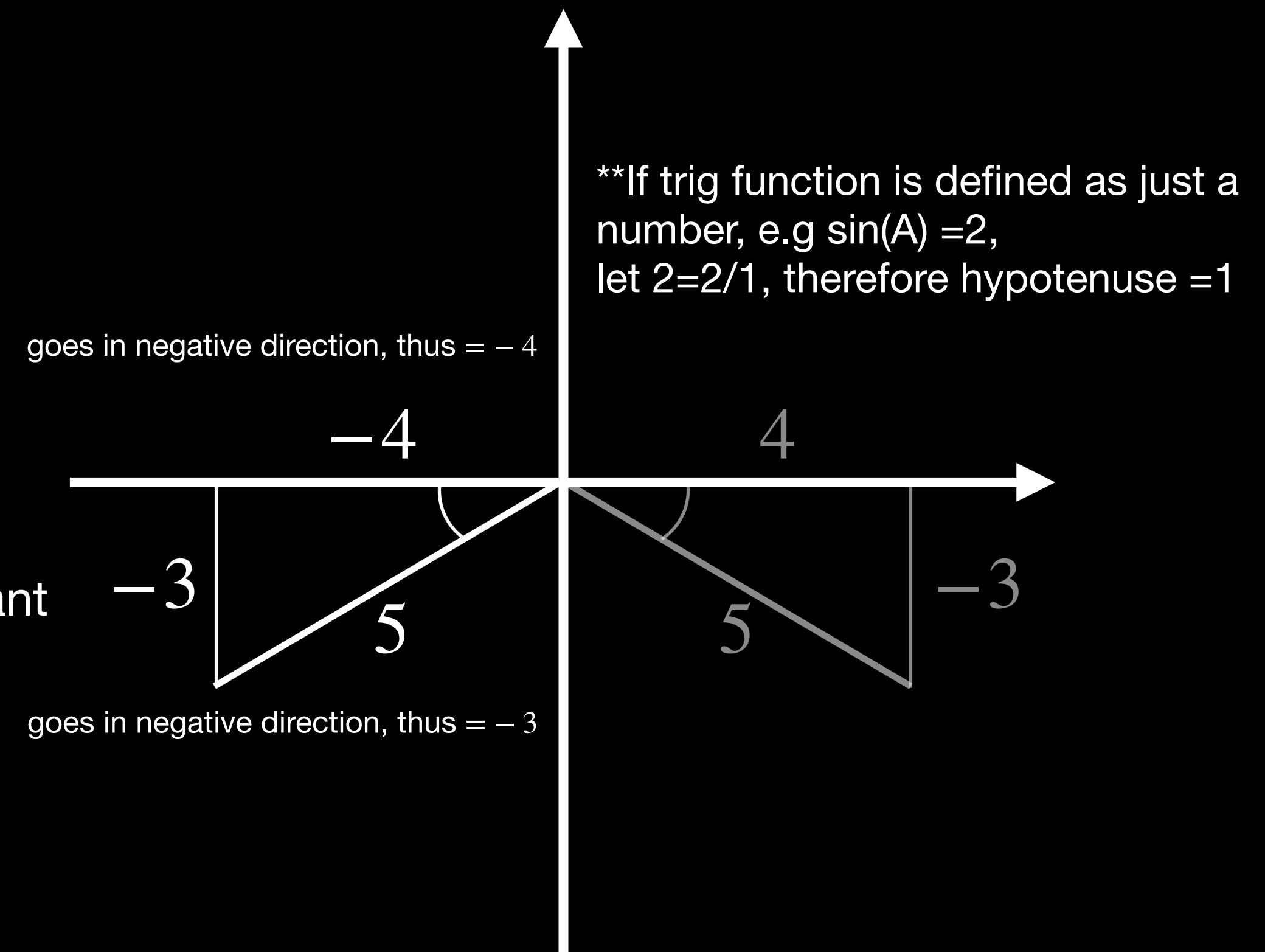
$\frac{\sin(A)}{\tan^2(A)}$

$$\tan(A) = \frac{-3}{-4} = \frac{opp}{adj}$$

$$\therefore hyp = \sqrt{(-3)^2 + (-4)^2}$$
$$hyp = 5$$

Since $\tan(A)$ is positive,
A is either in 1st or 3rd quadrant

however as $180^\circ \leq A \leq 360^\circ$,
A lies in the third quadrant



Trigonometry Unknowns

un knows

Given that $\tan(A) = \frac{3}{4}$, and that $180^\circ \leq A \leq 360^\circ$, find:

$$\cos(A)$$

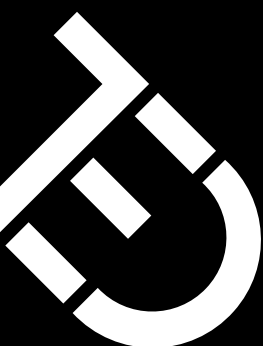
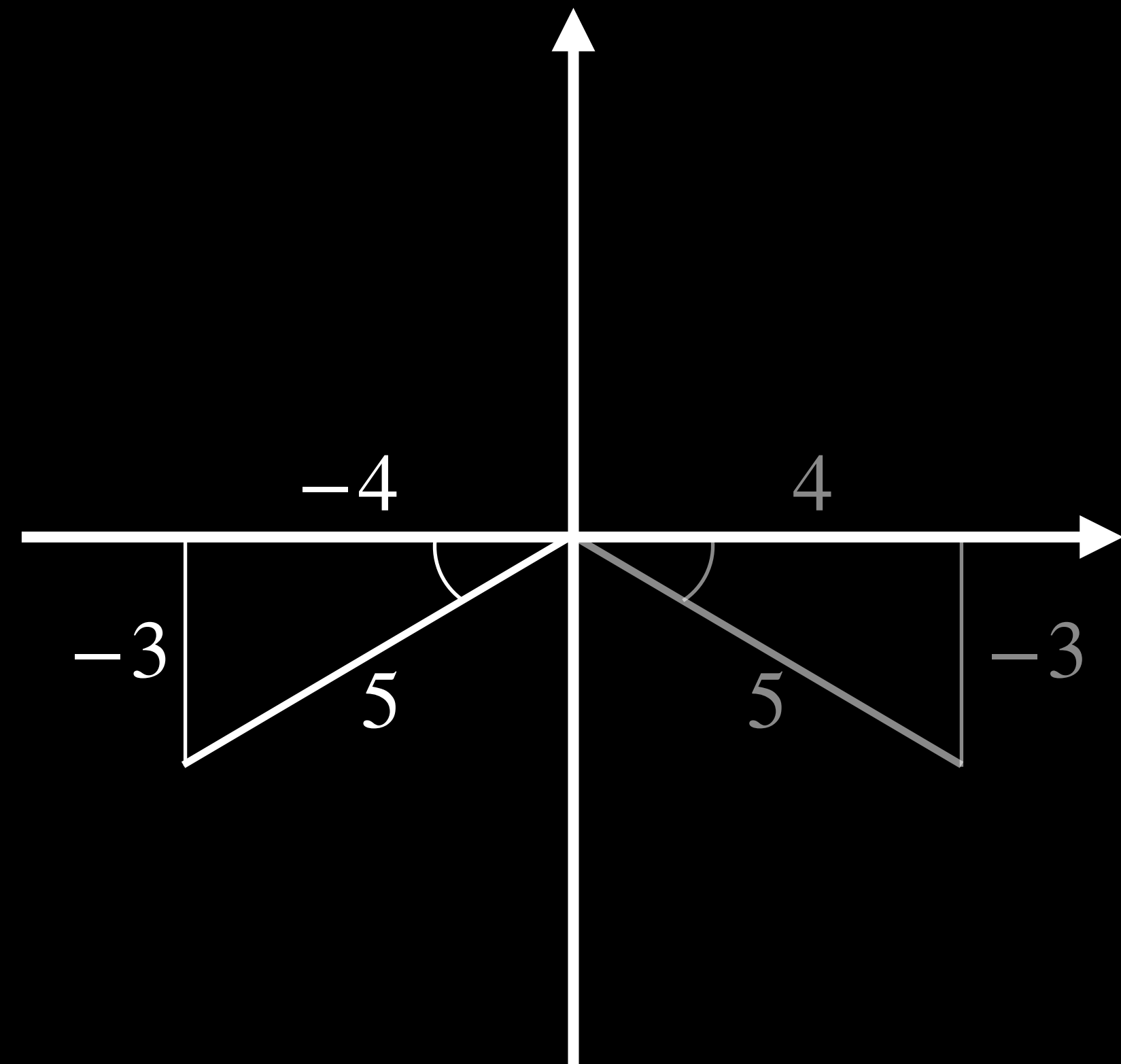
$$\cos(A) = \frac{adj}{hyp}$$

$$\cos(A) = \frac{-4}{5}$$

$$\cos(A) = -0.8$$

$$\tan(90 - A)$$

$$\frac{\sin(A)}{\tan^2(A)}$$



Trigonometry Unknowns

un knows

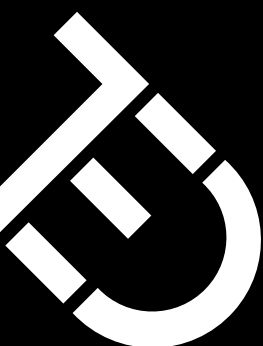
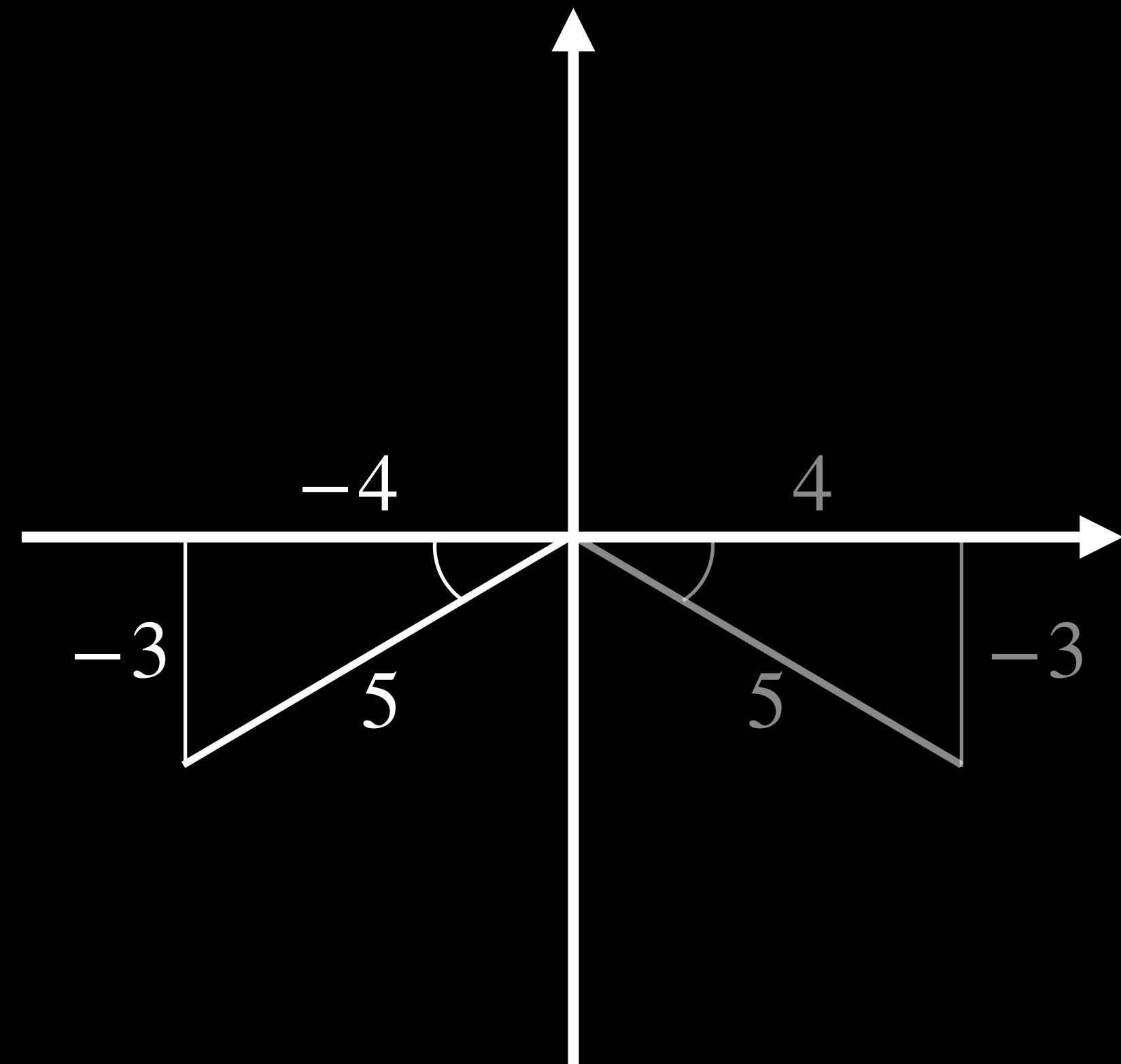
Given that $\tan(A) = \frac{3}{4}$, and that $180^\circ \leq A \leq 360^\circ$, find:

$$\cos(A)$$

$$\tan(90 - A)$$

$$\frac{\sin(A)}{\tan^2(A)}$$

$$\begin{aligned}\tan(90 - A) &= \frac{1}{\tan(A)} \\ &= \frac{1}{\frac{3}{4}} \\ &= \frac{4}{3}\end{aligned}$$



Trigonometry Unknowns

un knows

Given that $\tan(A) = \frac{3}{4}$, and that $180^\circ \leq A \leq 360^\circ$, find:

$\cos(A)$

$$\frac{\sin(A)}{\tan^2(A)} = \frac{-3}{5}$$

$\tan(90 - A)$

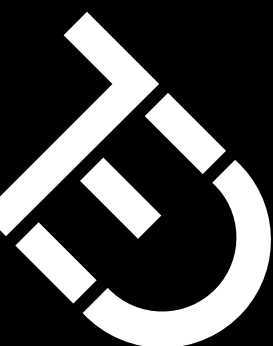
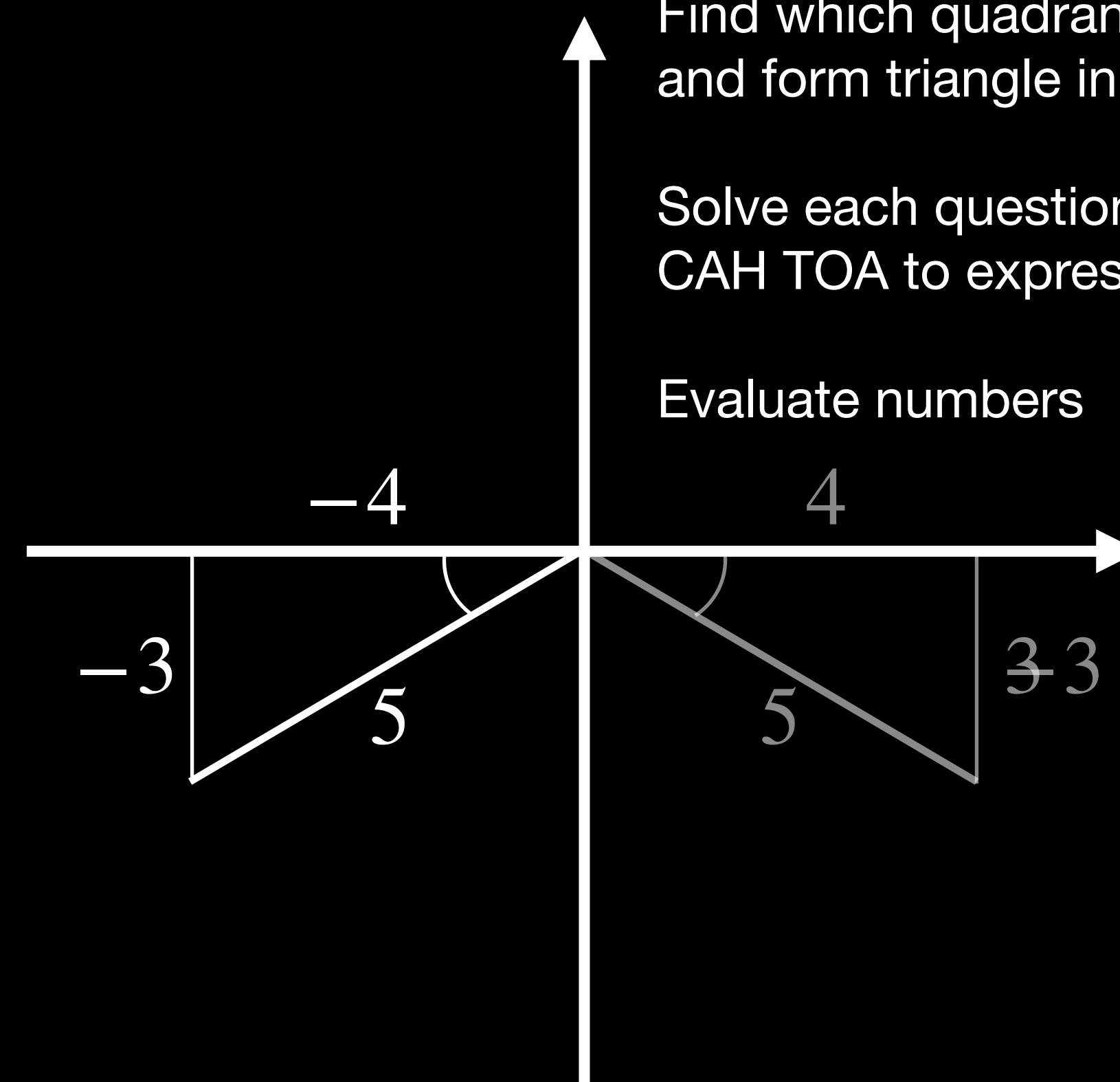
$$= -\frac{16}{15}$$

$$\frac{\sin(A)}{\tan^2(A)}$$

Find which quadrant angle lies in and form triangle in cartesian plane

Solve each question by using SOH CAH TOA to express as numbers

Evaluate numbers



More Trigonometrical Functions

n e w

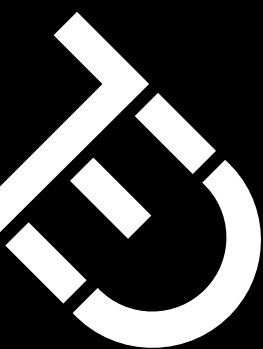
$$\frac{1}{\sin(x)} = \csc(x)$$

$$\frac{1}{\cos(x)} = \sec(x)$$

$$\frac{1}{\tan(x)} = \cot(x)$$

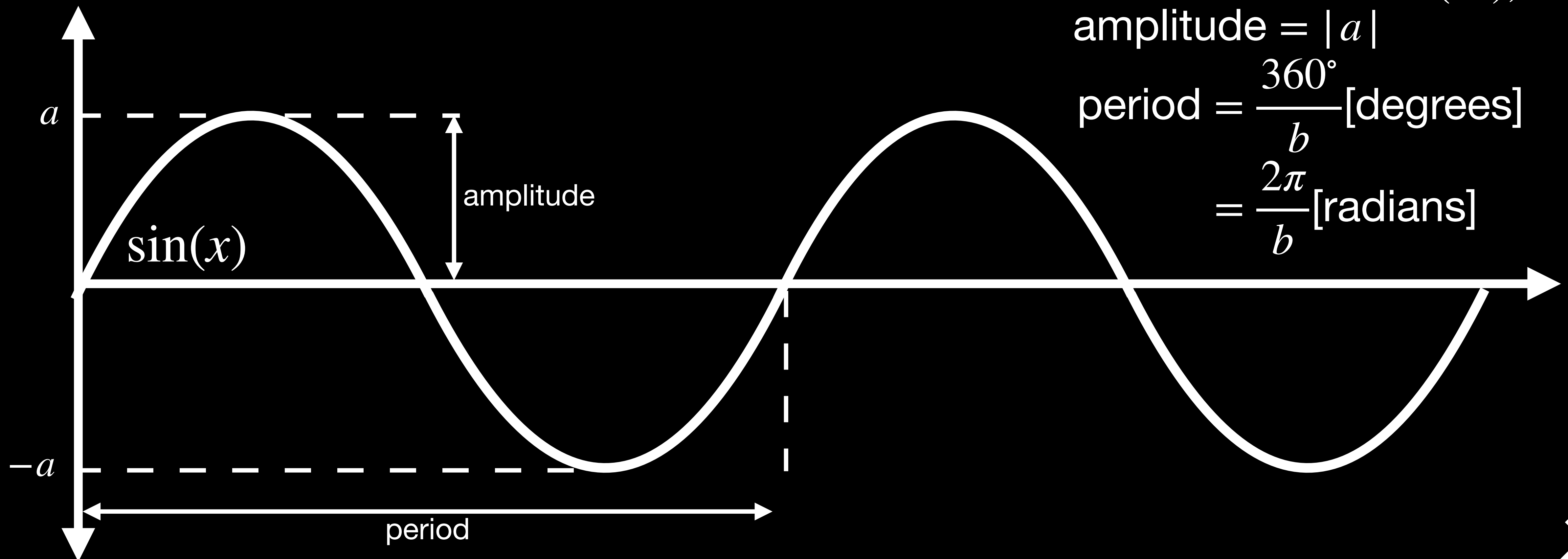
$$\sin^{-1}(x) \neq \sin(x)^{-1}$$

convert functions into base functions when finding unknowns, i.e $\csc(x) = \sin(x)^{-1}$



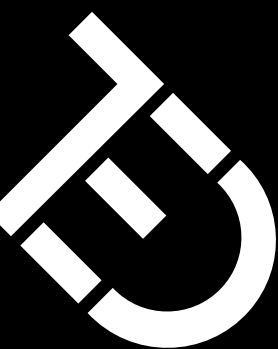
Trigonometric Functions

wa ter bal ;loon



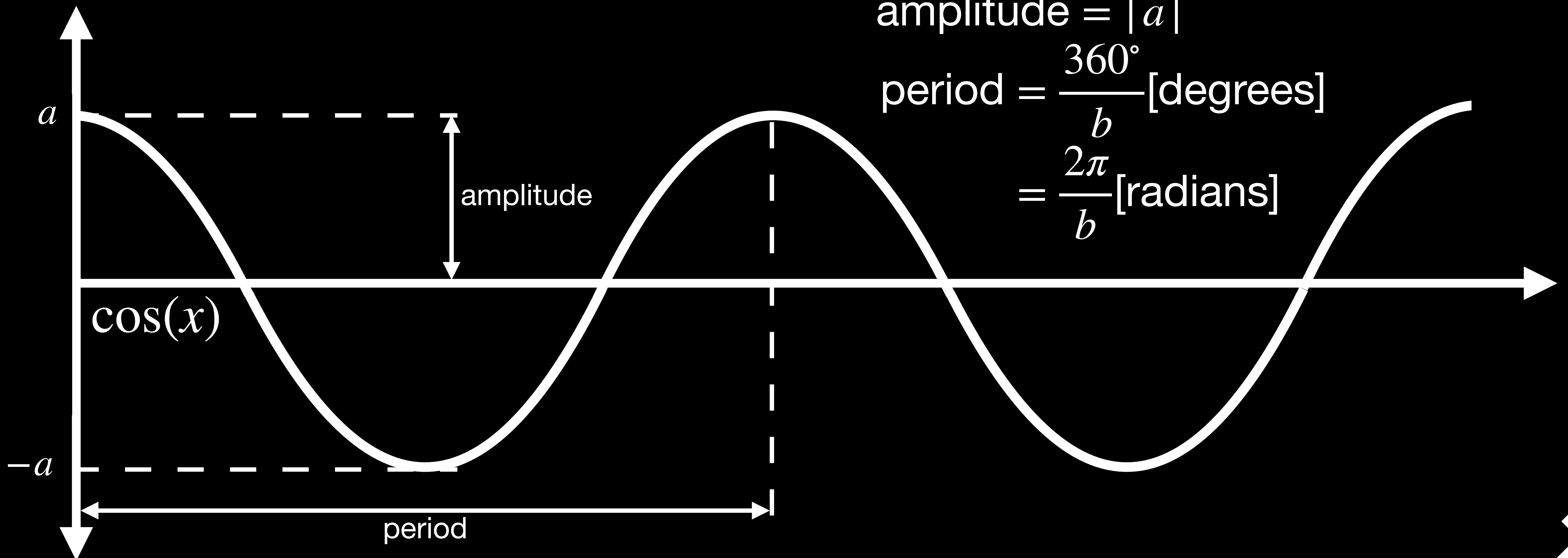
For a function $a \sin(bx)$,
amplitude = $|a|$

$$\text{period} = \frac{360^\circ}{b} [\text{degrees}]$$
$$= \frac{2\pi}{b} [\text{radians}]$$



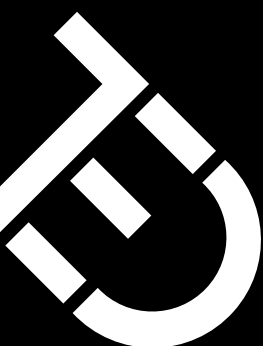
Trigonometric Functions

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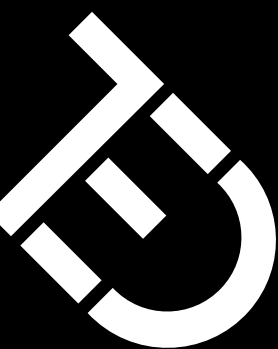
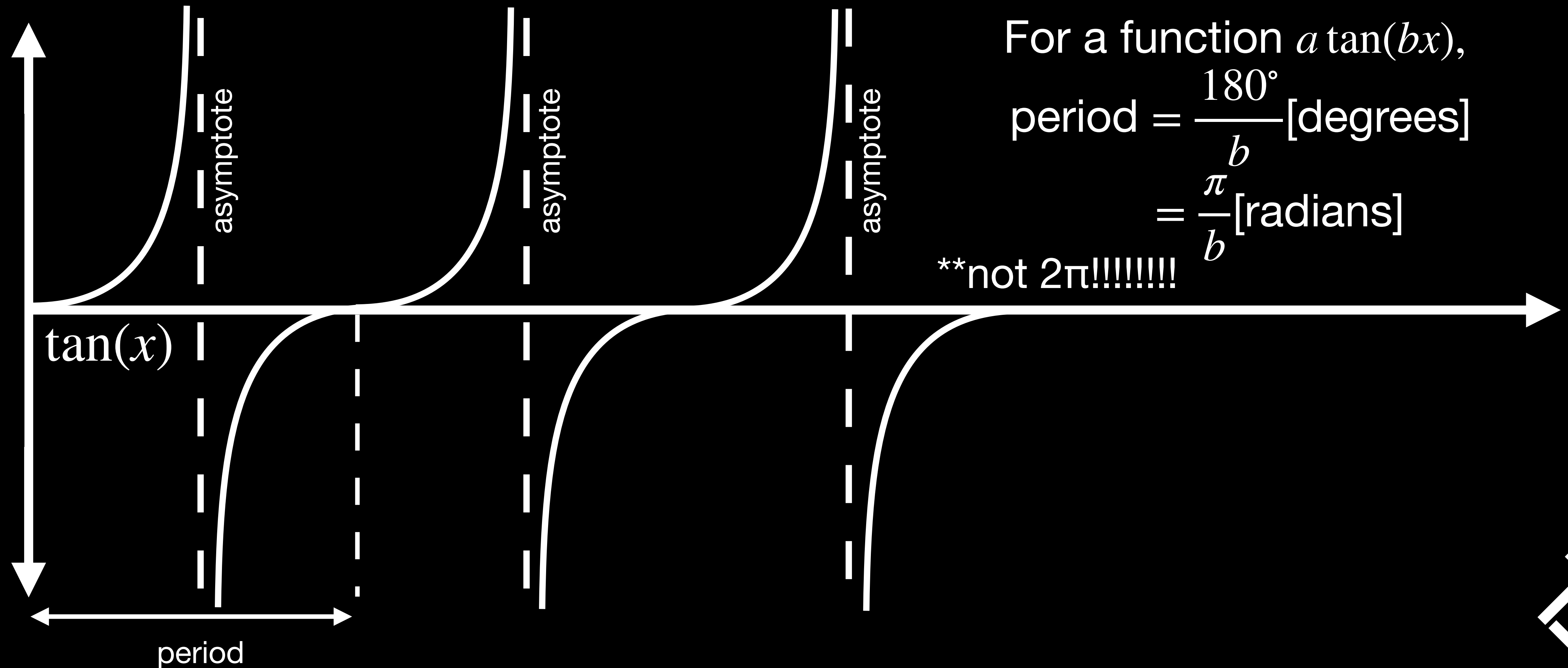
For a function $a \cos(bx)$,
amplitude = $|a|$

$$\text{period} = \frac{360^\circ}{b} [\text{degrees}]$$
$$= \frac{2\pi}{b} [\text{radians}]$$



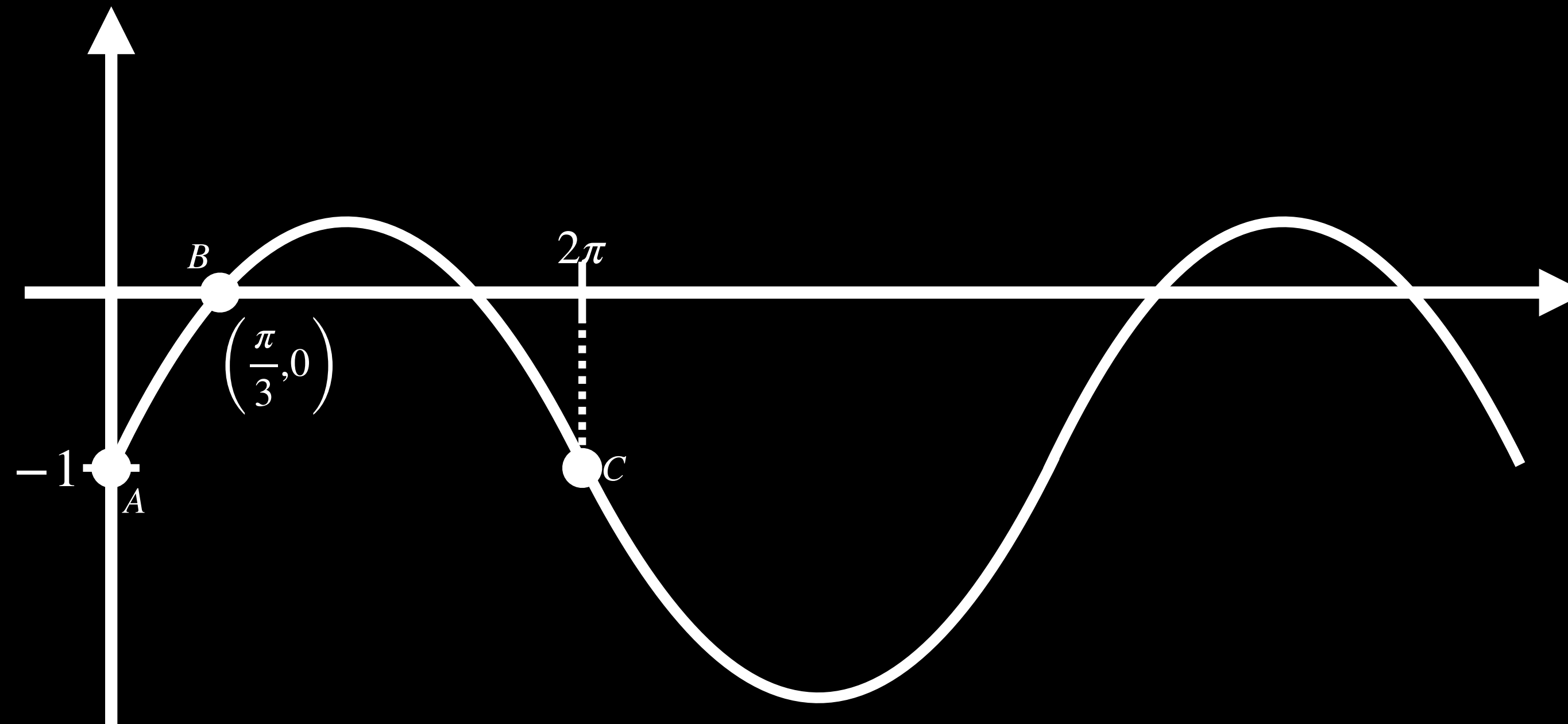
Trigonometric Functions

wa ter bal ;loon



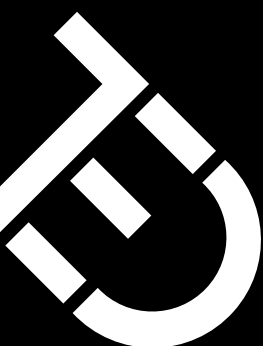
Trigonometric Functions

water balloon



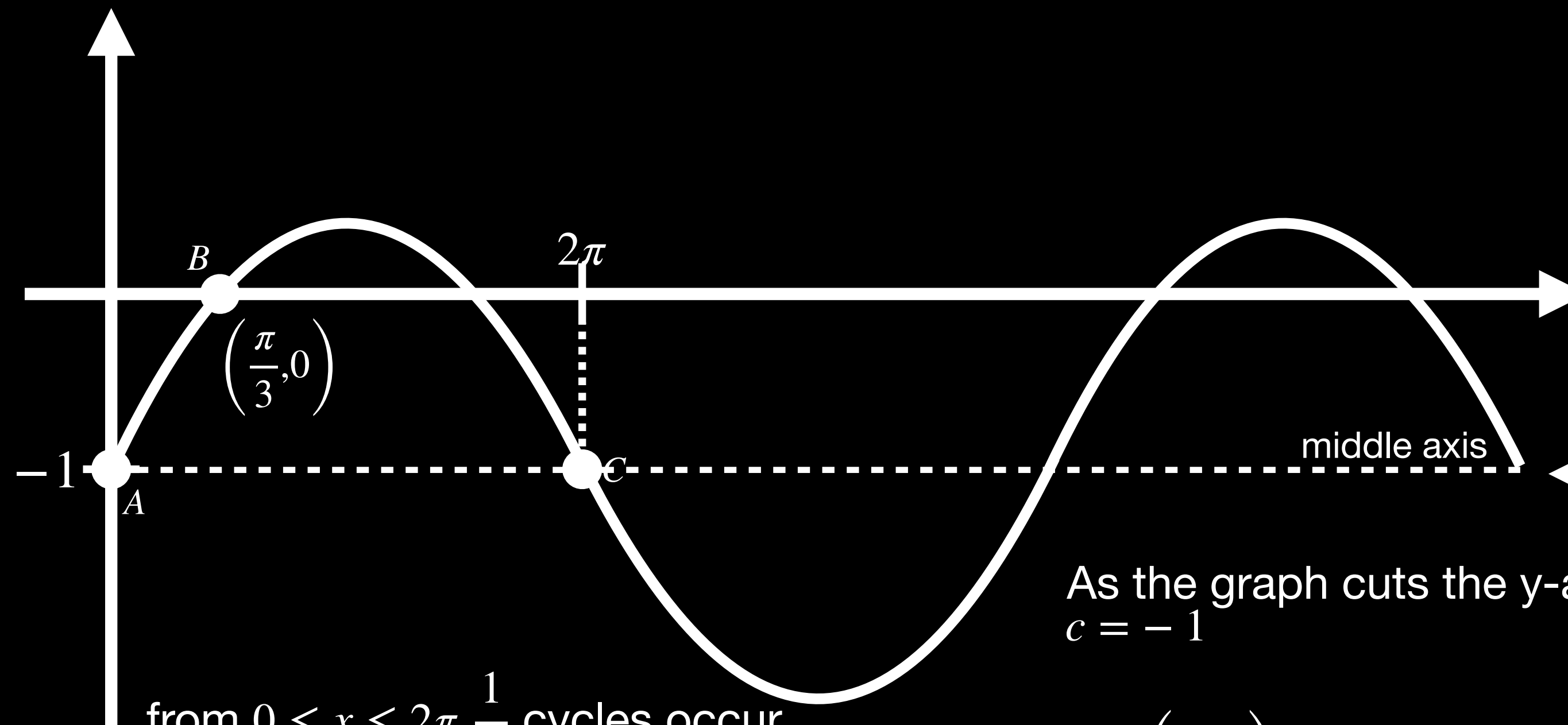
The diagram shows the graph of $y = a \sin(bx) + c$. Point A, B and C all lie on the curve, where A lies on the y axis and C shares the same y-coordinate as A.

Show & explain that $a = 2, b = 0.5, c = -1$, and hence find the equation of the graph



Trigonometric Functions

water balloon



The diagram shows the graph of $y = a \sin(bx) + c$. Point A, B and C all lie on the curve, where A lies on the y axis and C shares the same y-coordinate as A.

Show & explain that $a = 2, b = 0.5, c = -1$, and hence find the equation of the graph

for cosine, the constant term added to $\cos(x)$ is NOT the point where the curve cuts the y-axis, but is the middle axis of the graph(which can be found by taking the mean of the maximum and minimum values of the graph)

As the graph cuts the y-axis at $y = -1$, $c = -1$

As $\left(\frac{\pi}{3}, 0\right)$ lies on the curve, it satisfies the eqn.

$$\therefore 0 = a \sin\left(b \frac{\pi}{3}\right) + c$$

$$0 = a \sin\left(\frac{1}{2} \times \frac{\pi}{3}\right) - 1$$

$$0 = 0.5a - 1$$

$$1 = 0.5a$$

$$a = 2$$

from $0 \leq x \leq 2\pi$, $\frac{1}{2}$ cycles occur

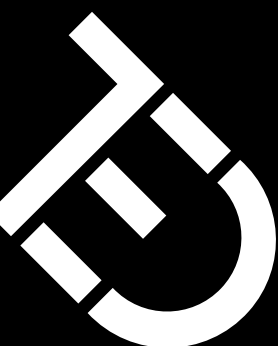
$\therefore$ One cycle is $2\pi \times 2 = 4\pi$

As period = $\frac{2\pi}{b}$, and period = 4π ,

$$4\pi = \frac{2\pi}{b}$$

$$b = \frac{2\pi}{4\pi} = 0.5$$

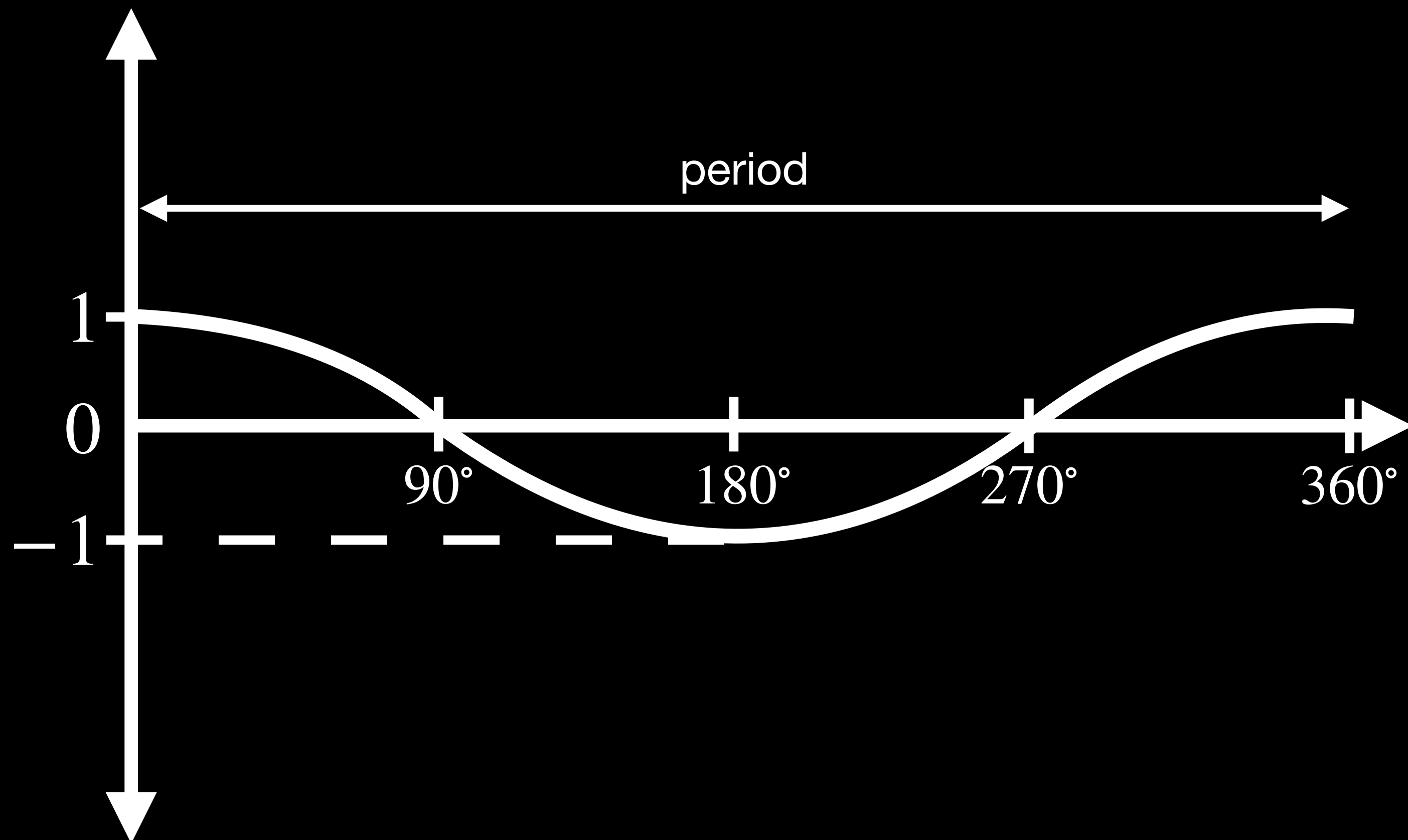
$$\therefore y = 2 \sin(0.5x) - 1$$



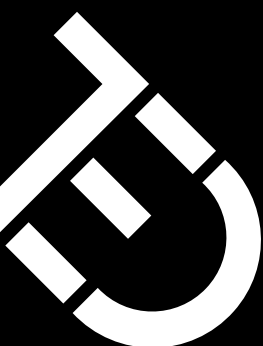
Sketching Trigonometric Functions

water balloon

Sketch $y = -1.5 \cos(2x) - 2$ from 0° to 360°



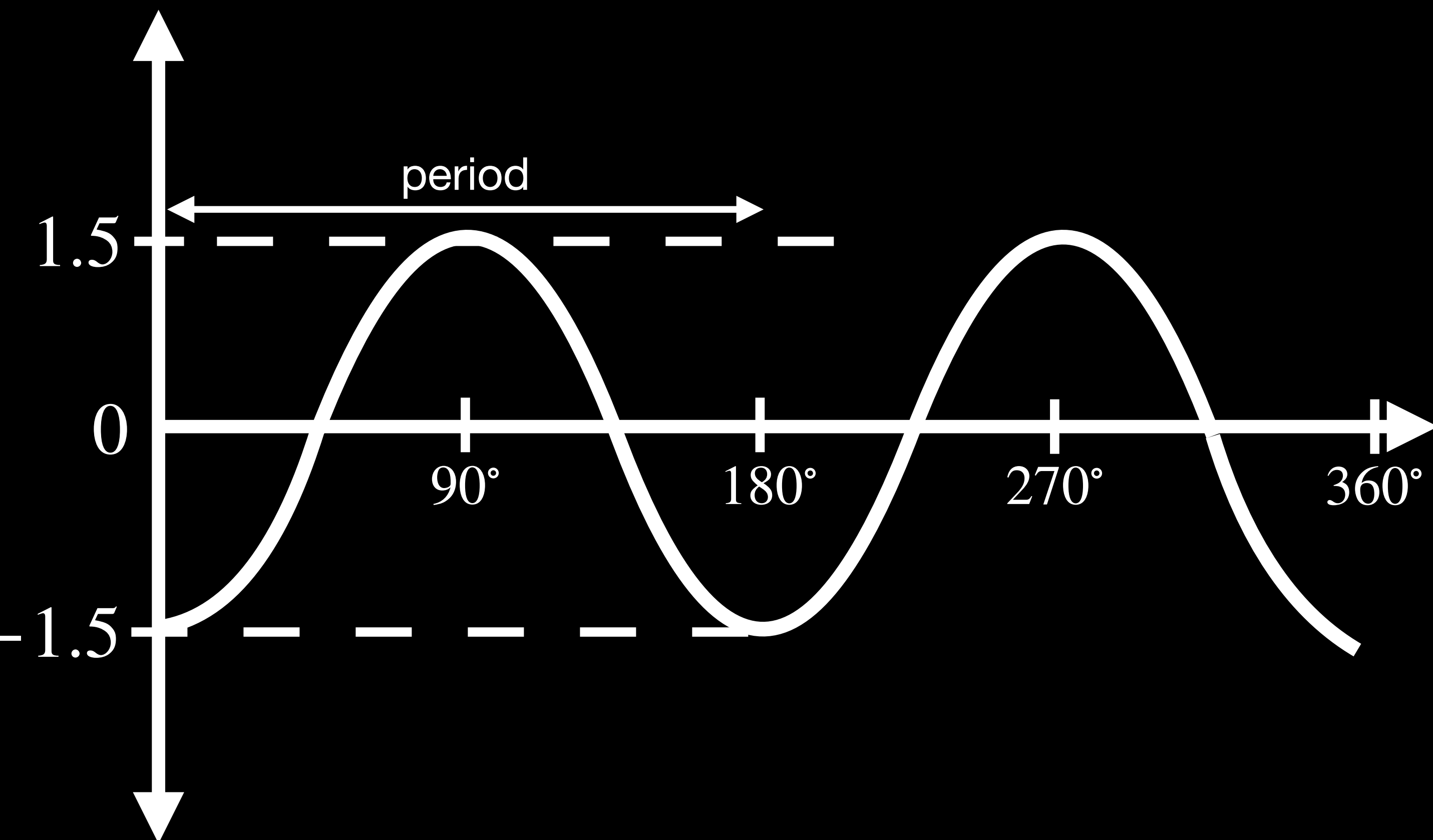
do not sketch these as final, only for working purposes
Start with base function (sin, cos, tan)



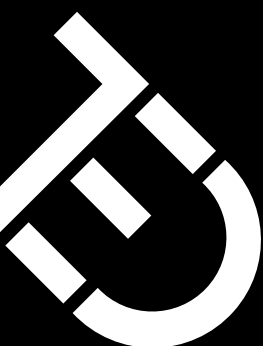
Sketching Trigonometric Functions

water balloon

Sketch $y = -1.5 \cos(2x) - 2$ from 0° to 360°



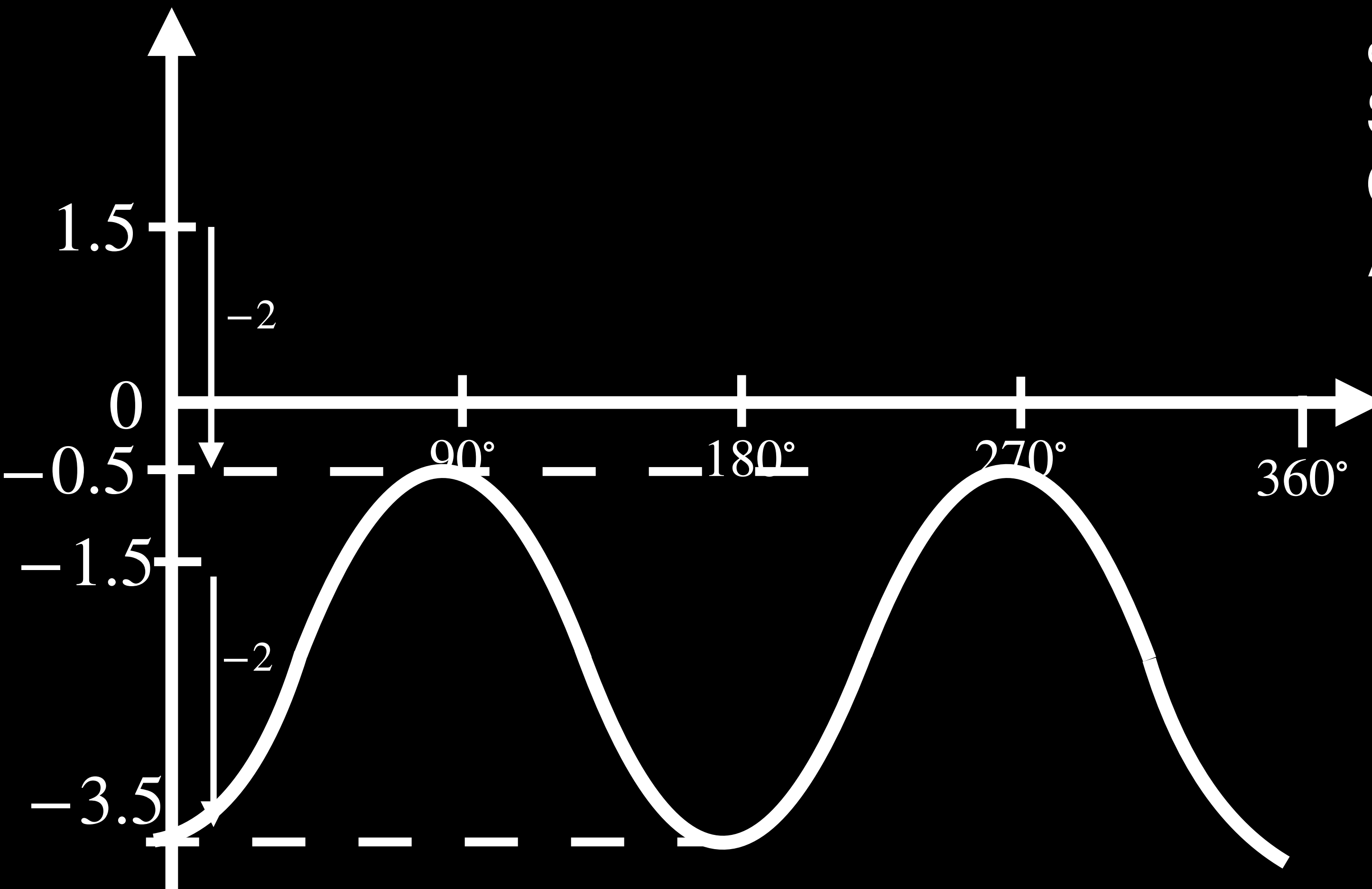
do not sketch these as final, only for working purposes
Start with base function (sin, cos, tan)
Change amplitude and period
(flip graph if negative)



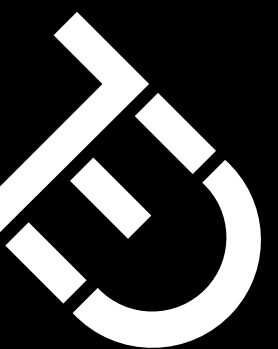
Sketching Trigonometric Functions

water balloon

Sketch $y = -1.5 \cos(2x) - 2$ from 0° to 360°



do not sketch these as final, only for working purposes
Start with base function (sin, cos, tan)
Change amplitude and period
Add back constant



Sketching Trigonometric Functions

water balloon

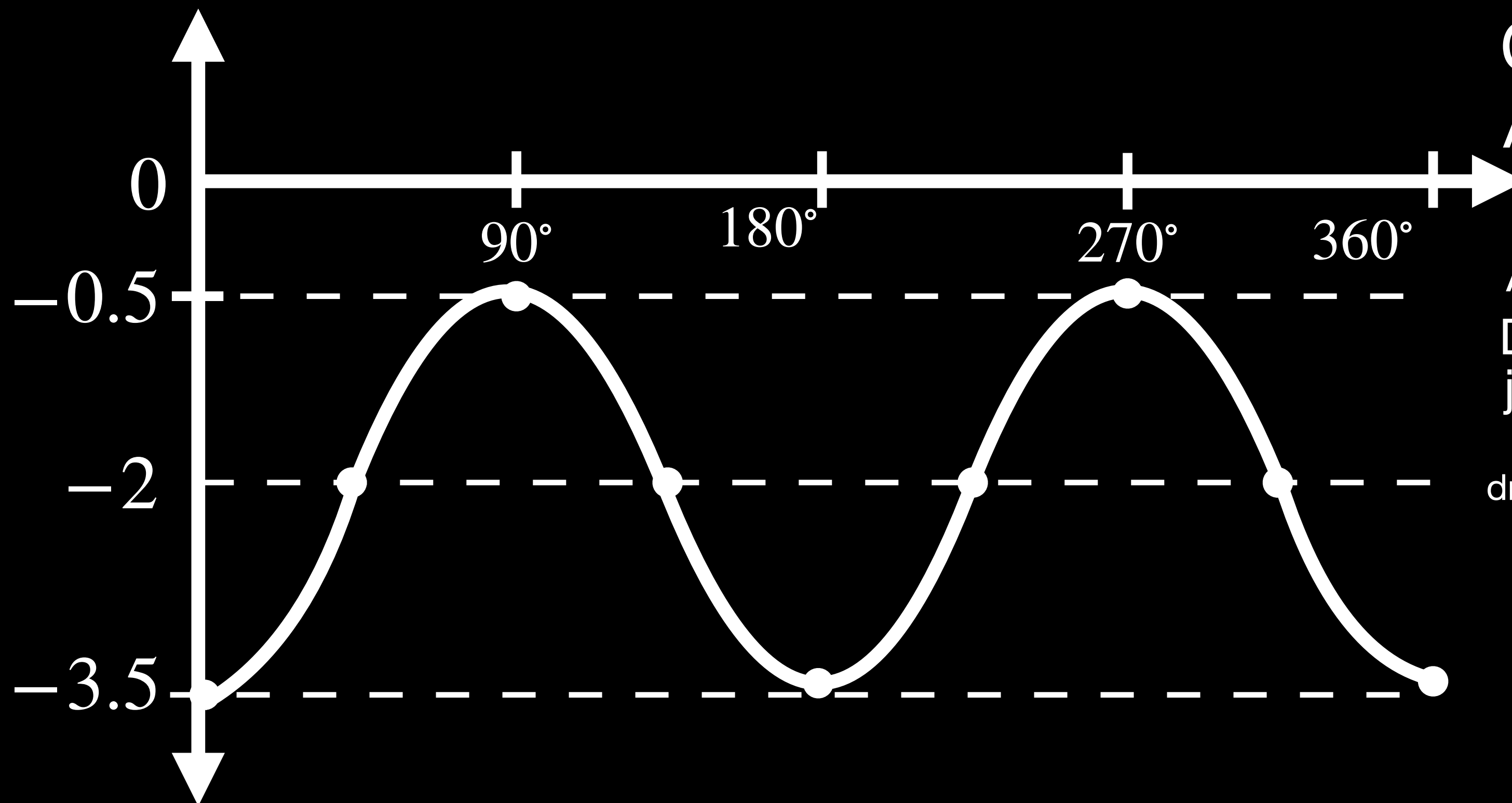
Sketch $y = 1.5 \cos(2x) - 2$ from 0° to 360°

do not sketch these as final, only for working purposes

Start with base function (sin, cos, tan)

Change amplitude and period

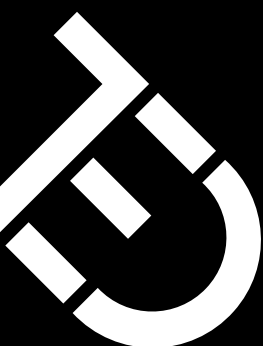
Add back constant



Actual sketching of graph

Draw axes and "plot" points (need not be accurate)
join points together

draw dotted lines on the maximum and minimum and also the axis



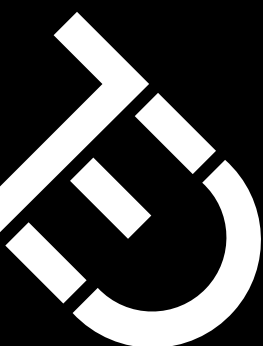
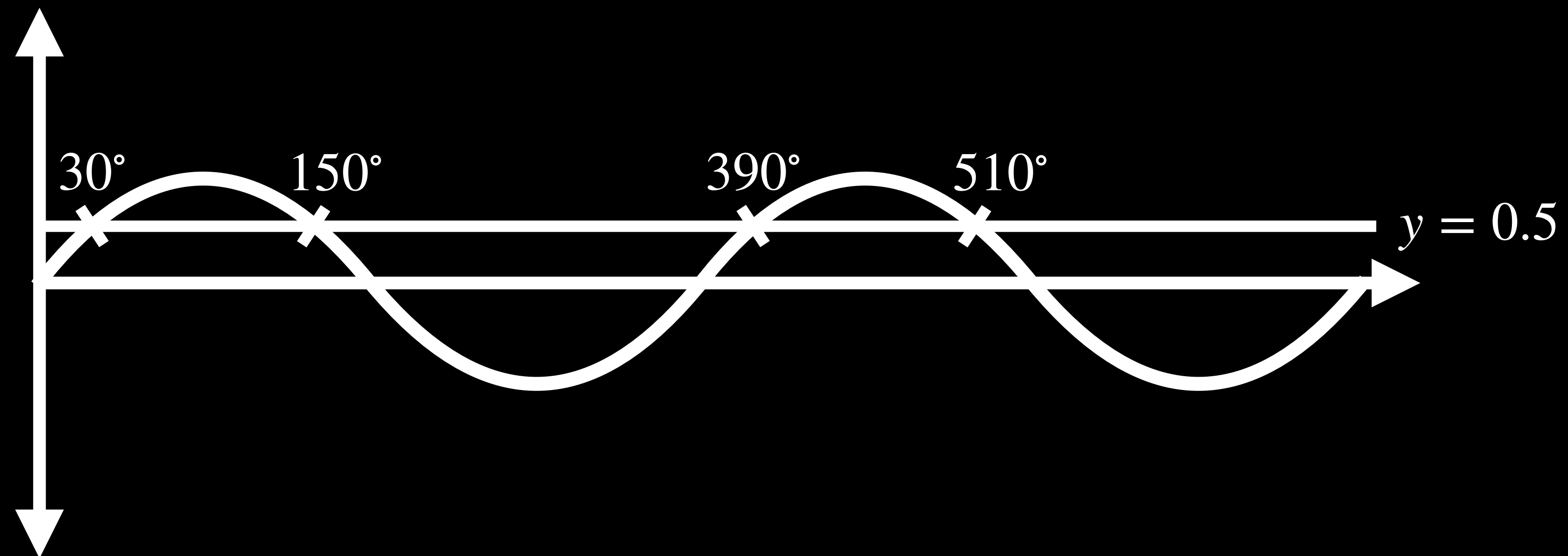
Solving Trigonometric Equations

epic

trigo functions have an infinite number of solutions

For $\sin(x) = 0.5$, $x = 30^\circ, 150^\circ, 390^\circ, 510^\circ, \dots$

Thus, we restrict to a domain e.g $0^\circ \leq x \leq 360^\circ$ and solve for all possible values



Solving Trigonometric Equations

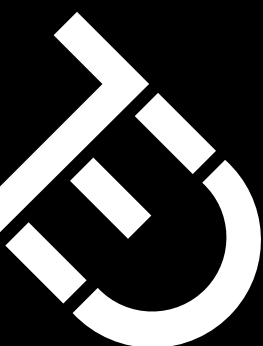
epic

Principal Value: first value that shows up when finding solutions of trig function
Both basic angle and principal value have the same value/number

What is the principal value of $\sin(x) = 0.5$?

$$x = \sin^{-1}(0.5)$$

$$x = 30^\circ$$



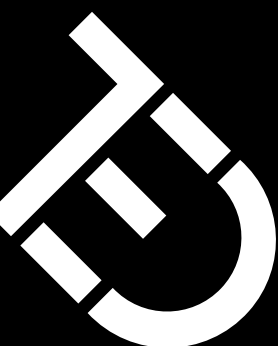
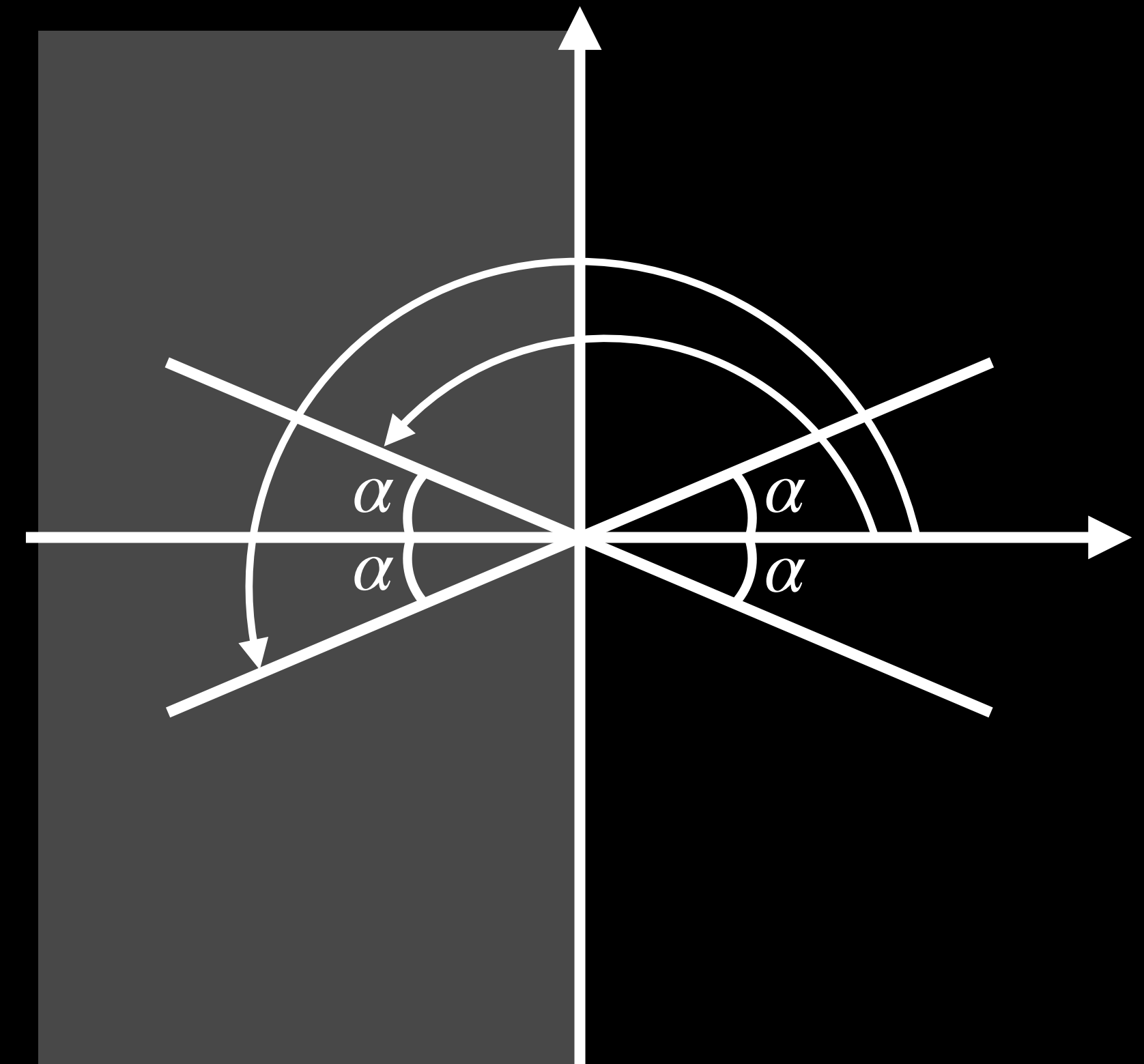
epic

$$\alpha = \cos^{-1}(0.75)$$
$$= 41.4^\circ(1.d.p)$$

Since $\cos(x)$ is negative,
x lies in 2nd and third quadrant

$$\therefore x = 180^\circ - \cos^{-1}(0.75), 180^\circ + \cos^{-1}(0.75)$$
$$= 138.6^\circ, 221.4^\circ$$

use exact value of α or 5.s.f version



Solving Trigonometric Equations

epic

Solve $4 \tan (2x + 20^\circ) = 1.2$, where $0^\circ \leq x \leq 180^\circ$

$$4 \tan (2x + 20^\circ) = 1.2$$

$$\tan (2x + 20^\circ) = 0.3$$

$$\alpha = \tan^{-1}(0.3)$$

$$= 16.7^\circ(1.d.p)$$

Reduce to basic trig equation i.e $f(x)=n$
Change domain accordingly based on expression inside trig function

$$0^\circ \leq x \leq 180^\circ$$

$$0^\circ \leq 2x \leq 360^\circ$$

$$20^\circ \leq 2x + 20^\circ \leq 380^\circ$$

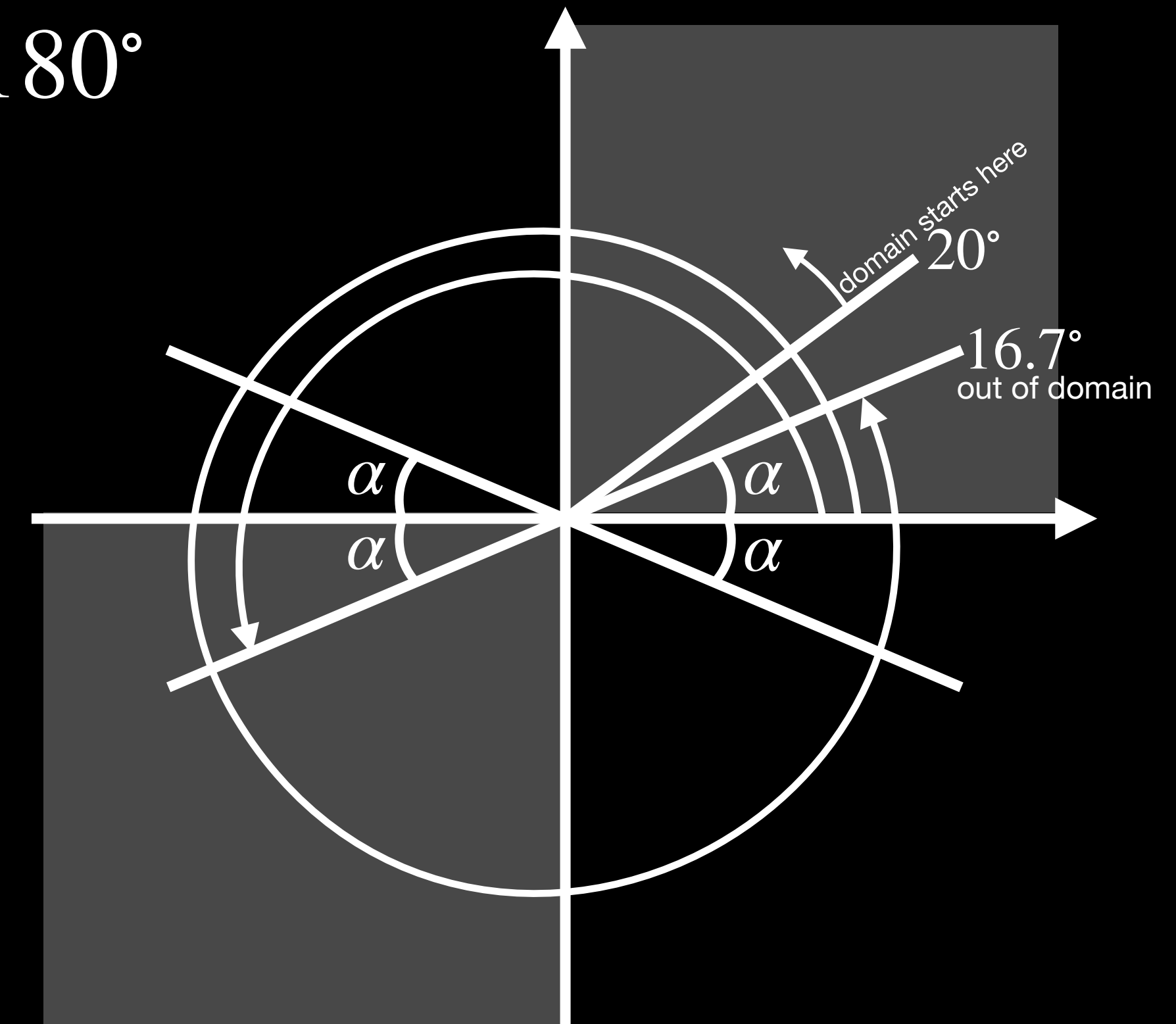
Since $\tan(2x + 20^\circ)$ is positive,
 $2x + 20^\circ$ lies in 1st & 3rd quadrant

$$2x + 20^\circ = 180^\circ + \tan^{-1}(0.3), 360^\circ + \tan^{-1}(0.3)$$

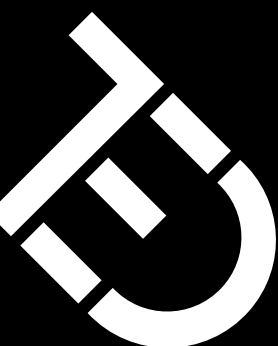
$$2x = 180^\circ + \tan^{-1}(0.3) - 20^\circ, 360^\circ + \tan^{-1}(0.3) - 20^\circ$$

$$x = \frac{160^\circ + \tan^{-1}(0.3)}{2}, \frac{340^\circ + \tan^{-1}(0.3)}{2} = 88.4^\circ, 178.4^\circ$$

use exact value of α or 5.s.f version



Undo changes after finding values (we do not include 16.7 as it is out of domain)



Solving Trigonometric Equations - Simplifying

no way

$$\sin^3(x) = \frac{1}{27}$$
$$\sqrt[3]{\sin^3(x)} = \sqrt[3]{\frac{1}{27}}$$

$$\sin(x) = \frac{\sqrt[3]{1}}{\sqrt[3]{27}}$$

$$\sin(x) = \frac{1}{3}$$

solve

$$2 \sin(x) = 5 \cos(x)$$

$$\frac{2 \sin(x)}{\cos(x)} = \frac{5 \cos(x)}{\cos(x)}$$

$$2 \tan(x) = 5$$

$$\tan(x) = 2.5$$

solve

If terms involves only $\sin(x)$ and $\cos(x)$, likely that $\sin(x)/\cos(x) = \tan(x)$ has to be used

(If sines and cosines are multiplied together, unlikely that $\sin/\cos = \tan$ will be used)

$$3 \csc(x) = 4$$

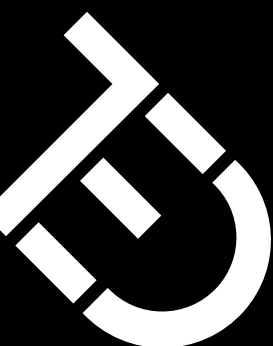
$$\csc(x) = \frac{4}{3}$$

$$\frac{1}{\sin(x)} = \frac{4}{3}$$

$$\sin(x) = 0.75$$

solve

Reduce to basic trig functions, e.g $\csc(x)$ to $1/\sin(x)$



Solving Trigonometric Equations - Factorisation

no way

$$\cos^2(x)\sin(x) = -2\sin(x)$$

$$\cos^2(x)\sin(x) + 2\sin(x) = 0$$

$$\sin(x)(\cos^2(x) + 2) = 0$$

$$\sin(x) = 0, (\cos^2(x) + 2) = 0$$

rej $\cos^2(x) + 2 = 0$ as no real sols.

solve $\sin(x)=0$

note that factors can be split only if RHS=0

Do not divide by $\sin(x)$ as $x=0$
when $\sin(x)=0$,
instead factorise
common term

For all trig equations,
simplify down to a
simple trig equation,
e.g. $\sin(x)=n$

$$2\tan(x) - 5 = 3\cot(x)$$

$$2\tan(x) - 5 = 3\frac{1}{\tan(x)} \quad \times \tan(x)$$

$$2\tan^2(x) - 5\tan(x) = 3$$

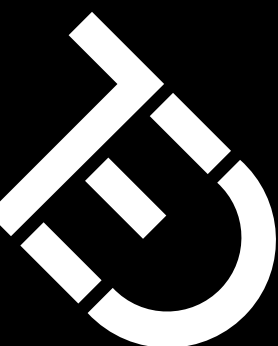
$$2\tan^2(x) - 5\tan(x) - 3 = 0$$

$$(2\tan(x) + 1)(\tan(x) - 3) = 0 \quad \text{form quadratic}$$

$$\tan(x) = -0.5, \tan(x) = 3$$

solve both, present both solutions

When there are 2 equations that need to be solved,
make sure to label the basic angle for each one
differently, e.g. $\alpha_1 = \tan^{-1}(0.5)$, $\alpha_2 = \tan^{-1}(3)$



Trigonometric Identities

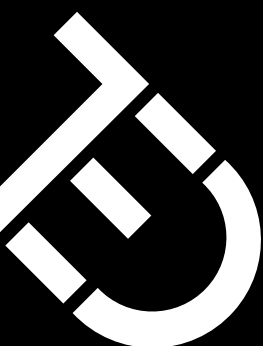
Me when I am at my lim $h \rightarrow 0$ ecks dee

$$\sin^2(x) + \cos^2(x) = 1$$

$$1 + \tan^2(x) = \sec^2(x)$$

$$1 + \cot^2(x) = \csc^2(x)$$

$$\frac{\sin(x)}{\cos(x)} = \tan(x) \quad \frac{\cos(x)}{\sin(x)} = \cot(x)$$



Trigonometric Identities

Me when I am at my lim $h \rightarrow 0$ ecks dee

$$\boxed{\sec^2(x)} + 3 \tan(x) = 6$$

$$\boxed{1 + \tan^2(x)} + 4 \tan(x) = 6$$

$$\tan^2(x) + 4 \tan(x) - 5 = 0$$

$$\tan(x) = -5, 1$$

solve

$$\cos(x) - \sin^2(x)\cos(x) = 0.125$$

$$\cos(x)\boxed{(1 - \sin^2(x))} = 0.125$$

$$\cos(x)\boxed{\cos^2(x)} = 0.125$$

$$\cos^3(x) = 0.125$$

$$\cos(x) = \sqrt[3]{0.125}$$

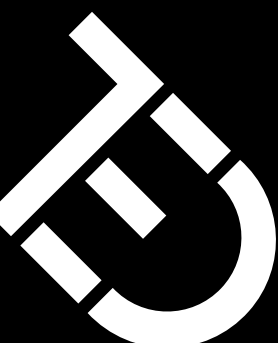
$$\cos(x) = 0.5$$

solve

Since most basic trigo identities involve squared functions, try and see if converting squared trigo function to identity will help

Other identities like $\cot(x) = \cos(x)/\sin(x)$ and $\tan(x) = \sin(x)/\cos(x)$ are also possible, so try converting those and see if you can solve via simplification/factorisation

Generally, you want to make your equation be in terms of only one trigo function (all sin, tan, etc.)

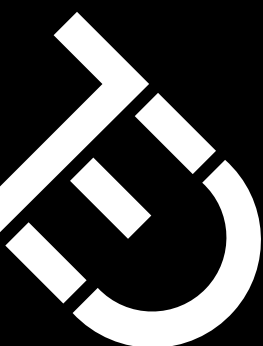


Proving Trigonometric Identities

brain break moment

$$\text{Prove } \frac{\sin(x) + 1}{\cos(x)\csc(x)} \times \frac{1 - \sin(x)}{\tan(x)} \equiv \cos^2(x)$$

$$\begin{aligned} LHS &= \frac{\sin(x) + 1}{\cos(x)\csc(x)} \times \frac{1 - \sin(x)}{\tan(x)} &&= (\sin(x) + 1)(1 - \sin(x)) \\ &= \frac{\sin(x) + 1}{\cos(x)\frac{1}{\sin(x)}} \times \frac{1 - \sin(x)}{\tan(x)} &&= (1 + \sin(x))(1 - \sin(x)) \\ &= \frac{\sin(x) + 1}{\cos(x)\frac{1}{\sin(x)}} \times \frac{1 - \sin(x)}{\tan(x)} &&= 1^2 - \sin^2(x) \\ &= \frac{\sin(x) + 1}{\frac{\cos(x)}{\sin(x)}} \times \frac{1 - \sin(x)}{\frac{\sin(x)}{\cos(x)}} &&= \cos^2(x) \\ &= \frac{(\sin(x) + 1)(1 - \sin(x))}{\frac{\cos(x)}{\sin(x)} \frac{\sin(x)}{\cos(x)}} &&= RHS \end{aligned}$$



Proving Trigonometric Identities

brain break moment

$$\text{Prove } \frac{2 \cos(x)}{\sin(x) + \sin(x)\cot^2(x)} \equiv \sin(2x)$$

$$LHS = \frac{2 \cos(x)}{\sin(x) + \sin(x)\cot^2(x)} = \frac{2 \cos(x)}{\sin(x)} \times \frac{1}{\frac{1}{\sin^2(x)}}$$

$$= \frac{2 \cos(x)}{\sin(x)(1 + \cot^2(x))} = \frac{2 \cos(x)}{\cancel{\sin(x)}} \times \sin^2(x)$$

$$= \frac{2 \cos(x)}{\sin(x)\csc^2(x)} \quad \text{identity}$$

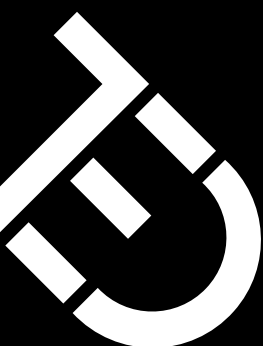
$$= \frac{2 \cos(x)}{\sin(x)} \times \frac{1}{\csc^2(x)}$$

$$= 2 \cos(x)\sin(x)$$

$$= \sin(2x) \text{ [double angle formula]}$$

$$= RHS$$

For more info/generalisations



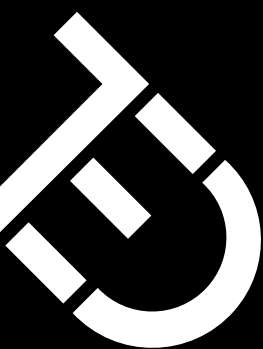
Compound Angle Formulae

Plus minus minus plus

$$\sin(A \pm B) = \sin(A)\cos(B) \pm \cos(A)\sin(B)$$

$$\cos(A \pm B) = \cos(A)\cos(B) \mp \sin(A)\sin(B)$$

$$\tan(A \pm B) = \frac{\tan(A) \pm \tan(B)}{1 \mp \tan(A)\tan(B)}$$



Compound Angle Formulae

Plus minus minus plus

Find $\tan(105^\circ)$ without the use of a calculator

$$\tan(105^\circ) = \tan(60^\circ + 45^\circ)$$

$$= \frac{\tan(60^\circ) + \tan(45^\circ)}{1 - \tan(60^\circ)\tan(45^\circ)}$$

$$= \frac{\sqrt{3} + 1}{1 - (\sqrt{3})(1)}$$

$$= \frac{1 + \sqrt{3}}{1 - \sqrt{3}}$$

$$= \frac{1 + \sqrt{3}}{1 - \sqrt{3}} * \frac{1 + \sqrt{3}}{1 + \sqrt{3}}$$

$$= \frac{(1 + \sqrt{3})^2}{1 - 3}$$

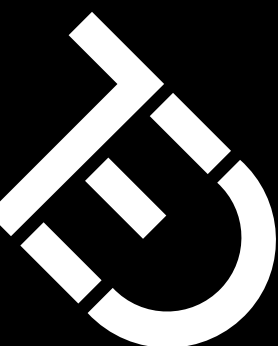
$$= \frac{1 + 2\sqrt{3} + 3}{1 - 3}$$

$$= \frac{4 + 2\sqrt{3}}{-2}$$

$$= \frac{2 + \sqrt{3}}{-1}$$

$$= -(2 + \sqrt{3})$$
$$= -2 - \sqrt{3}$$

split angle given into special angles, then use special angles



Compound Angle Formulae

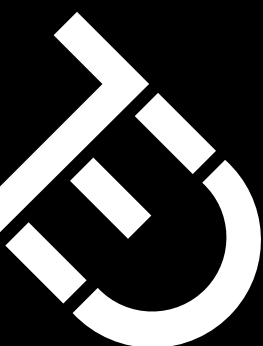
Plus minus minus plus

Find $\cos(37^\circ)\cos(23^\circ) - \sin(37^\circ)\sin(23^\circ)$ without the use of a calculator

$$\cos(37^\circ)\cos(23^\circ) - \sin(37^\circ)\sin(23^\circ) = \cos(37^\circ + 23^\circ)$$

$$= \cos(60^\circ)$$

$$= 0.5$$



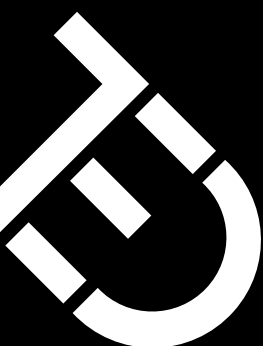
Double Angle Formulae

Compound angle but 2

$$\sin(2A) = 2 \sin(A)\cos(A)$$

$$\cos(2A) = \cos^2(A) - \sin^2(A) = 1 - 2 \sin^2(A) = 2 \cos^2(A) - 1$$

$$\tan(2A) = \frac{2 \tan(A)}{1 - \tan^2(A)}$$



Double Angle Formulae - Half Angle Formulae

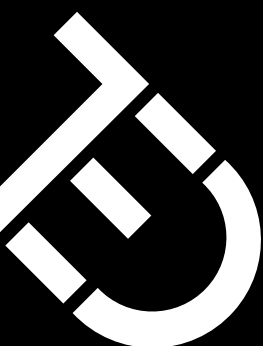
Compound angle but 2

Just let $2A$ be A and A be $0.5A$

$$\sin(A) = 2 \sin(0.5A) \cos(0.5A)$$

$$\cos(A) = \cos^2(0.5A) - \sin^2(0.5A) = 1 - 2 \sin^2(0.5A) = 2 \cos^2(0.5A) - 1$$

$$\tan(A) = \frac{2 \tan(0.5A)}{1 - \tan^2(0.5A)}$$



Double Angle Formulae

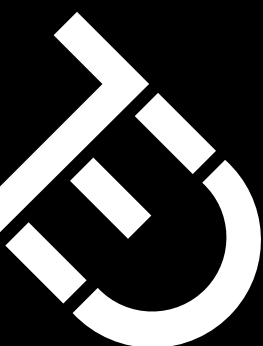
Compound angle but 2

$$\begin{aligned}\cos(3x) &= \cos(x + 2x) \\ &= \cos(x)\cos(2x) - \sin(x)\sin(2x) \\ &= \cos(x)(\cos^2(x) - \sin^2(x)) - 2\sin(x)\sin(x)\cos(x) \\ &= \cos^3(x) - \cos(x)\sin^2(x) - 2\sin^2(x)\cos(x)\end{aligned}$$

any trigo function of nx , where n is an integer, can be expressed like so

$$\begin{aligned}\cos(2x) &= 1 - 2\cos^2(x) \\ \cos(2x) - 1 &= -2\cos^2(x) \\ -\frac{\cos(2x) - 1}{2} &= \cos^2(x)\end{aligned}$$

for cosine double angle formulae, $\sin^2(x)$ & $\cos^2(x)$ can be expressed in terms of $\cos(2x)$
much rarer, but still possible (i.e when identity is expressed in terms of $\cos(2x)$)



Double Angle Formulae - Cosine

Compound angle but 2

When to use which form?

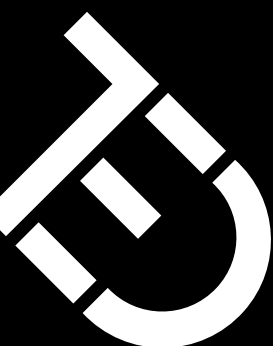
$$\cos(2A) = \boxed{\cos^2(A) - \sin^2(A)} = \boxed{1 - 2\sin^2(A)} = \boxed{2\cos^2(A) - 1}$$

Much rarer, barely
used for identities
but still possible

Usually when the equation
involves mainly $\sin(x)$ so
that it can factorised or
the expression contains
 $\sin(x)$ so that it can be
simplified e.g to form a
quadratic in terms of $\sin(x)$

Usually when the equation
involves mainly $\cos(x)$ so
that it can factorised or
the expression contains
 $\cos(x)$ so that it can be
simplified e.g to form a
quadratic in terms of
 $\cos(x)$

worse case scenario: use pencil and quickly write out expression with either form and test



R-Formulae

You r

if sine is in front, r formula will be in sine

for sine, sign stays the same

$$a \sin(\theta) \pm b \cos(\theta) = R \sin(\theta \pm \alpha)$$
$$a \cos(\theta) \pm b \sin(\theta) = R \cos(\theta \mp \alpha)$$

sign flips for cosine

if cosine is in front, r formula will be in cosine

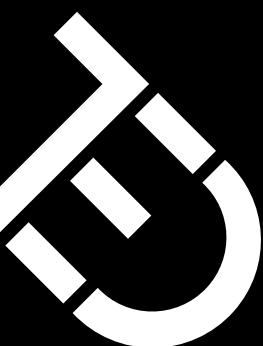
where:

$$R = \sqrt{a^2 + b^2}$$

$$\alpha = \tan^{-1} \left(\frac{b}{a} \right), \text{ where } \alpha \text{ is acute}$$

(dont really need to worry abt this as tangent inverse fixes this)

NOT PROVIDED IN FORMULA SHEET

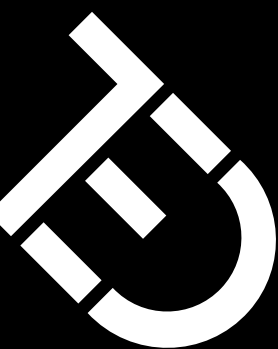
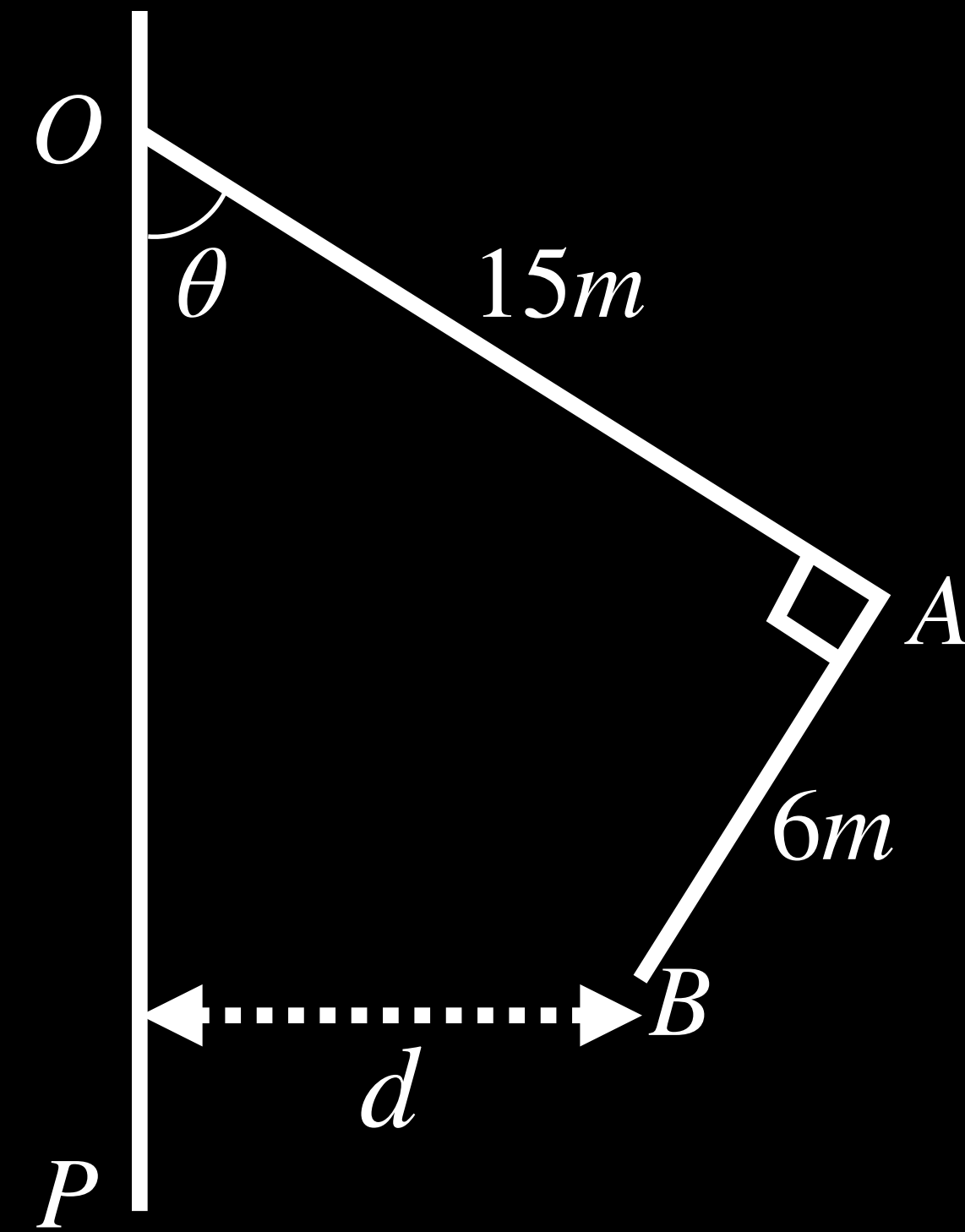


R-Formulae

You r

The figure shows 2 perpendicular line segments OA and AB of fixed lengths 15m and 6m, where OA makes an acute angle θ along OP that can vary from 0° to 180° . The distance d is the horizontal distance from point B to OP.

- a) show that $d = 15 \sin(\theta) - 6 \cos(\theta)$.
- b) Express d in the form of $R \sin(\theta - \alpha)$.
- c) When d is at 11m, what is the value of angle θ ?
- d) Find the maximum value of d .



R-Formulae

You r

The figure shows 2 perpendicular line segments OA and AB of fixed lengths 15m and 6m, where OA makes an acute angle θ along OP that can vary from 0° to 180° . The distance d is the horizontal distance from point B to OP.

a) show that $d = 15 \sin(\theta) - 6 \cos(\theta)$.

Let X be a point on OP such that OP is perpendicular to XA

Let Y be a point on XA such that XA is perpendicular to YB

$$\sin(\theta) = \frac{XA}{15} \quad \leftarrow \text{form trig ratios using SOHCAHTOA to find distances}$$

$$XA = 15 \sin(\theta)$$

$$\angle OAX = 90^\circ - \theta$$

$$\angle YAB = 90^\circ - (90^\circ - \theta)$$

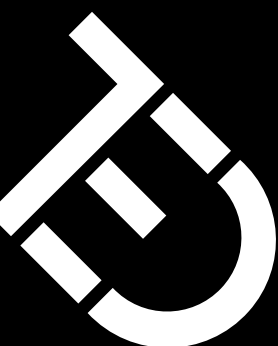
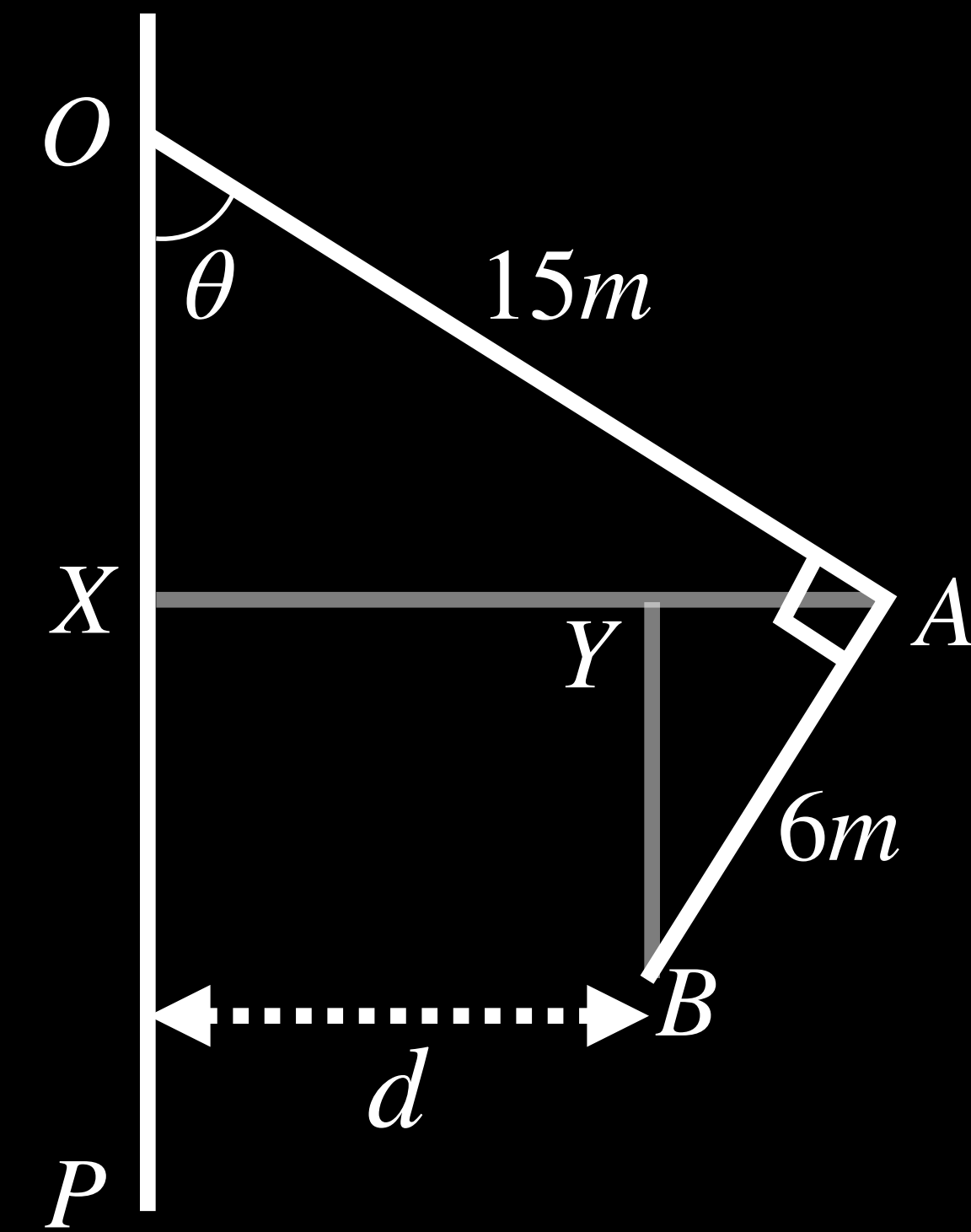
$$\angle YAB = \theta$$

$$\therefore \cos(\theta) = \frac{YA}{6}$$

$$YA = 6 \cos(\theta)$$

$$d = XA - YA$$

$$d = 15 \sin(\theta) - 6 \cos(\theta)$$



R-Formulae

You r

The figure shows 2 perpendicular line segments OA and AB of fixed lengths 15m and 6m, where OA makes an acute angle θ along OP that can vary from 0° to 180° . The distance d is the horizontal distance from point B to OP.

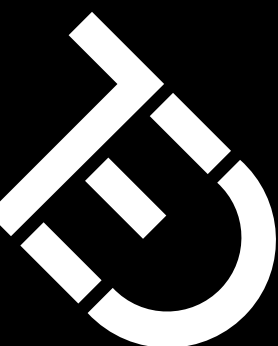
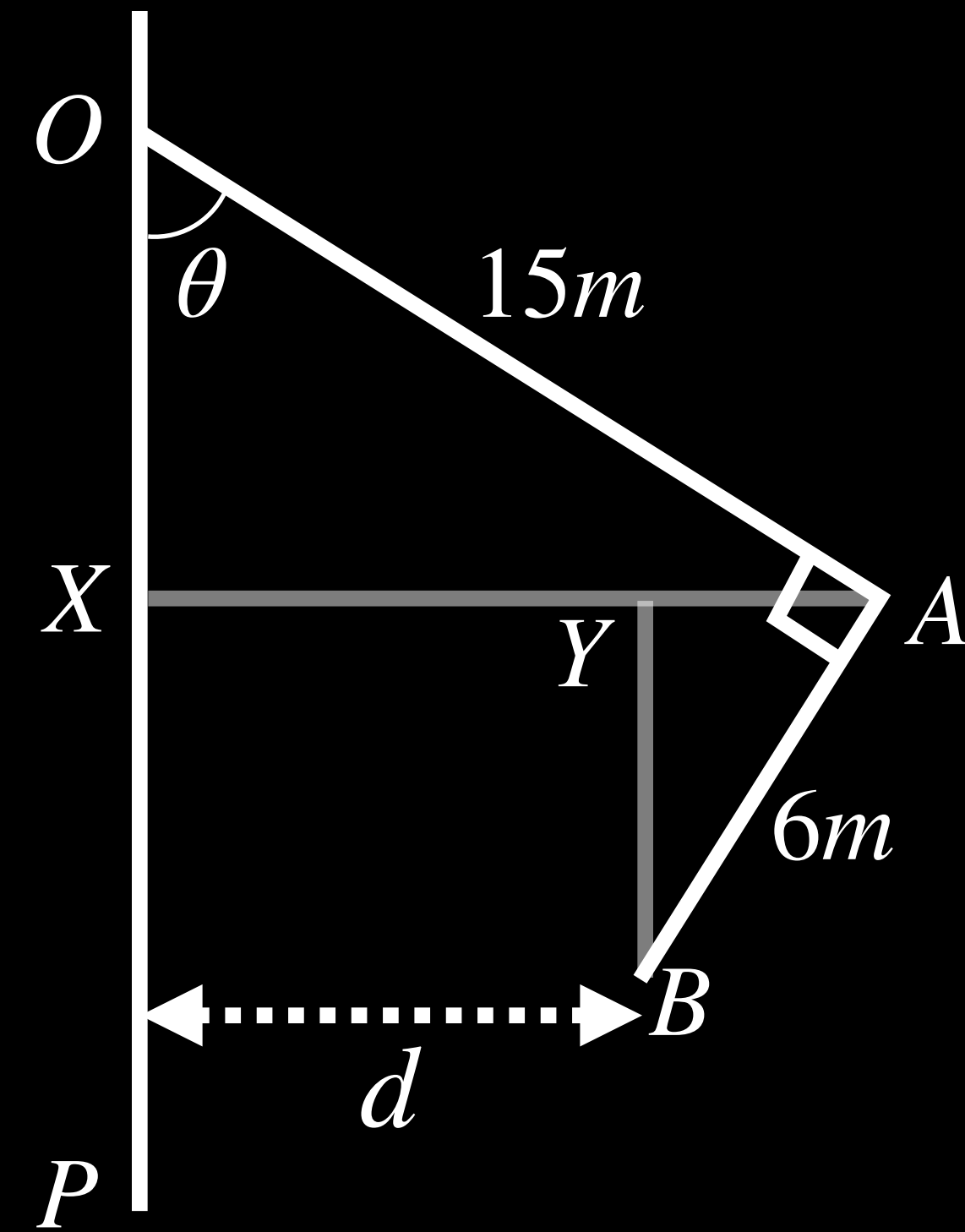
b) Express d in the form of $R \sin(\theta - \alpha)$.

$$d = 15 \sin(\theta) - 6 \cos(\theta)$$

$$R = \sqrt{15^2 + 6^2} = \sqrt{261}$$

$$\alpha = \tan^{-1}\left(\frac{6}{15}\right) = 21.8^\circ(1.d.p)$$

$$d = \sqrt{261} \sin(\theta - 21.8^\circ) \quad R \text{ can be expressed as a surd, however } \alpha \text{ must be in 1.d.p}$$



R-Formulae

You r

The figure shows 2 perpendicular line segments OA and AB of fixed lengths 15m and 6m, where OA makes an acute angle θ along OP that can vary from 0° to 180° . The distance d is the horizontal distance from point B to OP.

c) When $d=11$, what is the value of θ ?

$$d = \sqrt{261} \sin \left(\theta - \tan^{-1} \left(\frac{6}{15} \right) \right)$$

Exact value of α should be used or 5.s.f version

$$\sqrt{261} \sin \left(\theta - \tan^{-1} \left(\frac{6}{15} \right) \right) = 11$$

$$\sin \left(\theta - \tan^{-1} \left(\frac{6}{15} \right) \right) = \frac{11}{\sqrt{261}}$$

$$\text{Basic angle} = \sin^{-1} \left(\frac{11}{\sqrt{261}} \right)$$

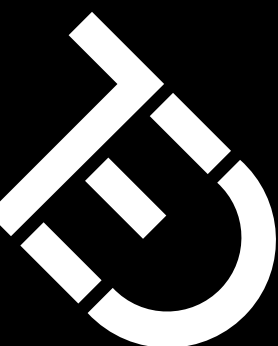
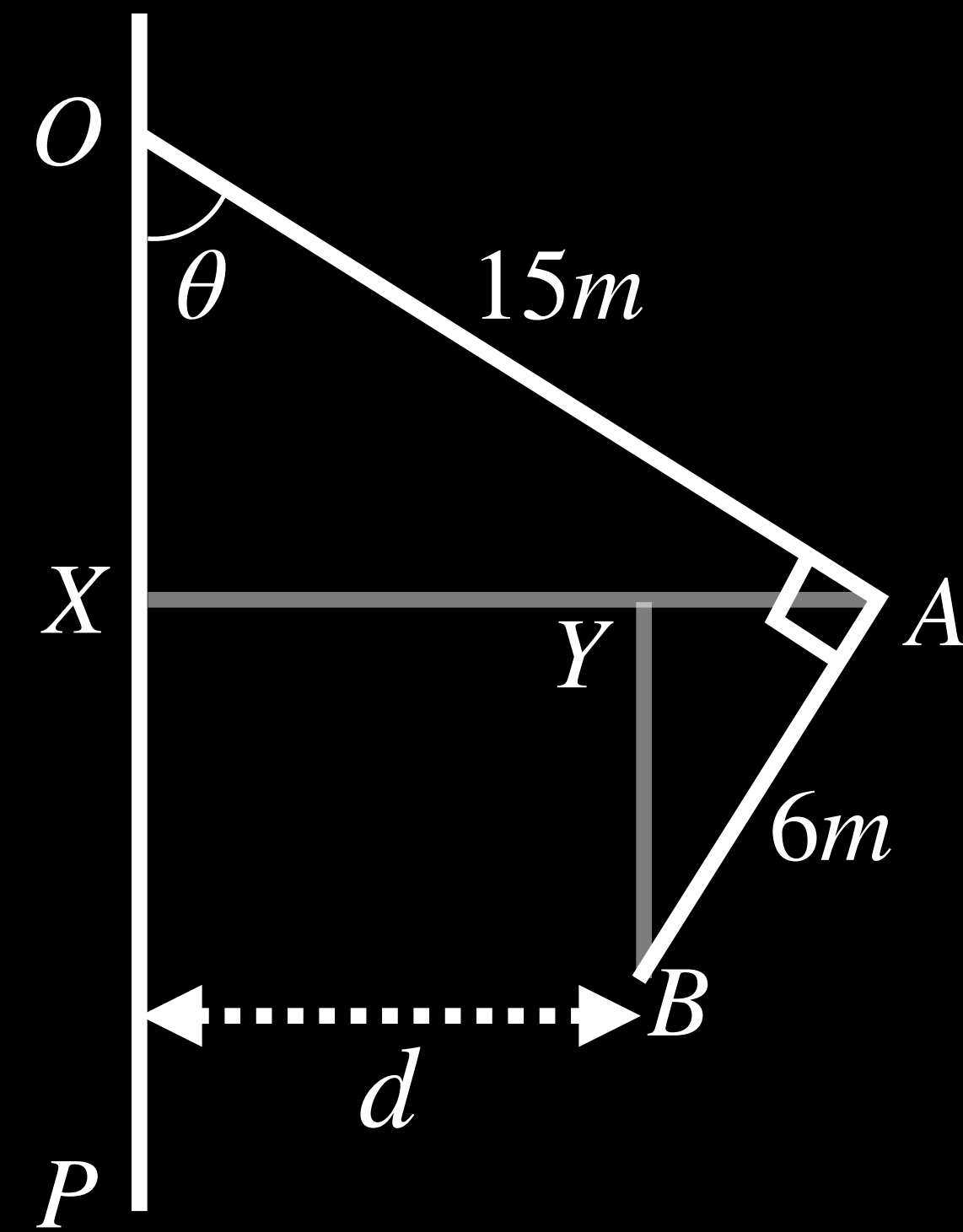
make sure to consider the domain to see if there are any values that need to be rejected or included.

$$\theta - \tan^{-1} \left(\frac{6}{15} \right) = \sin^{-1} \left(\frac{11}{\sqrt{261}} \right), 180 - \sin^{-1} \left(\frac{11}{\sqrt{261}} \right)$$

domain:

$$\left[-\tan^{-1} \left(\frac{6}{15} \right) \leq \theta - \tan^{-1} \left(\frac{6}{15} \right) \leq 180^\circ - \tan^{-1} \left(\frac{6}{15} \right) \right]$$

$$\theta = \sin^{-1} \left(\frac{11}{\sqrt{261}} \right) + \tan^{-1} \left(\frac{6}{15} \right), 180 - \sin^{-1} \left(\frac{11}{\sqrt{261}} \right) + \tan^{-1} \left(\frac{6}{15} \right) = 64.7^\circ, 158.9^\circ (1.d.p)$$



R-Formulae

You r

The figure shows 2 perpendicular line segments OA and AB of fixed lengths 15m and 6m, where OA makes an acute angle θ along OP that can vary from 0° to 180° . The distance d is the horizontal distance from point B to OP.

d) Find the maximum value of d .

$$d = \sqrt{261} \sin \left(\theta - \tan^{-1} \left(\frac{6}{15} \right) \right)$$

$$-1 \leq \sin \left(\theta - \tan^{-1} \left(\frac{6}{15} \right) \right) \leq 1$$

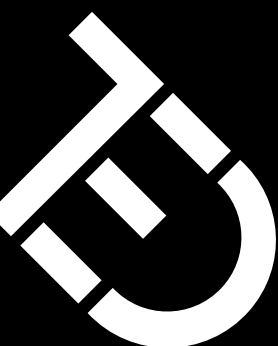
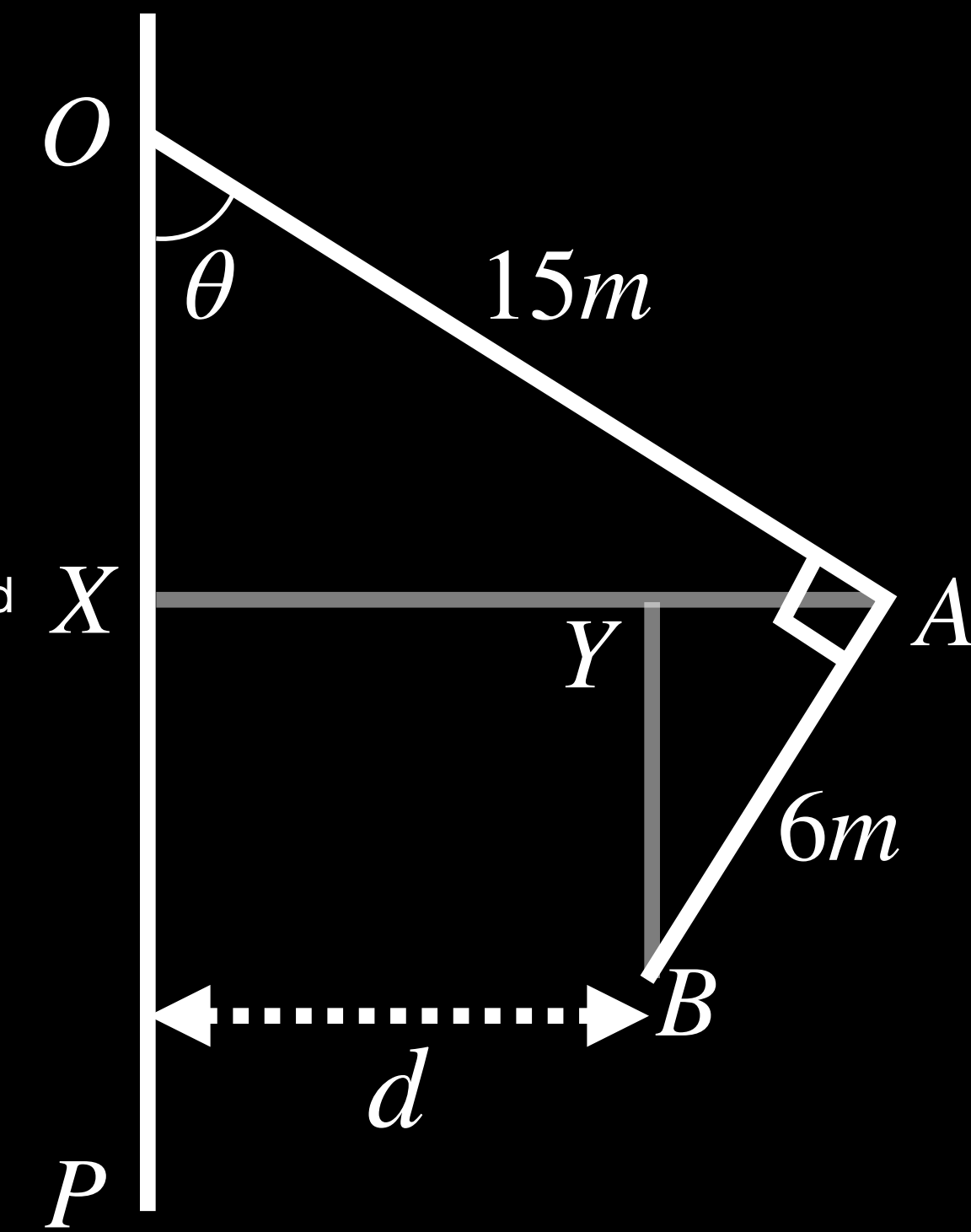
$$-\sqrt{261} \leq \sqrt{261} \sin \left(\theta - \tan^{-1} \left(\frac{6}{15} \right) \right) \leq \sqrt{261}$$

$$\therefore \text{maximum value of } d \text{ is } \sqrt{261} = 16.2(3.s.f)$$

Note that $-1 \leq \sin(ax + b) \leq 1$,
and $-1 \leq \cos(ax + b) \leq 1$

this allows us to change the middle expression until it is d
(sometimes, the r formula is added to a constant,
which has to be accounted for)

$$\begin{aligned} -1 &\leq \cos(ax + b) \leq 1 \\ -1 + c &\leq \cos(ax + b) + c \leq 1 + c \end{aligned}$$



Differentiation



Fundamentals

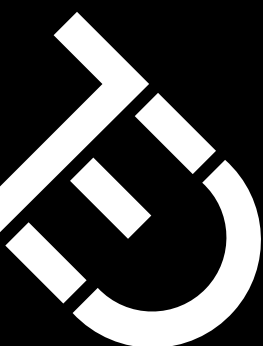
Basics

$$\frac{dy}{dx}$$

derivative

the change of value of y for a change of x

"derivative of y respective to x"



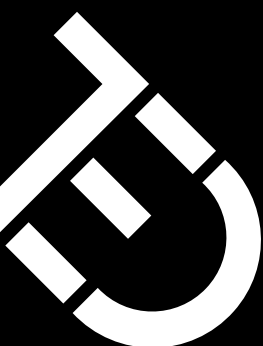
Fundamentals

Basics

$$\frac{d}{dx}f(x) = f'(x)$$

derivative

derivative of a function $f(x)$ respect to x is $f'(x)$

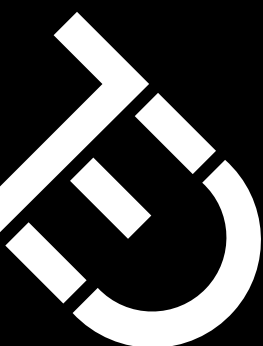


Derivative Rules - Power Rule

Basics

$$\frac{d}{dx}(x^n) = nx^{n-1}$$

power rule
derivative of x^n where n is a constant



Derivative Rules - Power Rule

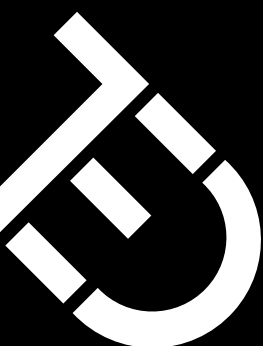
Basics

Find $\frac{dy}{dx}$ given that

$$y = x^5$$

$$y = \frac{1}{x}$$

$$y = \sqrt{x}$$



Derivative Rules - Power Rule

Basics

Find $\frac{dy}{dx}$ given that

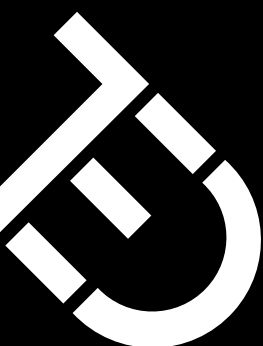
$$y = x^5$$

$$y = \frac{1}{x}$$

$$y = \sqrt{x}$$

$$y = x^5$$
$$\frac{dy}{dx} = 5x^{(5-1)}$$

$$\frac{dy}{dx} = 5x^4$$



Derivative Rules - Power Rule

Basics

Find $\frac{dy}{dx}$ given that

$$y = x^5$$

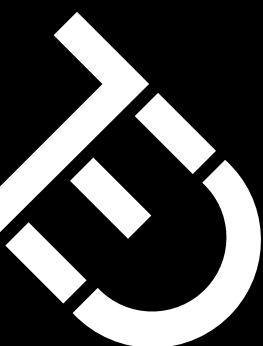
$$y = \frac{1}{x}$$

$$y = \sqrt{x}$$

$$y = \frac{1}{x} = x^{-1}$$

$$\frac{dy}{dx} = (-1)x^{(-1-1)}$$

$$\frac{dy}{dx} = -x^{-2}$$



Derivative Rules - Power Rule

Basics

Find $\frac{dy}{dx}$ given that

$$y = x^5$$

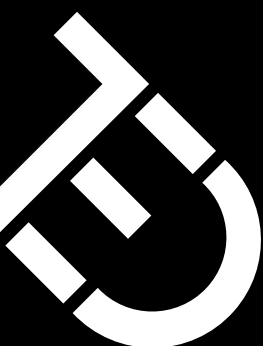
$$y = \frac{1}{x}$$

$$y = \sqrt{x}$$

$$y = \sqrt{x} = x^{0.5}$$

$$\frac{dy}{dx} = 0.5x^{(0.5-1)}$$

$$\frac{dy}{dx} = 0.5x^{-0.5}$$



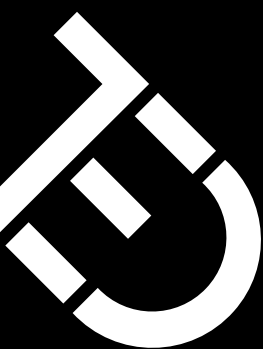
Derivative Rules - CM Rule

Basics

$$\frac{d}{dx} (kf(x)) = k \frac{d}{dx} (f(x))$$

constant multiple rule

derivative of $kf(x)$ where k is a constant and $f(x)$ is a function

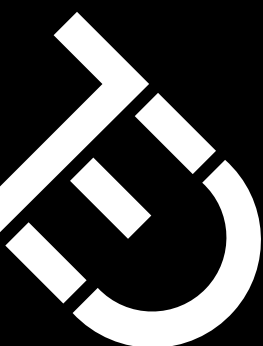


Derivative Rules - Constant Rule

Basics

$$\frac{d}{dx}k = 0$$

constant rule
derivative of k where k is a constant



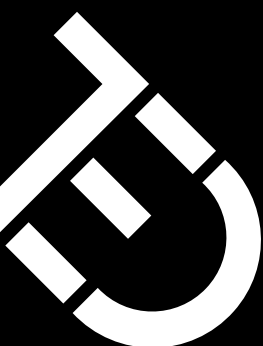
Derivative Rules - CM Rule

Basics

Find $\frac{dy}{dx}$ given that

$$y = 2x^3$$

$$y = \frac{4}{\sqrt{x}}$$



Derivative Rules - CM Rule

Basics

Find $\frac{dy}{dx}$ given that

$$y = 2x^3$$

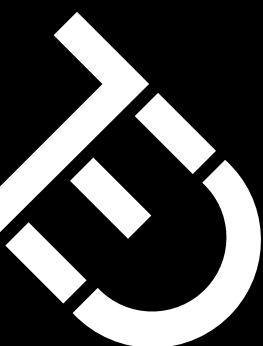
$$y = \frac{4}{\sqrt{x}}$$

$$y = 2x^3$$

$$\frac{dy}{dx} = 2 \frac{d}{dx}(x^3)$$

$$\frac{dy}{dx} = 2(3x^2)$$

$$= 6x^2$$



Derivative Rules - CM Rule

Basics

Find $\frac{dy}{dx}$ given that

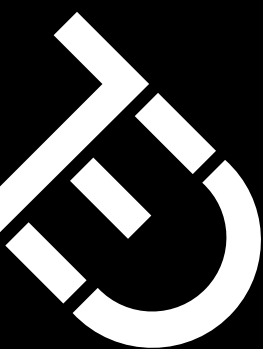
$$y = 2x^3$$

$$y = \frac{4}{\sqrt{x}}$$

$$\begin{aligned} y &= \frac{4}{\sqrt{x}} \\ &= 4 \left(\sqrt{x} \right)^{-1} \\ &= 4x^{-0.5} \end{aligned}$$

$$\frac{dy}{dx} = 4 \frac{d}{dx} x^{-0.5}$$

$$\begin{aligned} \frac{dy}{dx} &= 4(-0.5)x^{-0.5-1} = -2x^{-\frac{3}{2}} \\ &= -\frac{2}{\sqrt{x^3}} \end{aligned}$$



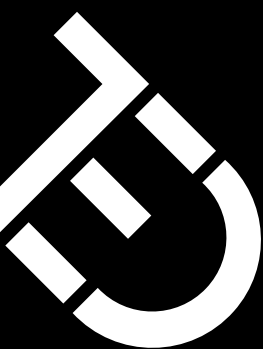
Derivative Rules - Sum/Difference Rule

Basics

$$\frac{d}{dx} (f(x) \pm g(x)) = f'(x) \pm g'(x)$$

sum/difference rule

derivative of sum/dif of two functions where $f(x)$ and $g(x)$ are functions



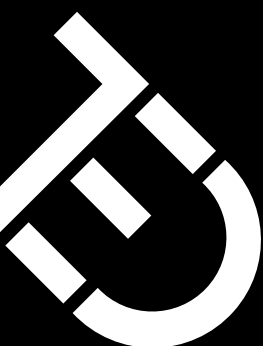
Derivative Rules - Product Rule

Basics

$$\frac{d}{dx} (f(x)g(x)) = f'(x)g(x) + g'(x)f(x)$$

product rule

derivative of product of two functions where $f(x)$ and $g(x)$ are functions



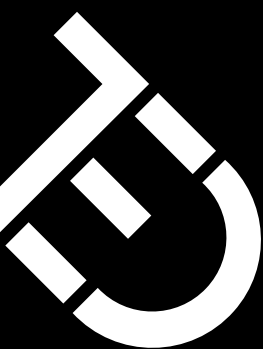
Derivative Rules - Quotient Rule

Basics

$$\frac{d}{dx} \left(\frac{f(x)}{g(x)} \right) = \frac{f'(x)g(x) - f(x)g'(x)}{g(x)^2}$$

quotient rule

derivative of division of two functions where $f(x)$ and $g(x)$ are functions



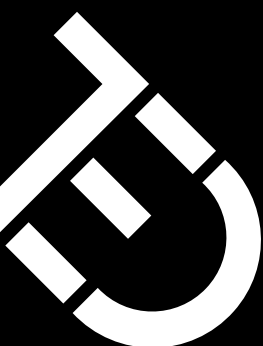
Derivative Rules - Chain Rule

Basics

$$\frac{d}{dx} (f(g(x))) = f'(g(x))g'(x)$$

chain rule

derivative of composition of two functions where $f(x)$ and $g(x)$ are functions



Derivative Rules - Chain Rule

Basics

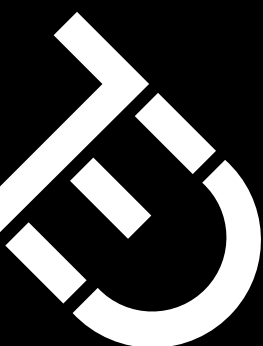
derivative of $f(g(x))$ in terms of $g(x)$

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

derivative of $g(x)$ in terms of x

chain rule

derivative of composition of two functions where $y = f(u)$ & $u = g(x)$



Derivative Rules

Basics

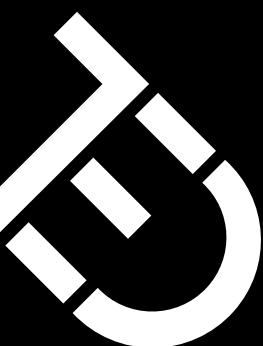
Find $\frac{dy}{dx}$ given that

$$y = 2x^2 - 5x + 3$$

$$y = (x^2 - 2)^3$$

$$y = (2x + x^3)\sqrt{x}$$

$$y = \frac{x^2 + 2}{x}$$



Derivative Rules

Basics

Find $\frac{dy}{dx}$ given that

$$y = 2x^2 - 5x + 3$$

$$y = (x^2 - 2)^3$$

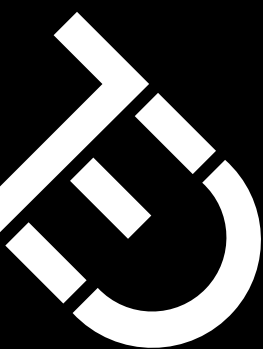
$$y = (2x + x^3)\sqrt{x}$$

$$y = \frac{x^2 + 2}{x}$$

$$y = 2x^2 - 5x + 3$$

$$\frac{dy}{dx} = 2(2x) - 5 + 0$$

$$\frac{dy}{dx} = 4x - 5$$



Derivative Rules

Basics

Find $\frac{dy}{dx}$ given that

$$y = 2x^2 - 5x + 3$$

$$y = (x^2 - 2)^3$$

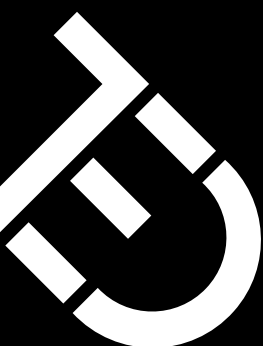
$$y = (2x + x^3)\sqrt{x}$$

$$y = \frac{x^2 + 2}{x}$$

$$y = (x^2 - 2)^3$$

$$\frac{dy}{dx} = 3(x^2 - 2)^2(2x - 0)$$

$$\frac{dy}{dx} = (6x)(x^2 - 2)^2$$



Derivative Rules

Basics

Find $\frac{dy}{dx}$ given that

$$y = 2x^2 - 5x + 3$$

$$y = (x^2 - 2)^3$$

$$y = (2x + x^3)\sqrt{x}$$

$$y = \frac{x^2 + 2}{x}$$

$$y = (2x + x^3)\sqrt{x}$$

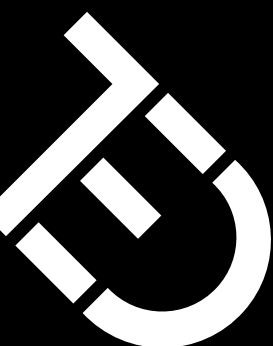
$$y = 2x\sqrt{x} + x^3\sqrt{x}$$

$$y = 2x^{1.5} + x^{3.5}$$

$$\frac{dy}{dx} = 2(1.5)x^{0.5} + 3.5x^{2.5}$$

$$\frac{dy}{dx} = 3\sqrt{x} + 3.5\sqrt{x^5}$$

If you have a function of sums of powers of x multiplied or divided by x^n where n is a constant, it is better to expand out first to make it more efficient



Derivative Rules

Basics

Find $\frac{dy}{dx}$ given that

$$y = 2x^2 - 5x + 3$$

$$y = (x^2 - 2)^3$$

$$y = (2x + x^3)\sqrt{x}$$

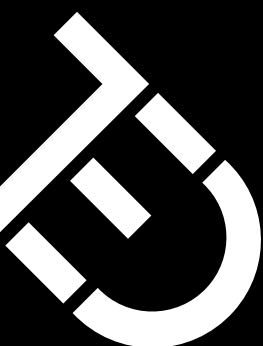
$$y = \frac{x^2 + 2}{x}$$

$$y = \frac{x^2 + 2}{x}$$

$$\frac{dy}{dx} = \frac{(2x + 0)x - 1(x^2 + 2)}{x^2}$$

$$\frac{dy}{dx} = \frac{2x^2 - x^2 - 2}{x^2}$$

$$\frac{dy}{dx} = \frac{x^2 - 2}{x^2}$$



Derivative Rules

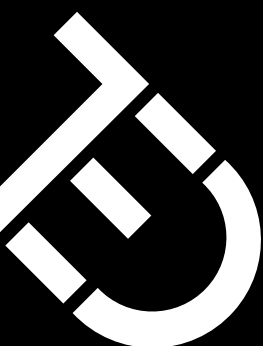
Basics

Find $\frac{dy}{dx}$ given that

$$y = x(3x^2 - 2x)^3$$

$$y = \frac{x^2 - 5x^3}{2x^3 + 5x}$$

$$y = (3x^2 - x^3) \sqrt{x^2 + 1}$$



Derivative Rules

Basics

Find $\frac{dy}{dx}$ given that

$$y = x(3x^2 - 2x)^3$$

$$y = \frac{x^2 - 5x^3}{2x^3 + 5x}$$

$$y = (3x^2 - x^3) \sqrt{x^2 + 1}$$

$$y = x(3x^2 - 2x)^3$$

$$\frac{dy}{dx} = (3x^2 - 2x)^3 + 3(3x^2 - 2x)^2(6x - 2)(x)$$

$$\frac{dy}{dx} = (3x^2 - 2x)^3 + 3(x) \left[(x)(3x - 2) \right]^2 (2)(3x - 1)$$

factor out as many common terms as possible

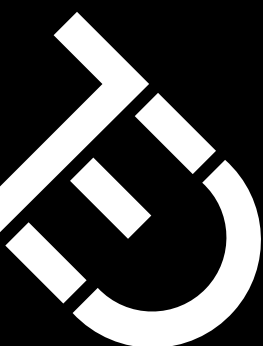
$$\frac{dy}{dx} = (3x^2 - 2x)^3 + 3(x)(x^2)((3x - 2))^2(2)(3x - 1)$$

$$\frac{dy}{dx} = x^3(3x - 2)^3 + 6x^3(3x - 2)^2(3x - 1)$$

$$\frac{dy}{dx} = x^3(3x - 2)^2((3x - 2) + 6(3x - 1))$$

factor out as many common terms as possible

$$\begin{aligned} \frac{dy}{dx} &= x^3(3x - 2)^2(3x - 2 + 18x - 6) \\ &= x^3(3x - 2)^2(21x - 8) \end{aligned}$$



Derivative Rules

Basics

Find $\frac{dy}{dx}$ given that

$$y = x(3x^2 - 2x)^3$$

$$y = \frac{x^2 - 5x^3}{2x^3 + 5x}$$

$$y = (3x^2 - x^3) \sqrt{x^2 + 1}$$

$$y = \frac{x^2 - 5x^3}{2x^3 + 5x}$$

when expanding, take note of max power, in this case its 5

$$\frac{dy}{dx} = \frac{\left((2x - 15x^2) (2x^3 + 5x) - (6x^2 + 5) (x^2 - 5x^3) \right)}{(2x^3 + 5x)^2}$$

$$\frac{dy}{dx} = \frac{(4x^4 + 10x^2 - 30x^5 - 75x^3 - 6x^4 + 30x^5 - 5x^2 + 25x^3)}{(2x^3 + 5x)^2}$$

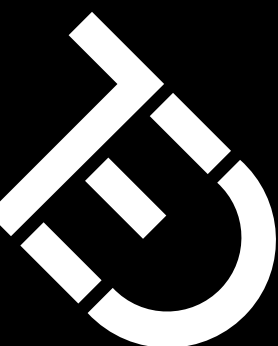
$$\frac{dy}{dx} = \frac{(30x^5 - 30x^5 + 4x^4 - 6x^4 - 75x^3 + 25x^3 + 10x^2 - 5x^2)}{(2x^3 + 5x)^2}$$

$$\frac{dy}{dx} = \frac{(-2x^4 - 50x^3 + 5x^2)}{(2x^3 + 5x)^2}$$

$$\frac{dy}{dx} = \frac{x^2 (-2x^2 - 50x + 5)}{x^2 (2x^2 + 5)^2}$$

factor out as many common terms as possible

$$\frac{dy}{dx} = \frac{(-2x^2 - 50x + 5)}{(2x^2 + 5)^2}$$



Derivative Rules

Basics

Find $\frac{dy}{dx}$ given that

$$y = x(3x^2 - 2x)^3$$

$$y = \frac{x^2 - 5x^3}{2x^3 + 5x}$$

$$y = (3x^2 - x^3) \sqrt{x^2 + 1}$$

$$y = (3x^2 - x^3) \sqrt{x^2 + 1}$$

$$\frac{dy}{dx} = (6x - 3x^2) \sqrt{x^2 + 1} + \frac{2x(3x^2 - x^3)}{2\sqrt{x^2 + 1}} \quad \text{good to memorise } \frac{d}{dx} \sqrt{x} = \frac{1}{2\sqrt{x}}$$

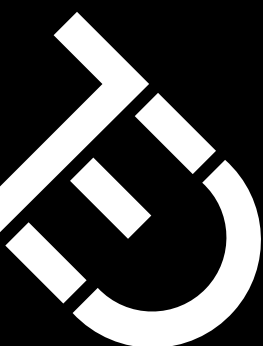
$$\frac{dy}{dx} = \frac{(6x - 3x^2)(x^2 + 1)}{\sqrt{x^2 + 1}} + \frac{x(3x^2 - x^3)}{\sqrt{x^2 + 1}}$$

$$\frac{dy}{dx} = \frac{(6x - 3x^2)(x^2 + 1) + x(3x^2 - x^3)}{\sqrt{x^2 + 1}} \quad \text{when expanding, take note of max power, in this case its 4}$$

$$\frac{dy}{dx} = \frac{6x^3 + 6x - 3x^4 - 3x^2 + 3x^3 - x^4}{\sqrt{x^2 + 1}}$$

$$\frac{dy}{dx} = \frac{-3x^4 - x^4 + 6x^3 + 3x^3 - 3x^2 + 6x}{\sqrt{x^2 + 1}}$$

$$\frac{dy}{dx} = \frac{-4x^4 + 9x^3 - 3x^2 + 6x}{\sqrt{x^2 + 1}}$$



Trigonometric Functions Derivatives

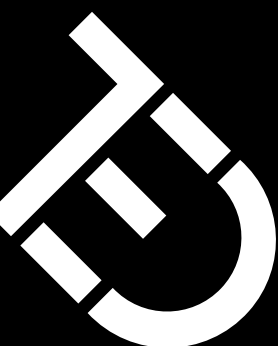
Longest chapter name

$$\frac{d}{dx} \sin(x) = \cos(x)$$

$$\frac{d}{dx} \cos(x) = -\sin(x)$$

$$\frac{d}{dx} \tan(x) = \sec^2(x)$$

derivatives of basic trigo functions
in calculus, radians are used instead of degrees



Trigonometric Functions Derivatives

Longest chapter name

Find $\frac{dy}{dx}$ given that

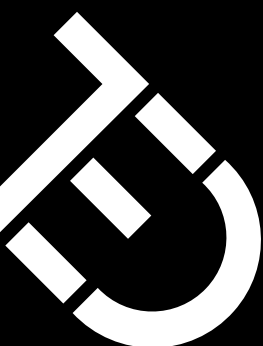
$$y = \sin(x)\cos(x)$$

$$y = \sin(2x + 3)$$

$$y = \csc(x)$$

$$y = \tan^3(x)$$

$$y = x^2 \cos(x)$$

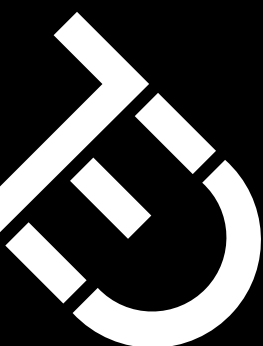


Trigonometric Functions Derivatives

Longest chapter name

Find $\frac{dy}{dx}$ given that

$$\begin{array}{lcl} y = \sin(x)\cos(x) & \frac{dy}{dx} = \cos(x)\cos(x) - \sin(x)\sin(x) & \\ y = \sin(2x + 3) & = \cos^2(x) - \sin^2(x) & \\ y = \csc(x) & = \cos(2x) & \text{[trig identity]} \\ y = \tan^3(x) & & \\ y = x^2 \cos(x) & & \end{array}$$

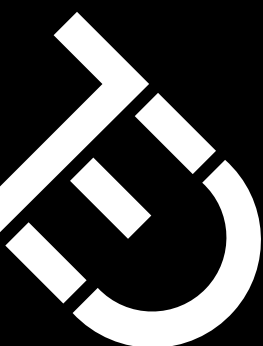


Trigonometric Functions Derivatives

Longest chapter name

Find $\frac{dy}{dx}$ given that

$$\begin{array}{ll} y = \sin(x)\cos(x) & \frac{dy}{dx} = \cos(2x + 3)(2) \\ y = \sin(2x + 3) & = 2 \cos(2x + 3) \\ y = \csc(x) & \\ y = \tan^3(x) & \\ y = x^2 \cos(x) & \end{array}$$

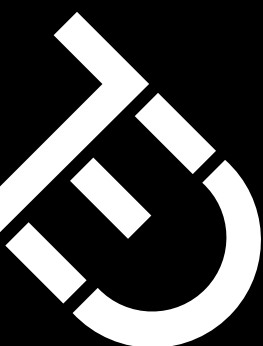


Trigonometric Functions Derivatives

Longest chapter name

Find $\frac{dy}{dx}$ given that

$$\begin{array}{ll} y = \sin(x)\cos(x) & \frac{dy}{dx} = (\sin(x))^{-1} \\ y = \sin(2x + 3) & = -(\sin(x))^{-2}(\cos(x)) \\ y = \csc(x) & = -\frac{\cos(x)}{\sin^2(x)} \\ y = \tan^3(x) & \\ y = x^2 \cos(x) & \end{array}$$

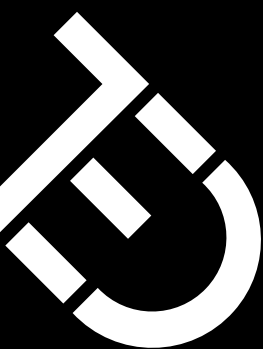


Trigonometric Functions Derivatives

Longest chapter name

Find $\frac{dy}{dx}$ given that

$y = \sin(x)\cos(x)$	$\frac{dy}{dx} = (\tan(x))^3$
$y = \sin(2x + 3)$	
$y = \csc(x)$	$= 3 (\tan(x))^2 (\sec^2(x))$
$y = \tan^3(x)$	$= 3 \tan^2(x) \sec^2(x)$
$y = x^2 \cos(x)$	

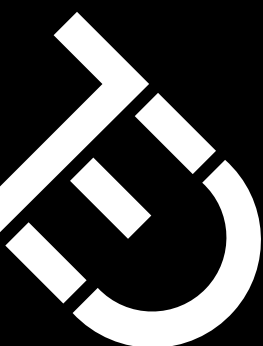


Trigonometric Functions Derivatives

Longest chapter name

Find $\frac{dy}{dx}$ given that

$$\begin{aligned} y &= \sin(x)\cos(x) & \frac{dy}{dx} &= 2x(\cos(x)) + (-\sin(x))(x^2) \\ y &= \sin(2x + 3) & \frac{dy}{dx} &= 2x \cos(x) - x^2 \sin(x) \\ y &= \csc(x) & &= x (2 \cos(x) - x \sin(x)) \\ y &= \tan^3(x) \\ y &= x^2 \cos(x) \end{aligned}$$

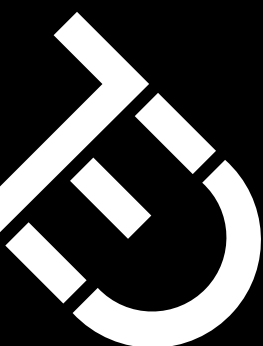


Exponential Derivatives

Longest chapter name

$$\frac{d}{dx}e^x = e^x$$

derivative of natural exponential



Exponential Functions Derivatives

Longest chapter name

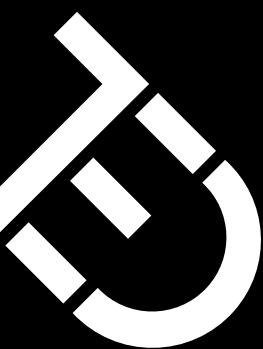
Find $\frac{dy}{dx}$ given that

$$y = \frac{e^x}{x^2}$$

$$y = 2e^{3x-1}$$

$$y = e^{x^e}$$

$$y = \frac{(e^x)(e^{x+4})}{e^{5x}}$$



Exponential Functions Derivatives

Longest chapter name

Find $\frac{dy}{dx}$ given that

$$y = \frac{e^x}{x^2}$$

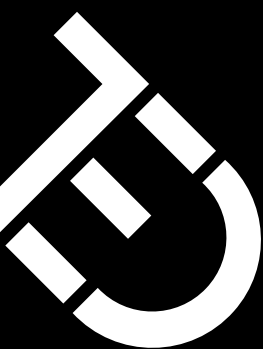
$$y = 2e^{3x-1}$$

$$y = \frac{(e^x)(e^{x+4})}{e^{5x}}$$

$$y = \frac{e^x}{x^2}$$

$$\frac{dy}{dx} = \frac{e^x x^2 - (2x)e^x}{x^4}$$

$$\frac{dy}{dx} = \frac{e^x(x^2 - 2x)}{x^4}$$



Exponential Functions Derivatives

Longest chapter name

Find $\frac{dy}{dx}$ given that

$$y = \frac{e^x}{x^2}$$

$$y = 2e^{3x-1}$$

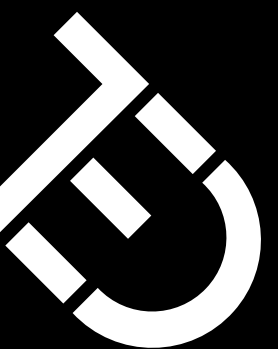
$$y = e^{x^e}$$

$$y = \frac{(e^x)(e^{x+4})}{e^{5x}}$$

$$y = 2e^{3x-1}$$

$$\frac{dy}{dx} = 2e^{3x-1} * (3)$$

$$\frac{dy}{dx} = 6e^{3x-1}$$



Exponential Functions Derivatives

Longest chapter name

Find $\frac{dy}{dx}$ given that

$$y = \frac{e^x}{x^2}$$

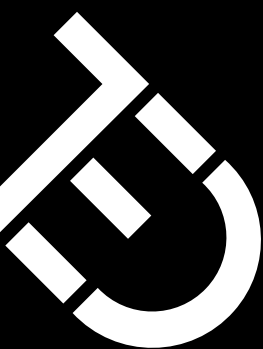
$$y = 2e^{3x-1}$$

$$y = e^{x^e}$$

$$y = \frac{(e^x)(e^{x+4})}{e^{5x}}$$

$$y = e^{x^e}$$

$$\frac{dy}{dx} = e^{x^e}(ex^{(e-1)})$$



Exponential Functions Derivatives

Longest chapter name

Find $\frac{dy}{dx}$ given that

$$y = \frac{e^x}{x^2}$$

$$y = 2e^{3x-1}$$

$$y = e^{x^e}$$

$$y = \frac{(e^x)(e^{x+4})}{e^{5x}}$$

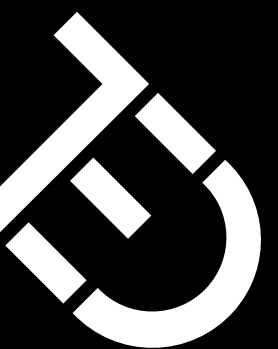
$$y = \frac{(e^x)(e^{x+4})}{e^{5x}}$$

$$= e^{x+x+4-5x}$$

$$= e^{4-3x}$$

use indices rules to simplify

$$\frac{dy}{dx} = -3e^{4-3x}$$

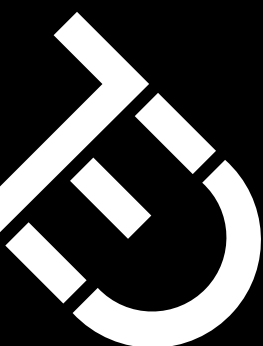


Logarithmic Derivatives

Longest chapter name

$$\frac{d}{dx} \ln(x) = \frac{1}{x}$$

derivative of natural logarithm



Logarithmic Functions Derivatives

Longest chapter name

Find $\frac{dy}{dx}$ given that

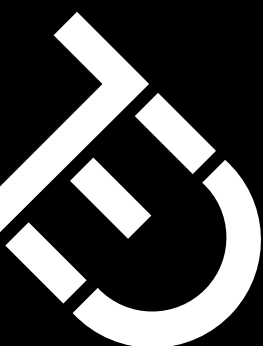
$$y = \ln(x)\sec(x)$$

$$y = \ln(\sqrt{x})$$

$$y = \ln(5x - 2)$$

$$y = \ln(\ln(x)\ln(x))$$

$$y = \ln\left(\frac{5x - 2}{x^2 - 1}\right)$$



Logarithmic Functions Derivatives

Longest chapter name

Find $\frac{dy}{dx}$ given that

$$y = \ln(x)\sec(x)$$

$$y = \ln(\sqrt{x})$$

$$y = \ln(5x - 2)$$

$$y = \ln(\ln(x)\ln(x))$$

$$y = \ln\left(\frac{5x - 2}{x^2 - 1}\right)$$

$$y = \ln(x)\sec(x)$$

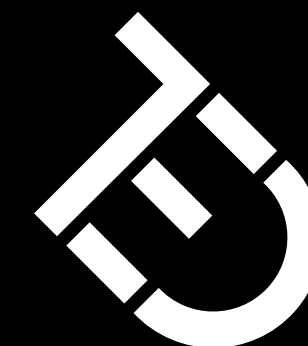
$$y = \frac{\ln(x)}{\cos(x)}$$

$$\frac{dy}{dx} = \frac{x^{-1}\cos(x) - (-\sin(x))(\ln(x))}{\cos^2(x)}$$

$$\frac{dy}{dx} = \frac{x^{-1}\cos(x) + \sin(x)\ln(x)}{\cos^2(x)}$$

$$\frac{dy}{dx} = \frac{\frac{\cos(x)}{x} + \frac{\sin(x)\ln(x)x}{x}}{\cos^2(x)}$$

$$\frac{dy}{dx} = \frac{\frac{\cos(x) + \sin(x)\ln(x)x}{x}}{\cos^2(x)} = \frac{\cos(x) + \sin(x)\ln(x)x}{x\cos^2(x)}$$



Logarithmic Functions Derivatives

Longest chapter name

Find $\frac{dy}{dx}$ given that

$$y = \ln(x)\sec(x)$$

$$y = \ln(\sqrt{x})$$

$$y = \ln(5x - 2)$$

$$y = \ln(\ln(x)\ln(x))$$

$$y = \ln\left(\frac{5x - 2}{x^2 - 1}\right)$$

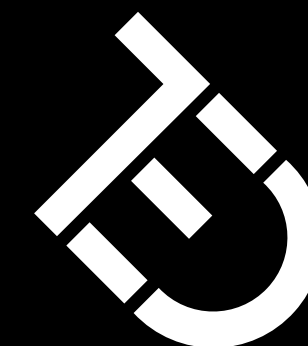
$$y = \ln(\sqrt{x})$$

$$y = \frac{1}{2} \ln(x)$$

$$\frac{dy}{dx} = \frac{1}{2} \times \frac{1}{x}$$

$$\frac{dy}{dx} = \frac{1}{2x}$$

make use of log rules to simplify expression before differentiating



Logarithmic Functions Derivatives

Longest chapter name

Find $\frac{dy}{dx}$ given that

$$y = \ln(x)\sec(x)$$

$$y = \ln(\sqrt{x})$$

$$y = \ln(5x - 2)$$

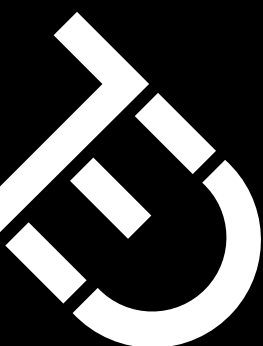
$$y = \ln(\ln(x)\ln(x))$$

$$y = \ln\left(\frac{5x - 2}{x^2 - 1}\right)$$

$$y = \ln(5x - 2)$$

$$\frac{dy}{dx} = \frac{1}{5x - 2} \times 5$$

$$\frac{dy}{dx} = \frac{5}{5x - 2}$$



Logarithmic Functions Derivatives

Longest chapter name

Find $\frac{dy}{dx}$ given that

$$y = \ln(x)\sec(x)$$

$$y = \ln(\sqrt{x})$$

$$y = \ln(5x - 2)$$

$$y = \ln(\ln(x)\ln(x))$$

$$y = \ln\left(\frac{5x - 2}{x^2 - 1}\right)$$

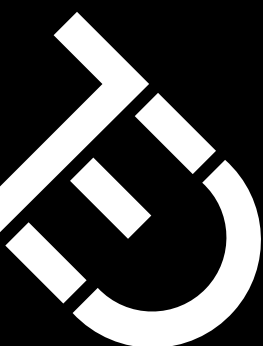
$$y = \ln(\ln(x)\ln(x))$$

$$y = \ln(\ln(x)^2)$$

$$\frac{dy}{dx} = \frac{1}{\ln(x)^2} \times \frac{d}{dx} \ln(x)^2$$

$$\frac{dy}{dx} = \frac{1}{\ln(x)^2} \times 2 \ln(x) \times \frac{1}{x}$$

$$\frac{dy}{dx} = \frac{2 \ln(x)}{x \ln(x)^2} = \frac{2}{x \ln(x)}$$



Logarithmic Functions Derivatives

Longest chapter name

Find $\frac{dy}{dx}$ given that

$$y = \ln(x)\sec(x)$$

$$y = \ln(\sqrt{x})$$

$$y = \ln(5x - 2)$$

$$y = \ln(\ln(x)\ln(x))$$

$$y = \ln\left(\frac{5x - 2}{x^2 - 1}\right)$$

$$y = \ln\left(\frac{5x - 2}{x^2 - 1}\right)$$

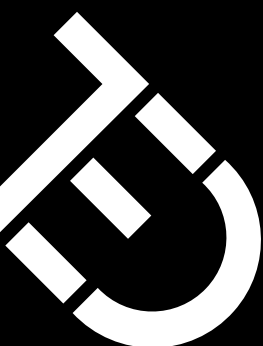
$$= \ln(5x - 2) - \ln(x^2 - 1)$$

$$\frac{dy}{dx} = (5)\frac{1}{5x - 2} - (2x)\frac{1}{x^2 - 1}$$

$$= \frac{5}{5x - 2} - \frac{2x}{x^2 - 1}$$

use log rules to simplify

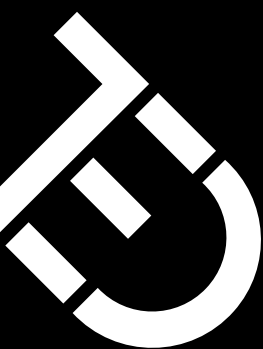
no need to use quotient rule, simplify into two logarithms



Differentiation Equations

Basics

Given that $y = \sin(x)\cos(x)$, show that $4y\frac{dy}{dx} = \sin(4x)$.



Differentiation Equations

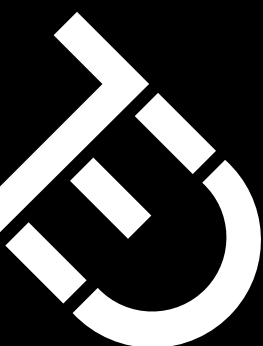
Basics

Given that $y = \sin(x)\cos(x)$, show that $4y\frac{dy}{dx} = \sin(4x)$.

$$\begin{aligned} y &= \sin(x)\cos(x) \\ \frac{dy}{dx} &= \cos(x)\cos(x) - \sin(x)\sin(x) \\ &= \cos^2(x) - \sin^2(x) \\ &= \cos(2x) \end{aligned} \quad \text{find } \frac{dy}{dx}$$

$$4y\frac{dy}{dx} = \sin(4x)$$

$$\begin{aligned} LHS &= 4 (\sin(x)\cos(x)) \cos(2x) && \text{sub in } y \text{ and } \frac{dy}{dx} \text{ into eqn} \\ &= 2 (2 \sin(x)\cos(x)) \cos(2x) \\ &= 2 (\sin(2x)) \cos(2x) \\ &= \sin(2(2x)) \\ &= \sin(4x) \\ &= RHS \end{aligned}$$

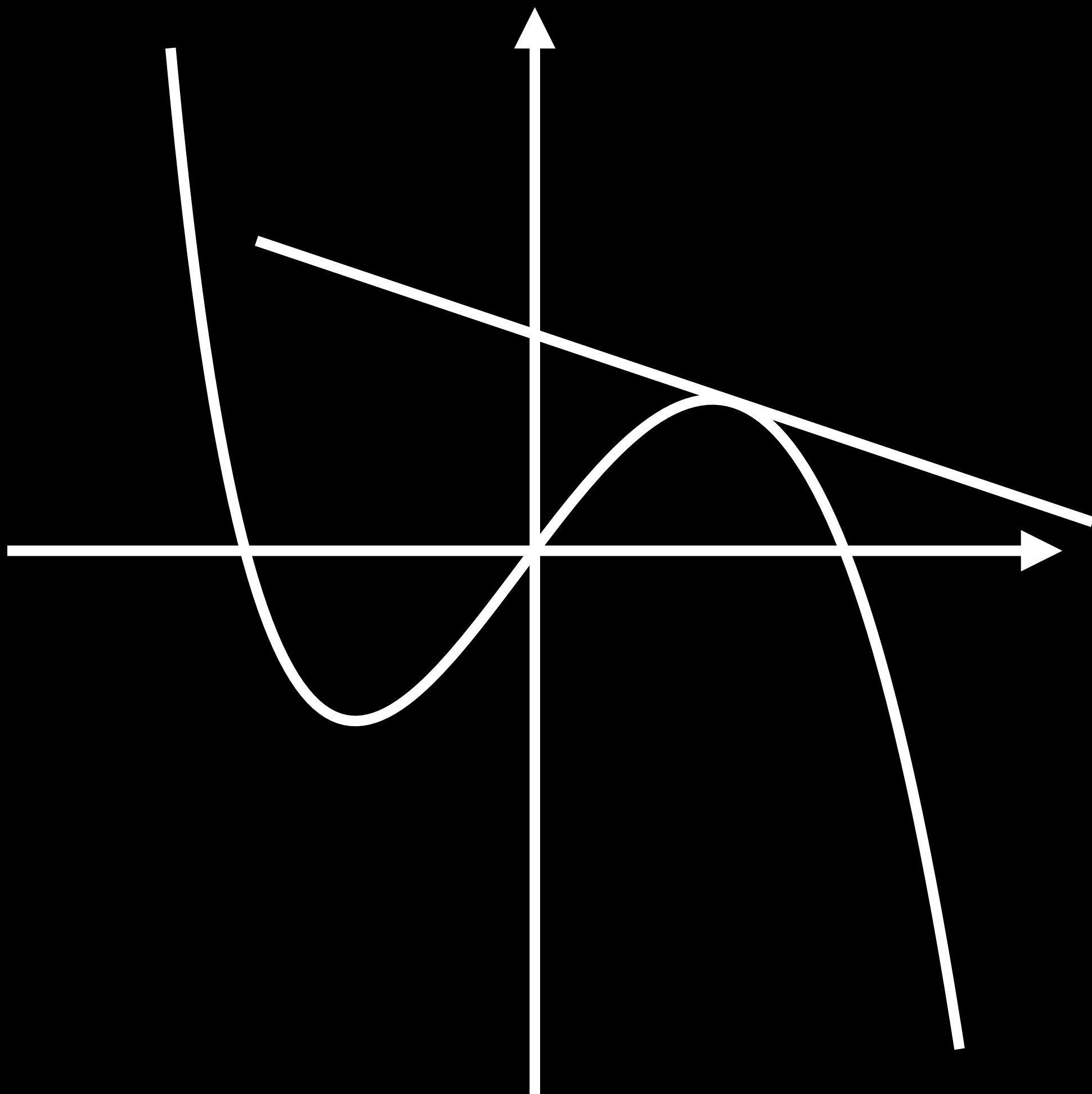


Applications of Differentiation



Tangents

emathtangents



To find gradient of tangent at $x=a$ for function $f(x)$,
Find $f'(x)$ and evaluate $f'(a)$

For example, given the function $f(x) = 3x - x^3$
Find $f'(x)$ and the gradient of the tangent at $x = -5$

$$f(x) = 3x - x^3$$

$$f'(x) = 3 - 3x^2$$

$$f'(-5) = 3 - 3(-5)^2$$

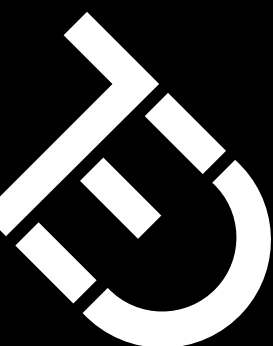
$$f'(-5) = -72$$

if it was $y = 3x - x^2$, we sub in -5 and write as so:

$$\frac{dy}{dx} \bigg|_{x=-5} = 3 - 3(-5)^2$$

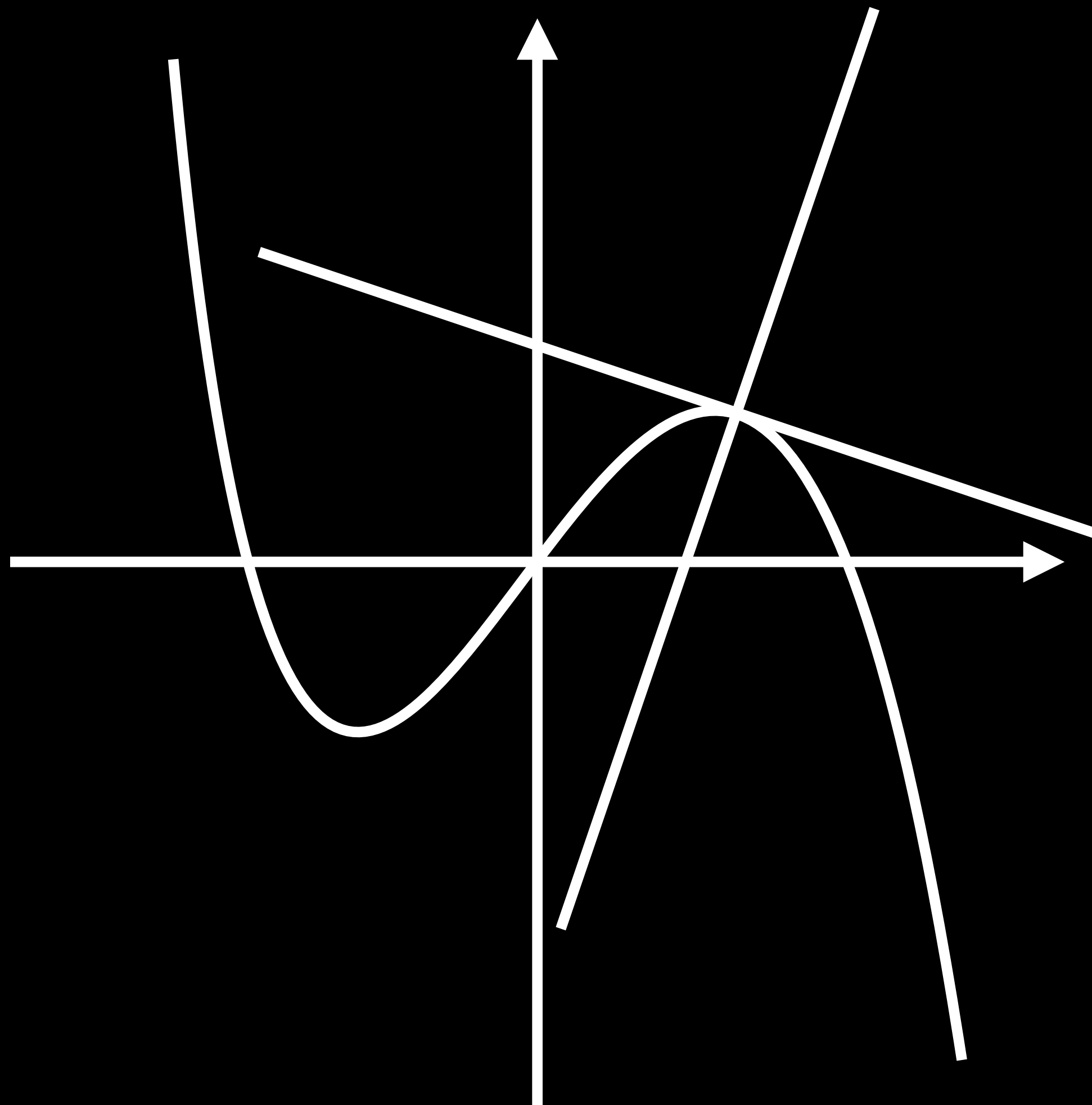
if question does not ask you to find derivative,
you do not need to simplify derivative

**you need to have the x value of where you want to find the gradient, not the y value



Tangents & Normals

emathtangents



To find tangent eqn at $x=a$ for function $f(x)$,
Find $f'(x)$ and evaluate $f'(a)$
Use point-gradient form to find eqn of tangent

$$y - y_1 = m(x - x_1)$$

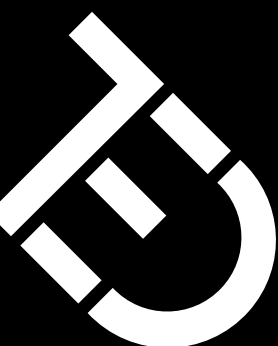
$$y - f(a) = f'(a)(x - a)$$

Likewise, to find normal, gradient is negative reciprocal

$$m_2 = -m_1^{-1}$$

$$m_2 = -f'(a)^{-1}$$

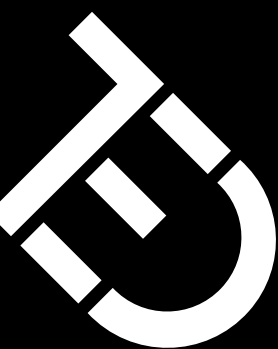
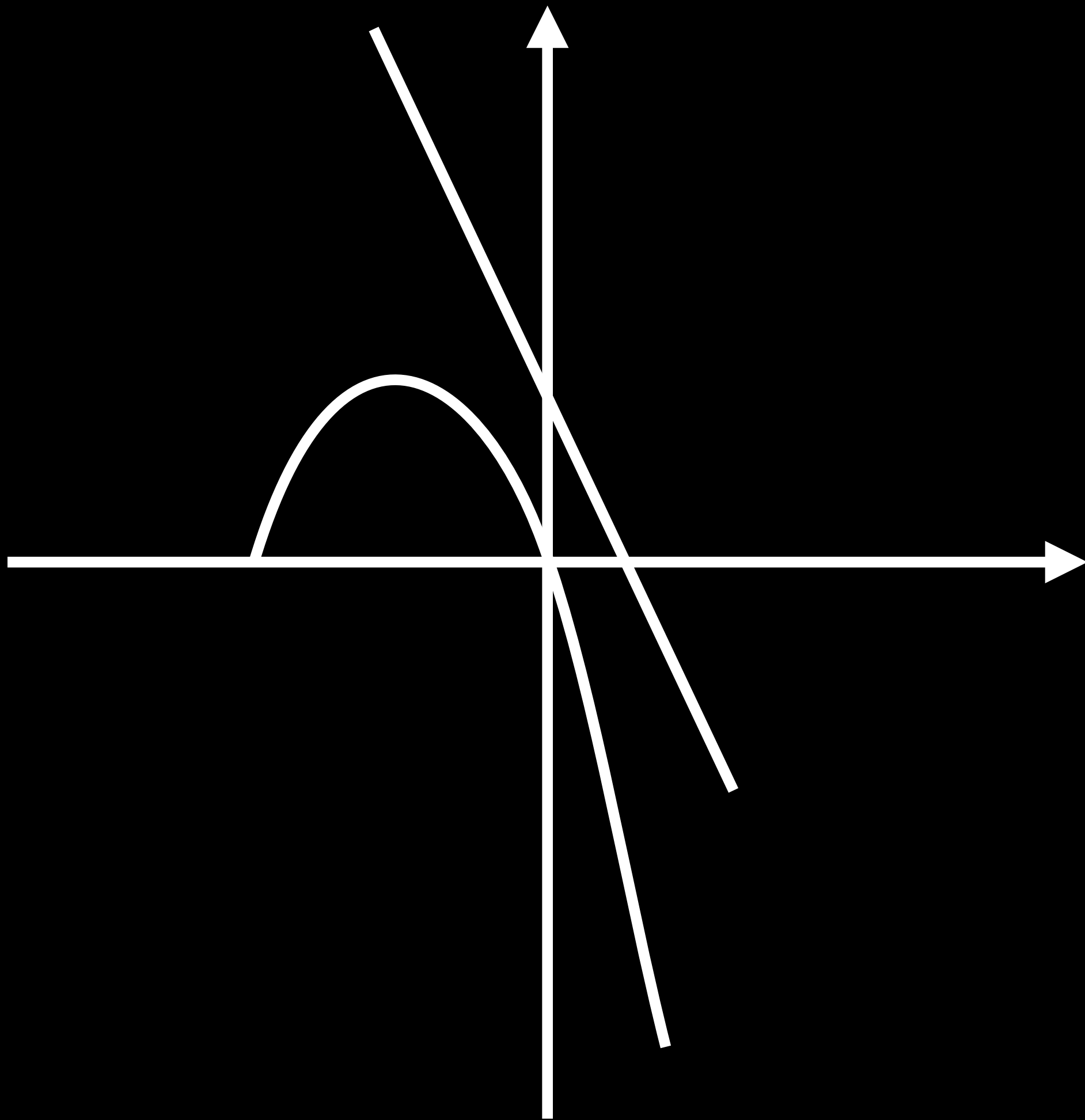
$$y - f(a) = -f'(a)^{-1}(x - a)$$



Tangents & Normals

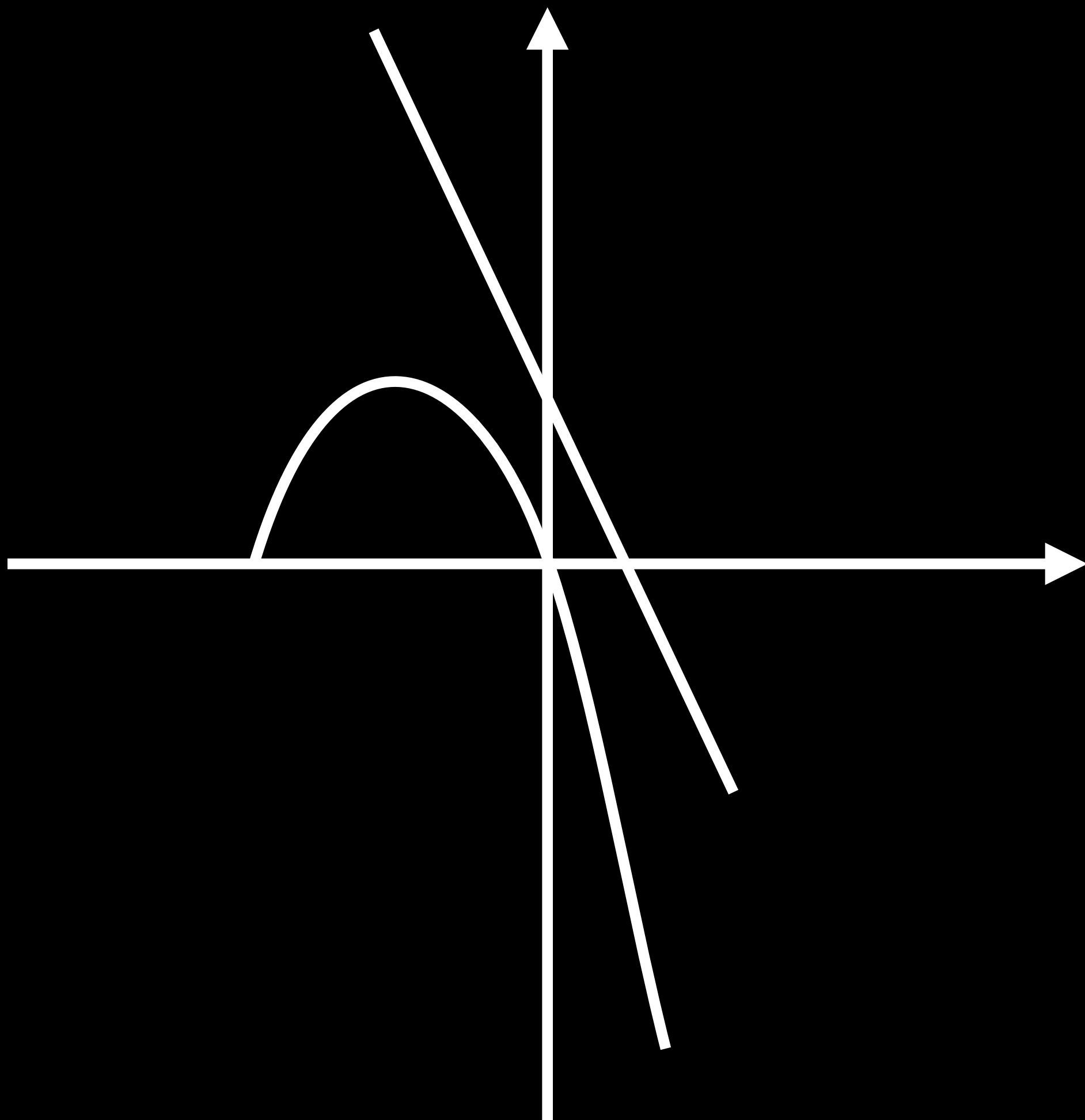
emathtangents

The graph shown is the function $y = -x\sqrt{x+4}$
The tangent line at $x = k$ is perpendicular to the line $4 - y = 2x$
a) Solve for k and b) find the eqn of the tangent at $x = k$



Tangents & Normals

emathtangents



The graph shown is the function $y = -x\sqrt{x+4}$

The tangent line at $x = k$ is perpendicular to the line $4 - y = 2x$

a) Solve for k and b) find the eqn of the tangent at $x = k$

$$\begin{aligned}
 \text{a) } y &= -x\sqrt{x+4} \\
 \frac{dy}{dx} &= (-1)\sqrt{x+4} + 0.5(x+4)^{-0.5}x \\
 &= -\sqrt{x+4} + \frac{x}{2\sqrt{x+4}} \\
 &= -\frac{2(x+4)}{2\sqrt{x+4}} + \frac{x}{2\sqrt{x+4}} \\
 &= -\frac{2(x+4) + x}{2\sqrt{x+4}} \\
 &= -\frac{3x+8}{2\sqrt{x+4}}
 \end{aligned}$$

$$\begin{aligned}
 4 - y &= 2x \\
 -y &= 2x - 4 \\
 y &= 4 - 2x
 \end{aligned}$$

$\therefore$ gradient of line perpendicular to tangent is -2

$$\begin{aligned}
 m_{\text{tangent}} &= -(-2)^{-1} \\
 &= 0.5 \\
 \text{let } \frac{dy}{dx} &= 0.5
 \end{aligned}$$

$$-\frac{3x+8}{2\sqrt{x+4}} = 0.5$$

$$-3x - 8 = \sqrt{x+4}$$

$$(-3x - 8)^2 = x + 4$$

$$9x^2 - 2(-3x)(8) + 8^2 = x + 4$$

$$9x^2 + 48x + 64 = x + 4$$

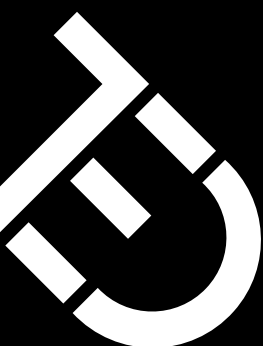
$$9x^2 + 47x + 60 = 0$$

recall surd eqns!!

$$x = -\frac{20}{9}, -3$$

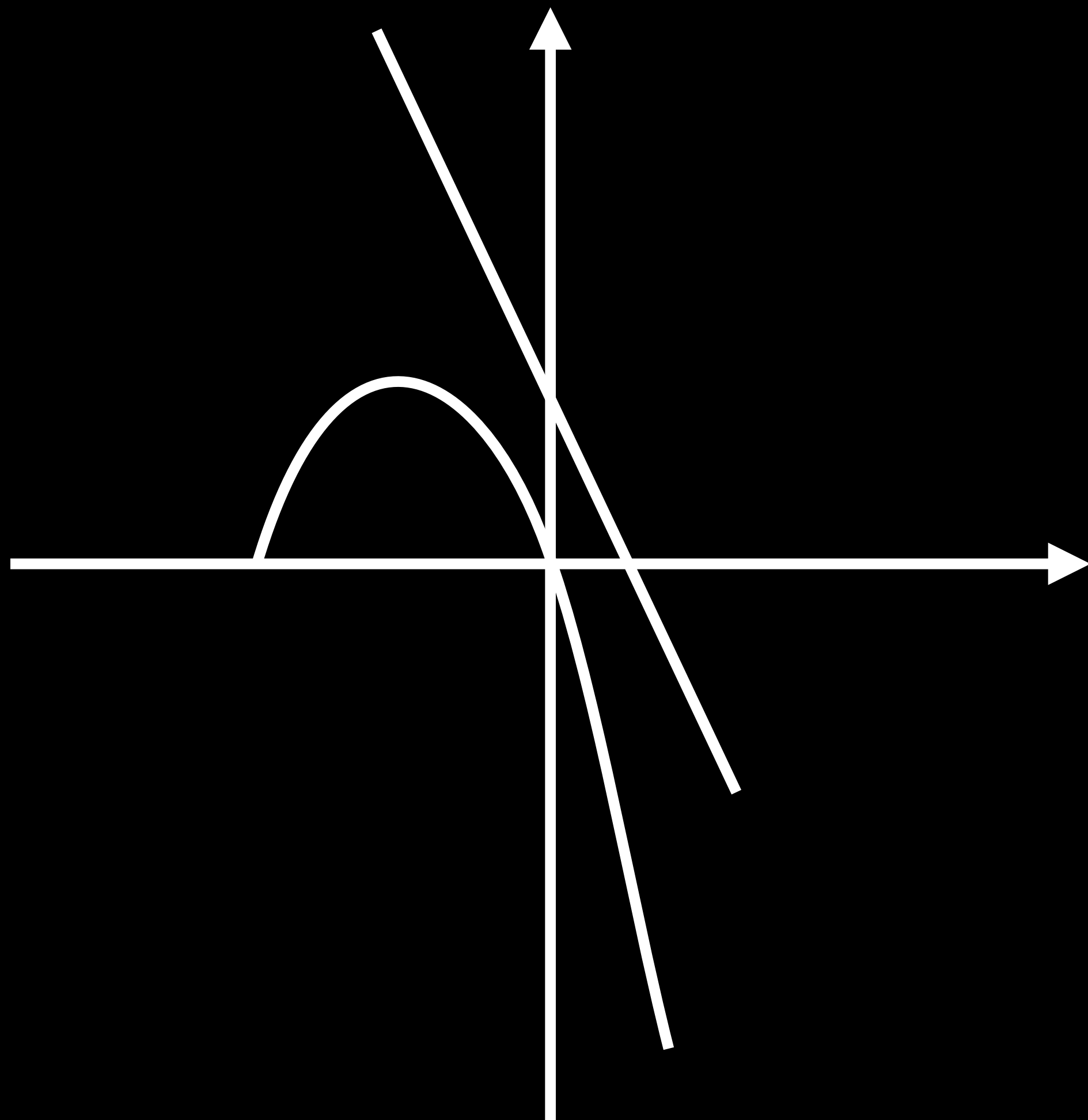
$$\text{rej. } x = -\frac{20}{9} \text{ as LHS at } x = -\frac{20}{9} \text{ is } -0.5$$

$$\therefore k = -3$$



Tangents & Normals

emathtangents



The graph shown is the function $y = -x\sqrt{x+4}$

The tangent line at $x = k$ is perpendicular to the line $4 - y = 2x$

a) Solve for k and b) find the eqn of the tangent at $x = k$

b)

at $x = -3$

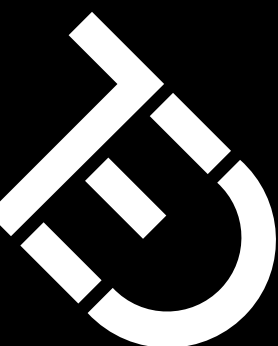
$$y - y_1 = m(x - x_1)$$

$$y - (-(-3)\sqrt{(-3)+4}) = 0.5(x - (-3))$$

$$y - 3 = 0.5(x + 3)$$

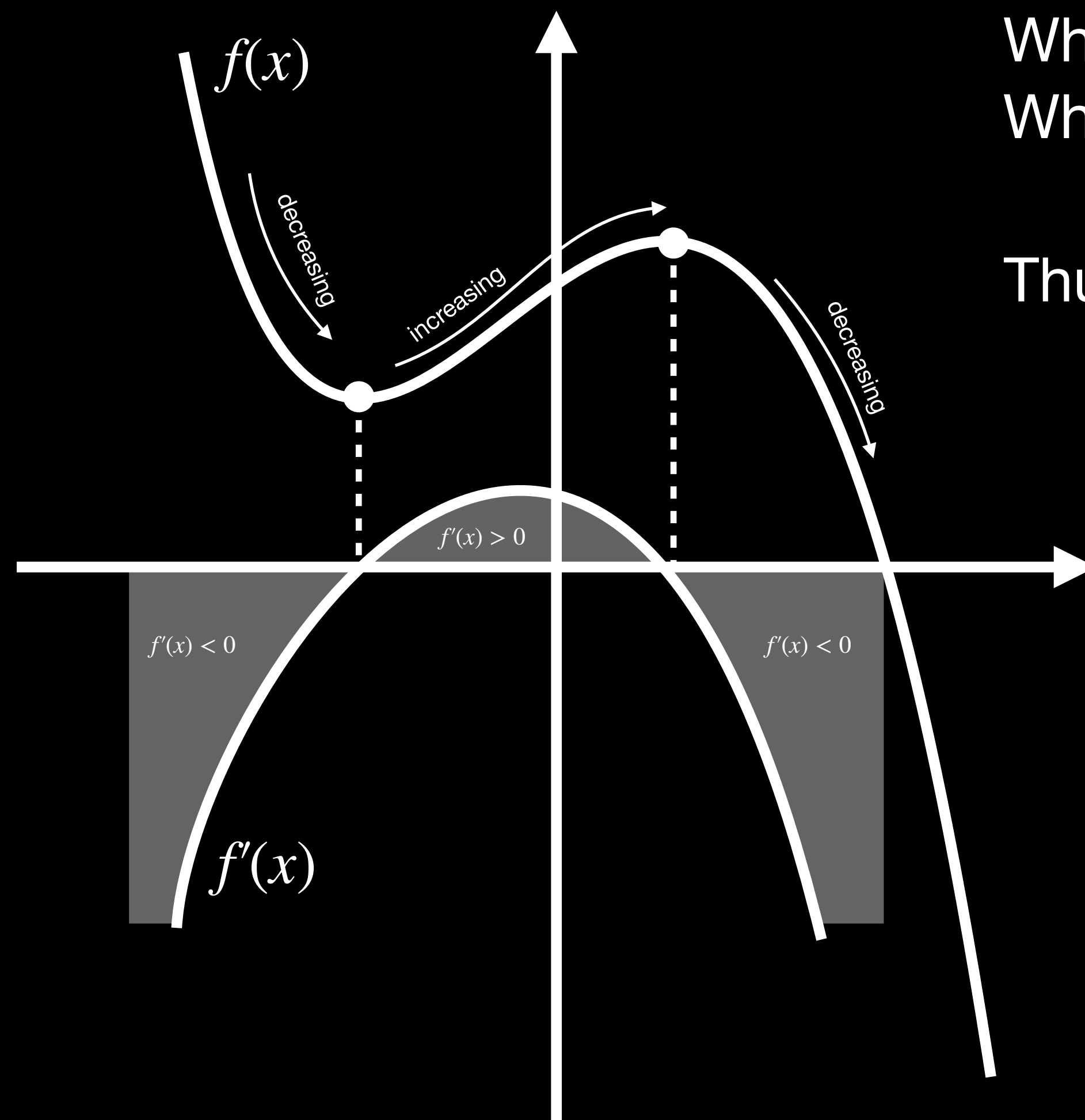
$$y - 3 = 0.5x + 1.5$$

$$y = 0.5x + 4.5$$



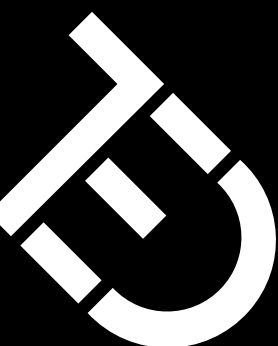
Increasing & Decreasing Functions

real



When a function $f(x)$ is increasing from a to b , $f'(a)$ to $f'(b) > 0$
When a function $f(x)$ is decreasing from a to b , $f'(a)$ to $f'(b) < 0$

Thus,
to find range of values where $f(x) \uparrow / \downarrow$
Find $f'(x)$ and let $f'(x) > 0$ or $f'(x) < 0$ respectively



Increasing & Decreasing Functions

real

Find the range of values of x such that the function

$$y = \frac{x+1}{\sqrt{2x-3}} \text{ is increasing.}$$

$$\frac{dy}{dx} = \frac{\sqrt{2x-3} - \left(\frac{x}{2\sqrt{2x-3}}\right)(x+1)}{2x-3}$$

$$= \frac{\sqrt{2x-3} - \frac{x+1}{\sqrt{2x-3}}}{2x-3}$$

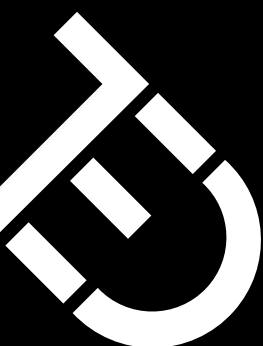
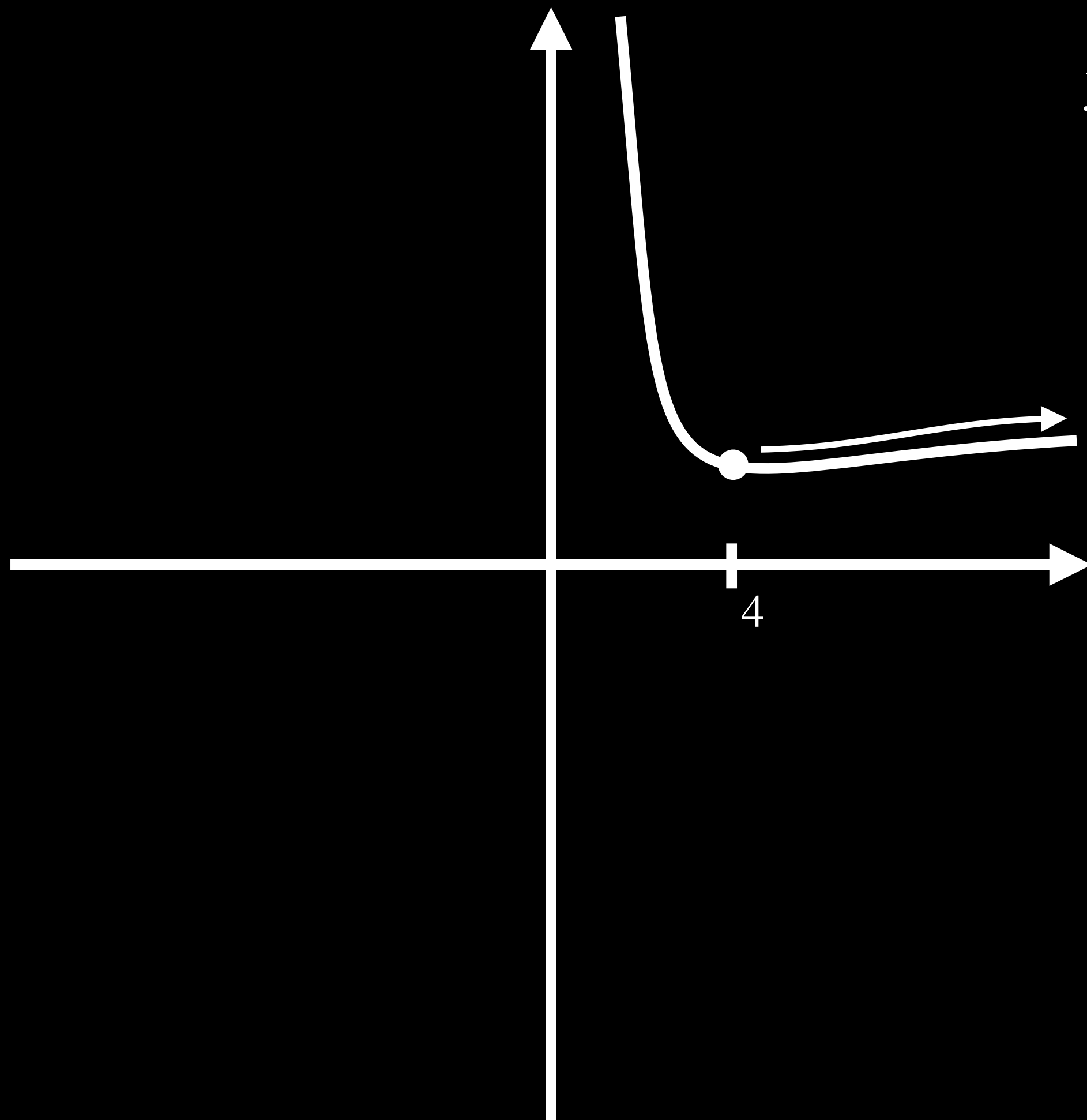
$$= \frac{\frac{2x-3}{\sqrt{2x-3}} - \frac{x+1}{\sqrt{2x-3}}}{2x-3}$$

$$= \frac{\frac{x-4}{\sqrt{2x-3}}}{2x-3}$$

$$= \frac{x-4}{(2x-3)^{1.5}}$$

When the function is increasing,

$$\begin{aligned} \frac{dy}{dx} &> 0 \\ \frac{x-4}{(2x-3)^{1.5}} &> 0 \\ x-4 &> 0 \\ x &> 4 \end{aligned}$$



Increasing & Decreasing Functions

real

Find the range of values of x such that the function $y = xe^{2x}$ is decreasing.

$$\begin{aligned}\frac{dy}{dx} &= e^{2x} + 2e^{2x}(x) \\ &= e^{2x}(1 + 2x)\end{aligned}$$

When the function is decreasing,

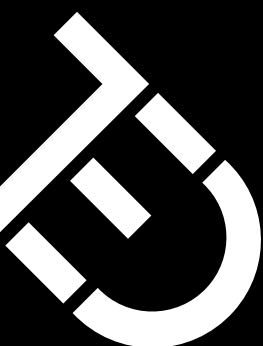
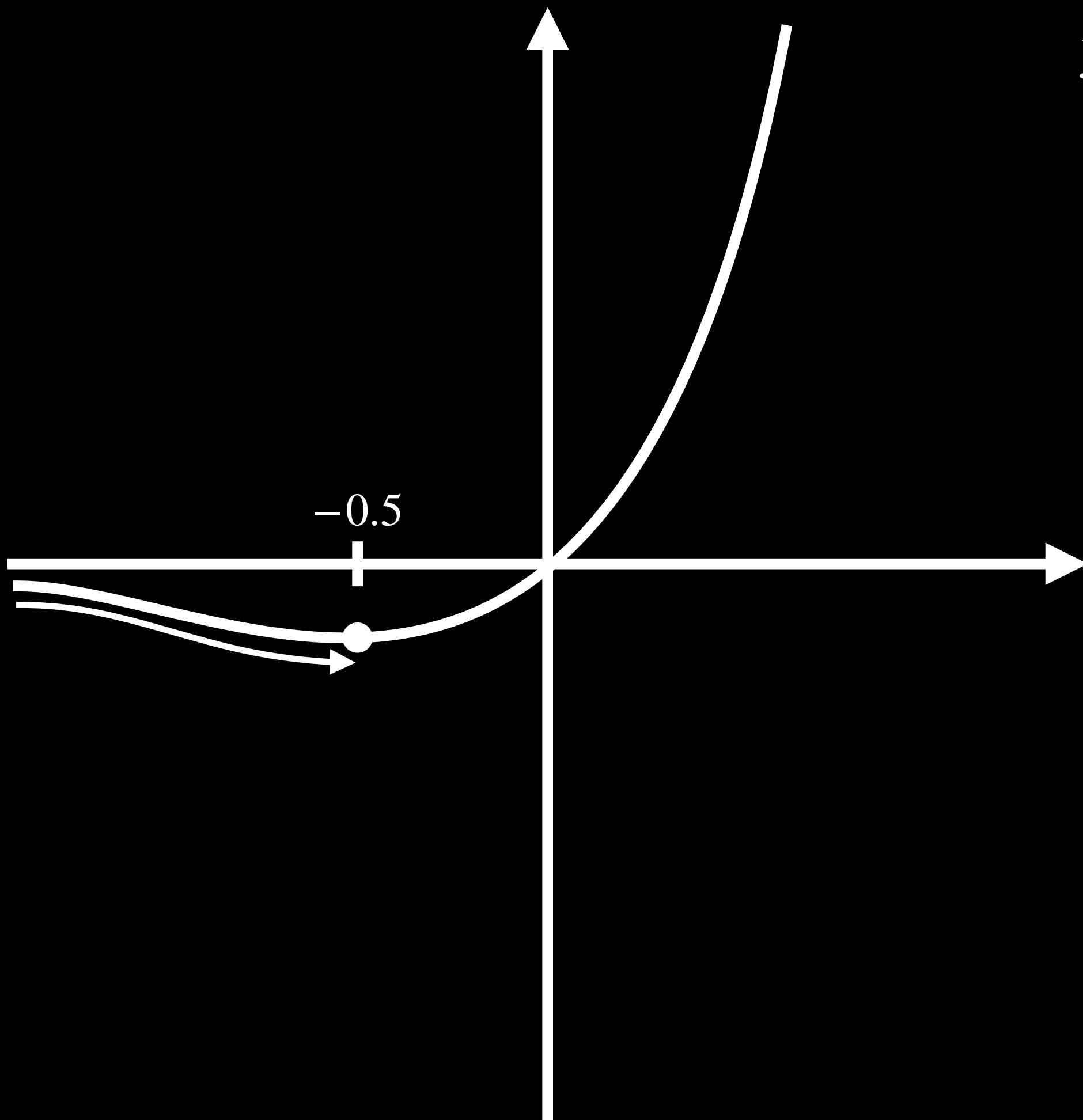
$$\begin{aligned}\frac{dy}{dx} &< 0 \\ e^{2x}(1 + 2x) &< 0\end{aligned}$$

As $e^{2x} > 0$, for $e^{2x}(1 + 2x) < 0$, $1 + 2x < 0$

$$\therefore 1 + 2x < 0$$

$$2x < -1$$

$$x < -0.5$$



Increasing & Decreasing Functions

real

Show that the function $y = \frac{3e^x}{5x - 2}$ is increasing when $x > 2$

$$\begin{aligned}\frac{dy}{dx} &= \frac{3e^x}{5x - 2} \\ &= \frac{3e^x(5x - 2) - 5(3e^x)}{(5x - 2)^2} \\ &= \frac{3e^x(5x - 2 - 5)}{(5x - 2)^2} \\ &= \frac{3e^x(5x - 7)}{(5x - 2)^2}\end{aligned}$$

As $3e^x > 0$, $(5x - 2)^2 > 0$ and $5x - 7 > 0$,

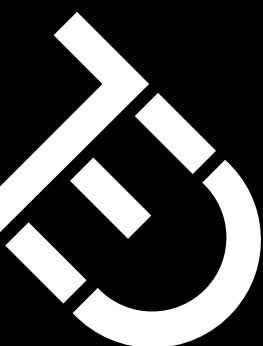
$$\frac{dy}{dx} > 0 \text{ for } x > 2$$

$\therefore y$ is increasing for $x > 2$

For all values of x , $e^x > 0$
 $3e^x > 0$
 For $x > 2$
 $5x > 10$
 $5x - 7 > 3$
 Thus, $5x - 7 > 0$

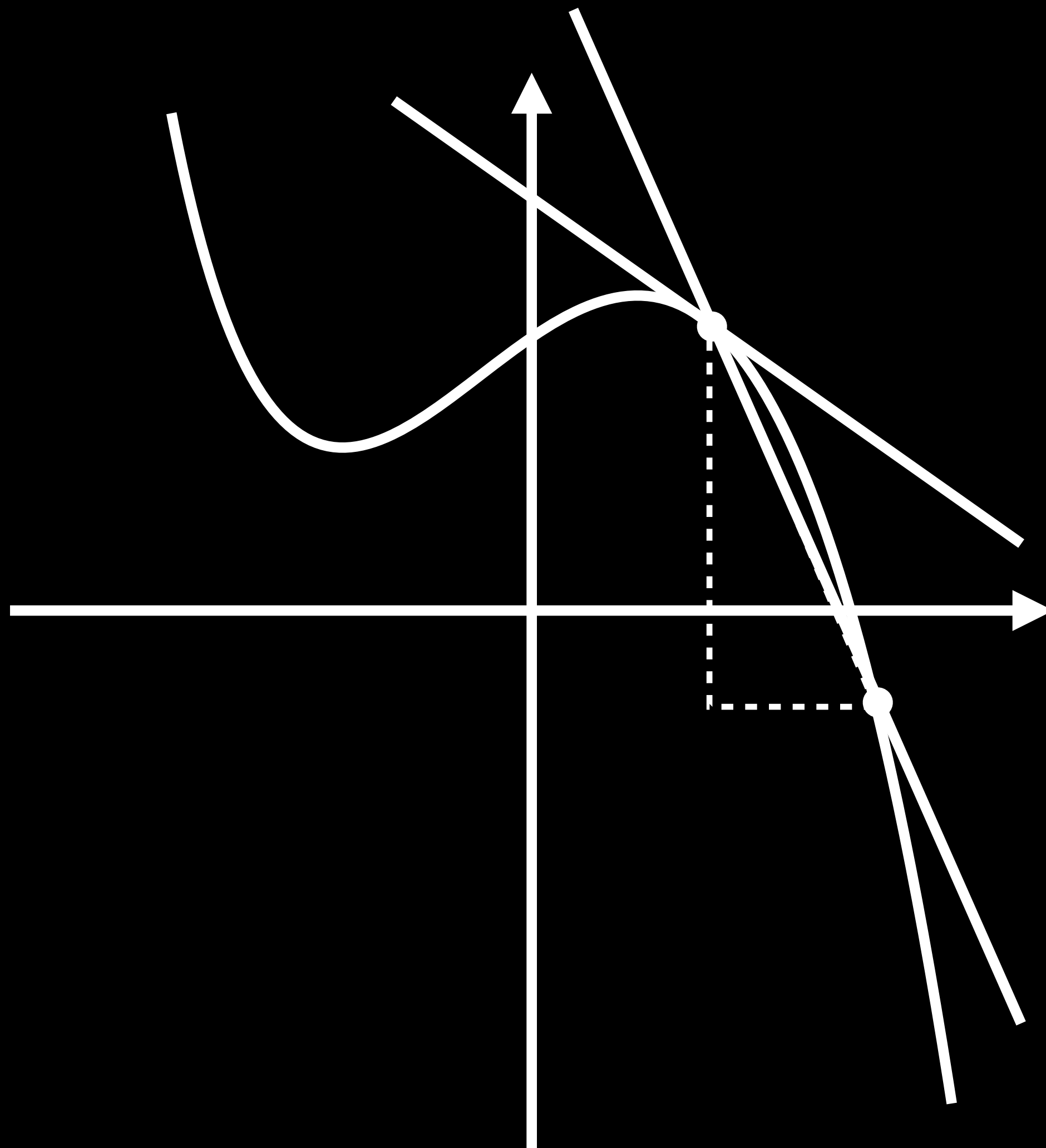
As $x \neq 0$, for all values of x ,
 $(5x - 2)^2 > 0$
 If the function is defined as $x \neq 0$
 anything squared is STRICTLY greater than 0

To determine if the derivative will be positive or negative, check if each multiplied/divided term is always positive or negative
 For example, if the derivative of a function is $\frac{(+)(-)}{(+)}$, we know that when we multiply the numerator we will get a negative value, and when we divide by the positive denominator, we will still get a negative value. Thus, the derivative will be negative.



Rate of Change

real



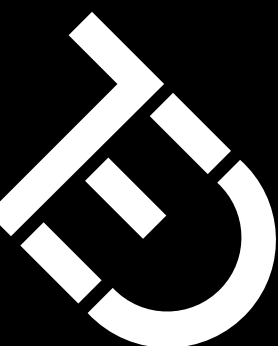
For a function of $y=f(x)$

Rate of change between points A and B = m_{AB}

If $f(x)$ is linear, $f(x)$ has a constant rate of change

else, gradient between AB is known as
average rate of change between AB

Instantaneous rate of change at $x=h = f'(h)$



Rate of Change

real

The height of a point on a ferris wheel h meters from the ground after t minutes is $h = -8 \cos \left(\frac{t}{2} \right) + 8$. Find

- a) the average rate of change of height in the first 4 minutes
- b) The instantaneous rate at which the water decreases at $t=8$

a)

$$\begin{aligned} \text{At } t = 0, \\ h &= -8 \cos \left(\frac{0}{2} \right) + 8 \\ &= 0 \end{aligned}$$

$$\begin{aligned} \text{At } t = 4, \\ h &= -8 \cos \left(\frac{4}{2} \right) + 8 \\ &= -8 \cos(2) + 8 \end{aligned}$$

$$\frac{-8 \cos(2) + 8 - 0}{4 - 0} = 2.83m/s(3.s.f)$$

$\therefore$ average rate of change in first 4 mins is $2.83m/s$

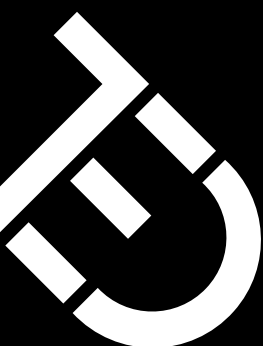
b)

$$h = -8 \cos \left(\frac{t}{2} \right) + 8$$

$$\begin{aligned} \frac{dh}{dt} &= -8(0.5) \left(-\sin \left(\frac{t}{2} \right) \right) \\ &= 4 \sin \left(\frac{t}{2} \right) \end{aligned}$$

$$\begin{aligned} \frac{dh}{dt} \Big|_{t=8} &= 4 \sin \left(\frac{8}{2} \right) \\ &= -3.02m/s(3.s.f) \end{aligned}$$

$\therefore$ instantaneous rate of decrease at $t=8$ is $-3.02m/s$



Connected Rates of Change

real

Connected rates of change happens when both x and y changes with another variable, like time
Usually, two functions are given: function of y against x and function of x against time
Thus, to find the rate of change of y against time:

$$\frac{dy}{dt} = \frac{dy}{dx} \times \frac{dx}{dt}$$

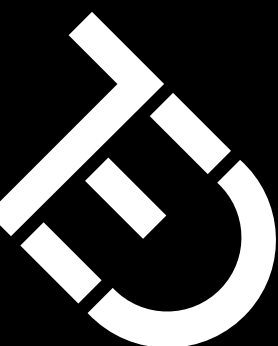
change of y against time

change of y against x

change of x against time

$$f(g(t)) = f'(g(t)) \times g'(t)$$

where $y=f(x)$ and $x=g(t)$

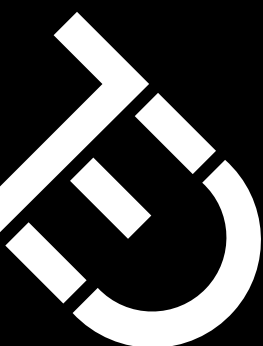


Connected Rates of Change

Real

To solve connected rates of change problems:

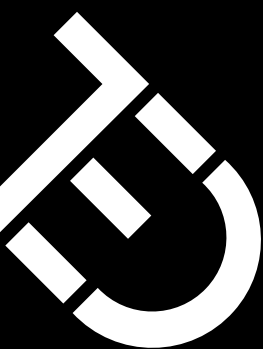
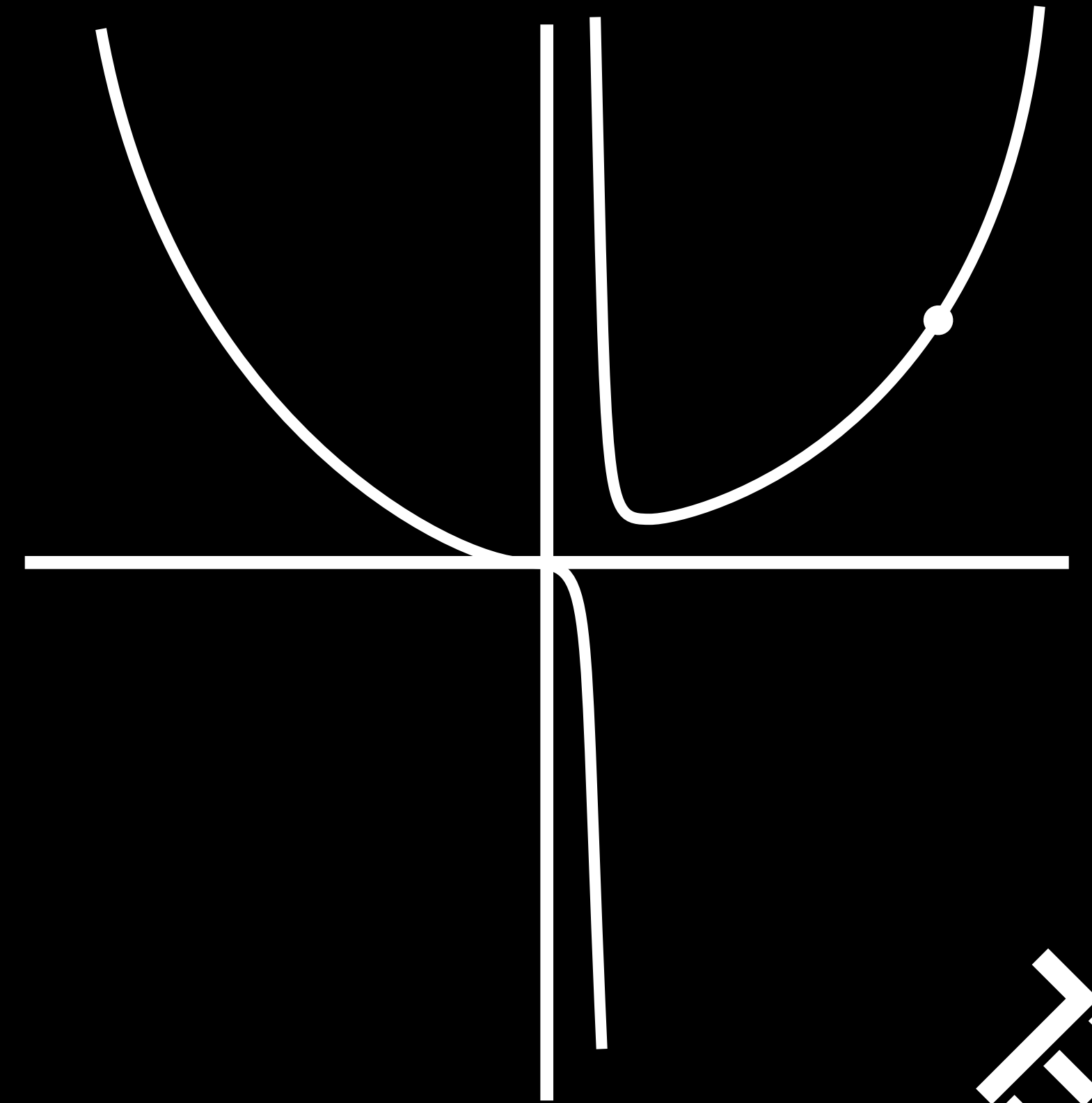
1. Find the variables used (one input, one output, one time) and the equation used to link the input and output together. If the equations has more than one input value, express in terms of the given input value
2. Differentiate $d(\text{output})/d(\text{input})$
3. Identify the rates of changes given and the rate of change you are solving for. Form the rate of change equation $\left[\frac{dy}{dt} = \frac{dy}{dx} \times \frac{dx}{dt} \right]$
4. Substitute in given input value and solve



Connected Rates of Change

real

A point travels along the curve $y = \frac{x^3}{3x - 1}$ where the x coordinate of the point is increasing at 2 units per second. Find the rate of change of the y -coordinate of the point at $x=4$.



Connected Rates of Change

real

equation
 x^3

$$\frac{dx}{dt} = 2 \text{ units/s}$$

A point travels along the curve $y = \frac{x^3}{3x - 1}$ where the x coordinate of the point is increasing at 2 units per second. Find the rate of change of the y-coordinate of the point at x=4.

Solve for $\frac{dy}{dt}$

x is input

$$y = \frac{x^3}{3x - 1}$$

Find and differentiate eqn

$$\frac{dy}{dx} = \frac{3x^2(3x - 1) - 3(x^3)}{(3x - 1)^2}$$

$$\frac{dy}{dx} = \frac{3x^2(3x - 1 - x)}{(3x - 1)^2}$$

$$\frac{dy}{dx} = \frac{3x^2(2x - 1)}{(3x - 1)^2}$$

$$\frac{dy}{dx} = \frac{3x^2(2x - 1)}{(3x - 1)^2}$$

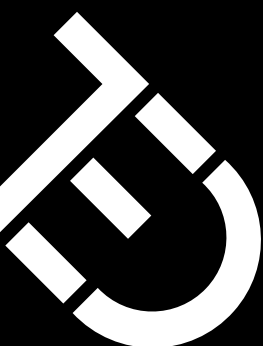
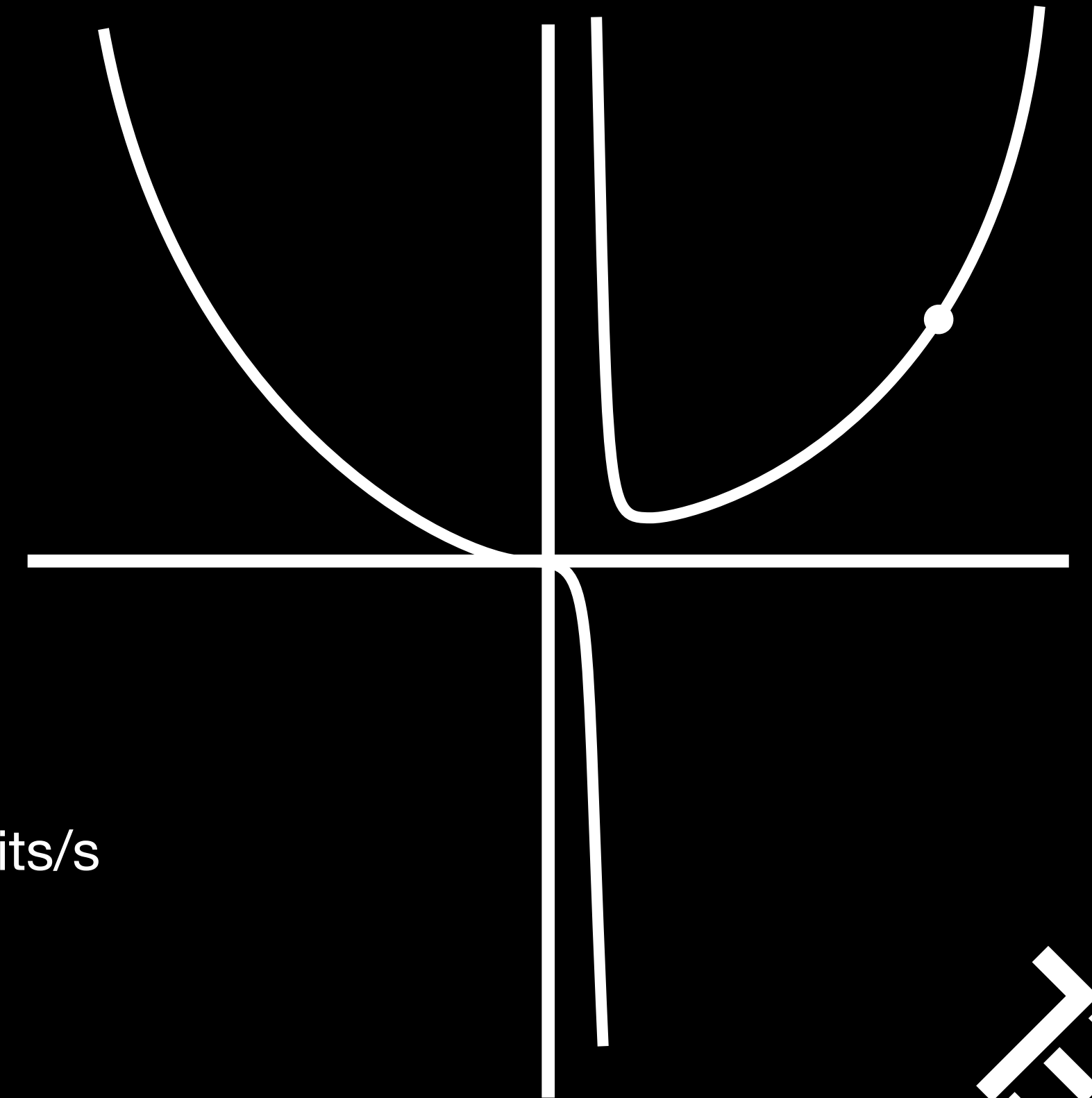
$$\frac{dy}{dt} = \frac{dy}{dx} \times \frac{dx}{dt} \quad \text{Form rate of change eqn}$$

$$\frac{dy}{dt} = \frac{3x^2(2x - 1)}{(3x - 1)^2} \times 2$$

$$\frac{dy}{dt} \bigg|_{x=4} = \frac{3(4)^2(2(4) - 1)}{(3(4) - 1)^2} \times 2 \quad \text{sub in input}$$

$$\frac{dy}{dt} = 5.55 \text{ units/s (3.s.f.)}$$

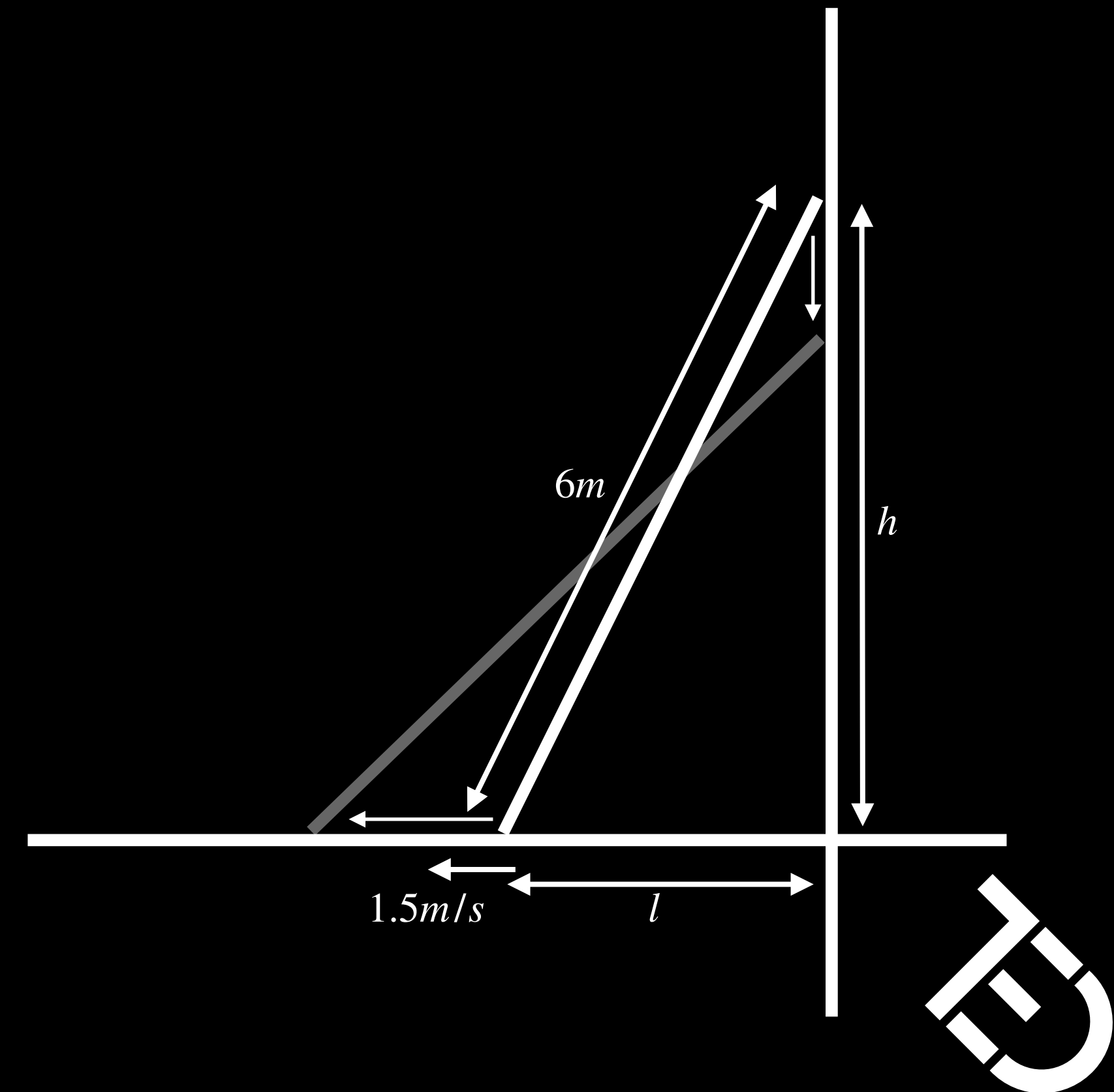
Thus, rate of change of y coord is 5.55 units/s



Connected Rates of Change

real

A ladder of 6m length slides against a vertical wall such that the horizontal length of the ladder l is increasing at 1.5m/s at $h=2m$. Find the rate of decrease of the height of the ladder h when $h = 2m$



Connected Rates of Change

real

A ladder of 6m length slides against a vertical wall such that the horizontal length of the ladder l is increasing at 1.5m/s at $h=2$ m. Find the rate of decrease of the height of the ladder h when $h = 2$ m

$$\frac{dl}{dt} = 1.5 \text{ m/s}$$

solve for $\frac{dh}{dt}$

Form and differentiate eqn

$$l^2 + h^2 = 6^2$$

$$l = \sqrt{36 - h^2}$$

$$\frac{dl}{dh} = (-2h)(0.5)(36 - h^2)^{-0.5}$$

$$= -h(36 - h^2)^{-0.5}$$

$$= -\frac{h}{\sqrt{36 - h^2}}$$

$$\frac{dl}{dt} = \frac{dl}{dh} \times \frac{dh}{dt}$$

$$1.5 = -\frac{h}{\sqrt{36 - h^2}} \times \frac{dh}{dt}$$

Form rate of change eqn

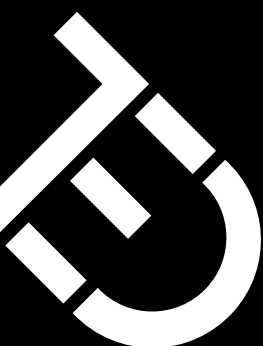
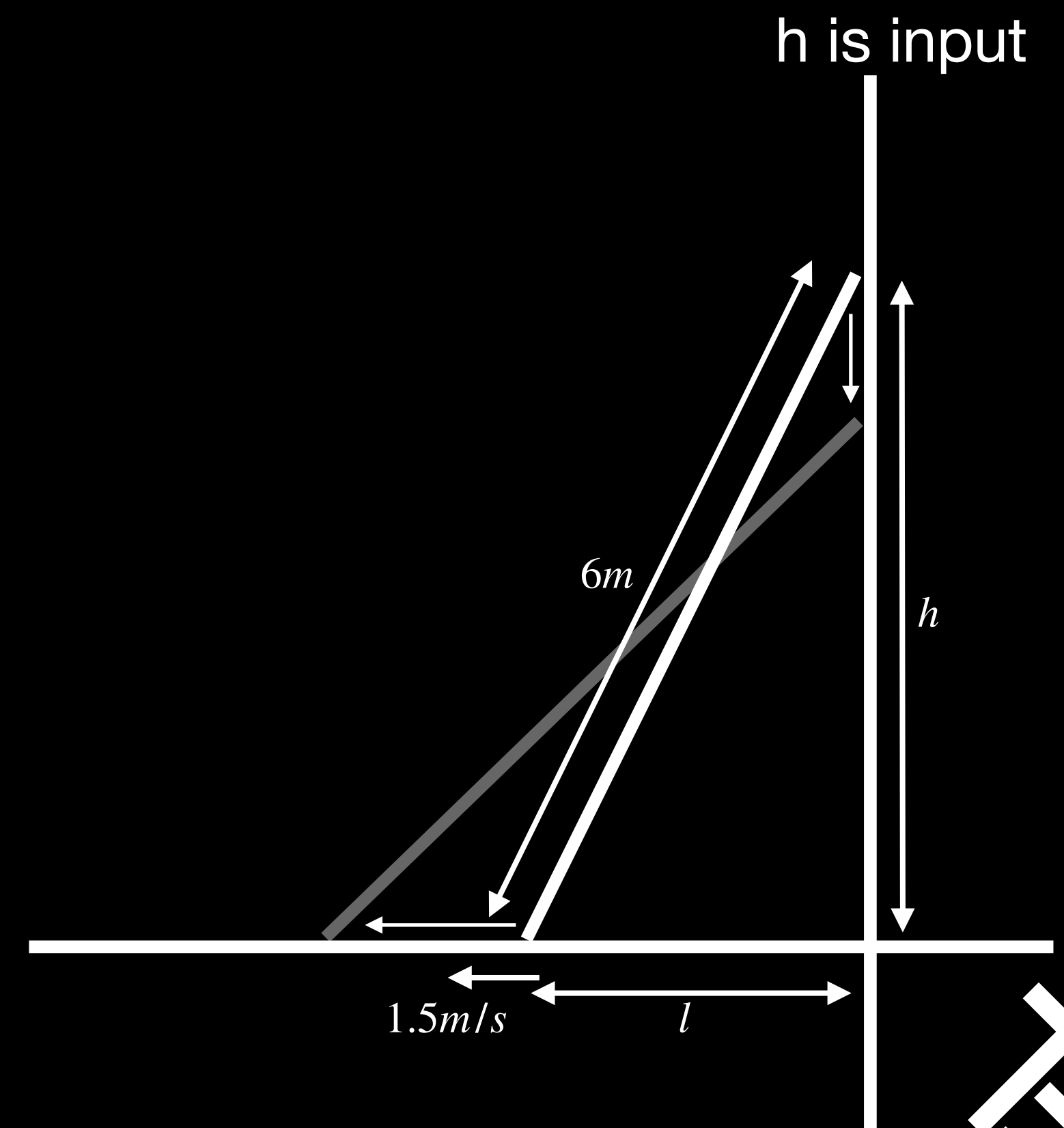
$$\frac{1.5}{-\frac{h}{\sqrt{36 - h^2}}} = \frac{dh}{dt}$$

$$\frac{dh}{dt} \bigg|_{h=2} = \frac{1.5}{-\frac{2}{\sqrt{36 - 2^2}}}$$

sub in input

$$\frac{dh}{dt} = -4.24 \text{ m/s}$$

Thus, rate of decrease of h is 4.24 units/s
when referring to rate of decrease, change sign

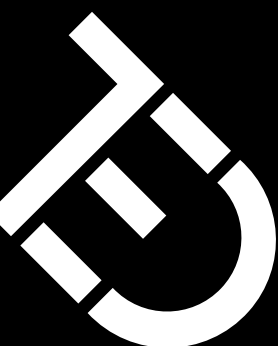
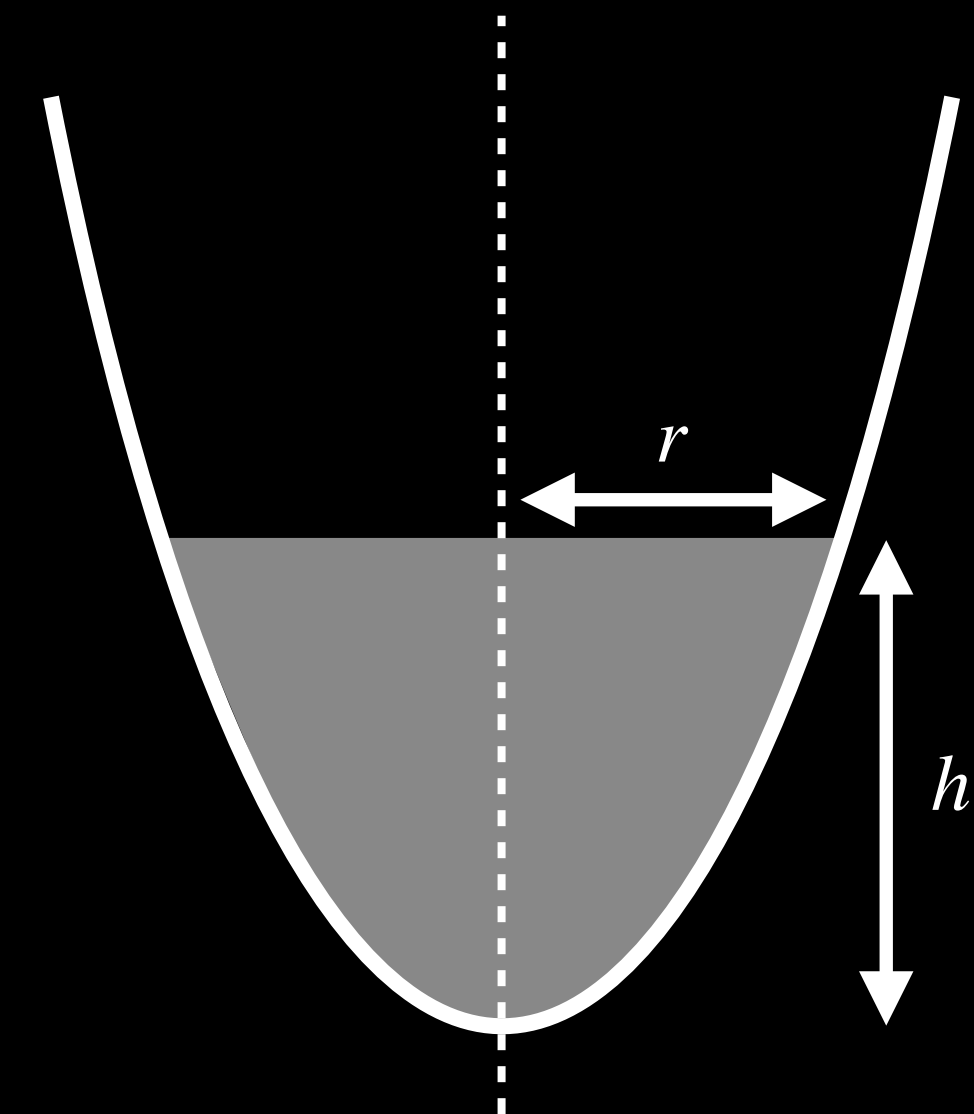


Connected Rates of Change

real

The diagram shows the cross section of a container whose shape follows $y = x^2$. The container is then filled with water.

The cross sectional area of the water is given to be $A = \frac{1}{3} (3h^2 - 2r^3)$ where h is the height of the water and r is the radius. Given that the water level increases at 2cm/s, find the rate of change of the cross-sectional area at $h = 9\text{cm}$



Connected Rates of Change

real

The diagram shows the cross section of a container whose shape follows $y = x^2$. The container is then filled with water.

The cross sectional area of the water is given to be $A = \frac{1}{3} (3h^2 - 2r^3)$ where h is the height of the water and r is the radius. Given that the water level increases at 2cm/s, find the rate of change of the cross-sectional area at $h = 9\text{cm}$

$$\frac{dh}{dt} = 2\text{cm/s}$$

solve for $\frac{dA}{dt}$
 h is input

As container follows curve $y = x^2$:

$$h = r^2$$

$$r = \sqrt{h}$$

Form eqn into terms of one variable

Sub $r = \sqrt{h}$ into $A = \frac{1}{3} (3h^2 - 2r^3)$:

$$A = \frac{1}{3} (3h^2 - 2\sqrt{h}^3)$$

$$= \frac{1}{3} (3h^2 - 2h^{1.5})$$

$$\begin{aligned} \frac{dA}{dh} &= \frac{1}{3} (6h - 3h^{0.5}) \text{ differentiate eqn} \\ &= 2h - \sqrt{h} \end{aligned}$$

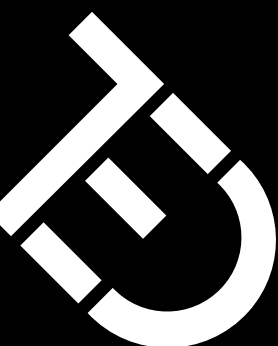
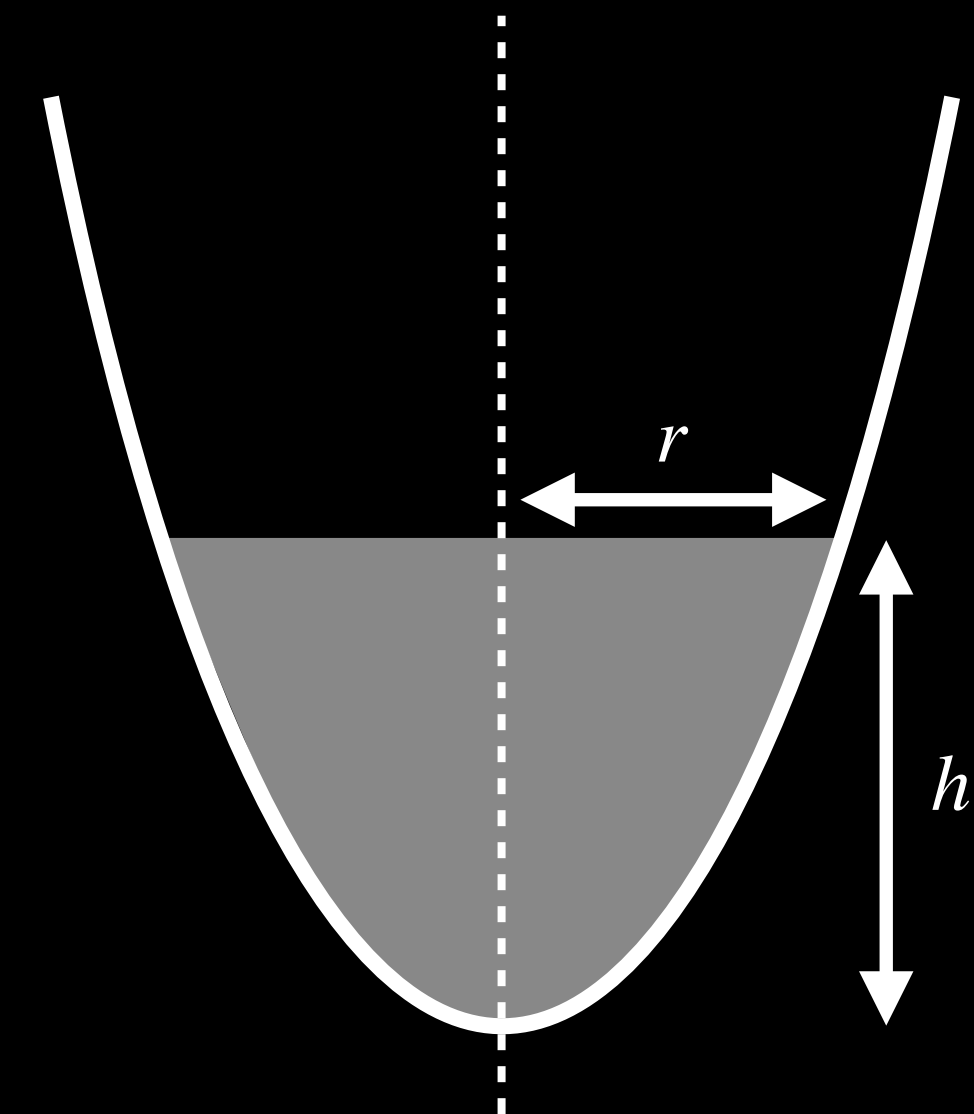
$$\frac{dA}{dt} = \frac{dA}{dh} \times \frac{dh}{dt} \text{ Form rate of change eqn}$$

$$\frac{dA}{dt} = (2h - \sqrt{h}) \times 2$$

$$\frac{dA}{dt} /_{h=9} = (2(9) - \sqrt{9}) \times 2 \text{ sub in input}$$

$$\frac{dA}{dt} = 30\text{cm}^2/\text{s}$$

Thus, rate of change of cross sectional area is 30 units/s



Types of Stationary Points

Yum yum

Stationary points are points such that their derivatives = 0

Points A, B and C lie on $f(x)$ such that their derivatives=0

A: Minimum point

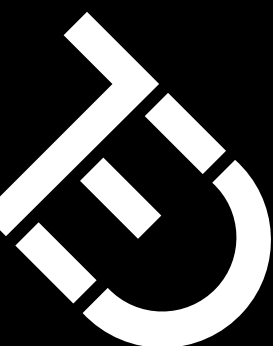
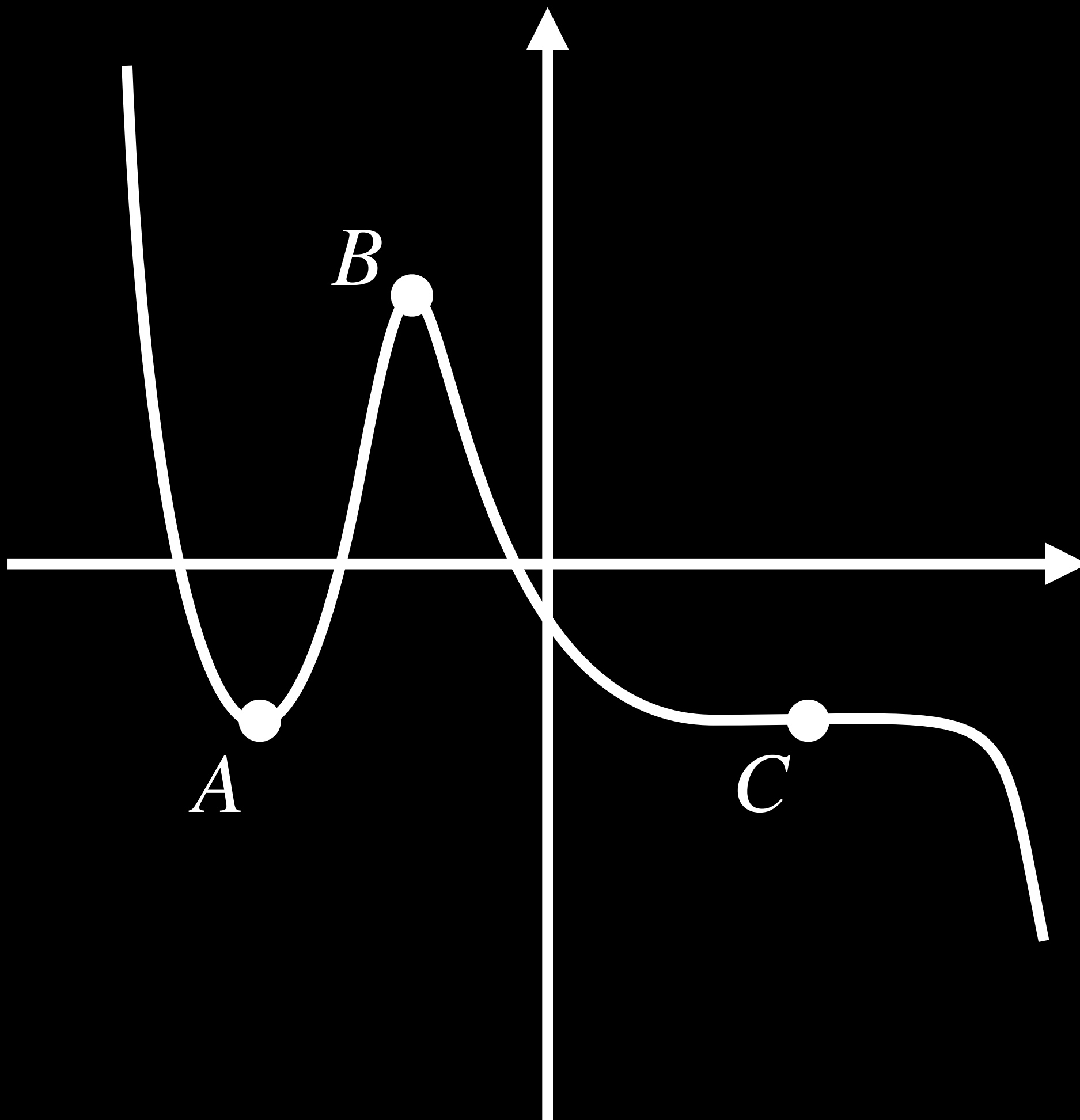
$f'(x)$ changes sign from negative to positive

B: Maximum point

$f'(x)$ changes sign from positive to negative

C: Point of inflexion

$f'(x)$ stays at positive sign/negative sign



Types of Stationary Points

Yum yum

Show that $y = \ln(5x + 3)$ has no stationary points.

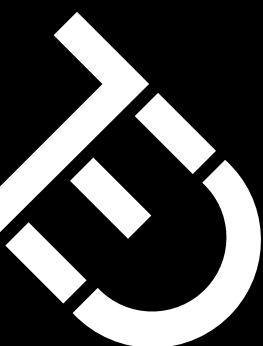
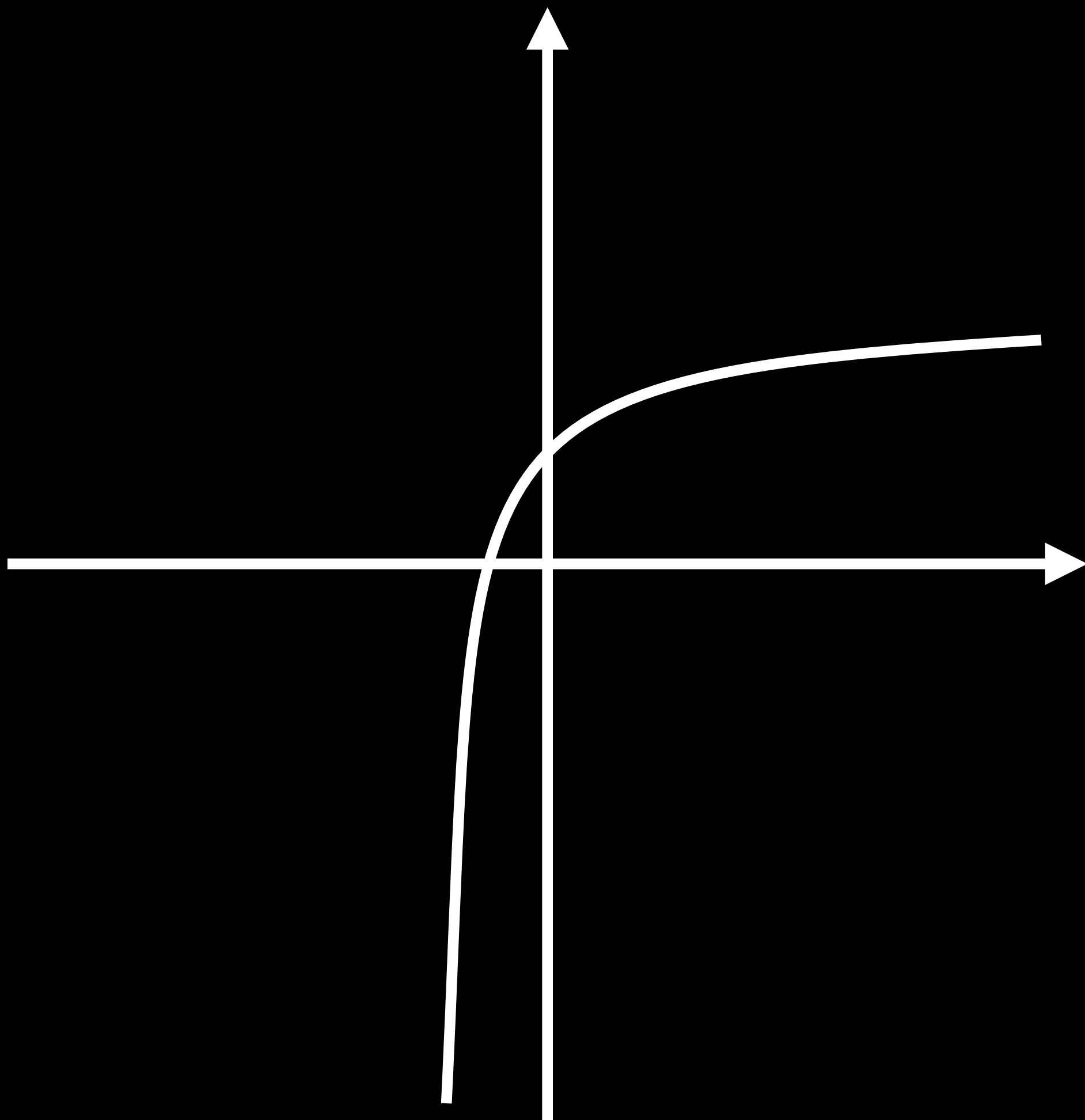
$$\frac{dy}{dx} = \frac{5}{5x + 3}$$

$$\text{let } \frac{dy}{dx} = 0$$

$$\frac{5}{5x + 3} = 0$$

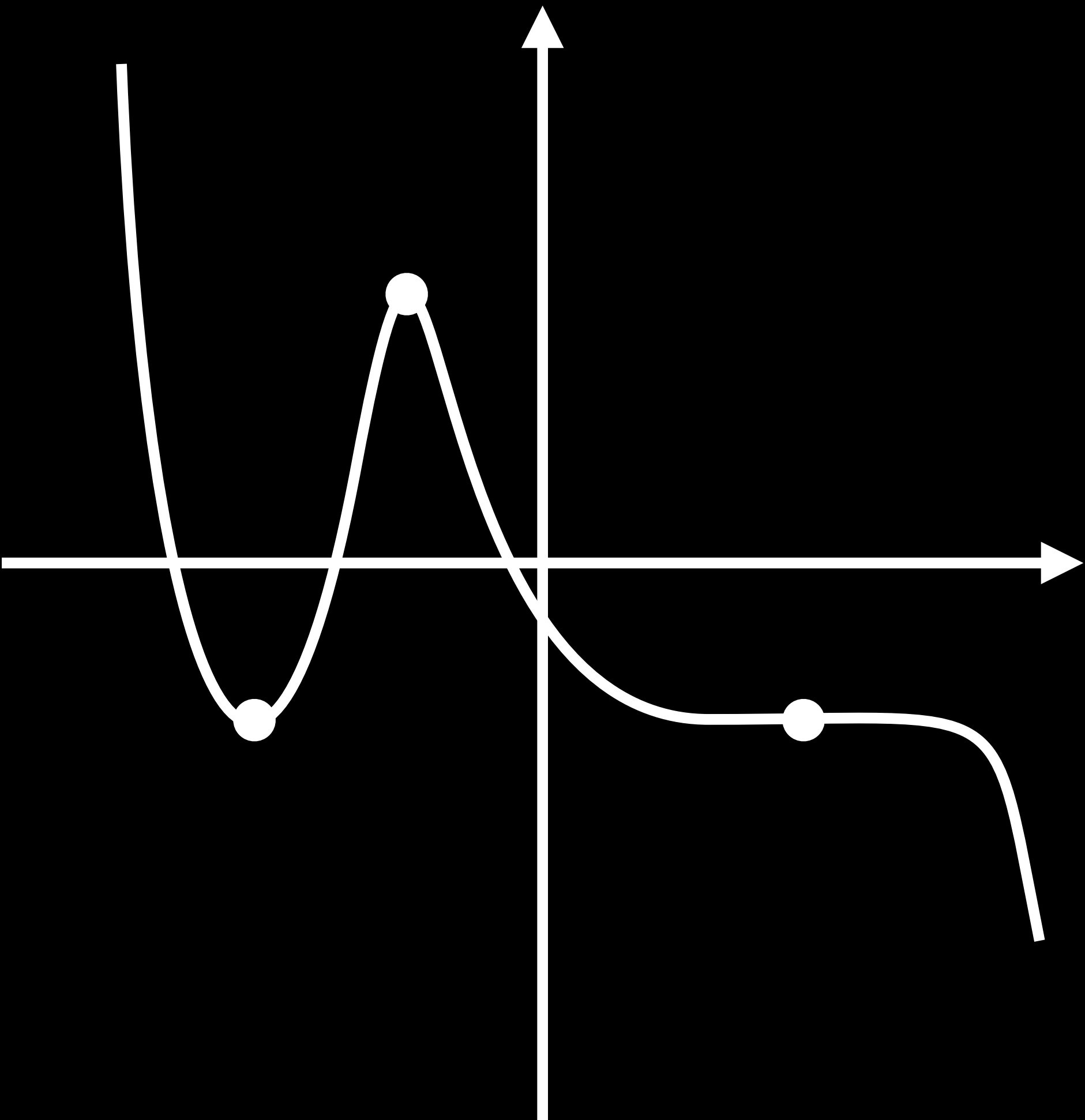
$$5 = 0 \text{ (not possible)}$$

$\therefore y$ has no stationary points



First Derivative Test

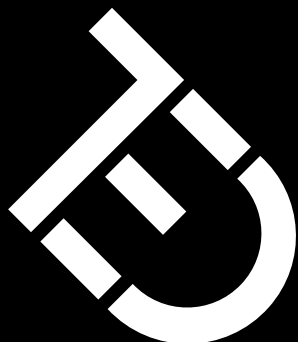
Yum yum



To find the nature of a stationary point of $f(x)$:
Find when $f'(x) = 0$ to find x value of stationary point (which we let it be a)
 a^- will be $x - h$ and a^+ will be $x + h$, where h is a small value like 0.1
Take $f'(a^-)$ and $f'(a^+)$ and write the signs of the results in the table shown. Draw the slope of the tangent at a^- , a and a^+ as well.

where $f'(a) = 0$

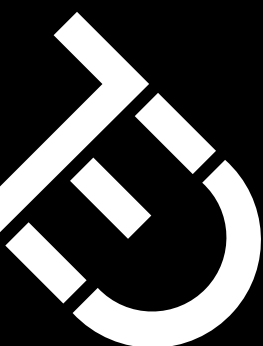
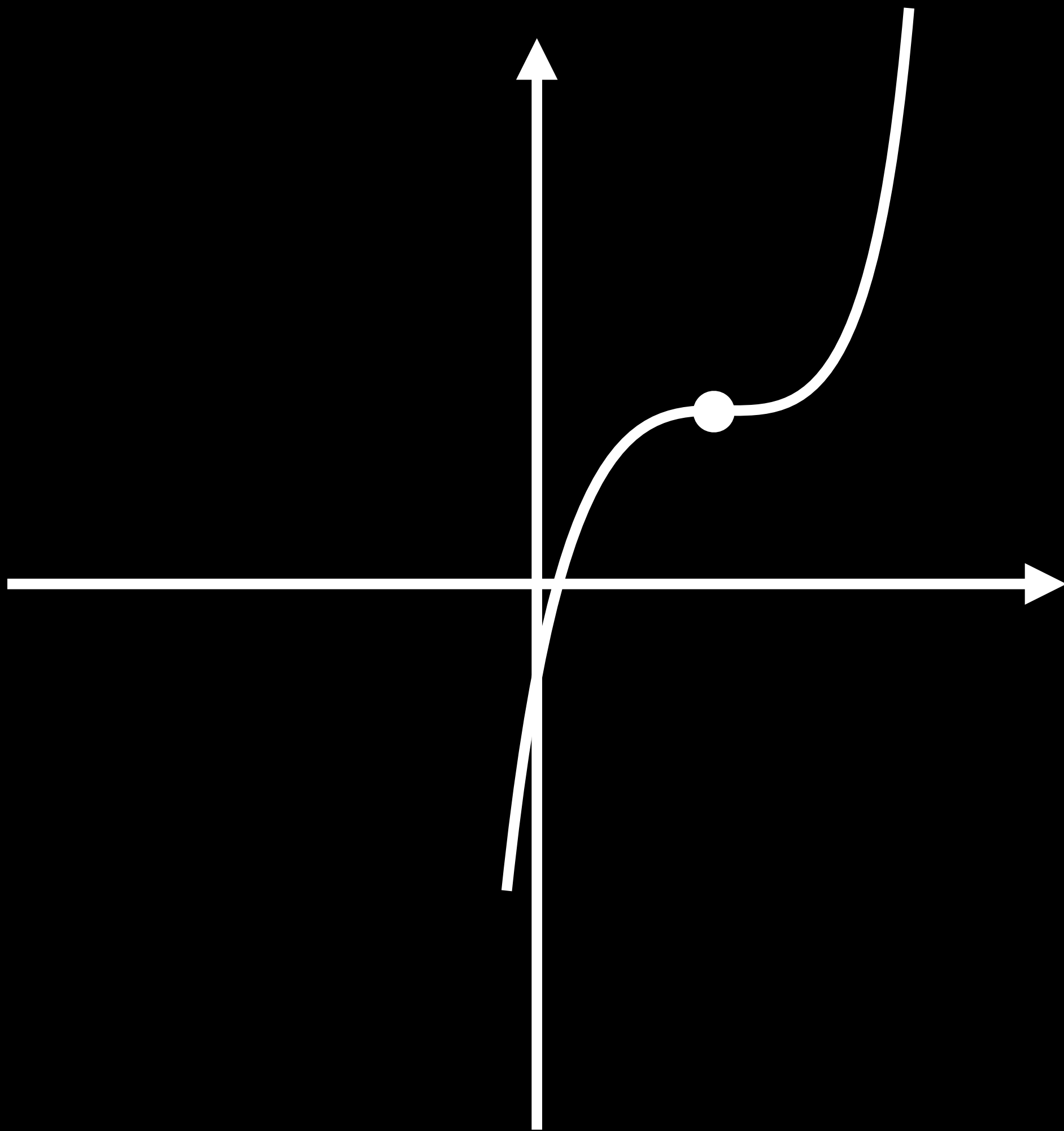
x	a^-	a	a^+
Sign of $\frac{dy}{dx}$	$\pm$	0	$\pm$
Slope		—	



First Derivative Test

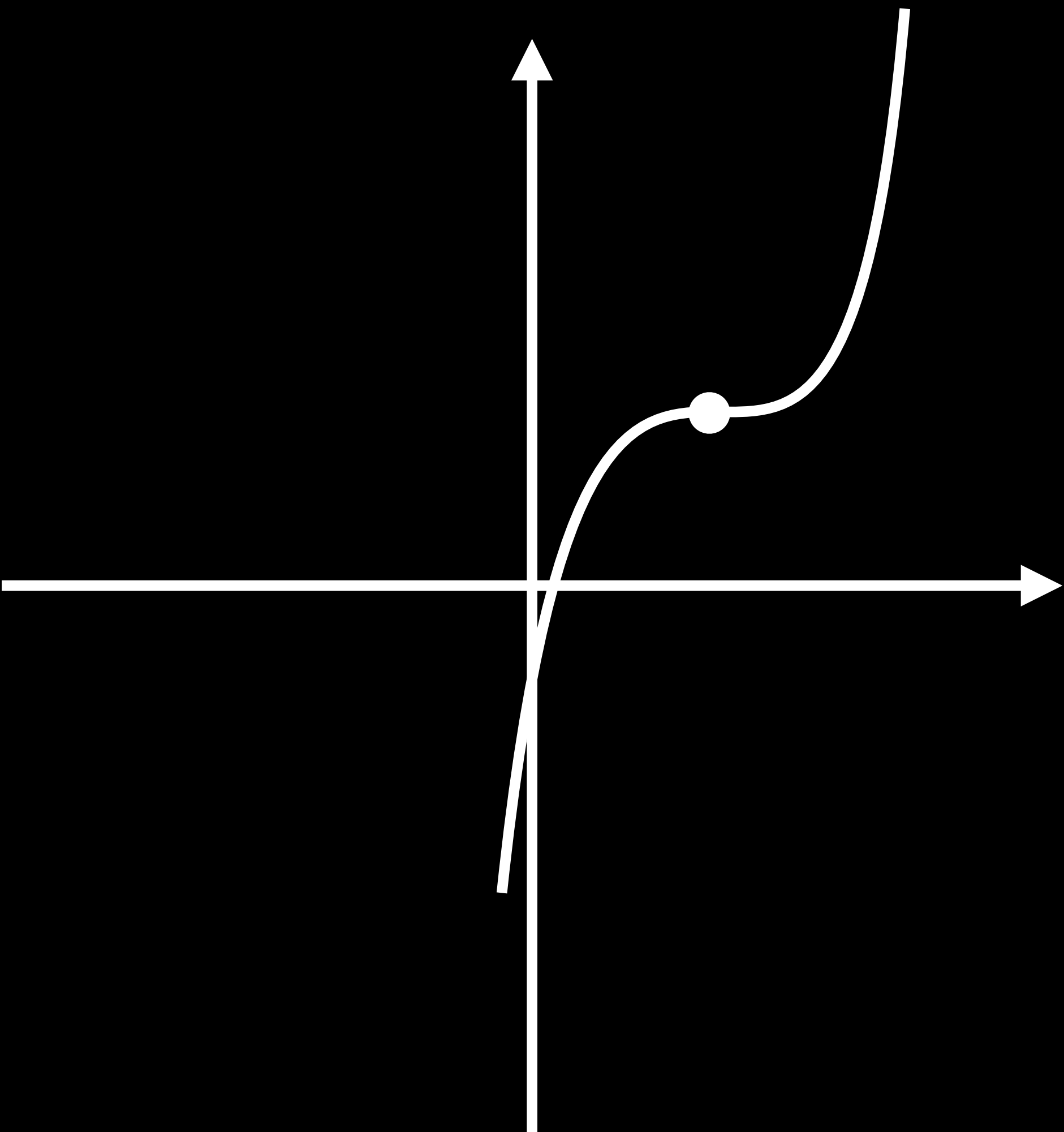
Yum yum

By considering the derivative of $f(x) = 0.5(x - 2)^3 + 3$, find the nature of the stationary point of the function



First Derivative Test

Yum yum



By considering the derivative of $f(x) = 0.5(x - 2)^3 + 3$, find the nature of the stationary point of the function

$$f'(x) = 0.5 \left(3(x - 2)^2(1) \right)$$

$$f'(x) = 1.5(x - 2)^2$$

$$1.5(x - 2)^2 = 0$$

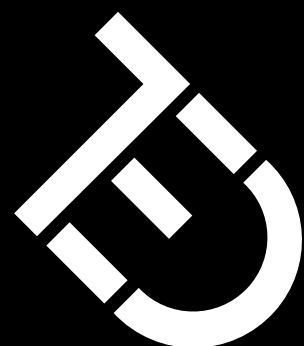
$$(x - 2)^2 = 0$$

$$x - 2 = 0$$

$$x = 2$$

x	2^-	2	2^+
Sign of $\frac{dy}{dx}$	$+$	0	$+$
Slope			

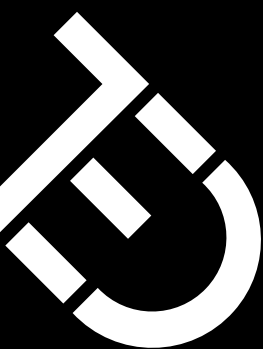
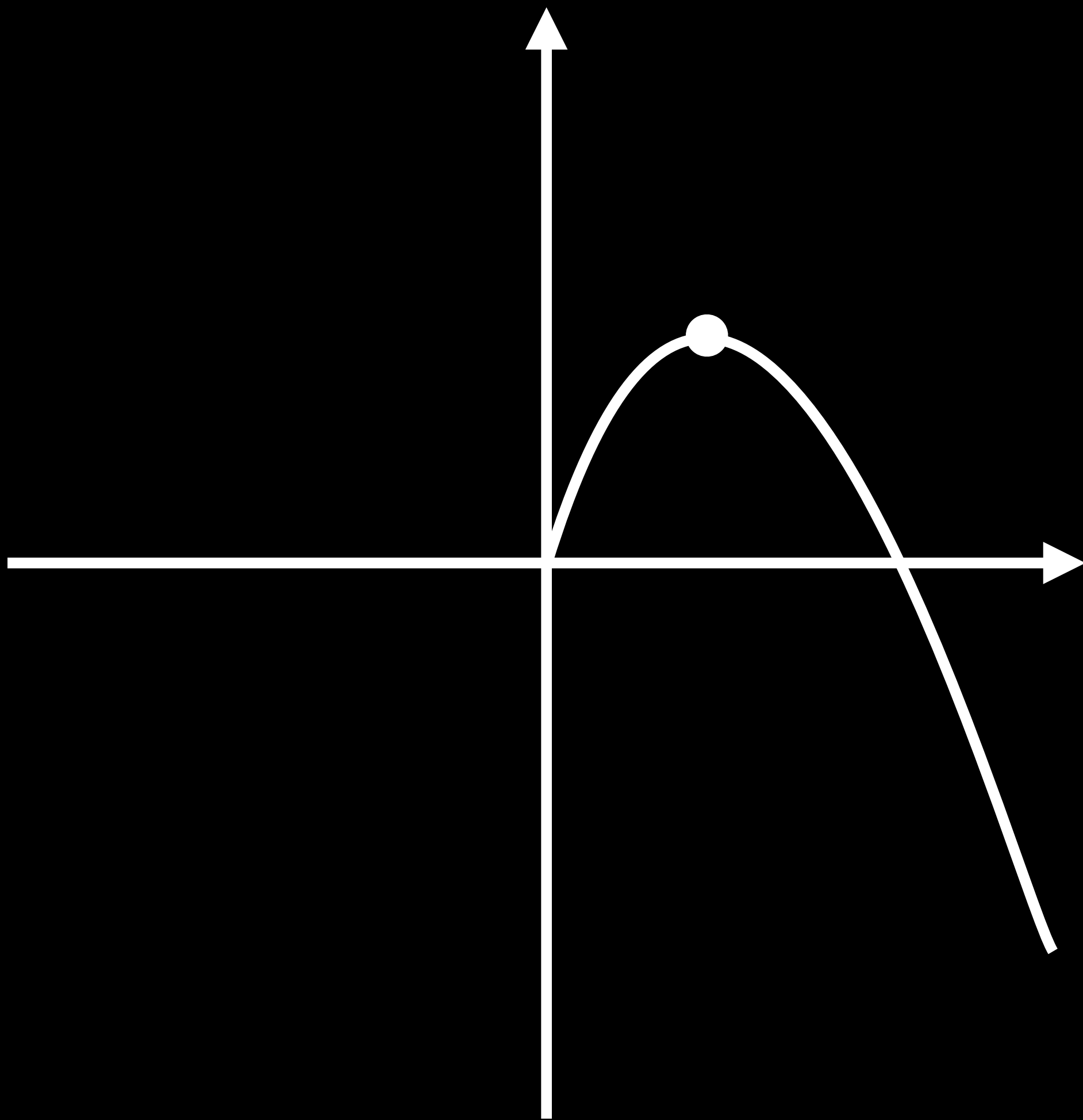
Thus, its stationary point is a point of inflexion



First Derivative Test

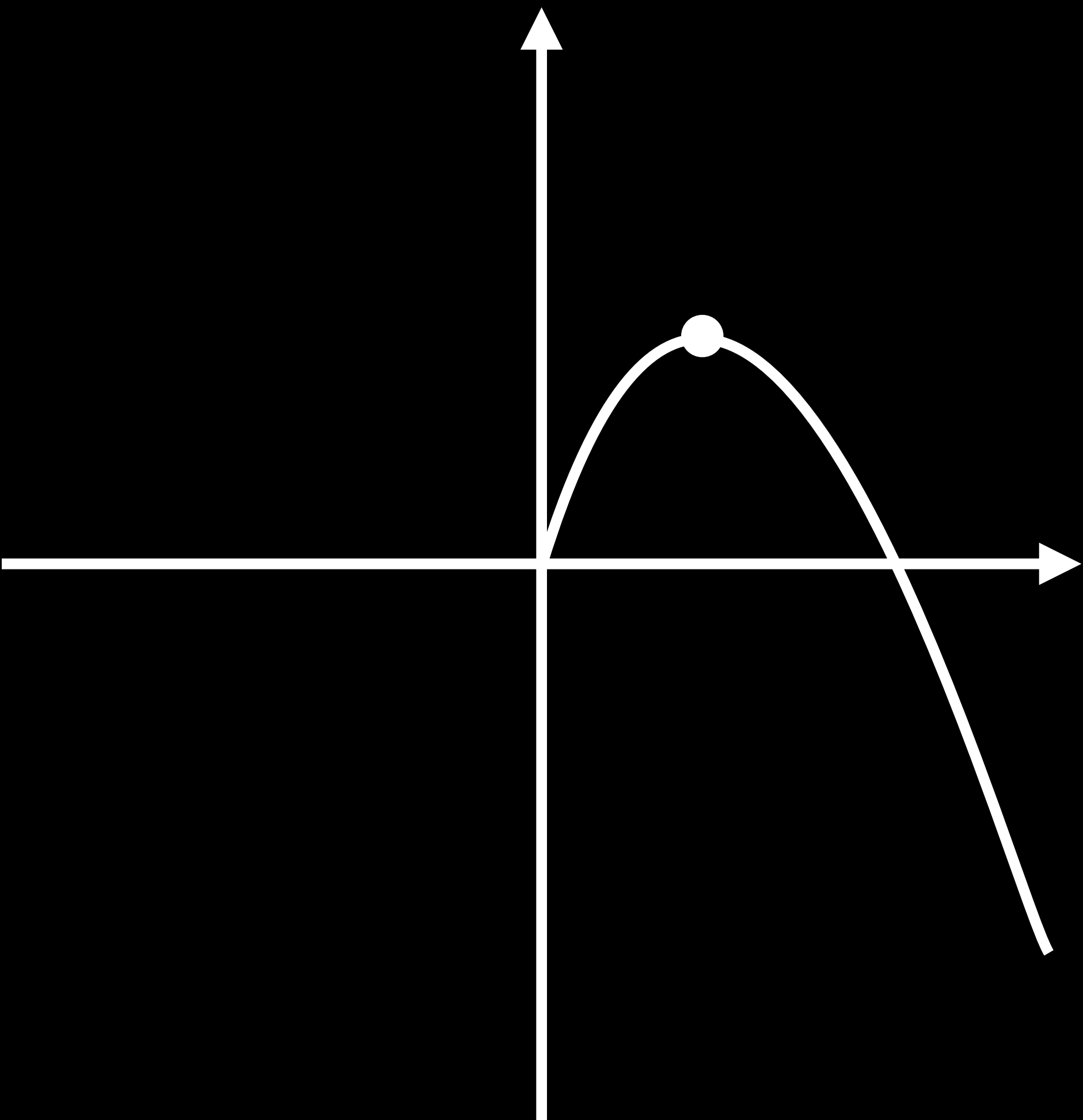
Yum yum

By considering the derivative of $f(x) = \sqrt{x}(5 - x)$, find the nature of the stationary point of the function



First Derivative Test

Yum yum

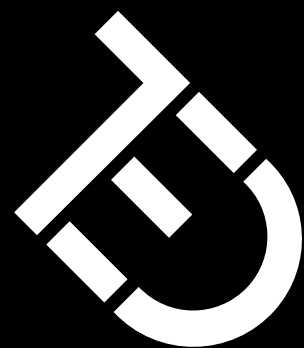


By considering the derivative of $f(x) = \sqrt{x}(5-x)$, find the nature of the stationary point of the function

$$\begin{aligned} f'(x) &= 0.5x^{(-0.5)}(5-x) + (-1)(\sqrt{x}) \\ f'(x) &= \frac{(2.5-0.5x)}{\sqrt{x}} - \sqrt{x} \\ f'(x) &= \frac{(2.5-0.5x)}{\sqrt{x}} - \sqrt{x} \\ f'(x) &= \frac{(2.5-0.5x)}{\sqrt{x}} - \frac{x}{\sqrt{x}} \\ f'(x) &= \frac{2.5-1.5x}{\sqrt{x}} \end{aligned} \qquad \begin{aligned} \frac{2.5-1.5x}{\sqrt{x}} &= 0 \\ 2.5-1.5x &= 0 \\ x &= 1.67(3.s.f) \end{aligned}$$

x	1.67^-	1.67	1.67^+
Sign of $\frac{dy}{dx}$	$+$	0	$-$
Slope			

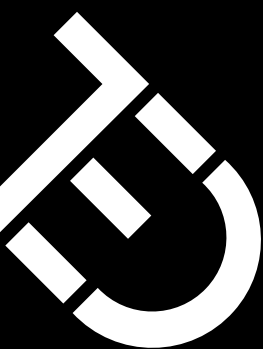
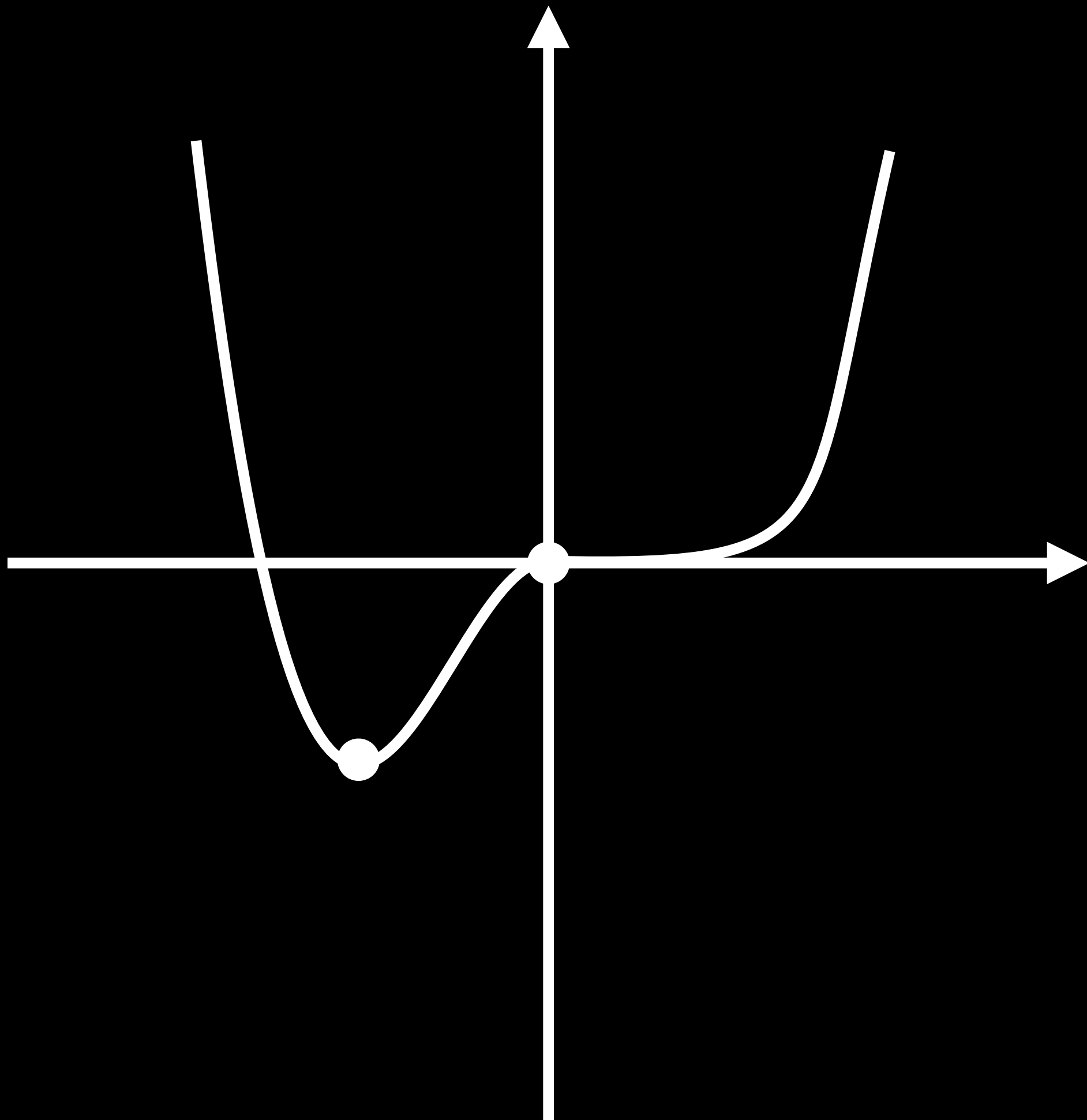
Thus, its stationary point is a maximum point



First Derivative Test

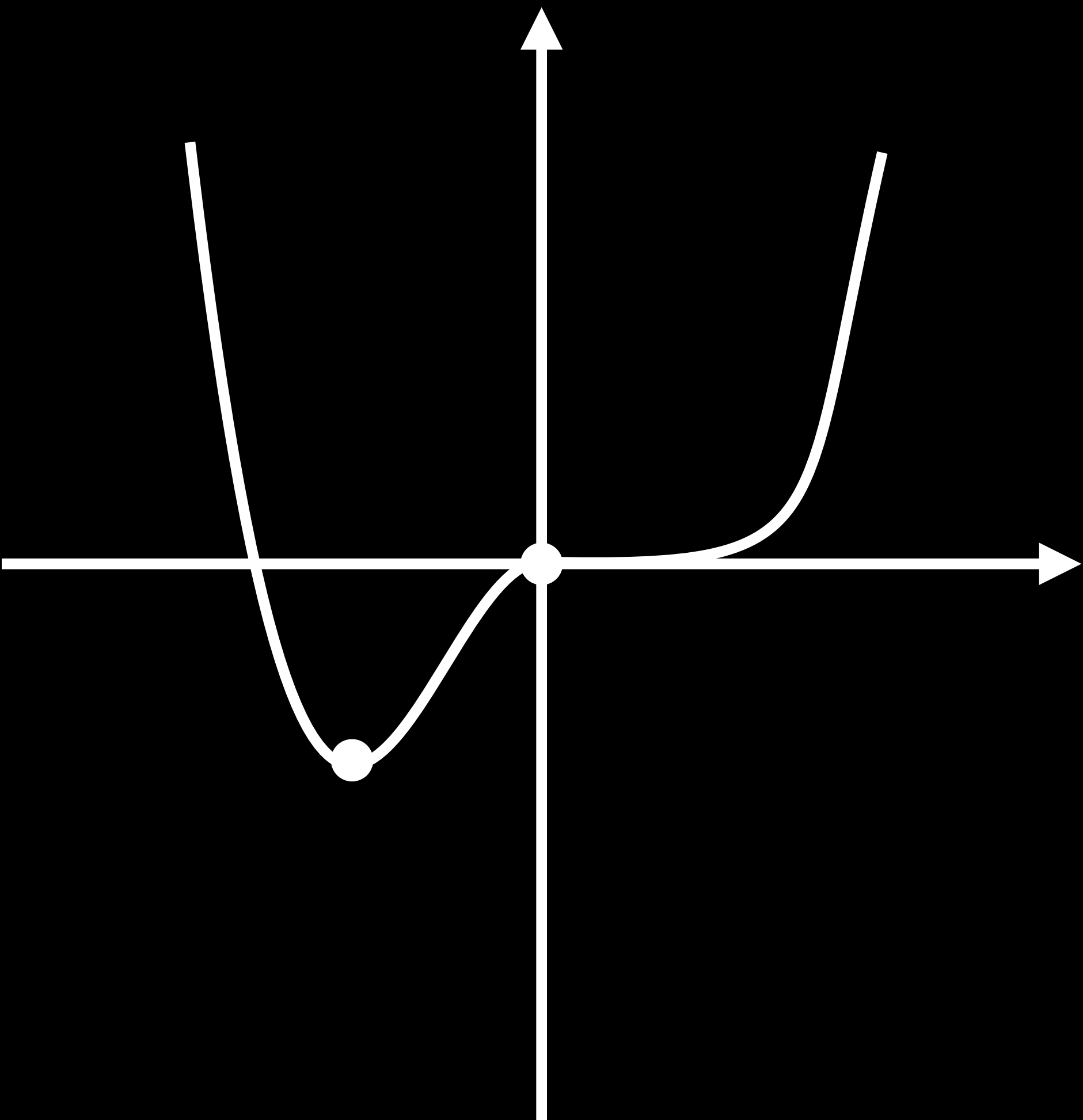
Yum yum

Given the function $f(x) = x^4 + 2x^3$, find its stationary point(s) and its nature.



First Derivative Test

Yum yum



Given the function $f(x) = x^4 + 2x^3$, find its stationary point(s) and its nature.

$$f'(x) = 4x^3 + 2(3x^2)$$

$$f'(x) = 4x^3 + 6x^2$$

$$f'(x) = x^2(4x + 6)$$

$$x^2(4x + 6) = 0$$

$$x^2 = 0, (4x + 6) = 0$$

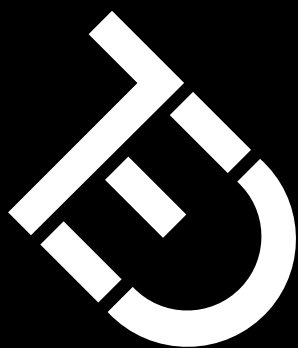
$$x = 0, -1.5$$

x	0^-	0	0^+
Sign of $\frac{dy}{dx}$.	$+$	0	$+$
Sketch of tangent			

Stationary point at $x = 0$ is inflexion point

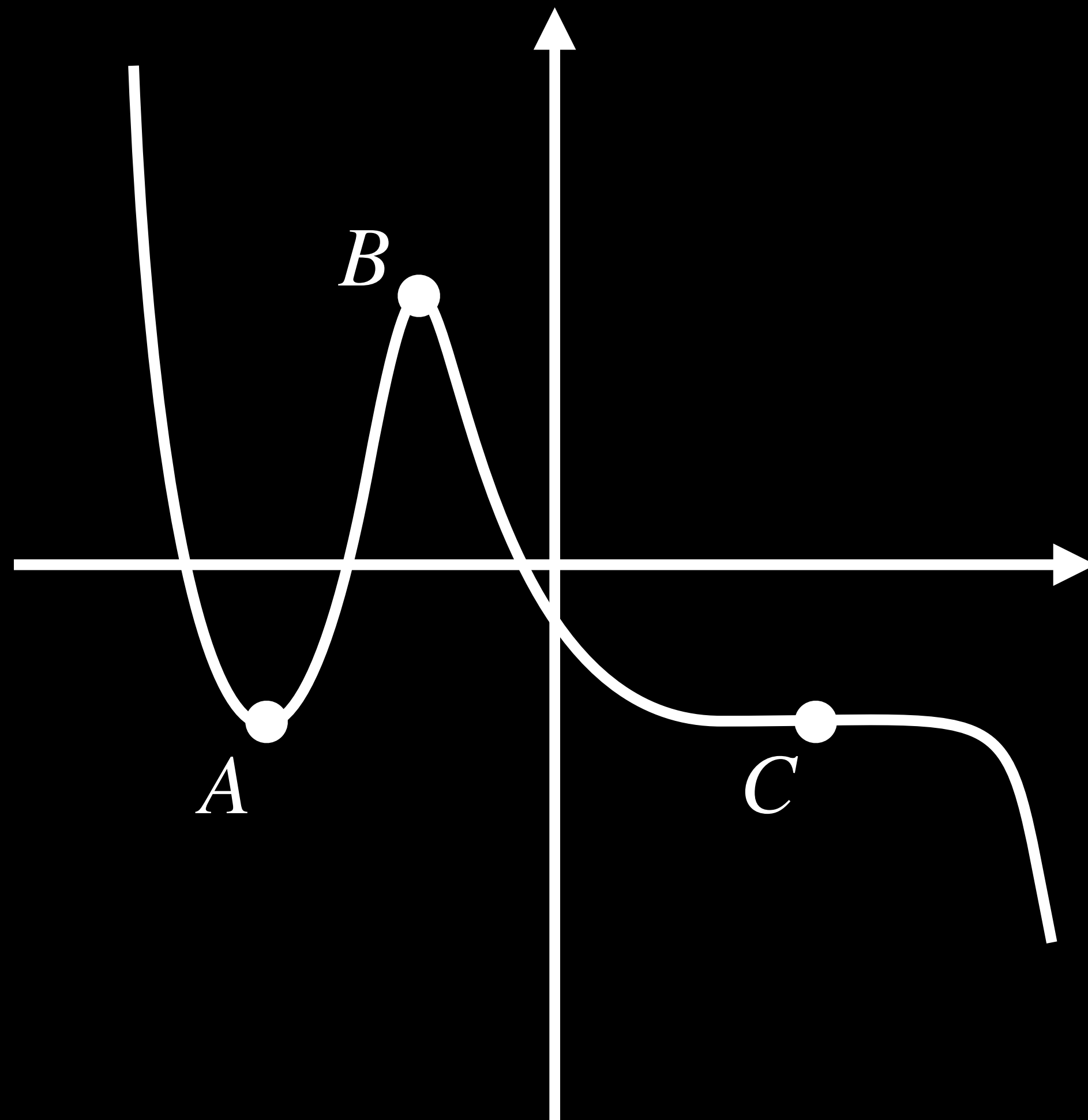
x	-1.5^-	-1.5	-1.5^+
Sign of $\frac{dy}{dx}$.	$-$	0	$+$
Sketch of tangent			

Stationary point at $x = -1.5$ is minimum point



Second Derivative Test

Yum yum



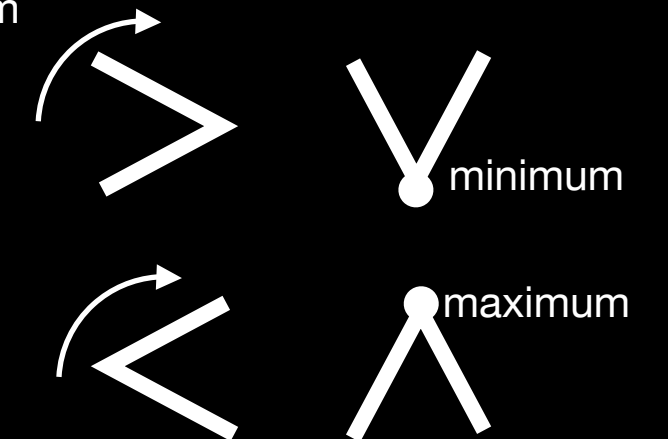
Second derivative is when the derivative of the derivative is taken

Second derivatives are notated as: $\frac{d^2y}{dx^2}$ or $f''(x)$

For a minimum point, $f''(x) > 0$ aka positive

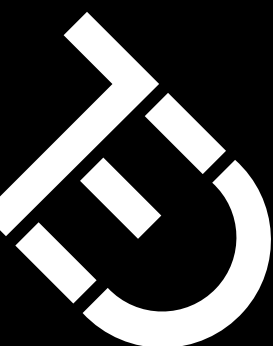
For a maximum point, $f''(x) < 0$ aka negative

One way to memorise: rotate inequality clockwise: shape of inequality represents if maximum or minimum



When $f''(x) = 0$, it may not be a point of inflexion. Although all points of inflexion have $f''(x) = 0$, a point of inflexion cannot be deduced by $f''(x) = 0$ [refer to x^4 , has minimum point but $f''(0) = 0$].

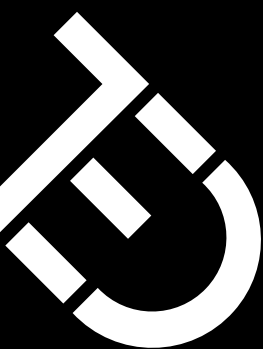
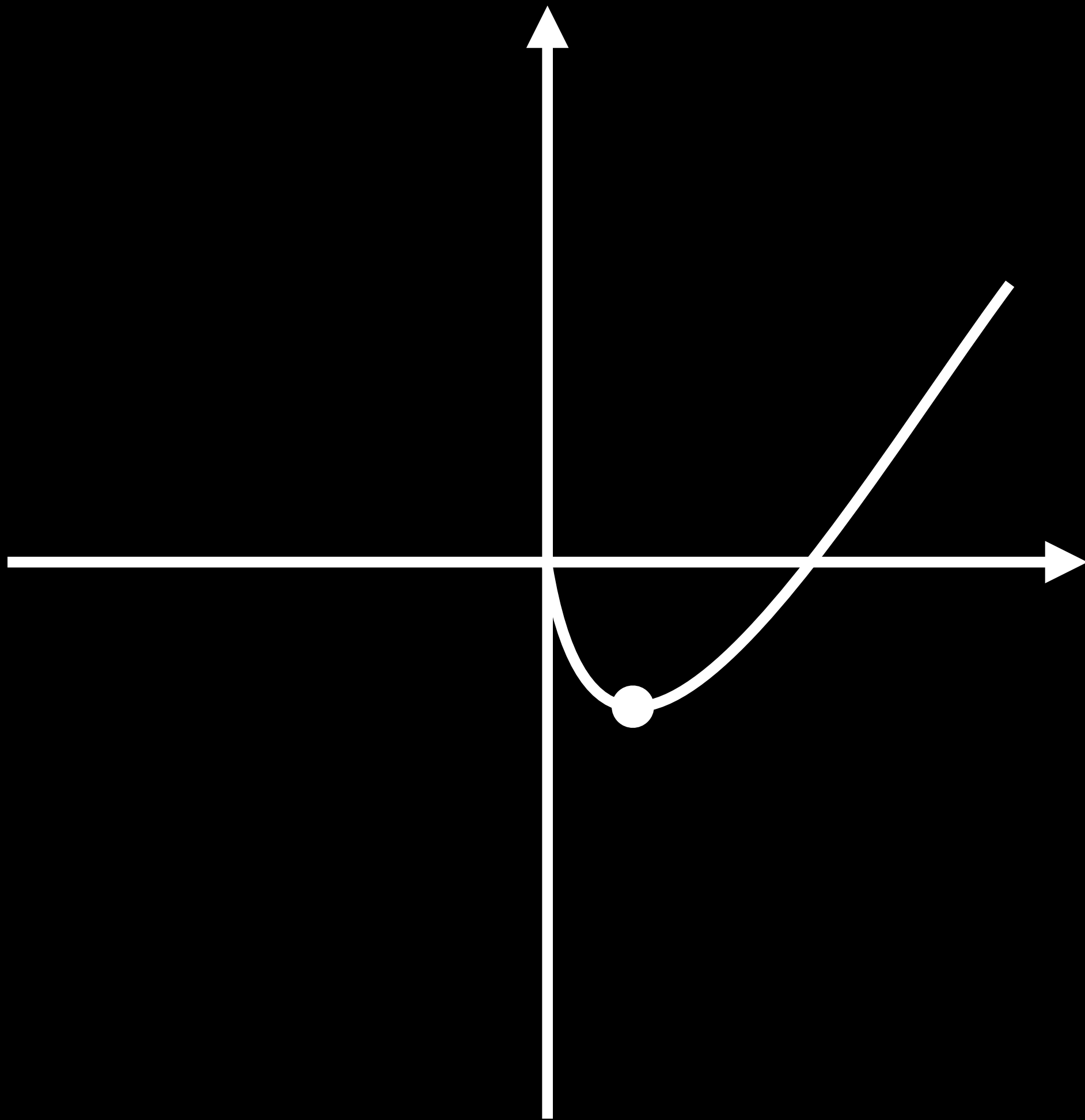
Thus, if your second derivative test gives $\frac{d^2y}{dx^2} = 0$, you have to use first derivative test.



Second Derivative Test

Yum yum

Use the second derivative test to find the coordinates of the stationary value of $y = (x - 2)\sqrt{x}$ and its nature.



Second Derivative Test

Yum yum

Use the second derivative test to find the coordinates of the stationary value of $y = (x - 2)\sqrt{x}$ and its nature.

$$y = (x - 2)\sqrt{x}$$

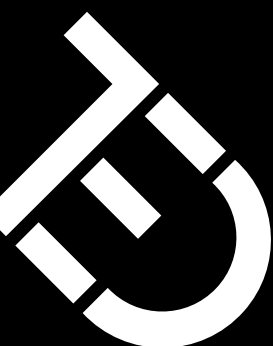
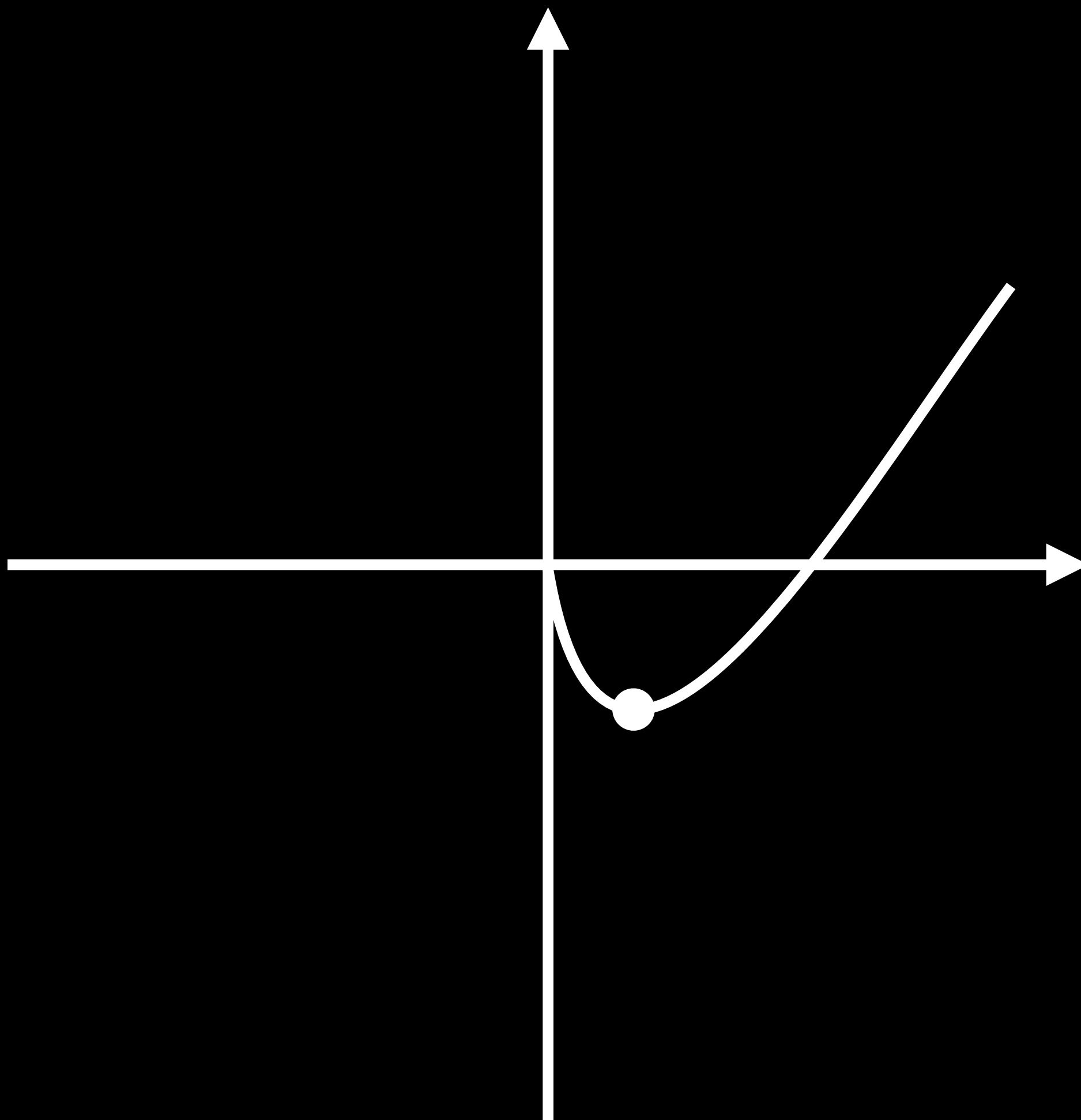
$$\begin{aligned}\frac{dy}{dx} &= (1)\sqrt{x} + (0.5\sqrt{x}^{-1})(x - 2) \\ &= \sqrt{x} + \frac{0.5(x - 2)}{\sqrt{x}} \\ &= \sqrt{x} + \frac{(x - 2)}{2\sqrt{x}} \\ &= \frac{2x}{2\sqrt{x}} + \frac{(x - 2)}{2\sqrt{x}} \\ &= \frac{3x - 2}{2\sqrt{x}}\end{aligned}$$

$$\begin{aligned}\frac{dy}{dx} &= 0 & \text{When } x &= \frac{2}{3}, \\ \frac{3x - 2}{2\sqrt{x}} &= 0 & y &= \left(\frac{2}{3} - 2\right)\sqrt{\frac{2}{3}} \\ 3x - 2 &= 0 & y &= -1.09(3.s.f) \\ x &= \frac{2}{3} \\ &= 0.67(3.s.f)\end{aligned}$$

$$\begin{aligned}\frac{d^2y}{dx^2} &= \frac{3(2\sqrt{x}) - (\sqrt{x}^{-1})(3x - 2)}{(2\sqrt{x})^2} \\ &= \frac{6\sqrt{x} - \frac{3x - 2}{\sqrt{x}}}{4x} \\ &= \frac{\frac{6x}{\sqrt{x}} - \frac{3x - 2}{\sqrt{x}}}{4x} \\ &= \frac{\frac{3x + 2}{\sqrt{x}}}{4x} \\ &= \frac{3x + 2}{4x\sqrt{x}}\end{aligned}$$

$$\begin{aligned}\frac{d^2y}{dx^2} \bigg|_{x=\frac{2}{3}} &= \frac{3(\frac{2}{3}) + 2}{4(\frac{2}{3})\sqrt{(\frac{2}{3})}} \\ \frac{d^2y}{dx^2} &= 1.84(3.s.f) \\ \frac{d^2y}{dx^2} &> 0\end{aligned}$$

$\therefore$ by the second derivative test, $(0.67, -1.09)$ is a minimum point



Maxima/Minima Problems

minimise

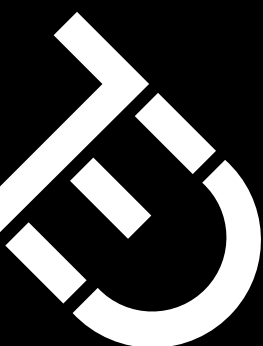
To solve maxima/minima problems:

1. Form a function relating the two variables in the qn (if there are three or more, express one in terms of another to get only two variables)
2. Differentiate and let derivative = 0 to find stationary values
3. Confirm stationary values' identity with first/second derivative test (minimum/maximum)
4. Solve question by finding unknown value with stationary value found. (Sub into original function or present stationary value as answer)

****Note, for first/second derivative test, only do second derivative test if it is easy to find, e.g. polynomials, else do first**

If your second derivative returns 0, you must do first derivative test.

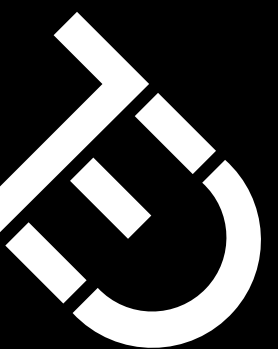
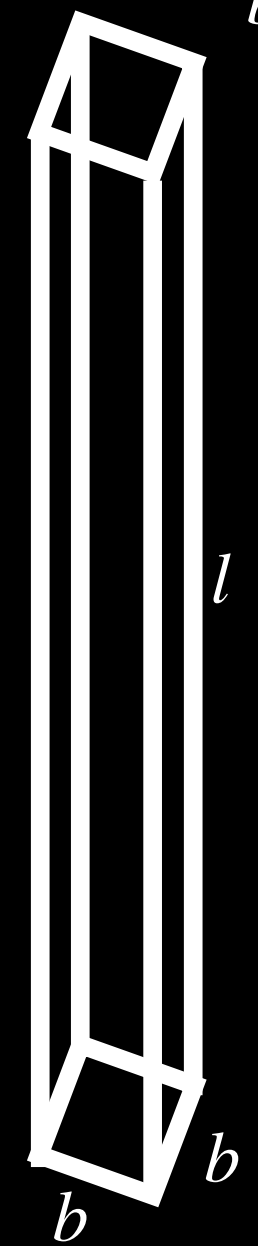
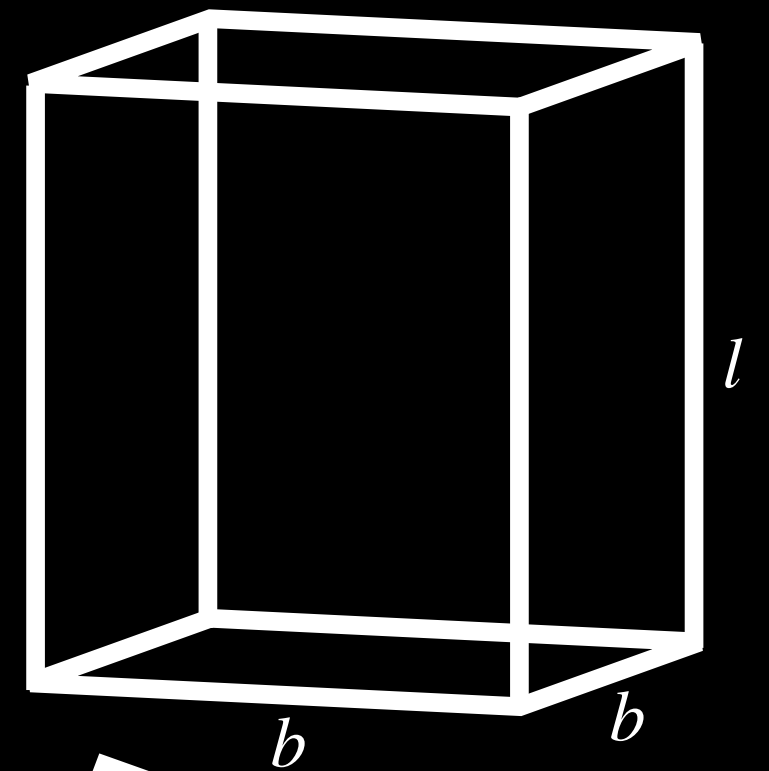
If you are unsure if the stationary value will be a maximum or minimum, look at the context: is it better to have more or less of the variable that will be a maximum or minimum? (e.g. cost should be minimised, land area or profit should be maximised)



Maxima/Minima Problems

minimise

A box has a square base of side length b and its surface area is 50cm^2 . What is the maximum possible volume it can have?



Maxima/Minima Problems

minimise

A box has a square base of side length b and its surface area is 50cm^2 . What is the maximum possible volume it can have?

differentiate and solve at 0

Let l be the height of the box

Surface Area:

$$2(b^2) + 4(bl) = 50$$

$$4bl = 50 - 2b^2$$

$$l = \frac{50 - 2b^2}{4b}$$

Volume:

$$V = b^2l$$

$$V = b^2 \frac{50 - 2b^2}{4b}$$

$$V = \frac{b^2(50 - 2b^2)}{4b} \quad V = \frac{b(50b - 2b^3)}{4b}$$

$$V = \frac{50b^2 - 2b^4}{4b} \quad V = 0.25(50b - 2b^3)$$

form eqn

$$V = 12.5b - 0.5b^3$$

$$\frac{dV}{db} = 12.5 - 3(0.5b^2)$$

$$\frac{dV}{db} = 12.5 - 1.5b^2$$

$$\frac{dV}{db} = 0$$

$$-1.5b^2 + 12.5 = 0$$

$$b = \sqrt{\frac{25}{3}} = 2.89(3.s.f)$$

rej negative value as $b > 0$

$$\frac{d^2V}{db^2} = -3b$$

$$\frac{d^2V}{db^2} /_{b=\sqrt{\frac{25}{3}}} = -3\sqrt{\frac{25}{3}}$$

$$\frac{d^2V}{db^2} = -5.77$$

$$\frac{d^2V}{db^2} < 0, \text{ thus } b=2.89 \text{ is a maximum}$$

check for stationary value's nature

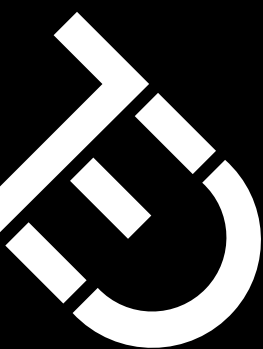
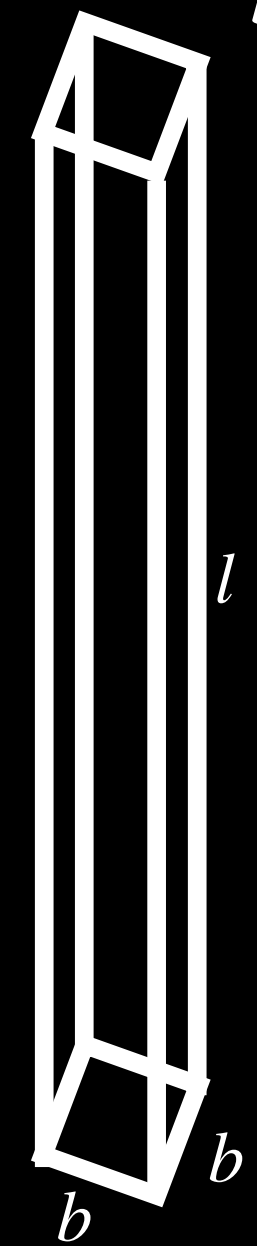
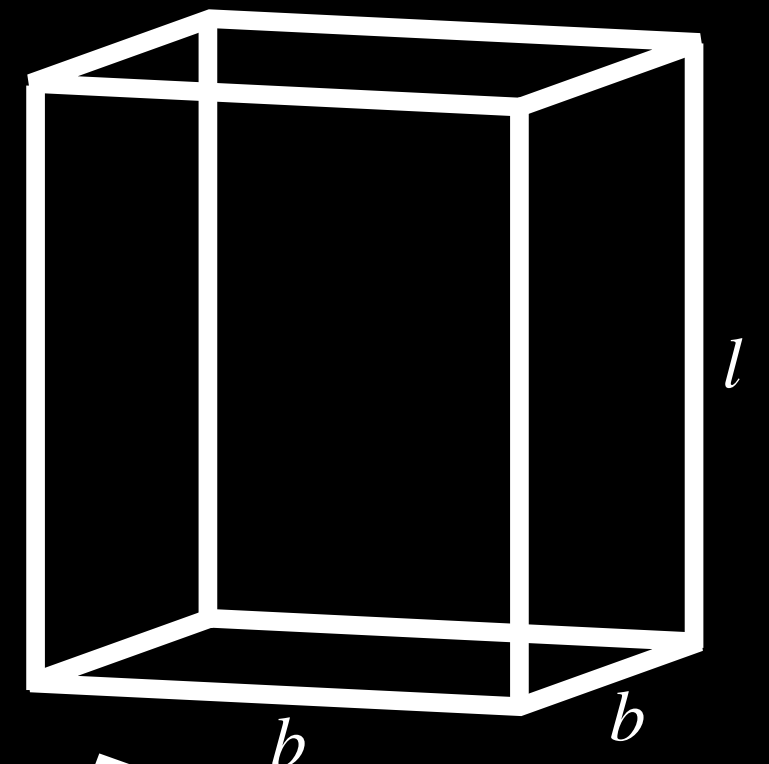
$$\text{For } b = \sqrt{\frac{25}{3}}$$

$$V = 12.5\sqrt{\frac{25}{3}} - 0.5\left(\sqrt{\frac{25}{3}}\right)^3$$

$$V = 24.1(3.s.f)$$

$$\therefore \text{max volume is } 24.1\text{cm}^3$$

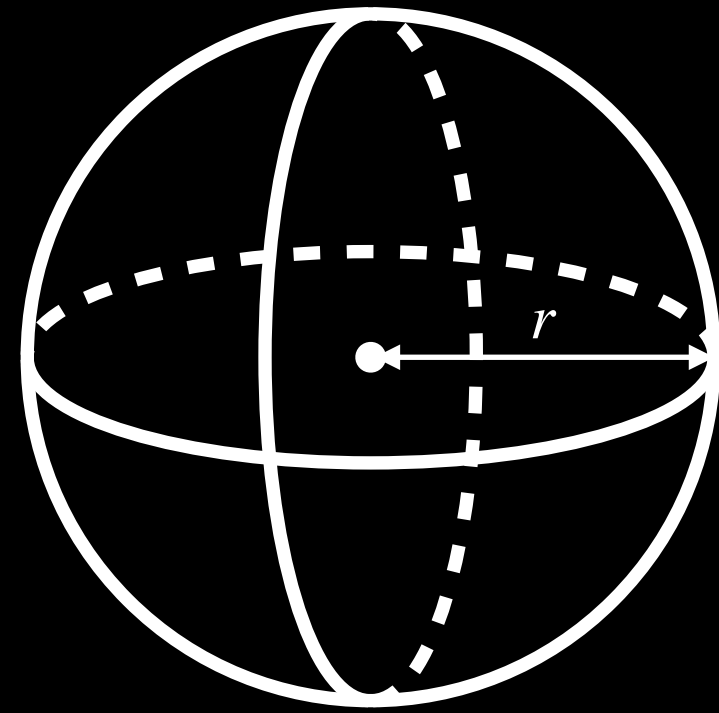
present answer



Maxima/Minima Problems

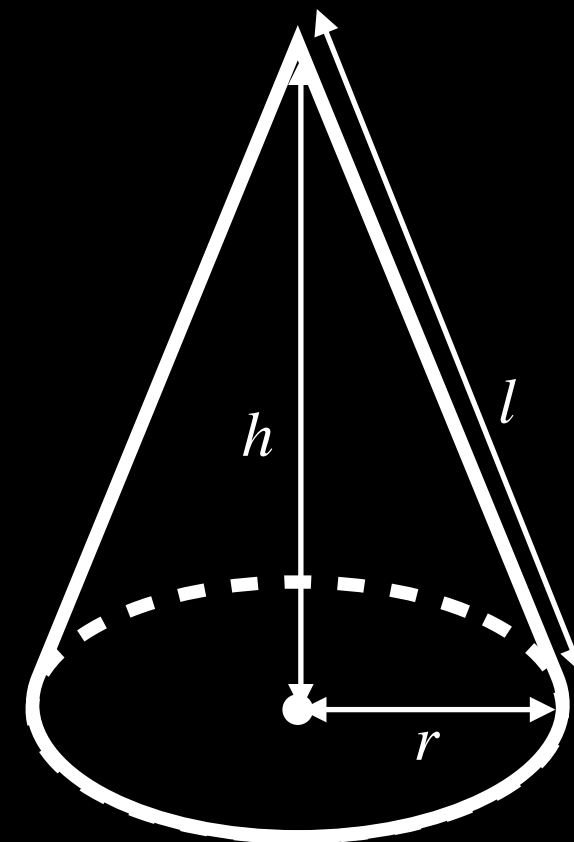
minimimimise

Maxima/Minima Problems tend to make use of shapes who's formulas are not in the AM formula sheet:



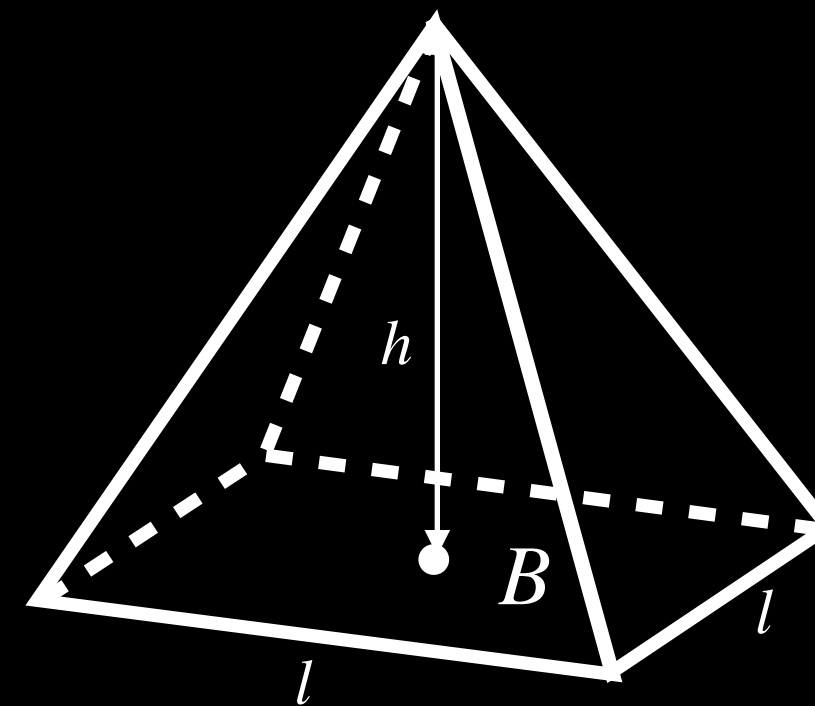
$$V = \frac{4}{3}\pi r^3$$

$$S.A = 4\pi r^2$$



$$V = \frac{1}{3}\pi r^2 h$$

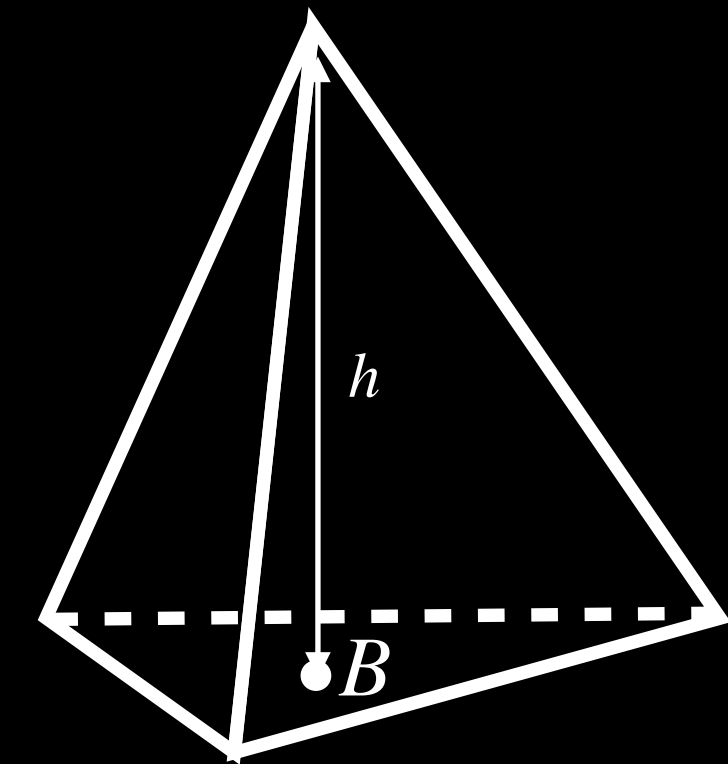
$$S.A = \pi r l + \pi r^2$$



$$V = \frac{1}{3}Bh = \frac{1}{3}l^2 h$$

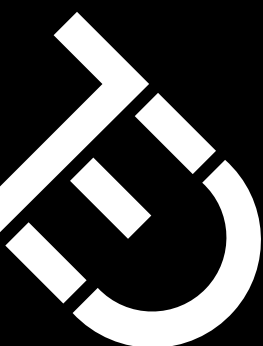
$$S.A = l^2 + 2l\sqrt{\left(\frac{l}{2}\right)^2 + h^2}$$

however can be found with pythagorean theorem intuitively



$$V = \frac{1}{3}Bh$$

S.A can be found intuitively



Integration

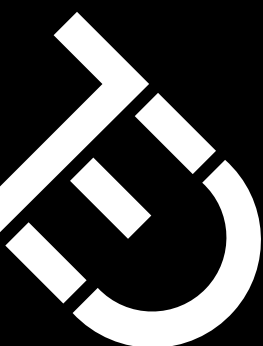


Integration

Crowbar

$$\int f(x) \, dx = F(x) + C$$

indefinite integral

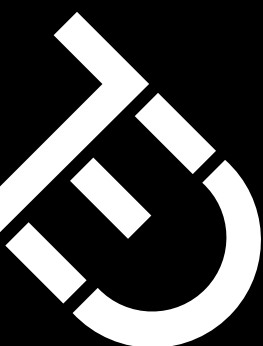


Fundamental Theorem of Calculus

Crowbar

$$\int \frac{d}{dx} f(x) \, dx = f(x) + C$$

fundamental theorem of calculus
integrals and derivatives are opposites



Integration

Crowbar

For a function $f(x)$ and its derivative $f'(x)$,

Taking the integral of $f'(x)$ gives us $f(x)$

integral of $f(x)$ can be written as $F(x)$

For example, since $\frac{d}{dx}x^2 = 2x$

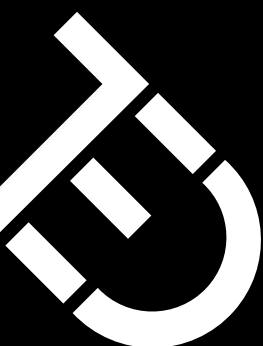
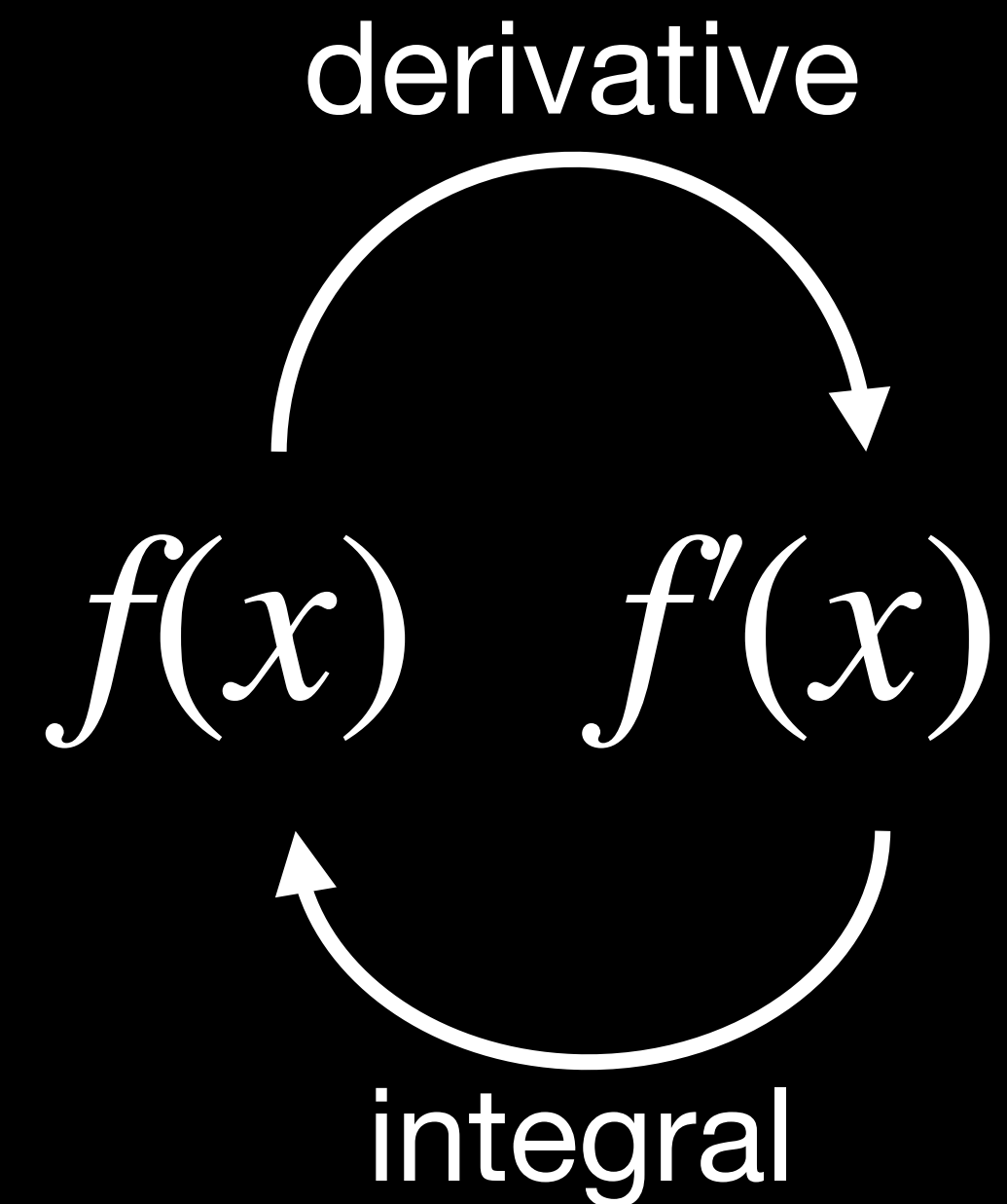
$$\int 2x \, dx = x^2 + C$$

dx is the integration variable, or $d(\text{something})$

Thus,

$$\int \frac{d}{dx} f(x) \, dx = f(x) + C$$

$$f(x) \xrightarrow{\text{integral}} F(x) \xrightarrow{\text{derivative}} f(x)$$



Crowbar

When x^2 is shifted up/down, its slope stays the same
Thus, we account for this by writing +C [constant]

****Do not forget the +C!!!!!!!**

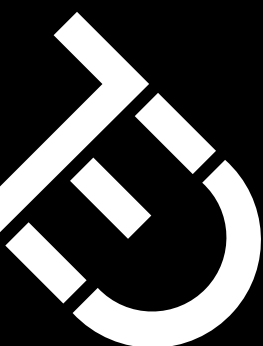


Integral Power Rule

The opposite

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

integral power rule
integral of x^n where n is a constant and is not -1



Integral Power Rule

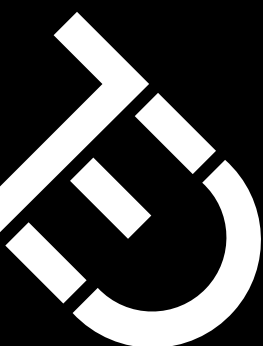
the opposite

Integrate the following:

$$\int x^5 dx$$

$$\int \sqrt[3]{x} dx$$

$$\int \frac{1}{\sqrt{x}} dx$$



Integral Power Rule

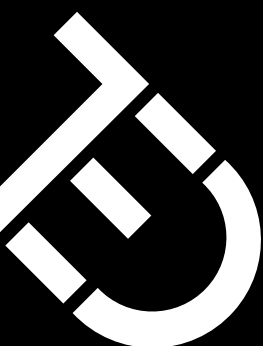
the opposite

Integrate the following:

$$\int x^5 dx \qquad \int x^5 dx = \frac{x^{5+1}}{5+1} + C$$

$$\int \sqrt[3]{x} dx \qquad = \frac{x^6}{6} + C$$

$$\int \frac{1}{\sqrt{x}} dx$$



Integral Power Rule

the opposite

Integrate the following:

$$\int x^5 dx$$

$$\int \sqrt[3]{x} dx = \int x^{\frac{1}{3}} dx$$

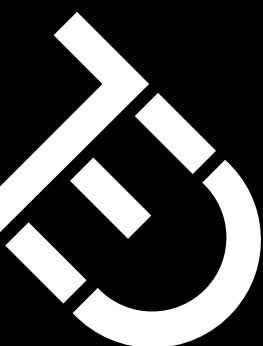
$$\int \sqrt[3]{x} dx$$

$$= \frac{x^{\frac{1}{3}+1}}{\frac{1}{3}+1} + C$$

$$\int \frac{1}{\sqrt{x}} dx$$

$$= \frac{x^{\frac{4}{3}}}{\frac{4}{3}} + C$$

$$= \frac{3x^{\frac{4}{3}}}{4} + C$$



Integral Power Rule

the opposite

Integrate the following:

$$\int x^5 dx$$

$$\int \frac{1}{\sqrt{x}} dx = \int x^{-0.5} dx$$

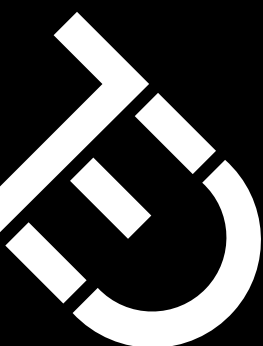
$$\int \sqrt[3]{x} dx$$

$$= \frac{x^{-0.5+1}}{-0.5+1} + C$$

$$\int \frac{1}{\sqrt{x}} dx$$

$$= \frac{x^{0.5}}{0.5} + C$$

$$= 2\sqrt{x} + C$$



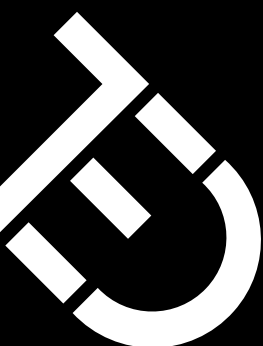
Integral Constant Rule

The opposite

$$\int k \, dx = kx + C$$

integral constant rule

integral of k where k is a constant



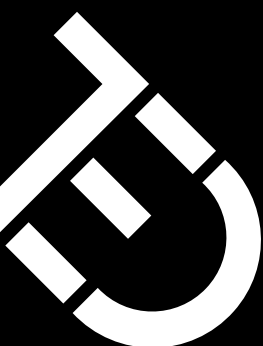
Integral CM Rule

The opposite

$$\int kf(x) \, dx = k \int f(x) \, dx$$

integral constant multiple rule

integral of $kf(x)$ where $f(x)$ is a function and k is a constant



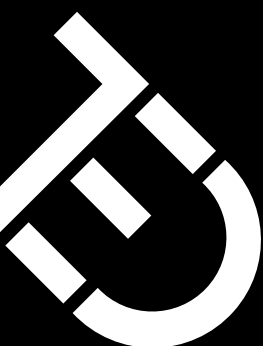
Integral Sum/Difference Rule

The opposite

$$\int (f(x) \pm g(x)) \, dx = \int f(x) \, dx \pm \int g(x) \, dx$$

sum/difference rule

integral of sum/dif of two functions where $f(x)$ and $g(x)$ are functions



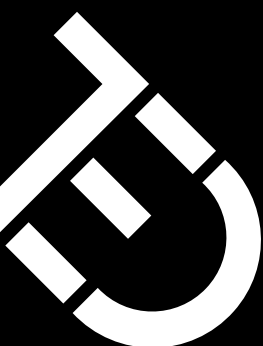
Integral CM/Sum/Difference Rule

the opposite

Integrate the following:

$$\int (x - 3)(x + 2) \, dx$$

$$\int x^3 - 3\sqrt{x} \, dx$$



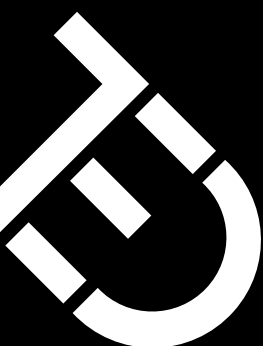
Integral CM/Sum/Difference Rule

the opposite

Integrate the following:

$$\int (x - 3)(x + 2) \, dx \qquad \int (x - 3)(x + 2) \, dx = \int x^2 - x - 6 \, dx$$

$$\int x^3 - 3\sqrt{x} \, dx \qquad = \frac{x^3}{3} - \frac{x^2}{2} - 6x + C$$

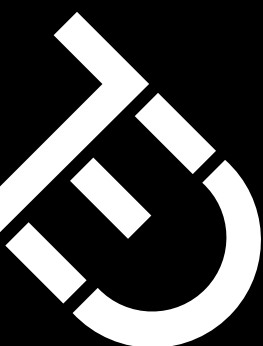


Integral CM/Sum/Difference Rule

the opposite

Integrate the following:

$$\begin{aligned}\int (x-3)(x+2) dx & \quad \int x^3 - 3\sqrt{x} dx = \int x^3 - 3x^{0.5} dx \\ \int x^3 - 3\sqrt{x} dx & \quad = \frac{x^4}{4} - 3\frac{x^{1.5}}{1.5} + C \\ & \quad = \frac{x^4}{4} - 2x^{1.5} + C\end{aligned}$$



Integral General Power Rule

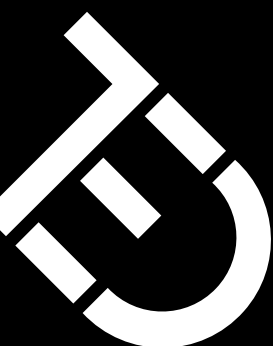
The opposite

$$\int (ax + b)^n dx = \frac{(ax + b)^{n+1}}{a(n+1)} + C$$

general power rule

integral of a linear function to a power n where a , b and n are constants
and $n \neq -1$

THIS ONLY WORKS FOR $AX+B$, NOT FOR ALL FUNCTIONS



Integral General Power Rule

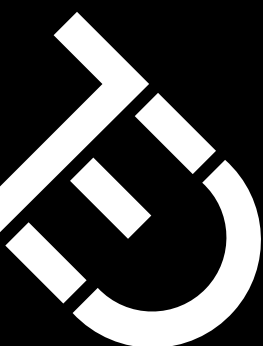
the opposite

Integrate the following:

$$\int (3x + 2)^3 dx$$

$$\int \sqrt{5x - 1} dx$$

$$\int \frac{1}{(5x - 1)^2} + x^2 dx$$



Integral General Power Rule

the opposite

Integrate the following:

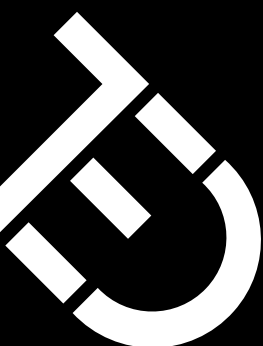
$$\int (3x + 2)^3 dx$$

$$\int (3x + 2)^3 dx = \frac{(3x + 2)^{3+1}}{3(3 + 1)} + C$$

$$\int \sqrt{5x - 1} dx$$

$$= \frac{(3x + 2)^4}{12} + C$$

$$\int \frac{1}{(5x - 1)^2} + x^2 dx$$



Integral General Power Rule

the opposite

Integrate the following:

$$\int (3x + 2)^3 dx$$

$$\int \sqrt{5x - 1} dx = \int (5x - 1)^{0.5} dx$$

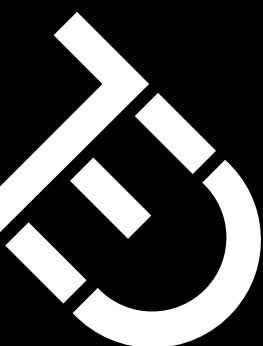
$$\int \sqrt{5x - 1} dx$$

$$= \frac{(5x - 1)^{0.5+1}}{5(0.5 + 1)} + C$$

$$\int \frac{1}{(5x - 1)^2} + x^2 dx$$

$$= \frac{(5x - 1)^{1.5}}{7.5} + C$$

$$= \frac{2(5x - 1)^{1.5}}{15} + C$$



Integral General Power Rule

the opposite

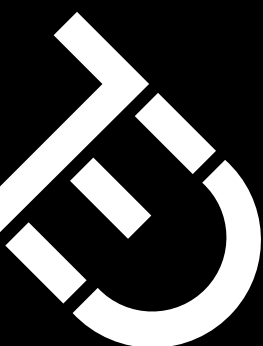
Integrate the following:

$$\int (3x + 2)^3 dx$$

$$\int \sqrt{5x - 1} dx$$

$$\int \frac{1}{(5x - 1)^2} + x^2 dx$$

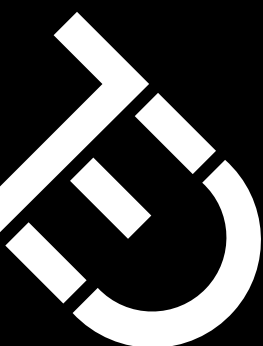
$$\begin{aligned} \int \frac{1}{(5x - 1)^2} + x^2 dx &= \int (5x - 1)^{-2} + x^2 dx \\ &= \frac{(5x - 1)^{-2+1}}{5(-2+1)} + \frac{x^3}{3} + C \\ &= \frac{(5x - 1)^{-1}}{-5} + \frac{x^3}{3} + C \\ &= -\frac{1}{5(5x - 1)} + \frac{x^3}{3} + C \end{aligned}$$



Indefinite Integral Applications

dropc

Given that $f'(x) = 6x^2 - 16x + 4$ and $f(x)$ passes through the point $(4,7)$. find $f(x)$.



Indefinite Integral Applications

dropc

Given that $f'(x) = 6x^2 - 16x + 4$ and $f(x)$ passes through the point $(4,7)$. find $f(x)$.

$$\begin{aligned} f(x) &= \int f'(x) \, dx \\ &= \int 6x^2 - 16x + 4 \, dx \\ &= \frac{6x^3}{3} - \frac{16x^2}{2} + 4x + C \\ f(x) &= 2x^3 - 8x^2 + 4x + C \end{aligned}$$

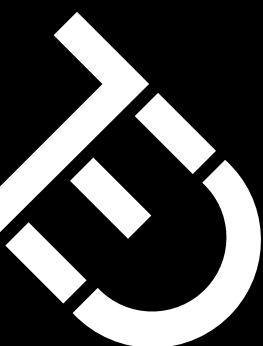
Sub in $(4,7)$ into $f(x)$

$$7 = 2(4)^3 - 8(4)^2 + 4(4) + C$$

$$7 = 16 + C$$

$$C = -9$$

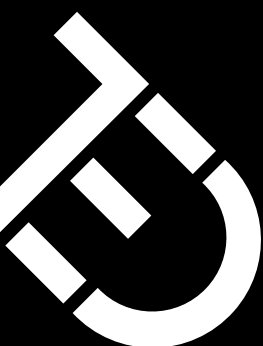
$$\therefore f(x) = 2x^3 - 8x^2 + 4x - 9$$



Indefinite Integral Applications

dropc

Given that $\frac{d^2y}{dx^2} = 6x + 6$, find y given that y passes through the points $(1,9)$ and $(-2,3)$.



Indefinite Integral Applications

dropc

Given that $\frac{d^2y}{dx^2} = 6x + 6$, find y given that y passes through the points (1,9) and (-2,3).

$$\begin{aligned}\frac{dy}{dx} &= \int \frac{d^2y}{dx^2} dx & y &= \int \frac{dy}{dx} dx \\ &= \int 6x + 6 dx & &= \int 3x^2 + 6x + C_1 dx \\ &= \frac{6x^2}{2} + 6x + C_1 & &= x^3 + \frac{6x^2}{2} + C_1x + C_2 \\ &= 3x^2 + 6x + C_1 & &= x^3 + 3x^2 + C_1x + C_2\end{aligned}$$

Sub in (1,9) into y

$$\begin{aligned}9 &= 1^3 + 3(1)^2 + C_1 + C_2 \\ 9 &= 4 + C_1 + C_2 \\ 5 &= C_1 + C_2 \\ C_2 &= 5 - C_1 \rightarrow (1)\end{aligned}$$

Sub in (-2,3) into y

$$\begin{aligned}3 &= (-2)^3 + 3(-2)^2 - 2C_1 + C_2 \\ 3 &= 4 - 2C_1 + C_2 \\ -1 &= -2C_1 + C_2 \rightarrow (2)\end{aligned}$$

Sub in (1) into (2):

$$\begin{aligned}-1 &= -2C_1 + (5 - C_1) \\ -1 &= -2C_1 + 5 - C_1 \\ -6 &= -3C_1 \\ C_1 &= 2\end{aligned}$$

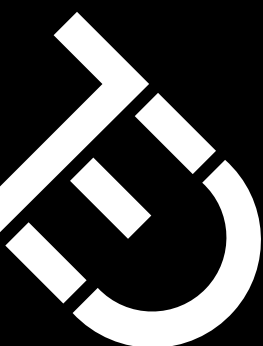
Sub in $C_1 = 2$ into (1)

$$\begin{aligned}C_2 &= 5 - 2 \\ C_2 &= 3\end{aligned}$$

We have to label the two different arbitrary constants as they may not necessarily be the same value (we can also label as different letters, like K or Q)

We also can only sub in the points once we have integrated twice as both points are on y , not dy/dx .

$$\therefore y = x^3 + 3x^2 + 2x + 3$$



Integration of Trigonometric Functions

trigomemotmetric

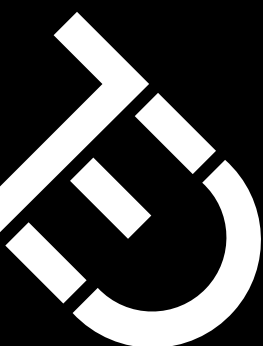
$$\int \cos(x) = \sin(x) + C$$

From $\int -\sin(x) = \cos(x) + C$,

$$\int \sin(x) = -\cos(x) + C$$

$$\int \sec^2(x) = \tan(x) + C$$

integrals of basic trigo functions



Integration of Trigonometric Functions

trigonometric

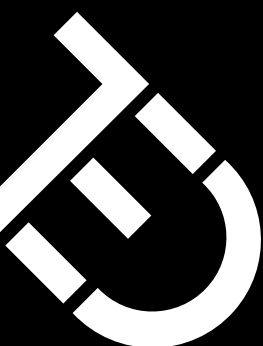
Integrate the following:

$$\int \sin(x) + \cos(x) \, dx$$

$$\int \frac{4}{3 - 3 \sin^2(x)} \, dx$$

$$\int \frac{\sin(x)\cos^2(x) - 5}{\cos^2(x)} \, dx$$

$$\int \sin^2(x) \, dx$$



Integration of Trigonometric Functions

trigonometric

Integrate the following:

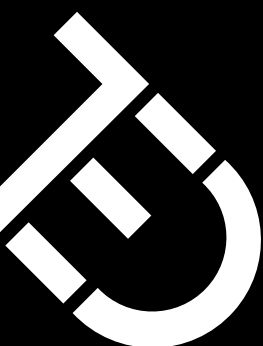
$$\int \sin(x) + \cos(x) \, dx$$

$$\begin{aligned} \int \sin(x) + \cos(x) \, dx &= -\cos(x) + \sin(x) + C \\ &= \sin(x) - \cos(x) + C \end{aligned}$$

$$\int \frac{4}{3 - 3 \sin^2(x)} \, dx$$

$$\int \frac{\sin(x) \cos^2(x) - 5}{\cos^2(x)} \, dx$$

$$\int \sin^2(x) \, dx$$

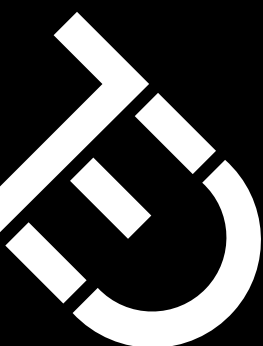


Integration of Trigonometric Functions

trigonometric

Integrate the following:

$$\begin{aligned} \int \sin(x) + \cos(x) \, dx & \quad \int \frac{4}{3 - 3 \sin^2(x)} \, dx = \int \frac{4}{3(1 - \sin^2(x))} \, dx \\ \int \frac{4}{3 - 3 \sin^2(x)} \, dx & \quad = \int \frac{4}{3 \cos^2(x)} \, dx \\ \int \frac{\sin(x) \cos^2(x) - 5}{\cos^2(x)} \, dx & \quad = \int \frac{4}{3} \sec^2(x) \, dx \\ \int \sin^2(x) \, dx & \quad = \frac{4}{3} \tan(x) + C \end{aligned}$$



Integration of Trigonometric Functions

trigonometric

Integrate the following:

$$\int \sin(x) + \cos(x) \, dx$$

$$\int \frac{4}{3 - 3 \sin^2(x)} \, dx$$

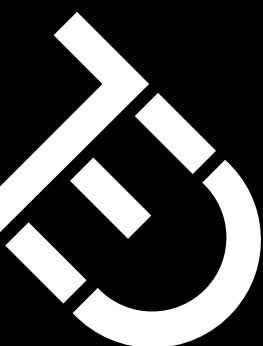
$$\int \frac{\sin(x)\cos^2(x) - 5}{\cos^2(x)} \, dx$$

$$\int \sin^2(x) \, dx$$

$$\int \frac{\sin(x)\cos^2(x) - 5}{\cos^2(x)} \, dx = \int \frac{\sin(x)\cos^2(x)}{\cos^2(x)} - \frac{5}{\cos^2(x)} \, dx$$

$$= \int \sin(x) - 5 \sec^2(x) \, dx$$

$$= -\cos(x) - 5 \tan(x) + C$$



Integration of Trigonometric Functions

trigonomotmetric

Integrate the following:

$$\int \sin(x) + \cos(x) \, dx$$

$$\int \frac{4}{3 - 3 \sin^2(x)} \, dx$$

$$\int \frac{\sin(x) \cos^2(x) - 5}{\cos^2(x)} \, dx$$

$$\int \sin^2(x) \, dx$$

cosine double angle can be used to integrate $\sin^2(x)$ and $\cos^2(x)$

$$\cos(2x) = 1 - 2 \sin^2(x)$$

$$\cos(2x) - 1 = -2 \sin^2(x)$$

$$-\frac{\cos(2x) - 1}{2} = \sin^2(x)$$

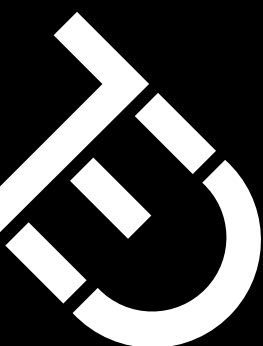
$$\int \sin^2(x) \, dx = \int -\frac{\cos(2x) - 1}{2} \, dx$$

$$= -\frac{1}{2} \int \cos(2x) - 1 \, dx$$

$$= -\frac{1}{2} \left(\frac{1}{2} \sin(2x) - x \right) + C$$

$$= -\frac{\sin(2x)}{4} + \frac{x}{2} + C$$

dont forget your trig identites as qns often use them!!!!



Integration of Trigonometric Functions

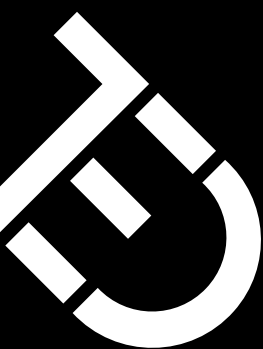
trigomemotmetric

$$\int \cos(ax + b) = \frac{1}{a} \sin(ax + b) + C$$

$$\int \sin(ax + b) = -\frac{1}{a} \cos(ax + b) + C$$

$$\int \sec^2(ax + b) = \frac{1}{a} \tan(ax + b) + C$$

integrals of basic trigo functions where $a \neq 0$



Integration of Trigonometric Functions

trigonometric

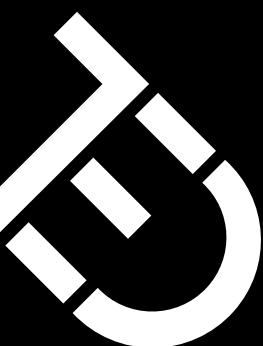
Integrate the following:

$$\int \cos(3x - 2) \, dx$$

$$\int \sin(x)(2 \cos(x) - 1) \, dx$$

$$\int \frac{3}{\cos(10x + 4) + \sin^2(5x + 2)} \, dx$$

$$\int (\sin(x) + \cos(x))^2 \, dx$$



Integration of Trigonometric Functions

trigonomotmetric

Integrate the following:

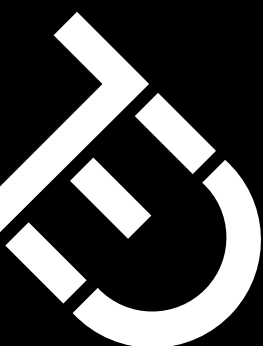
$$\int \cos(3x - 2) \, dx$$

$$\int \cos(3x - 2) \, dx = \frac{\sin(3x - 2)}{3} + C$$

$$\int \sin(x)(2 \cos(x) - 1) \, dx$$

$$\int \frac{3}{\cos(10x + 4) + \sin^2(5x + 2)} \, dx$$

$$\int (\sin(x) + \cos(x))^2 \, dx$$



Integration of Trigonometric Functions

trigonomotmetric

Integrate the following:

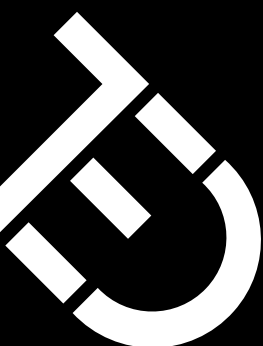
$$\int \cos(3x - 2) \, dx$$

$$\int \sin(x)(2 \cos(x) - 1) \, dx$$

$$\int \frac{3}{\cos(10x + 4) + \sin^2(5x + 2)} \, dx$$

$$\int (\sin(x) + \cos(x))^2 \, dx$$

$$\begin{aligned} \int \sin(x)(2 \cos(x) - 1) \, dx &= \int 2 \sin(x) \cos(x) - \sin(x) \, dx \\ &= \int \sin(2x) - \sin(x) \, dx \\ &= -\frac{\cos(2x)}{2} - (-\cos(x)) + C \\ &= -\frac{\cos(2x)}{2} + \cos(x) + C \end{aligned}$$



Integration of Trigonometric Functions

trigonomotmetric

Integrate the following:

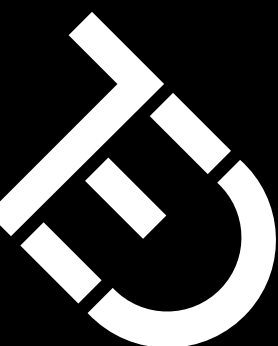
$$\int \cos(3x - 2) \, dx$$

$$\int \sin(x)(2 \cos(x) - 1) \, dx$$

$$\int \frac{3}{\cos(10x + 4) + \sin^2(5x + 2)} \, dx$$

$$\int (\sin(x) + \cos(x))^2 \, dx$$

$$\begin{aligned} \int \frac{3}{\cos(10x + 4) + \sin^2(5x + 2)} \, dx &= \int \frac{3}{\cos(2(5x + 2)) + \sin^2(5x + 2)} \, dx \\ &\stackrel{\substack{\cos(2x) \text{ identity} \\ \cos(2x) = \cos^2(x) - \sin^2(x)}}{=} \int \frac{3}{\cos^2(5x + 2) - \sin^2(5x + 2) + \sin^2(5x + 2)} \, dx \\ &= \int \frac{3}{\cos^2(5x + 2)} \, dx \\ &= \int 3 \sec^2(5x + 2) \, dx \\ &= \frac{3 \tan(5x + 2)}{5} + C \end{aligned}$$

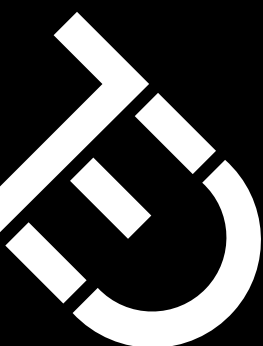


Integration of Trigonometric Functions

trigonomotmetric

Integrate the following:

$$\begin{aligned} &\int \cos(3x - 2) \, dx \\ &\int \sin(x)(2 \cos(x) - 1) \, dx \\ &\int \frac{3}{\cos(10x + 4) + \sin^2(5x + 2)} \, dx \\ &\int (\sin(x) + \cos(x))^2 \, dx \end{aligned}$$
$$\begin{aligned} \int (\sin(x) + \cos(x))^2 \, dx &= \int \sin^2(x) + 2 \sin(x)\cos(x) + \cos^2(x) \, dx \\ &= \int 2 \sin(x)\cos(x) + \sin^2(x) + \cos^2(x) \, dx \\ &= \int \sin(2x) + 1 \, dx \quad \begin{array}{l} \downarrow \sin(2x) = 2 \sin(x)\cos(x) \\ \swarrow \sin^2(x) + \cos^2(x) = 1 \end{array} \\ &= -\frac{\cos(2x)}{2} + x + C \end{aligned}$$

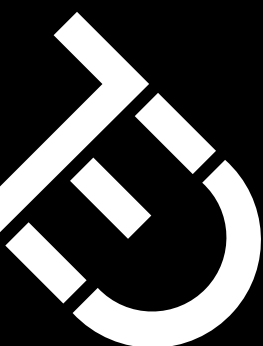


Integration of Exponentials

[back](#)

$$\int e^x dx = e^x + C$$
$$\int e^{ax+b} dx = \frac{1}{a} e^{ax+b} + C$$

integrals of exponentials where $a \neq 0$



Integration of Exponentials

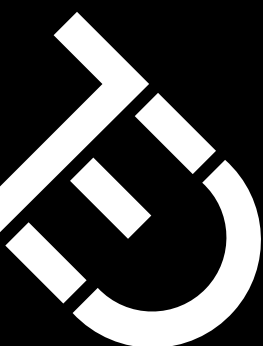
back

Integrate the following:

$$\int e^{3x-1} dx$$

$$\int \frac{7}{e^{2x}} dx$$

$$\int e^x(e^{2x} - 5) dx$$



Integration of Exponentials

back

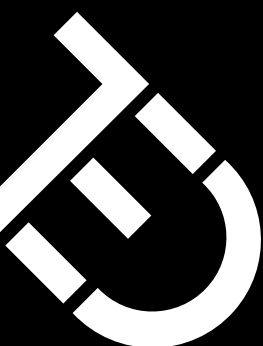
Integrate the following:

$$\int e^{3x-1} dx$$

$$\int e^{3x-1} dx = \frac{e^{3x-1}}{3} + C$$

$$\int \frac{7}{e^{2x}} dx$$

$$\int e^x(e^{2x} - 5) dx$$



Integration of Exponentials

back

Integrate the following:

$$\int e^{3x-1} dx$$

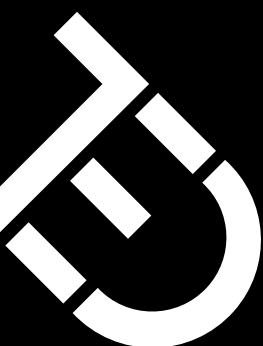
$$\int \frac{7}{e^{2x}} dx = \int 7e^{-2x} dx$$

$$\int \frac{7}{e^{2x}} dx$$

$$= 7 \frac{e^{-2x}}{-2} + C$$

$$\int e^x(e^{2x} - 5) dx$$

$$= -\frac{7e^{-2x}}{2} + C$$

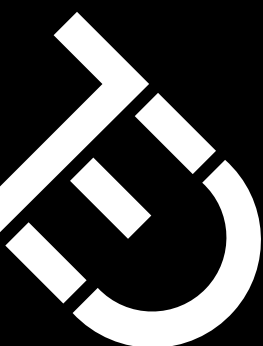


Integration of Exponentials

back

Integrate the following:

$$\begin{aligned}\int e^{3x-1} dx & \qquad \int e^x(e^{2x} - 5) dx = \int e^x * e^{2x} - 5e^x dx \\ \int \frac{7}{e^{2x}} dx & \qquad = \int e^{3x} - 5e^x dx \\ \int e^x(e^{2x} - 5) dx & \qquad = \frac{e^{3x}}{3} - 5e^x + C\end{aligned}$$



Antiderivative Questions

bruh!!

Find $\frac{d}{dx} 2\sqrt{x}(6-x)$ and hence or otherwise, find $\int \frac{2-x}{\sqrt{x}} dx$

$$\begin{aligned}\frac{d}{dx} 2\sqrt{x}(6-x) &= \frac{6-x}{\sqrt{x}} - 2\sqrt{x} \\ &= \frac{6-x}{\sqrt{x}} - \frac{2x}{\sqrt{x}} \\ &= \frac{6-3x}{\sqrt{x}}\end{aligned}$$

Antiderivative questions will always ask you to differentiate and then ask you to use your derivative to find the integral

$$= 3 \left(\frac{2-x}{\sqrt{x}} \right)$$

We can take out $\frac{1}{3}$ and evaluate this integral from the first part, as $\int \frac{d}{dx} f(x) dx = f(x)$

$$\frac{d}{dx} 2\sqrt{x}(6-x) = 3 \left(\frac{2-x}{\sqrt{x}} \right)$$

Make this the subj., then integrate on both sides.

$$\frac{1}{3} \frac{d}{dx} 2\sqrt{x}(6-x) = \left(\frac{2-x}{\sqrt{x}} \right)$$

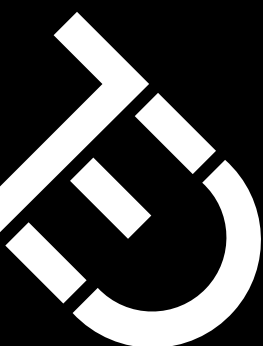
Multiply both sides by $\frac{1}{3}$ to get rid of 3

$$\int \frac{1}{3} \frac{d}{dx} 2\sqrt{x}(6-x) dx = \int \frac{2-x}{\sqrt{x}} dx$$

When we integrate both sides, we get the integral we want

$$\frac{1}{3} \int \frac{d}{dx} 2\sqrt{x}(6-x) dx = \int \frac{2-x}{\sqrt{x}} dx$$

$$\int \frac{2-x}{\sqrt{x}} dx = \frac{2}{3} \sqrt{x}(6-x) + C \quad \text{Add +C}$$



Antiderivative Questions

bruh!!

Find $\frac{d}{dx}xe^x$ and hence or otherwise, find $\int xe^x dx$

$$\begin{aligned}\frac{d}{dx}xe^x &= (1)e^x + xe^x \\ &= e^x + xe^x\end{aligned}$$

$$\frac{d}{dx}xe^x = e^x + \overbrace{xe^x}^{\text{Make this the subj., then integrate on both sides.}}$$

$$\frac{d}{dx}(xe^x) - e^x = xe^x \quad \text{Subtract } e^x \text{ to make } xe^x \text{ subj.}$$

$$\int \frac{d}{dx}(xe^x) - e^x dx = \boxed{\int xe^x dx} \quad \text{When we integrate both sides, we get the integral we want}$$

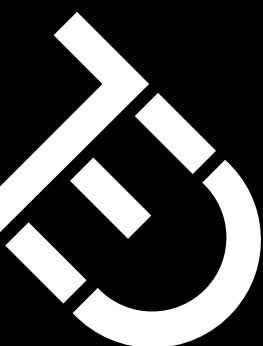
$$\int xe^x dx = \int \frac{d}{dx}(xe^x) dx - \int e^x dx$$

Integral and derivative cancel out Integral can be evaluated

$$\int xe^x dx = xe^x - e^x + C$$

Add +C

Sometimes, antiderivative qns are not explicitly stated, but can be spotted when you have differentiated something AND you have to integrate something that cannot be done with the formulas



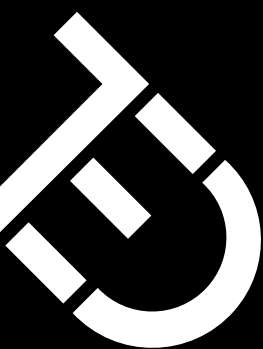
Integration of Reciprocals

under

$$\int \frac{1}{x} dx = \ln(x) + C, x > 0$$

$$\int \frac{1}{ax + b} dx = \frac{1}{a} \ln(ax + b) + C, ax + b > 0$$

integrals of reciprocals where $a \neq 0$



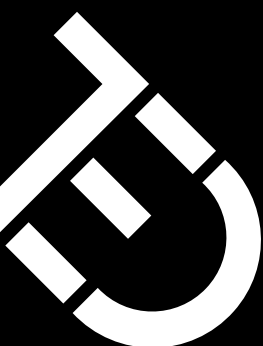
Integration of Reciprocals

Under

Integrate the following:

$$\int \frac{1}{6x - 5} dx$$

$$\int \frac{3}{2 - 5x} dx$$



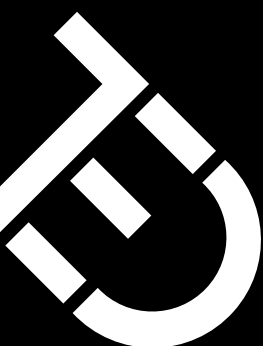
Integration of Reciprocals

Under

Integrate the following:

$$\int \frac{1}{6x-5} dx \qquad \int \frac{1}{6x-5} dx = \frac{\ln(6x-5)}{6} + C$$

$$\int \frac{3}{2-5x} dx$$



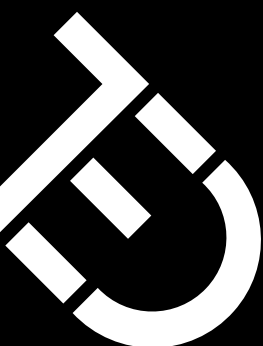
Integration of Reciprocals

Under

Integrate the following:

$$\int \frac{1}{6x-5} dx \qquad \int \frac{3}{2-5x} dx = -\frac{3 \ln(2-5x)}{5} + C$$

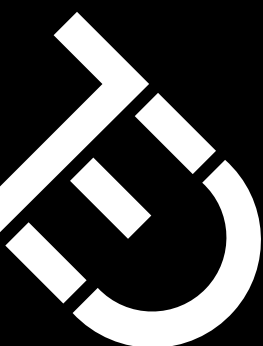
$$\int \frac{3}{2-5x} dx$$



Integration of Reciprocals

Under

Express $\frac{x^2 - 12x - 63}{(x + 2)^2(x - 5)}$ as partial fractions and hence find $\int \frac{x^2 - 12x - 63}{(x + 2)^2(x - 5)} dx$.



Integration of Reciprocals

Under

Express $\frac{x^2 - 12x - 63}{(x + 2)^2(x - 5)}$ as partial fractions and hence find $\int \frac{x^2 - 12x - 63}{(x + 2)^2(x - 5)} dx$.

$$\frac{x^2 - 12x - 63}{(x + 2)^2(x - 5)} \equiv \frac{A}{x + 2} + \frac{B}{(x + 2)^2} + \frac{C}{x - 5}$$

$$x^2 - 12x - 63 = A(x + 2)(x - 5) + B(x - 5) + C(x + 2)^2$$

Let $x = 5$

$$5^2 - 12(5) - 63 = C(5 + 2)^2$$

$$-98 = 49C$$

$$C = -2$$

Let $x = -2$

$$(-2)^2 - 12(-2) - 63 = B(-2 - 5)$$

$$-35 = -7B$$

$$B = 5$$

Let $x = 0$

$$-63 = A(2)(-5) + (5)(-5) + (-2)(2)^2$$

$$-63 = -10A - 25 - 8$$

$$-63 = -10A - 33$$

$$A = 3$$

$$\therefore \frac{x^2 - 12x - 63}{(x + 2)^2(x - 5)} = \frac{3}{x + 2} + \frac{5}{(x + 2)^2} - \frac{2}{x - 5}$$

$$\int \frac{x^2 - 12x - 63}{(x + 2)^2(x - 5)} dx = \int \frac{3}{x + 2} + \frac{5}{(x + 2)^2} - \frac{2}{x - 5} dx$$

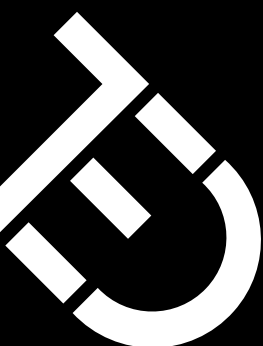
$$= \int \frac{3}{x + 2} + 5(x + 2)^{-2} - \frac{2}{x - 5} dx$$

$$= 3 \ln(x + 2) + \frac{5(x + 2)^{-1}}{-1} - 2 \ln(x - 5) + C$$

$$= 3 \ln(x + 2) - \frac{1}{5(x + 2)} - 2 \ln(x - 5) + C$$

not all of the partial fractions will be $\ln()$ when integrated !!

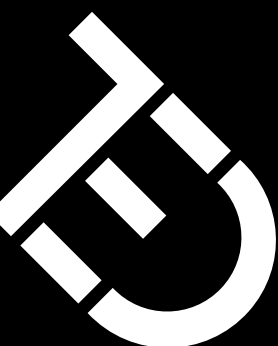
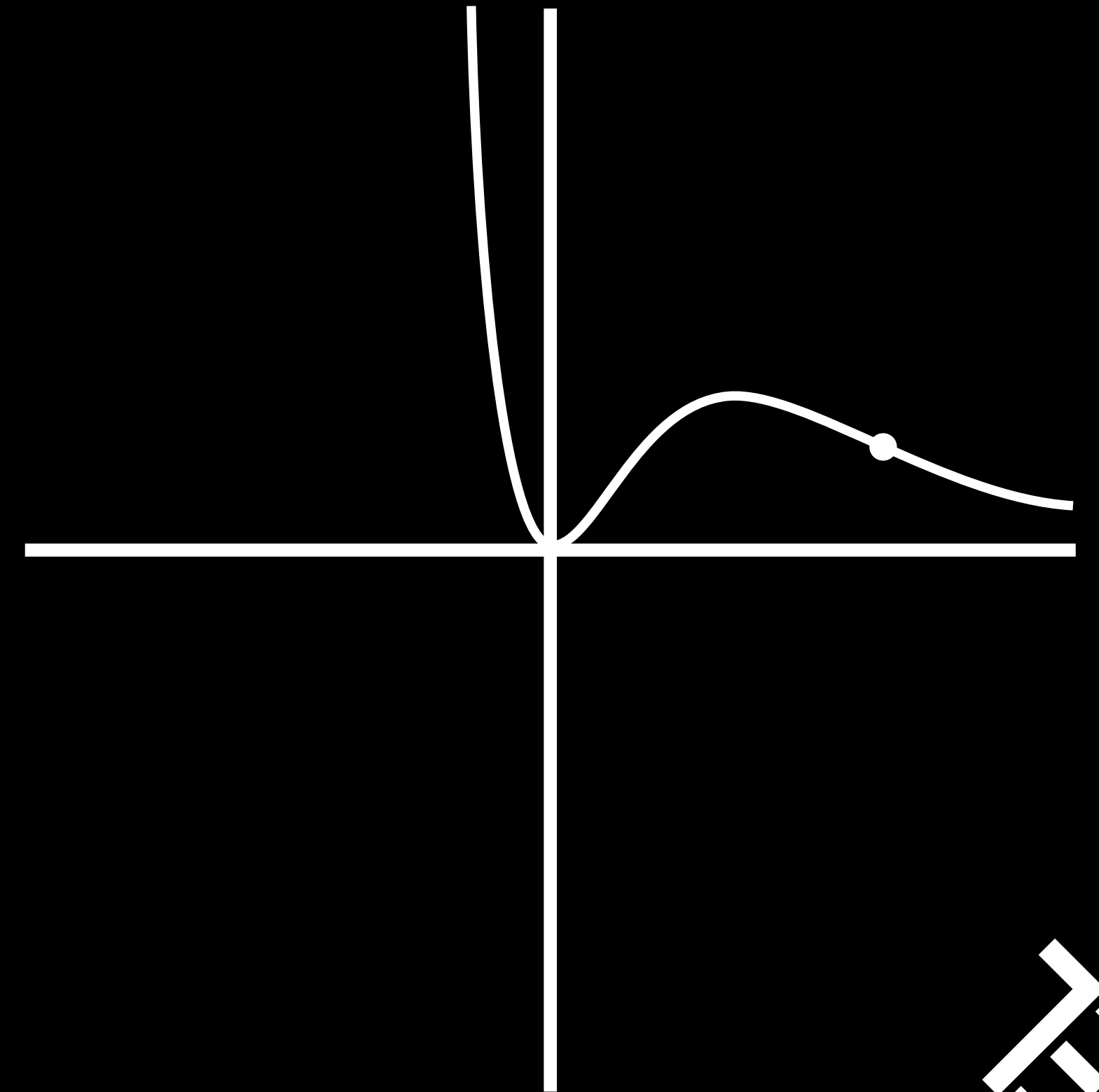
look out for any $\frac{1}{(ax + b)^2}$ as their integral has no $\ln()$



Connected Rate of Change with Integration

wh

A point travels along the curve $y = 3x^2e^{-x}$, such that the x coordinate of the point is increasing at t^{-1} units per second, where t is the number of seconds elapsed. Find the rate of change of the y-coordinate of the point at t=3.



Connected Rate of Change with Integration

wh

$$\frac{dx}{dt} = t^{-2} \text{ units/s}$$

A point travels along the curve $y = 3x^2e^{-x}$, such that the x coordinate of the point is increasing at t^{-2} units per second, where t is the number of seconds elapsed. Given that the particle starts from $x = 1$, find the rate of change of the y-coordinate of the point at t=2.

$$y = 3x^2e^{-x} \quad \text{Solve for } \frac{dy}{dt}$$

$$\begin{aligned} \frac{dy}{dx} &= 3 \left((2x)(e^{-x}) + (-e^{-x})(x^2) \right) \\ &= 3e^{-x} (2x - x^2) \\ &= 3xe^{-x} (2 - x) \end{aligned}$$

$$\begin{aligned} \frac{dy}{dt} &= \frac{dy}{dx} \times \frac{dx}{dt} \\ &= 3xe^{-x} (2 - x) \times t^{-2} \end{aligned}$$

Sub in $t = 2$ and $x = 0.5$:

$$\begin{aligned} \frac{dy}{dt} &= 3(0.5)e^{-0.5} (2 - 0.5) \times 2^{-2} \\ &= 0.341 \text{ units/s (3.s.f)} \end{aligned}$$

Notice how our rate of change equation has both x and t in it, as dx/dt is not a constant

Thus, as we only have t=2 that we can sub in, we need to find the value of x at t=2 by integrating dx/dt to find the function of x

Then, we can sub in both t=2 and x=0.5 into the rate of change equation to find dy/dt

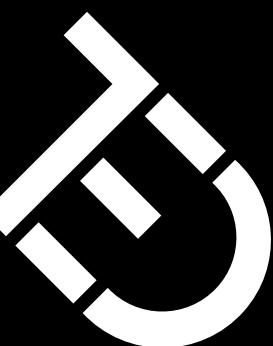
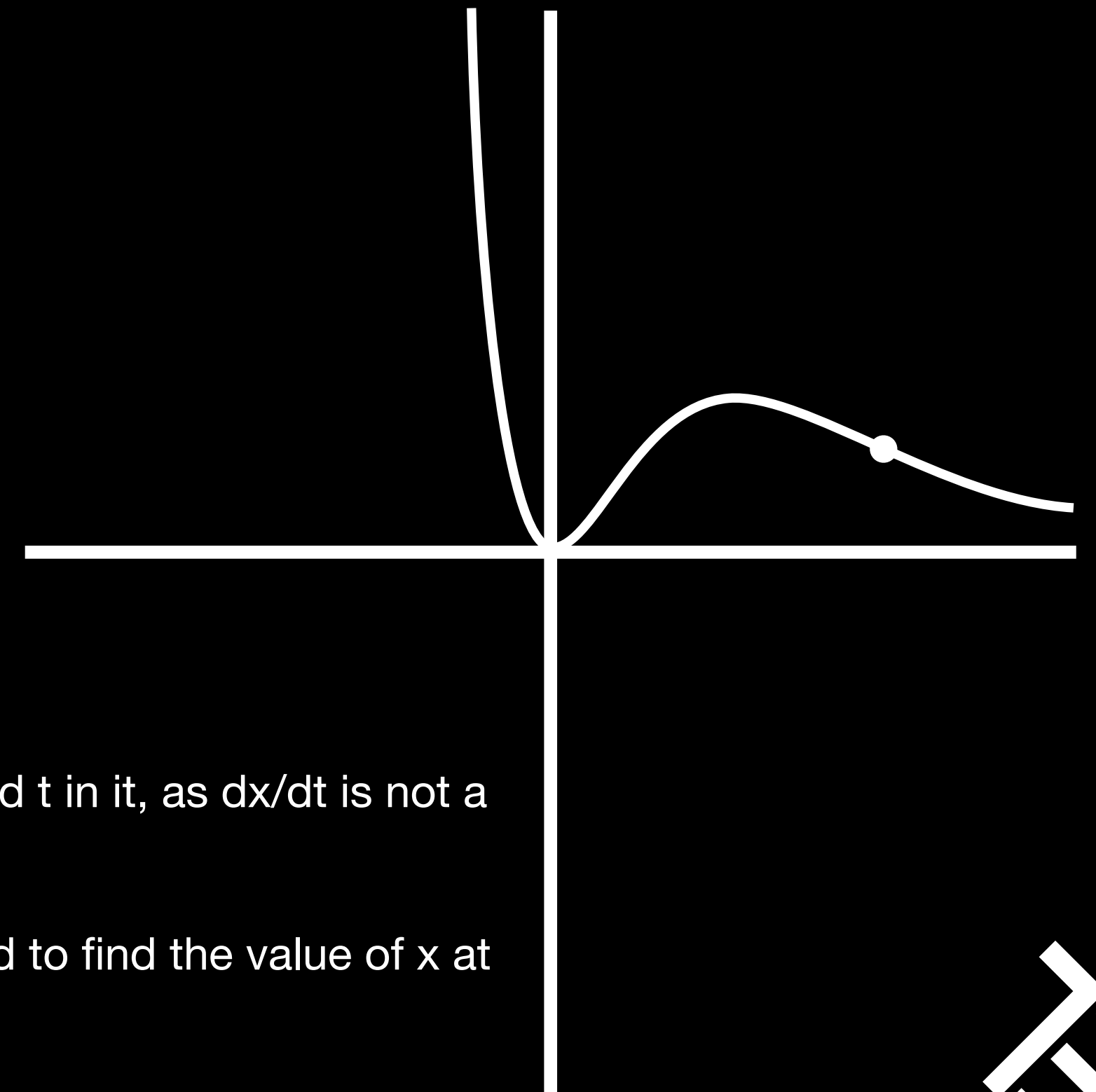
$$\begin{aligned} \frac{dx}{dt} &= t^{-2} \\ x &= \int \frac{dx}{dt} dt \\ &= \int t^{-2} dt \\ &= -t^{-1} + C \end{aligned}$$

At $t = 0, x = 1$. Thus,

$$\begin{aligned} 1 &= -0^{-1} + C \\ C &= 1 \\ x &= -t^{-1} + 1 \end{aligned}$$

At $t = 2$,

$$\begin{aligned} x &= -2^{-1} + 1 \\ x &= 0.5 \end{aligned}$$



Applications of Integration



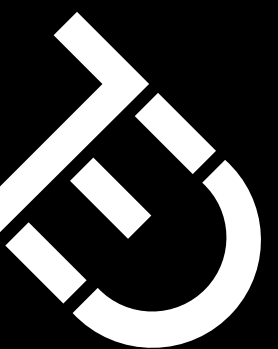
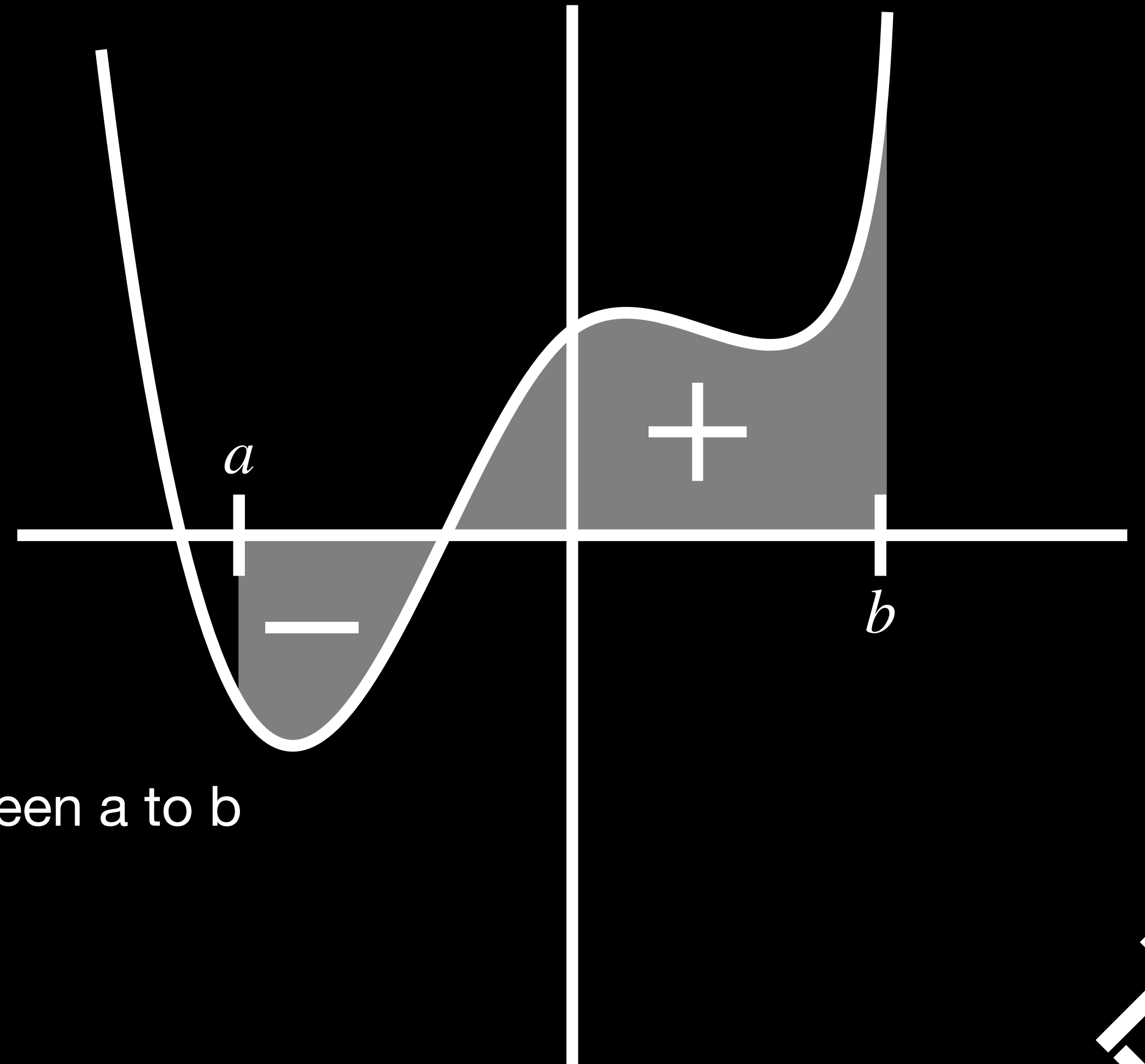
Definite Integrals

thank you for purchasing shoelace method+

$$\int_a^b f(x) \, dx$$

definite integrals

finds signed area bounded between curve and x-axis between a to b



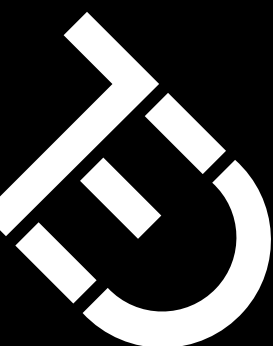
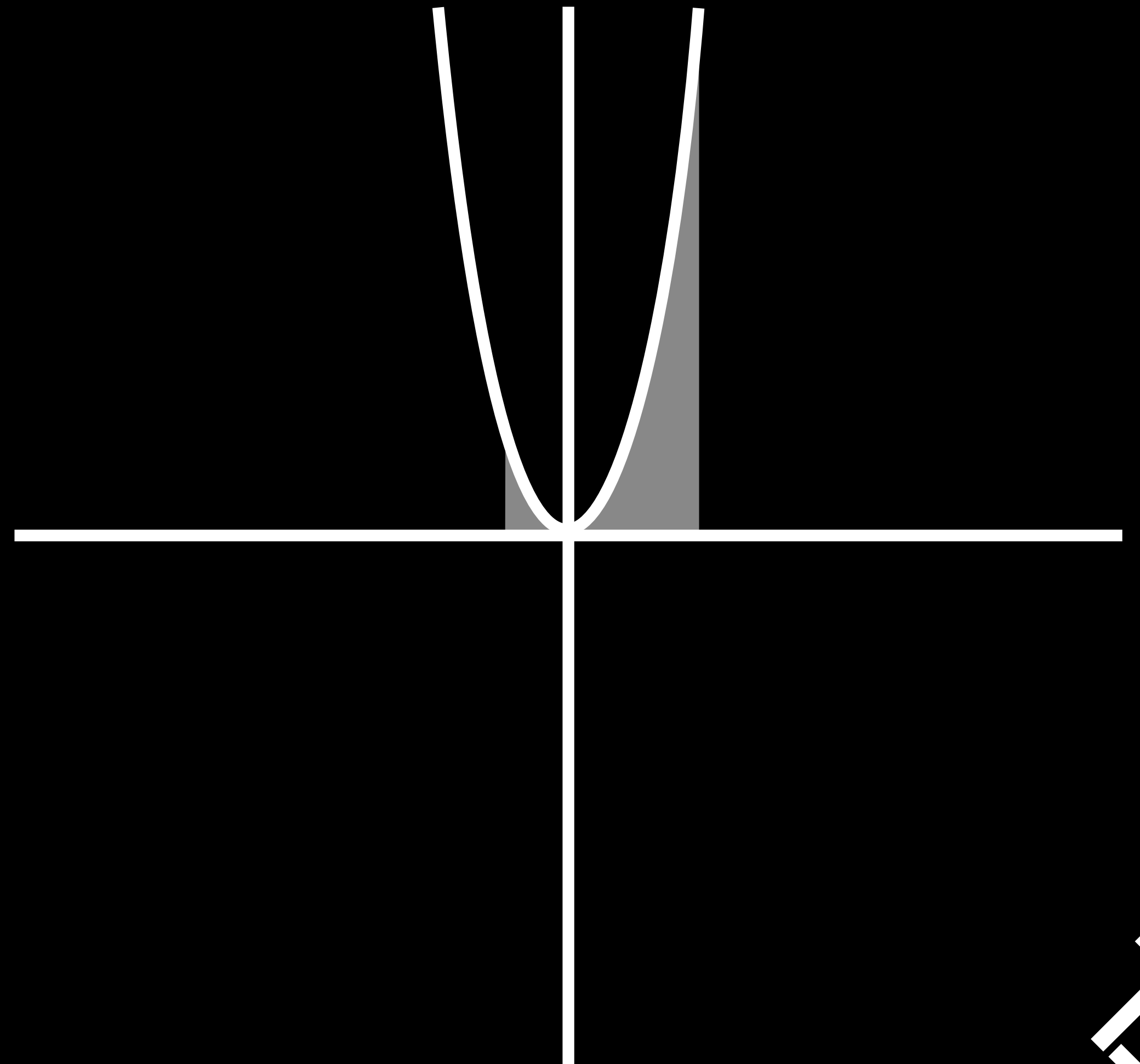
Definite Integrals

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To evaluate definite integrals:

$$\begin{aligned}\int_a^b f(x) \, dx &= [F(x)]_a^b \\ &= F(b) - F(a)\end{aligned}$$

$$\begin{aligned}\int_{-1}^4 3x^2 \, dx &= [x^3]_{-1}^4 \\ &= 4^3 - (-1)^3 \\ &= 65\end{aligned}$$



Definite Integrals

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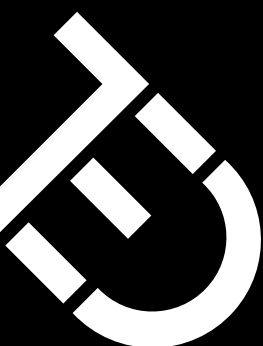
Evaluate the following:

$$\int_{-2}^3 x + (2x - 3)^2 \, dx$$

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{6}} \frac{\sin(x)}{\tan(x)} \, dx$$

$$\int_2^3 e^{2x-3} - \frac{1}{2x-3} \, dx$$

$$\int_{-1}^4 2 \cos(x) \sin(x) + \sqrt{3+2x} \, dx$$

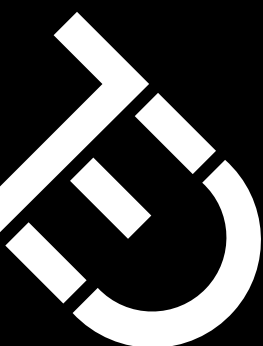


Definite Integrals

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Evaluate the following:

$$\int_{-2}^3 x + (2x - 3)^2 dx = \left[\frac{x^2}{2} + \frac{(2x - 3)^3}{6} \right]_{-2}^3$$
$$= \left[\frac{3x^2 + (2x - 3)^3}{6} \right]_{-2}^3$$
$$= \frac{3(3)^2 + (2(3) - 3)^3}{6} - \frac{3(-2)^2 + (2(-2) - 3)^3}{6}$$
$$= 9 - -\frac{331}{6}$$
$$= 64.2(3.s.f)$$
$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{6}} \frac{\sin(x)}{\tan(x)} dx$$
$$\int_2^3 e^{2x-3} - \frac{1}{2x-3} dx$$
$$\int_{-1}^4 2 \cos(x) \sin(x) + \sqrt{3+2x} dx$$



Definite Integrals

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Evaluate the following:

$$\int_{-2}^3 x + (2x - 3)^2 dx$$

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{6}} \frac{\sin(x)}{\tan(x)} dx$$

$$\int_2^3 e^{2x-3} - \frac{1}{2x-3} dx$$

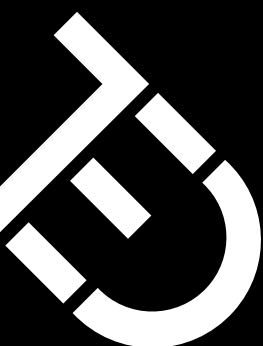
$$\int_{-1}^4 2 \cos(x) \sin(x) + \sqrt{3+2x} dx$$

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{6}} \frac{\sin(x)}{\tan(x)} dx = \int_{-\frac{\pi}{2}}^{\frac{\pi}{6}} \cos(x) dx$$

$$= [\sin(x)]_{-\frac{\pi}{2}}^{\frac{\pi}{6}}$$

$$= \sin\left(\frac{\pi}{6}\right) - \sin\left(-\frac{\pi}{2}\right)$$

$$= 1.5$$

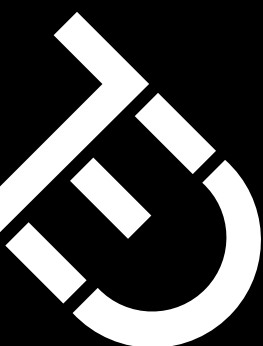


Definite Integrals

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Evaluate the following:

$$\begin{aligned} \int_{-2}^3 x + (2x - 3)^2 dx & \quad \int_2^3 e^{2x-3} - \frac{1}{2x-3} dx = \left[\frac{e^{2x-3}}{2} - \frac{\ln(2x-3)}{2} \right]_2^3 \\ \int_{-\frac{\pi}{2}}^{\frac{\pi}{6}} \frac{\sin(x)}{\tan(x)} dx & \quad = \left[\frac{e^{2x-3} - \ln(2x-3)}{2} \right]_2^3 \\ \int_2^3 e^{2x-3} - \frac{1}{2x-3} dx & \quad = \frac{e^{2(3)-3} - \ln(2(3)-3)}{2} - \frac{e^{2(2)-3} - \ln(2(2)-3)}{2} \\ \int_{-1}^4 2 \cos(x) \sin(x) + \sqrt{3+2x} dx & \quad = 8.13(3.s.f) \end{aligned}$$



Definite Integrals

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Evaluate the following:

$$\int_{-2}^3 x + (2x - 3)^2 dx$$

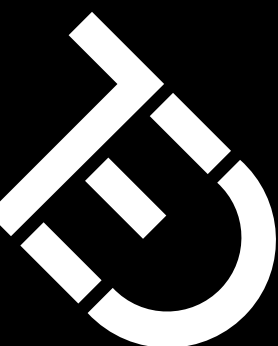
$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{6}} \frac{\sin(x)}{\tan(x)} dx$$

$$\int_2^3 e^{2x-3} - \frac{1}{2x-3} dx$$

$$\int_{-1}^4 2 \cos(x) \sin(x) + \sqrt{3+2x} dx$$

**in radians (all calc qns are in radians)

$$\begin{aligned} \int_{-1}^4 2 \cos(x) \sin(x) + \sqrt{3+2x} dx &= \int_{-1}^4 \sin(2x) + \sqrt{3+2x} dx \\ &= \left[-\frac{\cos(2x)}{2} + \frac{(3+2x)^{1.5}}{3} \right]_{-1}^4 \\ &= -\frac{\cos(2(4))}{2} + \frac{(3+2(4))^{1.5}}{3} - \left(-\frac{\cos(2(-1))}{2} + \frac{(3+2(-1))^{1.5}}{3} \right) \\ &= 11.7(3.s.f) \end{aligned}$$



Definite Integrals

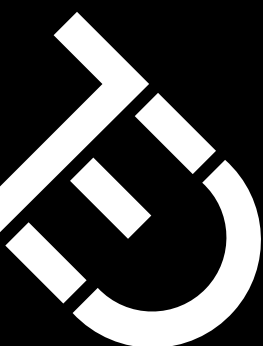
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Definite integrals have these properties:

$$\left(\int_b^a f(x) dx = -\int_a^b f(x) dx\right)$$

$$\int_a^{\textcircled{b}} f(x) dx + \int_{\textcircled{b}}^c f(x) dx = \int_a^c f(x) dx$$

$$\int_a^b f(x) dx \pm \int_a^b g(x) dx = \int_a^b f(x) \pm g(x) dx$$



Definite Integrals

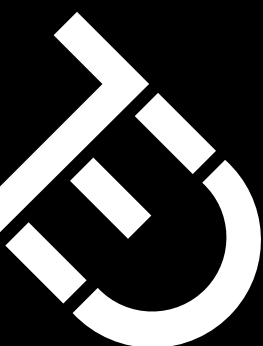
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Given that $\int_1^5 f(x) \, dx = k$, find

$$\int_5^1 2f(x) \, dx$$

$$\int_1^3 f(x) \, dx - \int_5^3 f(x)$$

$$\int_1^5 3f(x) + 4 \, dx$$



Definite Integrals

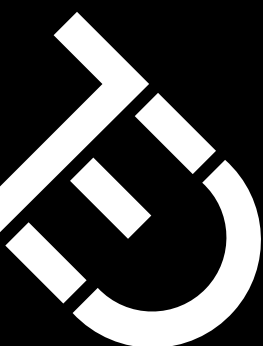
thank you for purchasing shoelace method+

Given that $\int_1^5 f(x) \, dx = k$, find

$$\int_5^1 2f(x) \, dx \qquad \int_5^1 2f(x) \, dx = -2 \int_1^5 f(x) \, dx = -2k$$

$$\begin{aligned} \int_1^3 f(x) \, dx - \int_5^3 f(x) \, dx & \quad \int_1^3 f(x) \, dx - \int_5^3 f(x) \, dx = \int_1^3 f(x) \, dx + \int_3^5 f(x) \\ & \quad = \int_1^5 f(x) \, dx = k \end{aligned}$$

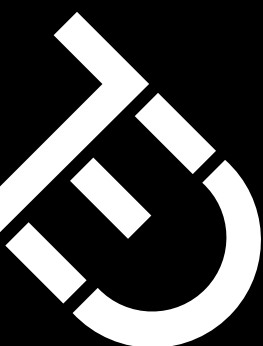
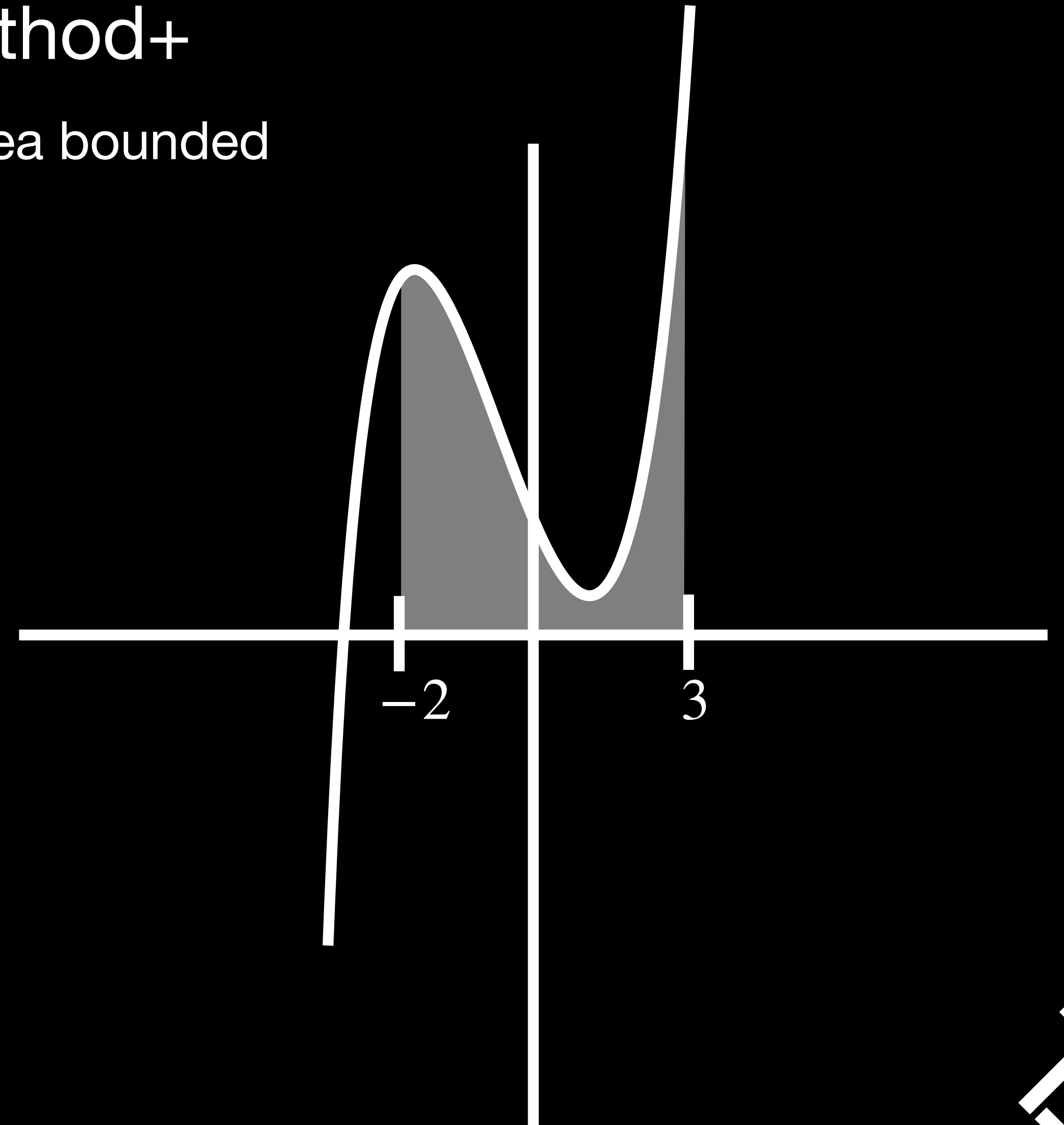
$$\begin{aligned} \int_1^5 3f(x) + 4 \, dx & \quad \int_1^5 3f(x) + 4 \, dx = 3 \int_1^5 f(x) \, dx + \int_1^5 4 \, dx \\ & \quad = 3k + [4x]_1^5 \\ & \quad = 3k + (4(5) - 4(1)) \\ & \quad = 3k + 16 \end{aligned}$$



Bounded Area - Curve & X axis

thank you for purchasing shoelace method+

Shown is the graph of $x^3 + 2x^2 - 3x + 3$. Find the area bounded by the curve and the x axis from -2 to 3



Bounded Area - Curve & X axis

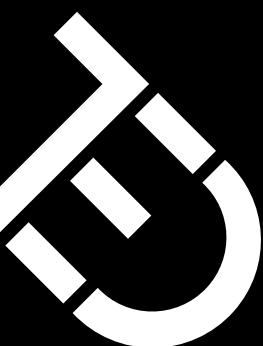
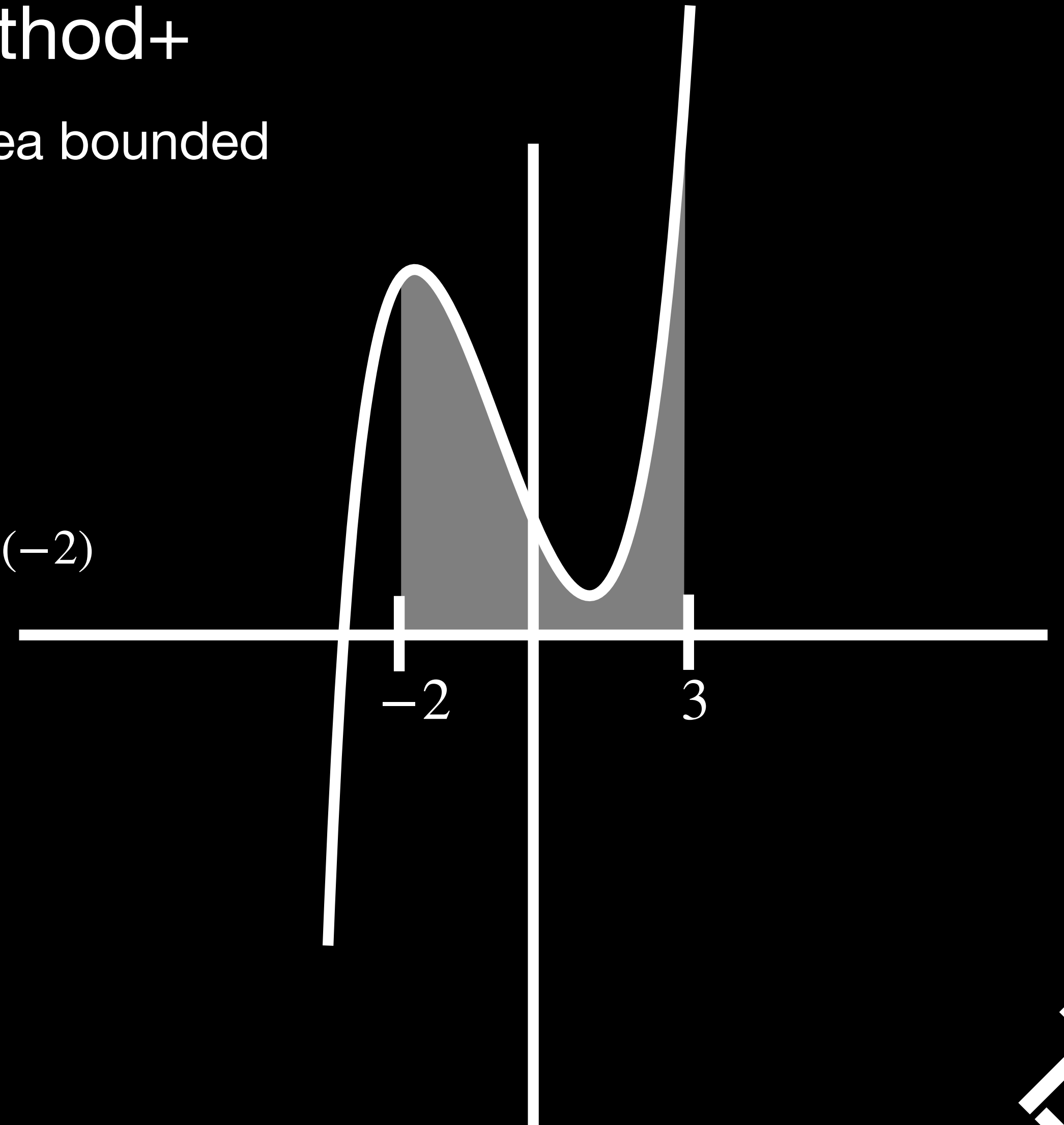
thank you for purchasing shoelace method+

Shown is the graph of $x^3 + 2x^2 - 3x + 3$. Find the area bounded by the curve and the x axis from -2 to 3

$$\int_{-2}^3 x^3 + 2x^2 - 3x + 3 \, dx = \left[\frac{x^4}{4} + \frac{2x^3}{3} - \frac{3x^2}{2} + 3x \right]_{-2}^3$$

$$= \frac{3^4}{4} + \frac{2(3)^3}{3} - \frac{3(3)^2}{2} + 3(3) - \left(\frac{(-2)^4}{4} + \frac{2(-2)^3}{3} - \frac{3(-2)^2}{2} + 3(-2) \right)$$

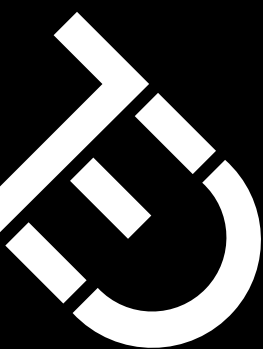
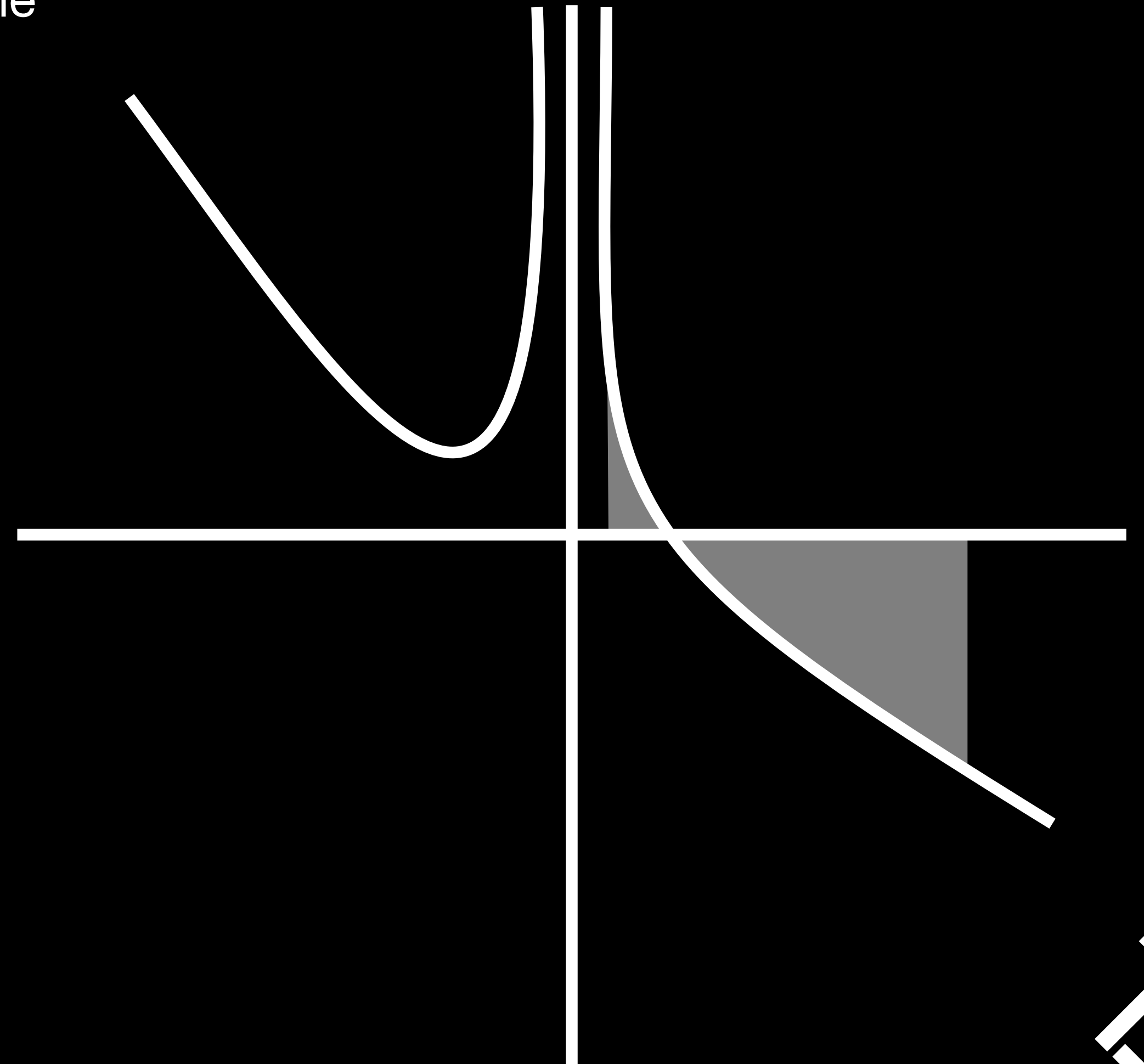
$$= 47.1 \text{ units}^2 (3.s.f.)$$



Bounded Area - Curve & X axis

thank you for purchasing shoelace method+

The function $\frac{1}{x^2} - x$ is sketched. Find the area bounded by the curve and the x-axis from 0.5 to 4



Bounded Area - Curve & X axis

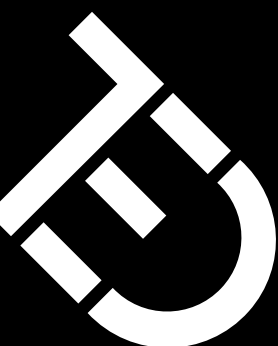
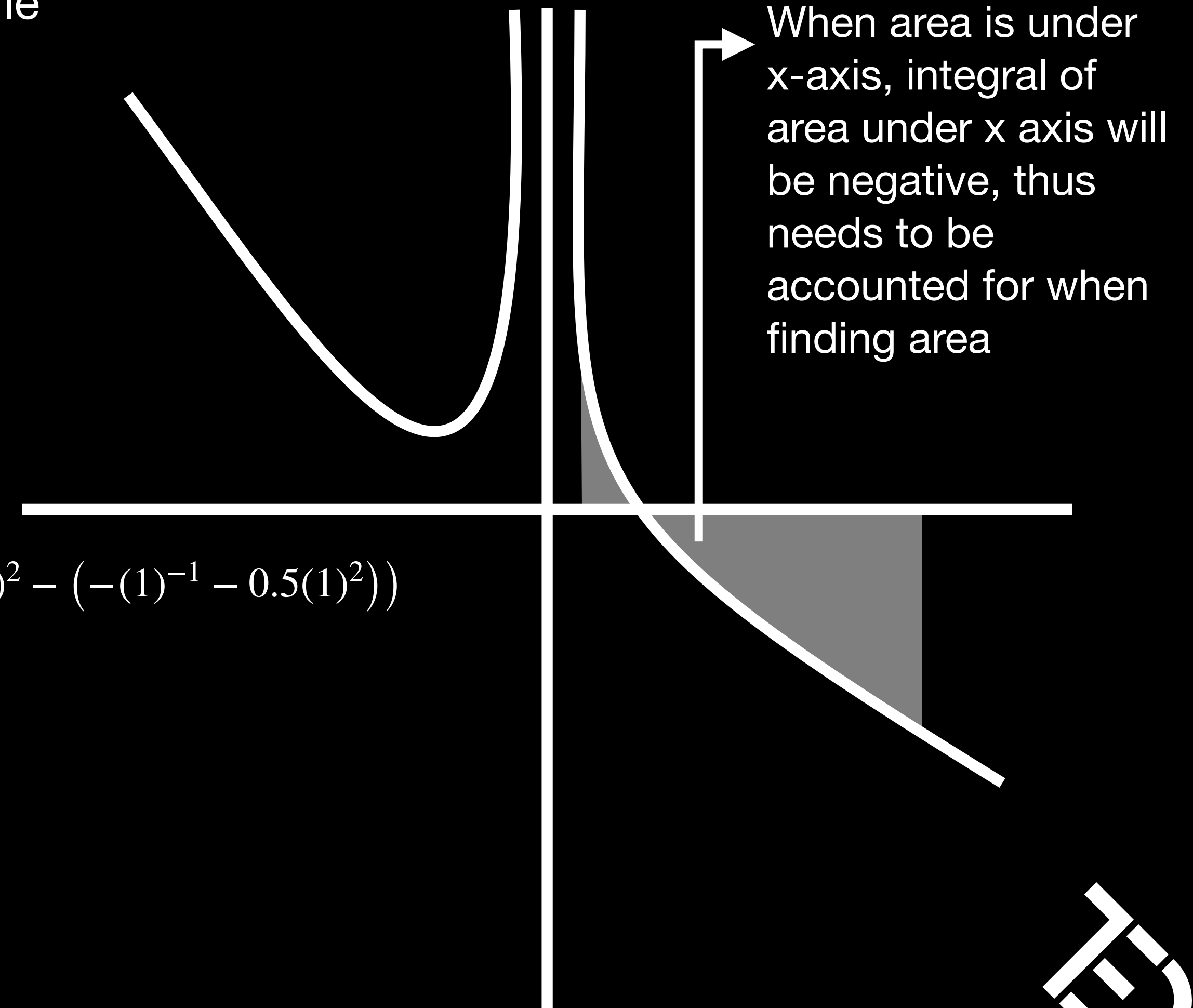
thank you for purchasing shoelace method+

The function $\frac{1}{x^2} - x$ is sketched. Find the area bounded by the curve and the x-axis from 0.5 to 4

$$\begin{aligned}\frac{1}{x^2} - x &= 0 \\ \frac{1}{x^2} &= x \\ 1 &= x^3 \\ x &= 1\end{aligned}$$

Thus, area is:

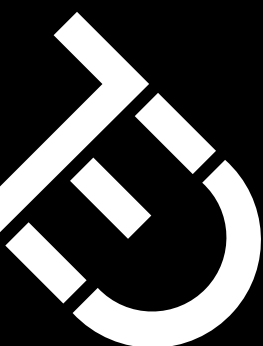
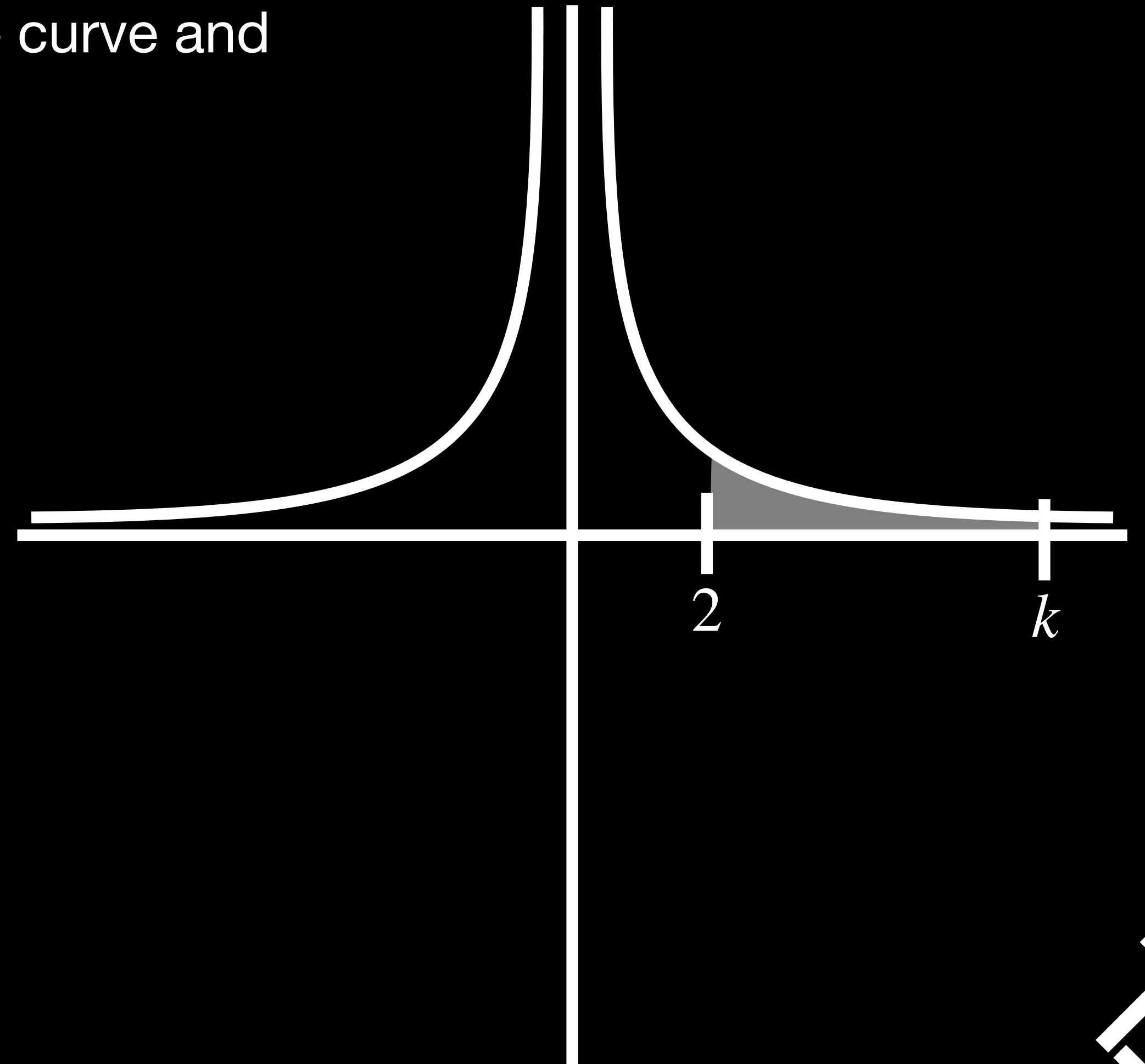
$$\begin{aligned}&\left(\int_{0.5}^1 \frac{1}{x^2} - x \, dx \right) + \left(- \int_1^4 \frac{1}{x^2} - x \, dx \right) \\&= \left[-x^{-1} - 0.5x^2 \right]_{0.5}^1 - \left[-x^{-1} - 0.5x^2 \right]_1^4 \\&= - (1)^{-1} - 0.5(1)^2 - \left(-(0.5)^{-1} - 0.5(0.5)^2 \right) - \left(-(4)^{-1} - 0.5(4)^2 - \left(-(1)^{-1} - 0.5(1)^2 \right) \right) \\&= 7.375 \text{ units}^2\end{aligned}$$



Bounded Area - Curve & X axis

thank you for purchasing shoelace method+

The function $\frac{3}{x^2}$ is sketched. The area bounded by the curve and the x axis is 1.2 units² from 2 to a constant k. Find k.



Bounded Area - Curve & X axis

thank you for purchasing shoelace method+

The function $\frac{3}{x^2}$ is sketched. The area bounded by the curve and the x axis is 1.2 units² from 2 to a constant k. Find k.

$$\int_2^k \frac{3}{x^2} dx = 1.2$$

$$[-3x^{-1}]_2^k = 1.2$$

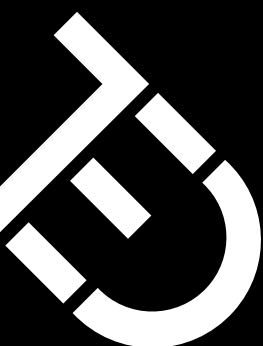
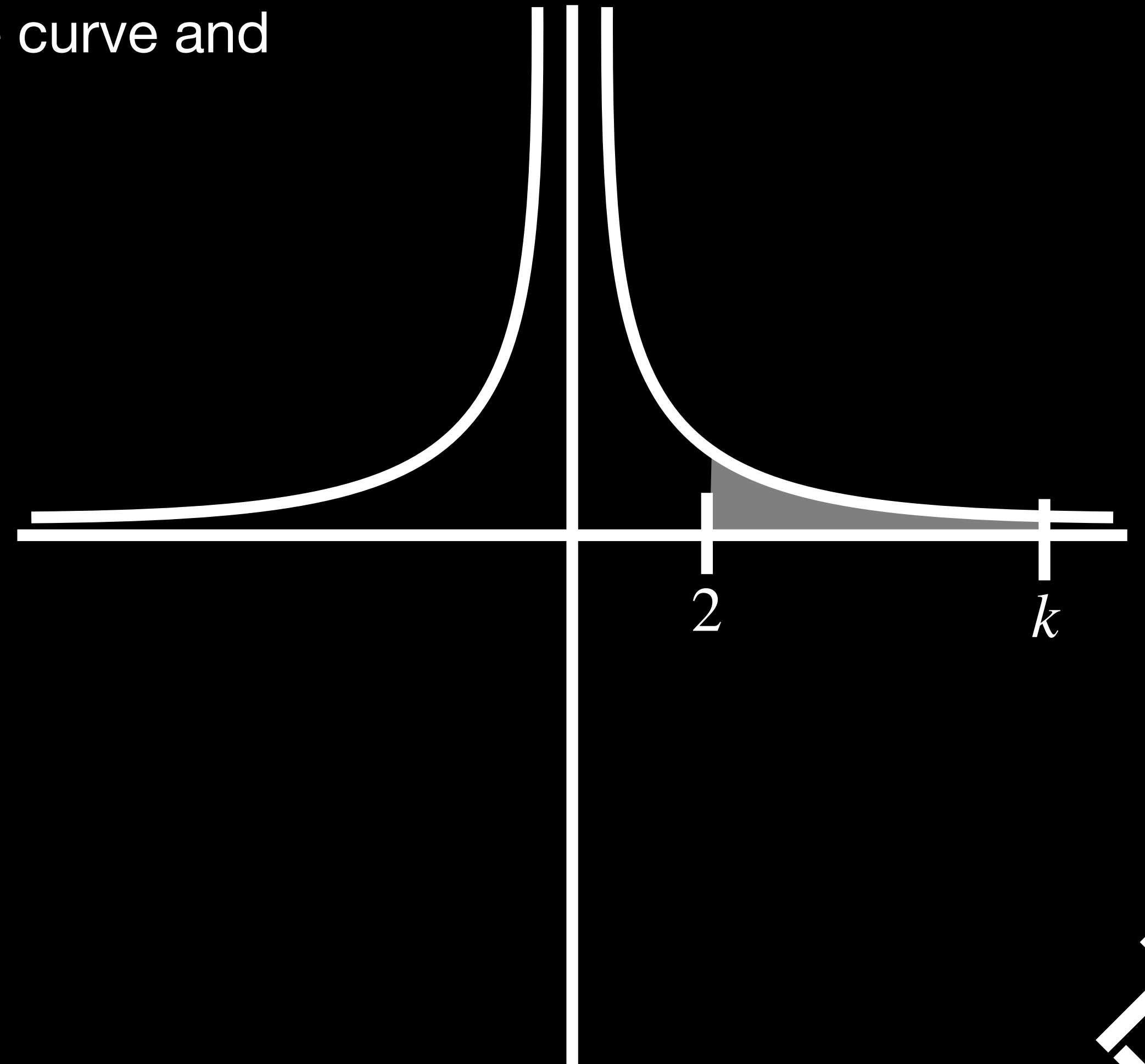
$$-3k^{-1} - (-3(2)^{-1}) = 1.2$$

$$-3k^{-1} + 1.5 = 1.2$$

$$-3k^{-1} = -0.3$$

$$k^{-1} = 0.1$$

$$k = 10$$

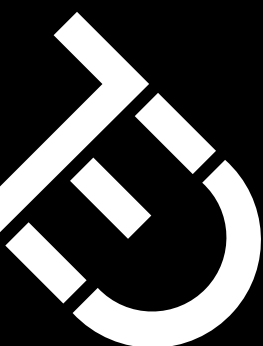


Bounded Area - Curve & X axis

thank you for purchasing shoelace method+

Find the area bounded by the x axis and the curve

$$y = \sin \left(x + \frac{\pi}{4} \right) \text{ from } -\frac{\pi}{2} \leq x \leq \pi$$



Bounded Area - Curve & X axis

thank you for purchasing shoelace method+

Find the area bounded by the x axis and the curve

$$y = \sin\left(x + \frac{\pi}{4}\right) \text{ from } -\frac{\pi}{2} \leq x \leq \pi$$

$$\sin\left(x + \frac{\pi}{4}\right) = 0$$

$$\alpha = \sin^{-1}(0)$$

$$\alpha = 0$$

$$x + \frac{\pi}{4} = 0, \pi$$

$$x = 0 - \frac{\pi}{4}, \pi - \frac{\pi}{4}$$

$$x = -\frac{\pi}{4}, \frac{3\pi}{4}$$

domain:

$$-\frac{\pi}{2} \leq x \leq \pi$$

$$-\frac{\pi}{2} + \frac{\pi}{4} \leq x + \frac{\pi}{4} \leq \pi + \frac{\pi}{4}$$

$$-\frac{\pi}{4} \leq x + \frac{\pi}{4} \leq \frac{5\pi}{4}$$

$$\int_{-\frac{\pi}{4}}^{\frac{3\pi}{4}} \sin\left(x + \frac{\pi}{4}\right) dx + \left(-\int_{-\frac{\pi}{2}}^{-\frac{\pi}{4}} \sin\left(x + \frac{\pi}{4}\right) dx\right) + \left(-\int_{\frac{3\pi}{4}}^{\pi} \sin\left(x + \frac{\pi}{4}\right) dx\right)$$

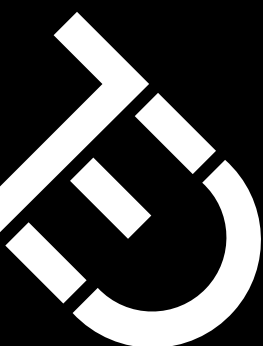
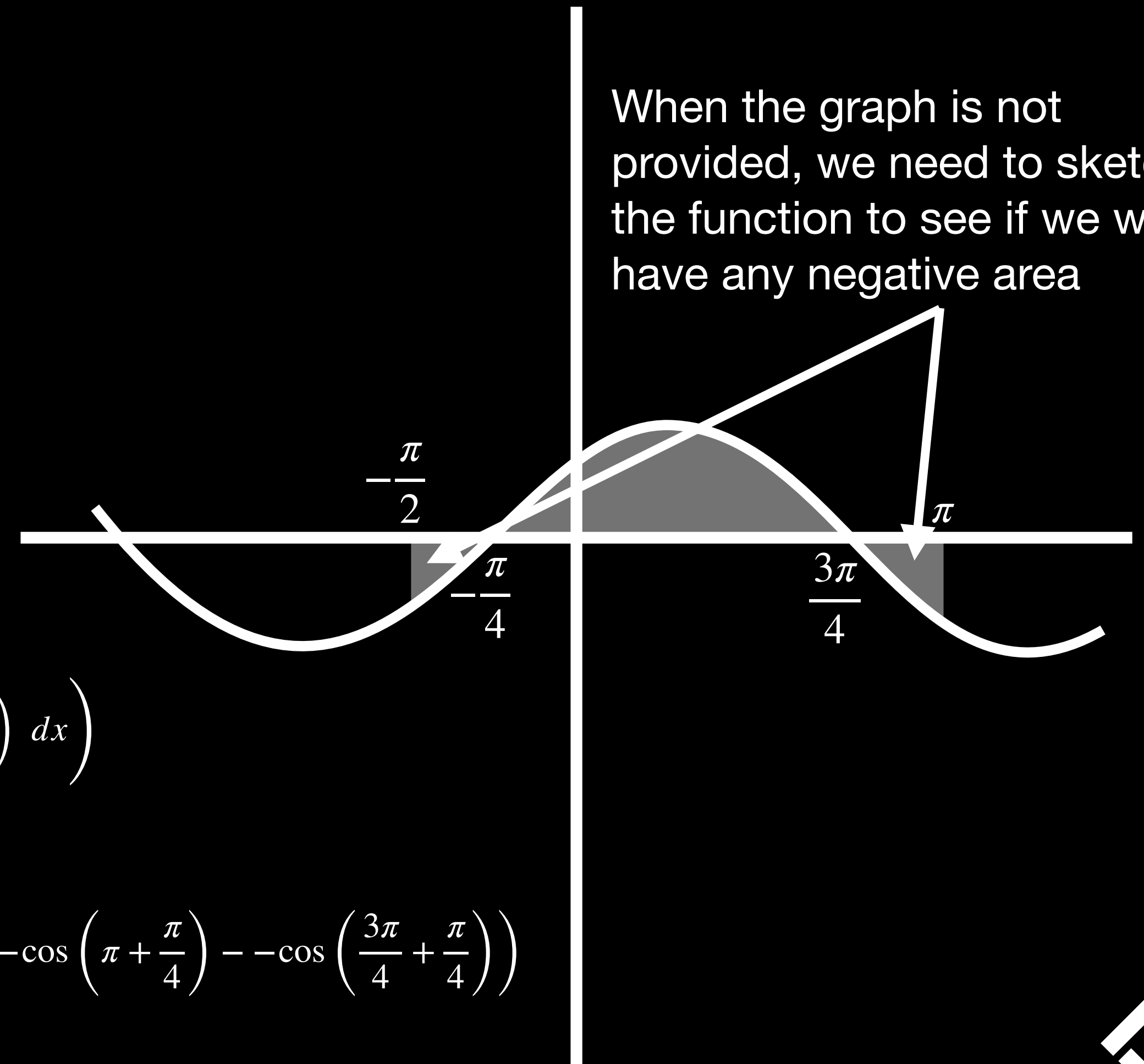
$$= \left[-\cos\left(x + \frac{\pi}{4}\right)\right]_{-\frac{\pi}{4}}^{\frac{3\pi}{4}} - \left[-\cos\left(x + \frac{\pi}{4}\right)\right]_{-\frac{\pi}{2}}^{-\frac{\pi}{4}} - \left[-\cos\left(x + \frac{\pi}{4}\right)\right]_{\frac{3\pi}{4}}^{\pi}$$

$$= \left(-\cos\left(\frac{3\pi}{4} + \frac{\pi}{4}\right) - -\cos\left(-\frac{\pi}{4} + \frac{\pi}{4}\right)\right) - \left(-\cos\left(-\frac{\pi}{4} + \frac{\pi}{4}\right) - -\cos\left(-\frac{\pi}{2} + \frac{\pi}{4}\right)\right) - \left(-\cos\left(\pi + \frac{\pi}{4}\right) - -\cos\left(\frac{3\pi}{4} + \frac{\pi}{4}\right)\right)$$

$$= 2 - (-0.29289) - (-0.29289)$$

$$= 2.59 \text{ units}^2(3.s.f)$$

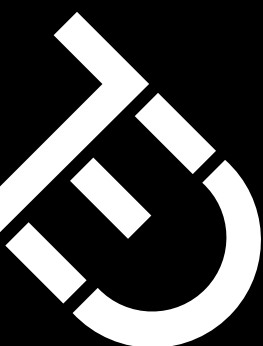
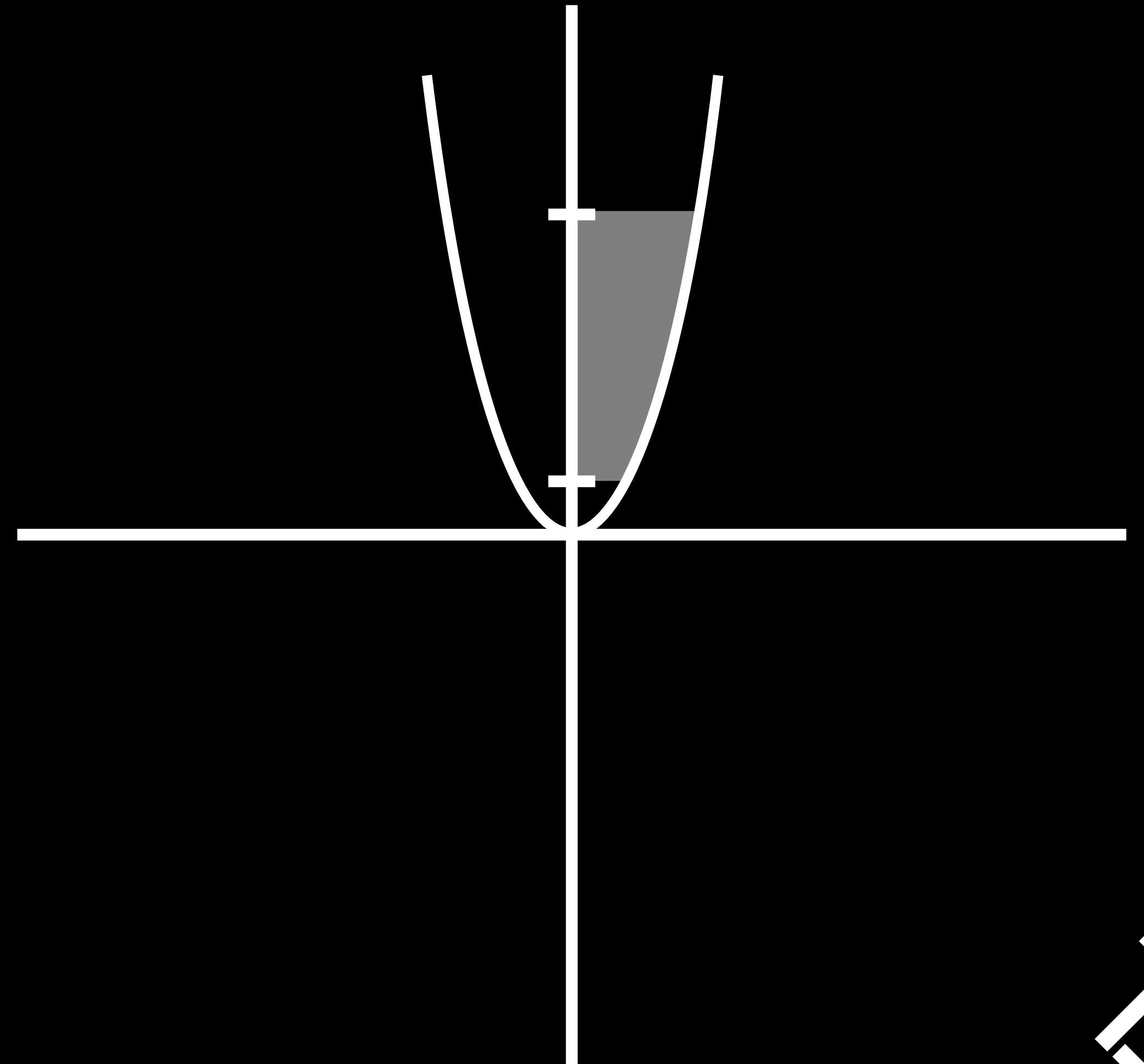
When the graph is not provided, we need to sketch the function to see if we will have any negative area



Bounded Area - Curve & Y axis

thank you for purchasing shoelace method+

The function $y = 2x^2$ is sketched. Find the area bounded by the curve and the y-axis from 1 to 4.



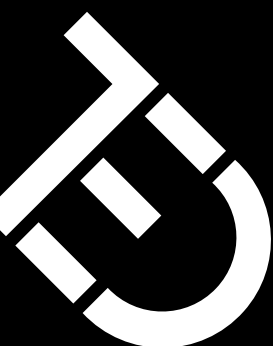
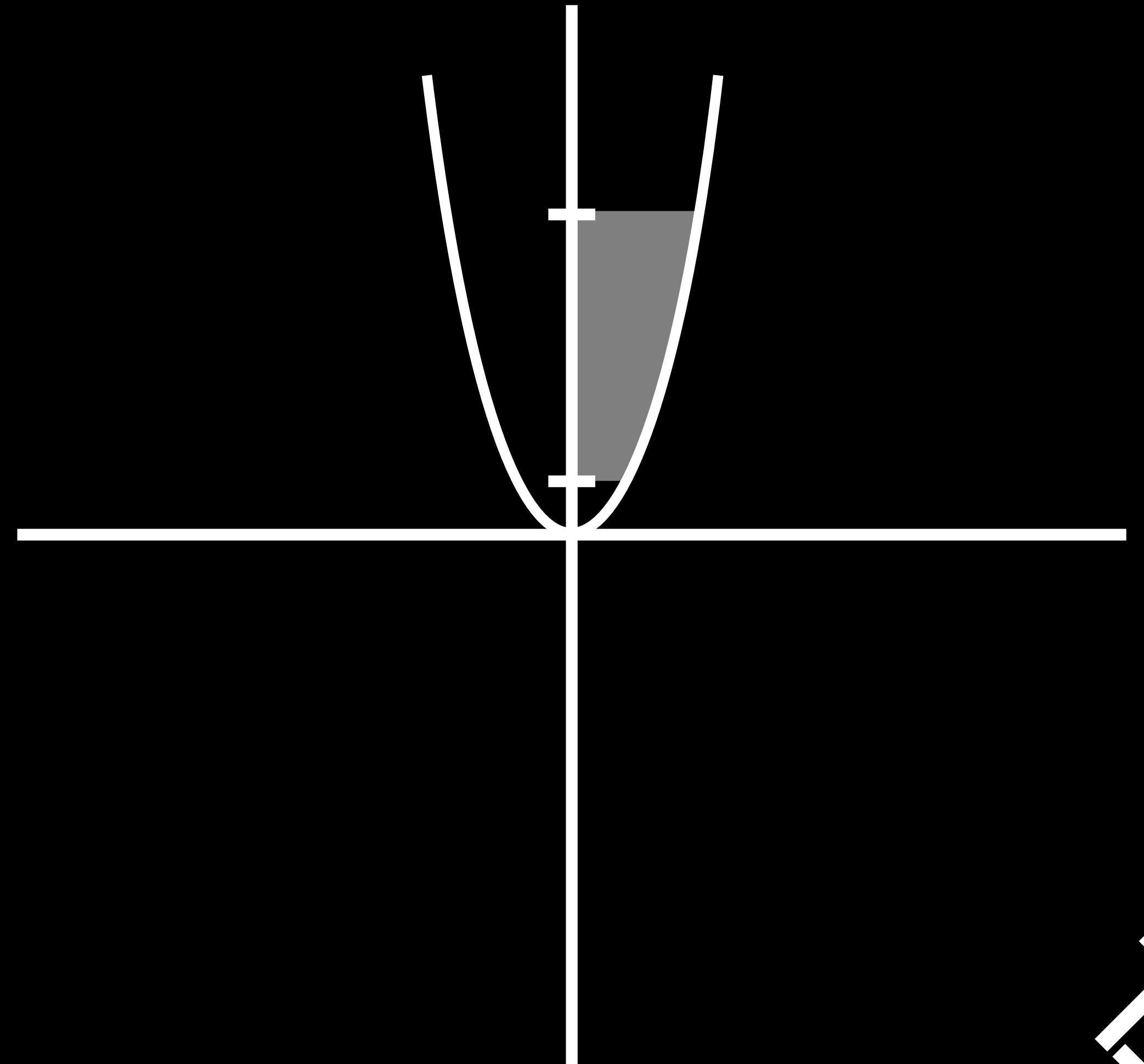
Bounded Area - Curve & Y axis

thank you for purchasing shoelace method+

The function $y = 2x^2$ is sketched. Find the area bounded by the curve and the y-axis from 1 to 4.

$$\begin{aligned} y &= 2x^2 \\ \frac{y}{2} &= x^2 \\ x &= \sqrt{\frac{y}{2}} \\ x &= \sqrt{0.5y} \end{aligned} \quad \begin{aligned} \int_1^4 \sqrt{0.5y} \, dy &= \left[\frac{(0.5y)^{1.5}}{0.5(1.5)} \right]_1^4 \\ &= \frac{(0.5(4))^{1.5}}{0.75} - \frac{(0.5(1))^{1.5}}{0.75} \\ &= 3.30 \text{ units}^2 (3.s.f) \end{aligned}$$

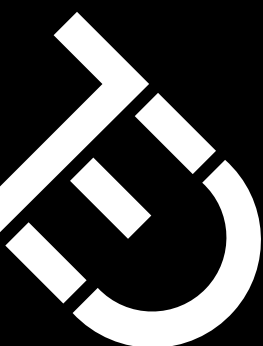
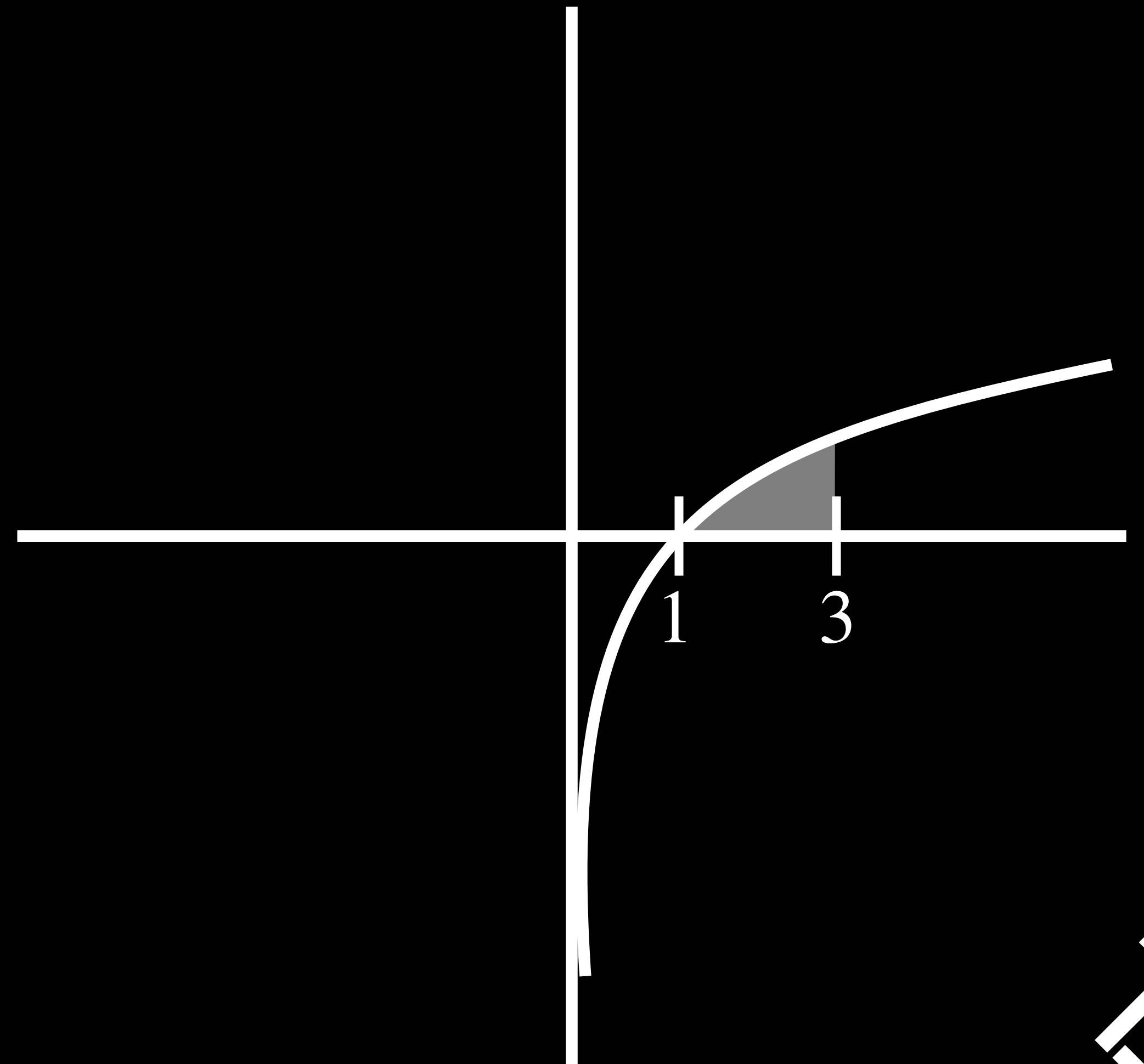
make x subj to integrate against y



Bounded Area - Curve & Y axis

thank you for purchasing shoelace method+

The function $y = \ln(x)$ is sketched. Find the area bounded by the curve and the x-axis from 1 to 3.



Bounded Area - Curve & Y axis

thank you for purchasing shoelace method+

The function $y = \ln(x)$ is sketched. Find the area bounded by the curve and the x-axis from 1 to 3.

$$y = \ln(x)$$

$$x = e^y$$

$$\text{Area bounded} = 3 \text{ by } \ln(3) \text{ rectangle} - \int_0^{\ln(3)} e^y dy$$

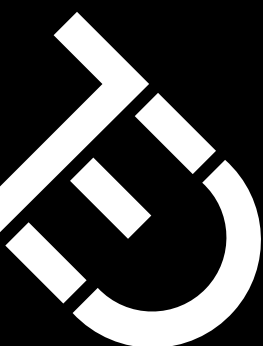
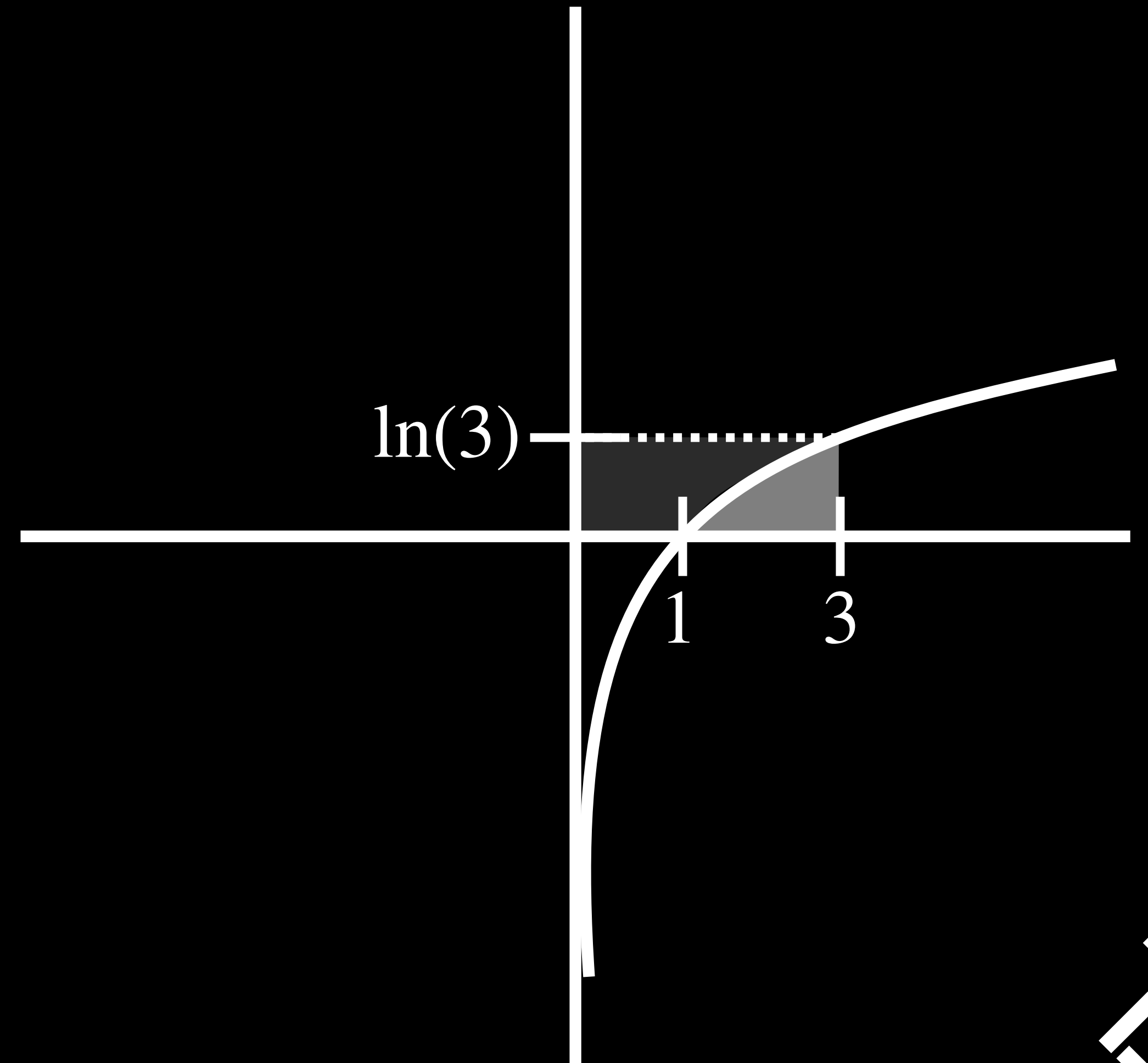
$$= 3 \ln(3) - [e^y]_0^{\ln(3)}$$

$$= 3 \ln(3) - (e^{\ln(3)} - e^0)$$

$$= 3 \ln(3) - (3 - 1)$$

$$= 3 \ln(3) - 2$$

$$= 1.30 \text{ units}^2 (3.s.f)$$

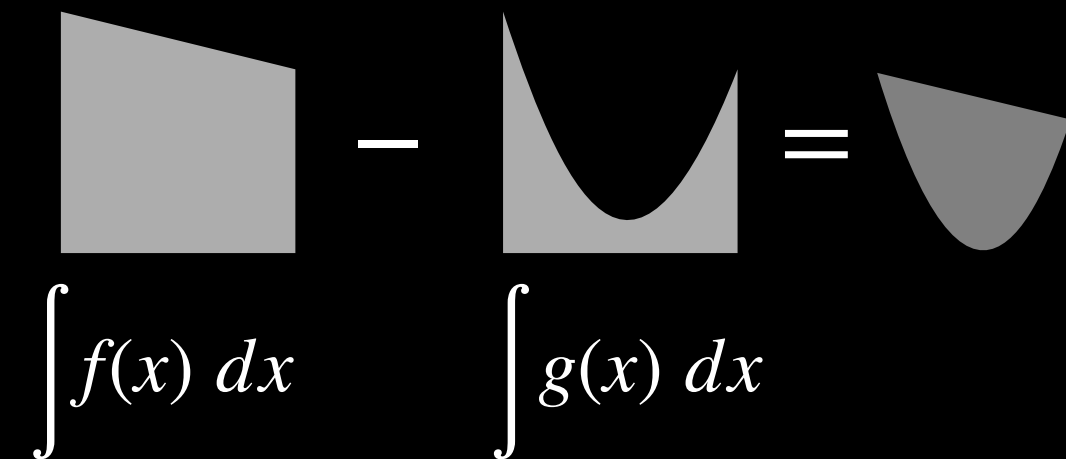


Bounded Area - Curve & Line

thank you for purchasing shoelace method+

When finding the area bounded between multiple functions, add and subtract integrals from each other to get the desired value

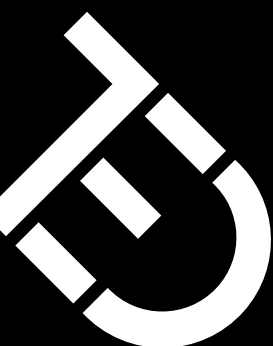
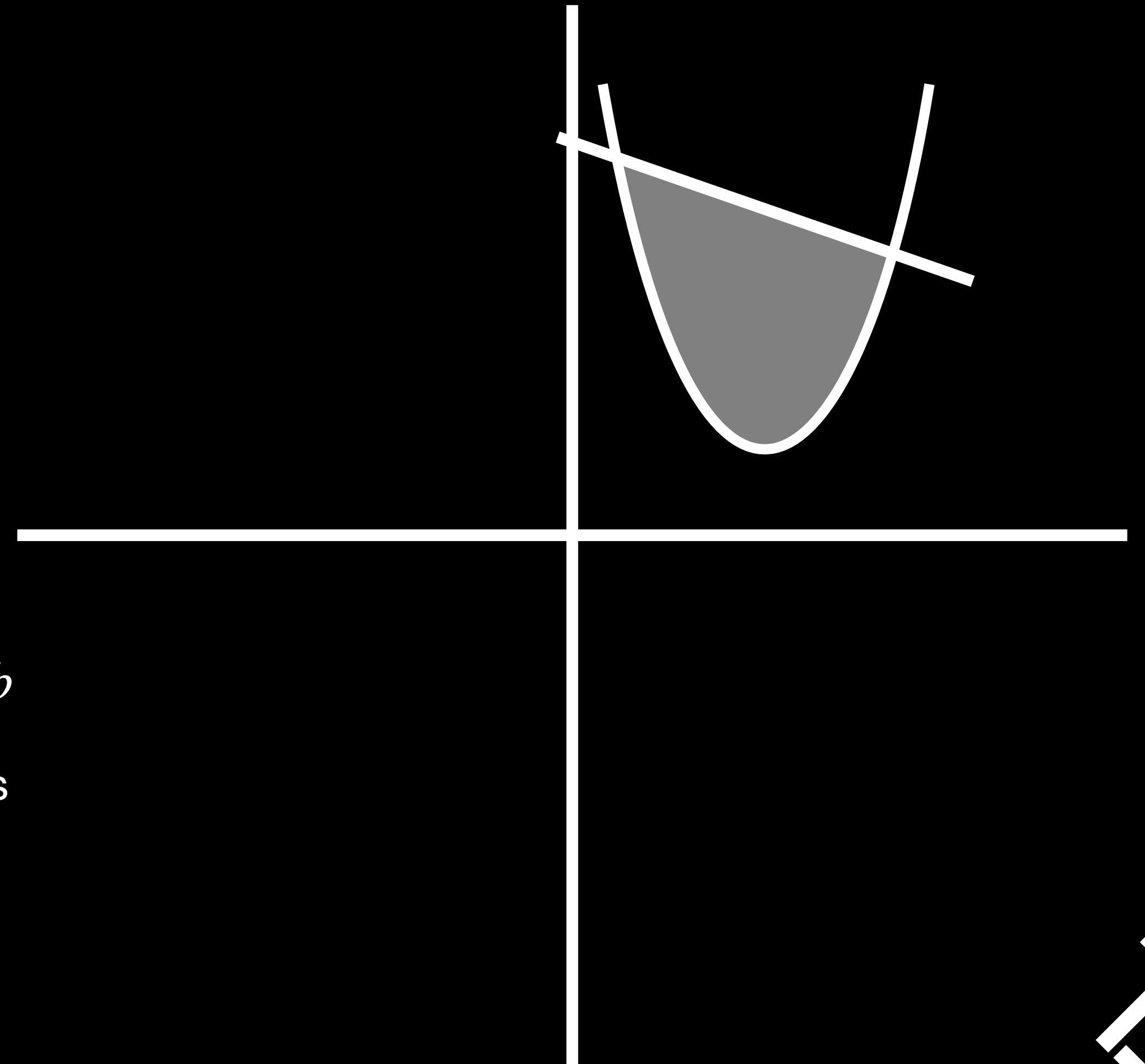
For example, to find this integral, we can find the integral of the line and subtract by the integral of the curve



The diagram illustrates the process of finding the area between two functions. It shows a gray trapezoid representing the area under a line, followed by a minus sign, then a gray region under a parabola, an equals sign, and finally a gray crescent-shaped region representing the area between the line and the parabola.

$$\int f(x) dx - \int g(x) dx =$$

If one function $f(x)$ lies completely above another function $g(x)$ for $a \leq x \leq b$, the area bonded between the two functions from $a \leq x \leq b$ is $\int_a^b f(x) - g(x) dx$, even when the shaded area is in the negative axis

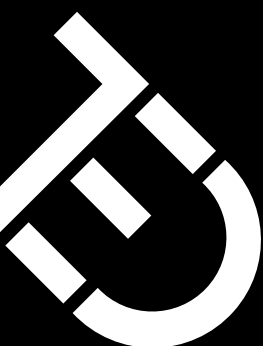
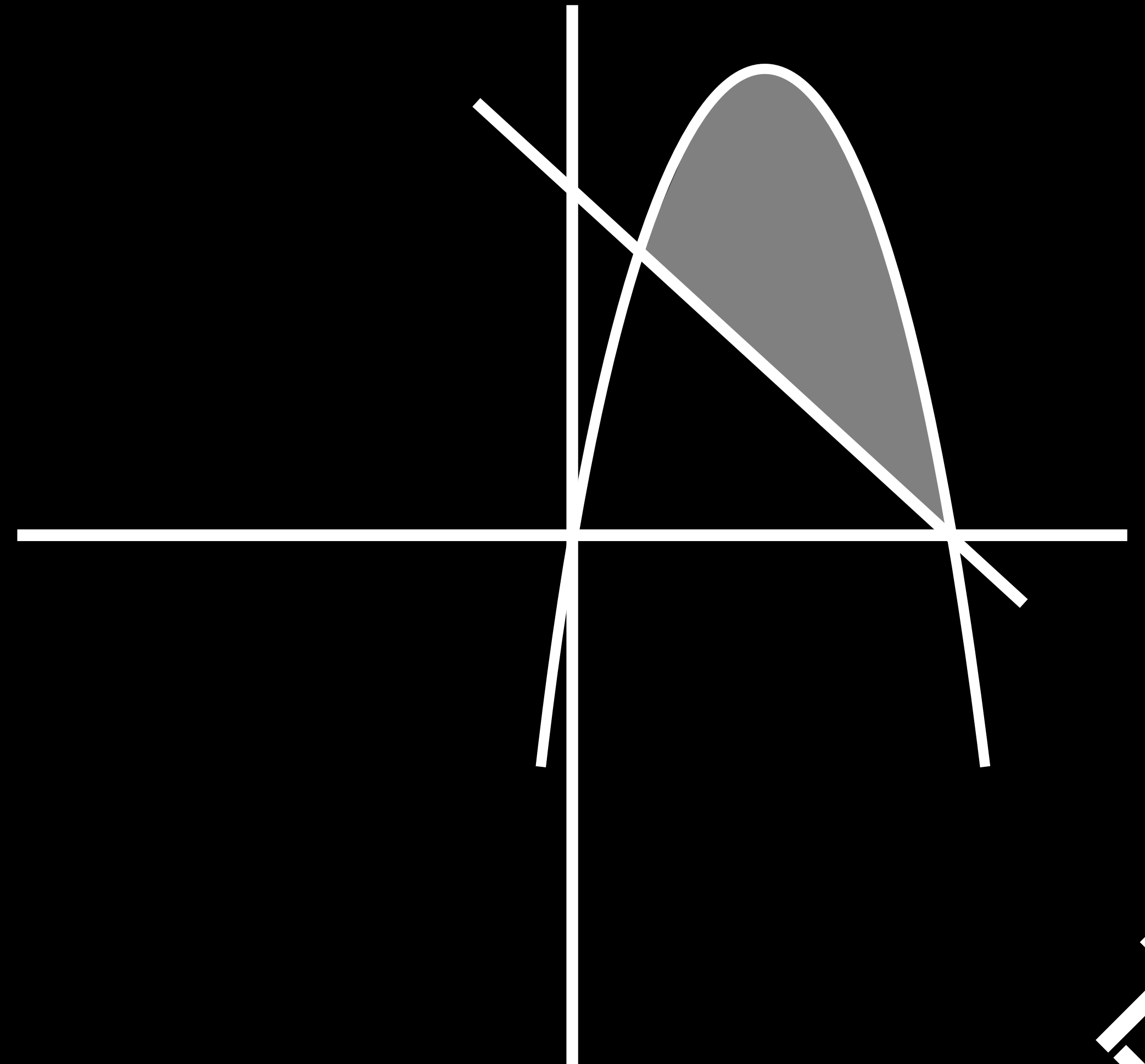


Bounded Area - Curve & Line

thank you for purchasing shoelace method+

The functions $y = -x^2 + 6x$ and $y = -x + 6$ is sketched.

Find the shaded area shown.



Bounded Area - Curve & Line

thank you for purchasing shoelace method+

The functions $y = -x^2 + 6x$ and $y = -x + 6$ is sketched.

Find the shaded area shown.

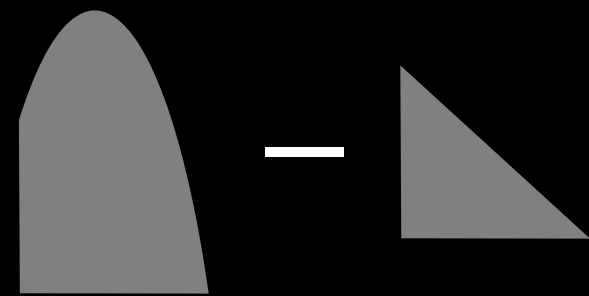
$$-x^2 + 6x = -x + 6$$

$$-x^2 + 7x - 6 = 0$$

$$-(x - 1)(x - 6) = 0$$

$$x = 1, x = 6$$

$$\int_1^6 -x^2 + 6x - (-x + 6) dx = \int_1^6 -x^2 + 6x + x - 6 dx$$

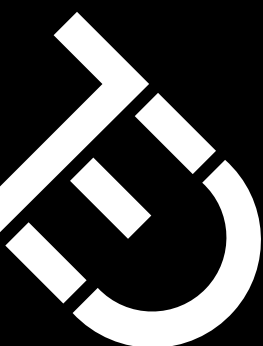
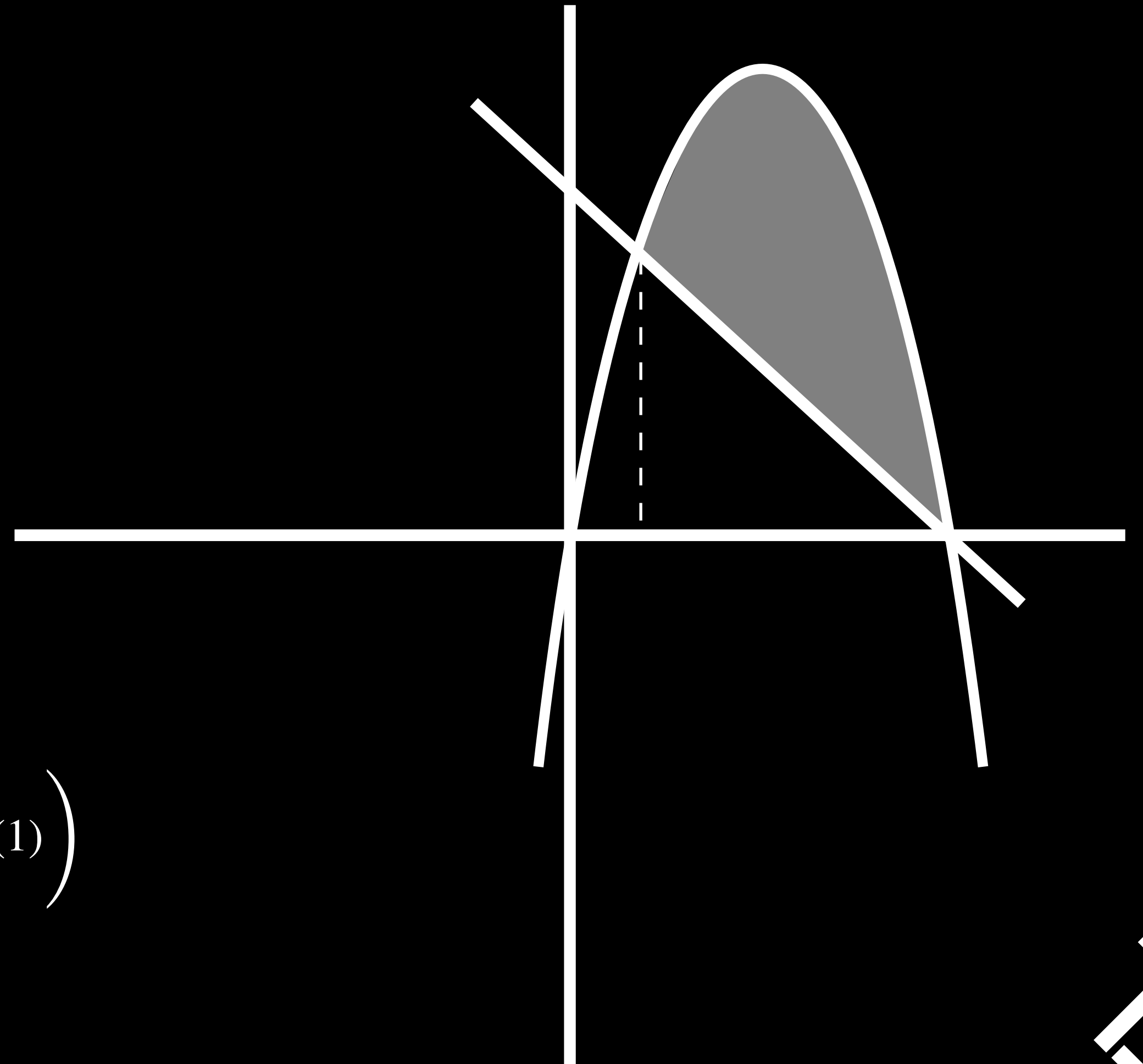


$$= \int_1^6 -x^2 + 7x - 6 dx$$

$$= \left[-\frac{x^3}{3} + \frac{7x^2}{2} - 6x \right]_1^6$$

$$= \left(-\frac{6^3}{3} + \frac{7(6)^2}{2} - 6(6) \right) - \left(-\frac{1^3}{3} + \frac{7(1)^2}{2} - 6(1) \right)$$

$$= \frac{125}{6} \text{ units}^2$$

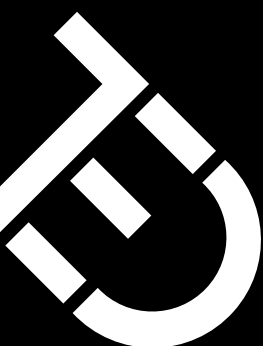
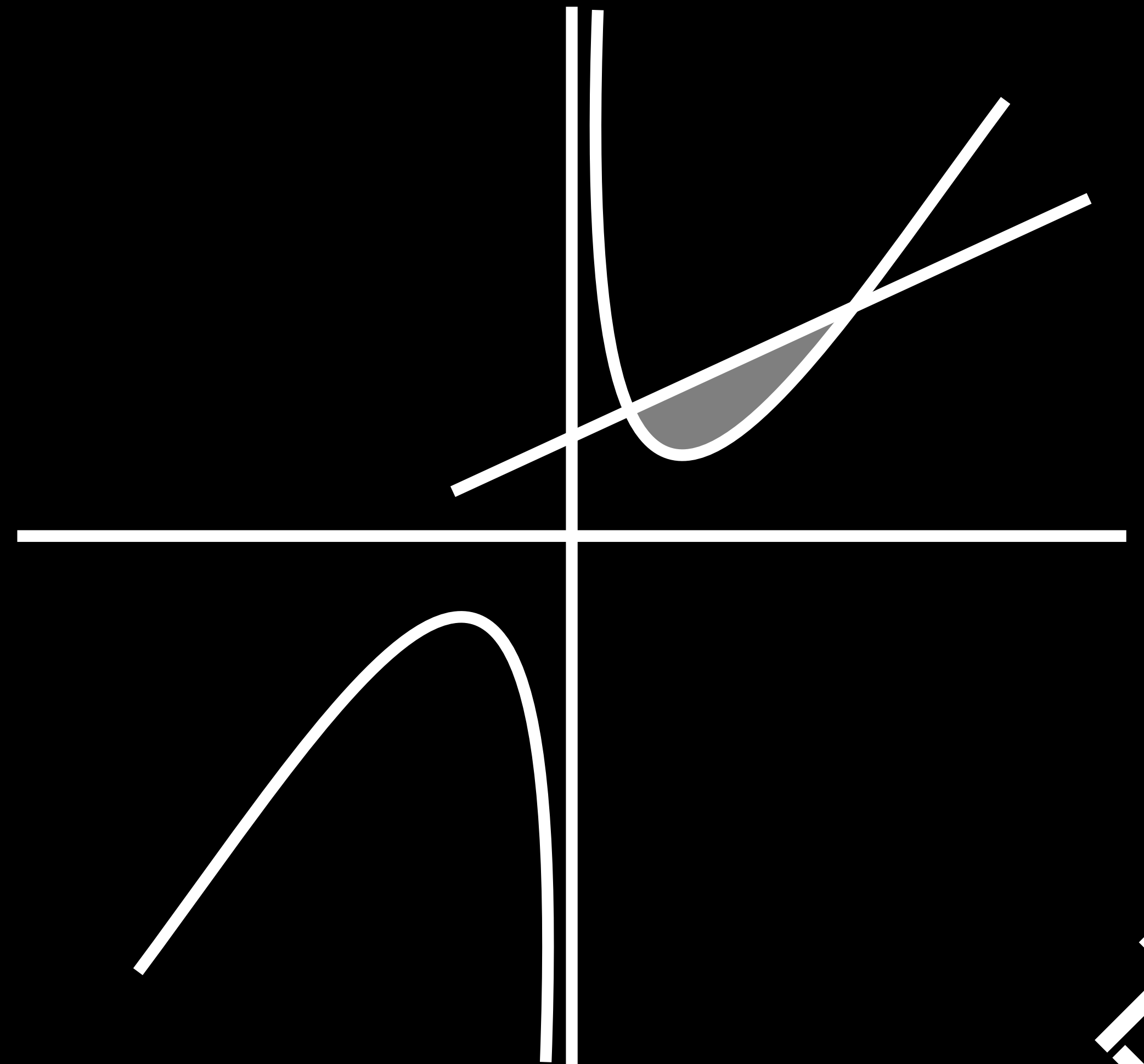


Bounded Area - Curve & Line

thank you for purchasing shoelace method+

The functions $y = \frac{1}{x} + x$ and $y = 0.5x + 2.25$ is sketched.

Find the shaded area shown.



Bounded Area - Curve & Line

thank you for purchasing shoelace method+

The functions $y = \frac{1}{x} + x$ and $y = 0.5x + 2.25$ is sketched.

Find the shaded area shown.

$$\frac{1}{x} + x = 0.5x + 2.25$$

$$\frac{1}{x} = -0.5x + 2.25$$

$$1 = -0.5x^2 + 2.25x$$

$$-0.5x^2 + 2.25x - 1 = 0$$

$$x = 0.5, 4$$

$$\int_{0.5}^4 0.5x + 2.25 - \left(\frac{1}{x} + x \right) dx = \int_{0.5}^4 0.5x + 2.25 - \frac{1}{x} - x dx$$

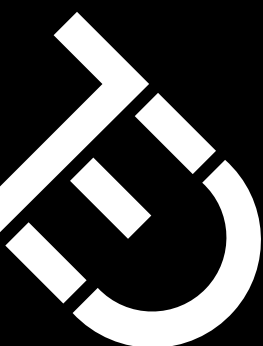
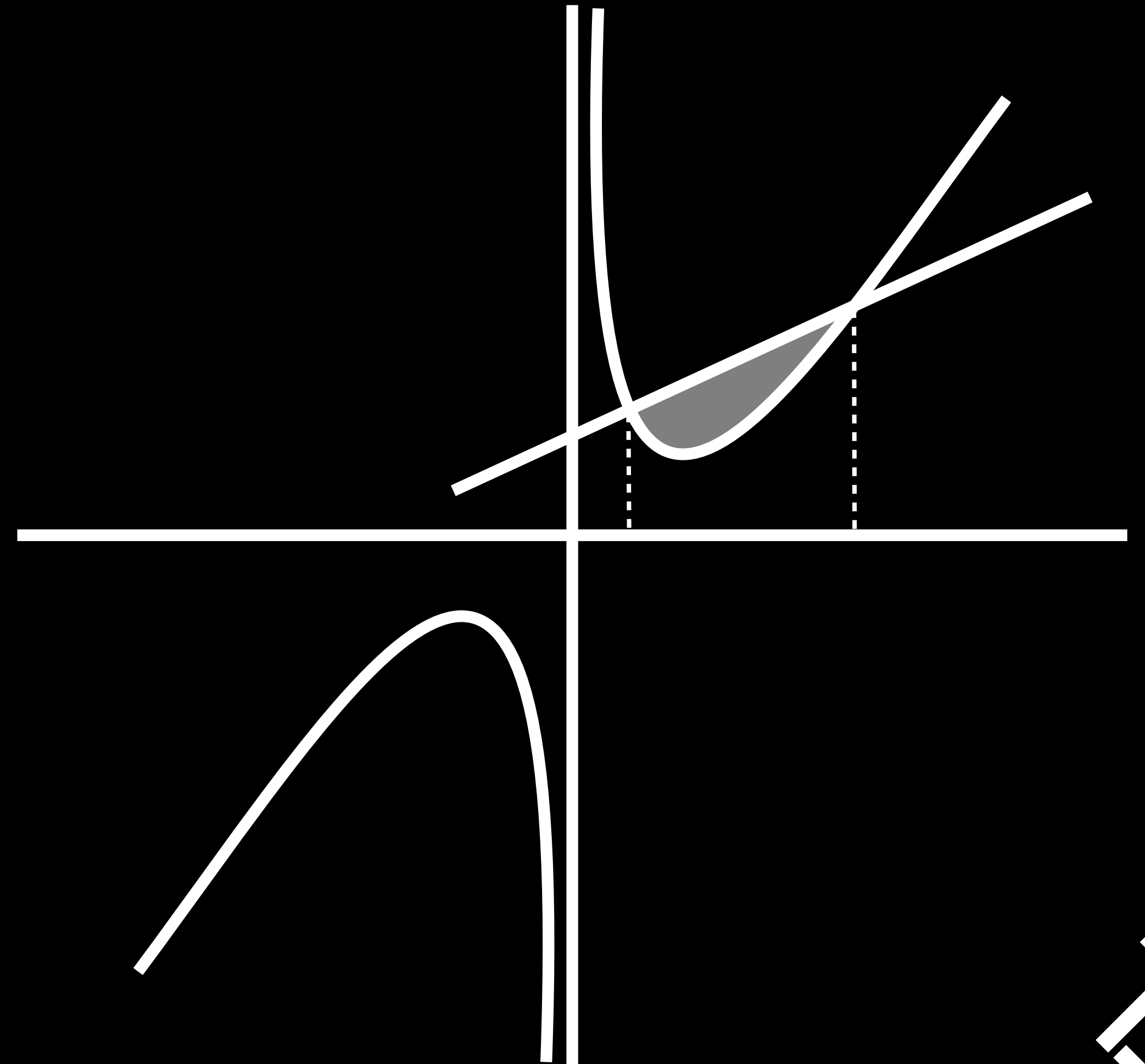


$$= \int_{0.5}^4 -0.5x + 2.25 - \frac{1}{x} dx$$

$$= \left[-\frac{0.5x^2}{2} + 2.25x - \ln(x) \right]_{0.5}^4$$

$$= \left(-\frac{0.5(4)^2}{2} + 2.25(4) - \ln(4) \right) - \left(-\frac{0.5(0.5)^2}{2} + 2.25(0.5) - \ln(0.5) \right)$$

$$= 1.86 \text{ units}^2 (3.s.f)$$

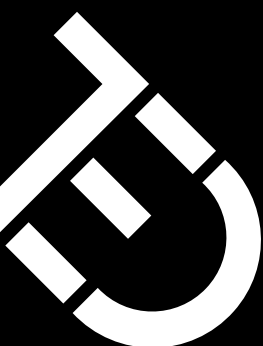
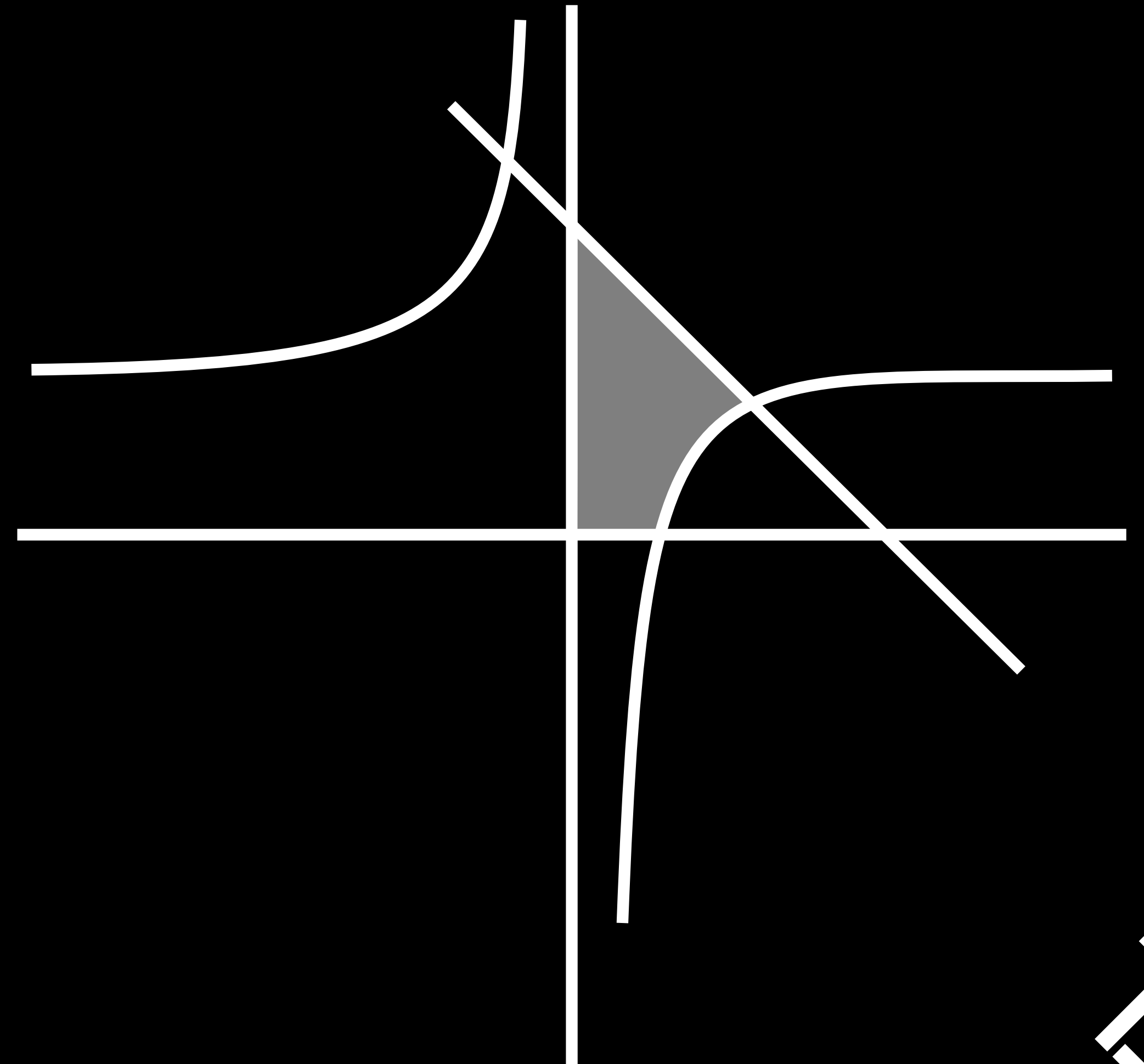


Bounded Area - Curve & Line

thank you for purchasing shoelace method+

The functions $y = -\frac{1}{x} + \frac{3}{2}$ and $y = 3 - x$ is sketched.

Find the shaded area shown.



Bounded Area - Curve & Line

thank you for purchasing shoelace method+

The functions $y = -\frac{1}{x} + \frac{3}{2}$ and $y = 3 - x$ is sketched.

Find the shaded area shown.

$$-\frac{1}{x} + \frac{3}{2} = 3 - x$$

$$-\frac{1}{x} = 3 - 1.5 - x$$

$$-\frac{1}{x} = 1.5 - x$$

$$-1 = x(1.5 - x)$$

$$-1 = 1.5x - x^2$$

$$-x^2 + 1.5x + 1 = 0$$

$$-2x^2 + 3x + 2 = 0$$

$$(x - 2)(2x + 1) = 0$$

$$x = -0.5, 2$$

rej. $x = -0.5$ as $x > 0$

$$-\frac{1}{x} + \frac{3}{2} = 0$$

$$-\frac{1}{x} = -1.5$$

$$-1 = -1.5x$$

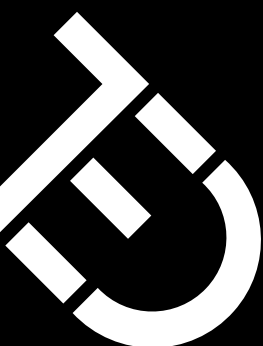
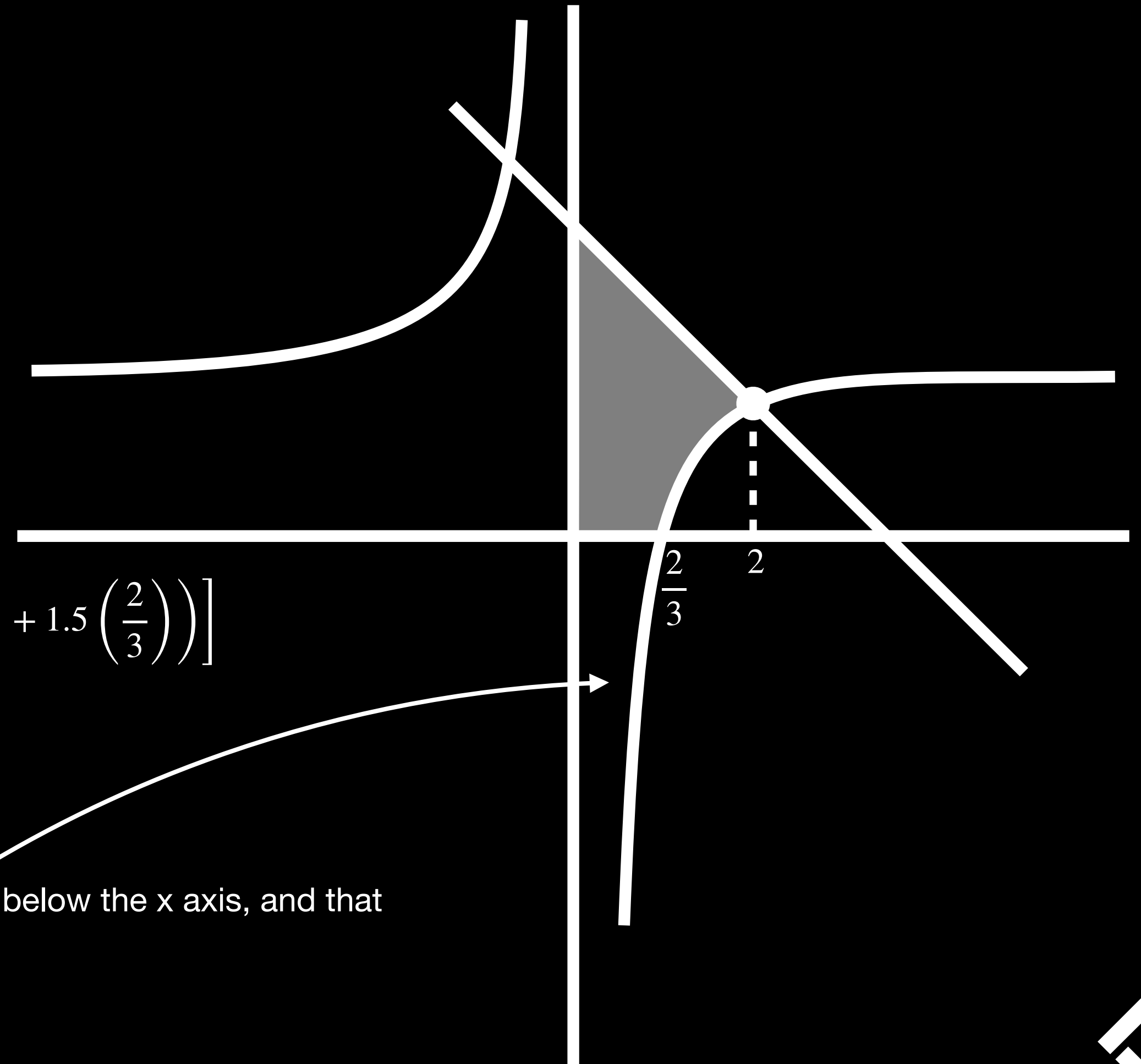
$$\frac{-1}{-1.5} = x$$

$$x = \frac{2}{3}$$



$$\begin{aligned} \int_0^2 3 - x \, dx - \int_{\frac{2}{3}}^2 -\frac{1}{x} + 1.5 \, dx &= \left[3x - \frac{x^2}{2} \right]_0^2 - \left[-\ln(x) + 1.5x \right]_{\frac{2}{3}}^2 \\ &= \left[\left(3(2) - \frac{2^2}{2} \right) - \left(3(0) - \frac{0^2}{2} \right) \right] - \left[(-\ln(2) + 1.5(2)) - \left(-\ln\left(\frac{2}{3}\right) + 1.5\left(\frac{2}{3}\right) \right) \right] \\ &= 3.10 \text{ units}^2 (3.s.f) \end{aligned}$$

We cannot take $\int_0^2 3 - x - \left(-\frac{1}{x} + \frac{3}{2} \right) dx$ as it will include the area below the x axis, and that the integral of $\frac{1}{x}$ from 0 will be undefined ($\ln(0)$ is undefined)

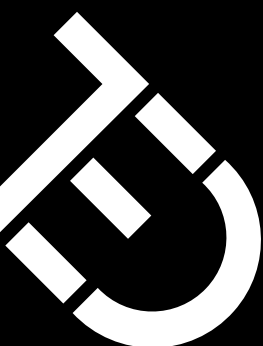
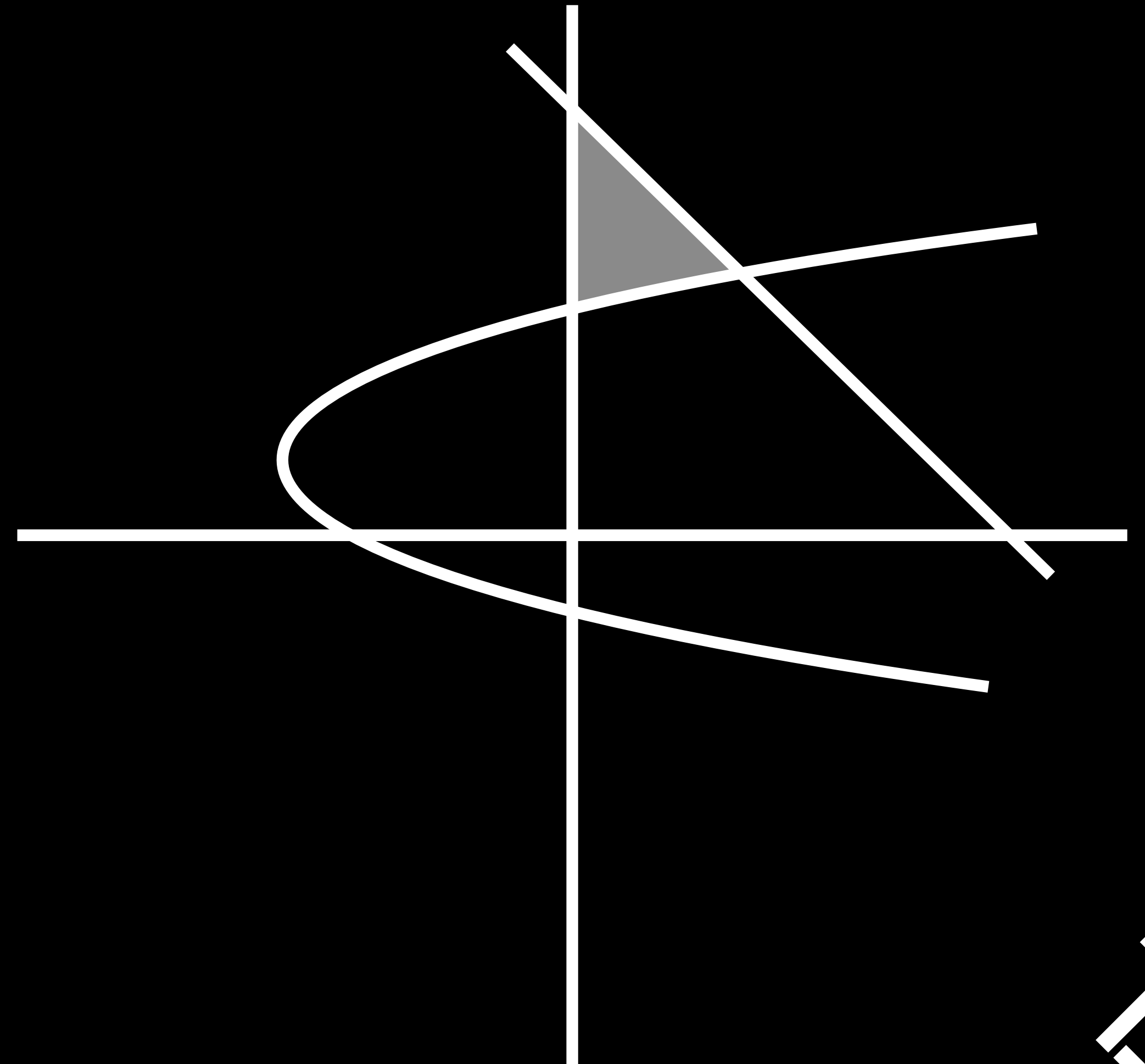


Bounded Area - Curve & Line

thank you for purchasing shoelace method+

The functions $x + 4 = (y - 1)^2$ and $x = -2y + 13$ is sketched.

Find the shaded area shown.



Bounded Area - Curve & Line

thank you for purchasing shoelace method+

The functions $x + 4 = (y - 1)^2$ and $x = -2y + 13$ is sketched.

Find the shaded area shown.

$$\begin{aligned}x + 4 &= (y - 1)^2 \\x &= (y - 1)^2 - 4 \\x &= y^2 - 2y + 1^2 - 4 \\x &= y^2 - 2y - 3\end{aligned}$$

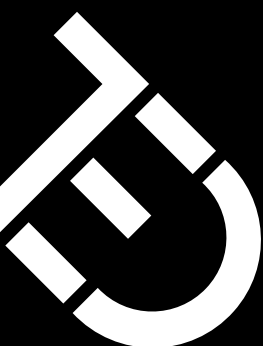
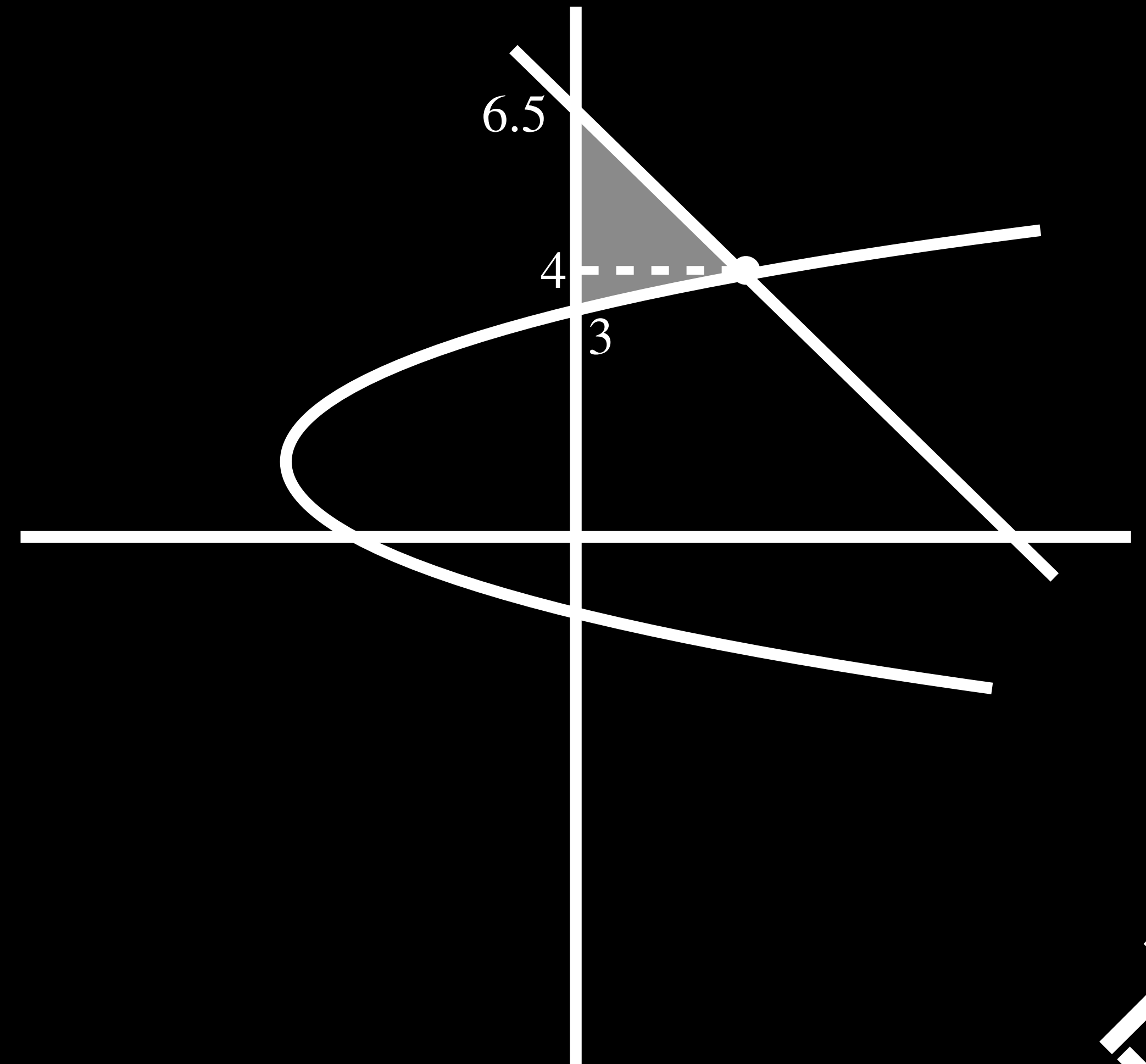
$$\begin{aligned}y^2 - 2y - 3 &= 0 \\y &= 3, -1 \\ \text{rej. } y &= -1 \text{ as } y > 0\end{aligned}$$

$$\begin{aligned}y^2 - 2y - 3 &= -2y + 13 \\y^2 - 3 &= 13 \\y^2 &= 13 + 3 \\y^2 &= 16 \\y &= \pm 4 \\&= 4 \\ \text{rej. } y &= -4 \text{ as } y > 0\end{aligned}$$

$$\begin{aligned}-2y + 13 &= 0 \\-2y &= -13 \\y &= \frac{-13}{-2} \\y &= 6.5\end{aligned}$$

As neither $f(y) \leq g(y)$ or $f(y) \geq g(y)$
for $3 \leq y \leq 6$, we cannot integrate
the difference of the two functions

$$\begin{aligned}&\int_3^4 y^2 - 2y - 3 \, dy + \int_4^{6.5} -2y + 13 \, dy = \left[\frac{y^3}{3} - y^2 - 3y \right]_3^4 + \left[-y^2 + 13y \right]_4^{6.5} \\&= \left[\left(\frac{4^3}{3} - 4^2 - 3(4) \right) - \left(\frac{3^3}{3} - 3^2 - 3(3) \right) \right] + \left[(-6.5^2 + 13(6.5)) - (-4^2 + 13(4)) \right] \\&= \frac{1117}{12} \text{units}^2\end{aligned}$$

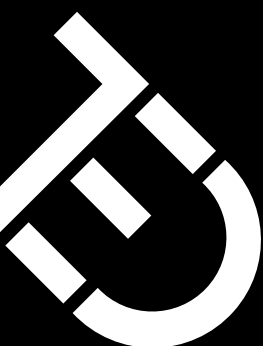
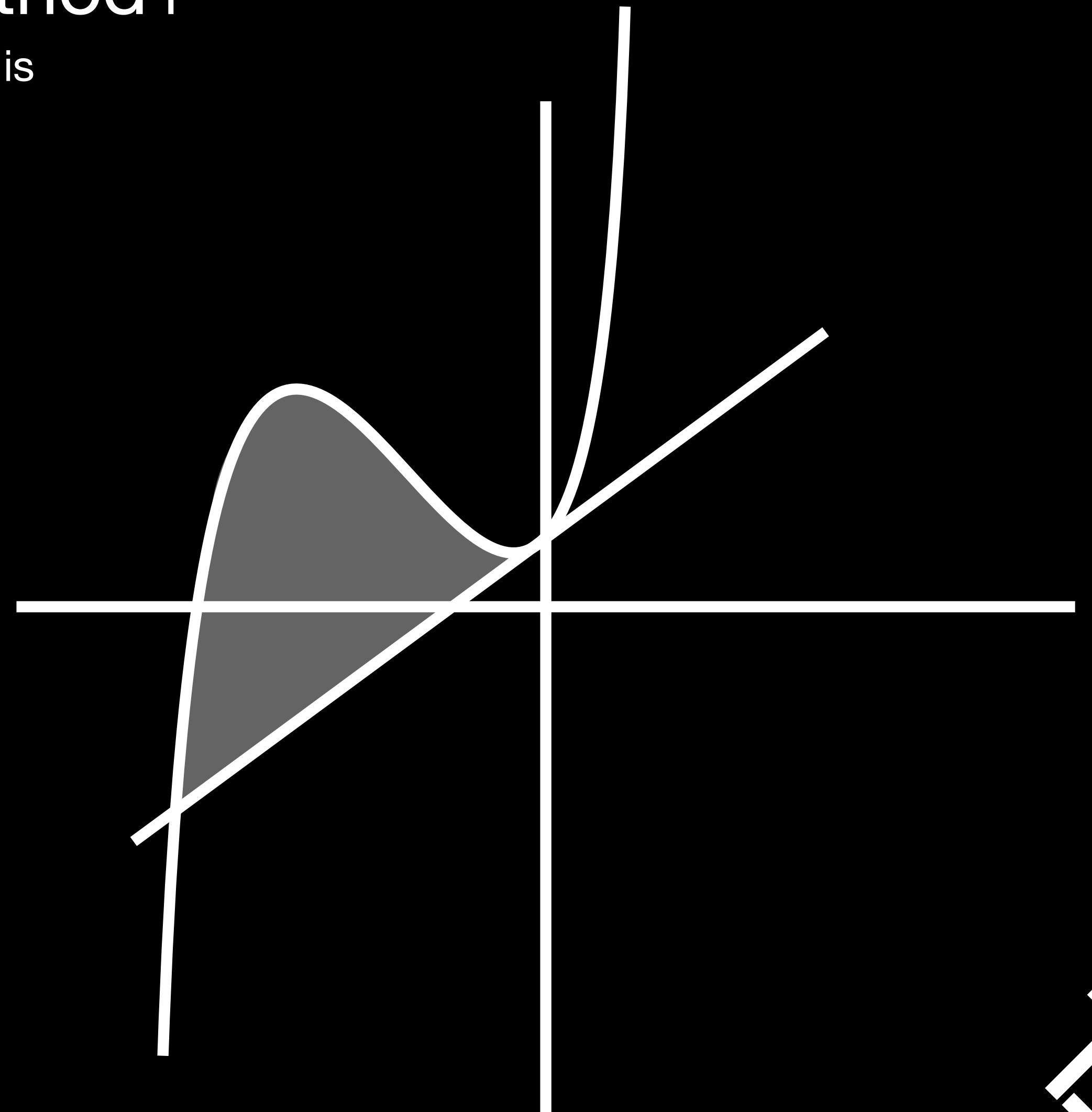


Bounded Area - Curve & Line

thank you for purchasing shoelace method+

The function $f(x) = x^3 + 3x^2 + x + 1$ and the tangent line at $x = 0$ is sketched.

Find the shaded area shown.



Bounded Area - Curve & Line

thank you for purchasing shoelace method+

The function $f(x) = x^3 + 3x^2 + x + 1$ and the tangent line at $x = 0$ is sketched.

Find the shaded area shown.

$$f'(x) = 3x^2 + 6x + 1$$

$$f'(0) = 3(0)^2 + 6(0) + 1 = 1$$

Eqn of tangent is

$$y - f(0) = 1(x - 0)$$

$$y - 1 = x$$

$$y = x + 1$$

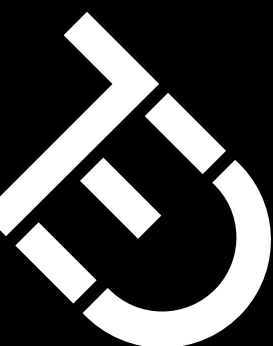
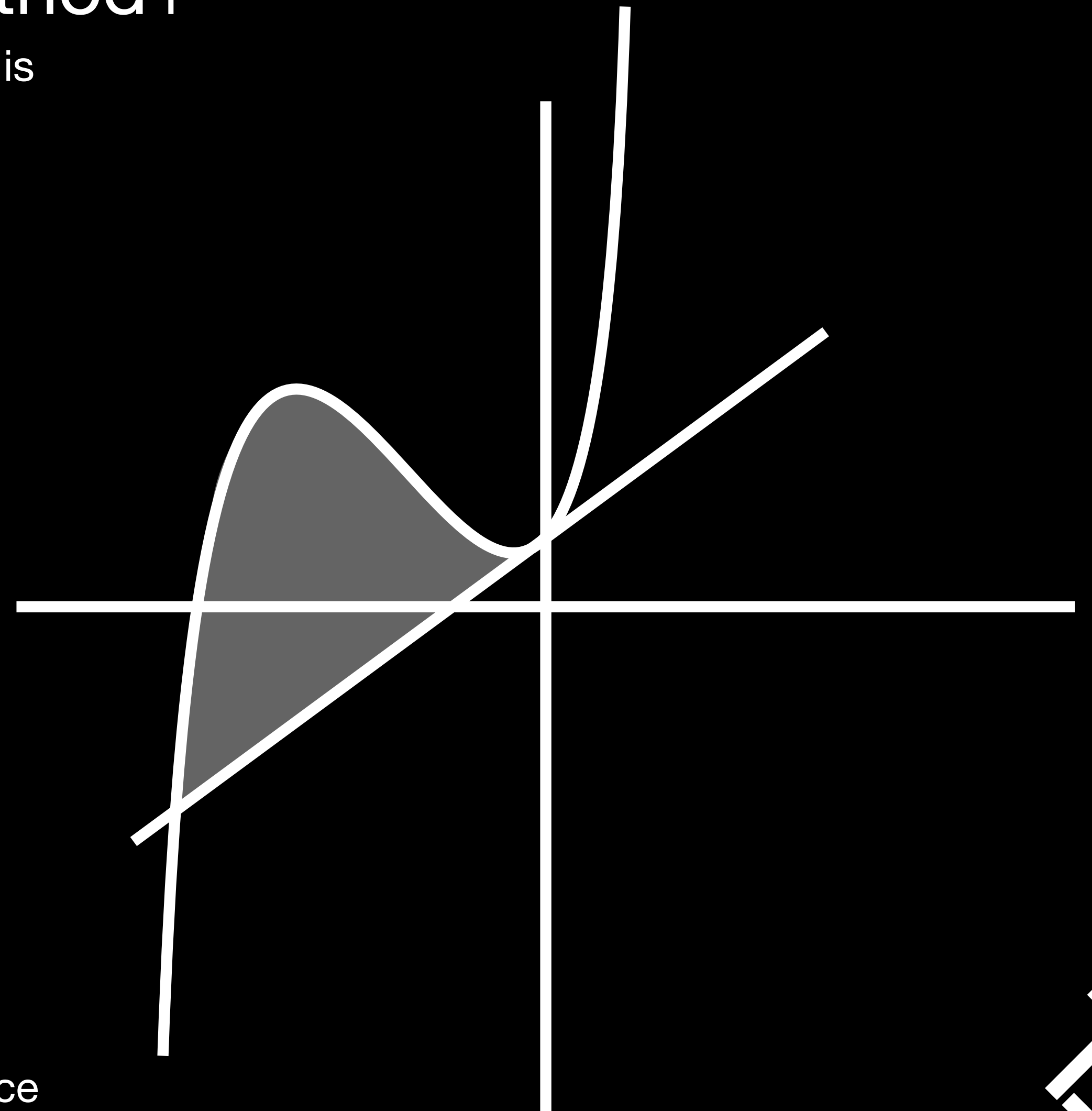
$$\int_{-3}^{-0} x^3 + 3x^2 + \cancel{x+1} - \cancel{(x+1)} dx = \int_{-3}^{-0} x^3 + 3x^2 dx$$

$$= \left[\frac{x^4}{4} + x^3 \right]_{-3}^0$$

$$= \left(\frac{0^4}{4} + 0^3 \right) - \left(\frac{(-3)^4}{4} + (-3)^3 \right)$$

$$= 6.75 \text{ units}^2$$

Even though the shaded area goes into the negative y axis region, the curve is still greater than the line, thus we can integrate the difference of the two.



Kinematics

physics jumpscare (!!!)

Definitions:

distance: how much the particle has moved (no direction, always positive)

displacement: how far a particle is from its starting position (has direction, can be negative)

speed: absolute change in distance per time (no direction, always positive)

velocity: change in displacement per change in time (has direction, can be negative)

acceleration: change in velocity per change in time (has direction, can be negative)

initial : $t = 0$

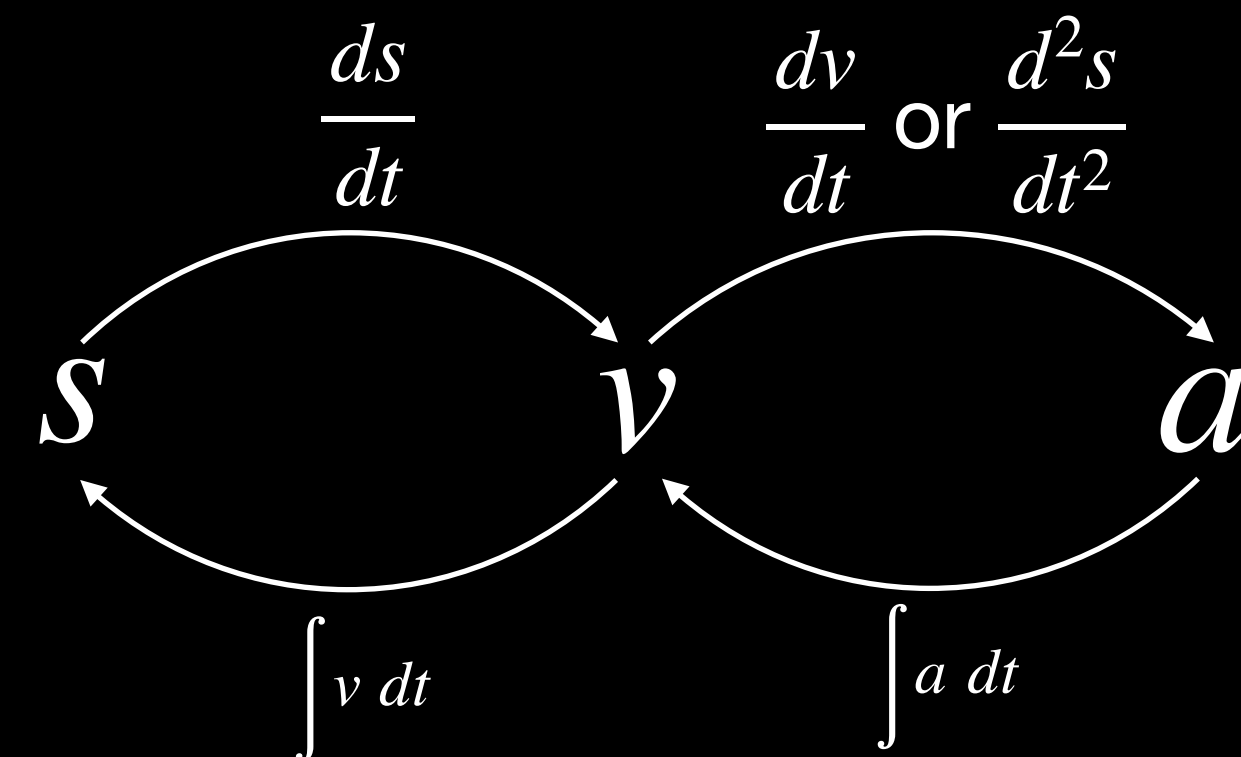
at rest: $v = 0$

turning: changing direction,

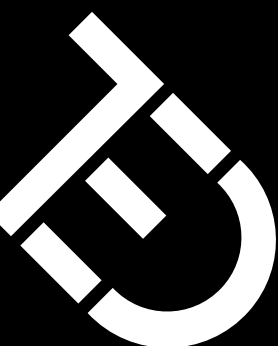
starting point: $s = 0$

in first n seconds: from 0 seconds to n seconds

in the n th second: from $n - 1$ seconds to n seconds



****RIGHTWARD DIRECTION IS POSITIVE, LEFTWARD DIRECTION IS NEGATIVE**

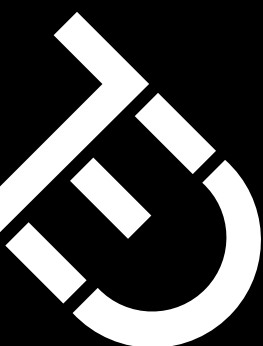


Kinematics

physics jumpscare (!!!)

A particle moves in a straight line so that its distance s meters after t seconds from a fixed point O is given by the function $s = -3t^2 + 12t + 15$. Find

- a) the initial distance the particle is away from O
- b) the distance of the particle away from O at $t=1$
- c) the displacement of the particle at $t=1$
- d) the distance travelled after 2 seconds
- e) the distance travelled in the 2nd second
- f) the value of t where the particle is at O
- g) the value of t where the particle is back at its starting point



Kinematics

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A particle moves in a straight line so that its distance s meters after t seconds from a fixed point O is given by the function $s = -3t^2 + 12t + 15$. Find

- the initial distance the particle is away from O
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- the displacement of the particle at $t=1$
- the distance travelled after 2 seconds
- the distance travelled in the 2nd second
- the value of t where the particle is at O
- the value of t where the particle is back at its starting point

$$a) s_{t=0} = -3(0)^2 + 12(0) + 15$$

$$s_{t=0} = 15m \quad \text{initial: } t=0$$

$$e) s_{t=2} - s_{t=1} = 27 - 24$$

$$\text{nth second is from } n-1 \text{ to } n = 3$$

$$b) s_{t=1} = -3(1)^2 + 12(1) + 15$$

$$s_{t=1} = 24m$$

$$f) -3t^2 + 12t + 15 = 0$$

$$\text{nth second is from } n-1 \text{ to } n$$

$$-3(t+1)(t-5) = 0$$

$$t = -1,5$$

$$c) 24 - 15 = 9m$$

$$\text{displacement is distance travelled from start}$$

$$\text{thus subtract original distance from } O$$

$$\text{rej. } t = -1 \text{ as } t \geq 0$$

$$d) s_{t=2} = -3(2)^2 + 12(2) + 15$$

$$s_{t=2} = 27$$

$$\therefore t = 5$$

$$g) -3t^2 + 12t + 15 = 15$$

$$-3t^2 + 12t = 0$$

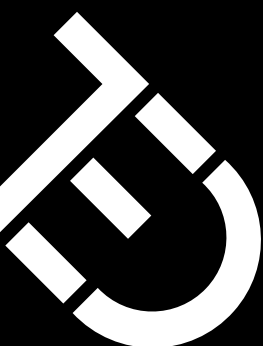
$$3t(-t + 4) = 0$$

$$-3t(t - 4) = 0$$

$$t = 0,4$$

rej. $t = 0$ as finding when particle returns back to start

$$\therefore t = 4$$

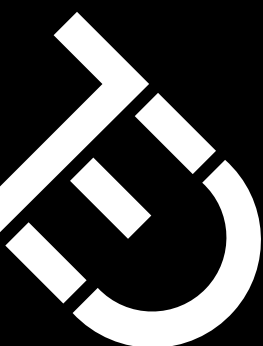


Kinematics

physics jumpscare (!!!)

A particle moves in a straight line so that its distance s meters after t seconds from a fixed point O is given by the function $s = t^3 - 7t^2 + 8t + 2$. Find

- a) an expression for the velocity in terms of t
- b) the velocity of the particle at $t=1$
- c) the values of t where the particle is at rest
- d) the acceleration of the particle at $t=2$



Kinematics

physics jumpscare (!!!)

A particle moves in a straight line so that its distance s meters after t seconds from a fixed point O is given by the function $s = t^3 - 7t^2 + 8t + 2$. Find

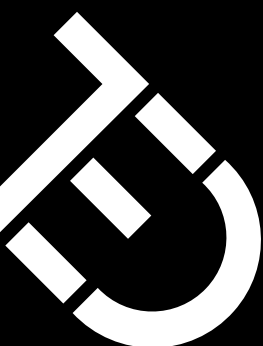
- a) an expression for the velocity in terms of t
- b) the velocity of the particle at $t=1$
- c) the values of t where the particle is at rest
- d) the acceleration of the particle at $t=2$

$$\begin{aligned} a) \quad v &= \frac{ds}{dt} \\ &= 3t^2 - 14t + 8 \end{aligned}$$

$$\begin{aligned} b) \quad v_{t=1} &= 3(1)^2 - 14(1) + 8 \\ &= -3 \end{aligned}$$

$$\begin{aligned} c) \quad 3t^2 - 14t + 8 &= 0 \\ (3t - 2)(t - 4) &= 0 \quad \text{at rest means when } v=0 \\ t &= \frac{2}{3}, 4 \end{aligned}$$

$$\begin{aligned} d) \quad a &= \frac{dv}{dt} \\ &= 6t - 14 \\ a_{t=2} &= 6(2) - 14 \\ &= -2 \end{aligned}$$

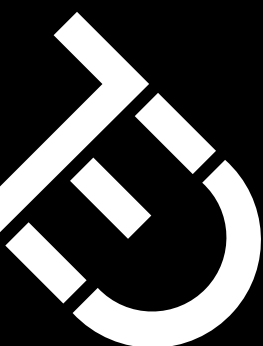


Kinematics

physics jumpscare (!!!)

A particle moves in a straight line such that t seconds after passing a fixed point O with a velocity of -5 m/s, its acceleration, a m/s², is given by the function $a = 2t + 4$. Find

- a) the velocity of the particle at $t=2$
- b) the value(s) of t when the particle is at rest
- c) the value of t when the particle is back at O



Kinematics

physics jumpscare (!!!)

A particle moves in a straight line such that t seconds ^{at $t = 0, s = 0$} after passing a fixed point O with a velocity of -5 m/s , its acceleration, $a \text{ m/s}^2$, is given by the function $a = 2t + 4$. Find

- the velocity of the particle at $t=2$
- the value(s) of t when the particle is at rest
- the value of t when the particle is back at O

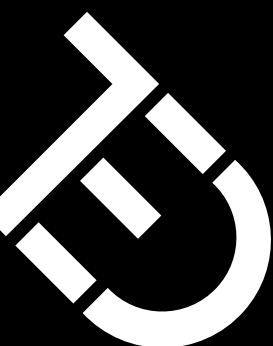
$$\begin{aligned}
 \text{a) } v &= \int a \, dt \\
 &= t^2 + 4t + C \\
 \text{at } t = 0, v &= -5 \\
 -5 &= 0^2 + 4(0) + C \\
 C &= -5 \\
 \therefore v &= t^2 + 4t - 5 \\
 v_{t=2} &= 2^2 + 4(2) - 5 \\
 v_{t=2} &= 7 \text{ m/s}
 \end{aligned}$$

$$\begin{aligned}
 \text{b) } v &= t^2 + 4t - 5 \\
 t^2 + 4t - 5 &= 0 \\
 (t + 5)(t - 1) &= 0 \\
 t &= -5, 1 \\
 &\text{rej. } t = -5 \text{ as } t \geq 0 \\
 t &= 1 \text{ s}
 \end{aligned}$$

$$\begin{aligned}
 \text{c) } s &= \int v \, dt \\
 &= \frac{t^3}{3} + 2t^2 - 5t + C \\
 \text{at } t = 0, s &= 0 \\
 0 &= \frac{0^3}{3} + 2(0)^2 - 5(0) + C \\
 C &= 0 \\
 \therefore s &= \frac{t^3}{3} + 2t^2 - 5t
 \end{aligned}$$

$$\begin{aligned}
 \frac{t^3}{3} + 2t^2 - 5t &= 0 \\
 t \left(\frac{t^2}{3} + 2t - 5 \right) &= 0 \\
 t &= 0, 1.90, -7.90(3.s.f) \\
 &\text{rej. } t = -7.89 \text{ as } t \geq 0 \\
 t &= 1.90 \text{ s} \quad \text{rej. } t = 0 \text{ as finding when particle returns}
 \end{aligned}$$

note that when $s=0$, C is not necessarily 0
e.g. for $e^x + C = 0$ when $x = 0$, $C = -1$

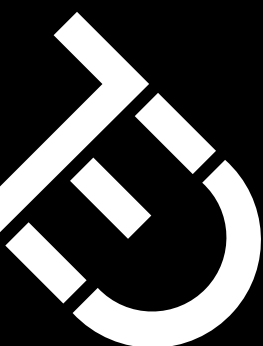


Kinematics

physics jumpscare (!!!)

A particle moves in a straight line such that its velocity v after t seconds from a fixed point O is $v = 3t^2 - 24t + 36$. Find

- a) the values of t where the particle is at rest
- b) the distance travelled by the particle in the first 7 seconds.



Kinematics

physics jumpscare (!!!)

A particle moves in a straight line such that its velocity v after t seconds from a fixed point is $v = 3t^2 - 24t + 36$. Find

- the values of t where the particle is at rest
- the distance travelled by the particle in the first 7 seconds.

$$a) 3t^2 - 24t + 36 = 0$$

$$3(t^2 - 8t + 12) = 0$$

$$3(t - 2)(t - 6) = 0$$

$$t = 2, t = 6$$

**note that particle at rest does not necessarily mean it turns, however

question will ask you to find if it turns at $v=0$ (first/second derivative test) if it does not turn when $v=0$

however for this case, as s will be a cubic and has 2 stationary points, neither of them are inflection points

As $3t^2$ is positive, the coeff of t^3 will be positive, thus first stationary value is maximum

alternate sign for each integral

$$b) \int_0^2 v \, dt + \left(- \int_2^6 v \, dt \right) + \int_6^7 v \, dt$$

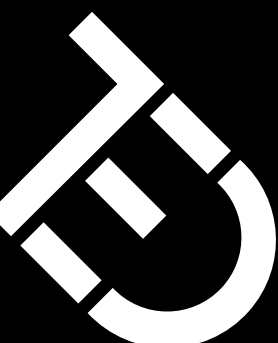
$$= [t^3 - 12t^2 + 36t]_0^2 - [t^3 - 12t^2 + 36t]_2^6 + [t^3 - 12t^2 + 36t]_6^7$$

$$= ((2^3 - 12(2)^2 + 36(2)) - 0) - ((6^3 - 12(6)^2 + 36(6)) - (2^3 - 12(2)^2 + 36(2))) + ((7^3 - 12(7)^2 + 36(7)) - (6^3 - 12(6)^2 + 36(6)))$$

$$= 32 - 32 + 7$$

$$= 7m$$

If the first stationary point is a minimum, the first integral should be negative and the next should be positive

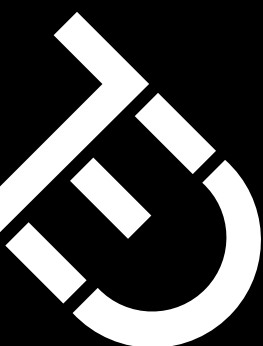


Kinematics

physics jumpscare (!!!)

A particle moves in a straight line such that its velocity v after t seconds from a fixed point O is $v = -e^{t-3} + 2$. Find

- a) the values of t where the particle is at rest and their natures
- b) if the particle returns back to O in the 6th second.
- c) the total distance travelled by the particle until it returns to O .



Kinematics

physics jumpscare (!!!)

A particle moves in a straight line such that its velocity v after t seconds from a fixed point O is $v = -e^{t-3} + 2$. Find

- the values of t where the particle is at rest and their natures
- if the particle returns back to O in the 6th second.
- the total distance travelled by the particle until it returns to O .

a) let $v = 0$

$$-e^{t-3} + 2 = 0$$

$$-e^{t-3} = -2$$

$$e^{t-3} = 2$$

$$t - 3 = \ln(2)$$

$$t = \ln(2) + 3 = 3.6931(5.s.f)$$

$$\frac{d^2s}{dt^2} = \frac{dv}{dt} = -e^{t-3}$$

Second derivative test to find nature of stationary values
(Second derivative of s is derivative of v)

$$\frac{d^2s}{dt^2} \Big|_{t=\ln(2)+3} = -e^{\ln(2)+3-3} = -e^{\ln(2)} = -2 < 0$$

$\therefore t = 3.6931$ gives a maximum

$$b) \int v dt = \int -e^{t-3} + 2 dt = -e^{t-3} + 2t + C$$

At $t = 0, s = 0$. Thus,

$$0 = -e^{0-3} + 2(0) + C$$

$$0 = -e^{-3} + C$$

$$e^{-3} = C$$

$$C = e^{-3} = 0.49787(5.s.f)$$

$$\therefore s = -e^{t-3} + 2t + e^{-3}$$

$$s_{t=5} = -e^{5-3} + 2(5) + e^{-3} = 2.66(3.s.f) > 0$$

$$s_{t=6} = -e^{6-3} + 2(6) + e^{-3} = -8.04(3.s.f) < 0$$

$\therefore$ the particle turns in the 6th second

$$c) s_{t=\ln(2)+3} = -e^{\ln(2)+3-3} + 2(\ln(2) + 3) + e^{-3} = 5.4361(5.s.f)$$

As the particle only turns at $t = \ln(2) + 3$,

$$\text{distance travelled} = 2 \times s_{t=\ln(2)+3}$$

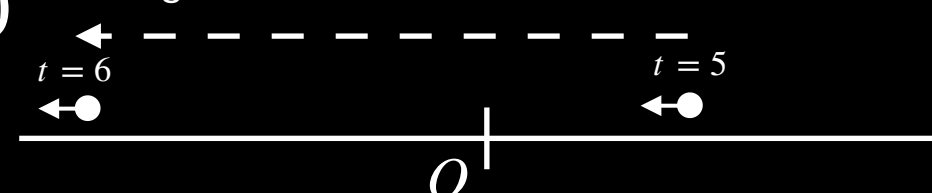
$$= 2 \times 5.4361$$

$$= 10.8722$$

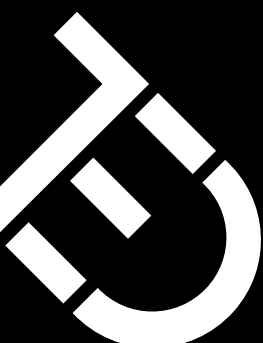
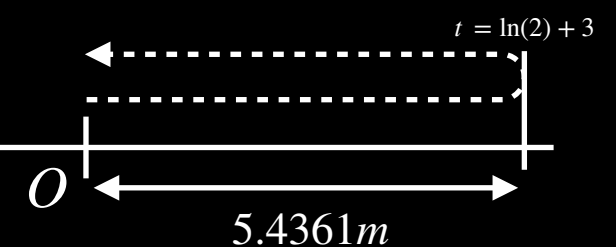
$$= 10.9m(3.s.f)$$

Note that even though $s=0$ when $t=0$, C is not 0

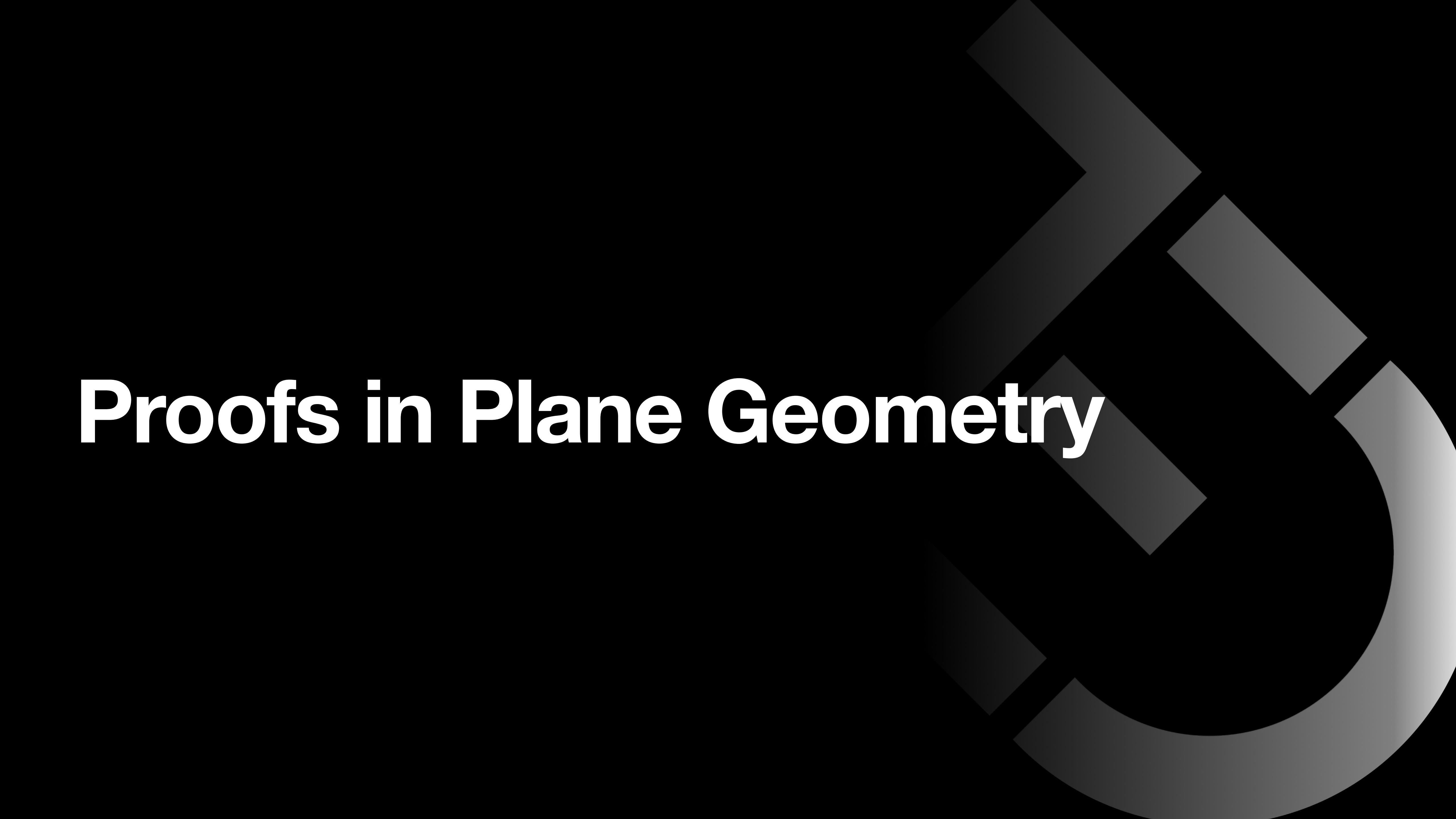
Any equation in the form of $ae^x + bx^n = 0$ where a, b and n are constants will be UNSOLVABLE, thus, we cannot solve for $s=0$ here. For this question, we just need to show the sign of s changes somewhere from $t=5$ to $t=6$.



As $s=0$ is unsolvable, we cannot use definite integrals as we do not know the integral bounds, however we know the particle only turns once before returning to O , so we can just double the distance from $t=0$ to $t=\ln(2)+3$.



Proofs in Plane Geometry



Symbol Notation

noted

The use of Δ , $\angle$ and $//$ is only allowed when there are defined points.

For proof abbreviations or when points are not defined, use “triangle”, “angle(s)” and “parallel”

Allowed:

$$\Delta ABC \equiv \Delta XYZ(SAS)$$

$$\angle ABC = \angle XYZ$$

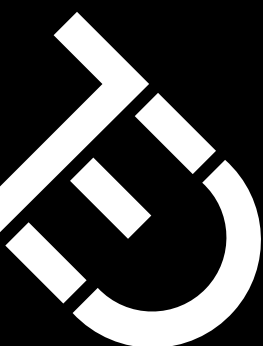
$$\angle PQR = \angle QRS \text{ (Alternate angles, } PQ // RS)$$

Not allowed:

...(∠s in same segment are equal)

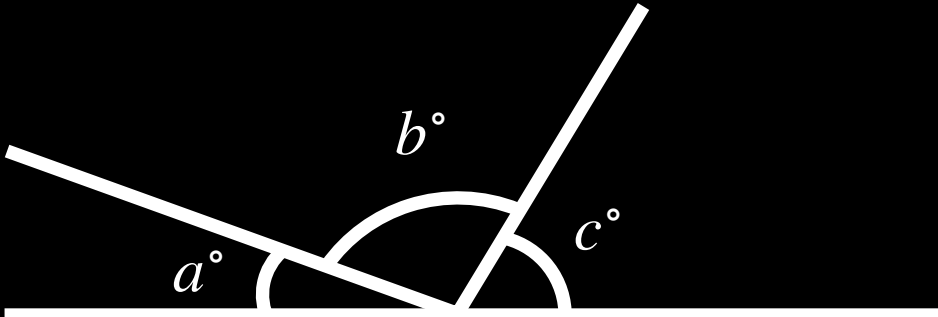
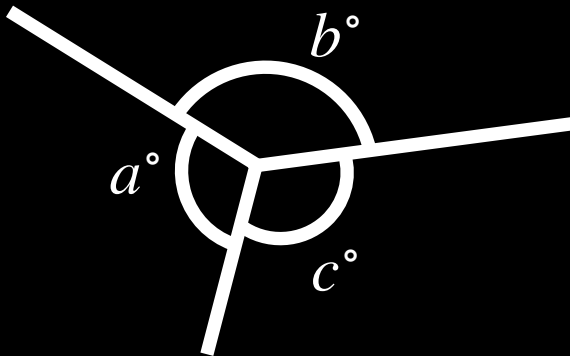
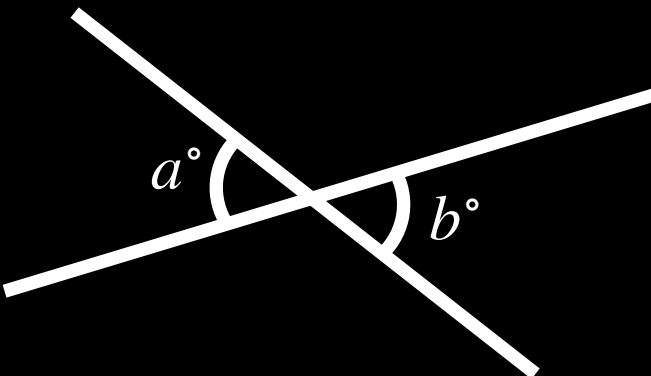
... (base angles of isosceles Δ are equal)

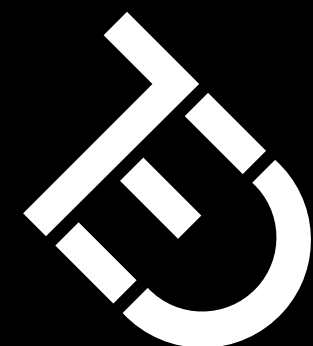
As the lines are $//$, ...



Geometric Properties - Angle Properties

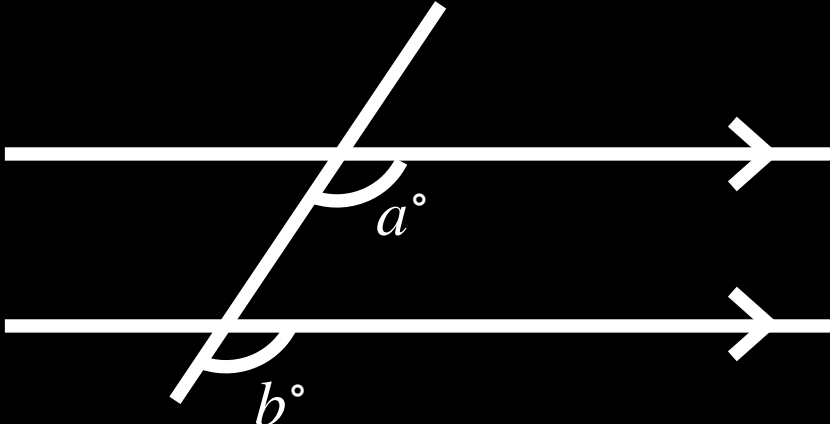
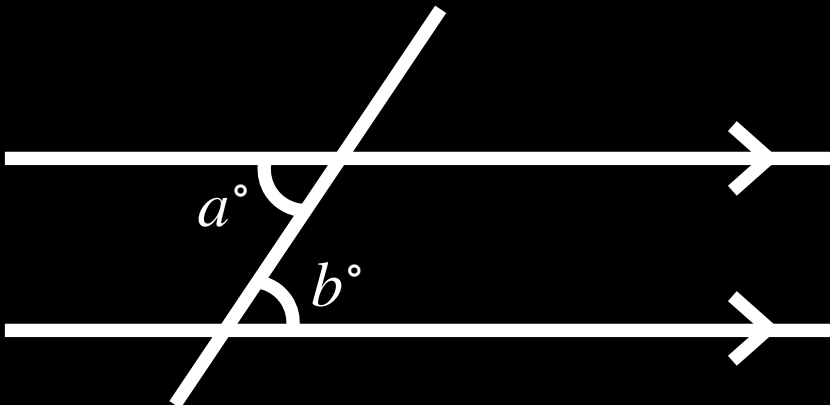
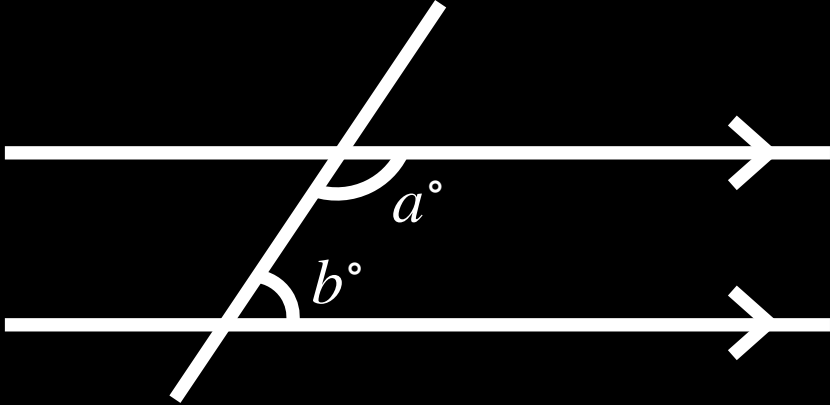
basics

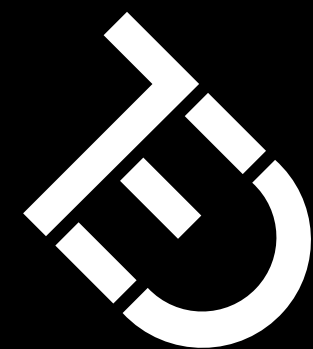
Property	Diagram	Abbreviation (what you write)
Sum of angles on straight line is 180° $a^\circ + b^\circ + c^\circ = 180^\circ$		adjacent angles on straight line
Sum of angles at a point is 360° $a^\circ + b^\circ + c^\circ = 360^\circ$		angles at point
Vertically opposite angles are equal $a^\circ = b^\circ$		vertically opposite angles



Geometric Properties - Parallel Lines

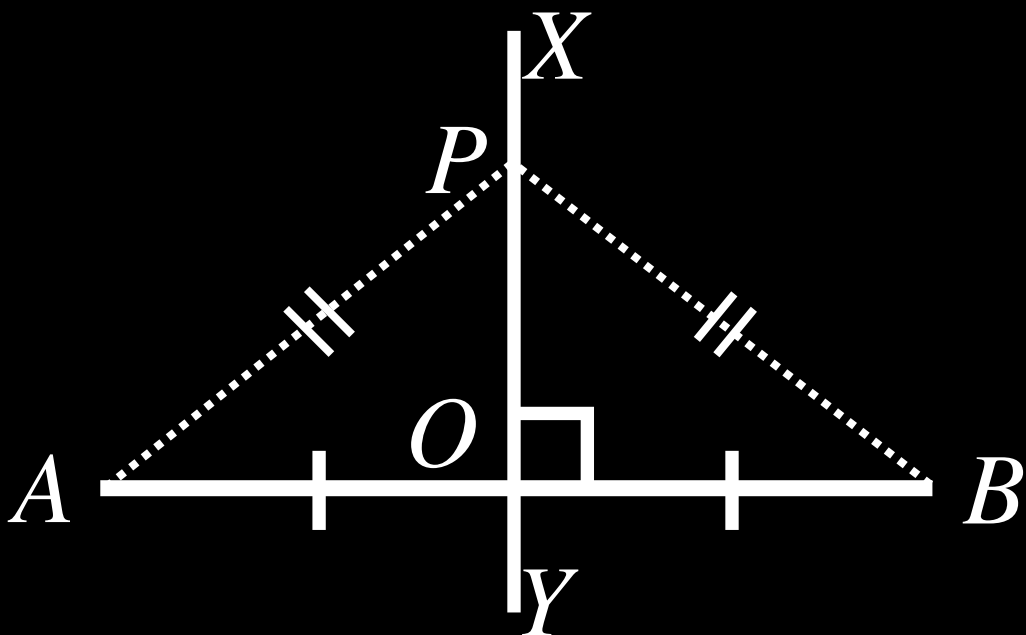
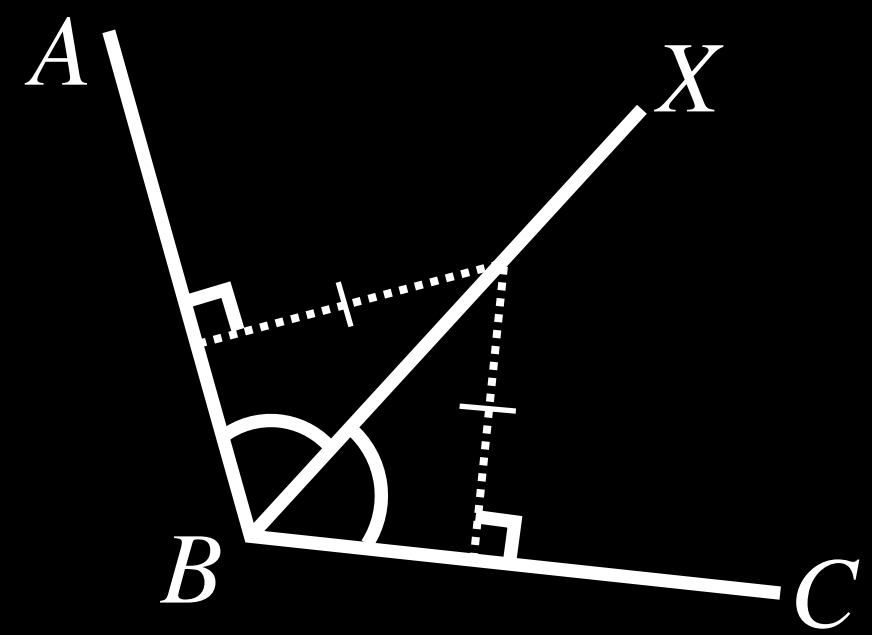
basics

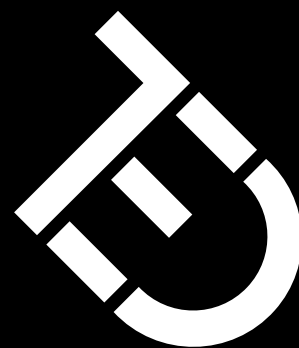
Property	Diagram	Abbreviation (what you write)
Corresponding angles are equal $a^{\circ}=b^{\circ}$		Corresponding angles, AB//CD (write parallel lines used)
Alternate angles are equal $a^{\circ}=b^{\circ}$		Alternate angles, AB//CD (write parallel lines used)
Sum of interior angles = 180° $a^{\circ}+b^{\circ}=180^{\circ}$		Interior angles, AB//CD (write parallel lines used)



Geometric Properties - Bisectors

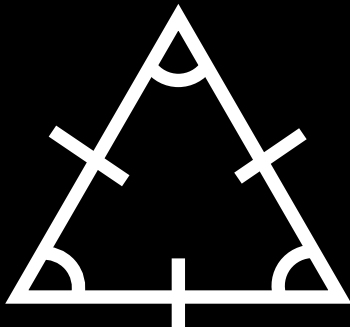
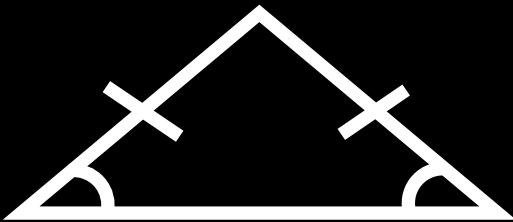
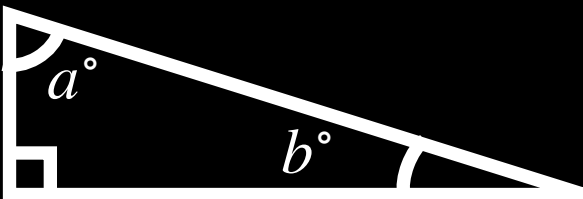
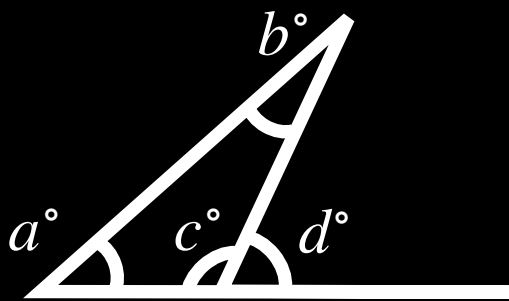
basics

Property	Diagram	Abbreviation (what you write)
Given that XY perpendicularly bisects AB at O and a point P along XY, XY is perpendicular to AB $AO = BO/O$ is the midpoint of AB $PA=PB$		XY perpendicularly bisects AB
Given that BX bisects the angle ABC and P is a point along BX, - angle $ABX =$ angle CBX P is equidistant from AB and AC		BX is an angle bisector of $\angle ABC$

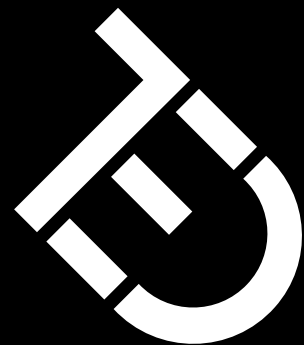


Geometric Properties - Triangles

basics

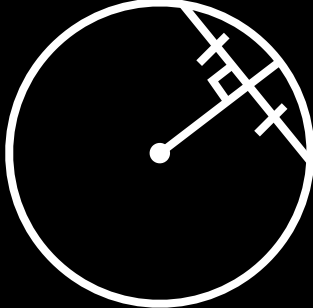
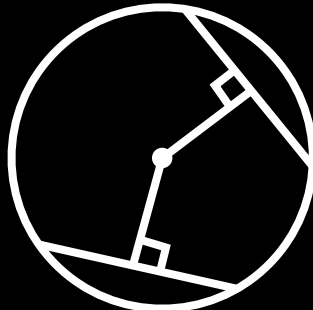
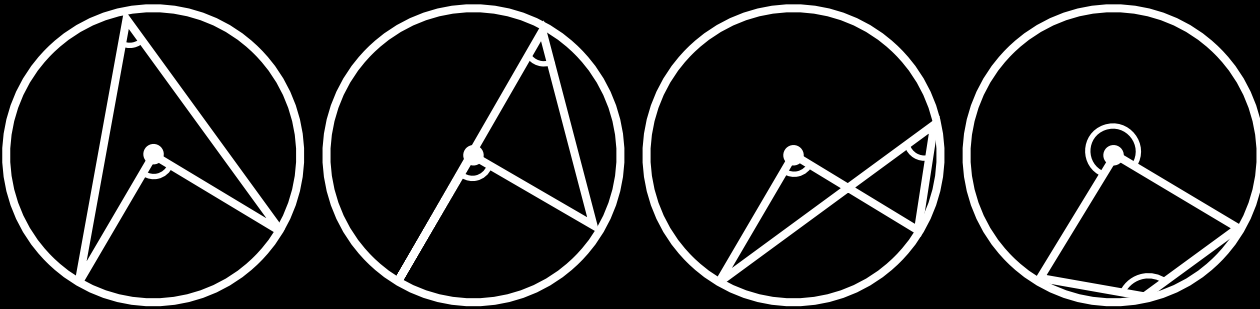
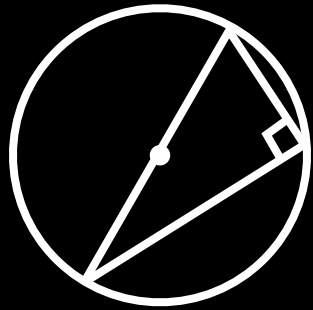
Property	Diagram	Abbreviation (what you write)
In an equilateral triangle, all sides are equal and all angles are equal (and are 60°)		Sides of equilateral triangle Angles of equilateral triangle
In an isosceles triangle, two sides are equal and base angles are equal		Equal sides of isosceles triangle Base angles of isosceles triangle
In a right triangle, Pythagoras theorem holds ($a^2 + b^2 = c^2$) $a^\circ = 90^\circ - b^\circ$		Pythagoras Theorem Interior angles of right triangle
Sum of interior angles of triangle = 180° ($a^\circ + b^\circ + c^\circ = 180^\circ$) Exterior angle is equal to sum of opposite interior angles. ($d^\circ = a^\circ + b^\circ$)		Angle sum of triangle Exterior angle equal sum of opposite interior angle of triangle

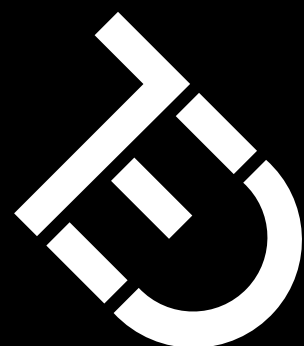
If a triangle's sides satisfies Pythagoras theorem, then it is a right triangle (CONVERSE of Pythagoras theorem)



Geometric Properties - Circles

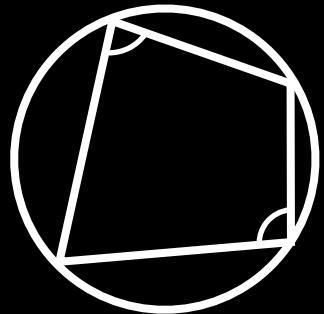
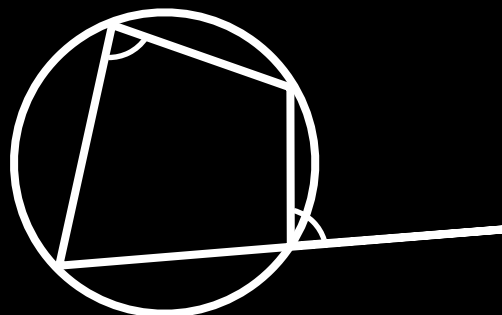
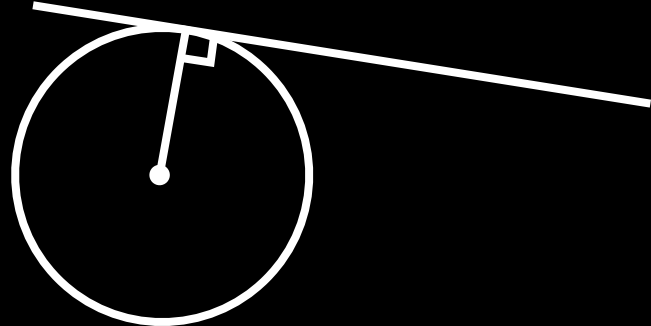
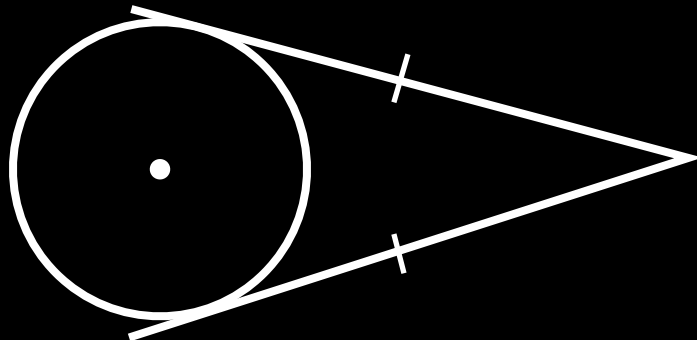
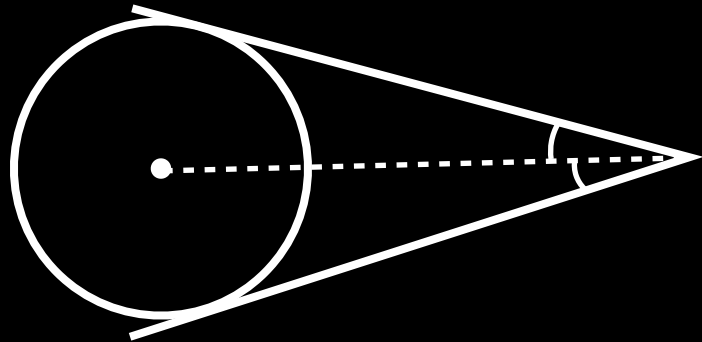
basics

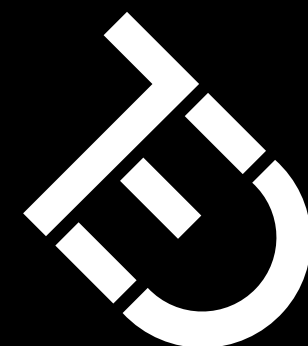
Property	Diagram	Abbreviation (what you write)
Perpendicular bisector of chord passes through centre		Perpendicular bisector of chord
Equal chords are equidistant from centre		Equal chords are equidistant to centre
Angle at centre is twice the angle at circumference		Angle at centre is twice angle at circumference
Angle in a semicircle is a right angle		Right angle in semicircle
Angles in the same segment are equal		Angles in same segment



Geometric Properties - Circles

basics

Property	Diagram	Abbreviation (what you write)
Angles in opposite segments add to 180°		Angles in opposite segments
Exterior angle of cyclic quadrilateral and interior opposite angle is equal		Exterior angle of cyclic quadrilateral
Tangent is perpendicular to radius		Tangent is perpendicular to radius
Tangents from external point are equal in length		Tangents from external point
Line joining external point to centre bisects angle between tangents		



Congruency and Similarity

basics

Congruency

SSS	$ABC \equiv XYZ$
SAS	Invalid Tests
SAA	AAA
ASA	SSA
RHS	(a type of SSA, where angle is 90°)

write congruency test name when writing $ABC \equiv XYZ$

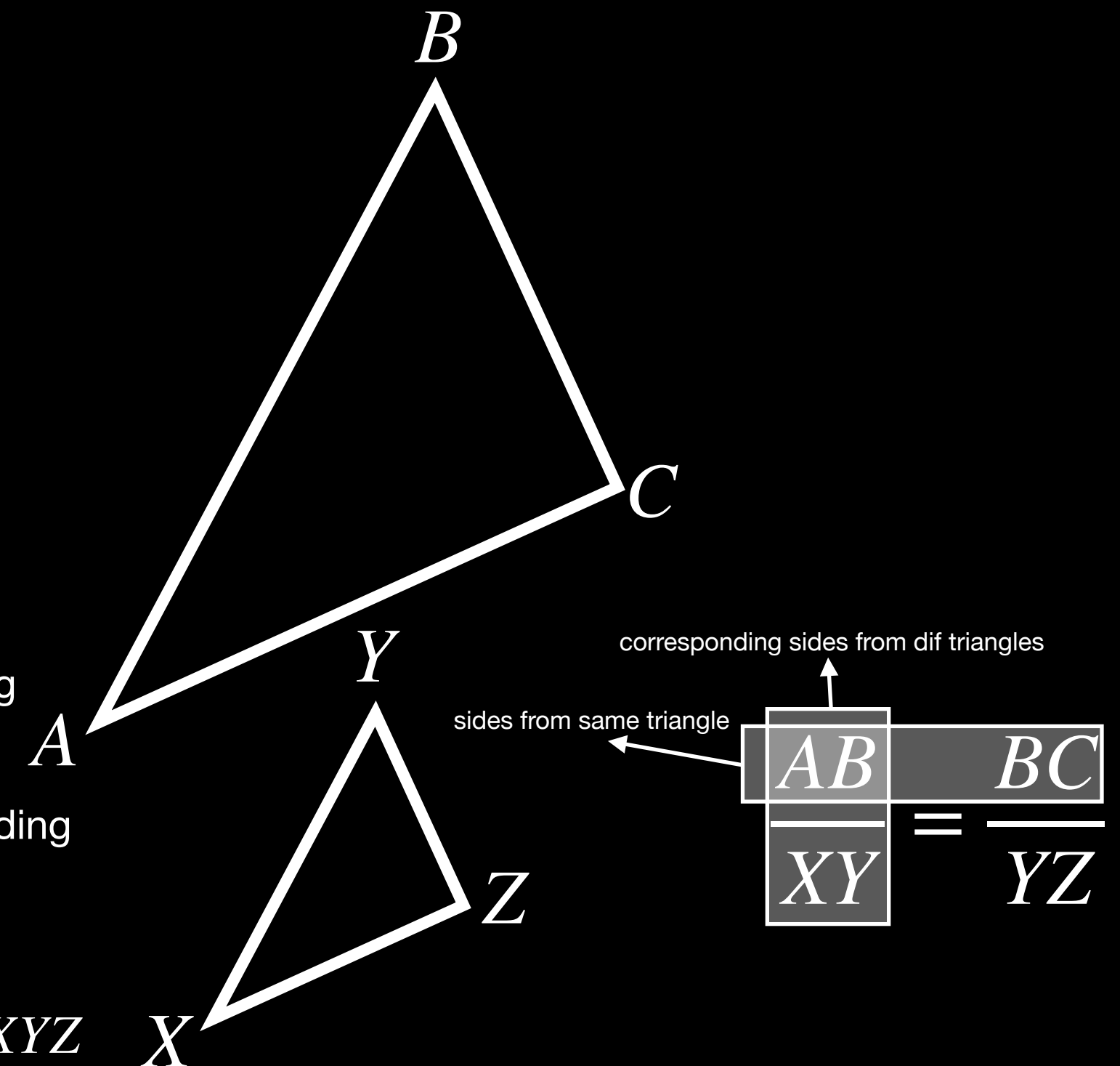
Similarity

SSS
SAS
AA

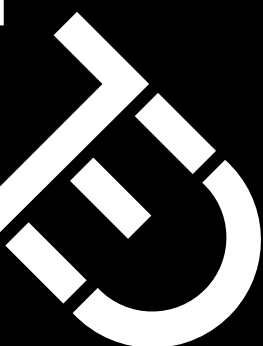
for tests with sides, check that corresponding sides have the same ratio

A has to similar to X, i.e corresponding

triangle ABC is similar to triangle XYZ



no need to write similarity test name

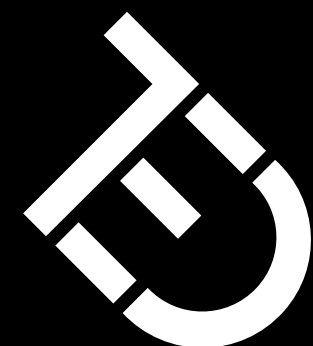


Midpoint & Tangent-Chord Theorem

new (extra fresh)

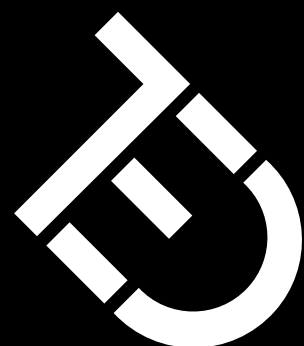
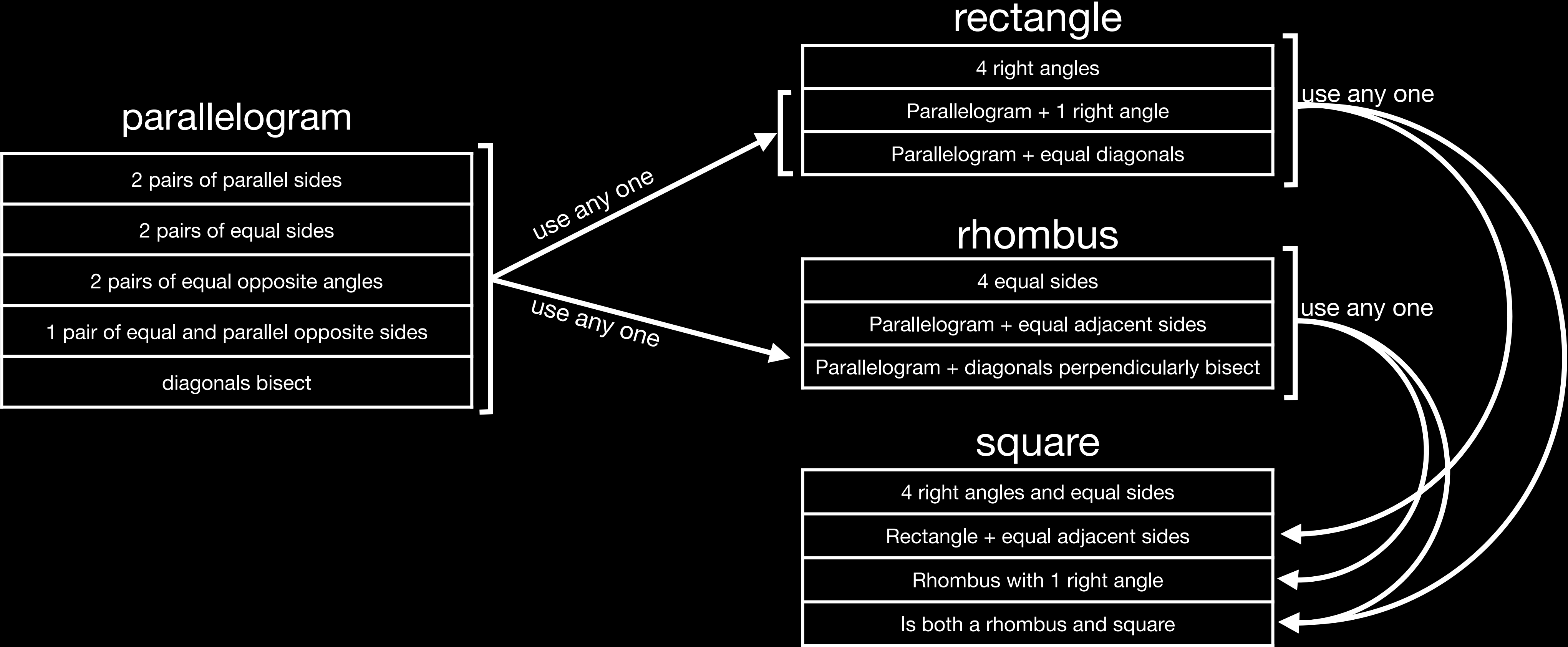
Likewise, if you find $XY \parallel BC$ AND $2XY = BC$,
X is a midpoint of AB OR Y is a midpoint
of AC (converse of midpoint theorem)

Property	Diagram	Abbreviation (what you write)
In a triangle ABC, if the midpoints of AB and AC are X and Y respectively, then $XY \parallel BC$ and $2XY = BC$		As X is a midpoint of AB and Y is a midpoint of AC, $XY \parallel BC$ OR $2XY = BC$ (midpoint theorem)
Angle between a tangent and chord is equal to angle subtended by the chord in the alternate segment		Tangent-chord theorem



Proving Quadrilaterals

new (extra fresh)

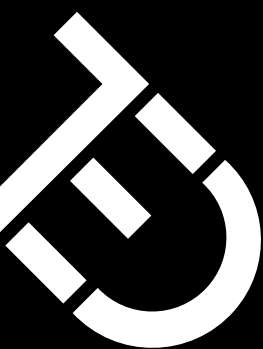
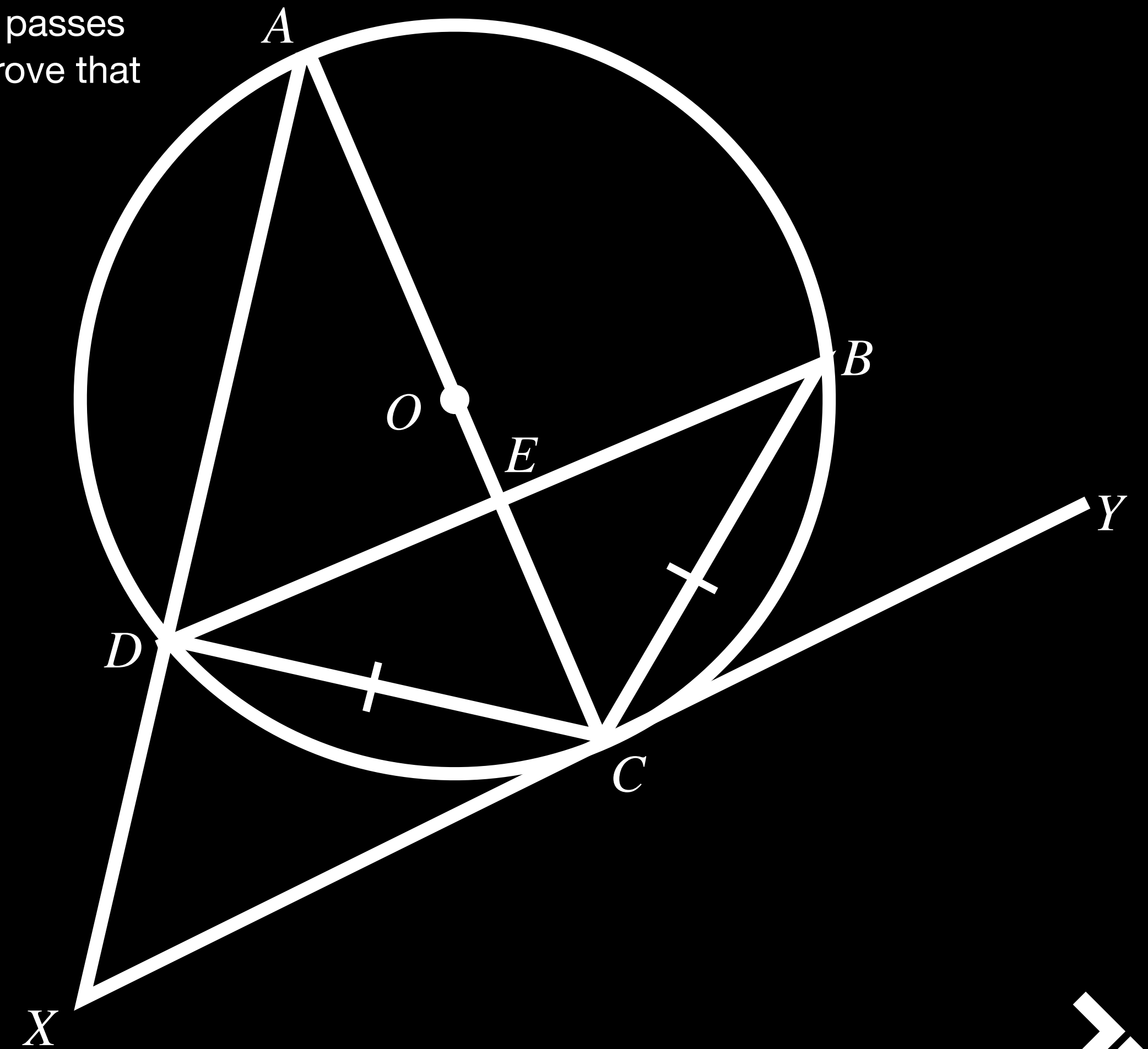


Sample Questions

saampull

The line XY is tangent to the circle at C . Line AX also intersects the circle at D , line AC passes through the centre of the circle O and line BD intersects AC at E . Given that $BC=CD$, prove that

- a) BD is parallel to XY .
- b) triangle ADC is similar to triangle DEC
- c) Show that $DC = \sqrt{AC \times EC}$



Sample Questions

saampull

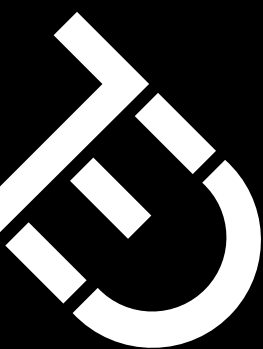
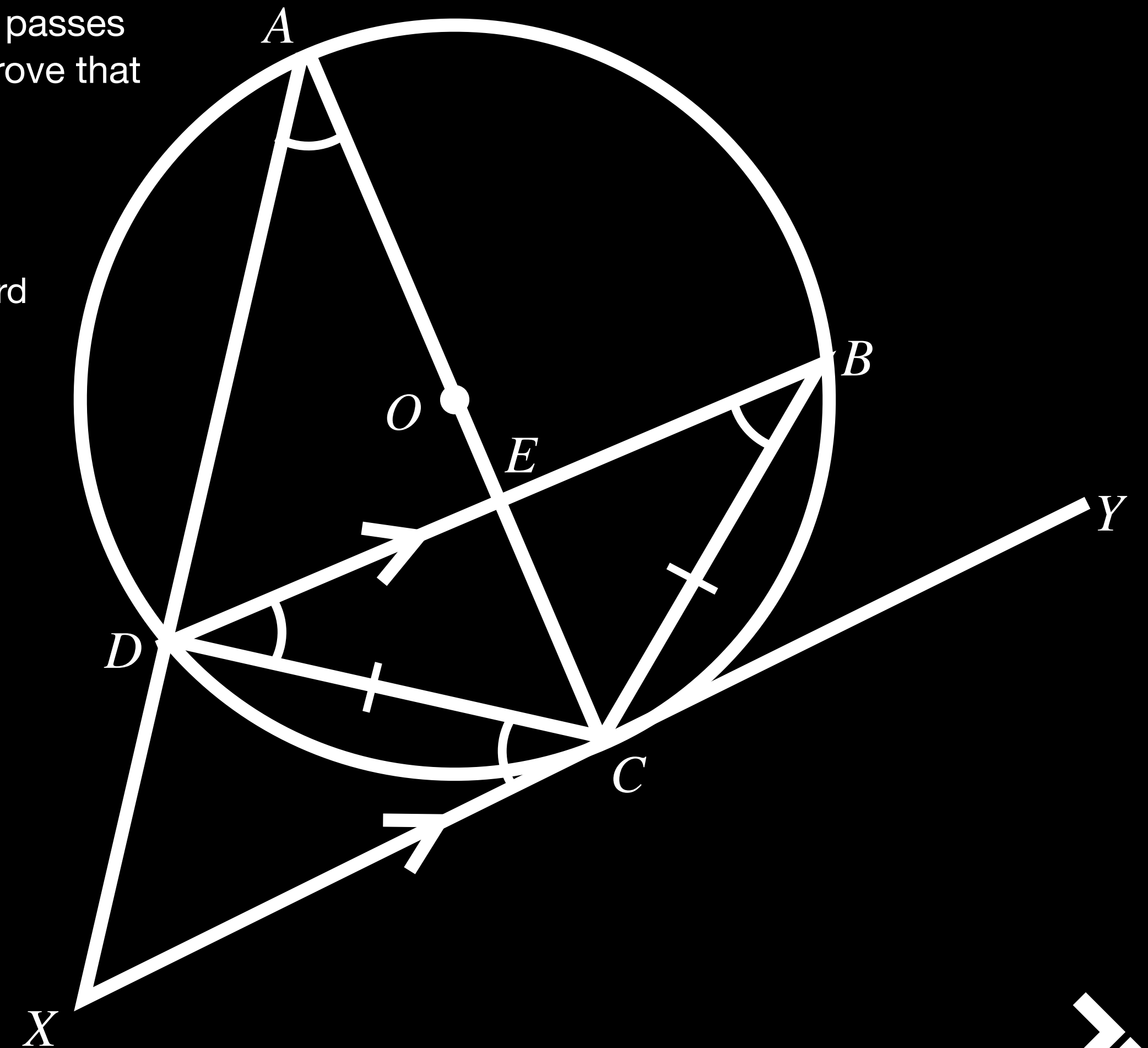
The line XY is tangent to the circle at C . Line AX also intersects the circle at D , line AC passes through the centre of the circle O and line BD intersects AC at E . Given that $BC=CD$, prove that

- a) BD is parallel to XY .
- b) triangle ADC is similar to triangle DEC
- c) Show that $DC = \sqrt{AC \times EC}$

For am plane geometry qns, you will almost always have to use tangent chord theorem/midpoint theorem

a)

$\angle DCX = \angle DAC$ (tangent chord theorem) ←
 $\angle DAC = \angle DBC$ (angles in same segment)
 $\angle DBC = \angle BDC$ (base isoscles triangle angles)
 $\therefore \angle DCX = \angle BDC$
Thus, by alternate angles, $BD \parallel XY$



Sample Questions

saampull

The line XY is tangent to the circle at C . Line AX also intersects the circle at D , line AC passes through the centre of the circle O and line BD intersects AC at E . Given that $BC=CD$, prove that

- a) BD is parallel to XY .
- b) triangle ADC is similar to triangle DEC
- c) Show that $DC = \sqrt{AC \times EC}$

b)

$$\angle ECD = \angle ACD \text{ (common)}$$

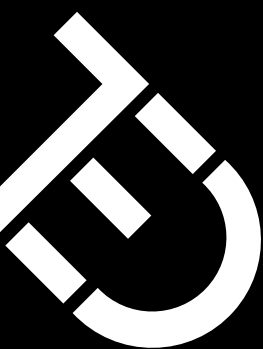
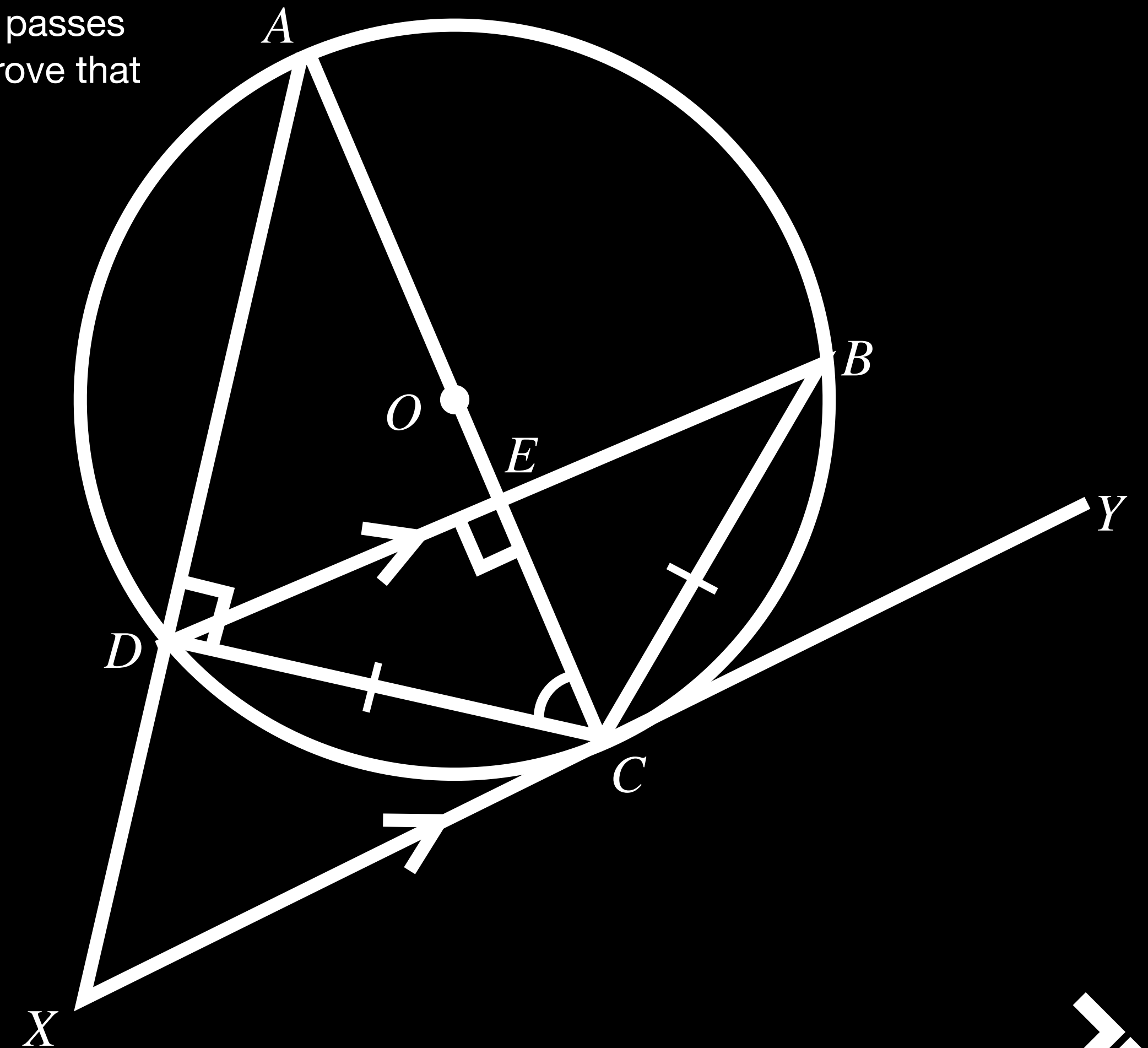
$$\angle ADC = 90^\circ \text{ (right angle in semicircle)}$$

$$\angle ACY = 90^\circ \text{ (tangent perpendicular to radius)}$$

$$= \angle DEC \text{ (alternate angles, } BD \parallel XY)$$

$$\therefore \angle ADC = \angle DEC$$

Thus, $\triangle ADC$ is similar to $\triangle DEC$.



Sample Questions

saampull

The line XY is tangent to the circle at C. Line AX also intersects the circle at D, line AC passes through the centre of the circle O and line BD intersects AC at E. Given that $BC=CD$, prove that

- a) BD is parallel to XY.
- b) triangle ADC is similar to triangle DEC
- c) Show that $DC = \sqrt{AC \times EC}$

c)

As $\triangle ADC$ is similar to $\triangle DEC$,

$$\frac{DC}{AC} = \frac{EC}{DC}$$

$$DC^2 = EC \times AC$$

$$DC = \sqrt{AC \times EC}$$

For similarity equation questions, its best to work backwards:

We know that the equation comes from two ratios of sides that are cross multiplied, thus,

$$DC = \sqrt{AC \times EC}$$

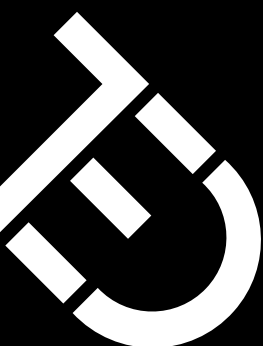
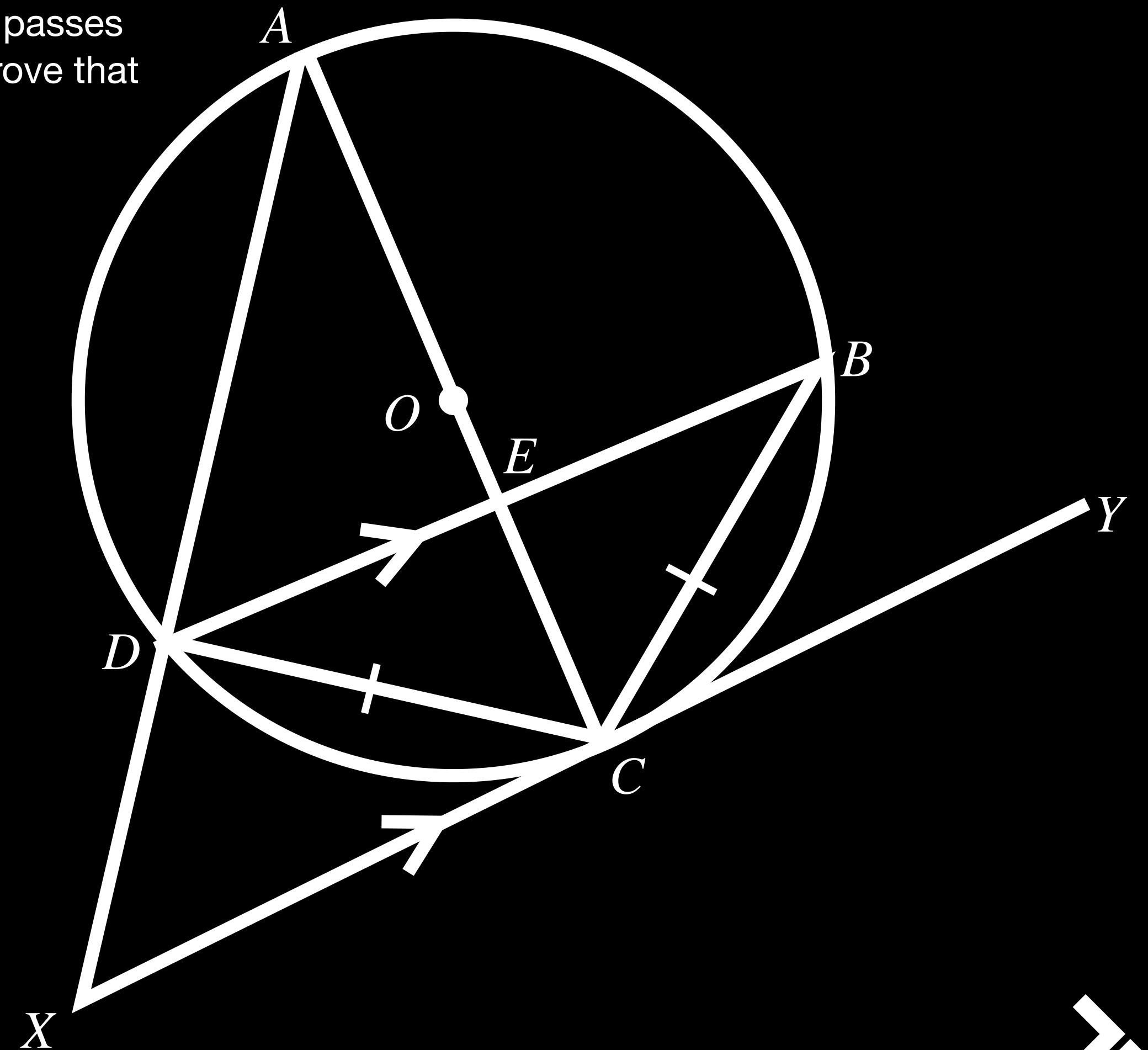
$DC^2 = AC \times EC$ (eqn where terms are only multiplied, thus before this ratios are formed)

We know DC will be the denominator of one of the ratios when we reverse the cross multiplication, so we must find which sides is corresponding to it (that side will be the numerator)

$$\frac{DC}{AC} = \frac{EC}{DC}$$

In this case, both sides are

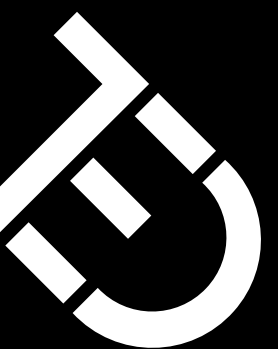
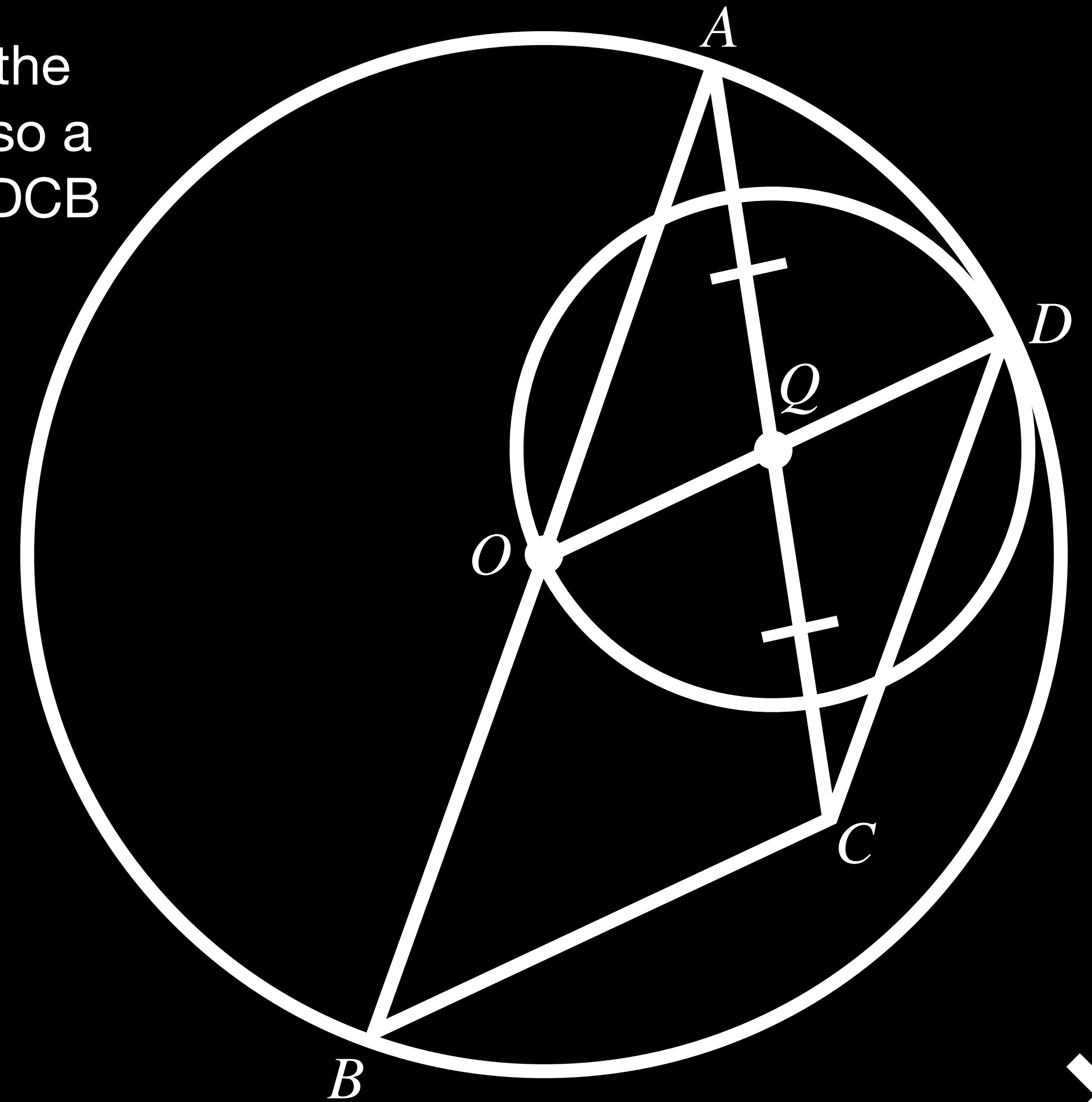
corresponding to DC, so it does not matter which way the ratio is written



Sample Questions

saampull

The straight line AB is a diameter of the circle with center O and the straight line OD is a diameter of the circle with center Q . AC is also a straight line that passes through Q , where $AQ = CQ$. Prove that $ODCB$ is a rhombus.



Sample Questions

saampull

The straight line AB is a diameter of the circle with center O and the straight line OD is a diameter of the circle with center Q. AC is also a straight line that passes through Q, where $AQ=CQ$. Prove that ODCB is a rhombus.

As AB is a diameter, O is the midpoint of AB

As $AQ=CQ$, Q is the midpoint of AC

$\therefore OD \parallel BC$ and $2OQ = BC$ (midpoint theorem)

As OD is a diameter, Q is the midpoint of OD

$\therefore OQ = DQ$

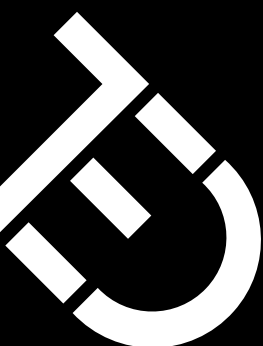
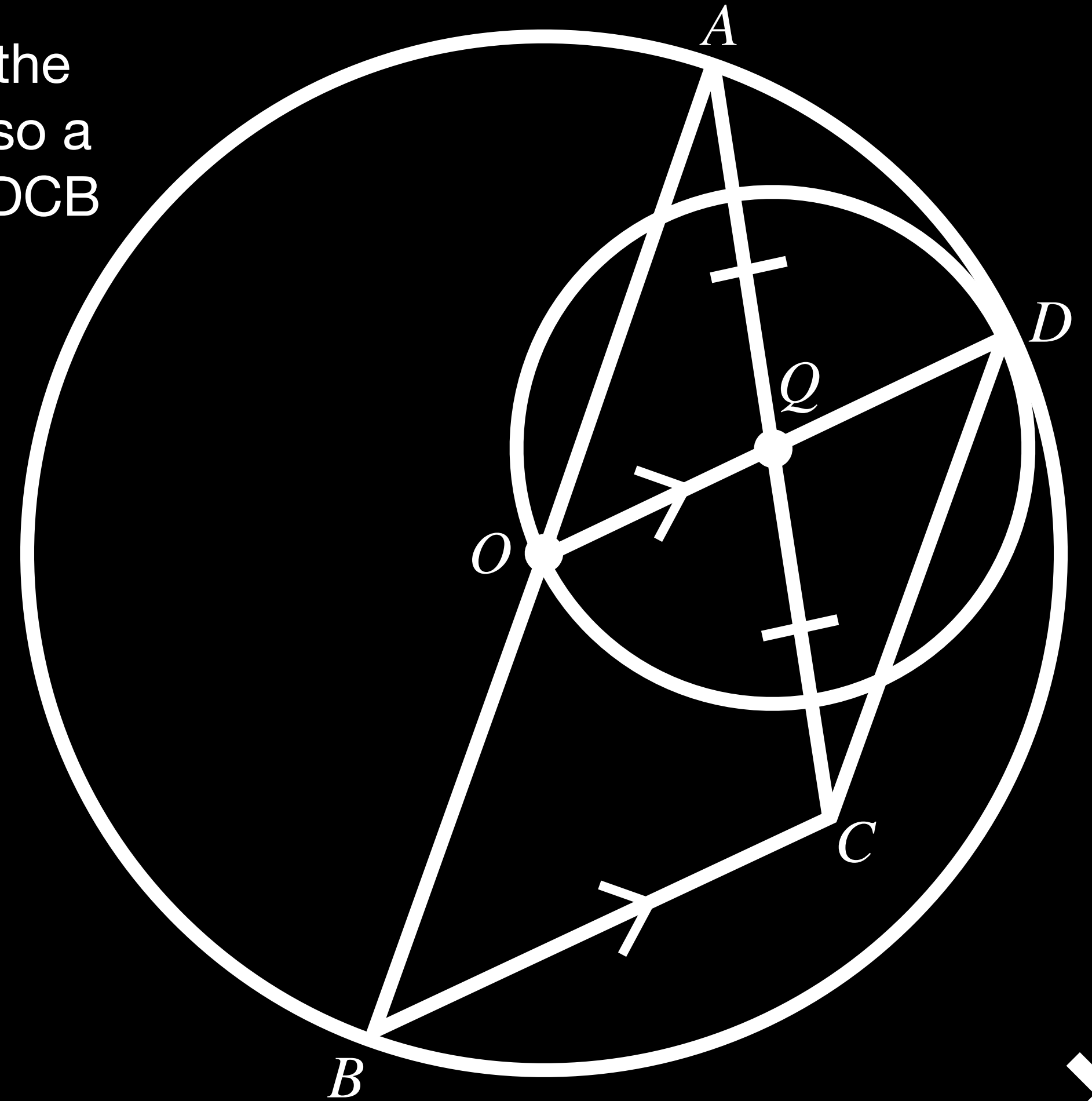
Thus, $2OQ = OQ + DQ$

$$2OQ = OD$$

$$BC = OD$$

As OD and OB are radii of the circle with centre O, $OD = OB$

$\therefore$ As OD and BC are equal and parallel sides,
and adjacent sides OD and OB are equal,
ODBC is a rhombus.



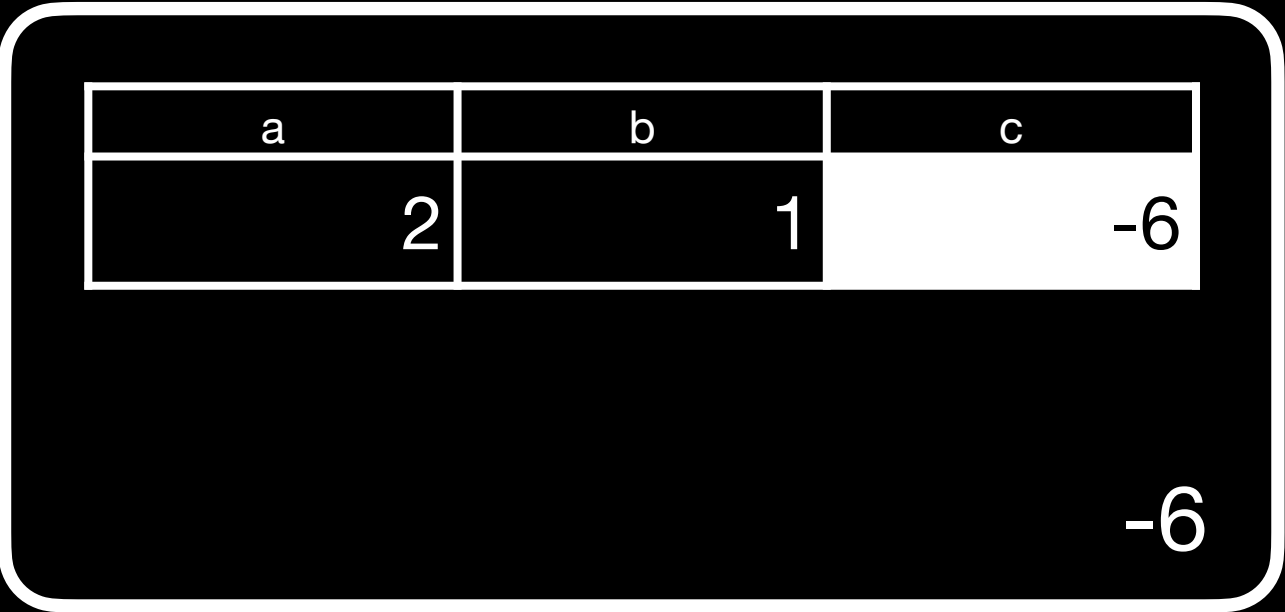
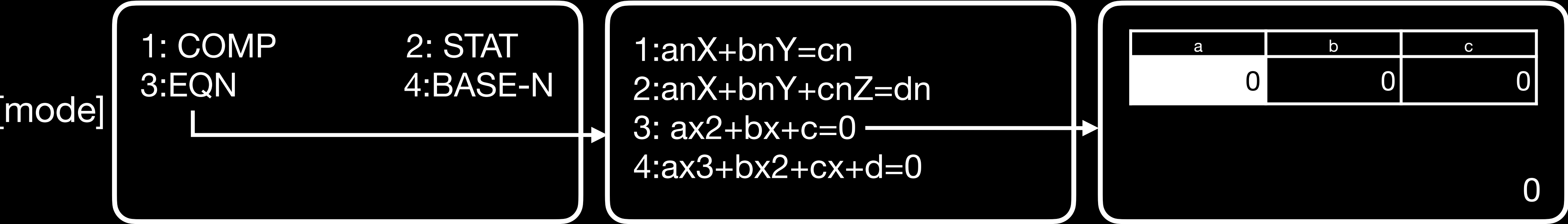
Exam Techniques & Answer Presentation

Quick Factorisation

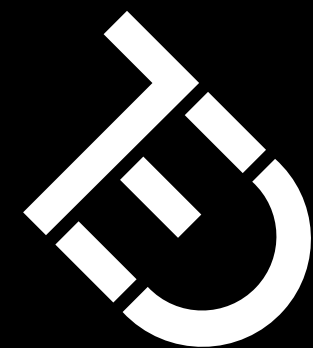
quick++

When dealing with quadratics(and also cubics), your calculator can help to evaluate the solutions of such equations like so

[mode] —> [3](or whatever number is under “equation”)—>[3](aka pick polynomial of desired degree)



Enter the coefficients of each term by entering them and pressing equals [=] to enter the next coefficient.
For example, these coefficients refer to the eqn: $2x^2 + x - 6 = 0$



Quick Factorisation

quick++

After entering the coefficients of the equation, pressing equals will show you the solutions of the equation

a	b	c
2	1	-6

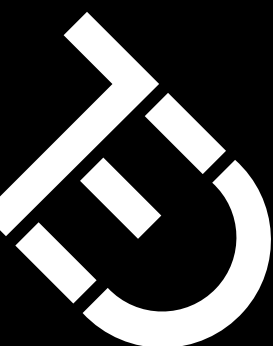
-6

X1= $\frac{3}{2}$

X2= -2

Thus, the solution of the eqn $2x^2 + x - 6 = 0$ has the solutions -2 and 1.5. However, when presenting solutions of quadratics, you must either write the quadratic formula or show that you have factorised the quadratic.

If the solutions of the quadratic are rational, it is easier to write the factorised form instead of the quadratic formula.



Quick Factorisation

quick++

X1=

$\frac{3}{2}$

X2=

-2

We know the second solution is an integer, thus we just need to flip the sign to write its corresponding factor $(x + 2)$

For fractions, we cannot write $(x - 1.5)$ as this would not be possible by factorisation. Instead, we make the coefficient of x be the denominator $[3x]$ and then subtract that by the numerator to get $3x - 2$ (or let $x = 1.5$ and make one side equal to 0)

Thus, the factorisation of the quadratic is $(x + 2)(3x - 2)$.

This allows us to write our working as:

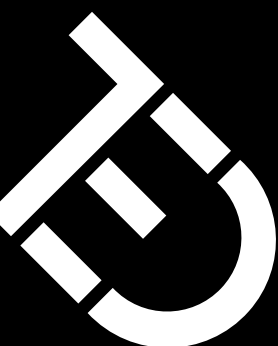
$$\begin{aligned} 2x^2 + x - 6 &= 0 \\ (x + 2)(3x - 2) &= 0 \\ x &= -2, 1.5 \end{aligned}$$

****note:** quadratics like $4x^2 + 2x - 12 = 0$ have the same roots as $2x^2 + x - 6 = 0$, however, the factorised forms are not the same. Thus, make sure to remove any common terms as so:

$$4x^2 + 2x - 12 = 2(2x^2 + x - 6)$$

then factorise the inner quadratic:

$$2(2x^2 + x - 6) = 2(x + 2)(3x - 2)$$



Quick Factorisation

quick++

X1=

$\frac{3}{2}$

X2=

-2

X3=

$-\frac{1}{2}$

For factorising cubics, we only need one integer solution to “fake our guess”, in this case we let $x = -2$ and divide the cubic by the respective factor of $(x+2)$

$$f(x) = 4x^3 + 4x^2 - 11x - 6$$

$$\text{Let } x = -2$$

$$f(-2) = 4(-2)^3 + 4(-2)^2 - 11(-2) - 6$$

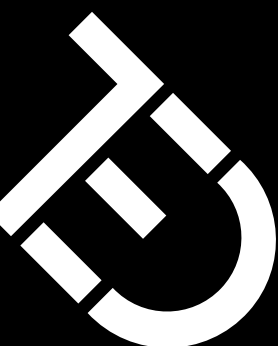
$$f(-2) = 0$$

$\therefore (x + 2)$ is a factor of $f(x)$

$$f(x) = 4x^3 + 4x^2 - 11x - 6$$

$$f(x) = (2x + 1)(2x^2 + x - 6)$$

$$f(x) = (x + 2)(2x - 3)(2x + 1)$$



Quick Factorisation

quick++

a	b	c
1	2	5

5

X1=

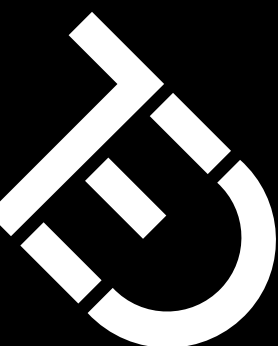
-1+2i

X2=

-1-2i

If your calculator outputs solutions with “ i ”, it means your equation has no real solutions, so you should check back that your equation is right/you inputted the right coefficients

**different calculator models will have different ways of changing to factorisation mode, refer to calculator manual if necessary



Time Conversion

t

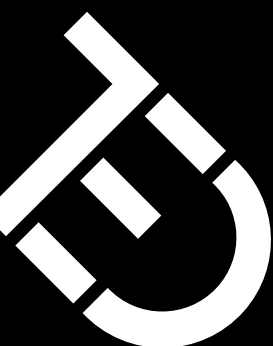
4.23°

4.23°

4° 13' 48"

Your calculator can convert hours into hours, minutes and seconds, by using the [° ' "] button.

Type in the number of hours in decimals then followed by the [° ' "] button to type out a °. The calculator will output the time in hours° minutes' seconds" (so for example, the time shown on the calculator is 4h, 13min and 48 seconds)



Substitution For Checking

Czech Republic

Almost all algebraic questions can be checked via substitution, where you replace variables with your answer/ arbitrary values to let your calculator check if your answer is right or wrong

Simplify $\frac{3^{x+2}}{(2^x)(6^x)}$ in positive index form

$$\begin{aligned}\frac{3^{x+2}}{(2^x)(6^x)} &= \frac{(3^x)(3^2)}{(2^x)(6^x)} \\ &= \frac{9(3^x)}{12^x} \\ &= \frac{9}{4^x}\end{aligned}$$

checking:

Let $x = 3$

$$\begin{aligned}\frac{3^{x+2}}{(2^x)(6^x)} &= \frac{3^{3+2}}{(2^3)(6^3)} = \frac{9}{64} \\ \frac{9}{4^x} &= \frac{9}{4^3} = \frac{9}{64}\end{aligned}$$

checking does not need to be written down, just type in your calculator

Solve $\log_x(3) + \log_3(9) = 0$

$$\frac{\log_3(3)}{\log_3(x)} + 2 = 0$$

$$\frac{1}{\log_3(x)} + 2 = 0$$

$$1 + 2\log_3(x) = 0$$

$$\log_3(x) = -0.5$$

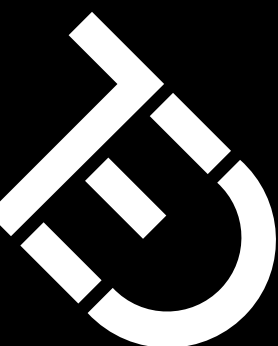
$$x = 3^{-0.5}$$

$$x = 0.577(3.s.f)$$

checking:

$\log_x(3) + \log_3(9)$ for logs that isn't base 10 or e, you have to convert base to check

$$\begin{aligned}&= \frac{\lg(3)}{\lg(x)} + \frac{\lg(9)}{\lg(3)} \\ &= \frac{\lg(3)}{\lg(3^{-0.5})} + \frac{\lg(9)}{\lg(3)} = 0\end{aligned}$$



Substitution For Checking

Czech Republic

Make a the subject of $ac^2 = b^2c - ab$

$$ac^2 + ab = b^2c$$

$$a(c^2 + b) = b^2c$$

$$a = \frac{b^2c}{c^2 + b}$$

checking:

Let $b=2, c=3$

$$a = \frac{(2)^2(3)}{(3)^2 + (2)}$$

$$a = \frac{12}{11}$$

$$ac^2 + ab = b^2c$$

$$LHS = \frac{12}{11}(3)^2 + \frac{12}{11}(2)$$

$$LHS = 12$$

$$RHS = (2)^2(3)$$

$$RHS = 12$$

Find the range of values of k where $kx^2 + 4x - 2$ has real sols.

$$kx^2 + 4x - 2$$

$$\text{Discriminant} \geq 0$$

$$4^2 - 4(k)(-2) \geq 0$$

$$16 + 8k \geq 0$$

$$8k \geq -16$$

$$k \geq -2$$

checking:

let $k=-2$

$$4^2 - 4(-2)(-2) = 0$$

let $k=2$

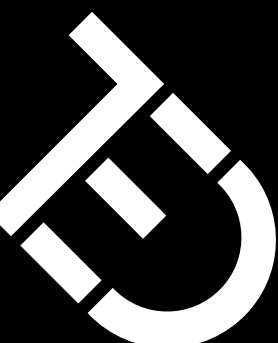
$$4^2 - 4(2)(-2) = 32$$

let $k=-4$

$$4^2 - 4(-4)(-2) = -16$$

check when k is equal, greater or less than the other side of the inequality
check that all substitutions follow inequality

checking does not need to be written down, just type in your calculator



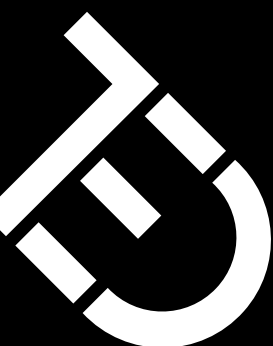
AM Specific Tips

Czech Republic

DO NOT forget EM formulas (e.g. sine/cosine rule, area of triangle with sine, mensuration and radian measure formulas), as they are commonly used (e.g. maximising area of cone), and are not in the formula sheet.

Some questions in AM do not need strictly AM to do, and EM may be more efficient (e.g using complete the square for quadratics to find maximum is more efficient than using derivatives)

AM concepts are strictly tested, thus if you don't know what concepts/theorems to use, check what strictly AM concepts have you not used (e.g. tangent chord theorem is strictly in AM, thus is commonly tested to check if you can use tangent chord theorem, so if you haven't used tangent chord theorem yet, you can see where to use it)



Answer Presentation

p

Labelling Calculations

Make sure that your calculations are labelled with what the value is/why you are doing said calculation: e.g. if qn asks for duration, write $\text{Duration} = a - b = c$, instead of just $a - b = c$

Rejecting answers

Make sure to write why certain answers are rejected, either if the value is strictly positive or negative, or if having the answer be 0 is illogical, or if the answer does not give any real solutions: e.g. For $x^2 - 4 = 0$ where $x > 0$, $x = \sqrt{4}$, not $\pm\sqrt{4}$.

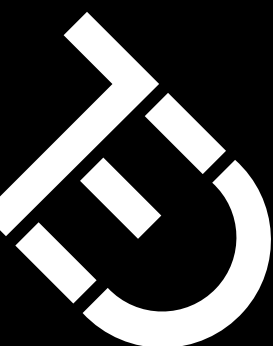
Show/Prove/Explain Questions

Questions that need you to show or prove a statement need you to agree with the statement without using the statement you need to prove: e.g. if you need to prove that $x = 3$, you cannot use that $x = 3$ to show that $x = 3$

Explain questions need you to state your reasoning for a statement (e.g. As 0.5 cycles occur from 0 to π , period = 2π)

Accuracy & Precision

REMEMBER: 3.s.f for final, 5.s.f for non exact intermediate, 1.d.p for degrees, other precisions used when asked



“Using math concepts is like peeling an onion: you have to deal with each layer at a time, and if it is too big you will start crying.”

me (Anonymous professor)

kah